

MetaTrader 5 trading platform

Account: 11856, Smith John, USD, 1:100

Overview Personal **Account** Security

Group: real\real

Color: ☐ None

Leverage: 1 : 100

Bank account:

Agent account:

☒ Enable this account

☒ Enable trading

☒ Enable to change password

☒ Enable trailing stops

☒ Enable one-time password

☒ Enable trading by Expert Advisors

☒ Enable Web API connections

☒ Enable daily reports

☐ Change password at next login

Trade accounts:

Gateway ID	Account
1, MetaTrader 4 Gateway	120002366

Synchronize

Add

Edit

Delete

Update Cancel Help

MetaTrader 5 trading platform

The ID is used as a magic number for all client orders (positions) passed to the external system. When a client's market transaction is sent to the MetaTrader 4 external server, the gateway additionally sends the command to close all opposite positions (Multiple close by) with the same magic number on the same symbol. Thus, only one resulting position remains at the external system for a single symbol.

MetaTrader 5 Trading Platform / Platform Components / Gateways / Integral

MetaTrader 5 Gateway to Integral

MetaTrader 5 Gateway to Integral is a simple, fast and secure integration solution for brokers. The gateway provides liquidity when working in MetaTrader 5. Integral ECN of Integral Development Corp. is one of the largest Electronic Communication Networks allowing to trade commodities. All orders entered in ECN are transferred to the unified requests database. The system selects appropriate orders automatically executing opposite orders with matching parameters (symbol, price etc.)

About Integral

Founded in 1993, Integral Development Corp. offers Integral multi-sided trading network (ECN, Electronic Communication Network) for foreign exchange. Integral technology unites various Forex market participants into the single trading network. Direct access to interbank liquidity via STP (Straight Through Processing) technology allows to maximize the speed of the participants orders execution. ECN Integral combines the liquidity of such financial institutions, as Citibank, Deutsche Bank, Bank of America, UBS, HSBC, NOMURA, RBS, BNP Paribas and many others.

Integral allows to trade more than 100 financial instruments including major currency pairs, cross rates, exotic currencies and precious metals.

Necessary Actions for Working with Integral

To be able to provide trading services using Integral, a brokerage company must first contact Integral Development Corp. for concluding the agreement. Select the closest Integral office:

- +1 (212) 252-2243 (North America),
- +44 203 514-2439 (UK, the EMEA region),
- +65 3158 0800 (Asia, Singapore)
- +852 80 903 855 (Hong Kong),
- +813 3242 6170 (Japan)

Or send an email to support@integral.com.

Further information can be requested on the official Integral website via a special form. After conclusion of an agreement, you will receive all necessary data for connection to the Integral server.

Order MetaTrader 5 Gateway to Integral

You can order the MetaTrader 5 Gateway to Integral through the "Buy" section of the technical support site. After you make the payment, the MetaTrader 5 platform

license will be automatically updated via the LiveUpdate system and you will be able to use MetaTrader 5 Integral Gateway without limitations. The executable module of the gateway is included into the platform package and can run in a demo mode, allowing you to perform 100 operations for a work session (until restart).

How the Gateway Works

MetaTrader 5 Gateway to Integral is a separate IntegralGateway64.exe module that uses the MetaTrader 5 Gateway API for operation.

The foundation of Integral's FX solutions is the FX Grid, a global inter-institutional connectivity and trading network, linking market making banks to Forex market participants. To connect to that network, the gateway uses FX Inside API, which is a special set of interfaces provided by Integral for direct access to FX Grid. The gateway operates as a mediator between two systems connecting Integral and MetaTrader 5 platform. All data between the Integral system and the gateway is transmitted using FIX protocol over the encrypted connection. The gateway sends encrypted FIX messages and returns them to the MetaTrader 5 platform using the MetaTrader 5 Gateway API.



Market Data

MetaTrader 5 Integral Gateway automatically imports all the necessary symbols and processes their properties. An administrator only needs to perform primary setup.

Price data is transmitted in real time. The gateway can narrow or widen prices: convert quotes in accordance according with the settings when sending them to the platform, and then perform the inverse conversion to the original state when sending trading operations to Integral. Detailed information on prices correction is available below.

Trading Operations

Orders are sent the the MetaTrader 5 Gateway to Integral in accordance with the configured routing rules. Processing of requests depends on the type of the order, as well as in the gateway configuration.

Order type	Execution
Market Order	Delivered directly to Integral as a market order.
Take Profit Buy Limit Sell Limit	Depends on <u>Limit Orders Coverage</u>: Limit Orders Coverage Mode=Gateway (default) Limit orders are processed on the side of Integral. Once a limit order has been placed by a client, an appropriate order is sent to Integral. Take Profit orders are processed the same way as in the Limit mode. The Integral system checks the availability of the required amount of funds to cover any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the operation is carried out. <u>Margin reservation</u> of the clients should be configured for the appropriate order types in case Limit orders are directly delivered to an external system. By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client.

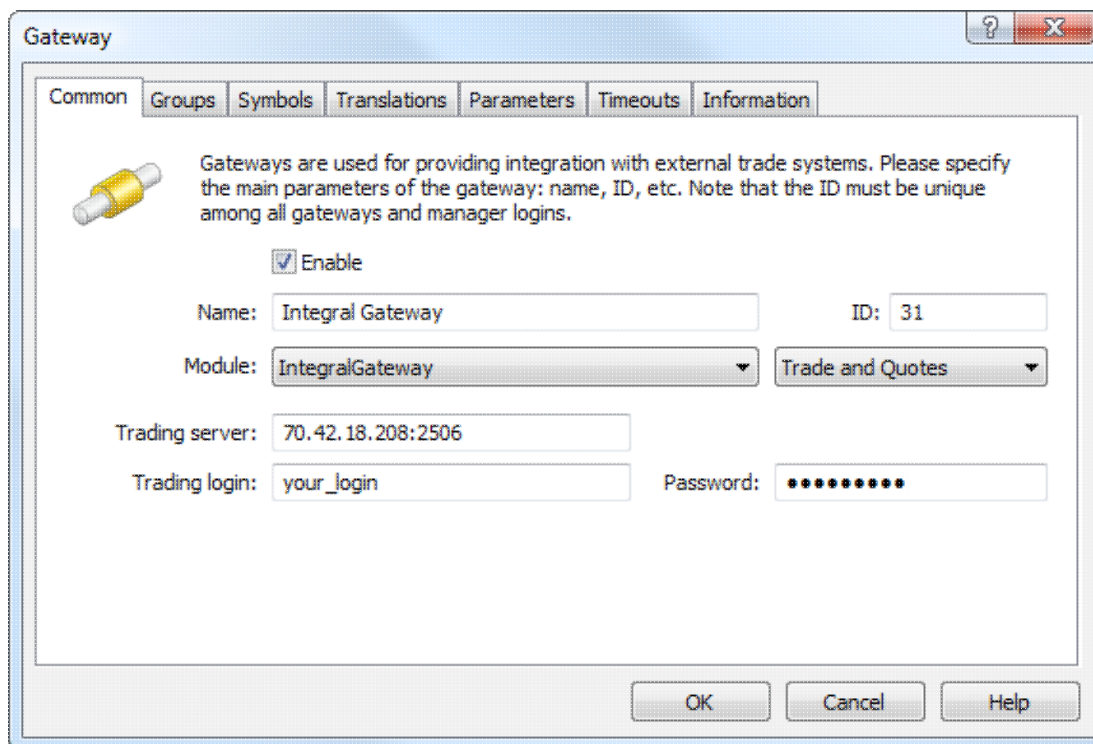
Order type	Execution
	Limit Orders Coverage Mode=Limit Limit Orders and Take Profit orders are processed on the side of the MetaTrader 5. Once a limit order is triggered, an equivalent limit order is sent to Integral. That order has a short action time specified in Limit Orders Coverage Timeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Integral. Thus, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5. A limit order with the price equal to a Take Profit level is sent to Integral at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Integral. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on MetaTrader 5 side. Limit mode allows to protect against slippage, as a limit order is sent to Integral system with a specified price rather than a market order for execution by the current price. Limit Orders Coverage Mode=Market Limit orders and Take Profit orders are processed on the MetaTrader 5 platform side. Once they trigger, an appropriate market order is sent to Integral.
Buy Stop	Depends on <u>Stop Orders Coverage</u> :
Sell Stop	Stop Orders Coverage=Y

MetaTrader 5 trading platform

Order type	Execution
	Delivered directly to Integral. Like with limit orders, margin reservation of the clients should also be configured for the appropriate order types on the MetaTrader 5 side in case Stop orders are directly delivered to Integral. Stop Orders Coverage=N Processed on the MetaTrader 5 platform side until their stop price is reached. After a stop order is activated, an appropriate market order will be sent to Integral system.
Buy Stop Limit Sell Stop Limit	Depends on Stop Orders Coverage: Stop Orders Coverage=Y Delivered directly to Integral. Like with limit orders, margin reservation of the clients should also be configured for the appropriate order types on the MetaTrader 5 side in case Stop-Limit orders are directly delivered to Integral. Stop Orders Coverage=N Processed on the MetaTrader 5 side. Upon order order activation, an appropriate limit order is created in MetaTrader 5, and this order is then processed in accordance with the value of the Limit Orders Coverage Mode parameter.
Stop Loss Stop Out	Processed on the MetaTrader 5 side. Once they trigger, an appropriate market order us sent to Integral.
• In case connection to Integral server is lost, the application will try to restore it repeatedly. In case pending orders have been executed at that, they will be updated in the MetaTrader 5 platform after connection is restored. • The Integral system allows a broker to cancel orders in case of connection loss. Integral system does not cancel orders by default in that case but, nevertheless, you should notify the Integral technical support about the necessity to disable that option for your account (pending orders should not be deleted in case of connection loss).	

Gateway Setup

To start working, add the new gateway configuration:



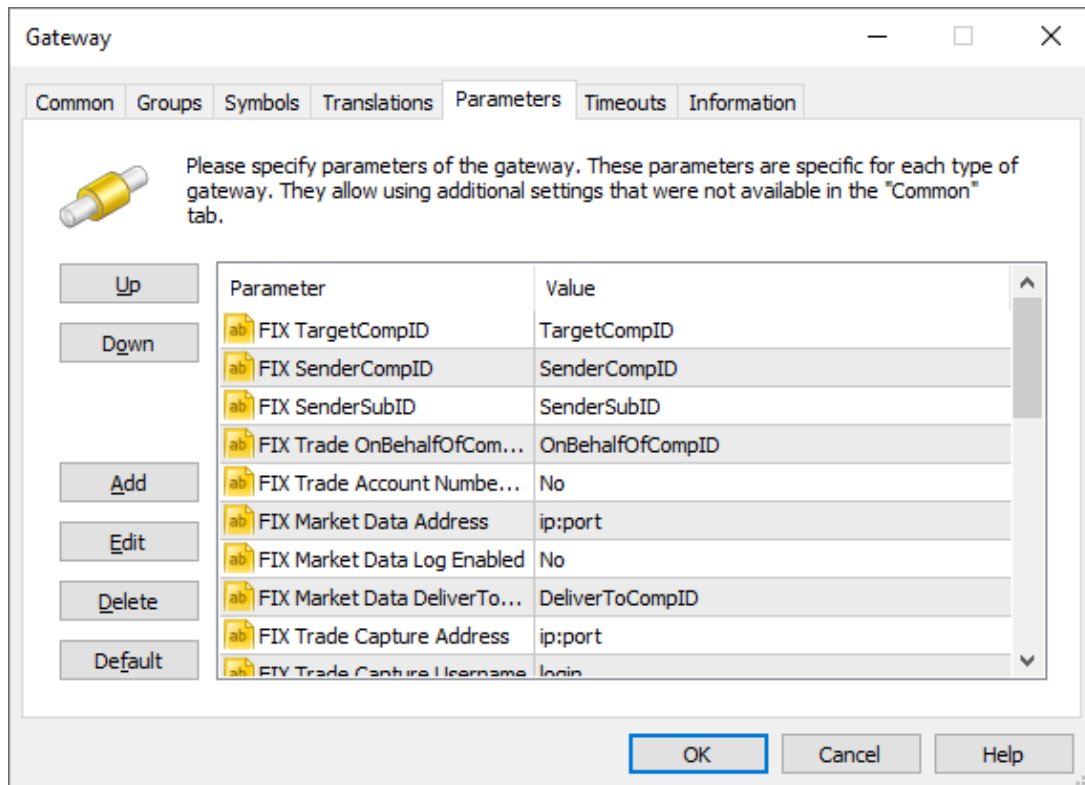
Set the following parameters on the "Common" tab:

- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Module** specify IntegralGateway64 and accept the default settings after the module selection.
- **Trading server** Integral server IP-address and the port, where trade requests are processed. This information is provided by Integral.
- **Trading login** Integral server connection login (equal to SenderCompID).
- **Password** password for connection to the Integral server (equal to SenderCompPasswd).

- ID value must be unique in the field of manager logins and gateway identifiers.
- The details for connection to the Integral server, where trade requests are processed, are provided during the agreement conclusion.

Other parameters are set similarly to other gateways. Default values are used in most cases.

Now, go to the "Parameters" tab.



Specify the following parameters values here:

- **FIX TargetCompID** a standard parameter of the FIX messages heading used for trading messages recipient identification. This parameter is provided by Integral as the value of TargetCompID.
- **FIX SenderSubID** a standard parameter of the FIX messages heading used for a trading participant (a legal entity) identification. Provided by Integral as the value of FIX_SenderSubID.
- **FIX SenderCompID** a standard parameter of the FIX messages heading used for a data sender identification. Provided by Integral as the value of senderCompID without the 'quote' or 'trader' prefixes. For example, if Integral has submitted to you sendercompID = quote.9898AKD.10, the value 9898AKD.10 should be used for FIX SenderCompID parameter.
- **FIX Trade OnBehalfOfCompID** is a standard parameter of the FIX messages heading used for a trade participant identification. Provided by Integral as the value of OnBehalfOfCompID. If the parameter value is specified, the gateway will additionally fill the OnBehalfOfCompID (115) tag in outgoing messages of the trading channel. An optional parameter.
- **FIX Trade Account Number Set** if you set "Yes" for this parameter, the gateway will additionally fill the OnBehalfOfSubID (116) tag in all outgoing FIX messages. The MetaTrader 5 login of the client who performed this operation is written in this tag. When default No is used, the tag is not filled.
- **FIX Trade Capture Address** IP address and port of the Trade Capture server, provided by Integral. The gateway uses a separate FIX connection to the Trade Capture (Drop Copy) Exchange service. This connection is used by the gateway to receive the stream of deals performed through other terminals (not MetaTrader 5). If this parameter is not filled, the gateway will not connect to

Trade Capture and will only work with the trading/quoting server of the exchange.

- **FIX Trade Capture Username** login for connection to the Trade Capture server.
- **FIX Trade Capture Password** password for connection to the Trade Capture server.
- **FIX Trade Capture TargetCompID** FIX messages heading standard parameter used for trading messages recipient identification. The parameter is provided by Integral as the TargetCompID value.
- **FIX Trade Capture SenderCompID** FIX messages heading standard parameter used for a data sender identification. Provided by Integral as the value of senderCompID without 'quote' or 'trader' prefixes.
- **FIX Trade Capture SenderSubID** FIX messages header standard parameter used for a trading participant (a legal entity) identification. Provided by Integral as the value of FIX_SenderSubID.
- **FIX Trade Capture Login MT** account number on the MetaTrader 5 platform side, to which all deals from Trade Capture will be transmitted.
- **FIX Trade Capture Stop Out Enabled** allow closing of positions upon reaching Stop Out. If set to "No" (default), positions opened via the gateway will not be closed automatically, if Stop Out is reached on the trading platform side. If a position is opened with FIX Trade Capture Stop Out Enabled = Yes, then closing by Stop Out can occur even if you change the parameter value to "No". It is only guaranteed that the ban to close will be effective for positions opened after changing the parameter value to "No".
- **FIX Trade Spread Set** transmit information about the spread difference values in the trader group to Integral. Such information can be required by the exchange. If set to "No" (default), this data is not transmitted. If set to "Yes", the gateway will transmit the current spread value in tag 7547 for each request. For market orders, the current symbol price will be additionally provided in tag 44.
- **FIX Market Data Address** IP address and port of the Integral server, from which the quotes are provided. By default, the gateways receives information about trading requests and price data from one server, which is specified in the "Trade server" field. By using the FIX Market Data Address parameter, you can configure the gateway to receive quotes in a separate stream. To connect to the quoting server you should use the same login and password, which are used for the trade request processing server.
- **FIX Market Data Log Enabled** if "YES" is set, the gateway will save to disk the full quoting connection log. This can be useful in operation debugging. The log is not saved by default (value "No").
- **FIX Market Data DeliverToCompID** indicates liquidity provider from which quotes should be requested. If the parameter is set, the specified value will be added to the DeliverToCompID (128) tag of FIX messages of the quoting session subscription. If the parameter is not set, "All" will be used for the DeliverToCompID tag, which means subscription to quotes from all available liquidity providers.
- **Limit Orders Coverage Mode** the mode of processing of Limit and Take Profit orders by the gateway. This parameter simplifies configuration of trade requests routing to the gateway, since there is no need to create a separate rule for routing the appropriate order types. Three processing modes are available:

- ◆ **Market** limit and Take Profit orders are processed on the MetaTrader 5 platform side. An appropriate market order is sent to Integral after an order has been triggered.
- ◆ **Limit** limit and Take Profit orders are processed on the MetaTrader 5 side.
Once a limit order is triggered, an equivalent limit order is sent to Integral. That order has a short action time specified in Limit Orders Coverage ModeTimeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Integral. Thus, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5.
A limit order with the price equal to a Take Profit level is sent to Integral at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Integral. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on the MetaTrader 5 side.
Limit mode allows to protect against slippage, as a limit order is sent to Integral system with a specified price rather than a market order for execution by the current price. In addition, this mode allows you not to reserve margin on a client's account before sending an order to Integral.
- ◆ **Gateway** limit orders are processed on the Integral side. Once a limit order has been placed by a client, an appropriate order is sent to Integral. Take Profit orders are processed the same way as in the Limit mode.
- **Stop Orders Coverage** Stop and Stop Limit order processing mode. In case of 'Y' value these order types will be delivered to Integral directly. In case of 'N' value the orders will be processed inside MetaTrader 5 platform until their stop price is reached. After a stop order is activated, an appropriate market order will be sent to Integral system. After a stop limit order has been activated, a limit order is created, which will be processed according to Limit Orders Coverage Mode parameter value.
- **Limit Orders Coverage Timeout** expiry of Limit Orders that are sent to Integral in the Limit mode. Specified in seconds. The default value is 5.
- **Week Time Begin** gateway operation start time on Sunday. The value is specified in HH:MM format for Eastern Standard Time (EST). For example, 2:00.
- **Week Time End** gateway operation end time in Friday. The value is specified in HH:MM format for Eastern Standard Time (EST). For example, 23:00.
- **Weekend Trading Enabled** allow the gateway to trade on weekends. The default value is "No", which means that trading begins on Sunday at "Week

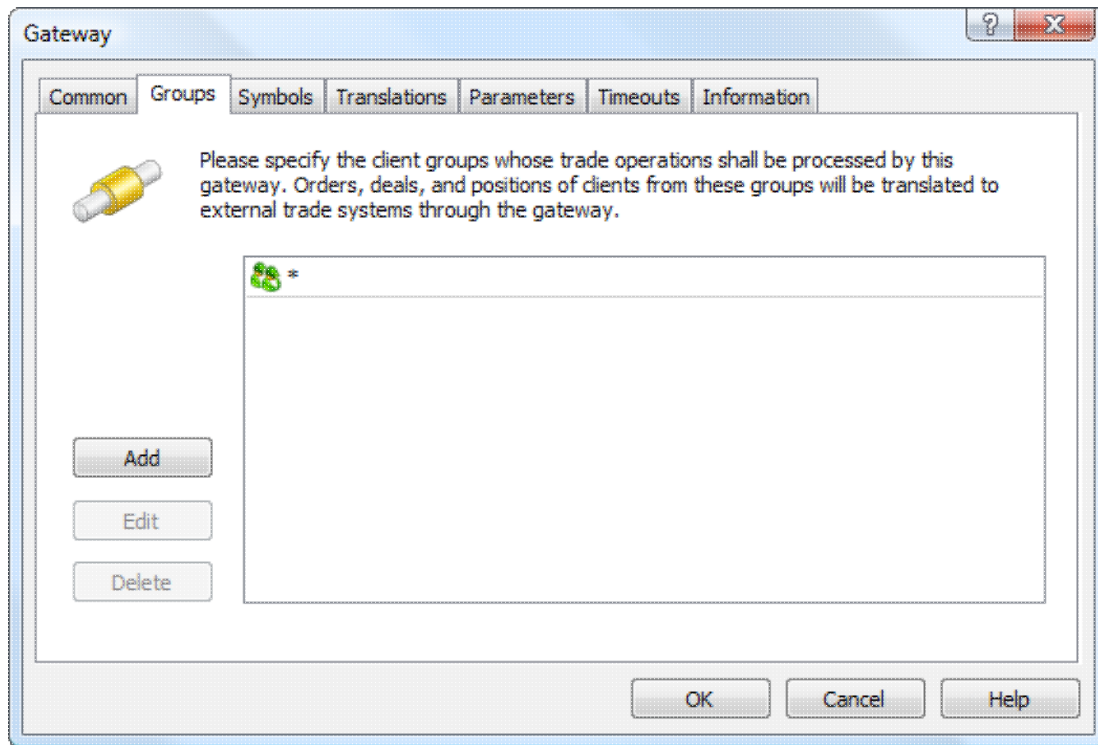
Time Begin" and ends on Friday at "Week Time End". If set to "Yes", the gateway will operate from "Week Time Begin" on Sunday until "Week Time End" on the next Sunday. To keep the gateway running 24/7, enable "Weekend Trading Enabled" and set "Week Time Begin" to a time earlier than "Week Time End".

- **Min Quantity Set** enables/disables minimum volume checks for symbols when executing orders in Integral. The default "No" value instructs Integral to execute orders using deals of any size. The "Yes" value informs Integral that orders can be filled using deals with the volume no less than the "Minimum volume" parameter in symbol settings on the trading platform side.
- **Slippage Allowable** maximum allowable slippage for market orders. The parameter corresponds to tag 211 in the FIX protocol. The default value is 0, i.e. no tag is passed.
- **Orders Full Fill Only** enables/disables the filling of tag 110 in the FIX protocol when forwarding all orders. The tag sets the minimum volume (equal to the requested volume) in which the order can be executed. The flag ensures the correct implementation of the Fill or Kill execution. Supported values are "Yes" and "No" (default).
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

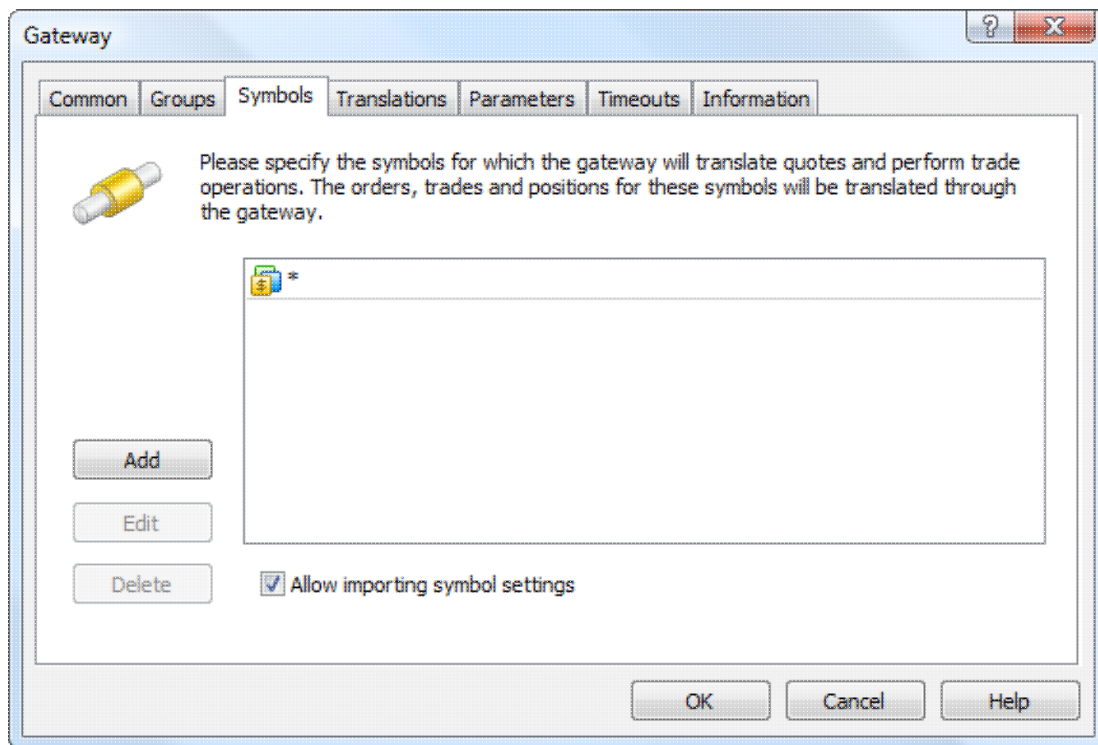
- In no circumstances it is allowed to change operations transfer mode while in operation. Changing the mode is allowed only after all client positions are closed.
- When Limit, Stop and/or Stop Limit orders are directly delivered to an external system, margin reservation of the clients should be configured for the appropriate order types.

The next stage is to specify the groups of the clients, whose requests will be processed via the MetaTrader 5 Integral Gateway. All groups are configured on the screenshot below, but you can configure groups according to your business logic.

MetaTrader 5 trading platform



Then configure the list of symbols, according to which the gateway will process trade operations and feed quotes.



MetaTrader 5 trading platform

Make sure to enable "Allow importing symbol settings" option. The symbols available to Integral will be imported to Symbols/Preliminary/Integral directory of the MetaTrader 5 platform. Besides, that will allow the gateway to manage the settings of the symbols used in trading via Integral.

- The symbols imported by the gateway are put to the "\Preliminary" symbols subgroup. All symbols have trading ability disabled. System administrator must relocate imported symbols to the proper subgroup and allow trading for them.
- After symbols are relocated and trading abilities are enabled, the main trading server must be restarted.
- In case the gateway transmits configuration for the symbol that is already present in the platform, the configuration is updated. In this case the symbol is not transferred and its trading ability is not turned off.
- In case some changes are implemented to the Depth of Market parameter of the symbol settings, Integral gateway and a history server must be restarted to let the changes take effect. In fact, restart is required after any change in the symbol settings.
- The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

Margin Setup

Margin reservation of the clients should be configured for the appropriate order types in case Limit, Stop and/or Stop Limit orders are directly transferred to an external system.

The external system checks sufficiency of the funds that are necessary to provide any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the operation is carried out.

By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client.

After an order has been transferred to the external system, MetaTrader 5 platform is not able to check the client's margin sufficiency any more. After the order has been executed in the external system, the gateway cannot ignore that fact. Therefore, the appropriate trading operation is performed in the platform.

Set non-zero coefficients for the orders directly transferred to the external trading system in symbol settings for the appropriate symbols to configure margin collection:

MetaTrader 5 trading platform

Symbol: EURUSD

Common Quotes Trade Execution Margin **Margin Rates** Swaps Sessions

 Please specify margin rates for different types of trade operations.

Liquidity margin rate:

Margin rates:

	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
 Buy	1.0000000	1.0000000	0.0000000	0.0000000
 Sell	1.0000000	1.0000000	0.0000000	0.0000000
Maintenance margin				
 Buy	0.0000000	0.0000000	0.0000000	0.0000000
 Sell	0.0000000	0.0000000	0.0000000	0.0000000

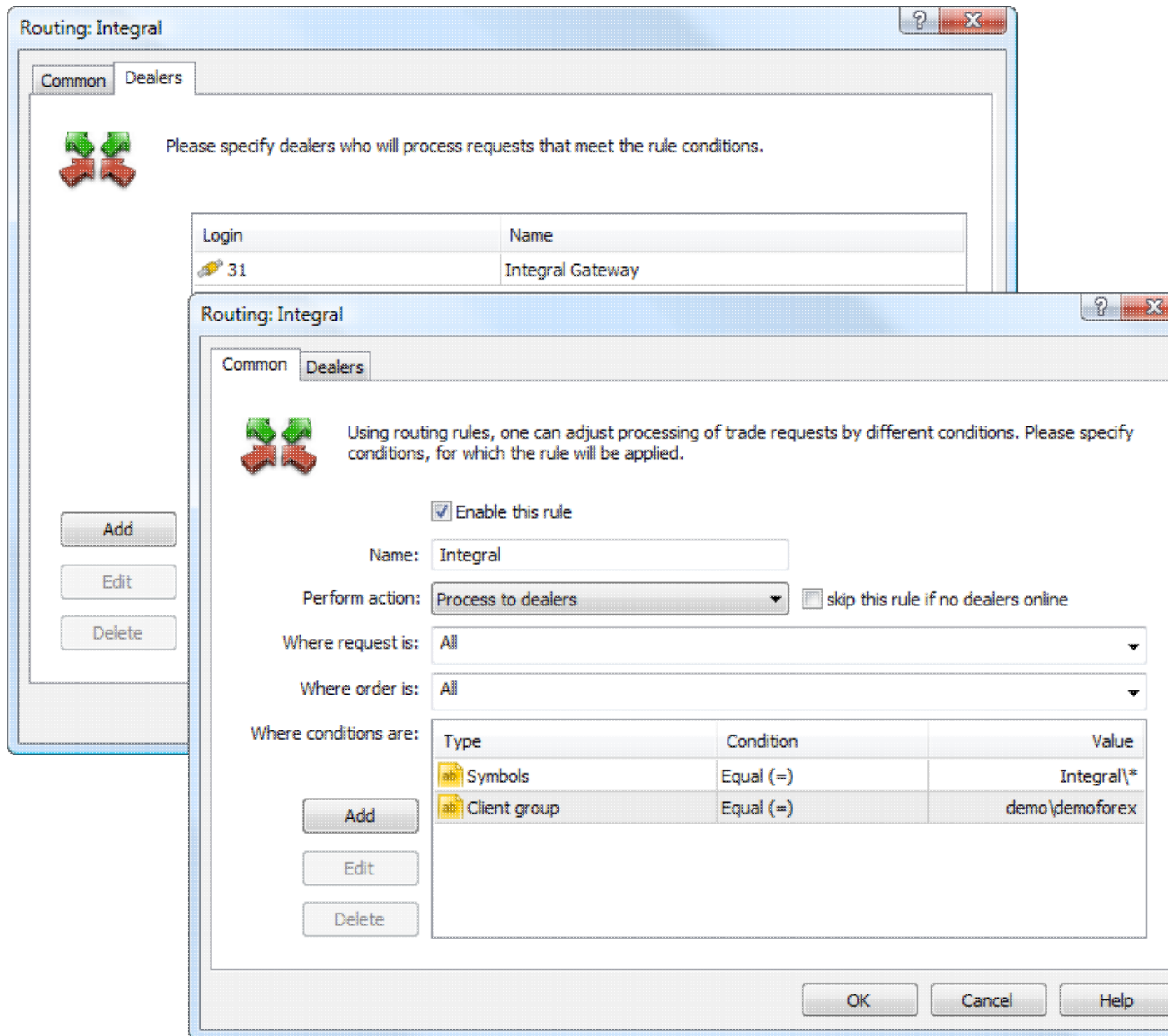
OK Cancel Help

Configuring trade requests routing

Configure the routing to let the clients requests to be transmitted to the Integral gateway. To do this, add a routing rule to the relevant MetaTrader 5 Administrator section.

Select "Process to dealers" as an action in common settings. Assign this routing rule for all requests and orders. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway.

In the figure below all client orders created by users in the demo\demoforex group having symbols from Integral\ group will be sent to the gateway for processing.



MetaTrader 5 trading platform

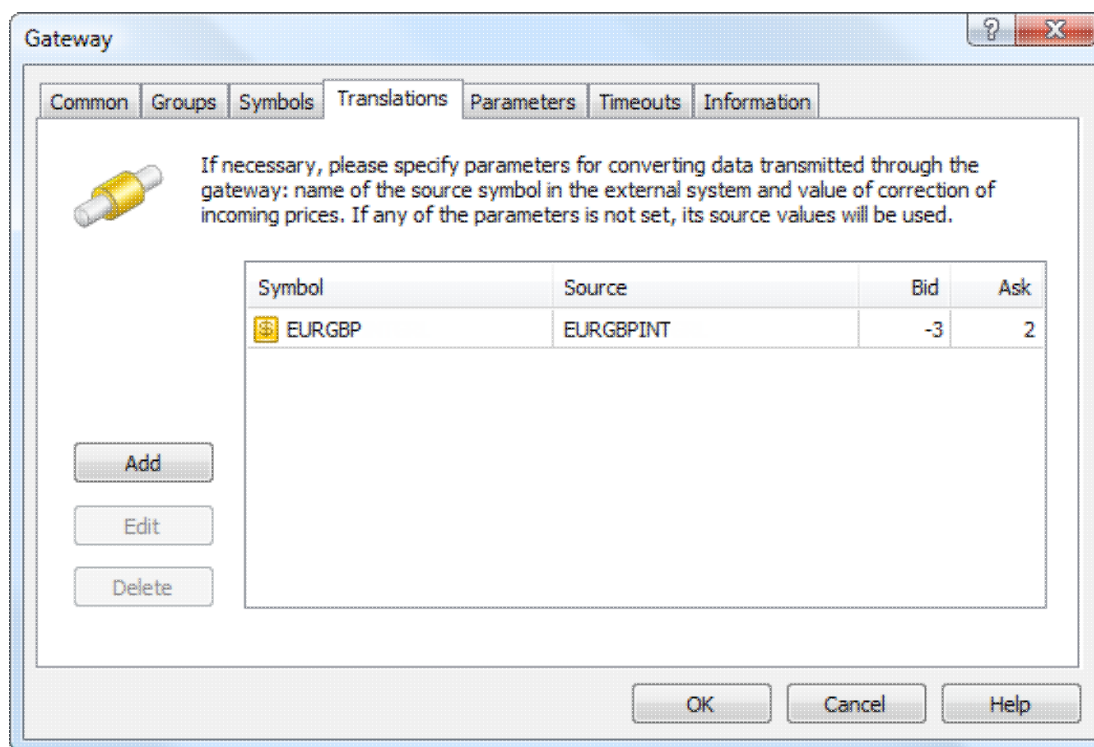
After the correct execution of the steps described above, the gateway will be ready for work.

Changing symbols names and prices correction

MetaTrader 5 Gateway to Integral feeds the price flow from an external trading system to the MetaTrader 5 platform and controls settings of appropriate symbols. In addition, the gateway allows you to edit quotes and Market Depth data transmitted to clients from an external system.

The gateway receives prices from Integral and delivers them to clients taking into account conversion settings. Clients perform trading operations using converted prices. However, while processing trading operations on the gateway and their transmission to Integral, initial, not converted prices are automatically used.

Thus, by increasing the selling price and reducing the purchase price ("price spreading") a brokerage company receives its profit share from each deal performed at Integral. The correction value is set separately for each symbol on the "Translations" tab:



MetaTrader 5 trading platform

Here you can configure matching of symbol names used in Integral with the names used in your MetaTrader 5 platform. For example, if the symbol name in Integral is EURGBPINT, and it is called EURGBP in the MetaTrader 5, enter EURGBP in the "Symbol" field and EURGBPINT in the "Source" field.

The above screenshot shows price conversion: at every tick, the Bid price will be reduced by 3 points, and the Ask price will be increased by 2 points. Below is a schematic example of the conversion:

Integral	>>>	ask price	EURGBP 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	ask price	EURGBP 0.83006	>>>	Client terminal
Client terminal	>>>	buy limit	EURGBP 0.83006	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit	EURGBP 0.83004	>>>	Integral
Integral	>>>	buy limit execution	EURGBP 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit execution	EURGBP 0.83006	>>>	Client terminal

A broker gains 2 pips of profit in this example. The price is sent to the client terminal only after the correction, so the clients work only with the corrected prices. If no correction is set for a symbol, the client will work with the original prices submitted by Integral.

Prices Round Off

During the gateway's operation, accuracy of quotes (decimal places) passed for some symbol may change in the external trading system. Decrease in price accuracy at the external trading system's side does not affect the gateway's operation. It still transmits prices with less accuracy. However, if the number of decimal places at the external system's side increases, the gateway starts rounding off the passed prices.

Suppose that the accuracy of quotes has changed from 4 to 5 digits. Obtained five-digit quotes are rounded up by the gateway and used for creating the Market Depth. The round off is always performed in broker's favor. Thus, buy requests of 1.23447, 1.23441 are rounded up to 1.2345, while sell ones of 1.23447, 1.23441 are rounded down to 1.23440.

Changes in symbol price accuracy are recorded in the gateway journal.

Fill Policy and Order Expiration

While sending an order from the MetaTrader 5 client terminal, the trader can additionally specify the Fill Policy (FOK, IOC or Return) and expiry of the order (Good till canceled, Today, Date and time, Time). In the Integral trading system, the fill

policy and expiration are set in a single parameter - Time in Force. The gateway performs the appropriate changes when sending orders:

- If FOK or IOC fill policy is set for an order, the same policy is applied in Integral system.
- If a market order with Return fill policy is placed, it will be passed to Integral system in Good Till Canceled (GTC) mode.
- If a limit order with Return fill policy is placed, expiration time from MetaTrader 5 (Good Till Canceled, Today, Specified, Specified Day) is inserted into the appropriate order parameter in Integral system.
- If a limit order is activated in MetaTrader 5 platform and the gateway works in the mode of passing limit orders (Limit Orders Coverage Mode = Limit), a limit order with Good Till Date fill time policy is passed to Integral system. Limit Orders Coverage Timeout parameter also affects the order's lifetime. The gateway removes the placed order upon expiration of the specified time.
- If a take profit position is activated in MetaTrader 5 platform and the gateway works in the mode of passing limit orders (Limit Orders Coverage Mode = Limit or Limit Orders Coverage Mode = Gateway), a limit order with Good Till Date fill time policy is passed to Integral system. Limit Orders Coverage Timeout parameter also affects the order's lifetime. The gateway removes the placed order upon expiration of the specified time.

MetaTrader 5 Trading Platform / Platform Components / Gateways / Currenex

MetaTrader 5 Gateway to Currenex

MetaTrader 5 Gateway to Currenex is a simple, fast and secure integration solution for brokers. The gateway provides liquidity when working in MetaTrader 5. Currenex forex means the best quotations, sound liquidity and the most efficient orders execution with use of hi-tech solutions which were applied in the platform development and which allow to create the quickest electronic trading system.

About the company

Founded in 1999, Currenex, offers corporate and institutional buyers and sellers in the FX and money markets reliable, low-cost and secure electronic access to the \$1+ trillion a day global FX market.

Currenex platform represents one of the largest ECN systems operating with forex instruments and getting feed from more than seventy largest financial organizations. Direct quotations provide users with the ability of complete independent, transparent and reliable control over the prices which precisely reflect the market situation in every point of time.

What Currenex offers:

- 28 currency pairs available for trading
- instant orders execution
- an access to level2 and to real market volumes
- narrow spreads
- no re-quotation
- Non Dealing Desk (NDD) technology
- expanded list of orders (Standard, Conditional)
- all transactions information is protected, absolute confidentiality

Getting started with Currenex

In order to be able to provide trading services using Currenex, a brokerage company must first contact Currenex Sales Department for concluding the agreement: Contact details are given in the table below:

	Sales department	Friendly support
New York	+1 212 340 1780	+1 212 340 1780
London	+44 (0) 20 3395 7930	+44 (0) 20 3395 7930
Tokyo	+81 3 4530 7555	+81 3 4530 7555
Singapore	+65 6826 7476	+65 6826 7476
Australia	+612 8429 1204	+612 8429 1204

Sales department

Friendly support

sales@currenex.com

support@currenex.com

www.currenex.com

www.currenex.com

After conclusion of an agreement, all necessary data for connection to the Currenex server will be provided to the brokerage company. Additional information is available on the official website www.currenex.com.

Order MetaTrader 5 Currenex Gateway

You can order the MetaTrader 5 Gateway to Currenex through the "Buy" section of the technical support site. After you make the payment, the MetaTrader 5 platform license will be automatically updated via the LiveUpdate system and you will be able to use MetaTrader 5 Gateway to Currenex without limitations. The executable module of the gateway is included into the platform package and can run in a demo mode, allowing you to perform 100 operations for a work session (until restart).

How the Gateway Works

MetaTrader 5 Gateway to Currenex is a separate CurrenexGateway64.exe module that uses the MetaTrader 5 Gateway API for operation. The gateway operates as a mediator between two systems connecting Currenex and MetaTrader 5 platform. All data between Currenex and the gateway is transmitted using FIX protocol over the encrypted connection. The gateway sends encrypted FIX messages and returns them to the MetaTrader 5 platform using MetaTrader 5 Gateway API.

Market Data

MetaTrader 5 Gateway to Currenex automatically imports all the necessary symbols and processes their properties. An administrator only needs to perform primary setup as described below. Price data is transmitted in real time. The gateway allows to transmit the initial price data provided by Currenex or transform them. In the latter case quotes, reports and orders prices, that are directed to the client terminals, will be transformed according to the applied settings. Detailed information on prices conversion is available below.

Trading Operations

Orders will be sent to MetaTrader 5 Currenex Gateway for processing in accordance with the configured routing rules. Processing of requests depends on the type of the order, as well as in the gateway configuration.

Order type	Execution
Market Order	Delivered directly to Currenex as a market order.

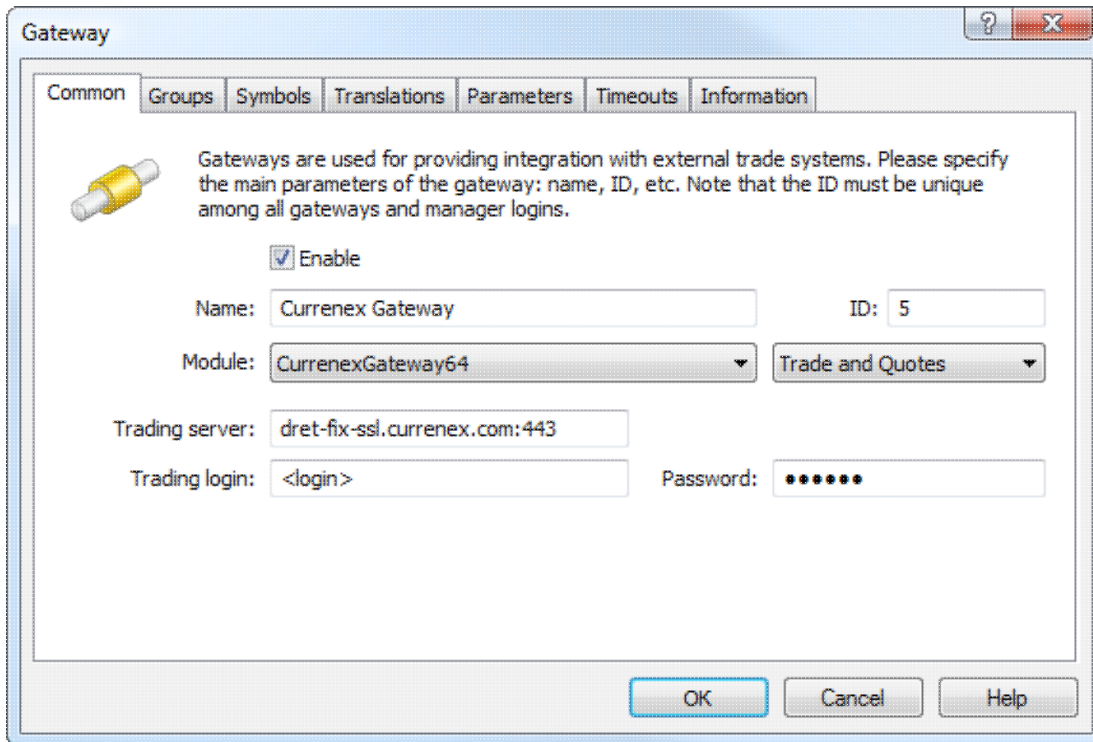
Order type	Execution
Take Profit Buy Limit Sell Limit	Depends on <u>LimitOrdersCoverage</u>: LimitOrdersCoverage=Gateway (default) Limit orders are processed on the side of Currenex. Once a limit order has been placed by a client, an appropriate order is sent to Currenex. Take Profit orders are processed the same way as in the Limit mode. The Currenex system checks the availability of the required amount of funds to cover any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the operation is carried out. <u>Margin reservation</u> of the clients should be configured for the appropriate order types in case Limit orders are directly delivered to an external system. By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client. LimitOrdersCoverage=Limit Limit Orders and Take Profit orders are processed on the side of the MetaTrader 5. Once a limit order is triggered, an equivalent limit order is sent to Currenex. That order has a short action time specified in LimitOrdersCoverageTimeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Currenex. Thus, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5.

Order type	Execution
	A limit order with the price equal to a Take Profit level is sent to Currenex at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Currenex. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on MetaTrader 5 side. Limit mode allows to protect against slippage, as a limit order is sent to Currenex system with a specified price rather than a market order for execution by the current price. LimitOrdersCoverage=Market Limit orders and Take Profit orders are processed on the MetaTrader 5 platform side. Once they trigger, an appropriate market order is sent to Currenex.
Buy Stop Sell Stop Stop Loss Stop Out	Processed on the MetaTrader 5 platform side until their stop price is reached. After a stop order is activated, an appropriate market order will be sent to Currenex system.
Buy Stop Limit Sell Stop Limit	Processed on the MetaTrader 5 side. Upon order activation, an appropriate limit order is created in MetaTrader 5, and this order is then processed in accordance with the value of the LimitOrdersCoverage parameter.
• In case connection to Currenex server is lost, the application will try to restore it repeatedly. In case pending orders have been executed at that, they will be updated in the MetaTrader 5 platform after connection is restored.	

- The Currenex system allows a broker to cancel orders in case of connection loss. The Currenex system does not cancel orders by default in that case but, nevertheless, you should notify the Currenex technical support about the necessity to disable that option for your account (pending orders should not be deleted in case of connection loss).
- The gateway supports multiple modes of trade operation delivery to Currenex: on behalf of the broker's general account and on behalf of the individual accounts used for trading in MetaTrader 5 platform.

Gateway Setup

To start working, add the new gateway configuration:



The screenshot shows the 'Gateway' dialog box with the 'Common' tab selected. The dialog contains the following fields and options:

- Enable:** A checked checkbox.
- Name:** A text field containing 'Currenex Gateway'.
- ID:** A text field containing '5'.
- Module:** A dropdown menu showing 'CurrenexGateway64'.
- Trade and Quotes:** A dropdown menu showing 'Trade and Quotes'.
- Trading server:** A text field containing 'dret-fix-ssl.currenex.com:443'.
- Trading login:** A text field containing '<login>'.
- Password:** A text field containing seven dots.

At the bottom of the dialog are three buttons: 'OK', 'Cancel', and 'Help'.

MetaTrader 5 trading platform

Set the following parameters on the "Common" tab:

- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Module** specify CurrenexGateway64 and accept the default settings after the module selection.
- **Trading server** Currenex server IP-address and the port, where trade requests are processed. This information is provided by Currenex. An encrypted SSL connection to a server is established by default. If you need to establish an unencrypted connection, additionally specify the /nssl key in the address bar. For example: 10.123.100.17:10219/nssl.
- **Trading login** a login for connection to the Currenex server. It is provided by Currenex as the 'trading comp id' parameter.
- **Password** a password for connection to the Currenex server. It is also provided by Currenex.

- ID value must be unique in the field of manager logins and gateway identifiers.
- The data for connection to the Currenex server, where trade requests are processed, is submitted during the agreement conclusion.

Other parameters are set similarly to other gateways. Default values are used in most cases.

Now, go to the "Parameters" tab.

Gateway

Common Groups Symbols Translations Parameters Timeouts Information

Please specify parameters of the gateway. These parameters are specific for each type of gateway. They allow using additional settings that were not available in the "Common" tab.

Parameter	Value
FIX TargetCompID	CNX
FIX Market Data Address	ip:port
FIX Market Data ClientCompID	ClientCompID
Account Mapping Mode	Omnibus
Limit Orders Coverage Mode	Limit
Limit Orders Coverage Time...	5

Up Down Add Edit Delete Default

OK Cancel Help

Specify the following parameters values here:

- **FIX TargetCompID** a standard parameter of the FIX messages heading used for trading messages recipient identification. This parameter is provided by Currenex and is usually equal to CNX.
- **FIX Market Data ClientCompID** a standard parameter of the FIX messages heading used for a data sender identification. It is provided by Currenex as the 'market data comp id' parameter.
- **FIX Market Data Log Enabled** if "YES" is set, the gateway will save to disk the full quoting connection log. This can be useful in operation debugging. The log is not saved by default (value "No").
- **FIX Market Data Password** password for connecting to the market data stream. If the parameter is not specified or has an empty value, the trading connection password is used. The parameter is empty by default.
- **Account Mapping Mode** the gateway supports multiple modes of trade operation delivery to Currenex: on behalf of the broker's general account and on behalf of the individual accounts used for trading in MetaTrader 5 platform. Three modes of trades operations transfer are available:
 - ◆ **omnibus** all orders will be transferred to Currenex on behalf of the broker's main account, specified in the "Trading login" parameter;
 - ◆ **one-to-one** all orders will be transferred to Currenex on behalf of the individual accounts, at which they are set in MetaTrader 5 (account in the external system is displayed in each account's settings);
 - ◆ **conversion** combination of the previous two modes: orders of the accounts that have specified external system account will be transferred on their own behalf, while all other orders will be transferred on behalf of the broker's general account.
- **Limit Orders Coverage Mode** the mode of processing of Limit and Take Profit orders by the gateway. This parameter simplifies configuration of trade requests routing to the gateway, since there is no need to create a separate rule for routing the appropriate order types. Three processing modes are available:
 - ◆ **Market** limit and Take Profit orders are processed on the MetaTrader 5 platform side. Once the order triggers, an appropriate market order is sent to Currenex.
 - ◆ **Limit** limit and Take Profit orders are processed on the MetaTrader 5 side.

Once a limit order is triggered, an equivalent limit order is sent to Currenex. That order has a short action time specified in Limit Orders Coverage ModeTimeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Currenex. Thus, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5.

A limit order with the price equal to a Take Profit level is sent to Currenex at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Currenex. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on the MetaTrader 5 side.

The Limit mode allows to protect against slippage, as a limit order is sent to the Currenex system with a specified price rather than a market order for execution by the current price. In addition, this mode allows you not to reserve margin on a client's account before sending an order to Currenex.

- ♦ **Gateway** limit orders are processed on the Currenex side. Once a limit order has been placed by a client, an appropriate order is sent to Currenex. Take Profit orders are processed the same way as in the Limit mode.
- **Limit Orders Coverage Timeout** expiry of Limit Orders that are sent to Currenex in the Limit mode. Specified in seconds. The default value is 5. The minimal value is 2.
- **FIX Market Data Address** the IP address and port of the Currenex server, from which market data are transmitted. This information is provided by Currenex. An encrypted SSL connection to a server is established by default. If you need to establish an unencrypted connection, additionally specify the /noss key in the address bar. For example: 10.123.100.17:10219/noss.
- **Weekend Trading Enabled** allow the gateway to process trading operations on weekends. The parameter can be set to Yes or No (default). If the parameter is absent or is set to No, trading on weekends is prohibited. For further details please see [Operation on Weekend](#).
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

For connection to the system, Currenex requires authentication with a certificate. To obtain the certificate, use the [dedicated Currenex service](#), and then configure the corresponding settings in your gateway:

- **FIX Trade Certificate Path** the path to the client certificate store for FIX trading connections. If this parameter is missing or empty, the client certificate for trading connections will not be used. Therefore, this update will not affect brokers already using the gateway.

If set to Local Machine (the default), the gateway will use the certificate from the Local Machine personal store of the operating system where it is installed. For proper operation, the certificate must be imported into this store.

If a path is specified, the certificate will be loaded from a file in PFX (P12) format.

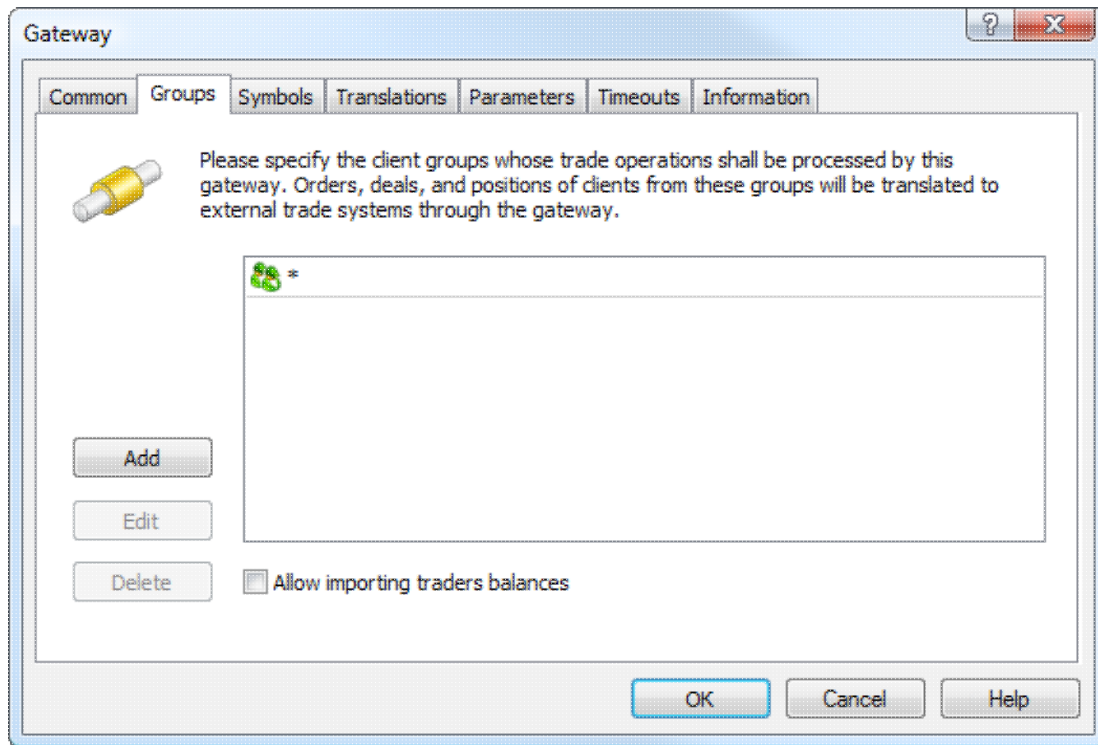
After adding the parameter, the gateway will retrieve the certificate from the specified store (system or file). The search is conducted using the CN (Common Name) field, which must match the trading connection login (ClientCompID) specified in the Trade Login parameter.

- **FIX Trade Certificate Password** password to open the PFX certificate file for the trade connection.
- **FIX Market Data Certificate Path** works similarly to FIX Trade Certificate Path but is used for the quote connection. The certificate search in the specified store is performed by the CN (Common Name) field, which must match the quoting connection login (ClientCompID) specified in the 'FIX Market Data ClientCompID' parameter.
- **FIX Market Data Certificate Password** password to open the PFX certificate file for the quoting connection.

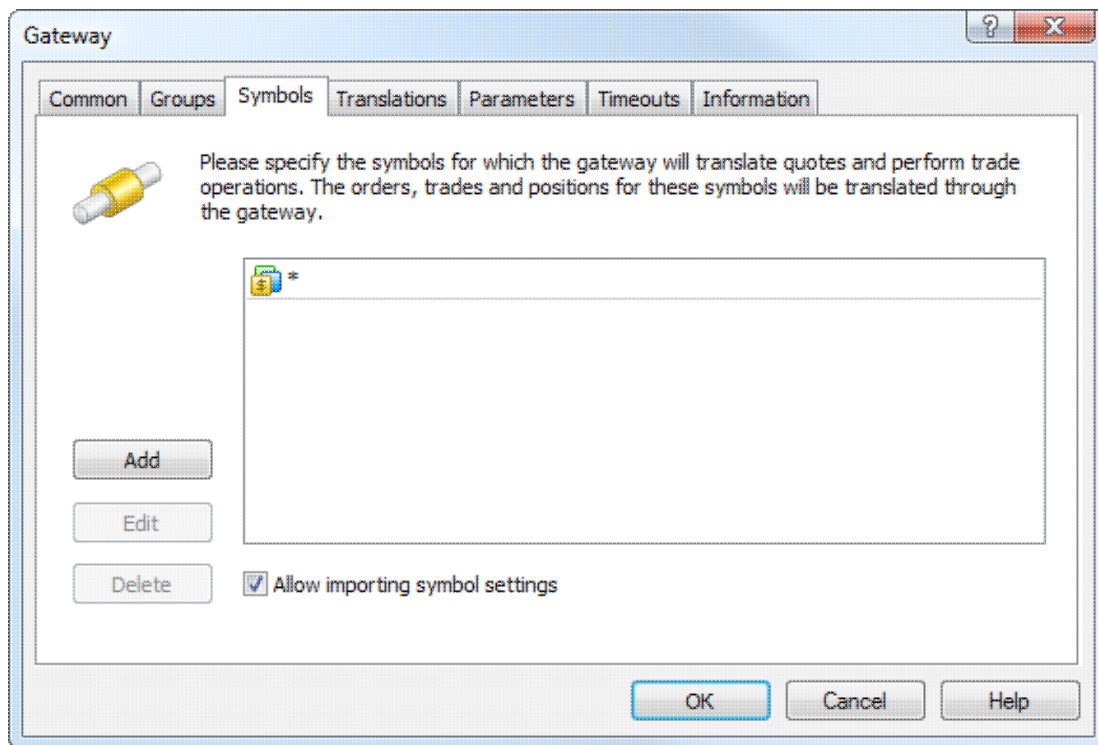
- In no circumstances it is allowed to change operations transfer mode while in operation. Changing the mode is allowed only after all client positions are closed.
 - When Limit orders are directly delivered to an external system, [margin reservation](#) of the clients should be configured for the appropriate order type.

The next stage is to specify the groups of the clients, whose requests will be processed via the gateway. All groups are configured on the screenshot below, but you can configure groups according to your business logic.

MetaTrader 5 trading platform



Then configure the list of symbols, according to which the gateway will process trade operations and feed quotes.



MetaTrader 5 trading platform

Make sure to enable "Allow importing symbol settings" option. The symbols available to Currenex will be imported to Symbols/Preliminary/Currenex directory of the MetaTrader 5 platform. Besides, that will allow the gateway to manage the settings of the symbols used in trading via Currenex.

- The symbols imported by the gateway are put to the "\Preliminary" symbols subgroup. All symbols have trading ability disabled. System administrator must relocate imported symbols to the proper subgroup and allow trading for them.
- After symbols are relocated and trading abilities are enabled, the main trading server must be restarted.
- In case the gateway transmits configuration for the symbol that is already present in the platform, the configuration is updated. In this case the symbol is not transferred and its trading ability is not turned off.
- In case some changes are implemented to the Depth of Market parameter of the symbol settings, Currenex gateway and a history server must be restarted to let the changes take effect. In fact, restart is required after any change in the symbol settings.
- The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

Margin Setup

When Limit orders are directly delivered to an external system, margin reservation of the clients should be configured for the appropriate order type.

The external system checks sufficiency of the funds that are necessary to provide any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the operation is carried out.

By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client.

After an order has been transferred to the external system, MetaTrader 5 platform is not able to check the client's margin sufficiency any more. After the order has been executed in the external system, the gateway cannot ignore that fact. Therefore, the appropriate trading operation is performed in the platform.

Set non-zero coefficients for the orders directly transferred to the external trading system in symbol settings for the appropriate symbols to configure margin collection:

MetaTrader 5 trading platform

Symbol: EURUSD

Common Quotes Trade Execution Margin **Margin Rates** Swaps Sessions

 Please specify margin rates for different types of trade operations.

Liquidity margin rate:

Margin rates:

	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
 Buy	1.0000000	1.0000000	0.0000000	0.0000000
 Sell	1.0000000	1.0000000	0.0000000	0.0000000
Maintenance margin				
 Buy	0.0000000	0.0000000	0.0000000	0.0000000
 Sell	0.0000000	0.0000000	0.0000000	0.0000000

OK Cancel Help

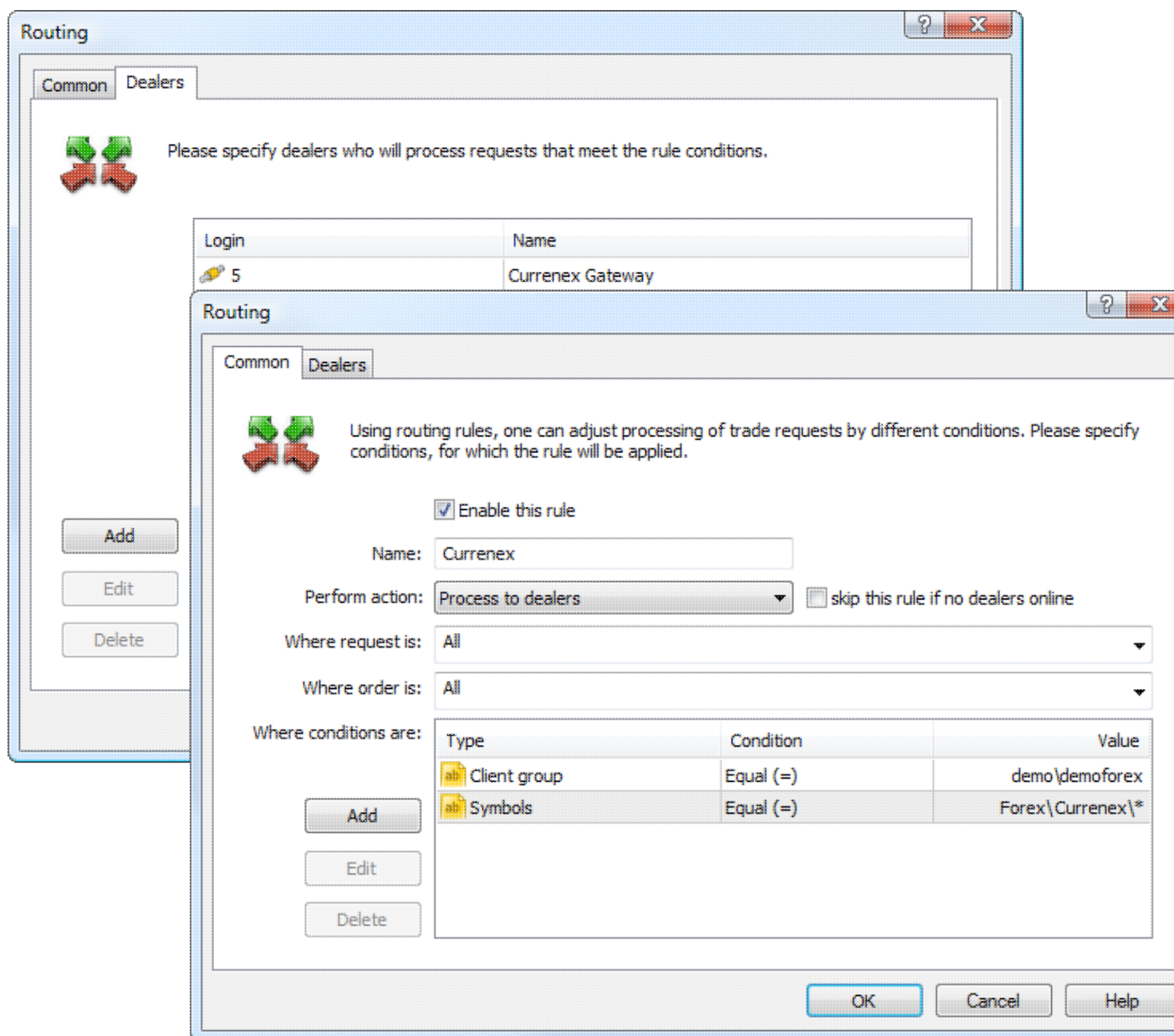
Configuring trade requests routing

Configure the routing to let the clients requests to be transmitted to the Currenex gateway. To do this, add a routing rule to the relevant MetaTrader 5 Administrator section.

Select "Process to dealers" as an action in common settings. Assign this routing rule for all requests and orders. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway.

In the figure below all client orders created by users in the demo\demoforex group having symbols from Forex\Currenex group will be sent to the gateway for processing.

MetaTrader 5 trading platform



MetaTrader 5 trading platform

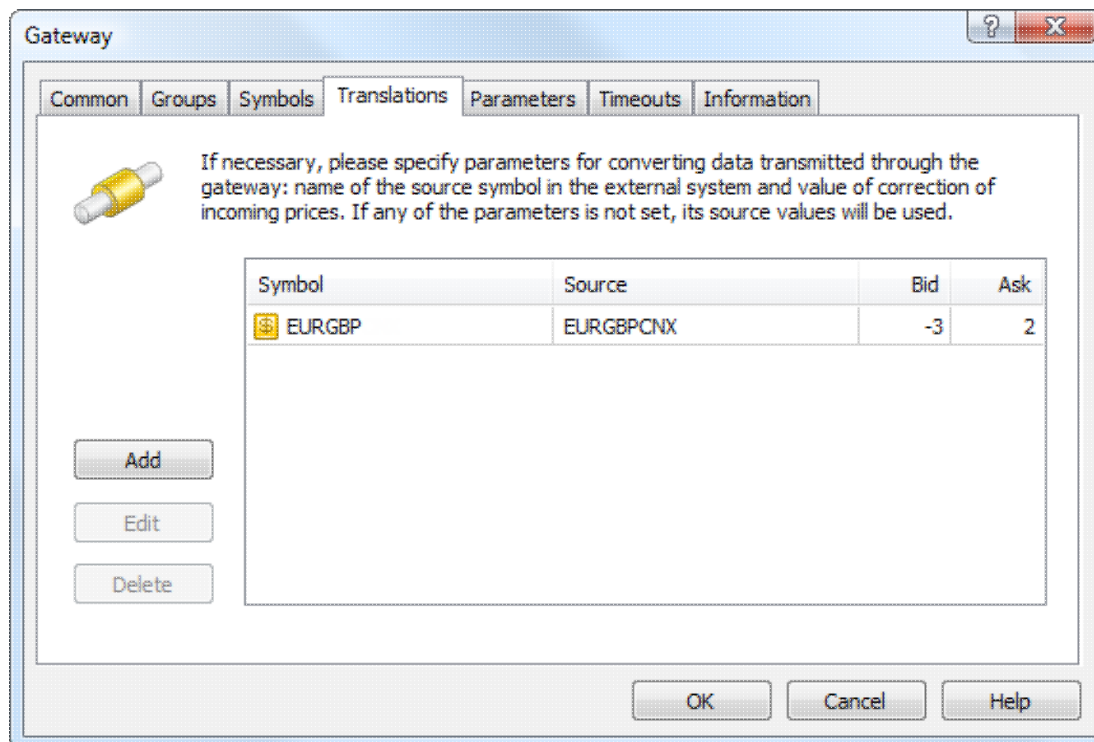
After the correct execution of the steps described above, the gateway will be ready for work.

Changing symbols names and prices correction

The gateway the price flow from the Currenex system to the MetaTrader 5 platform and controls settings of appropriate symbols. In addition, the gateway allows you to edit quotes and Market Depth data transmitted to clients from an external system.

The gateway receives prices from Currenex and delivers them to clients taking into account conversion settings. Clients perform trading operations using converted prices. However, while processing trading operations on the gateway and their transmission to Currenex , initial, not converted prices are automatically used.

Thus, by increasing the selling price and reducing the purchase price ("price spreading") a brokerage company receives its profit share from each deal performed at Currenex . The correction value is set separately for each symbol on the "Translations" tab:



MetaTrader 5 trading platform

Here you can configure matching of symbol names used in Currenex with the names used in your MetaTrader 5 platform. For example, if the symbol name in Currenex is EURGBPCNX, and it is called EURGBP in the MetaTrader 5, enter EURGBP in the "Symbol" field and EURGBPCNX in the "Source" field.

The above screenshot shows price conversion: at every tick, the Bid price will be reduced by 3 points, and the Ask price will be increased by 2 points. Below is a schematic example of the conversion:

Currenex	>>>	ask price	EURGBP 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	ask price	EURGBP 0.83006	>>>	Client terminal
Client terminal	>>>	buy limit	EURGBP 0.83006	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit	EURGBP 0.83004	>>>	Currenex
Currenex	>>>	buy limit execution	EURGBP 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit execution	EURGBP 0.83006	>>>	Client terminal

A broker gains 2 pips of profit in this example. The price is sent to the client terminal only after the correction, so the clients work only with the corrected prices. If no correction is set for a symbol, the client will work with the original prices submitted by Currenex.

Prices Round Off

During the gateway's operation, accuracy of quotes (decimal places) passed for some symbol may change in the external trading system. Decrease in price accuracy at the external trading system's side does not affect the gateway's operation. It still transmits prices with less accuracy. However, if the number of decimal places at the external system's side increases, the gateway starts rounding off the passed prices.

Suppose that the accuracy of quotes has changed from 4 to 5 digits. Obtained five-digit quotes are rounded up by the gateway and used for creating the Market Depth. The round off is always performed in broker's favor. Thus, buy requests of 1.23447, 1.23441 are rounded up to 1.2345, while sell ones of 1.23447, 1.23441 are rounded down to 1.23440.

Changes in symbol price accuracy are recorded in the gateway journal.

Trade Operations Transfer Modes

The MetaTrader 5 Gateway to Currenex allows to send trade operations to Currenex using different modes. Trading orders placed by clients in the MetaTrader 5 platform can be sent to on behalf of the broker's general account (specified in the "Trading

login" parameter of the gateway settings) or on behalf of the clients' individual accounts. In the latter case, gateway connection to Currenex is performed via broker's general account. However, clients' trade operations are executed on Currenex on individual accounts.

- Order sending mode is controlled by the Account Mapping Mode parameter in the gateway configuration.
- Trade operations transfer conditions are determined while concluding an agreement between a brokerage company and Currenex.
- In no circumstances it is allowed to change operations transfer mode while in operation. Changing the mode is allowed only after all client positions are closed.

If you send trading operations using individual client accounts, the appropriate Currenex account number must be specified in each account's settings. This can be done in the Administrator or Manager terminal:

Account: 3541, Johnson Adrian, USD, 1:100

Overview Personal **Account** Security

Group: demo\demo forex

Color: ☐ None Leverage: 1 : 100

Bank account: Agent account:

☒ Enable this account ☒ Enable trading by Expert Advisors

☒ Enable trading ☒ Enable Web API connections

☒ Enable to change password ☒ Enable daily reports

☒ Enable trailing stops ☐ Change password at next login

Trade accounts:

Gateway ID	Account
5, Currenex Gateway	123456

Synchronize

Add

Edit

Delete

Update Cancel Help

MetaTrader 5 trading platform

In "Trade accounts" section, select Currenex gateway configuration and specify the client's account in the external system. That is the account, from which client trade operations will be transferred to Currenex.

Clients' account numbers are submitted by Currenex.

MetaTrader 5 Trading Platform / Platform Components / Gateways /
Euronext FX

MetaTrader 5 Euronext FX Gateway

Euronext FX is a heavy-duty matching system of foreign exchange, Euronext FX offers its customers access to a large pool of diversified liquidity at unparalleled speed, complete transparency, and excellent customer service. MetaTrader 5 Euronext FX Gateway is a simple, fast and secure integration solution for brokers.

Integration provides:

- **Liquidity.** Euronext FX provides services to various clients including brokerage companies, banks, hedge funds and other organizations. Together they form a unique pool of liquidity.
- **Speed.** Euronext FX offers a trading environment with ultra-low latency. The Euronext FX system's average delay of the full data sending and receiving cycle equals to 200 microseconds with a standard deviation of 50 microseconds. The delay in the full cycle is measured as the time required to deliver a client's message to Euronext FX, to process it and to send it back to the client.
- **Transparency.** Customers can see quotes and trades with corresponding prices and volumes without delays in real time. The execution price is selected in accordance with a strict priority order: Price/Volume/Placing time. As soon as a better price appears, it is immediately sent to the liquidity recipient.

Getting started with Euronext FX

In order to be able to provide trading services using Euronext FX, you should first contact the sales department for concluding the agreement:

Sales department

+442079033826

sales@fastmatchfx.com

www.euronextfx.com

Friendly support

+16464322940

support@fastmatchfx.com

Once you have an agreement with Euronext FX, you will be provided with further instructions on how to connect to the Euronext FX server. For more details please visit the official Euronext FX site at www.euronextfx.com.

Order the MetaTrader 5 Euronext FX Gateway

You can order the MetaTrader 5 Euronext FX Gateway through the "[Buy](#)" section of the technical support site. After you make the payment, the MetaTrader 5 platform license will be automatically updated via the LiveUpdate system and you will be able to use MetaTrader 5 Euronext FX Gateway without limitations. The executable module of the gateway is included into the platform package and can run in a demo

mode, allowing you to perform 100 operations for a work session (until restart).

How the Gateway Works

MetaTrader 5 Euronext FX Gateway is a separate module that uses the MetaTrader 5 Gateway API for operation. The gateway operates as a mediator between two systems: Euronext FX and MetaTrader 5 platform. Data is transmitted using FIX protocol over the encrypted connection. The gateway translates FIX messages and communicates them back to MetaTrader 5 platform using MetaTrader 5 Gateway API.

There are two main types of data that flow through the gateway:

- market data (quotes, reports);
- trading messages.



Market Data

MetaTrader 5 Euronext FX Gateway automatically imports all the necessary symbols and processes their properties. An administrator only needs to perform primary setup described below. Price data is transmitted in real time. The gateway allows to transmit original price data provided by Euronext FX, as well as to convert them. In the latter case quotes, reports and orders prices, that are directed to the client terminals, will be transformed according to the applied settings. Detailed information on prices conversion is available below.

Trading Operations

Orders will be sent to MetaTrader 5 Euronext FX Gateway for processing in accordance with the configured routing rules. Processing of requests depends on the type of the order, as well as in the gateway configuration.

Order type	Execution
Market Order	Delivered directly to Currenex as a market order.
Take Profit	Depends on <u>Limit Orders Coverage Mode</u> :

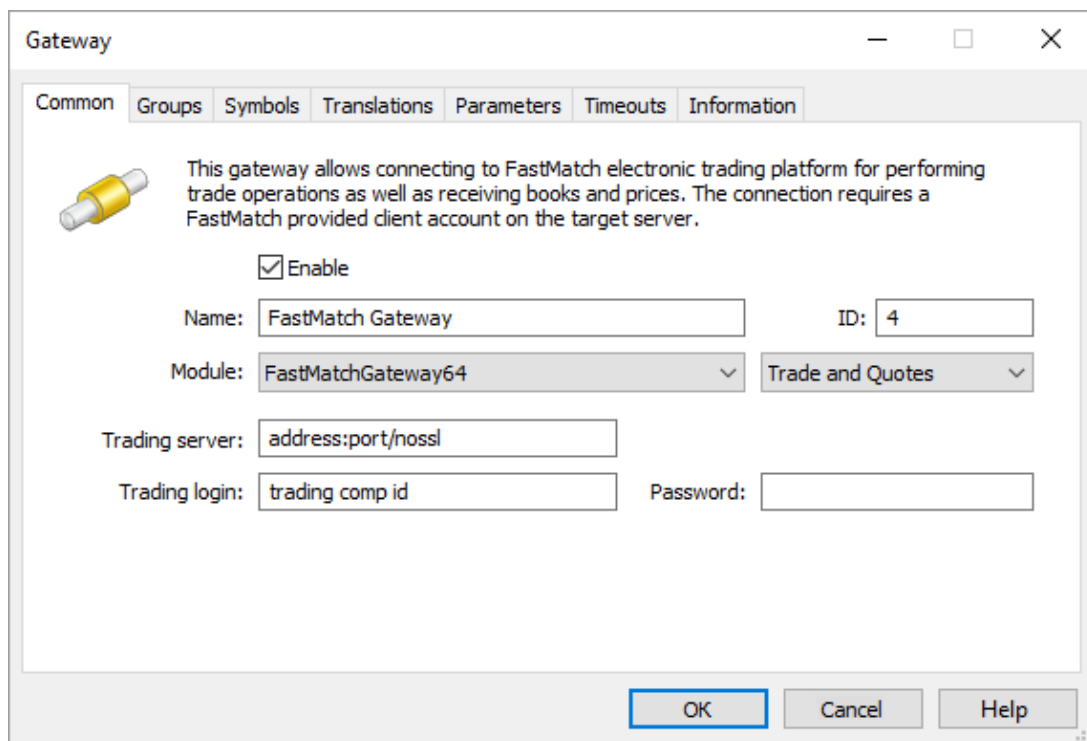
Order type	Execution
Buy Limit Sell Limit	Limit Orders Coverage Mode=Gateway (default) Limit orders are processed on the side of Euronext FX. Once a limit order has been placed by a client, an appropriate order is sent to Euronext FX. There it is placed in the aggregate Depth of Market awaiting for an opposite request with the same price to appear. Thus, clients can see their requests in the Depth of Market in the client terminal in real time. Take Profit orders are processed the same way as in the Limit mode. The Euronext FX system checks the availability of the required amount of funds to cover any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the operation is carried out. <u>Margin reservation</u> of the clients should be configured for the appropriate order types in case Limit orders are directly delivered to an external system. By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client. Limit Orders Coverage Mode=Limit Limit Orders and Take Profit orders are processed on the side of the MetaTrader 5. Once a limit order is triggered, an equivalent limit order is sent to Euronext FX. That order has a short validity time specified in Limit Orders Coverage Timeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Euronext FX. Thus, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5.

Order type	Execution
	A limit order with the price equal to a Take Profit level is sent to Integral at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Euronext FX. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on MetaTrader 5 side. The Limit mode allows to protect against slippage, as a limit order is sent to the Euronext FX system with a specified price rather than a market order for execution by the current price. Limit Orders Coverage Mode=Market Limit orders and Take Profit orders are processed on the MetaTrader 5 platform side. Once they trigger, an appropriate market order is sent to Euronext FX.
Buy Stop Sell Stop Stop Loss Stop Out	Processed on the MetaTrader 5 platform side until their stop price is reached. After a stop order is activated, an appropriate market order will be sent to the Euronext FX system.
Buy Stop Limit Sell Stop Limit	Processed on the MetaTrader 5 side. Upon order activation, an appropriate limit order is created in MetaTrader 5, and this order is then processed in accordance with the value of the Limit Orders Coverage Mode parameter.

Gateway Setup

To start working, add the new gateway configuration:

MetaTrader 5 trading platform



The screenshot shows the 'Gateway' dialog box with the 'Common' tab selected. The dialog has a title bar with standard window controls. Below the title bar are tabs: 'Common', 'Groups', 'Symbols', 'Translations', 'Parameters', 'Timeouts', and 'Information'. The 'Common' tab contains a description of the gateway, an 'Enable' checkbox, and several input fields: 'Name' (FastMatch Gateway), 'ID' (4), 'Module' (FastMatchGateway64), 'Trade and Quotes' (Trade and Quotes), 'Trading server' (address:port/nossl), 'Trading login' (trading comp id), and 'Password' (empty). At the bottom are 'OK', 'Cancel', and 'Help' buttons.

Gateway

Common Groups Symbols Translations Parameters Timeouts Information

This gateway allows connecting to FastMatch electronic trading platform for performing trade operations as well as receiving books and prices. The connection requires a FastMatch provided client account on the target server.

☒ Enable

Name: FastMatch Gateway ID: 4

Module: FastMatchGateway64 Trade and Quotes

Trading server: address:port/nossl

Trading login: trading comp id Password:

OK Cancel Help

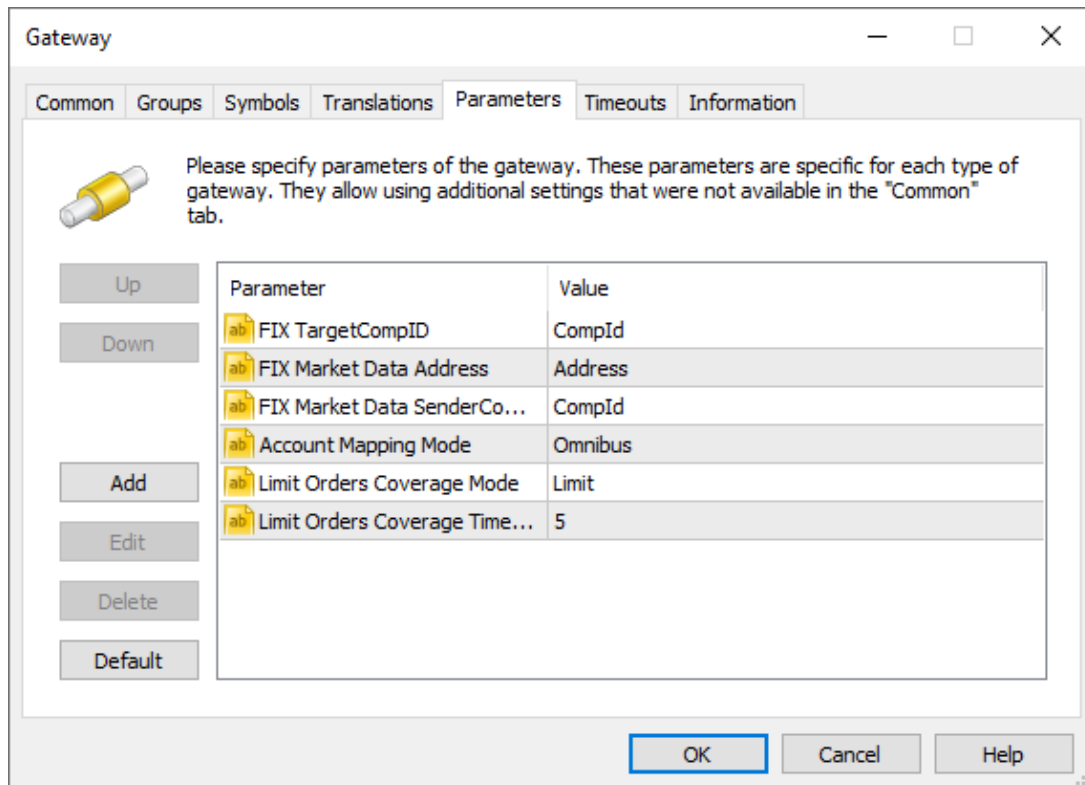
Set the following parameters on the "Common" tab:

- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Module** specify EuronextFXGateway64 and accept the default settings after the module selection.
- **Trading server** Euronext FX server IP-address and the port, where trade requests are processed. This information is provided by Euronext FX. An encrypted SSL connection to a server is established by default. If you need to establish an unencrypted connection, additionally specify the /nossl key in the address bar. For example: 10.123.100.17:10219/nossl.
- **Trading login** a login for connection to the Euronext FX server. It is provided by Euronext FX as the "trading comp id" parameter.
- **Password** password for connection to the Euronext FX server. It is also provided by Euronext FX.

- ID value must be unique in the field of manager logins and gateway identifiers.
- The details for connection to the Euronext FX server, where trade requests are processed, will be provided with the agreement.

Other parameters are set similarly to other gateways. Default values are used in most cases.

Now, go to the "Parameters" tab.



Specify the following parameters values here:

- **FIX TargetCompID** a standard parameter of the FIX messages heading used for trading messages recipient identification. Provided by Euronext FX.
- **FIX Market Data SenderCompID** a standard parameter of the FIX messages heading used for a data sender identification. Provided by Euronext FX.
- **Account Mapping Mode** the gateway supports multiple modes of trade operation delivery to Euronext FX: on behalf of the broker's general account and on behalf of the individual accounts used for trading in MetaTrader 5 platform. Three modes of trades operations transfer are available:
 - ◆ **omnibus** all orders will be transferred to Euronext FX on behalf of the broker's main account, specified in the "Trading login" parameter.
 - ◆ **one-to-one** all orders will be transferred to Euronext FX on behalf of the individual accounts, at which they are set in MetaTrader 5 (account in the external system is displayed in each account's settings).
 - ◆ **conversion** combination of the previous two modes: orders of the accounts that have specified external system account will be transferred on their own behalf, while all other orders will be transferred on behalf of the broker's general account.
- **Limit Orders Coverage Mode** the mode of processing of Limit and Take Profit orders by the gateway. This parameter simplifies configuration of trade requests routing to the gateway, since there is no need to create a separate rule for routing the appropriate order types. Three processing modes are available:
 - ◆ **Market** limit and Take Profit orders are processed on the MetaTrader 5 platform side. Once a limit order triggers, an appropriate market order is sent to Euronext FX.

- ◆ **Limit** limit and Take Profit orders are processed on the MetaTrader 5 side.

Once a limit order is activated, an appropriate limit order is sent to Euronext FX. The order has a short lifetime specified in the Limit Orders Coverage Timeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed.

If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Euronext FX. Thus, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5.

A limit order with the price equal to a Take Profit level is sent to Euronext FX at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Euronext FX. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on the MetaTrader 5 side.

Limit mode allows to protect against slippage, as a limit order is sent to the Euronext FX system with a specified price rather than a market order for execution by the current price. In addition, this mode allows you not to reserve margin on a client's account before sending an order to Euronext FX.

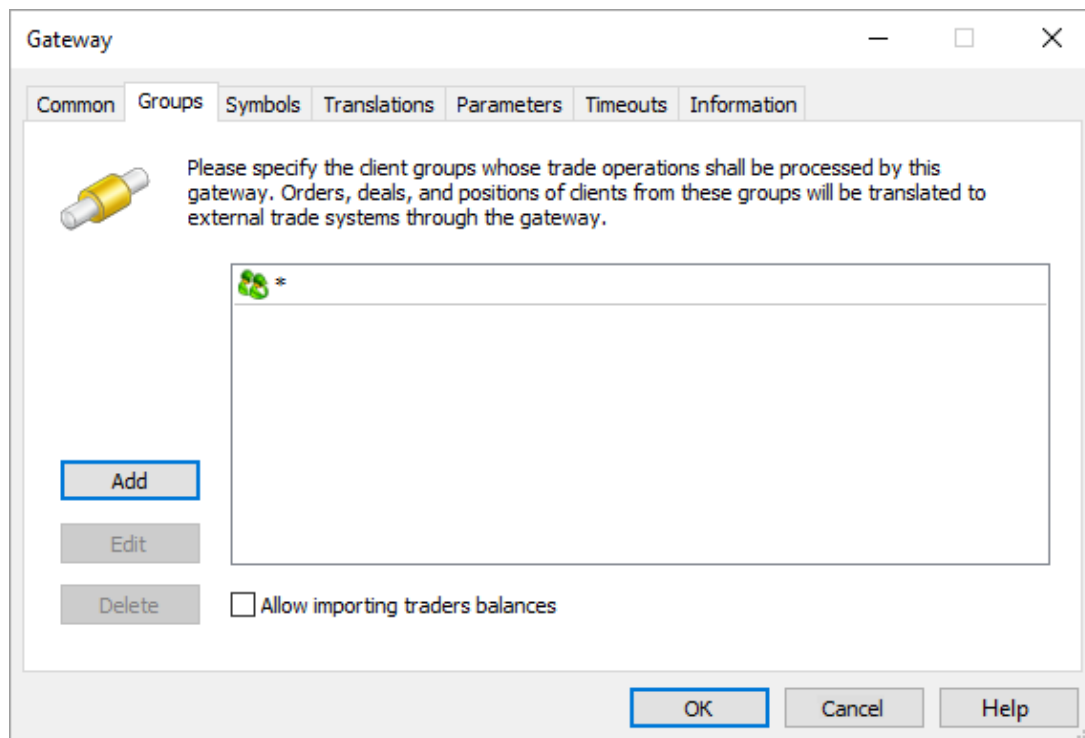
- ◆ **Gateway** limit orders are processed on the Euronext FX side. Once a limit order has been placed by a client, an appropriate order is sent to Euronext FX. Take Profit orders are processed the same way as in the Limit mode.
- **Limit Orders Coverage Timeout** expiry of Limit Orders that are sent to Euronext FX in the Limit mode. Specified in seconds. The default value is 5.
- **FIX Market Data Address** the address of the server with market data in the format of ip:port. Provided by Euronext FX. An encrypted SSL connection to a server is established by default. If you need to establish an unencrypted connection, additionally specify the /nossll key in the address bar. For example, 10.123.100.18:10219/nossll.
- **FIX Market Data Log Enabled** if "YES" is set, the gateway will save to disk the full quoting connection log. This can be useful in operation debugging. The log is not saved by default (value "No").
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay

is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.

- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

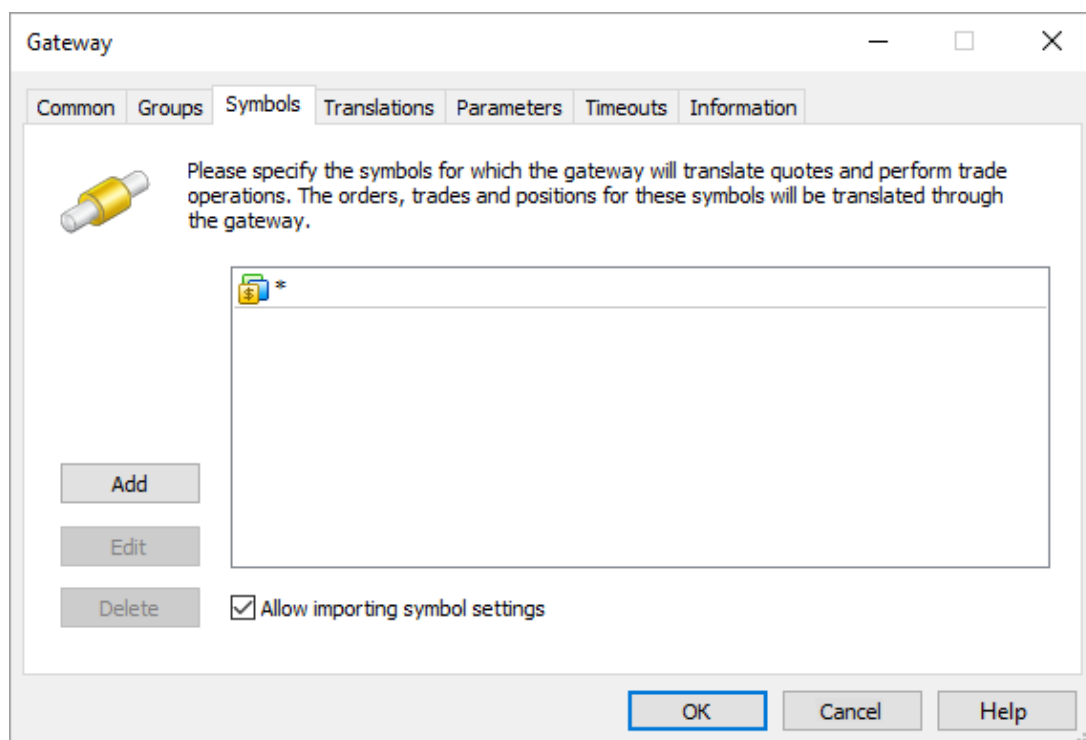
- It is not allowed to change operations transfer mode while in operation. Changing the mode is allowed only after all client positions are closed.
- When Limit orders are directly delivered to an external system, margin reservation of the clients should be configured for the appropriate order type.
- For correct operation, the gateway needs to receive messages regarding the trading session states. Request from the data provider 'Trading Session Status (35=h)' events which should be sent in your quoting connection.

The next stage is to specify the groups of the clients, whose requests will be processed via the MetaTrader 5 Euronext FX Gateway. All groups are configured on the screenshot below, but you can configure groups according to your business logic.



MetaTrader 5 trading platform

Then configure the list of symbols, according to which the gateway will process trade operations and feed quotes.



Make sure to enable "Allow importing symbol settings" option. Symbols available to Euronext FX will be imported to the folder 'Symbols\Preliminary\EuronextFX' of the MetaTrader 5 platform. Besides, that will allow the gateway to manage the settings of the symbols used in trading via Euronext FXI.

- The symbols imported by the gateway are put to the "\Preliminary" symbols subgroup. All symbols have trading ability disabled. System administrator must relocate imported symbols to the proper subgroup and allow trading for them.
- After symbols are relocated and trading abilities are enabled, the main trading server must be restarted.
- In case the gateway transmits configuration for the symbol that is already present in the platform, the configuration is updated. In this case the symbol is not transferred and its trading ability is not turned off.
- In case some changes are implemented to the Depth of Market parameter of the symbol settings, the Euronext FX gateway and a history server must be restarted to let the changes take effect. Restart is required after any change in symbol settings.
- The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

Margin Setup

MetaTrader 5 trading platform

When Limit orders are directly delivered to an external system, margin reservation of the clients should be configured for the appropriate order type. The external system checks sufficiency of the funds that are necessary to provide any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the operation is carried out.

By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client.

After an order has been transferred to the external system, MetaTrader 5 platform is not able to check the client's margin sufficiency any more. After the order has been executed in the external system, the gateway cannot ignore that fact. Therefore, the appropriate trading operation is performed in the platform.

Set non-zero coefficients for the orders directly transferred to the external trading system in order to configure margin collection:

Symbol: EURUSD

Common Quotes Trade Execution Margin **Margin Rates** Swaps Sessions

 Please specify margin rates for different types of trade operations.

Liquidity margin rate:

Margin rates:

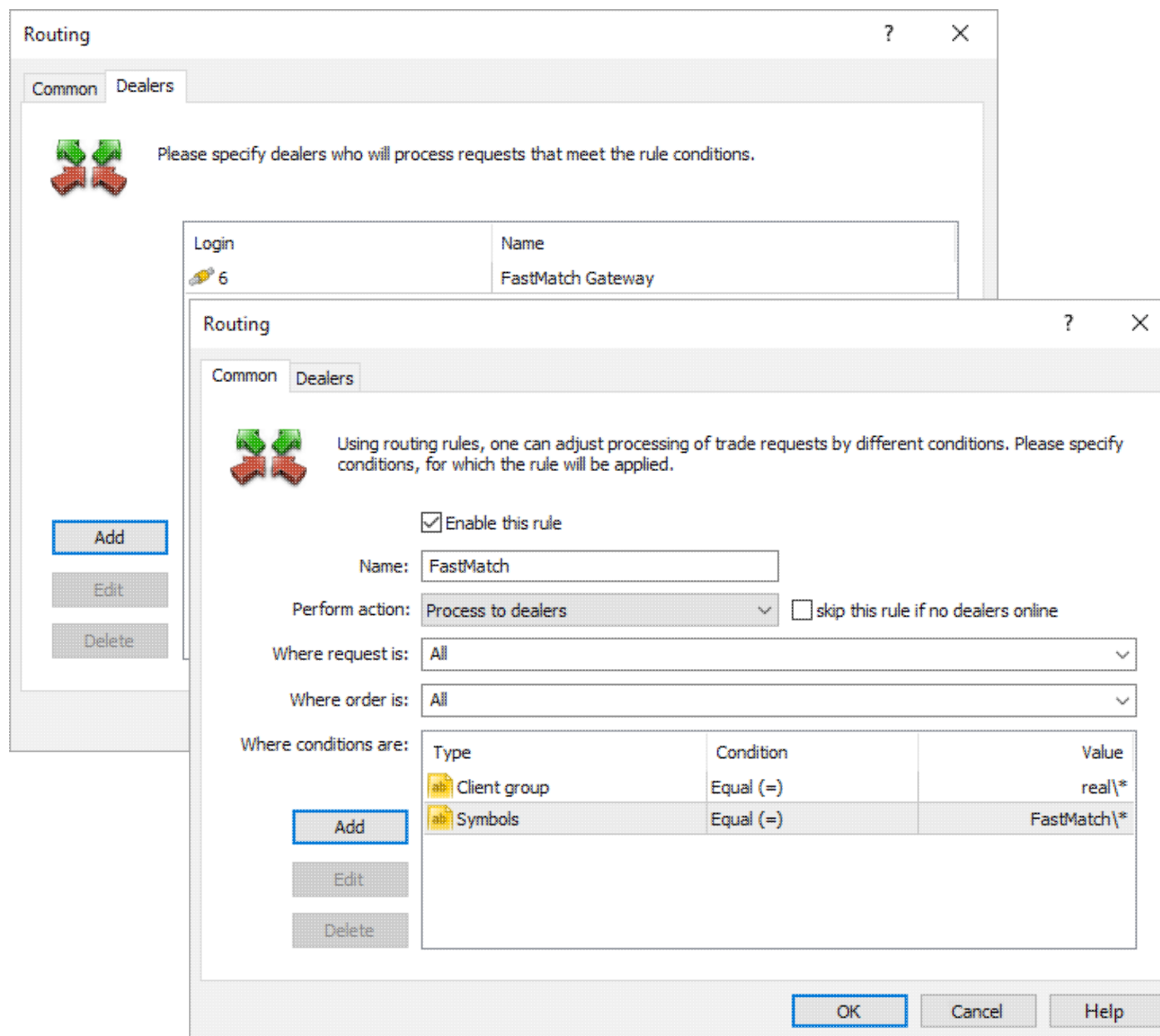
	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
 Buy	1.0000000	1.0000000	0.0000000	0.0000000
 Sell	1.0000000	1.0000000	0.0000000	0.0000000
Maintenance margin				
 Buy	0.0000000	0.0000000	0.0000000	0.0000000
 Sell	0.0000000	0.0000000	0.0000000	0.0000000

OK Cancel Help

Configuring trade requests routing

Configure the routing to let the clients requests to be transmitted to the Euronext FX gateway. To do this, add a routing rule to the relevant MetaTrader 5 Administrator section.

Select "Process to dealers" as an action in common settings. Assign this routing rule for all requests and orders. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway. In the picture below all orders for symbols from the "Euronext FX" group will be routed to the gateway for processing.



Routing

Common Dealers

Please specify dealers who will process requests that meet the rule conditions.

Login	Name
6	FastMatch Gateway

Routing

Common Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: FastMatch

Perform action: Process to dealers ☐ skip this rule if no dealers online

Where request is: All

Where order is: All

Where conditions are:

Type	Condition	Value
Client group	Equal (=)	real*
Symbols	Equal (=)	FastMatch*

Buttons: Add, Edit, Delete, OK, Cancel, Help

asd

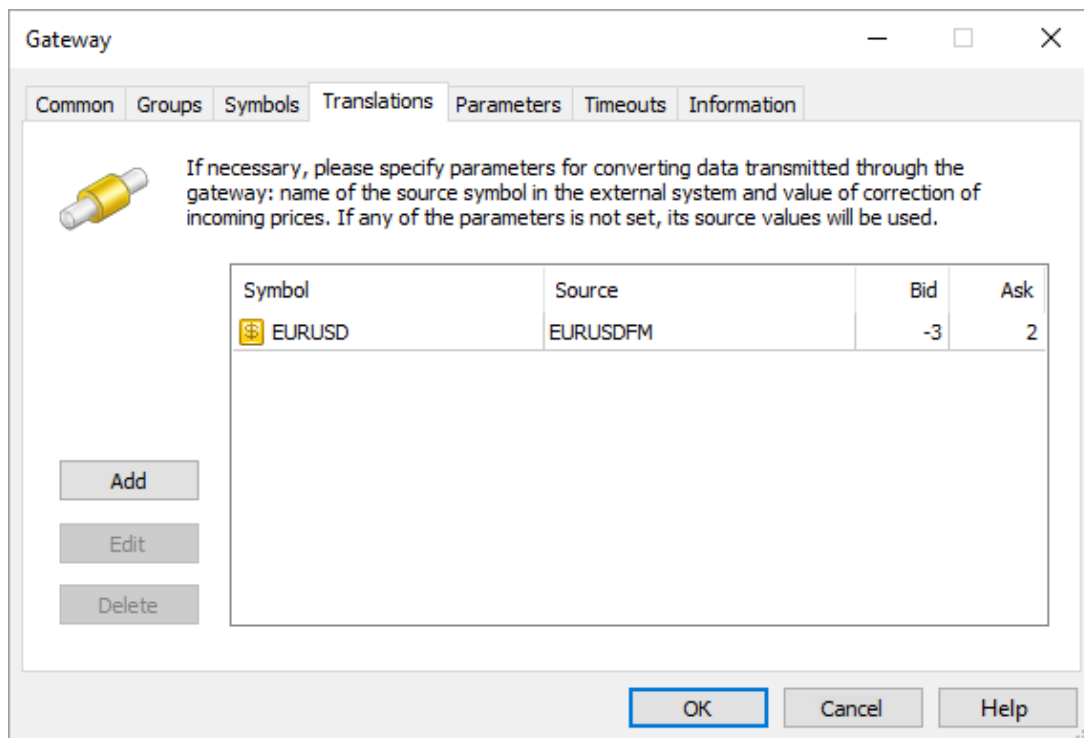
After the correct execution of the steps described above, Euronext FX Gateway will be ready for use.

Changing symbols names and prices correction

The gateway the price flow from the Euronext FX system to the MetaTrader 5 platform and controls settings of appropriate symbols. The standard feature of gateways is the ability to change the quotes and market depth transmitted to the clients from an external system.

The gateway receives prices from Euronext FX and delivers them to clients taking into account conversion settings. Clients perform trading operations using converted prices. However, while processing trading operations on the gateway and their transmission to Euronext FX, initial, not converted prices are automatically used.

Thus, by increasing the selling price and reducing the purchase price ("price spreading") a brokerage company receives its profit share from each deal performed at Euronext FX. The correction value is set separately for each symbol on the "Translations" tab:



MetaTrader 5 trading platform

Here you can configure matching of symbol names used in Euronext FX with the names used in your MetaTrader 5 platform. For example, if the symbol name in Euronext FX is EURUSD FM, and it is called EURUSD in the MetaTrader 5, enter EURUSD in the "Symbol" field and EURUSD FM in the "Source" field.

The above screenshot shows price conversion: at every tick, the Bid price will be reduced by 3 points, and the Ask price will be increased by 2 points. Below is a schematic example of the conversion:

Euronext FX	>>>	ask price	EURUSD 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	ask price	EURUSD 0.83006	>>>	Client terminal
Client terminal	>>>	buy limit	EURUSD 0.83006	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit	EURUSD 0.83004	>>>	Euronext FX
Euronext FX	>>>	buy limit execution	EURUSD 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit execution	EURUSD 0.83006	>>>	Client terminal

A broker gains 2 pips of profit in the provided example. The price is sent to the client terminal only after the correction, so the clients work only with the corrected prices. If no correction is set for a symbol, the client will work with the original prices submitted by Euronext FX.

Prices Round Off

During the gateway's operation, accuracy of quotes (decimal places) passed for some symbol may change in the external trading system. Decrease in price accuracy at the external trading system's side does not affect the gateway's operation. It still transmits prices with less accuracy. However, if the number of decimal places at the external system's side increases, the gateway starts rounding off the passed prices.

Suppose that the accuracy of quotes has changed from 4 to 5 digits. Obtained five-digit quotes are rounded up by the gateway and used for creating the Market Depth. The round off is always performed in broker's favor. Thus, buy requests of 1.23447, 1.23441 are rounded up to 1.2345, while sell ones of 1.23447, 1.23441 are rounded down to 1.23440.

Changes in symbol price accuracy are recorded in the gateway journal.

Trade Operations Transfer Modes

The MetaTrader 5 Euronext FX Gateway allows to send trade operations to Euronext FX using different modes. Client trading orders that are set in MetaTrader 5 platform can be sent to Euronext FX on behalf of the broker's general account (specified in the

"Trading login" parameter of the gateway settings) or on behalf of the clients' individual accounts. In the latter case gateway connection to Euronext FX is performed via broker's general account. However, clients' trade operations are transferred to Euronext FX using individual accounts.

- Order sending mode is controlled by the Account Mapping Mode parameter in the gateway configuration.
- Trade operations transfer conditions are determined while concluding an agreement between a brokerage company and Euronext FX.
- In no circumstances it is allowed to change operations transfer mode while in operation. Changing the mode is allowed only after all client positions are closed.

If you send trading operations using individual client accounts, the appropriate Euronext FX account number must be specified in each account's settings. This can be done via the administrator or manager terminal:

Account: 9105, James Jayden, USD, 1:100

Overview Personal **Account** Security

Group: real\FastMatch

Color: ☐ None Leverage: 1 : 100

Bank account: Agent account:

☒ Enable this account ☒ Enable trading by Expert Advisors

☒ Enable trading ☒ Enable Web API connections

☒ Enable to change password ☒ Enable daily reports

☒ Enable trailing stops ☐ Change password at next login

☒ Enable one-time password

Trade accounts:

Gateway ID	Account
6, FastMatch Gateway	741128789

Synchronize

Add

Edit

Delete

Update Отмена Справка

MetaTrader 5 trading platform

In "Trade accounts" section, select Euronext FX gateway configuration and specify the client's account in the external system. That is the account, on which client's trade operations will be processed in Euronext FX.

Client account numbers are provided by Euronext FX.

MetaTrader 5 Trading Platform / Platform Components / Gateways / Cboe FX

MetaTrader 5 Cboe FX Gateway

Cboe FX is one of the largest ECNs for the institutional FX market. The key advantages of Cboe FX are:

- independence and transparency
- complete Market Depth
- centralized price setting system (defining the best price of all providers)
- direct and anonymous market access
- high trade execution speed
- real-time quotes stream

The following financial assets can be traded via Cboe FX:

- Currency pairs: 53 basic currency pairs and 13 additional ones.
- Precious metals: XAGUSD, XAUUSD, LPDUSD, LPTUSD, XPDUSD, XPTUSD.

The MetaTrader 5 Gateway to Cboe FX enables brokers to transmit trading operations from the MetaTrader 5 platform to a ECN system. This will attract many customers who trade via Cboe FX. In addition, the gateway provides the ability for brokerage companies to earn additional income by automatically rearing the prices (spread increase) transmitted from ECN to the platform.

Getting started with Cboe FX

First of all a broker should contact Cboe FX using the specified contact details, and to sign an agreement with them.

For sales inquiries:

Technical support:

Americas: +1 212 209 1420

Europe: +44 (0)20 7131 3450 +1 212 378 8558

Asia: +65 6911 6688

fxtradedesk@cboe.com

fxtradedesk@cboe.com

When this stage is completed, authorization data for connection to ECN will be provided to the brokerage company. Additional information is available on the official Cboe FX website at <https://fx.cboe.com>.

Order the MetaTrader 5 Gateway to Cboe FX

You can order the MetaTrader 5 Gateway to Cboe FX through the "[App Store](#)" section of the technical support site. After you make the payment, the MetaTrader 5 platform license will be automatically updated via the LiveUpdate system and you will be able to use MetaTrader 5 Gateway to Cboe FX without limitations. The executable module of the gateway is included into the platform package and can run in a demo mode, allowing you to perform 100 operations for a work session (until restart).

How the Gateway Works

MetaTrader 5 Gateway to Cboe FX is a separate module that uses the MetaTrader 5 Gateway API for operation. It streams quotes and the Market depth, as well as it transmits clients' trade requests to the Cboe FX ECN system and back.

All orders entered in ECN are transferred to the unified requests database. The system selects appropriate orders automatically executing opposite orders with matching parameters (symbol, price etc.)

The broker's servers are configured in such a way that the clients' market requests are sent to MetaTrader 5 Cboe FX Gateway. The gateway verifies correctness of the client requests. In case validation is successful, the relevant trade request is directed to Cboe FX ECN server via a standard protocol for exchanging financial information - FIX. After receiving a response from the Cboe FX server, the gateway saves the trade request processing result in the platform, which in its turn reports this result to the trader.

Orders will be sent to MetaTrader 5 Cboe FX Gateway for processing in accordance with the configured [routing rules](#). Processing of requests depends on the type of the order, as well as in the gateway configuration.

Order type	Execution
Market Order	Delivered directly to Cboe FX as a market order.
Take Profit	Depends on LimitOrdersCoverage :
Buy Limit	LimitOrdersCoverage=Gateway (default)
Sell Limit	Limit orders are processed on the side of Cboe FX. Once a limit order has been placed by a client, an appropriate order is sent to Cboe FX. Take Profit orders are processed the same way as in the Limit mode.
	The Cboe FX system checks the availability of the required amount of funds to cover any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the

Order type	Execution
	operation is carried out. Margin reservation of the clients should be configured for the appropriate order types in case Limit orders are directly delivered to an external system. By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client. LimitOrdersCoverage=Limit Limit Orders and Take Profit orders are processed on the side of the MetaTrader 5. Once a limit order is triggered, an equivalent limit order is sent to Cboe FX. That order has a short action time specified in LimitOrdersCoverageTimeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Cboe FX. Thus, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5. A limit order with the price equal to a Take Profit level is sent to Cboe FX at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Cboe FX. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on MetaTrader 5 side. Limit mode allows to protect against slippage, as a limit order is sent to Cboe FX system with a specified price rather than a market order for execution by the current price.

Order type	Execution
	LimitOrdersCoverage=Market Limit orders and Take Profit orders are processed on the MetaTrader 5 platform side. Once they trigger, an appropriate market order is sent to Cboe FX.
Buy Stop Sell Stop Stop Loss Stop Out	Processed on the MetaTrader 5 platform side until their stop price is reached. After a stop order is activated, an appropriate market order will be sent to Cboe FX system.
Buy Stop Limit Sell Stop Limit	Processed on the MetaTrader 5 side. Upon order activation, an appropriate limit order is created in MetaTrader 5, and this order is then processed in accordance with the value of the LimitOrdersCoverage parameter.
• Slippages are possible when transmitting market orders to ECN as a result of pending orders activation. The price on the Cboe FX server may change at the moment when the price has reached an order activation level but the market transaction has not yet been performed. • Due to the fact that pending orders are not transmitted to ECN, they will not be visible in the Depth of Market.	

Gateway Setup

To start working, add the new gateway configuration:

MetaTrader 5 trading platform

Gateway

Common Groups Symbols Translations Parameters Timeouts Information

 This gateway allows connecting to a remote Hotspot server

☒ Enable

Name: Cboe FX Gateway ID: 230

Module: CboeFXGateway64 Trade and Quotes

Trading server: 209.191.250.157:8029

Trading login: 16699 Password:

[Show additional network settings](#)

OK Cancel Help

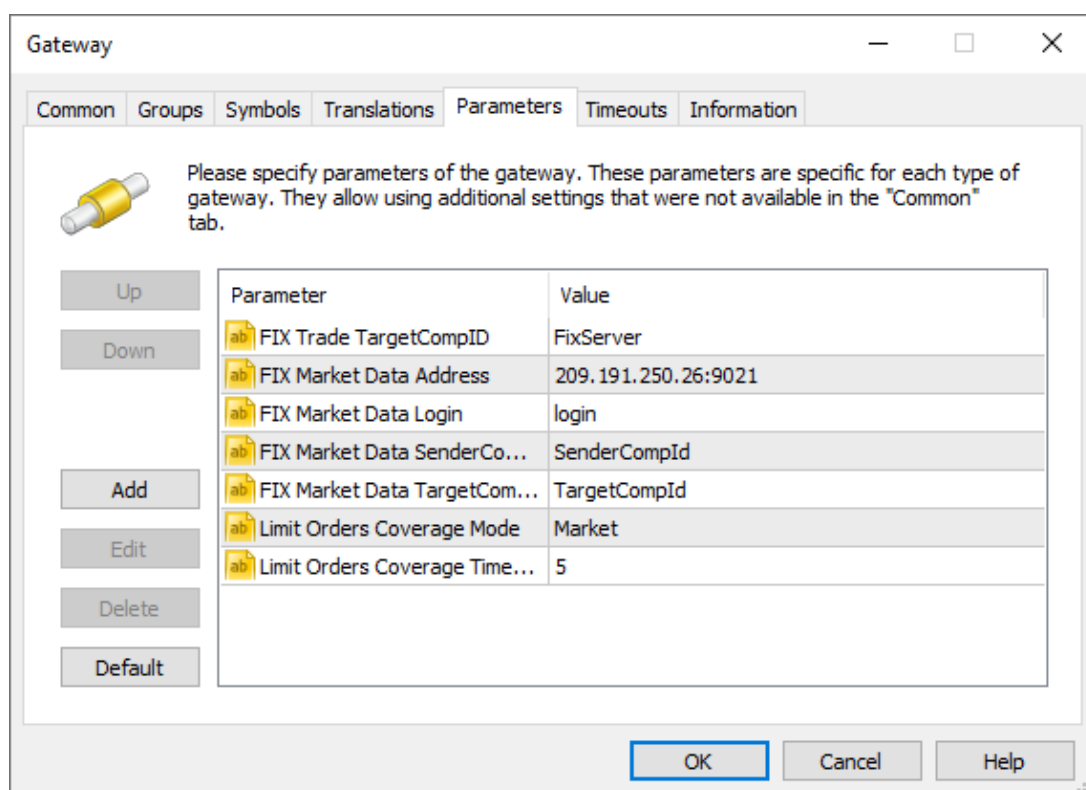
Set the following parameters on the "Common" tab:

- **Module** Cboe FXGateway64. Agree to set the default settings after selecting the module.
- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Trade server** Cboe FX server IP-address and port, where trade requests are processed. They are designated as "ip" and "port" parameters in "Orders" section in the connection data provided by Cboe FX.
- **Trading login** a login for connection to the Cboe FX server. Designated as "username" parameter in "Orders" section in the connection data provided by Cboe FX.
- **Trading password** a password for connection to the Cboe FX server. Designated as "password" parameter in "Orders" section in the connection data provided by Cboe FX.

- ID value must be unique in the field of manager logins and gateway identifiers.
- The data for connection to the Cboe FX server, where trade requests are processed, is provided with the agreement.

Other parameters are set similarly to other gateways. Default values are used in most cases.

Now, go to the "Parameters" tab.



Specify the following parameters values here:

- **FIX Trade TargetCompID** standard parameter of FIX messages header used for trading messages recipient identification. This is Cboe FX company identifier in the trade flow.
- **FIX Market Data Address** address and port of the Cboe FX server that streams quotes. They are provided as "ip" and "port" parameters in "Market Data" section in the connection data provided by Cboe FX.
- **FIX Market Data Login** login for connection to the quoting server. Provided as "username" parameter in "Market Data" section in the connection data provided by Cboe FX.
- **FIX Market Data SenderCompID** standard parameter of FIX message header used for the identification of the quote message sender. This is your company identifier in the quote stream.
- **FIX Market Data TargetCompID** standard parameter of FIX message header used for the identification of the quote message recipient. This is the Cboe FX identifier in the quote stream.
- **Limit Orders Coverage Mode** the mode of processing of Limit and Take Profit orders by the gateway. This parameter simplifies configuration of trade requests routing to the gateway, since there is no need to create a separate rule for routing the appropriate order types. Three processing modes are available:
 - ◆ **Market** limit and Take Profit orders are processed on the MetaTrader 5 platform side. Once the order triggers, an appropriate market order is sent to Cboe FX.
 - ◆ **Limit** limit and Take Profit orders are processed on the MetaTrader 5 side.

Once a limit order is triggered, an equivalent limit order is sent to Cboe FX. That order has a short action time specified in Limit Orders Coverage ModeTimeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Cboe FX. Thus, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5.

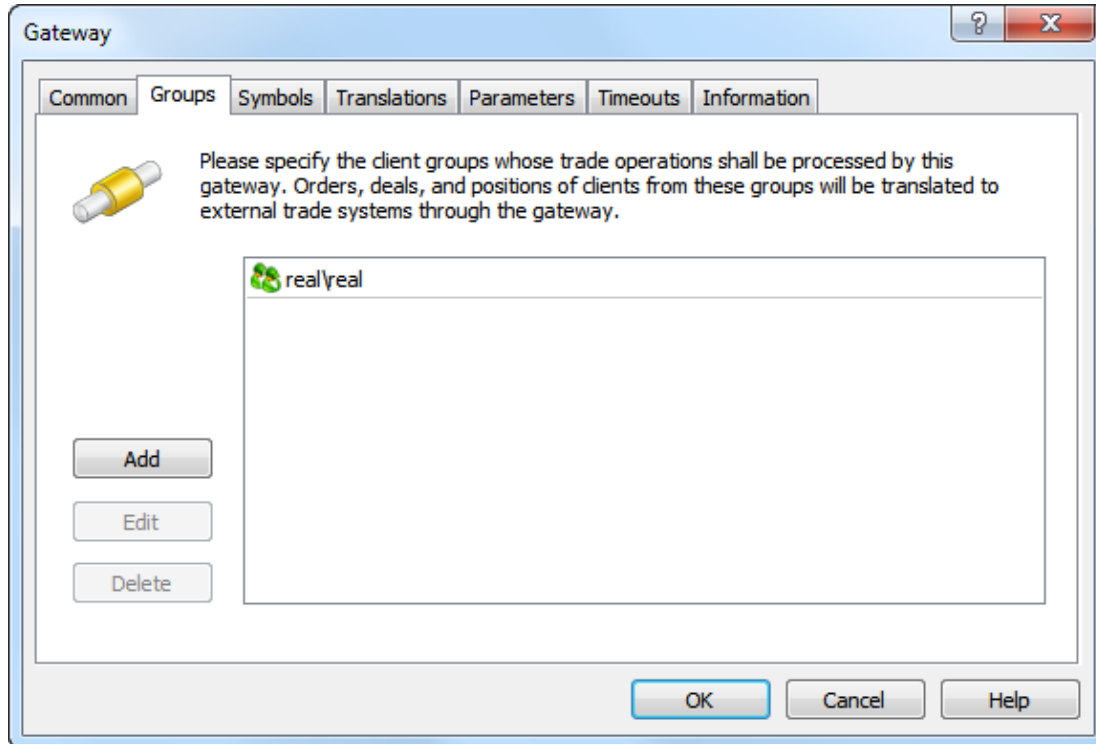
A limit order with the price equal to a Take Profit level is sent to Cboe FX at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from Cboe FX. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on the MetaTrader 5 side.

The Limit mode allows to protect against slippage, as a limit order is sent to the Cboe FX system with a specified price rather than a market order for execution by the current price. In addition, this mode allows you not to reserve margin on a client's account before sending an order to Cboe FX.

- ♦ **Gateway** limit orders are processed on the Cboe FX side. Once a limit order has been placed by a client, an appropriate order is sent to Cboe FX. Take Profit orders are processed the same way as in the Limit mode.
- **Limit Orders Coverage Timeout** expiry of Limit Orders that are sent to Cboe FX in the Limit mode. Specified in seconds. The default value is 5.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

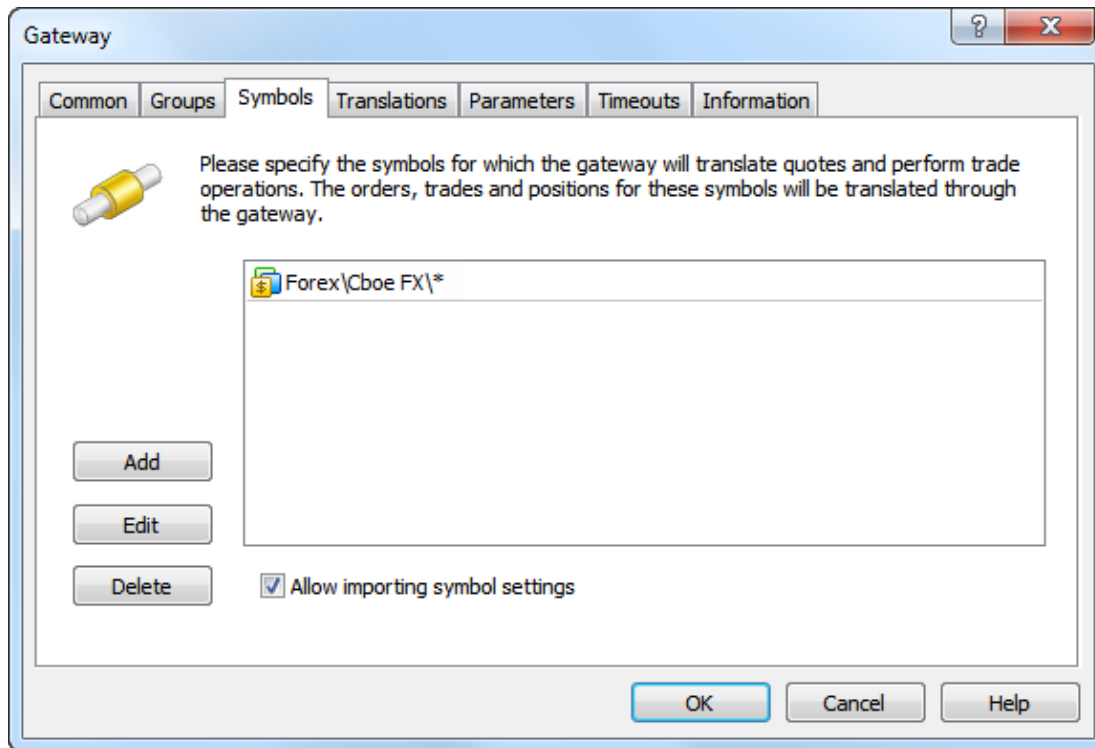
- The data for connection to the Cboe FX quote server, as well as identifiers for trading and quoting connection are provided by Cboe FX with the agreement.
- After editing the market depth settings the main trading server must be restarted to let the changes take effect.
- The maximum market depth provided by Cboe FX is equal to 11.

The next stage is to specify the groups of the clients, whose requests will be processed via the MetaTrader 5 Cboe FX Gateway.



MetaTrader 5 trading platform

Then configure the list of symbols, according to which the gateway will process trade operations and feed quotes.



Make sure to enable the option "Allow importing symbol settings", as the gateway adds necessary symbols and changes their settings by itself.

- The symbols imported by the gateway are put to the "\Preliminary\CboeFX" symbols subgroup. All symbols have trading ability disabled. System administrator must relocate imported symbols to the proper subgroup and allow trading for them.
- After symbols are transferred and trading is enabled, the main trading server must be restarted.
- In case the gateway transmits configuration for the symbol that is already present in the platform, the configuration is updated. In this case the symbol is not transferred and its trading ability is not turned off.
- The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

Configuring trade requests routing

Configure the routing to let the clients requests to be transmitted to the Cboe FX gateway. To do this, add a routing rule to the relevant [MetaTrader 5 Administrator](#) section.

Select "Process to dealers" as an action in common settings. Assign this routing rule for all requests and orders. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway. Add the previously created gateway at the Dealers tab.

MetaTrader 5 trading platform

Routing: HotspotFXi Gateway Execution

Common

Dealers



Please specify dealers who will process requests that meet the rule conditions.

Login	Name
 2	Cboe FX Gateway

Add

Edit

Delete

Routing

Common

Dealers



Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: Cboe FX Gateway Executionon

Perform action: Process to dealers ☐ skip this rule if no dealers online

Where request is: All

Where order is: All

Where conditions are:

Type	Condition	Value
 Client group	Equal (=)	real\real

Add

Edit

Delete

OK

Cancel

Help

MetaTrader 5 trading platform

After the correct execution of the steps described above, Cboe FX gateway will be ready for use.

Getting Started

After it has been launched, the gateway will connect to the liquidity provider:

- to receive symbols and the price stream
- to transmit trade requests and to receive notifications about executed trades

Connection to the liquidity provider is performed via FIX protocol. All incoming and outgoing FIX messages are stored on the disk at the \gateway\Cboe FX\fix\ directory of the history server. The gateway follows the Cboe FX ECN schedule – it will close connection with the ECN after the end of the working session, and will establish the new connection after the beginning of the next session.

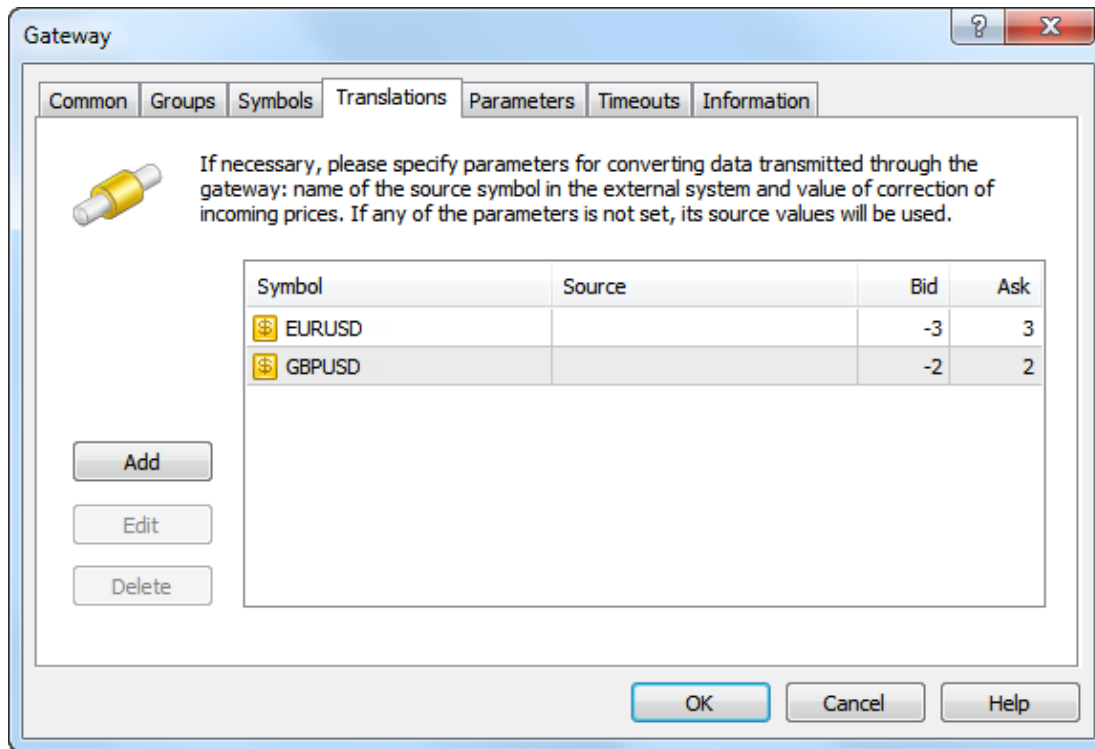
The result of the gateway operation is reflected in its journal.

Getting Profit from Agency Work

The gateway streams prices from the external trading system and controls symbol settings. The standard feature of gateways is the ability to change the quotes and market depth transmitted to the clients from an external system.

The gateway received prices from Cboe FX ECN and transmits them to clients taking into account transformation settings. Clients perform trading operations using converted prices. However, while processing trading operations on the gateway and their transmission to ECN, initial, not converted prices are automatically used.

Thus, by increasing the selling price and reducing the purchase price ("price spreading") a brokerage company receives its profit share from each deal performed at Cboe FX. The correction value is set separately for each symbol on the "Translations" tab:



Here you can configure matching of symbol names used in Cboe FX with the names used in your MetaTrader 5 platform. For example, if the symbol name in Cboe FX is EURUSDHS, and it is called EURUSD in the MetaTrader 5, enter EURUSD in the "Symbol" field and EURUSDHS in the "Source" field.

The above screenshot shows price conversion: at every tick, the Bid price will be reduced by 3 points, and the Ask price will be increased by 2 points. For GBPUSD Bid, price is decreased by 2 points, and Ask is increased by 2 points.

Prices Round Off

During the gateway's operation, accuracy of quotes (decimal places) passed for some symbol may change in the external trading system. Decrease in price accuracy at the external trading system's side does not affect the gateway's operation. It still transmits prices with less accuracy. However, if the number of decimal places at the external system's side increases, the gateway starts rounding off the passed prices.

Suppose that the accuracy of quotes has changed from 4 to 5 digits. Obtained five-digit quotes are rounded up by the gateway and used for creating the Market Depth. The round off is always performed in broker's favor. Thus, buy requests of 1.23447, 1.23441 are rounded up to 1.2345, while sell ones of 1.23447, 1.23441 are rounded down to 1.23440.

Changes in symbol price accuracy are recorded in the gateway journal.

MetaTrader 5 Trading Platform / Platform Components / Gateways /
LMAX Global

MetaTrader 5 Gateway to LMAX Global

LMAX Global is a large London-based multilateral trading facility (MTF) providing access to the institutional level liquidity. Being registered as MFT, LMAX applies exchange style trading model – the market depth is formed only by limit orders of major financial institutions. LMAX provides reliable execution within three milliseconds at more than 70 Forex and CFD symbols, transparent operation, anonymous trading and access to twenty market depth levels.

MetaTrader 5 Gateway to LMAX is a simple, fast and secure integration solution for brokers. The gateway provides institutional level liquidity when working in MetaTrader 5.

LMAX Global key advantages:

- Over 70 Forex and CFD symbols
- Metals, indices and commodities
- Average trading operation execution time – 4 ms
- Processing up to 40 000 orders per second
- No deviations, requotes and "last look" (canceling order at the last moment)
- Commission discounts depending on a broker's turnover
- LMAX Global trading systems (matching engine) are located in London (Equinix LD4 data center) and Tokyo (Equinix TY3 data center). The system can be accessed both via the Internet and via cross connect cable in the data center where LMAX's servers are hosted.

The complete list of advantages and available trading instruments can be found on the [official website](#).

Getting started with LMAX Global

First, connect LMAX Global using the contact details on the official website: <https://www.lmax.com/global/contact>. After concluding an agreement, you will receive all necessary data for connecting to LMAX Global trading system.

Order MetaTrader 5 Gateway to LMAX Global

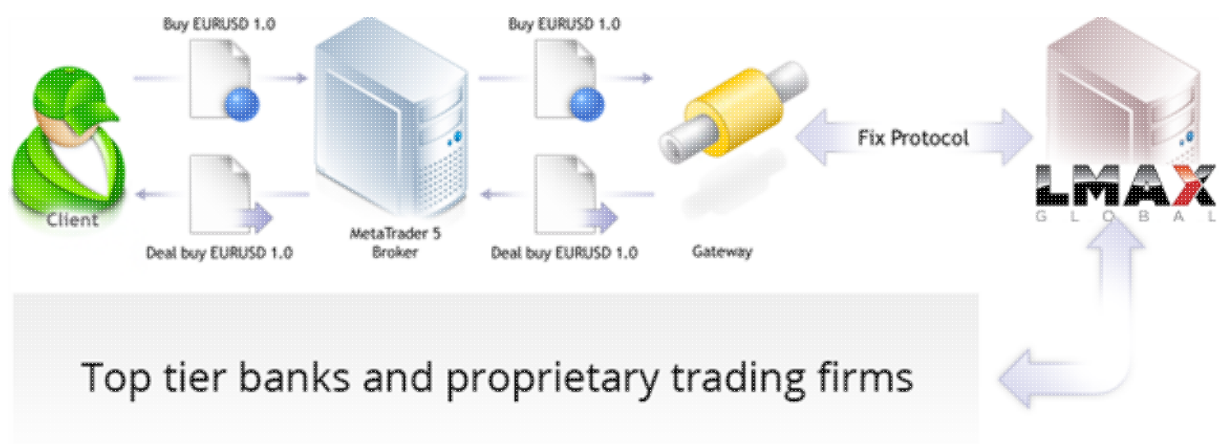
You can order the MetaTrader 5 LMAX Global Gateway through the "Buy" section of the technical support site. After you make the payment, the MetaTrader 5 platform license will be automatically updated via the LiveUpdate system and you will be able to use MetaTrader 5 LMAX Global Gateway without limitations. The executable module of the gateway is included into the platform package and can run in a demo mode, allowing you to perform 100 operations for a work session (until restart).

How the gateway works

MetaTrader 5 trading platform

MetaTrader 5 Gateway to LMAX Global is a separate LMAXGateway64.exe module that uses the MetaTrader 5 Gateway API for operation. The gateway works via two FIX channels: the first one is for trading, while the second one is for market data.

All orders entered in LMAX Global are transferred to the unified requests database. The system selects appropriate orders automatically executing opposite orders with matching parameters (symbol, price etc.)



Market data

MetaTrader 5 Gateway to LMAX Global automatically imports all the necessary symbols and processes their properties. The administrator only needs to perform a primary setup.

Price data is transmitted in real time. The gateway is capable of narrowing or expanding prices on the go: quotes can be converted according to the settings when passing them to the platform and then re-converted back to their original state when passing trade operations to LMAX Global. Detailed information on prices conversion is available below.

Trading operations

Orders are sent to MetaTrader 5 Gateway to LMAX for processing in accordance with the set routing rules. Processing of requests depends on the type of the order, as well as on the gateway configuration.

All traders' orders are sent to LMAX Global through a single broker account.

Order type	Execution
Market order	Delivered directly to LMAX Global as a market order.
Take Profit	Depends on <u>Limit Orders Coverage Mode</u> :

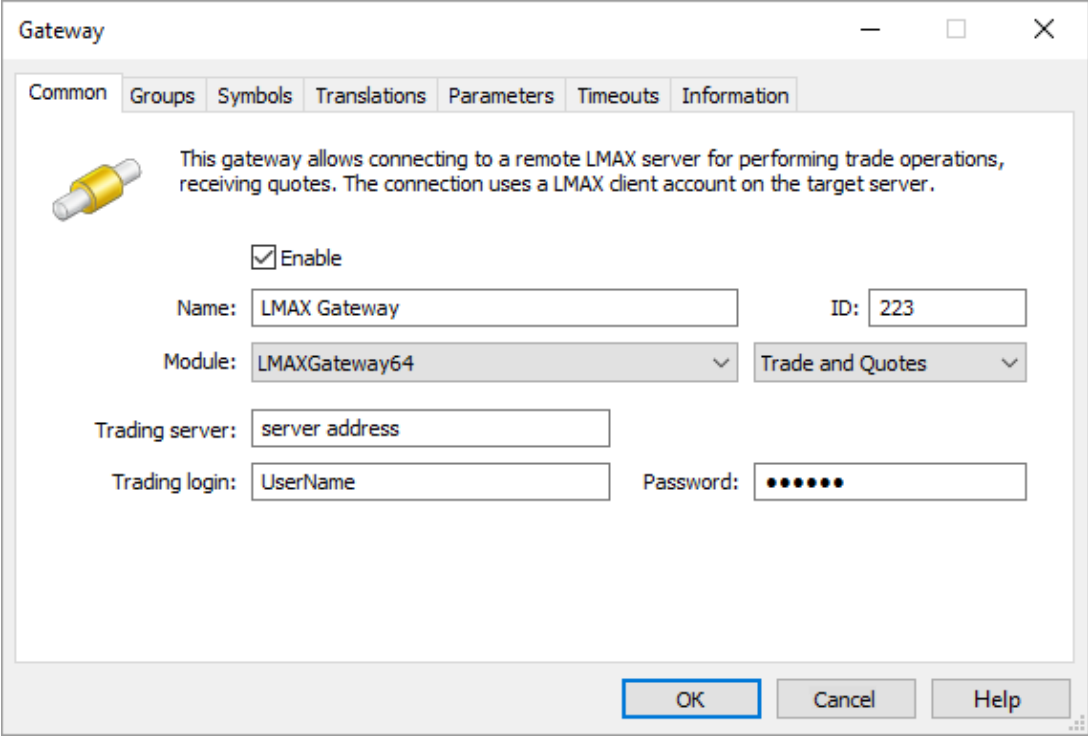
Order type	Execution
Buy Limit Sell Limit	Limit Orders Coverage Mode=Gateway (default) Limit orders are processed on the side of LMAX Exchnage. Once a limit order has been placed by a client, an appropriate order is sent to LMAX Global. There it is placed in the aggregate Depth of Market awaiting for an opposite request with the same price to appear. Thus, clients can see their requests in the Depth of Market in the client terminal in real time. Take Profit orders are processed the same way as in Limit mode. The LMAX Global system checks the availability of the required amount of funds to cover any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the work is carried out. <u>Margin reservation</u> of the clients should be configured for the appropriate order types in case Limit orders are directly delivered to an external system. By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client. Limit Orders Coverage Mode=Limit Limit Orders and Take Profit orders are processed on the side of the MetaTrader 5. Once a limit order is triggered, an equivalent limit order is sent with the expiration time at the end of the current trading day (intraday). Since the price specified in the order is already present in the market, the order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. In this case, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5. A limit order with the price equal to a Take Profit level is sent to LMAX

Order type	Execution
	Echange at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. The client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on MetaTrader 5 side. Limit mode allows you to protect against slippage, as a limit order is sent to LMAX Global system with a specified price rather than a market order for execution by the current price. Limit Orders Coverage Mode=Market Limit orders and Take Profit orders are processed on the MetaTrader 5 platform side. Once they trigger, an appropriate market order is sent to LMAX Global.
Buy Stop Sell Stop	Depends on <u>Stop Orders Coverage</u>: Stop Orders Coverage=Y Delivered directly to LMAX Global. Similarly to limit orders, margin reservation of the clients should be configured for the appropriate order types on MetaTrader 5 side in case of direct delivery of stop orders to LMAX Global. Stop Orders Coverage=N Processed on the MetaTrader 5 platform side until their stop price is reached. After a stop order is activated, an appropriate market order will be sent to the LMAX Global system.
Buy Stop Limit Sell Stop Limit	Depends on Stop Orders Coverage: Stop Orders Coverage=Y Delivered directly to LMAX Global. Similarly to limit orders, margin reservation of the clients should be configured for the appropriate order

Order type	Execution
	types on MetaTrader 5 side in case of direct delivery of stop limit orders to LMAX Global. Stop Orders Coverage=N Processed on the MetaTrader 5 side. Upon order activation, an appropriate limit order is created in MetaTrader 5, and this order is then processed in accordance with the value of the Limit Orders Coverage Mode parameter.
Stop Loss Stop Out	Processed on the MetaTrader 5 side. Once they trigger, an appropriate market order is sent to LMAX Global.
In case connection to LMAX Global server is lost, the application will try to restore it repeatedly. If pending orders were executed during the disconnection period, they will be updated in the MetaTrader 5 platform after successful reconnection.	

Gateway setup

To start working, add the new gateway configuration:



Gateway

Common Groups Symbols Translations Parameters Timeouts Information

This gateway allows connecting to a remote LMAX server for performing trade operations, receiving quotes. The connection uses a LMAX client account on the target server.

☒ Enable

Name: LMAX Gateway ID: 223

Module: LMAXGateway64 Trade and Quotes

Trading server: server address

Trading login: UserName Password:

OK Cancel Help

MetaTrader 5 trading platform

Set the following parameters on the "Common" tab:

- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Module** specify LMAXGateway64 and accept the default settings after the module selection.
- **Trading server** LMAX Global server IP-address and the port, where trade requests are processed. This information is provided by LMAX Global as parameters DNS and Port. An encrypted SSL connection to a server is established by default. If you need to establish an unencrypted connection, additionally specify the /nossll key in the address bar. For example:
10.123.100.17:10219/nossll.
If connection to a trade server is not required (the gateway only receives quotes from LMAX Global), leave this field empty.
- **Trading login** login for connecting the trade flow corresponding to UserName tag (553) in FIX protocol. Provided by LMAX Global.
- **Password** password for connecting the trade flow corresponding to Password tag (554) in FIX protocol. Provided by LMAX Global.

- ID value must be unique in the field of manager logins and gateway identifiers.
- The details for connection to the LMAX Global server where trade requests are processed will be provided during the agreement conclusion.

Now, go to the "Parameters" tab.

Gateway

Common Groups Symbols Translations Parameters Timeouts Information

Please specify parameters of the gateway. These parameters are specific for each type of gateway. They allow using additional settings that were not available in the "Common" tab.

Parameter	Value
FIX Trade TargetCompID	LMXBD
FIX Trade SenderCompID	CompId
FIX Market Data Address	fix-marketdata.london-demo.lmax.com:443
FIX Market Data Username	Username
FIX Market Data Password	Password
FIX Market Data TargetCo...	LMXBDM
FIX Market Data SenderCo...	CompId
FIX Market Data Log Enabled	No
Limit Orders Coverage Mode	Gateway
Stop Orders Coverage	No

Up Down Add Edit Delete Default OK Cancel Help

Specify the following parameters values here:

- **FIX Trade TargetCompID** standard parameter of the FIX messages heading used for trading messages recipient identification. This parameter is provided by LMAX Global as the value of TargetCompID (56).
- **FIX Trade SenderCompID** standard parameter of the FIX messages heading used for a data sender identification. This parameter is provided by LMAX Global as the value of SenderCompID (49).
- **FIX Market Data Address** IP address and port of the LMAX Global server, from which market (price) data is transmitted. This information is provided by LMAX Global as parameters DNS and Port. An encrypted SSL connection to a server is established by default. If you need to establish an unencrypted connection, additionally specify the /nossll key in the address bar. For example: 10.123.100.17:10219/nossll.
If connection to a quoting server is not required (the gateway only passes trading operations), leave this field empty.
- **FIX Market Data Username** login for connecting the trade flow corresponding to UserName tag (553) in FIX protocol. Provided by LMAX Global.
- **FIX Market Data Password** password for connecting the trade flow corresponding to Password tag (554) in FIX protocol. Provided by LMAX Global.
- **FIX Market Data TargetCompID** standard parameter of the FIX messages heading used for market messages recipient identification. This parameter is provided by LMAX Global as the value of TargetCompID (56).
- **FIX Market Data SenderCompID** standard parameter of the FIX messages heading used for a market data sender identification. This parameter is provided by LMAX Global as the value of SenderCompID (49).
- **FIX Market Data Log Enabled** if Yes, the gateway saves FIX connection (market flow) logs. Enable the parameter only in case of the gateway operation issues. The default value is No.
- **Limit Orders Coverage Mode** the mode of processing of Limit and Take Profit orders by the gateway. This parameter simplifies configuration of trade requests routing to the gateway, since there is no need to create a separate rule for routing the appropriate order types. Three processing modes are available:
 - ◆ **Market** limit and Take Profit orders are processed on the MetaTrader 5 platform side. If an order is activated, an appropriate market order is sent to the LMAX Global system.
 - ◆ **Limit** limit and Take Profit orders are processed on the MetaTrader 5 side.
Once a limit order is triggered, an equivalent limit order is sent to LMAX Global. The order remains valid till the end of the current trading day (intraday). Since the price specified in the order is already present in the market, the order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. In this case, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5.
A limit order with the price equal to a Take Profit level is sent to LMAX Global at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order

will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. The client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on the MetaTrader 5 side.

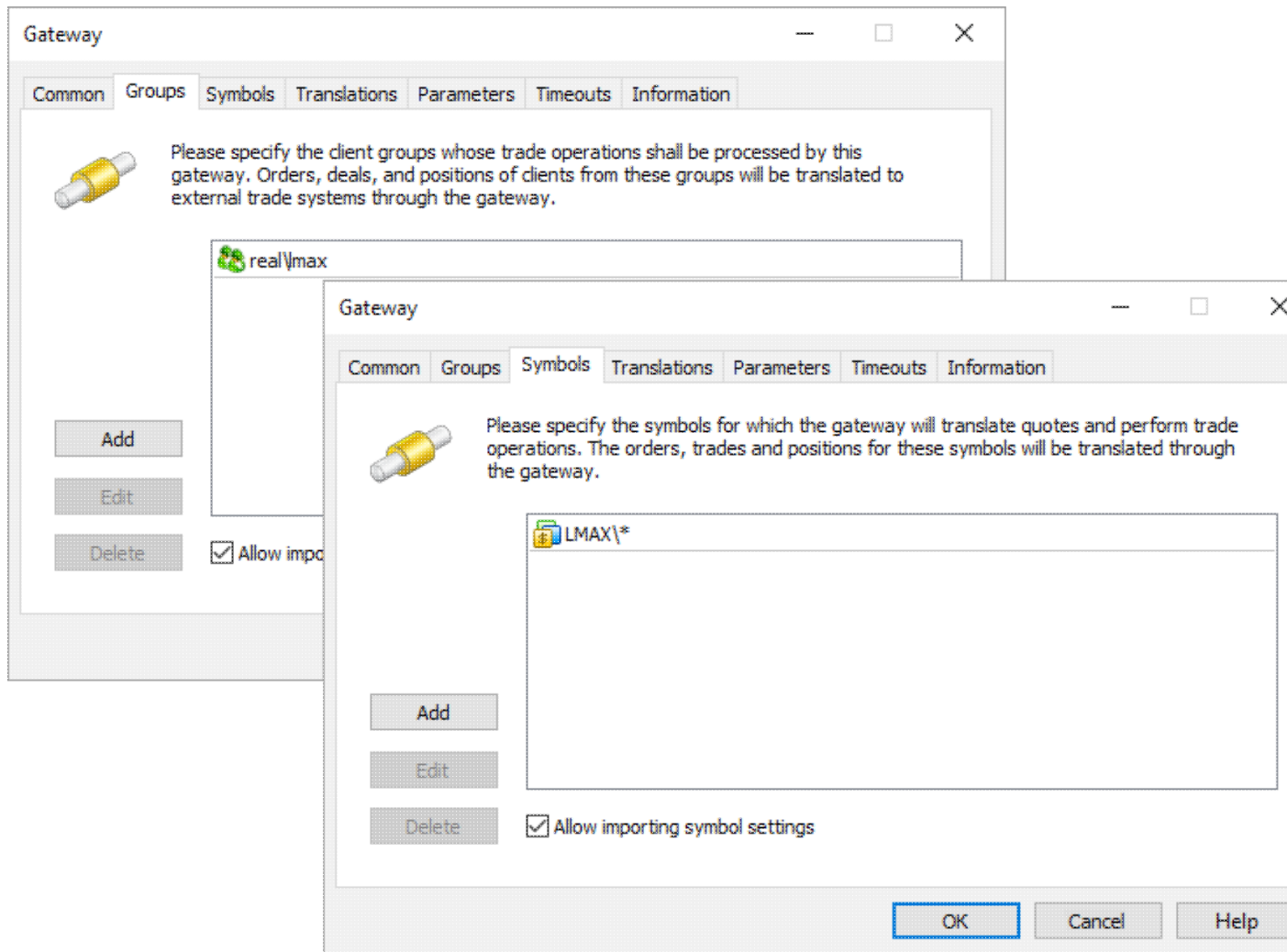
Limit mode allows to protect against slippage, as a limit order is sent to the LMAX Global system with a specified price rather than a market order for execution by the current price. In addition, this mode allows you not to reserve margin on a client's account before sending an order to LMAX Global.

- ◆ **Gateway** limit orders are processed on the LMAX Global side. Once a limit order has been placed by a client, an appropriate order is sent to LMAX Global. Take Profit orders are processed the same way as in the Limit mode.
- **Stop Orders Coverage** mode of handling stop and stop limit orders. In case of 'Y' value, these order types will be transferred to LMAX Global directly. In case of 'N' value, the orders will be processed inside MetaTrader 5 platform until their stop price is reached. After a stop order is activated, an appropriate market order will be sent to the LMAX Global system. After a stop limit order has been activated, a limit order is created, which will be processed according to Limit Orders Coverage Mode parameter value.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

- In no circumstances it is allowed to change operations transfer mode while in operation. Changing the mode is allowed only after all client positions are closed.
- Margin reservation of the clients should be configured for the appropriate order types in case Limit, Stop and/or Stop Limit orders are directly transferred to an external system.

The next stage is to specify the groups of the clients, whose requests will be processed via the MetaTrader 5 Gateway to LMAX, as well as symbols, by which the gateway processes trading operations and broadcasts quotes.

MetaTrader 5 trading platform



MetaTrader 5 trading platform

The gateway supports more than 100 LMAX instruments. Make sure to enable "Allow importing symbol settings" option. The gateway imports symbols from LMAX Global to Symbols/Preliminary/LMAX directory of the MetaTrader 5 and manages their parameters.

- Initially, all imported symbols have trading ability disabled. System administrator must relocate imported symbols to the proper subgroup and allow trading for them.
- After the symbols are relocated and trading abilities are enabled, the main trading server must be restarted.
- Only Market or Exchange execution can be used for the symbols. In all other cases, operations will be rejected.
- In case the gateway transmits configuration for the symbol that is already present in the platform, the configuration is updated. In this case the symbol is not transferred and its trading ability is not turned off.
- The gateway supports conversion of symbols and quotes. For details, please view the Symbol and Price Translation section.

Margin setup

Margin reservation of the clients should be configured for the appropriate order types in case Limit, Stop and/or Stop Limit orders are directly transferred to an external system.

The external system checks the funds that are necessary to provide any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the work is carried out.

By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client.

After an order has been transferred to the external system, MetaTrader 5 platform is not able to check the client's margin sufficiency any more. After the order has been executed in the external system, the gateway cannot ignore that fact. Therefore, the appropriate trading operation is performed in the platform.

Set non-zero coefficients for the orders directly transferred to the external trading system in symbol settings for the appropriate symbols:

MetaTrader 5 trading platform

Symbol: EURUSD

Common Quotes Trade Execution Margin **Margin Rates** Swaps Sessions

 Please specify margin rates for different types of trade operations.

Liquidity margin rate:

Margin rates:

	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
 Buy	1.0000000	1.0000000	0.0000000	0.0000000
 Sell	1.0000000	1.0000000	0.0000000	0.0000000
Maintenance margin				
 Buy	0.0000000	0.0000000	0.0000000	0.0000000
 Sell	0.0000000	0.0000000	0.0000000	0.0000000

OK Cancel Help

Configuring trade requests routing

Configure the routing to let the client requests to be transmitted to the LMAX Global gateway. To do this, add a routing rule to the relevant MetaTrader 5 Administrator section.

Select "Process to dealers" as an action in common settings. Assign this routing rule for all requests and orders. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway.

The screenshots below all orders created by users in the real\lmax* groups by symbols from the LMAX* section will be sent to the gateway for processing.

MetaTrader 5 trading platform

Routing

Common

Dealers



Please specify dealers who will process requests that meet the rule conditions.

Login

223

Name

LMAX Exchange

Add

Edit

Delete

Routing

Common

Dealers



Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name:

LMAX Exchange

Perform action:

Process to dealers

☐ skip this rule if no dealers online

Where request is:

All

Where order is:

All

Where conditions are:

Type	Condition	Value
 Symbols	Equal (=)	LMAX*
 Client group	Equal (=)	real\max*

Add

Edit

Delete

OK

Cancel

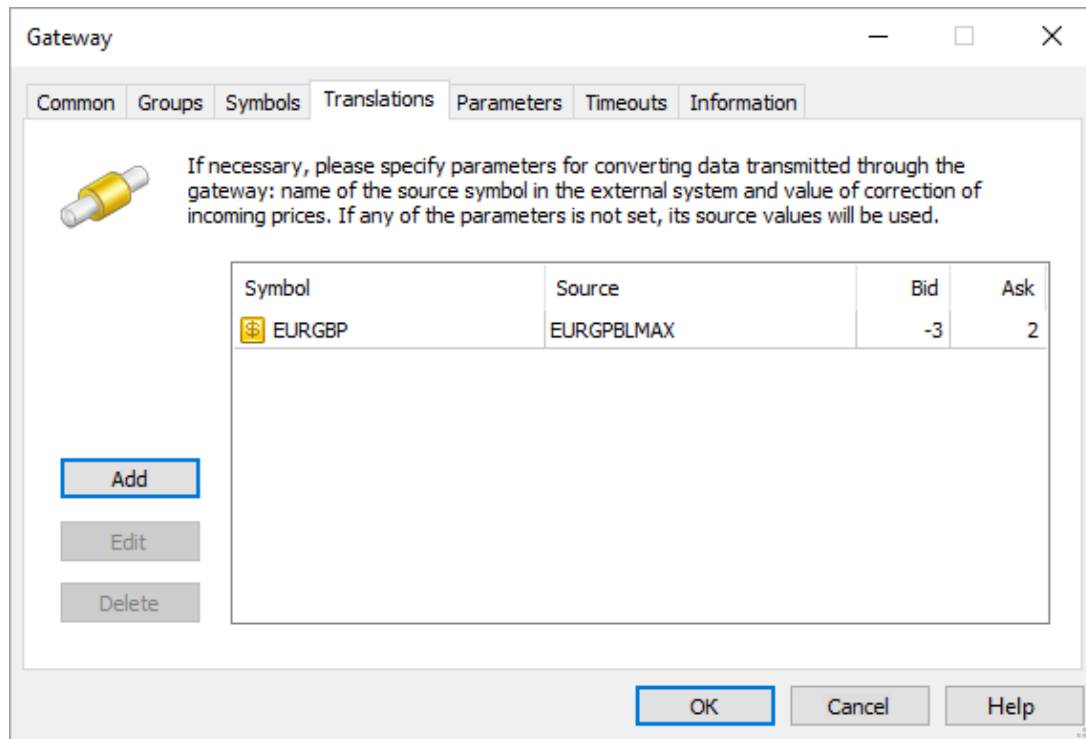
Help

MetaTrader 5 trading platform

After the correct execution of the steps described above, the gateway will be ready for work.

Changing Symbols Names and Markups

The gateway receives the prices from LMAX Global and transmits them to clients considering markups. Thus, a brokerage company receives its profit share from each deal performed at LMAX Global. Markup values are set separately for Bid and Ask prices by each symbol:



MetaTrader 5 trading platform

Besides, you can configure matching of symbol names used in LMAX Global with the names used in your MetaTrader 5 platform. For example, if the symbol name in LMAX Global is EURGBPLMAX, and it is called EURGBP in the MetaTrader 5, enter EURGBP in the "Symbol" field and EURGBPLMAX in the "Source" field.

The above screenshot shows price conversion: at every tick, the Bid price will be reduced by 3 points, and the Ask price will be increased by 2 points. Below is a schematic example of the correction:

LMAX Global	>>>	ask price	EURGBP 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	ask price	EURGBP 0.83006	>>>	Client terminal
Client terminal	>>>	buy limit	EURGBP 0.83006	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit	EURGBP 0.83004	>>>	LMAX Global
LMAX Global	>>>	buy limit execution	EURGBP 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit execution	EURGBP 0.83006	>>>	Client terminal

A broker gains 2 pips of profit in this example. The price is sent to the client terminal only after the correction, so the clients work only with the corrected prices. If no correction is set for a symbol, the client will work with the original prices submitted by LMAX Global.

Prices Round Off

During the gateway's operation, accuracy of quotes (decimal places) passed for some symbol may change in the external trading system. Decrease in price accuracy at the external trading system's side does not affect the gateway's operation. It still transmits prices with less accuracy. However, if the number of decimal places at the external system's side increases, the gateway starts rounding off the passed prices.

Suppose that the accuracy of quotes has changed from 4 to 5 digits. Obtained five-digit quotes are rounded up by the gateway and used for creating the Market Depth. The round off is always performed in broker's favor. Thus, buy requests of 1.23447, 1.23441 are rounded up to 1.2345, while sell ones of 1.23447, 1.23441 are rounded down to 1.23440.

Changes in symbol price accuracy are recorded in the gateway journal.

Volume conversion for different contract sizes

The gateway automatically converts the volume of orders sent to the exchange if the symbol's contract size in MetaTrader 5 differs from its contract size on the LMAX side. For example, if the contract size on the platform side is 10,000 while LMAX provides

MetaTrader 5 trading platform

the size of 100,000, the 1-lot order will be sent to the exchange as a 0.1-lot order.

Before changing the contract size on the platform side, make sure that you do not have open positions or orders for LMAX symbols. Restart the gateway after changing the contract size. After that an entry about mismatching contract sizes will be printed to the gateway journal, such as:

Gateway symbol AUDCHF LMAX contract sizes are different at MT5 (500000.00) and LMAX

Fill Policy and Order Expiration

When sending an order from the MetaTrader 5 client terminal, traders can set the order execution policy (FOK, IOC or Return) and expiration time (Good Till Canceled or Today, expiration by a certain date or time is not supported). In LMAX Global trading system, these parameters are combined into one Time in Force. Therefore, the gateway performs the following changes when passing the orders:

- If FOK or IOC fill policy is set for an order, the same policy is applied in LMAX Global system.
- If a market order with Return fill policy is placed, it will be passed to LMAX Global system in Good Till Canceled (GTC) mode.
- If a limit order with Return fill policy is placed, expiration time from MetaTrader 5 (Good Till Canceled or Today) is inserted into the appropriate order parameter in LMAX Global system.
- If a limit order is activated in MetaTrader 5 platform and the gateway works in the mode of passing limit orders (Limit Orders Coverage Mode = Limit), a limit order with Today execution policy is passed to LMAX Global system.
- If a position take profit is activated in MetaTrader 5 platform and the gateway works in the mode of passing limit orders (Limit Orders Coverage Mode = Limit or Limit Orders Coverage Mode = Gateway), a limit order with Today execution policy is passed to LMAX Global system.

Additional

For the proper operation of the gateway, the time on the computer where the gateway is running, should be synchronized with the master time source server. If time is not configured, the gateway will print the following error into the journal:

2016.09.30 13:34:34.576 Gateway session logout (SendingTime accuracy problem.)
--

If your computer is not a member of a domain, you can synchronize your computer clock with an Internet time server in Control Panel - Date and Time - Internet Time. Otherwise you can start time synchronization using the command line:

net time /domain

Supported Assets

Currently, the gateway supports the following trading symbols:

Imported symbol name	Symbol name on the LMAX side
----------------------	------------------------------

MetaTrader 5 trading platform

AUDCADLMAX	AUD/CAD
AUDCHFLMAX	AUD/CHF
AUDJPYLMAX	AUD/JPY
AUDNZDLMAX	AUD/NZD
AUDUSDLMAX	AUD/USD
CADCHFLMAX	CAD/CHF
CADJPYLMAX	CAD/JPY
CHFJPYLMAX	CHF/JPY
EURAUDLMAX	EUR/AUD
EURCADLMAX	EUR/CAD
EURCHFLMAX	EUR/CHF
EURCZKLMAX	EUR/CZK
EURDKKLMAX	EUR/DKK
EURGBPLMAX	EUR/GBP
EURHKDLMAX	EUR/HKD
EURHUFLMAX	EUR/HUF
EURJPYLMAX	EUR/JPY
EURMXNLMAX	EUR/MXN
EURNOKLMAX	EUR/NOK
EURNZDLMAX	EUR/NZD
EURPLNLMAX	EUR/PLN
EURRUBLMAX	EUR/RUB
EURSEKLMAX	EUR/SEK
EURSGDLMAX	EUR/SGD
EURTRYLMAX	EUR/TRY
EURUSDLMAX	EUR/USD
EURZARLMAX	EUR/ZAR
GBPAUDLMAX	GBP/AUD
GBPCADLMAX	GBP/CAD
GBPCHFLMAX	GBP/CHF

Imported symbol name	Symbol name on the LMAX side
GBPCZKLMAX	GBP/CZK
GBPDKKLMAX	GBP/DKK
GBPHKDLMAX	GBP/HKD
GBPHUFLMAX	GBP/HUF
GBPJPYLMAX	GBP/JPY
GBPMXNLMAX	GBP/MXN
GBPNOKLMAX	GBP/NOK
GBPNZDLMAX	GBP/NZD
GBPPLNLMAX	GBP/PLN
GBPSEKLMAX	GBP/SEK
GBPSGDLMAX	GBP/SGD
GBPTRYLMAX	GBP/TRY
GBPUSDLMAX	GBP/USD
GBPZARLMAX	GBP/ZAR
NOKSEKLMAX	NOK/SEK
NZDCADLMAX	NZD/CAD
NZDCHFLMAX	NZD/CHF
NZDJPYLMAX	NZD/JPY
NZDSGDLMAX	NZD/SGD
NZDUSDLMAX	NZD/USD
USDCADLMAX	USD/CAD
USDCHFLMAX	USD/CHF
USDCNHLMAX	USD/CNH
USDCZKLMAX	USD/CZK
USDDKKLMAX	USD/DKK
USDHKDLMAX	USD/HKD
USDHUFLMAX	USD/HUF
USDILSLMAX	USD/ILS
USDJPYLMAX	USD/JPY
USDMXNLMAX	USD/MXN

HSILMAX	Hong Kong 50
J225LMAX	Japan 225
SPN35LMAX	MetaTrader 5 trading platform Spain 35
Imported symbol name	Symbol name on the LMAX side
UK100LMAX	UK 100
SPXLMAX	US SPX 500
SPXmLMAX	US SPX 500 (Mini)
NDXLMAX	US Tech 100
NDXmLMAX	US Tech 100 (Mini)
WS30LMAX	Wall Street 30
WS30mLMAX	Wall Street 30 (Mini)
XBTUSDLMAX	XBT/USD
XETUSDLMAX	XET/USD
XBNUSDLMAX	XBN/USD
XLCUSDLMAX	XLC/USD
XRPUSDLMAX	XRP/USD
XBNJPYLMAX	XBN/JPY
XBTJPYLMAX	XBT/JPY
XETJPYLMAX	XET/JPY
XLCJPYLMAX	XLC/JPY
XRPJPYLMAX	XRP/JPY

MetaTrader 5 Trading Platform / Platform Components / Gateways /
FXCM PRO

MetaTrader 5 Gateway to FXCM PRO

FXCM PRO is the institutional arm of FXCM, the largest broker in the United States. FXCM Pro trading system serves 200 000 traders all over the world conducting over 500 000 trades per day with the average daily volume of \$14 billion.

FXCM Pro offers brokers the transparent service system based on agency commission with fully anonymous trading. The system provides access to **40 trading symbols** including currency pairs, indices, metals, energy and Treasury bonds. FXCM Pro offers customized pricing for each instrument and account. Based on customer mandate, FXCM's liquidity management team can source the most efficient liquidity providers and partnering venues to match your needs, be it single tickets on large orders, stickier pricing intraday on metals, and much more.

FXCM Pro trading systems (matching engine) are located in New York (Equinix NY7 data center) and Tokyo (Equinix TY3 data center). The system can be accessed both via the Internet and cross connect in FXCM data center.

Two Operation Modes: Wholesale and Liquidity Solutions

You can select one of the two operation modes depending on your needs.

Wholesale	Liquidity Solutions
Wholesale is optimal for retail brokers looking to leverage FXCM's scale. Brokers are able to trade in FXCM Pro using large volumes, thus increasing the acceptable volume of non-hedged positions of their traders. Cross Collateralization of FX and CFD Trading All customer positions are netted into one account, allowing for more efficient use of corporate cash. Credit line Through a formal review process FXCM can extend credit in the form of NOP (Net Open Position) in a	Liquidity Solutions mode provides liquidity for your traders and ability to send their trading operations to FXCM Pro. There are two distinctly different liquidity solutions trading via a single account in FXCM Core system and trading in FXCM Pro ECN where an individual account is allocated to each trader. FXCM Core This technology suite supports nearly 200 000 traders globally, with over 500 000 trades done daily as well as over \$14 billion in daily volumes. • NDD execution model is agency based and truly anonymous. Banks and financial institutions comprising

Wholesale	Liquidity Solutions
traditional bank Prime Brokerage manner with flexible settlement terms. No fees on small tickets FXCM provides all currency pairs on offer at 1K ticket sizes with no additional fees. Execution suitable for automated trading FXCM's FX NDD model, and enhanced index and commodity CFD model is ideal for automated traders seeking a better execution experience.	FXCM's matching engine have no information about their counterparty. Pending orders are stored on FXCM servers and sent to the matching system only if their execution conditions are fulfilled. • Small ticket advantage FXCM provides all currency pairs on offer at 1K ticket sizes with no additional fees. • Multi asset clearing all assets are traded on one trading account via a single API. FXCM Pro ECN institutional API service This technology offers professional and institutional users the ability to tailor pricing on a per instrument, per account basis. Based on customer mandate, FXCM's liquidity management team can source the most efficient liquidity providers and partnering venues to match your needs, be it single tickets on large orders, stickier pricing intraday on metals, and much more. FXCM also provides cross collateralization capabilities for all trading accounts.

Getting Started with FXCM Pro

First, connect FXCM Pro to find out the details and conclude an agreement.

Brandon Mulvihill
Global Head, FXCM Pro

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Head of EMEA, FXCM Pro

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MetaTrader 5 trading platform

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Mobile: +44 7 540 789
656

Email:
chossain@fxcmpro.com

After concluding an agreement, you will receive all necessary data for connecting to FXCM Pro trading system.

Order MetaTrader 5 Gateway to FXCM Pro

You can order the MetaTrader 5 FXCM Pro Gateway through the "Buy" section of the technical support site. After you make the payment, the MetaTrader 5 platform license will be automatically updated via the LiveUpdate system and you will be able to use MetaTrader 5 Gateway to FXCM Pro without limitations. The executable module of the gateway is included into the platform package and can run in a demo mode, allowing you to perform 100 operations for a work session (until restart).

How the Gateway Works

MetaTrader 5 Gateway to FXCM Pro is a separate FXCMProGateway64.exe module that uses the MetaTrader 5 Gateway API for operation. The gateway works via two FIX channels: the first one is for trading, while the second one is for market data.

All orders entered in FCXM Pro are transferred to the unified requests database. The system selects appropriate orders automatically executing opposite orders with matching parameters (symbol, price etc.)



Market Data

MetaTrader 5 Gateway to FXCM Pro automatically imports all the necessary symbols and processes their properties. The administrator only needs to perform a primary setup.

Price data is transmitted in real time. The gateway is capable of narrowing or expanding prices on the go: quotes can be converted according to the settings when passing them to the platform and then re-converted back to their original state when passing trade operations to FXCM Pro. Detailed information on prices conversion is available below.

Trading Operations

Orders are sent to MetaTrader 5 Gateway to FXCM Pro for processing in accordance with the set routing rules. Processing of requests depends on the type of the order, as well as the gateway configuration.

Order type	Execution
Market order	Delivered directly to FXCM Pro as a market order.
Take Profit	Depends on <u>Limit Orders Coverage Mode</u> :
Buy Limit	Limit Orders Coverage Mode=Gateway (default)
Sell Limit	Limit orders are processed on FXCM Pro side. Once a limit order has been placed by a client, an appropriate order is sent to FXCM Pro. There it is placed to the general queue awaiting for an opposite request with the same price to appear.
	Take Profit orders are processed the same way as in the Limit mode.
	The FXCM Pro system checks the availability of the required amount of funds to cover any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the work is carried out. <u>Margin reservation</u> of the clients should be configured for the appropriate order types in case Limit orders are directly delivered to an external system.

Order type	Execution
	By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client. Limit Orders Coverage Mode=Limit Limit Orders and Take Profit orders are processed on the side of the MetaTrader 5. Once a limit order is triggered, an equivalent limit order is sent to FXCM Pro. That order has a short action time specified in Limit Orders Coverage Timeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from FXCM Pro. Thus, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5. A limit order with the price equal to a Take Profit level is sent to FXCM Pro at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a limit order with a residual volume will be removed from FXCM Pro. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on MetaTrader 5 side. Limit mode allows to protect against slippage, as a limit order is sent to FXCM Pro system with a specified price rather than a market order for execution by the current price. Limit Orders Coverage Mode=Market Limit orders and Take Profit orders are processed on the MetaTrader 5 platform side. Once they trigger, an appropriate market order is sent to

Order type	Execution
	FXCM Pro.
Buy Stop Sell Stop	Depends on <u>Stop Orders Coverage</u>: Stop Orders Coverage=Y Delivered directly to FXCM Pro. Similarly to limit orders, margin reservation of the clients should be configured for the appropriate order types on MetaTrader 5 side in case of direct delivery of stop orders to FXCM Pro. Stop Orders Coverage=N Processed on the MetaTrader 5 platform side until their stop price is reached. After a stop order is activated, an appropriate market order will be sent to the FXCM Pro system.
Buy Stop Limit Sell Stop Limit	Depends on Stop Orders Coverage: Stop Orders Coverage=Y Delivered directly to FXCM Pro. Similarly to limit orders, margin reservation of the clients should be configured for the appropriate order types on MetaTrader 5 side in case of direct delivery of stop limit orders to FXCM Pro. Stop Orders Coverage=N Processed on the MetaTrader 5 side. Upon order activation, an appropriate limit order is created in MetaTrader 5, and this order is then processed in accordance with the value of the Limit Orders Coverage Mode parameter.
Stop Loss Stop Out	Processed on the MetaTrader 5 side. Once they trigger, an appropriate market order is sent to FXCM Pro.
In case connection to FXCM Pro server is lost, the application will try to restore it repeatedly. If pending orders were executed during the disconnection period, they	

will be updated in the MetaTrader 5 platform after successful reconnection.

Gateway Setup

To start working, add the new gateway configuration:

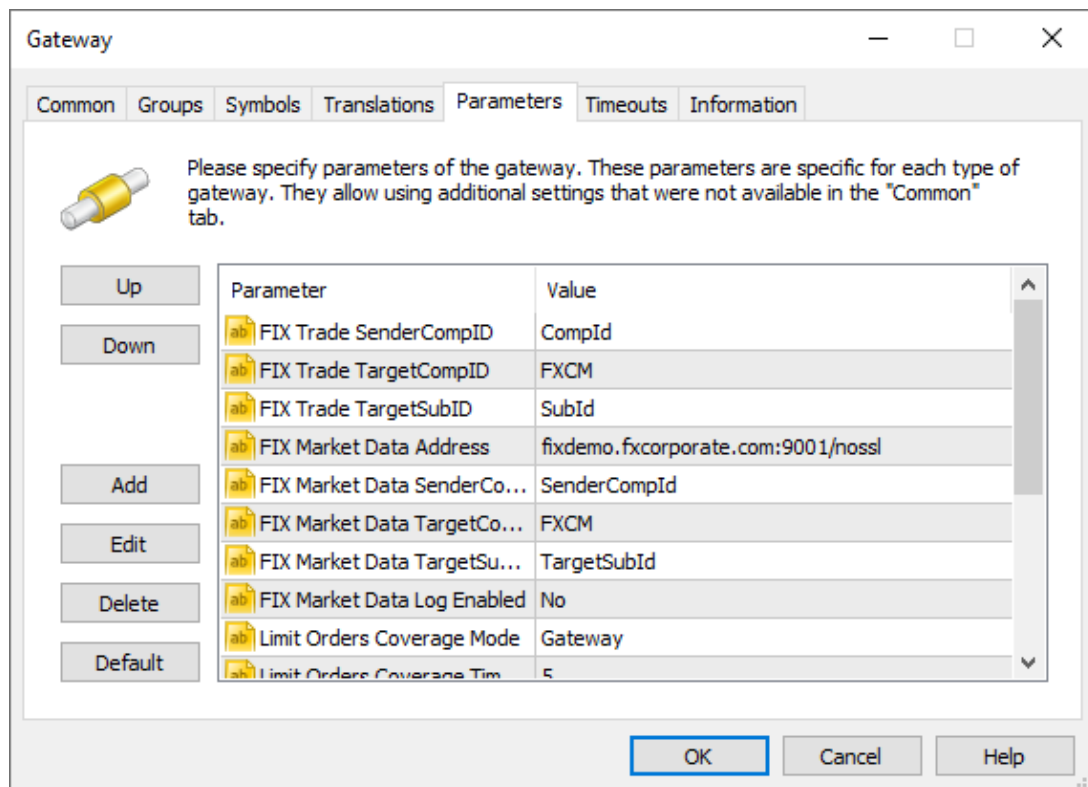
The screenshot shows the 'Gateway' configuration window with the 'Common' tab active. The window title is 'Gateway'. Below the title bar are tabs: 'Common', 'Groups', 'Symbols', 'Translations', 'Parameters', 'Timeouts', and 'Information'. The 'Common' tab contains a description: 'This gateway allows connecting to a remote FXCM Pro server for performing trade operations, receiving quotes. The connection uses a FXCM client account on the target server.' Below this is an 'Enable' checkbox which is checked. There are two rows of input fields: the first row has 'Name: FXCM Pro Gateway' and 'ID: 4'; the second row has 'Module: FXCMProGateway64' (selected from a dropdown) and 'Trade and Quotes' (selected from a dropdown). Below these are 'Trading server: trade server address' and 'Trading login: login'. The 'Password:' field is masked with dots. At the bottom right are 'OK', 'Cancel', and 'Help' buttons.

Set the following parameters on the "Common" tab:

- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Module** specify FXCMProGateway64 and accept the default settings after the module selection.
- **Trading server** FXCM Pro server IP-address and the port, where trade requests are processed. This information is provided by FXCM Pro as the SocketConnectHost and SocketConnectPort parameters. An encrypted SSL connection to a server is established by default. If you need to establish an unencrypted connection, additionally specify the /noss1 key in the address bar. For example: 10.123.100.17:10219/noss1.
- **Trading login** login for connecting the trade flow corresponding to UserName tag (553) in FIX protocol. Provided by FXCM Pro.
- **Password** password for connecting the trade and quote flow corresponding to Password tag (554) in FIX protocol. Provided by FXCM Pro.

- ID value must be unique in the field of manager logins and gateway identifiers.
- The details for connection to the FXCM Pro server where trade requests are processed will be provided during the agreement conclusion.

Now, go to the "Parameters" tab.



Gateway

Common Groups Symbols Translations **Parameters** Timeouts Information

Please specify parameters of the gateway. These parameters are specific for each type of gateway. They allow using additional settings that were not available in the "Common" tab.

Up Down Add Edit Delete Default

Parameter	Value
FIX Trade SenderCompID	CompId
FIX Trade TargetCompID	FXCM
FIX Trade TargetSubID	SubId
FIX Market Data Address	fixdemo.fxcorporate.com:9001/nossl
FIX Market Data SenderCompID	SenderCompId
FIX Market Data TargetCompID	FXCM
FIX Market Data TargetSubID	TargetSubId
FIX Market Data Log Enabled	No
Limit Orders Coverage Mode	Gateway
Limit Orders Coverage Time	5

OK Cancel Help

Specify the following parameters values here:

- **FIX Trade SenderCompID** standard parameter of the FIX messages heading used for a data sender identification. This parameter is provided by FXCM Pro as the value of SenderCompID (49).
- **FIX Trade TargetCompID** standard parameter of the FIX messages heading used for trading messages recipient identification. This parameter is provided by FXCM Pro as the value of TargetCompID (56).
- **FIX Trade TargetSubID** standard parameter of the FIX messages heading used for trading messages recipient identification. This parameter is provided by FXCM Pro as the value of TargetSubID (57).
- **FIX Market Data Address** IP address and port of the FXCM Pro server, from which market (price) data is transmitted. This information is provided by FXCM Pro as the SocketConnectHost and SocketConnectPort parameters. An encrypted SSL connection to a server is established by default. If you need to establish an unencrypted connection, additionally specify the /nossl key in the address bar. For example: 10.123.100.17:10219/nossl.
- **FIX Market Data SenderCompID** standard parameter of the FIX messages heading used for a market data sender identification. This parameter is provided by FXCM Pro as the value of SenderCompID (49).
- **FIX Market Data TargetCompID** standard parameter of the FIX messages heading used for market messages recipient identification. This parameter is provided by FXCM Pro as the value of TargetCompID (56).
- **FIX Market Data TargetSubID** standard parameter of the FIX messages

heading used for market messages recipient identification. This parameter is provided by FXCM Pro as the value of TargetSubID (57).

- **FIX Market Data Log Enabled** if Yes, the gateway saves FIX connection (market flow) logs. Enable the parameter only in case of the gateway operation issues. The default value is No.
- **Limit Orders Coverage Mode** mode of processing Limit and Take Profit orders by the gateway. This parameter simplifies configuration of trade requests routing to the gateway, since there is no need to create a separate rule for routing the appropriate order types. Three processing modes are available:
 - **♦ Market** Limit and Take Profit orders are processed on the MetaTrader 5 platform side. If an order is activated, an appropriate market order is sent to FXCM Pro.
 - **♦ Limit** Limit and Take Profit orders are processed on the MetaTrader 5 side.
 Once a Limit order is triggered, an equivalent Limit order is sent to FXCM Pro. That order has a short action time specified in Limit Orders Coverage Timeout parameter. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a Limit order with a residual volume will be removed from FXCM Pro. Thus, a client will have a market position as well as a Limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5.
 A Limit order with the price equal to a Take Profit level is sent to FXCM Pro at the moment a Take Profit order has been activated. Since the price specified in the order is already present in the market, the order will be executed with that price a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. Thanks to a short expiration time, a Limit order with a residual volume will be removed from FXCM Pro. Therefore, a client's position will be closed partially. Control over the Take Profit position level with the remaining volume will then be carried out on the MetaTrader 5 side.
 Limit mode allows to protect against slippage, as a Limit order is sent to FXCM Pro system with a specified price rather than a market order for execution by the current price. In addition, this mode allows you not to reserve margin on a client's account before sending an order to FXCM Pro.
 - **♦ Gateway** Limit orders are processed on FXCM Pro side. Once a Limit order has been placed by a client, an appropriate order is sent to FXCM Pro. Take Profit orders are processed the same way as in the Limit mode.
- **Limit Orders Coverage Timeout** duration of Limit orders sent to FXCM Pro in Limit mode. Specified in seconds. The default value is 5.
- **Stop Orders Coverage** mode of handling Stop and Stop Limit orders. In case of 'Y' value, these order types will be transferred to FXCM Pro directly. In case of 'N' value, the orders will be processed inside MetaTrader 5 platform until their stop price is reached. After a Stop order is activated, an appropriate market order will be sent to the FXCM Pro system. After a Stop Limit order has

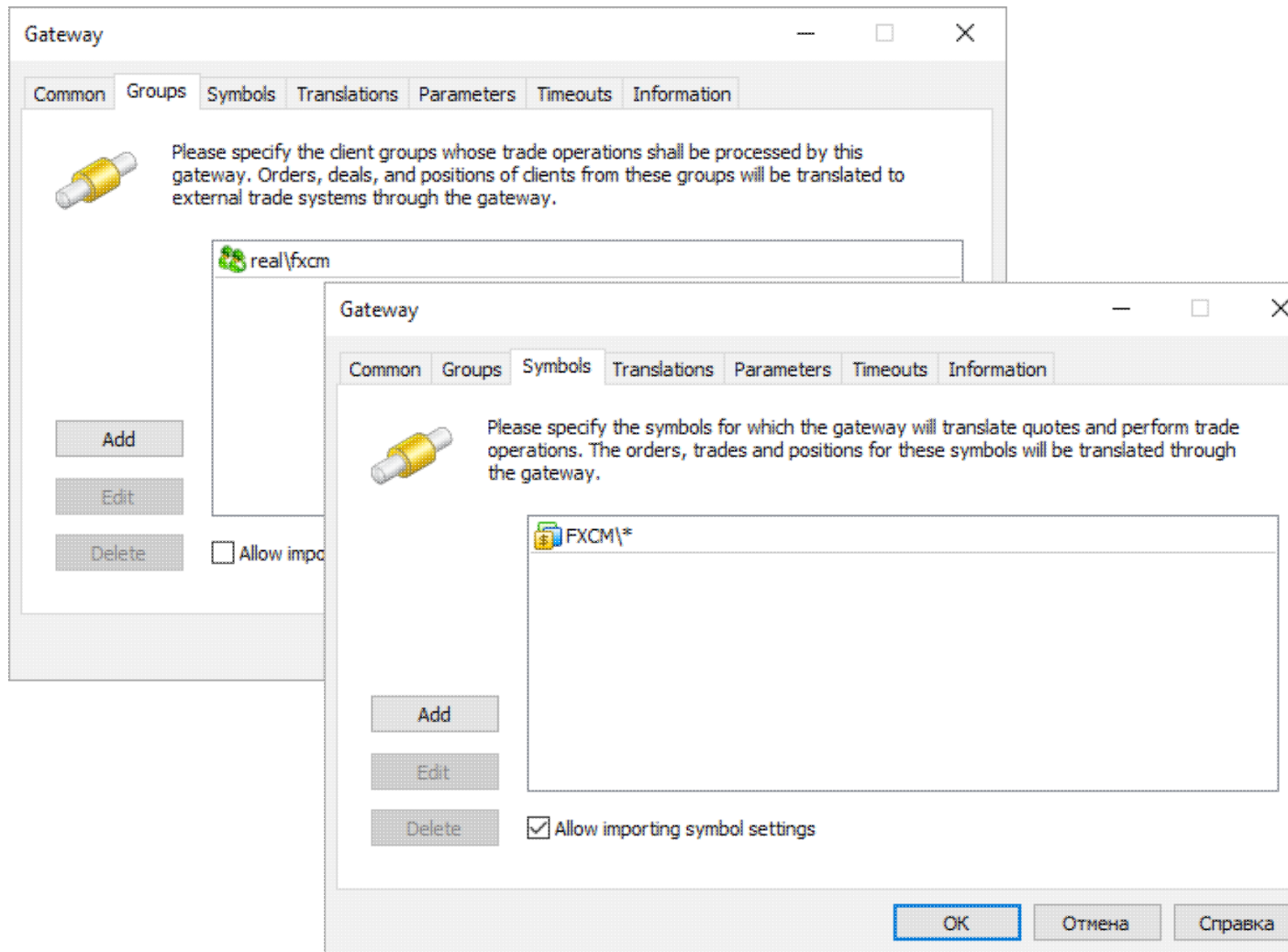
been activated, a Limit order is created, which will be processed according to Limit Orders Coverage Mode parameter value.

- **Account Mapping Mode** the gateway supports several modes of trade operation transferring to FXCM PRO: on behalf of the broker's general account and on behalf of the individual accounts used for trading in the MetaTrader 5 platform. Three modes of trades operations transfer are available:
 - ♦ **omnibus** all orders will be sent to FXCM PRO on behalf of the broker's main account specified in the "Account Mapping" parameter.
 - ♦ **one-to-one** all orders will be sent to FXCM PRO on behalf of individual accounts on which the orders are placed in MetaTrader 5 (an external system account is specified in the settings of each account).
 - ♦ **conversion** combination of the previous two modes: orders of the accounts that have specified external system account will be sent on their own behalf, while all other orders will be transferred on behalf of the broker's general account.
- **Account Mapping** the number of the single account in the FXCM PRO system, on behalf of which clients' orders will be sent if the "Account Mapping Mode" parameter is set to "omnibus" or "conversion".
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

- In no circumstances it is allowed to change operations transfer mode while in operation. Changing the mode is allowed only after all client positions are closed.
- Margin reservation of the clients should be configured for the appropriate order types in case Limit, Stop and/or Stop Limit orders are directly transferred to an external system.

The next stage is to specify the groups of the clients, whose requests will be processed via the MetaTrader 5 Gateway to FXCM Pro, as well as symbols, by which the gateway processes trading operations and broadcasts quotes.

MetaTrader 5 trading platform



MetaTrader 5 trading platform

Make sure to enable "Allow importing symbol settings" option. The gateway imports symbols from FXCM Pro to Symbols/Preliminary/FXCM directory of the MetaTrader 5 and manages their parameters.

- Initially, all imported symbols have trading ability disabled. System administrator must relocate imported symbols to the proper subgroup and allow trading for them.
- After symbols are relocated and trading abilities are enabled, the main trading server must be restarted.
- In case the gateway transmits configuration for the symbol that is already present in the platform, the configuration is updated. In this case, the symbol is not transferred and its trading ability is not turned off.
- The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

Margin Setup

Margin reservation of the clients should be configured for the appropriate order types in case Limit, Stop and/or Stop Limit orders are directly transferred to an external system.

The external system checks the funds that are necessary to provide any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the work is carried out.

By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client.

After an order has been transferred to the external system, MetaTrader 5 platform is not able to check the client's margin sufficiency any more. After the order has been executed in the external system, the gateway cannot ignore that fact. Therefore, the appropriate trading operation is performed in the platform.

Set non-zero coefficients for the orders directly transferred to the external trading system in symbol settings for the appropriate symbols:

Symbol: EURUSD

Common Quotes Trade Execution Margin **Margin Rates** Swaps Sessions

 Please specify margin rates for different types of trade operations.

Liquidity margin rate:

Margin rates:

	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
 Buy	1.0000000	1.0000000	0.0000000	0.0000000
 Sell	1.0000000	1.0000000	0.0000000	0.0000000
Maintenance margin				
 Buy	0.0000000	0.0000000	0.0000000	0.0000000
 Sell	0.0000000	0.0000000	0.0000000	0.0000000

OK Cancel Help

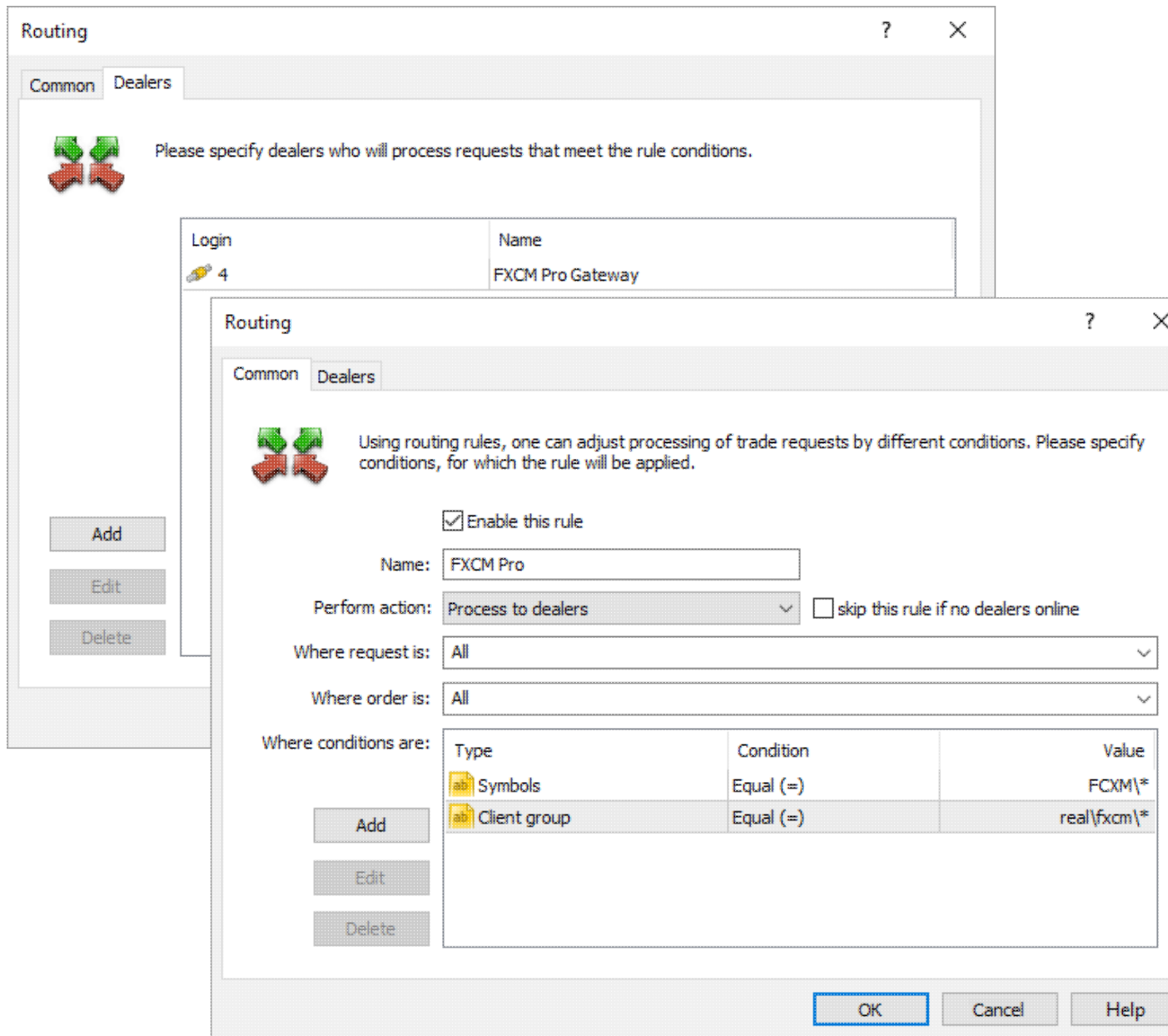
Configuring Trade Requests Routing

Configure the routing to let the clients requests to be transmitted to the FXCM Pro gateway. To do this, add a routing rule to the relevant MetaTrader 5 Administrator section.

Select "Process to dealers" as an action in common settings. Assign this routing rule for all requests and orders. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway.

In the screenshots below all orders created by users in the real\fxcm* groups by symbols from the FXCM* section will be sent to the gateway for processing.

MetaTrader 5 trading platform



After the correct execution of the steps described above, the gateway will be ready for work.

Configuring Trading Accounts

MetaTrader 5 GateWay to FXCM PRO allows transferring trading operations to an external system in different modes. The trading orders that are placed by clients in the MetaTrader 5 platform, can be sent to FXCM PRO on behalf of the broker's general account (specified in the Account Mapping" parameter of the gateway settings) or on behalf of the clients' individual accounts. In the latter case, the gateway connects to FXCM PRO via the broker's general account, but clients' trading operations are sent to FXCM PRO using individual accounts.

- Order transferring mode is controlled by the Account Mapping Mode parameter in the gateway configuration.
- Trading operations transfer conditions are determined when concluding an agreement between a brokerage company and FXCM PRO.
- In no case should you change the operation transfer mode during operation. Changing the mode is allowed only when all client positions have been closed.

If you send trading operations using individual client accounts, the appropriate FXCM PRO account number must be specified in each account's settings. This can be done via the administrator or manager terminal:

MetaTrader 5 trading platform

Account: 3580, John Smith, USD, 1:100

Overview Personal **Account** Security Certificate

Group:

Color: Leverage:

Bank account: Agent account:

☒ Enable this account ☒ Enable trading by Expert Advisors

☐ Enable trading ☒ Enable Web API connections

☒ Enable to change password ☒ Enable daily reports

☒ Enable trailing stops ☐ Change password at next login

☒ Enable one-time password

Trade accounts:

Gateway ID	Account
 4, FXCM Pro Gateway	245822311

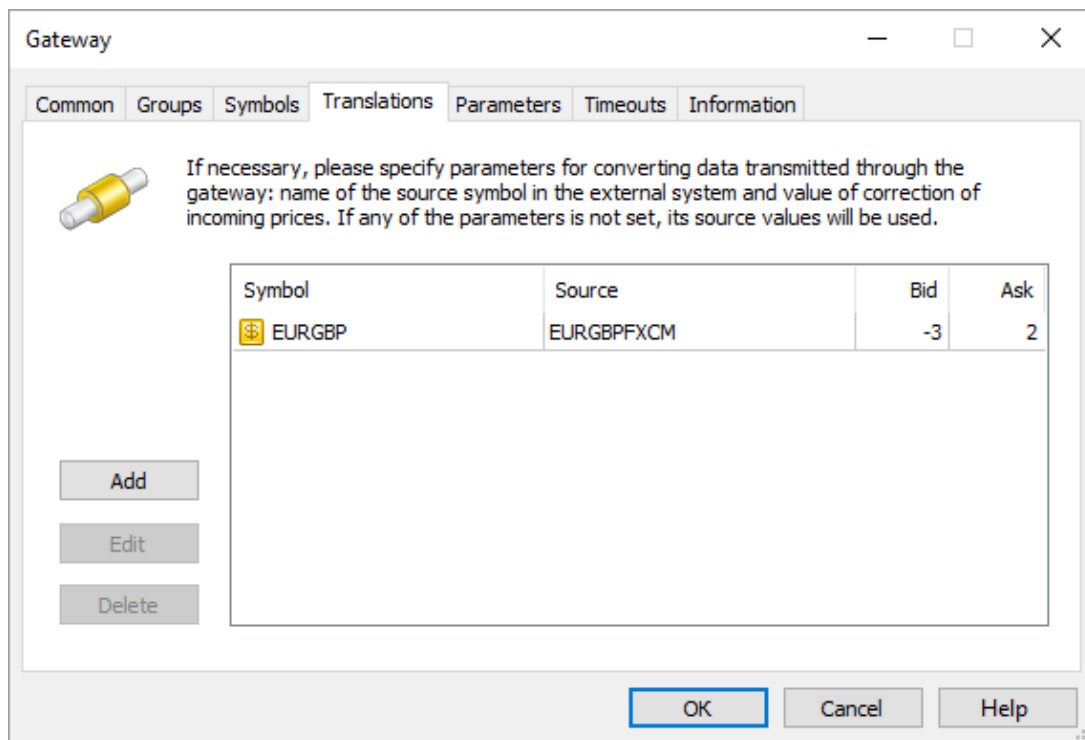
MetaTrader 5 trading platform

In "Trade accounts" section, select FXCM PRO gateway configuration and specify the client's account in the external system. That is the account, from which client trade operations will be transferred to FXCM PRO.

Client account numbers are provided by FXCM PRO.

Changing Symbols Names and Markups

The gateway receives the prices from FXCM PRO and transmits them to clients considering markups. Thus, a brokerage company receives its profit share from each deal performed at FXCM Pro. Markup values are set separately for Bid and Ask prices by each symbol:



Gateway

Common Groups Symbols Translations Parameters Timeouts Information

If necessary, please specify parameters for converting data transmitted through the gateway: name of the source symbol in the external system and value of correction of incoming prices. If any of the parameters is not set, its source values will be used.

Symbol	Source	Bid	Ask
EURGBP	EURGBPFXCM	-3	2

Add Edit Delete

OK Cancel Help

MetaTrader 5 trading platform

Besides, you can configure matching of symbol names used in FXCM Pro with the names used in your MetaTrader 5 platform. For example, if the symbol name in FXCM Pro is EURGBPFXCM, and it is called EURGBP in MetaTrader 5, enter EURGBP in the "Symbol" field and EURGBPFXCM in the "Source" field.

The above screenshot shows price conversion: at every tick, the Bid price will be reduced by 3 points, and the Ask price will be increased by 2 points. Below is a schematic example of the conversion:

FXCM Pro	>>>	ask price	EURGBP 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	ask price	EURGBP 0.83006	>>>	Client terminal
Client terminal	>>>	buy limit	EURGBP 0.83006	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit	EURGBP 0.83004	>>>	FXCM Pro
FXCM Pro	>>>	buy limit execution	EURGBP 0.83004	>>>	MetaTrader 5 server
MetaTrader 5 server	>>>	buy limit execution	EURGBP 0.83006	>>>	Client terminal

A broker gains 2 pips of profit in this example. The price is sent to the client terminal only after the correction, so the clients work only with the corrected prices. If no correction is set for a symbol, the client will work with the original prices submitted by FXCM Pro.

Prices Round Off

During the gateway's operation, accuracy of quotes (decimal places) passed for some symbol may change in the external trading system. Decrease in price accuracy at the external trading system's side does not affect the gateway's operation. It still transmits prices with less accuracy. However, if the number of decimal places at the external system's side increases, the gateway starts rounding off the passed prices.

Suppose that the accuracy of quotes has changed from 4 to 5 digits. Obtained five-digit quotes are rounded up in a broker's favor. Thus, buy requests of 1.23447, 1.23441 are rounded up to 1.2345, while sell ones of 1.23447, 1.23441 are rounded down to 1.23440.

Changes in symbol price accuracy are recorded in the gateway journal.

Fill Policy and Order Expiration

When sending an order from the MetaTrader 5 client terminal, traders can set the order execution policy (FOK, IOC or Return) and expiration time (Good Till Canceled, Today, Date and Time, Date). In FXCM Pro trading system, the fill policy and expiration are set in a single parameter - Time in Force. Therefore, the gateway

MetaTrader 5 trading platform

performs the following changes when passing the orders:

- If FOK or IOC fill policy is set for an order, the same policy is applied in FXCM Pro system.
- If a market order with Return fill policy is placed, it will be passed to FXCM Pro system in Good Till Canceled (GTC) mode.
- If a limit order with Return fill policy is placed, expiration time from MetaTrader 5 (Good Till Canceled, Today, Specified, Specified Day) is inserted into the appropriate order parameter in FXCM Pro system.
- If a limit order is activated in MetaTrader 5 platform and the gateway works in the mode of passing limit orders (Limit Orders Coverage Mode = Limit), a limit order with Good Till Date fill time policy is passed to FXCM Pro system. Limit Orders Coverage Timeout parameter also affects the order's lifetime. The gateway removes the placed order upon expiration of the specified time.
- If a take profit position is activated in MetaTrader 5 platform and the gateway works in the mode of passing limit orders (Limit Orders Coverage Mode = Limit or Limit Orders Coverage Mode = Gateway), a limit order with Good Till Date fill time policy is passed to FXCM Pro system. Limit Orders Coverage Timeout parameter also affects the order's lifetime. The gateway removes the placed order upon expiration of the specified time.

MetaTrader 5 Trading Platform / Platform Components / Gateways / Borsa Istanbul

MetaTrader 5 Gateway to Borsa Istanbul Futures

Borsa İstanbul brings together all exchanges operating in the Turkish capital markets under a single roof. Established on the basis of Capital Markets Law no. 6362, Borsa İstanbul is an internal entity under private law. Borsa İstanbul is a self-regulatory entity.

Borsa İstanbul offers the opportunity to invest in various products in an organized, transparent and reliable trading environment to local and international investors with its modern technological capabilities. On all markets of Borsa İstanbul, transactions are conducted electronically, and market information is disseminated on a real-time basis.

Borsa İstanbul markets are organized under five main categories: Equity Market, Emerging Companies Market, Debt Securities Market, Foreign Securities Market and Futures and Options Market.

MetaTrader 5 Gateway to Borsa Istanbul Futures allows futures trading at the Main Board of Futures and Options Market at Borsa Istanbul:

- Equity Futures (Main Board)
- Equity Index Futures (Main Board)
- Currency Futures (Main Board)
- Precious Metals Futures (Main Board)
- Commodity Futures (Main Board)
- Power Futures (Main Board)

Request MetaTrader 5 Gateway to Borsa Istanbul Futures

You can request the MetaTrader 5 Gateway to Borsa Istanbul from the "Buy" section of the technical support site. After you make the payment, the MetaTrader 5 platform license will be automatically updated via the LiveUpdate system and you will be able to use the MetaTrader 5 Gateway to Borsa Istanbul Futures without limitations. The executable module of the gateway is included into the platform package and can run in a demo mode, allowing you to perform 100 operations for a work session (until restart).

Preparing for the Gateway Launch

Before setting a brokerage company gateway, configure the virtual network channel to be able to connect to Borsa Istanbul servers.

Configuration of a virtual network channel is performed together with Borsa Istanbul technical specialists. To create a virtual channel to the exchange, a broker must request technical requirements and recommendations for connection to the Borsa

MetaTrader 5 trading platform

Istanbul virtual private network (VPN) from the mercantile exchange technical support team. At the next stage the mercantile exchange technical support team will submit to the brokerage company description of the equipment settings on SMX side: tunnel type, data encryption method, data encryption keys etc.

The programmed virtual channel can be created for connection to Borsa Istanbul test environment.

How the Gateway works

MetaTrader 5 Gateway to Borsa Istanbul Futures is a separate BorsaIstanbulVIOPGateway64.exe module that uses the MetaTrader 5 Gateway API. The gateway operates over 5 FIX channels. One FIX connection is used for passing Reference Data (symbol settings), two more are used for trading and the other two for receiving data on trades performed via other trading systems (Drop Copy). Some symbols are processed on the one and Drop Copy connection, the rest symbols are handled on the other one. Information about what servers process specified symbols is transmitted via the Reference Data channel.

Orders and positions are synchronized over the SOAP service, and quotes are delivered via a TIP channel.

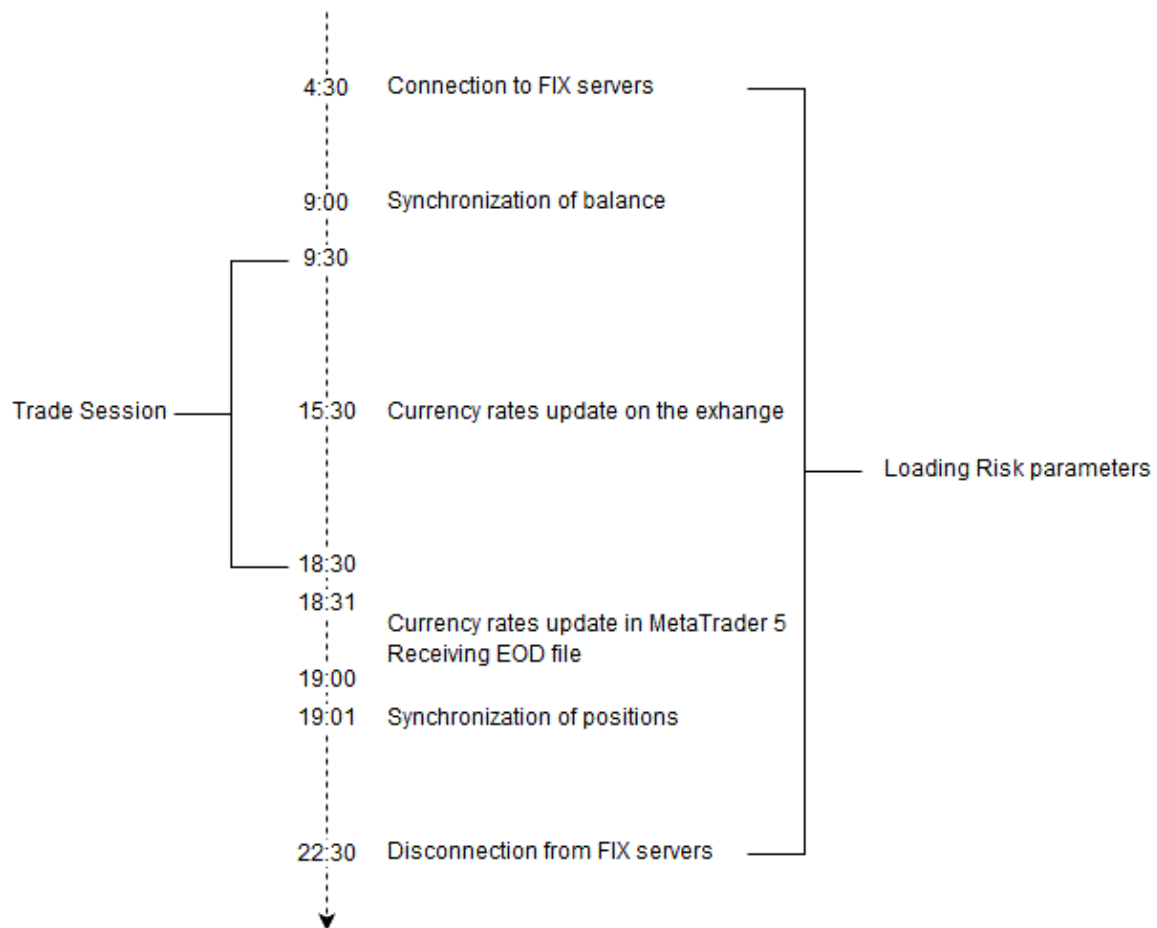
Orders are sent to the MetaTrader 5 Gateway to Borsa Istanbul for processing in accordance with configured routing rules. Processing of requests depends on the type of the order:

Order Type	Execution
Market Orders	Delivered directly to the exchange as a market order.
Buy Limit Sell Limit	Delivered directly to the exchange as limit orders.
Buy Stop Sell Stop	Delivered directly to the exchange as stop orders.
Buy Stop Limit Sell Stop Limit	Delivered directly to the exchange as stop limit orders.
Take Profit	Processed on the MetaTrader 5 side. After activation in the MetaTrader 5 terminal, a market order to close a client's position is sent to the exchange.

MetaTrader 5 trading platform

Order Type	Execution
Stop Loss	
Stop Out	Processed on the MetaTrader 5 side. In case a limit order must be removed after a Stop out level has been reached, an appropriate request for its removal is sent to Borsa Istanbul server. In case a position must be closed because of Stop out, a market request to close a client's position is sent to the exchange.

The below diagram shows the interaction of the gateway with the exchange (the exchange time UTC+3 is specified):

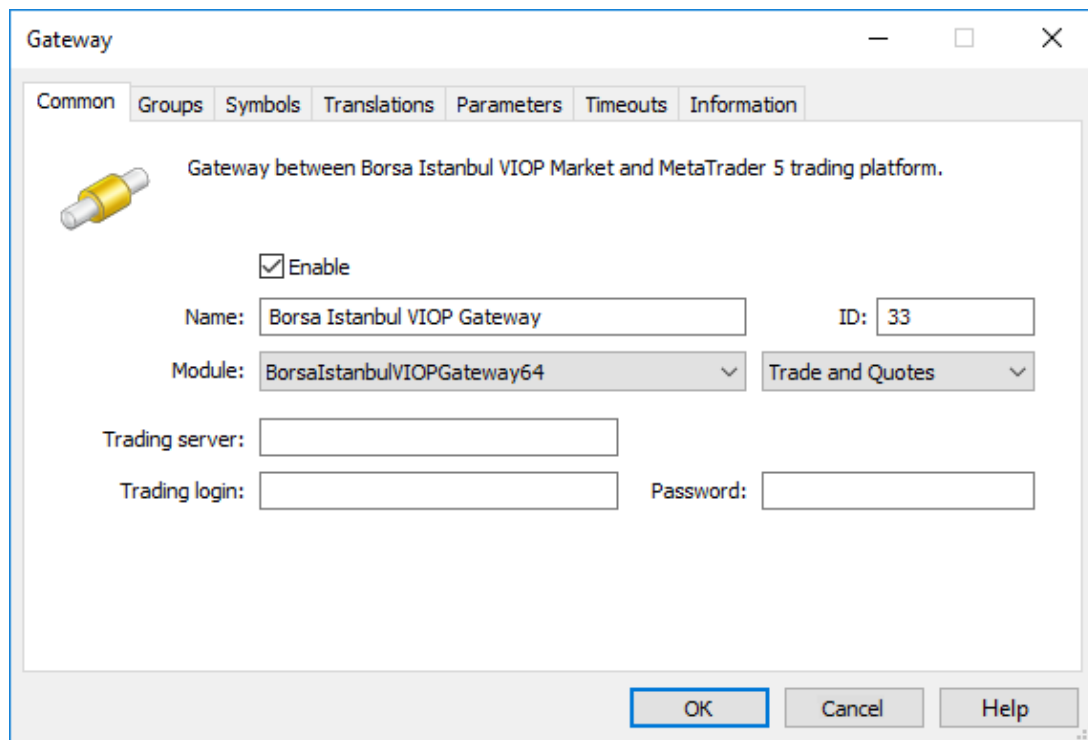


MetaTrader 5 trading platform

- The gateway starts to connect to FIX servers of the exchange at 4:30 . It will try to obtain from the exchange actual files with risk parameters till 22:30. When new files appear, the gateway downloads them and updates symbol settings in MetaTrader 5.
- Clients' balances are synchronized with the exchange at 9:00 by default. This time can be changed using the BalancesSyncTime parameter.
- The trading session lasts from 9:30 to 18:30.
- The exchange updates official exchange rates at 15:30. These rates are used to recalculate the value of contracts that are not expressed in Turkish liras: the symbol's tick value is calculated and set based on these rates. New rates are not applied until the end of the trading session (previous rates are used).
- During the time specified in the EOD Time parameter, the gateway downloads new rates from the exchange and recalculates results of all trades executed during this session using the new rates.
- Immediately after this, the gateway receives the EOD file from the exchange. Results of all trades executed from 15:30 until the end of the session are recalculated using new currency rates. Update of risk parameters of instruments, as well as of inter-month and inter-contract spreads, calculation of variation margin, delivery (closing of positions of expired futures) and unblocking of accumulated profits are performed based on the EOD file.
- If PositionsSync parameter is enabled, positions are synchronized at 19:01 (after applying the EOD file).
- At 22:30, the gateway is disconnected from the exchange's FIX servers.

Configuring the gateway

To start working, add a new gateway configuration:



Gateway

Common Groups Symbols Translations Parameters Timeouts Information

Gateway between Borsa Istanbul VIOP Market and MetaTrader 5 trading platform.

☒ Enable

Name: ID:

Module:

Trading server:

Trading login: Password:

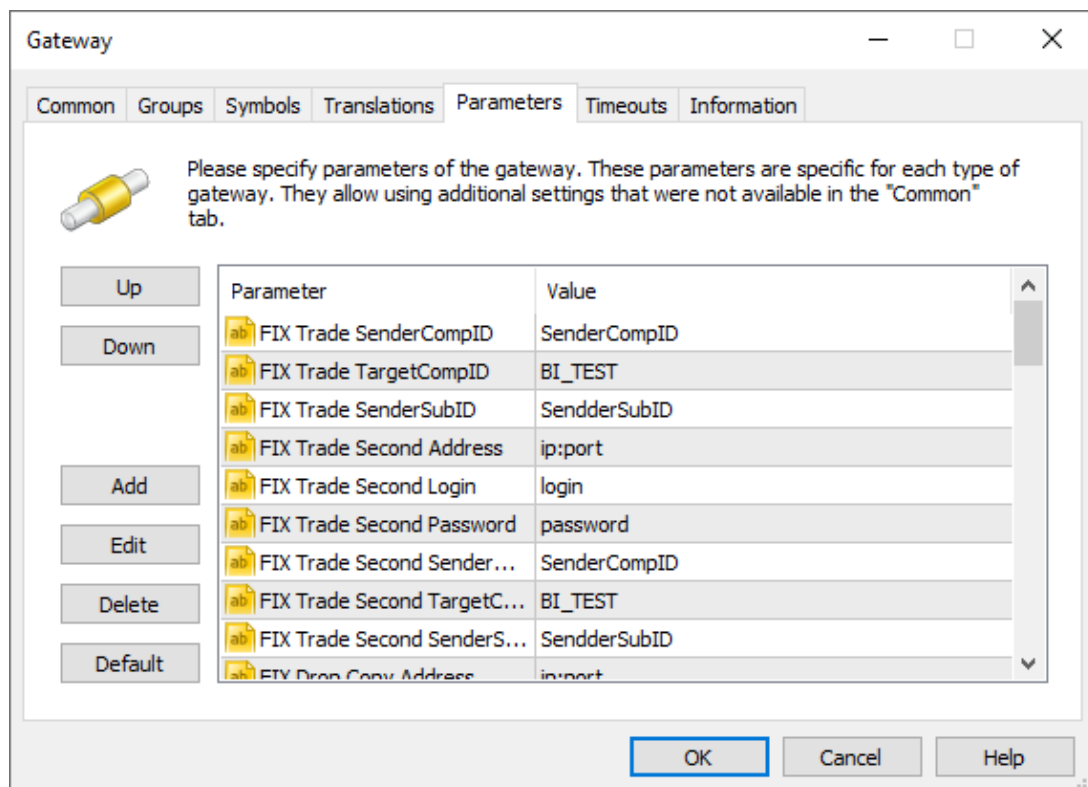
OK Cancel Help

Set the following parameters on the "Common" tab:

- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Module** specify BorsaIstanbulVIOPGateway64 and accept the default settings after the module selection.
- **Trading server** IP address and the port of the exchange's first FIX server, where trade requests are processed.
- **Trading login** a login for connection to the first trade server of the exchange. Corresponds to the UserName (553) tag in the FIX protocol.
- **Password** a password for connection to the first trade server of the exchange. Corresponding to the Password tag (554) in FIX protocol. The exchange requires that the password used for connection be changed every 90 days. The gateway automates this process by changing the password in advance, every 70 days, in case of unforeseen issues. To activate this feature, change the password manually once. This action will provide the gateway with the last password change date, from which further tracking will be performed.

- ID value must be unique in the field of manager logins and gateway identifiers.
- Connection details are provided by the exchange.

Now, go to the "Parameters" tab.



Gateway

Common Groups Symbols Translations **Parameters** Timeouts Information

Please specify parameters of the gateway. These parameters are specific for each type of gateway. They allow using additional settings that were not available in the "Common" tab.

Parameter	Value
FIX Trade SenderCompID	SenderCompID
FIX Trade TargetCompID	BI_TEST
FIX Trade SenderSubID	SenderSubID
FIX Trade Second Address	ip:port
FIX Trade Second Login	login
FIX Trade Second Password	password
FIX Trade Second Sender...	SenderCompID
FIX Trade Second TargetC...	BI_TEST
FIX Trade Second SenderS...	SenderSubID
FIX Trade Second Address	ip:port

Up Down Add Edit Delete Default OK Cancel Help

Specify the following parameters values here:

- **FIX Trade SenderCompID** a standard parameter of the FIX messages header used for a data sender identification. This parameter is provided by the exchange as the value of SenderCompID (49).
- **FIX Trade TargetCompID** a standard parameter of the FIX messages header used for trading messages recipient identification. This parameter is provided by the exchange as the value of TargetCompID (56).
- **FIX Trade SenderSubID** FIX messages header standard parameter used for a trading participant (a legal entity) identification. This parameter is provided by the exchange as the value of SenderSubID.
- **FIX Trade Second Address** IP address and the port of the exchange's second FIX server, where trade requests are processed.
- **FIX Trade Second Login** a login for connection to the second trade server of the exchange. Corresponds to the UserName (553) tag in the FIX protocol.
- **FIX Trade Second Password** a password for connection to the second trade server of the exchange. Corresponding to the Password tag (554) in FIX protocol.
- **FIX Trade Second SenderCompID** a standard parameter of the FIX messages header used for a data sender identification. This parameter is provided by the exchange as the value of SenderCompID (49).
- **FIX Trade Second TargetCompID** a standard parameter of the FIX messages header used for trading messages recipient identification. This parameter is provided by the exchange as the value of TargetCompID (56).
- **FIX Trade Second SenderSubID** FIX messages header standard parameter used for a trading participant (a legal entity) identification. This parameter is provided by the exchange as the value of SenderSubID.
- **FIX Drop Copy Address** the IP and port of the first Drop Copy server. Operations performed via other trading systems are processed on this FIX server.
- **FIX Drop Copy Login** login for connection to the first Drop Copy server of the exchange. Corresponds to the UserName (553) tag in the FIX protocol.
- **FIX Drop Copy Password** password for connection to the first Drop Copy server of the exchange. Corresponding to the Password tag (554) in FIX protocol.
- **FIX Drop Copy SenderCompID** a standard parameter for the FIX message header, which is used for identifying a data sender in the Drop Copy service. This parameter is provided by the exchange as the value of SenderCompID (49).
- **FIX Drop Copy TargetCompID** a standard parameter for the FIX message header, which is used for identifying a message recipient in the Drop Copy service. This parameter is provided by the exchange as the value of TargetCompID (56).
- **FIX Drop Copy SenderSubID** a standard parameter for the FIX messages header, which is used for a trading participant (a legal entity) identification in the Drop Copy service. This parameter is provided by the exchange as the value of SenderSubID.
- **FIX Drop Copy Third Address** IP-address and port of the third Drop Copy server. This is the exchange's FIX server which processes operations executed through other trading systems.
- **FIX Drop Copy Third Login** login for connection to the exchange's third Drop Copy server. Corresponds to the UserName tag (553) in the FIX protocol.

- **FIX Drop Copy Third Password** password for connection to the exchange's third Drop Copy server. Corresponds to the Password tag (554) in the FIX protocol.
- **FIX Drop Copy Third SenderCompID** standard FIX message header parameter which is used to identify the trading data sender. This parameter is provided by the exchange as the SenderCompId (49) value.
- **FIX Drop Copy Third TargetCompID** standard FIX message header parameter which is used to identify the recipient of trading messages. This parameter is provided by the exchange as the TargetCompId (56) value.
- **FIX Drop Copy Third SenderSubID** standard FIX message header parameter which is used to identify a trading participant (legal entity). This parameter is provided by the exchange as the SenderSubId value.
- **FIX Drop Copy Fourth *** a set of parameters for the fourth Drop Copy server: Address, Login, Password, SenderCompID, TargetCompID and SenderSubID. They are set similarly to the third server parameters.
- **FIX Drop Copy Fifth *** a set of parameters for the fifth Drop Copy server: Address, Login, Password, SenderCompID, TargetCompID and SenderSubID. They are set similarly to the third server parameters.
- **FIX Drop Copy Sixth *** a set of parameters for the sixth Drop Copy server: Address, Login, Password, SenderCompID, TargetCompID and SenderSubID. They are set similarly to the third server parameters.
- **FIX Reference Data Address** IP address and port of the exchange's FIX server, through which trade symbol settings are transmitted (the Reference Data server).
- **FIX Reference Data Login** login for connection to the Reference Data server. Corresponds to the UserName (553) tag in the FIX protocol.
- **FIX Reference Data Password** password for connection to the Reference Data server. Corresponding to the Password tag (554) in FIX protocol.
- **FIX Reference Data SenderCompID** a standard parameter for the FIX message header, which is used for identifying a data sender in the Reference Data service. This parameter is provided by the exchange as the value of SenderCompID (49).
- **FIX Reference Data TargetCompID** a standard parameter for the FIX message header, which is used for identifying a message recipient in the Reference Data service. This parameter is provided by the exchange as the value of TargetCompID (56).
- **FIX Reference Data SenderSubID** a standard parameter for the FIX messages header, which is used for a trading participant (a legal entity) identification in the Reference Data service. This parameter is provided by the exchange as the value of SenderSubID.
- **FIX Reference Data Detached** by default, when any of the FIX connections is interrupted, the gateway completely disconnects from the exchange to avoid data integrity violation. The settings of trading instruments transmitted through the FIX Reference channel are updated not very often. Therefore, in some cases, the loss of this connection can be disregarded. To prevent the gateway from disconnecting from the exchange upon losing the FIX Reference Data connection, set "FIX Reference Data Detached" to "No".
- **FIX Connect Attempts** the maximum number of attempts to connect to the current IP address of the FIX channel (Trade, Drop Copy, or Reference). The

default value is three. After the specified number of unsuccessful attempts, the gateway will try to connect to the next address.

- **FIX Trade SMPLevel** value of tag 21114 in the FIX protocol. Possible values: 'Within' (default) and 'Across'. The value is used along with FIX Trade SMPMethod and FIX Trade SMPID for Self-Match Prevention Feature, an algorithm that prevents the matching of orders of the same client. Please contact the exchange to find out the required parameter values.
- **FIX Trade SMPMethod** value of tag 21115 in the FIX protocol. Possible values: Aggressive (default), Passive, Both.
- **FIX Trade SMPID** value of tag 21116 in the FIX protocol. If this parameter is not filled, the gateway will not transmit the SMPLevel (21114) and SMPMethod (21115) values.
- **TIP Address** IP address and port of the exchange server, through which quotes are transmitted over the TIP protocol.
- **Nasdaq Identifiers Path** path to the file containing identifiers of instruments in the TIP channel. It is used for the gateway to exchange connection in the middle of the trading session.
- **Nasdaq Identifiers Enabled** enable/disable processing of the file specified in the Nasdaq Identifiers Path parameter. Default is No. At the beginning of the day, the exchange send over the TIP protocol information about symbol identifiers (identifier/symbol name pairs). Further quotes are provided only with an identifier (without a name). To enable the gateway to receive quotes after connecting to Borsa Istanbul in the middle of the trading session, enable this option. The gateway will load the identifiers from the file specified in Nasdaq Identifiers Path.
- **Symbols Path** path for importing the exchange symbols. By default, if this parameter is absent, the gateway imports the exchange's trading symbols to \Preliminary subdirectory with trading ability disabled. A system administrator should manually relocate imported symbols to the proper group and allow trading for them. If this parameter is present, the gateway imports trading symbols following the specified path. The ability to trade the symbols is enabled immediately. Thus, the administrator has no need to additionally configure the symbols. For example, if "BorsaIstanbul" is specified in the parameter, working symbols will be automatically added to the BorsaIstanbul* subdirectory, and expired symbols will be transferred to BorsaIstanbul\Expired*.
- **Takas Web Service Address** the address of the server that transmits data for synchronizing client balance positions on the MetaTrader 5 side (the SOAP service).
- **Takas Web Service Login** login for connection to the SOAP service (provide by the Exchange).
- **Takas Web Service Password** password for connection to the SOAP service (provide by the Exchange).
- **Takas Web Service Member Code** trading participant code, which should be specified for connecting to the SOAP service (provide by the Exchange).
- **Genium Risk Parameters Url** the address for downloading files with risk parameters. These files are used for obtaining current settlement prices, for updating risk parameters of instruments, for calculating the variation margin and for updating the inter-month and inter-contract spreads of instruments.

- **Positions Sync** enable/disable synchronization of positions with the SOAP service at the end of the trading session. Enabled by default (set to Yes).
- **Balances Sync** enable/disable synchronization of client balances with the SOAP service at the end of the trading session. Enabled by default (set to Yes).
- **Balances SyncTime** the start time of balance synchronization with SOAP. Synchronization starts at 09:00 by default.
- **Daily Rollover Positions** enable (Yes)/disable (No) rollover operations (charging variation margin) at the end of the trading session. Enabled by default (set to Yes).
- **Settlement Prices File Path** full path to the CSV file with settlement prices. The file must be UTF-8 encoded and prices must be in X,YYY format (for example, 1,234). By default, the path is not specified, and the gateway receives prices from the FIX Reference Data channel or from the 'span' file. If the path is specified, the gateway checks the date when the file was last modified. If the file has been modified on the current day, its contents are read and applied. If the file has not been modified, the gateway will use prices from the FIX Reference Data channel or from the 'span' file. If the file does not contain data for certain instruments, the relevant data will be taken from the FIX Reference Data channel or from the 'span' file.
- **EOD Time** end-of-day time on the exchange. Variation margin operations are performed at this time, and not upon the receipt of the "EOD span" file. The parameter is set as "HH:MM". Valid values are from 18:30 (end of the afternoon session) to 18:59 (beginning of the evening session). The default time is 18:52.
- **Trade Requests Limit** the maximum total number of requests sent to the exchange per second. Default is 0, which means there is no limit.
- **Client Trade Requests Limit** the maximum number of requests per second sent by each separate client to the exchange. Default is 0, which means there is no limit.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.
- **Stop Out Allow** allow closing of positions upon reaching Stop Out. If set to "No" (default), positions opened via the gateway will not be closed automatically, if Stop Out is reached on the trading platform side. If a position is opened with Stop Out Allow = Yes, then closing by Stop Out can occur even if you change the parameter value to "No". It is only guaranteed that the ban to

close will be effective for positions opened after changing the parameter value to "No".

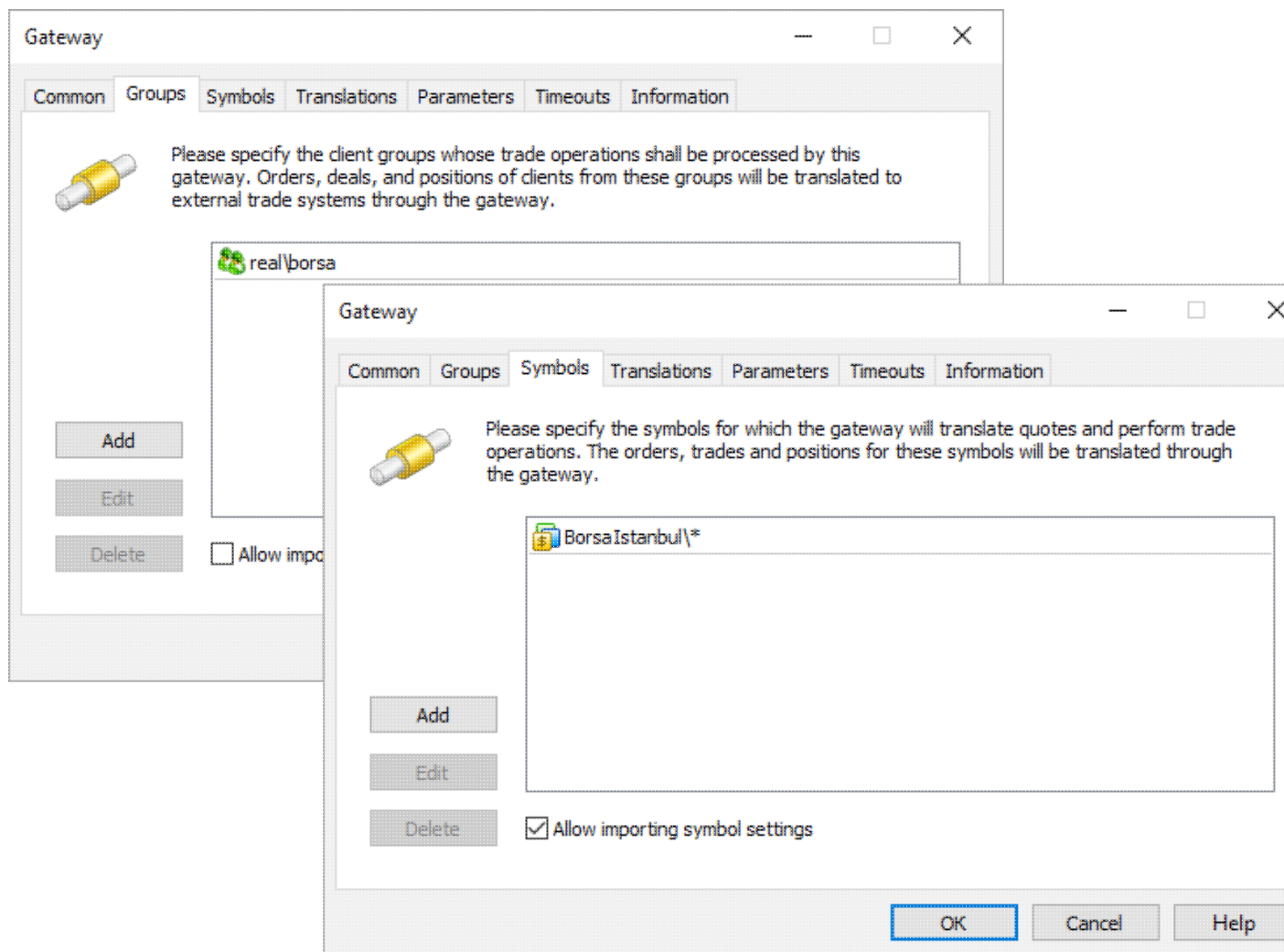
Two addresses are supported for each FIX channel: primary and secondary. You can specify them separating by commas. For example: FIX Drop Copy Address = 31.145.34.115:30187,31.145.34.115:10030. If the gateway fails to connect to the first address, it will automatically switch to the second one.

The gateway supports the old and new exchange environments: BISTECH 2 and BIST 3.0, respectively. The parameters below are only used to connect to the old environment. If they are filled, the following parameter groups will be ignored: FIX Drop Copy Third *, FIX Drop Copy Fourth *, FIX Drop Copy Fifth * and FIX Drop Copy Sixth *.

- **FIX Drop Copy Second Address** IP address and port of the second Drop Copy sever. This is the exchange's FIX server which processes operations executed through other trading systems.
- **FIX Drop Copy Second Login** login for connection to the exchange's second Drop Copy server. Corresponds to the UserName tag (553) in the FIX protocol.
- **FIX Drop Copy Second Password** password for connection to the exchange's second Drop Copy server. Corresponds to the Password tag (554) in the FIX protocol.
- **FIX Drop Copy Second SenderCompID** standard FIX message header parameter which is used to identify the trading data sender. This parameter is provided by the exchange as the SenderCompId (49) value.
- **FIX Drop Copy Second TargetCompID** standard FIX message header parameter, which is used to identify the recipient of trading messages. This parameter is provided by the exchange as the TargetCompId (56) value.
- **FIX Drop Copy Second SenderSubID** standard FIX message header parameter which is used to identify a trading participant (legal entity). This parameter is provided by the exchange as the SenderSubId value.

The next stage is to specify the groups of clients, whose requests will be processed via the MetaTrader 5 Gateway to Borsa Istanbul, as well as symbols, by which the gateway processes trading operations and broadcasts quotes.

MetaTrader 5 trading platform



MetaTrader 5 trading platform

Make sure to enable the "Allow importing symbol settings" option. The gateway imports symbols from the exchange to the MetaTrader 5 platform in accordance with the Symbols Path parameter and further controls parameters of these symbols.

The gateway supports conversion of symbols and quotes. For details, please view the Symbol and Price Translation section.

Groups Configuration and Using Profit/Loss for the Current Trading Session

According to the exchange rules, profit obtained during a trading day cannot be used before clearing is performed. In fact, only clearing determines the financial results of all transactions during the day. The profit received by a client during the current trading session is accumulated during the trading session and released only after the clearing becoming available for use by the client for further trade.

In order to avoid a situation in which a client who suffered losses will try to continue trading, fixed loss is withdrawn immediately.

In this regard, the following parameters should be specified in "Profit/loss in free margin" block of Margin tab in the settings of the group that will operate at Borsa Istanbul:

The screenshot shows the 'Group: real\borsa' settings window with the 'Margin' tab selected. The window has a title bar with standard Windows controls. Below the title bar is a tabbed interface with tabs for 'Common', 'Company', 'News & Mail', 'Permissions', 'Margin', 'Symbols', 'Commissions', and 'Reports'. The 'Margin' tab is active, displaying a green icon of two hands shaking and a message: 'Please set up the mechanism of margin calculation for the group: choose the method of calculation and margin requirements.' Below this message are several settings: 'Risk management:' with a dropdown menu set to 'for Retail Forex, CFD, Futures'; 'Margin call level:' with a text box containing '50'; 'Stop out level:' with a text box containing '30' and a unit dropdown set to '%, percent'; a section titled 'Profit/loss in free margin' containing 'Unrealized profit:' with a dropdown set to 'use unrealized loss', 'Daily fixed profit:' with a dropdown set to 'use daily fixed loss', and a checked checkbox labeled 'Release fixed profit at the end of day'; and 'Virtual credit:' with an empty text box and a note '(applies only to opening new positions)'. At the bottom of the window are three buttons: 'OK', 'Cancel', and 'Help'.

Group: real\borsa

Common Company News & Mail Permissions **Margin** Symbols Commissions Reports

Please set up the mechanism of margin calculation for the group: choose the method of calculation and margin requirements.

Risk management: for Retail Forex, CFD, Futures

Margin call level: 50 Stop out level: 30 in %, percent

Profit/loss in free margin

Unrealized profit: use unrealized loss

Daily fixed profit: use daily fixed loss

☒ Release fixed profit at the end of day

Virtual credit: (applies only to opening new positions)

OK Cancel Help

MetaTrader 5 trading platform

All financial accounting on Borsa Istanbul is done in Turkish liras. Thus, Turkish liras (TRY) should be specified as deposit currency in the group settings:

Group: real\borsa

Common Company News & Mail Permissions Margin Symbols Commissions Reports

Group is a set of users that have the same permission settings and service conditions. Please specify name of group, deposit currency, trade server, and authentication type.

Name: real\borsa Currency: TRY

Trade server: Trade Main, 1

Authentication: Normal Minimum password length: 8

One-time password: Disabled ☐ Force one-time password usage

☒ Enable connections

☐ Enable certificate confirmation

☐ Change password at first login

☐ Show the risk warning window after connection

OK Cancel Help

Configuring Trade Requests Routing

Configure the routing to let the clients requests to be transmitted to the Borsa Istanbul Futures gateway. To do this, add a routing rule to the relevant MetaTrader 5 Administrator section.

Select "Process to dealers" as an action in common settings. Assign this routing rule for all requests and orders. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway.

In the screenshots below all orders created by users in the real\borsa* groups by symbols from the BorsaIstanbul* section will be sent to the gateway for processing.

MetaTrader 5 trading platform

Routing

Common

Dealers



Please specify dealers who will process requests that meet the rule conditions.

Login

33

Name

Borsa Istanbul Gateway

Add

Edit

Delete

Routing

Common

Dealers



Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name:

Borsa Istanbul

Perform action:

Process to dealers

☐ skip this rule if no dealers online

Where request is:

All

Where order is:

All

Where conditions are:

Type	Condition	Value
 Symbols	Equal (=)	BorsaIstanbul*
 Client group	Equal (=)	real\borsa*

Add

Edit

Delete

OK

Cancel

Help

Configuring Trading Accounts

Each client on MetaTrader 5 side should be provided with a corresponding separate trading account on the exchange. After an account is opened at the exchange, it should be connected with the account within the platform. Open the appropriate client account and move to Account tab:

Account: 1812010, Young Christian, TRY, 1:100

Overview Personal **Account** Security

Group: real\borsa

Color: ☐ None Leverage: 1 : 100

Bank account: Agent account:

☒ Enable this account ☒ Enable trading by Expert Advisors

☒ Enable trading ☒ Enable Web API connections

☒ Enable to change password ☒ Enable daily reports

☒ Enable trailing stops ☐ Change password at next login

☒ Enable one-time password

Trade accounts:

Gateway ID	Account
33, Borsa Istanbul Gateway	00231021312

Synchronize Add Edit Delete

Update Cancel Help

MetaTrader 5 trading platform

A new entry should be added in "Trade accounts" section. Select MetaTrader 5 Gateway to Borsa Istanbul in Gateway ID field. Specify the Borsa Istanbul client account in "Account" field.

Gateway Launch

After the application of all settings described above, your gateway is almost ready to go. It will connect to Borsa Istanbul server.

Borsa Istanbul submits a default password for the first connection (specified in the Trading Password field). Borsa Istanbul security policy requires the password to be changed right after the first connection. The gateway automatically generates and installs the new password. You can change the password later.

The password should meet Borsa Istanbul safety requirements:

- the password should consist of 8 characters;
- the password should contain 3 types of symbols: lowercase Latin characters (a-z), upper case ones (A-Z), numbers (0-9);
- the password must be different from the previous ones (last 6 used passwords are checked).

The following entries appear in the gateway journal when the password is changed automatically:

12:23:48	Gateway	new password has generated due to previous logon response
----------	---------	---

The following entry means that the gateway has successfully connected to Borsa Istanbul and is now ready for operation:

12:24:19	Gateway	synchronized with Borsa Istanbul exchange
----------	---------	---

In case of connection failure, the entries concerning the failure reasons appear in the gateway journal.

New symbols are imported after the first connection. As noted above, after adding new symbols, trading should be enabled for them. After that the MetaTrader 5 platform main trade server must be restarted. These actions should be performed only during the gateway first launch.

After the main trading server is restarted, the gateway installation and configuration is complete. Traders can perform transactions on Borsa Istanbul using their MetaTrader 5 client terminal.

Updating the Symbols Settings

The gateway automatically updates the symbols settings during its operation, if necessary. Appropriate entries appear in the journal in that case:

12:25:36	SymbolsBase	symbol 'F_EURUSD1213S0' config updated
----------	-------------	--

Changing the Password

MetaTrader 5 trading platform

In case you need to change the password used by the gateway for connection to Borsa Istanbul server, simply specify the new password in Trading password field. The password must comply with security requirements. Once the settings are saved, the password at Borsa Istanbul server will be changed automatically.

12:25:48	Gateway	password has been changed manually
----------	---------	------------------------------------

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Gateways](#) / Interactive Brokers

MetaTrader 5 Gateway to Interactive Brokers

[Interactive Brokers](#) provides trading access to the largest Exchanges in Americas, Europe, Africa and Asia-Pacific region. One account features access to 125 markets in 31 countries, including NASDAQ, NYSE, CHX, MSE and LSE, among others. With Interactive Brokers, you can offer multi-asset trading to your traders: stocks, options, futures, currencies, ETF and CFD.

Using the MetaTrader 5 Gateway to Interactive Brokers, the integration can be performed with the minimum investment of time and money. The gateway launch requires only a few steps:

- [Open an account with Interactive Brokers](#)
- [Purchase a gateway from the App Store](#) (temporarily not available for purchasing)
- [Perform a simple gateway configuration](#)
- [Specify appropriate Interactive Brokers accounts in MetaTrader 5 trading accounts](#)

After that, your traders will be able to trade on the largest global exchanges using the MetaTrader 5 desktop, mobile and web terminals. You do not need to set up trading for each market and each instrument separately. The gateway automatically controls settings of all symbols.

[Request MetaTrader 5 Gateway to Interactive Brokers](#)

To integrate with Interactive Brokers, request the gateway from the [App Store](#) section of the technical support site. After you make the payment, the MetaTrader 5 platform license will be automatically updated via LiveUpdate and you will be able to use gateway without limitations. The executable module of the gateway is included into the platform package and can run in a demo mode, allowing you to perform 100 operations per working session (up to gateway restart).

How the Gateway operates

MetaTrader 5 Gateway to Interactive Brokers is a separate InteractiveBrokersGateway64.exe module which operates using the MetaTrader 5 Gateway API. To connect to a trading system, the gateway uses an interlayer, which can be the IB Gateway or The Trader Workstation (TWS) application provided by Interactive Brokers. Please [see below](#) for details.

Trading operations are forwarded via the gateway in accordance with the [routing rules](#). Request processing depends on the order type:

	Execution
--	------------------

MetaTrader 5 trading platform

Order Type	
Market Order	Forwarded directly to the Interactive Brokers system as a market order.
Buy Limit Sell Limit	Forwarded directly to the Interactive Brokers system as a limit order.
Buy Stop Sell Stop	Processed on the MetaTrader 5 platform side until their stop price is reached. After a stop order is activated, an appropriate market order is sent to the Interactive Brokers system.
Buy Stop Limit Sell Stop Limit	Processed on the MetaTrader 5 side. Upon activation in MetaTrader 5, an appropriate limit order is sent to the Interactive Brokers system.
Take Profit	Processed on the MetaTrader 5 side. Upon activation in MetaTrader 5, a limit order with the Take Profit activation price is sent to the Interactive Brokers system.
Stop Loss	Processed on the MetaTrader 5 side. Upon activation, an appropriate market order is forwarded to the Interactive Brokers system.
Stop Out	Processed on the MetaTrader 5 side. Upon activation, market orders to close all client's positions are sent to the Interactive Brokers system.

The following parameters are supported by the gateway:

- Trading server address and port for connection to IB Gateway or TWS.
- Login and Password authorization data for logging into the Interactive Brokers system.
- Trading Mode account type at Interactive Brokers.
- Start IBGateway IB Gateway application launch mode when started by the gateway.
- Command Port port for managing IB Gateway.
- Client ID identifier of connection to IB Gateway.
- Import Default Symbols mode of operation with trading instruments.
- Rate Limit allowable frequency of requests, sent by the gateway.
- Flex Request Token token to request deals history.
- Flex Request Query report template to request deals history.

Detailed descriptions of each of the parameters are available at the links.

Connection to Interactive Brokers

Firstly, you need access to the Interactive Brokers trading system. Please contact the company through their official website <https://www.interactivebrokers.com>. If you already have the system login and password, this step can be skipped.

MetaTrader 5 trading platform

API Interactive Brokers runs as an interface to their trading applications IB Gateway and The Trader Workstation (TWS). These are two different applications: TWS is a full-fledged trading terminal with a graphical interface, while IB Gateway is a simplified windows application specifically designed for integration with other systems.

The MetaTrader 5 gateway also operates through these applications. The gateway can either connect to installed applications or it can deploy IB Gateway on a local computer, if an appropriate application is not found.

IB Gateway and TWS have some differences in terms of interaction with the gateway. We strongly recommend using IB Gateway, because it supports all of the available Interactive Brokers API features.

Connection with a local application launch

By default, the gateway searches for an installed IB Gateway of a compatible version on the local server and launches it in automatic mode, so there is no need to authenticate manually and to confirm warnings by the operator. If an appropriate IB Gateway installation is not found, the gateway will download and install it.

To launch the gateway in this mode, set up the following parameters in the gateway configuration:

MetaTrader 5 trading platform

Gateway

Common Groups Symbols Translations Parameters Timeouts Information

 This gateway allows connecting to Interactive Brokers Services

☒ Enable

Name: ID:

Module:

Trading server:

Trading login: Password:

[Show add](#)

Gateway

Common Groups Symbols Translations Parameters Timeouts Information

 Please specify parameters of the gateway. These parameters are specific for each type of gateway. They allow using additional settings that were not available in the "Common" tab.

Parameter	Value
 Trading Mode	paper or live
 Start IBGateway	Yes
 Command Port	7462
 Client ID	0
 Import Default Symbols	Yes
 Rate Limit	40
 Flex Request Token	1op29sf907rnuu9
 Flex Request Query ID	123456

In the "Common" section:

- Trading server address 127.0.0.1 and port, which will be used by the gateway to establish an API-connection with IB Gateway. The gateway receives and sends data at this address. Address and port must be separated by a colon. Any free system port can be used. The default is 127.0.0.1:4002.
- Trading login user name for authorization in the Interactive Brokers system via IB Gateway.
- Password password for authorization.

In the "Parameters" section:

- Trading Mode your Interactive Brokers account type. Can be 'live' or 'paper'.
- Start IBGateway IB Gateway or TWS operation mode. To enable automatic IB Gateway launch by the gateway, set "Yes" (default). If the appropriate application is not found, the gateway will automatically deploy IB Gateway and will connect to it.
- Command Port command port via which connection to control IB Gateway will be established. Any free system port can be used. The default port is 7462.
- Client ID identifier of client API connection with which the gateway will connect to IB Gateway. You can use any value from 0 to 31. If several clients connect to one IB Gateway, you should ensure their Client IDs are unique. Value 0 is used by default.

To enable the gateway to connect IB Gateway to Interactive Brokers servers, please disable the two-factor authentication in your account settings. This can be done in your personal account on the Interactive Brokers site. Navigate to Settings User Settings Secure Login System.

User Settings

Login

Password

Online Features

Security

IP Restrictions

Secure Login System

Security Questions

Communication

Email Address

SMS Address

Mobile Number

Alert Notification

Marketing Preferences

Reporting

PortfolioAnalyst Delivery

Secure Login System

User Settings / Secure Login System

Secure Login Devices

	Security Code Card	Active
	IB Key Security via IBKR Mobile	Active

SHARE DEVICE

Secure Login Settings

Secure Login not required for trading

Seamless Authentication: Enabled

Click the gear icon next to "Secure Login required for trading". If this section is not available, please contact the Interactive Brokers technical support. Then select "I only want to use my Secure Login Device when logging into Account Management".

Secure Login Device Opt Out

User Settings / Secure Login System / Secure Login Device Opt Out

Secure Login System Opt Out

You can change your Secure Login configuration to only require the use of your Secure Login Device when logging into certain applications.

☒ I want to always use my Secure Login Device when logging in.

☐ I only want to use my Secure Login Device when logging into Account Management.

CONTINUE

Connection without local application launch

If for any reason you need to start IB Gateway (or TWS) manually, set the following parameters in the gateway configuration:

- Trading server address and port, which will be used by the gateway to establish an API-connection with IB Gateway, separated with a colon. The port must be previously specified in application settings.
- Client ID identifier of client API connection with which the gateway will connect to IB Gateway (or TWS). You can use any value from 0 to 31. If several clients connect to one IB Gateway, you should ensure their Client IDs are unique. Value 0 is used by default.
- Start IBGateway set "No", for the gateway to use connection to a remote IB Gateway (or TWS).

The port on which IB Gateway listens to incoming connections from the gateway should be configured under 'Configure' 'Settings':

MetaTrader 5 trading platform

IB GATEWAY. API ACCOUNT: File Configure Help

Connection Status

Purpose	Status
Interactive Brokers API Server	connected

DU1001550 Trader Workstation Configuration (Simulated Trading)

Client 0

Configuration

- General
- Lock and Exit
- Messages
- API
 - Settings
 - Precautions
 - News Configuration
- Presets
- Orders

Although you defined option exercise request, it will not follow this rule and will remain revocable

General

- ☐ Enable DDE clients
- ☐ Read-Only API
- ☐ TotalQuantity field may be used to transmit market data
- ☒ Download open orders on connection
- ☒ Include virtual FX positions when sending portfolio
- ☒ PrepareDailyPnL when downloading positions
- ☒ Send status updates for EFP and Volatility orders

Socket port: 4002

- ☒ Use negative numbers to bind automatic orders
- ☐ Create API message log file
- ☐ Include market data in API log file
- ☒ Expose entire trading schedule to API
- ☐ Split Insured Deposit from Cash Balance

Logging Level: Error

Master API client ID: 0

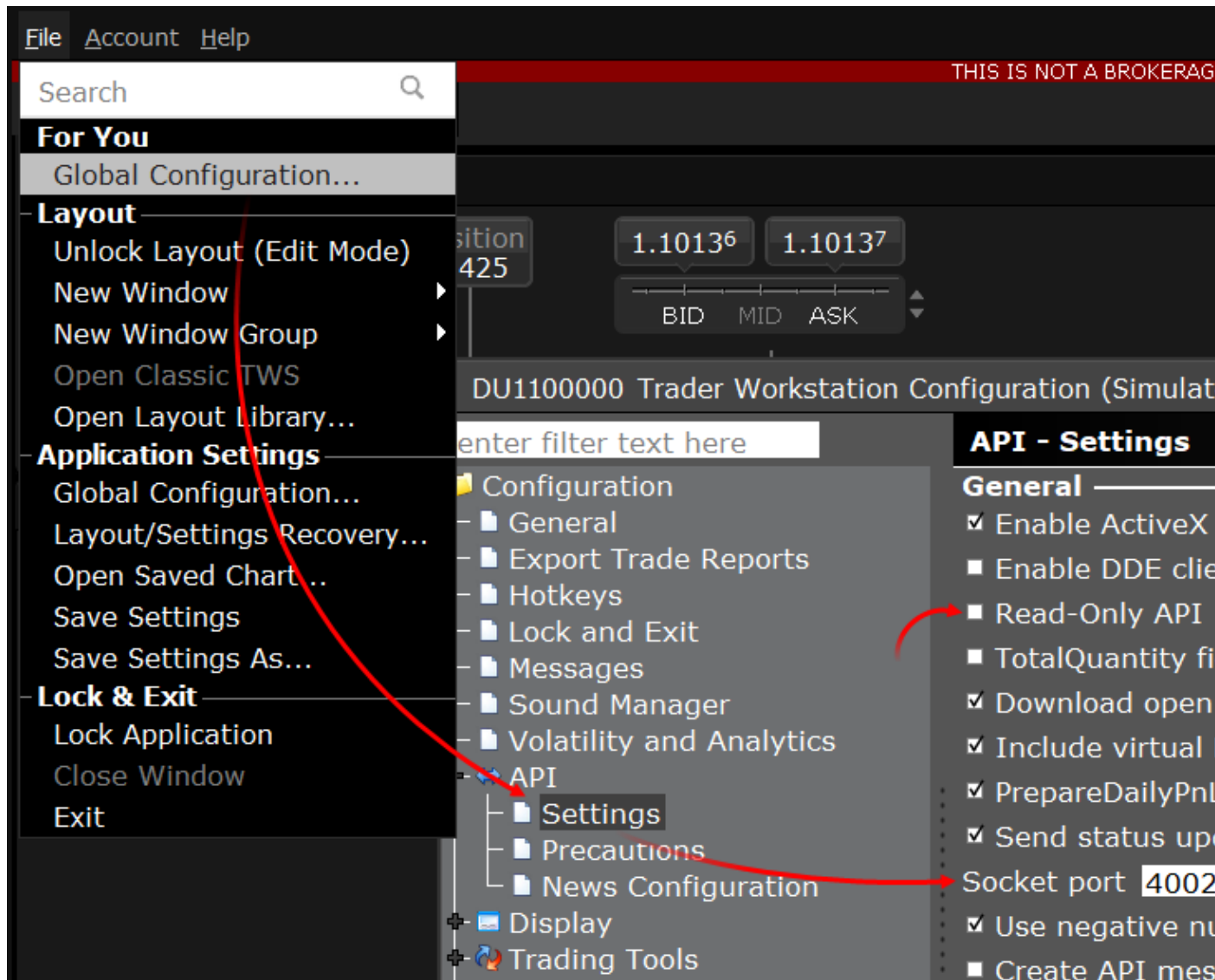
Timeout to send bulk data to API: 30 seconds

Component Exch. Separator:

- ☒ Show Forex data in 1/10 pips
- ☒ Allow Forex trading in 1/10 pips

MetaTrader 5 trading platform

The port on which TWS listens to incoming connections from the gateway should be configured under 'Files' 'Global Configuration':



Any free system port can be used. Also it is necessary to disable the "Read-Only API" option in this section.

Configuring financial instruments

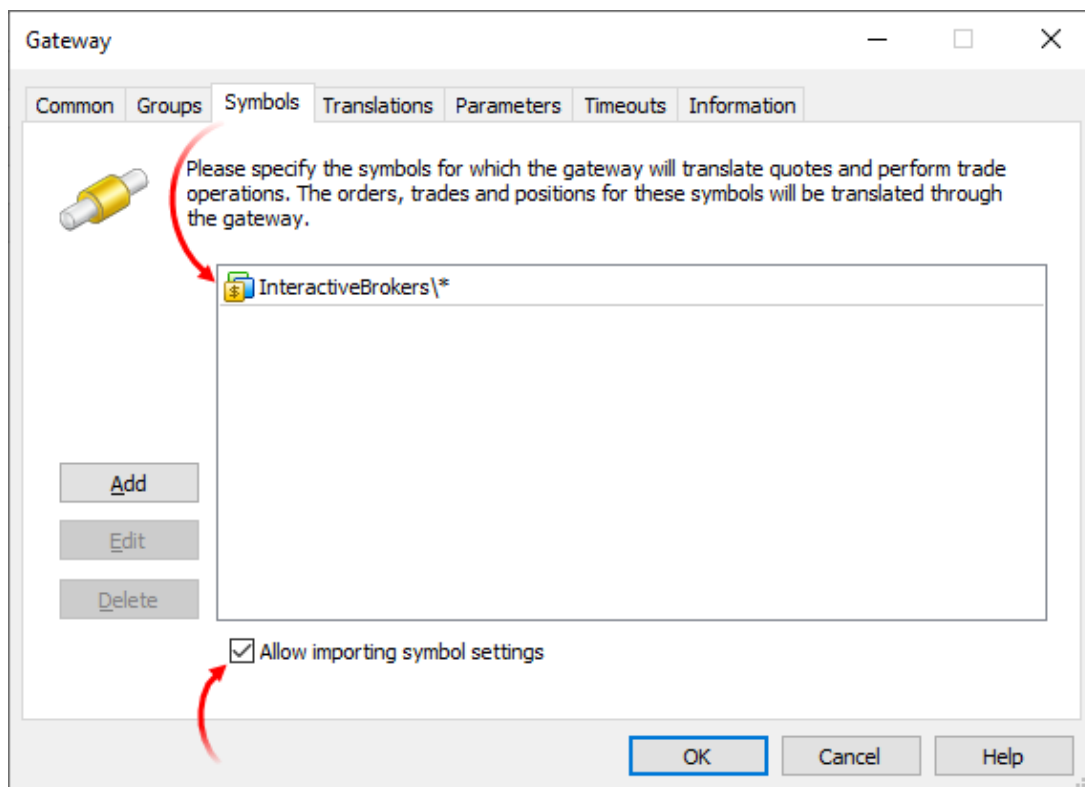
Interactive Brokers provides access to a variety of trading exchanges. Please visit the company's website for the full list of trading symbols. API Interactive Brokers does not provide the ability to automatically download all available trading instruments, therefore, work with symbols can be organized in two ways:

Built-in list of symbols

You can use the built-in list of symbols, which was created based on publicly available data provided by exchanges. This is the default method used in the gateway.

At start, the gateway creates the file [history server directory]\Gateway\IBGateway\<Gateway configuration name>\bases\instruments.csv with a predefined list of financial instruments. By default, the list contains symbols from the three most popular exchanges: NYSE, NASDAQ and AMEX. The file can be edited, and thus new symbols can be added or unused instruments can be deleted from it. According to this list, the gateway requests symbol specifications on the Interactive Brokers side and imports them to the trading platform, into InteractiveBrokers\<Exchange> symbol groups.

To enable this mode, allow the gateway access to the appropriate group of symbols on the trading platform side. Add it to the "Symbol" section of the gateway settings:



Also, make sure to enable "Allow importing symbol settings" option.

Each symbol description is contained in the instruments.csv file. Each symbol description starts with a new line and has the following format:

"<Symbol>,<Type>,<Exchange>". The following values can be used for the instrument type:

- STK stock or ETF
- OPT option
- FUT futures
- IND index
- FOP futures option
- CASH Forex
- BAG combo
- WAR warrant
- BOND bond
- CMDTY commodity
- FUND mutual fund

Example of instruments.csv contents:

```
AAPL,STK,NASDAQ
MSFT,STK,NASDAQ
MSFT,STK,MEXI
LE,FUT,GLOBEX
ES,FUT,GLOBEX
NQ,FUT,GLOBEX
EUR,FUT,GLOBEX
GBL,FUT,DTB
DAX,FUT,DTB
EUR.USD,CASH,IDEALPRO
ES,FUT,GLOBEX,201912
ES,FOP,GLOBEX,201912
```

This file can be edited and thus you can add new symbols or delete unused ones.

A symbol name used by Interactive Brokers can differ from its traditional name used on the exchange. The same applies to the exchange name. To check symbol and exchange names, please visit the Interactive Brokers website, section [Product Listings](#). For example, to find Microsoft stocks traded on the NASDAQ exchange, perform the following:

1. Select the NASDAQ exchange (NASDAQ) from the list:

Product Listings

Stocks	Options	Futures	FOPs
Structured Products	SSFs	Forex	Metals
Funds			
North America	Europe	Asia-Pacific	

Stocks - North America

Click on any *Market Center Details* link below to find details on products traded, order type

Country/Region	Market Center Details	Products
 United States	ArcaEdge (ARCAEDGE)	Stocks (OTCBB)
	Bats BYX (BYX)	Stocks
	Bats Global Markets (BATS)	Stocks
	Chicago Stock Exchange (CHX)	Stocks
	Direct Edge (DRCTEDGE)	Stocks
	Direct Edge (EDGEA)	Stocks
	IB VWAP Dealing Network (VWAP)	Stocks (VWAPs)
	IEX	Stocks
	Knight Securities	Stocks (OTCBB)
	NASDAQ (NASDAQ)	Stocks 629

2. Find the company name by specifying Microsoft in the search bar and find the symbol name in the table. Next, open the page with the detailed symbol description: it features the names of the exchange in which the instrument is traded:

Products

Indices

Stocks ☒

IB Symbol	Product Description (click link for more details)
MSFT	MICROSOFT CORP

Interactive Brokers
The Professional's Gateway to the World's Markets

Underlying Information	
Description/Name	Microsoft Corp (MSFT@NASDAQ)

Contract Information	
Description/Name	MICROSOFT CORP
Symbol	MSFT
Exchange	NASDAQ, CHX, IEX, NMS, NYSE
Contract Type	Stock
Country/Region	United States
Closing Price	137.63
	U.S. Dollar (USD)

MetaTrader 5 trading platform

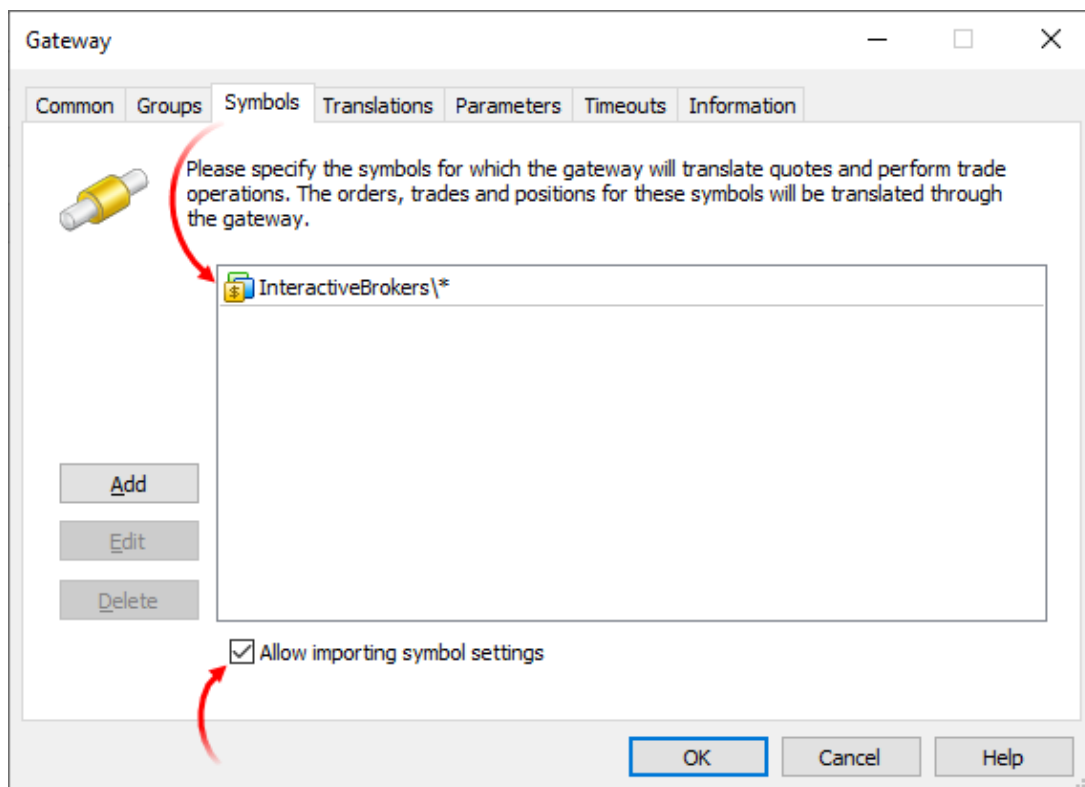
In this example, the MSFT security is traded on five exchanges: NASDAQ, CHX, IEX, NMS, NYSE, however the main exchange is the one marked in bold, i.e. NASDAQ. Thus, the following should be specified in file instruments.csv: MSFT,STK,NASDAQ.

Creating symbols manually

Instead of using the built-in list, you can create your own list of required symbols. You only need to create a configuration with proper naming and leave default settings. Using the platform list, the gateway will request appropriate configurations from Interactive Brokers and will update them if the symbol is found.

We recommend creating all symbols in a separate subcategory, for example InteractiveBrokers. This will enable efficient control of traders' access to liquidity using group settings. All symbols should be named according to the following rule: <Symbol>.<Exchange>. For example, AAPL.NASDAQ, MSFT.MEXI etc.

After creating all required symbols allow the gateway access to them. This can be done by adding the relevant symbol category to the "Symbols" section under gateway settings:

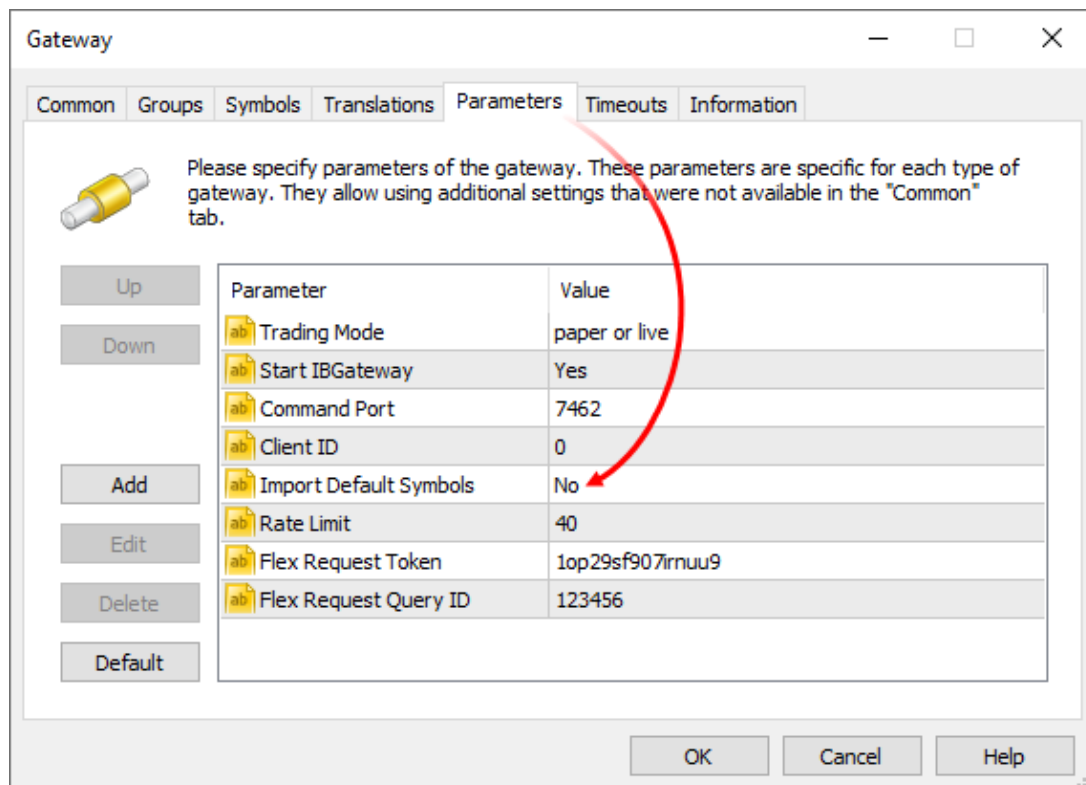


MetaTrader 5 trading platform

Also, make sure to enable "Allow importing symbol settings" option.

The gateway will regularly (at least once a day) request symbol specifications on the Interactive Brokers side and, if necessary, update settings for these symbols in the MetaTrader 5 platform. If any of the symbols is not available for trading in InteractiveBrokers, the gateway will move it to the InteractiveBrokers\Deleted group.

To prevent the gateway from using the built-in list of symbols, set "No" for the parameter "Import Default Symbols".



Obtaining the history of trades

One of the main gateway purposes is to ensure a complete and correct trading history for each account on the MetaTrader 5 side. This is also required for a proper calculation of current open positions.

The gateway transmits all executed deals to the platform in real time. This includes results of operations requested directly from the platform as well as deals executed via other clients (for example, TWS). If the gateway is stopped for some time, it will need to retrieve from Interactive Brokers the history of all deals which were performed within this period. The gateway has two mechanisms, which are used depending on the downtime duration:

- If the gateway was stops for no more than one trading day, the history of deals for the current trading day is requested.
- When the gateway was down for a longer period, the history for the last year is requested.

No additional setup is required for requesting the last day deals, while the gateway can automatically retrieve them if necessary. The history of deals for the last year is requested from special Flex reports provided by Interactive Brokers. To enable access to the reports, you should set up a template and receive a token via a personal account.

The history of deals can only be obtained when connected via IB Gateway. This functionality is not supported for TWS.

Creating a Flex Token

Go to your [personal account on the Interactive Brokers website](#) and select Settings Account Settings section from the main menu. Then click the gear icon next to the Flex Web Service:

Account Settings

Profile(s)

Mr.

Financial Information

Annual Net Income	250,000
Net Worth	5,000,000
Liquid Net Worth	> 5,000,000

Warrants

France

Options

2 Years

Configuration

Customer Type: Individual

Legal Residence:

Account Alias

Account Type: Margin

Base Currency: USD

Commissions Pricing Structure

Dividend Reinvestment: Disabled

Excess Funds Sweep: Do not Sweep

Paper Trading Account:

Reporting

Flex Web Service : Enabled

Third-party Services

The "Configure Flex Web Service" window will open. Check the box next to "Flex Web Service Status" and click "GENERATE NEW TOKEN":

Configure Flex Web Service

Use this page to activate or deactivate the Flex Web Service. You will need to generate a token to initiate Flex Query requests and retrieve Flex Queries using this service.

☒ Flex Web Service Status

Current Token

Current Token

Activation Period:2019-07-09, 08:29:29 EDT — 2020-06-09, 08:29:29 EDT

Valid for IP Address---

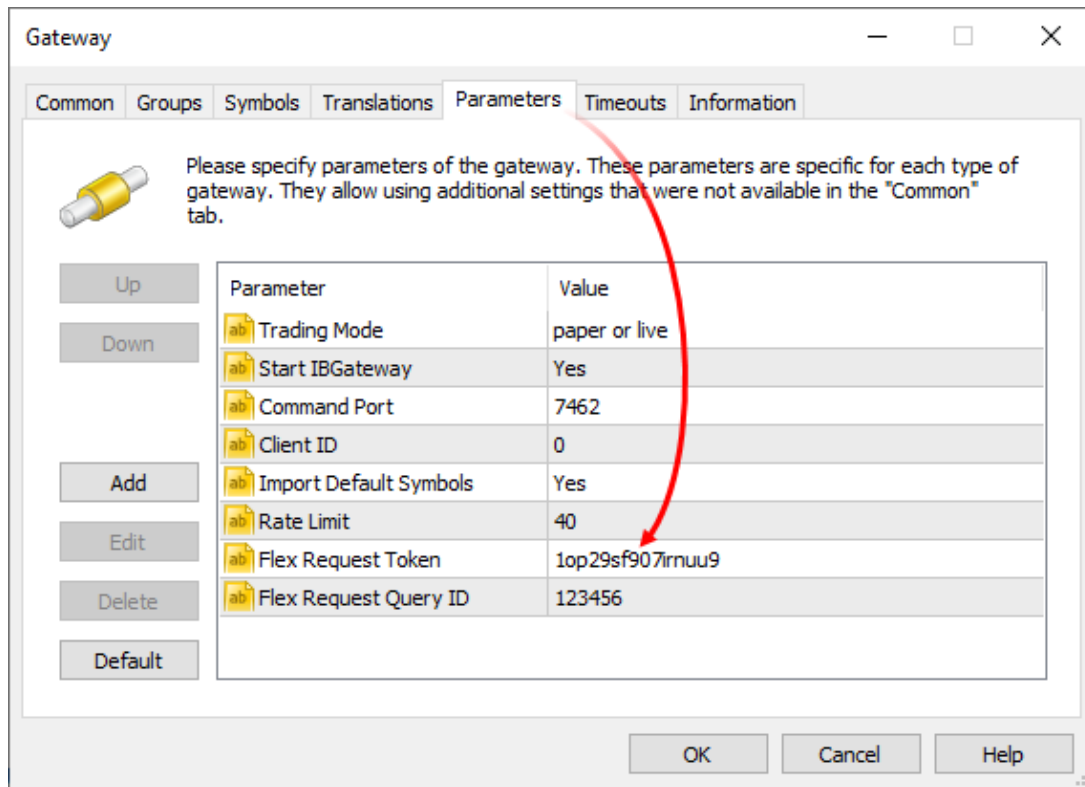
GENERATE NEW TOKEN

CANCEL

SAVE

MetaTrader 5 trading platform

In the next step, select the token validity period of 1 year and click "GENERATE NEW TOKEN". Specify the the generated token in the "Flex Request Token" gateway parameter.



When the token expires, generate a new one and update the corresponding gateway parameter.

Creating a Flex Report template

Open the menu in your personal account and go to the Reports Tax Docs section:

×

TOOLS & SERVICES

Home

Portfolio

Orders & Trades

Order Ticket

Transfer & Pay

Watchlist

Market Overview

Scanners

Events Calendar

Forum

Investors' Marketplace

PortfolioAnalyst

Reports / Tax Docs

FYI Notifications

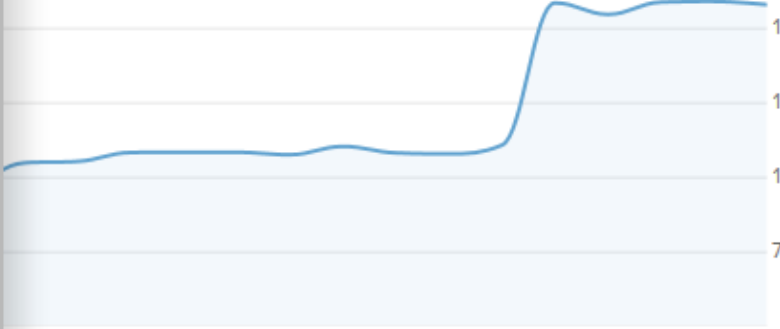
Client Portal

Trading/Market Data permissions

on another platform. You can only trade and view real-time
can still manage your account but can't trade. Another
connected your session.

Change (1M)

\$8,644,195

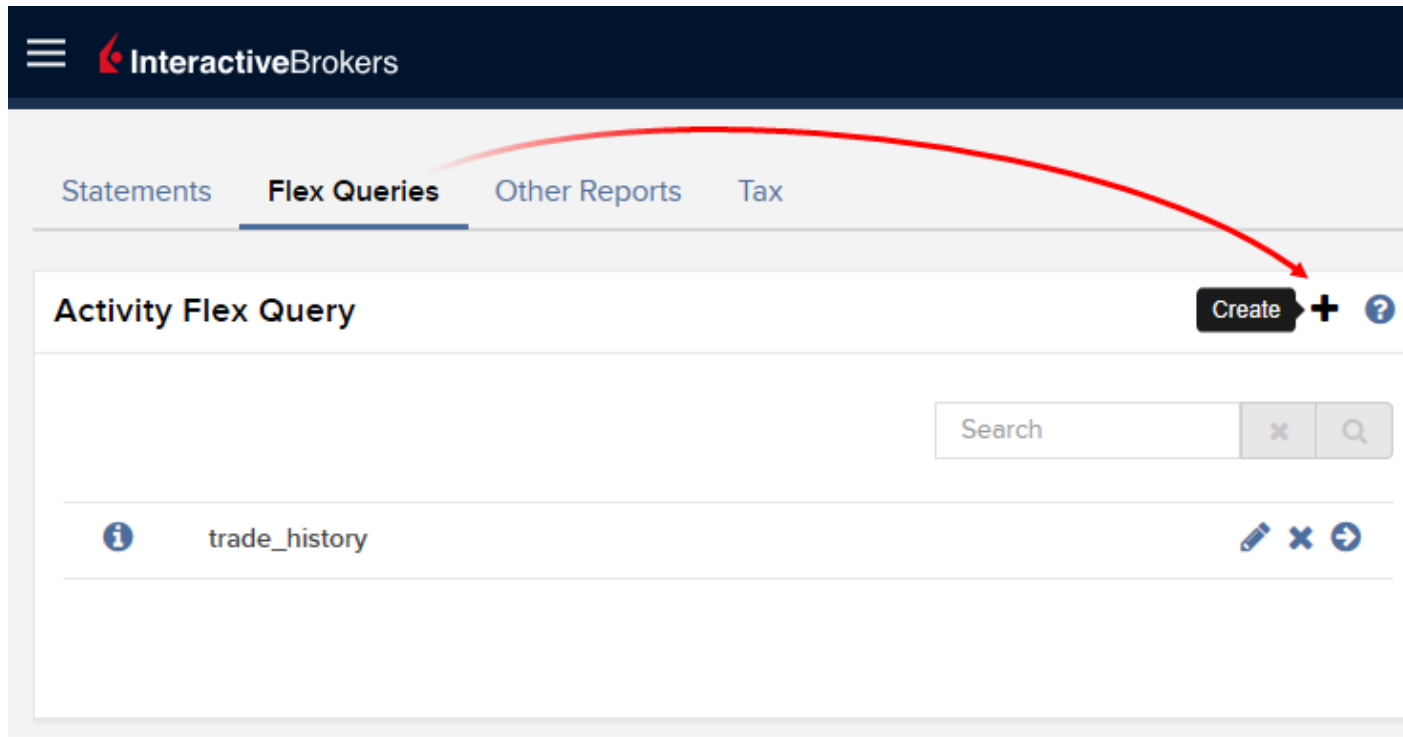


The chart displays a blue line representing a value over time. The line starts at a low level, remains relatively flat with minor fluctuations, and then experiences a sharp, significant increase, reaching a peak before slightly declining and then leveling off again. The background of the chart area is light blue with horizontal grid lines.

7D MTD 1M YTD 1Y

Watchlist

Add a new report on the "Flex Queries" tab. Click the plus icon in the "Activity Flex Query" section:



Specify the desired name for the report, and then select "Trades" under Sections.

Create Activity Flex Query

Activity Flex Query Details

Query Name

trade_history

Sections (Select Multiple)

Account Information

Cash Report

Cash Transactions

CFD Charges

HK IPO Open Subscriptions

HK IPO Subscription Activity

IBG Notes

Mark-to-Market Performance Summary in Base

Month & Year to Date Performance Summary in Base

Net Asset Value (NAV) in Base

Net Stock Position Summary

Statement of Funds

Trades

Transaction Fees

MetaTrader 5 trading platform

In the next window, select "Executions" and the following fields:

- AccountID
- Symbol
- Currency
- Conid
- Date/Time
- ListingExchange
- Exchange
- TradeID
- IBExecutionID
- Quantity
- TradePrice
- IbCommission

Trades

Options

Symbol Summary

Asset Class

Orders

Executions 

Closed Lots

Wash Sales



Select All



Account ID



Currency



Symbol



Conid



Listing Exchange



Trade ID



Date/Time



Exchange



Quantity



TradePrice



IB Commission



IB Execution ID



Move to Top



Account Alias



Model



MetaTrader 5 trading platform

Then click "Save". Select the following options under the "Delivery Configuration" section:

- Accounts all accounts in Interactive Brokers which are available to the gateway
- Models Optional
- Format XML
- Period Last 365 Calendar Days

InteractiveBrokers

Delivery Configuration (Applicable for Email, FTP and Flex Web Service)

Accounts U3796350, U2442157
[ADD/EDIT ACCOUNT\(S\)](#)

Models Optional

Format XML

Period Last Business Day

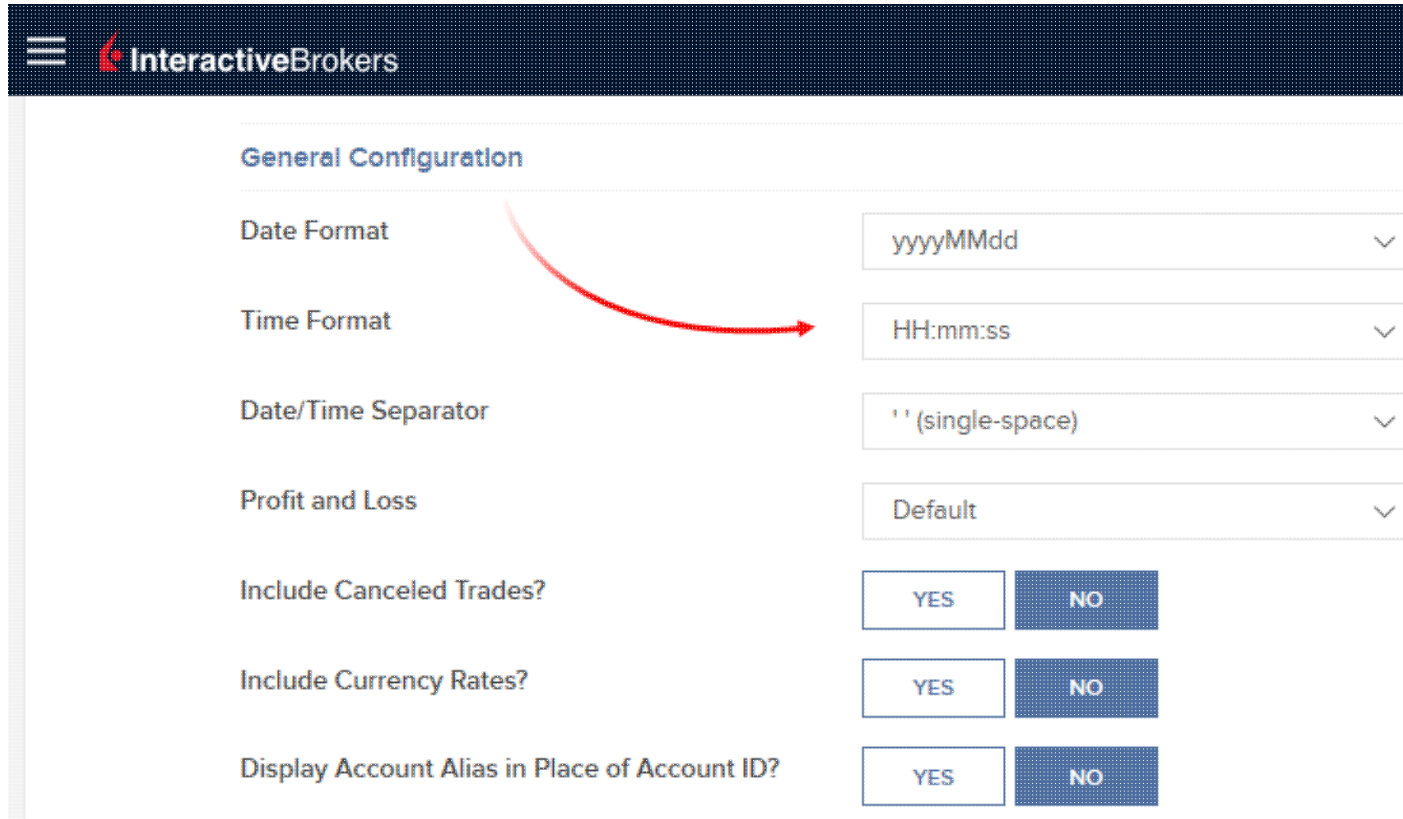
Account Customer Type

✓ U2442157	Individual
✓ U3796350	Individual

[RESET](#) [CONTINUE](#)

Set the required data format under the "General Configuration" section:

- DateFormat yyyMMdd
- TimeFormat HH:mm:ss
- Date/Time Separator single-space



InteractiveBrokers

General Configuration

Date Format	yyyyMMdd
Time Format	HH:mm:ss
Date/Time Separator	' ' (single-space)
Profit and Loss	Default
Include Canceled Trades?	YES <input checked="" type="button" value="NO"/>
Include Currency Rates?	YES <input checked="" type="button" value="NO"/>
Display Account Alias in Place of Account ID?	YES <input checked="" type="button" value="NO"/>

Then click "Continue". Continue settings in the next window and click "Create".



Review Your Activity Flex Query

Activity Flex Query Details

Query Name	trade_history
------------	---------------

Sections

Trades

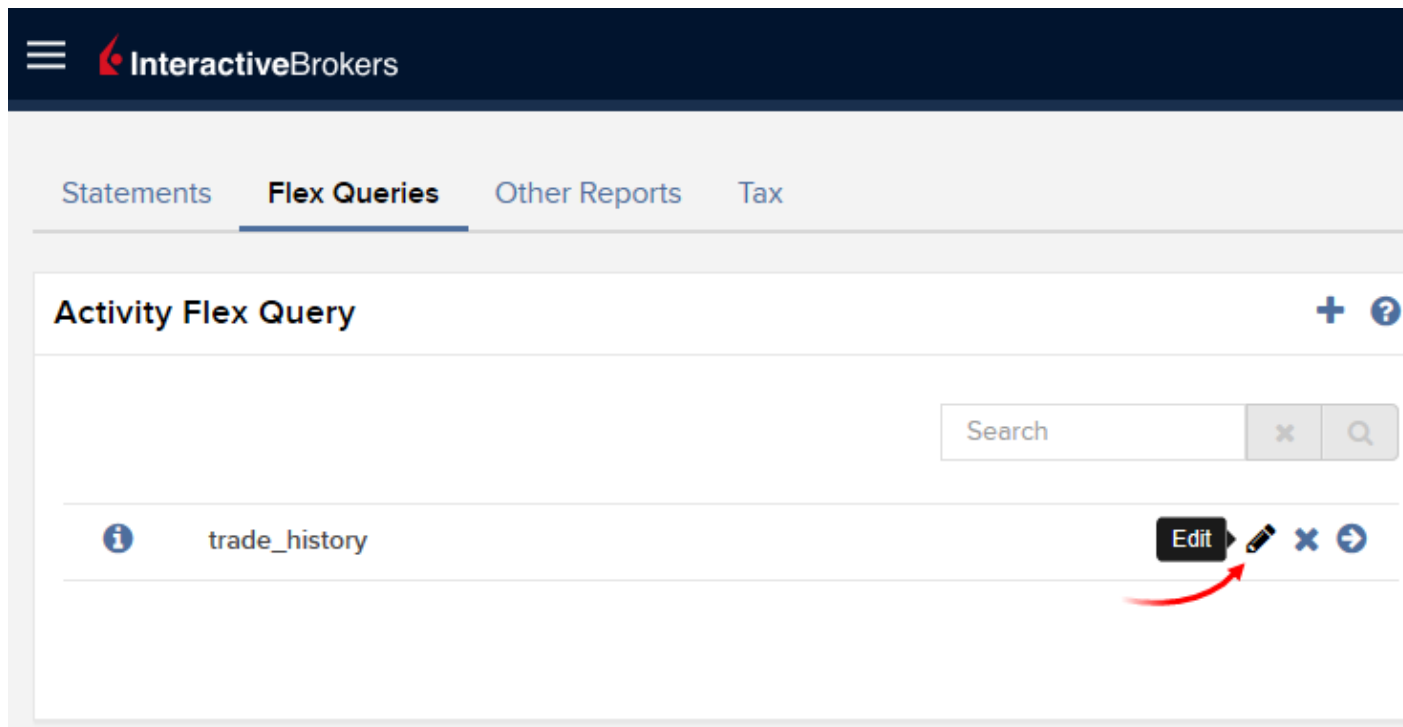
Options: Executions

1. ClientAccountID	3. Symbol
2. CurrencyPrimary	4. Conid
7. DateTime	9. Quantity
8. Exchange	10. TradePrice

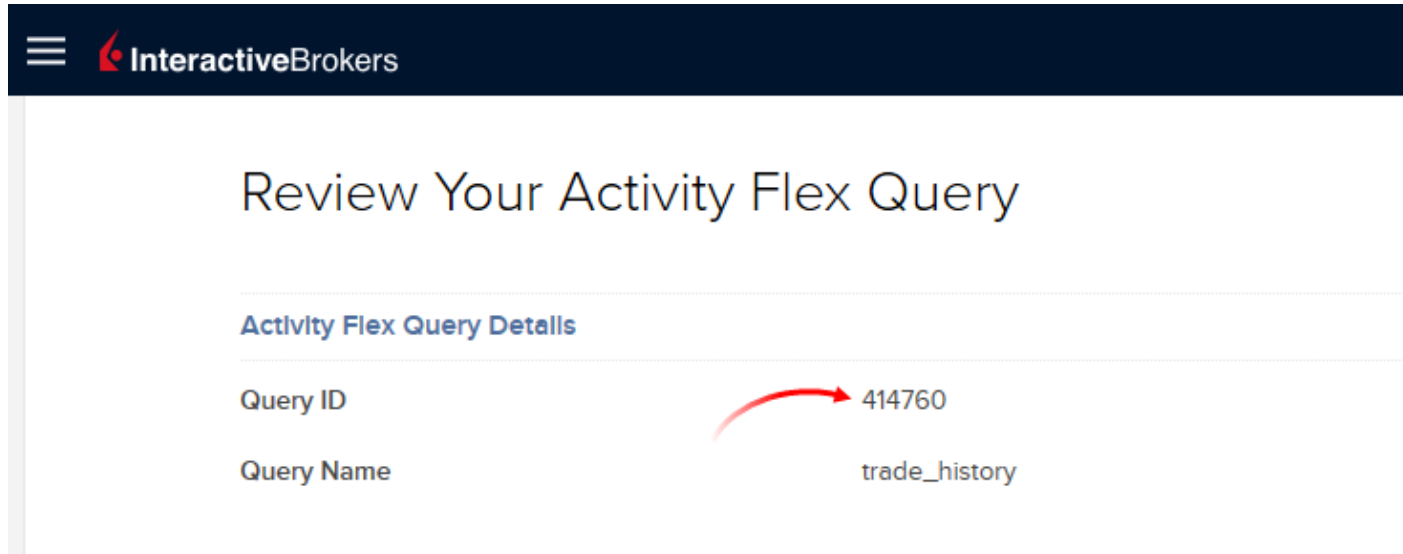
BACK

CREATE

You can find out the new report ID by clicking on the edit button:

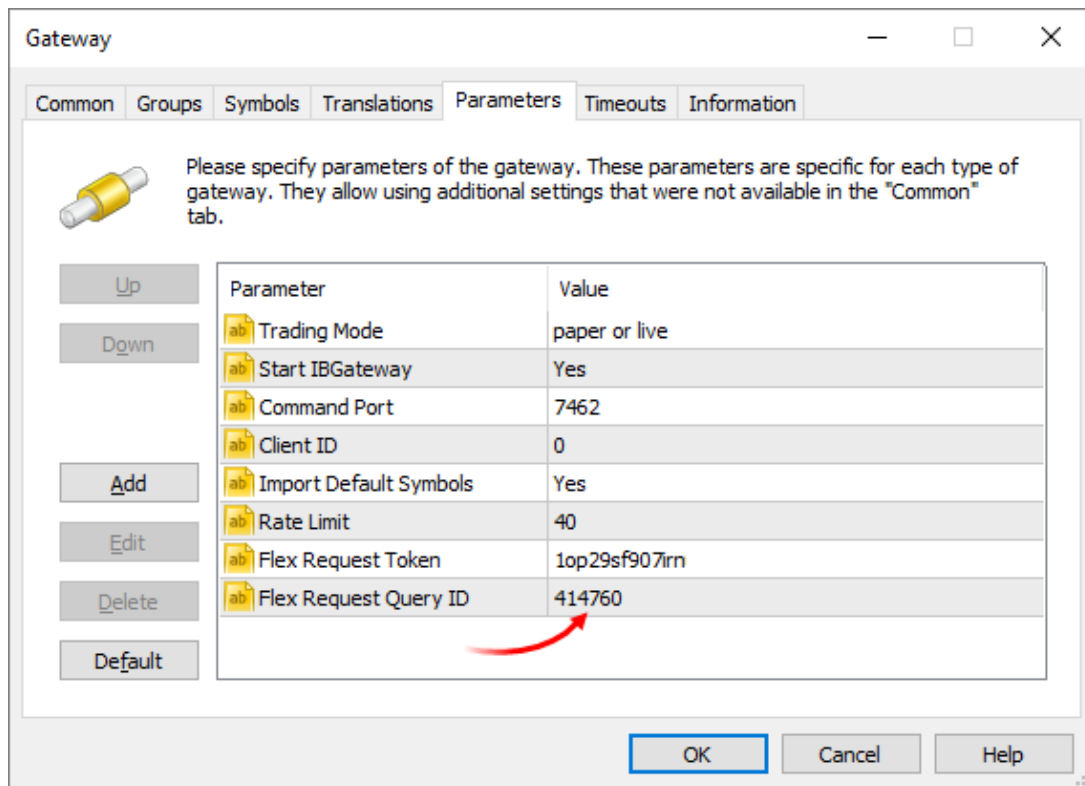


The report ID will be shown in the upper part of the window. Copy this text.



MetaTrader 5 trading platform

Paste this ID as value of the "Flex Request Query ID" parameter in gateway settings:



Interactive Brokers API request limit

Interactive limits request sending frequency to 50 requests per second. Depending on the IB Gateway version used by the gateway, the exceeding the limit may have different consequences:

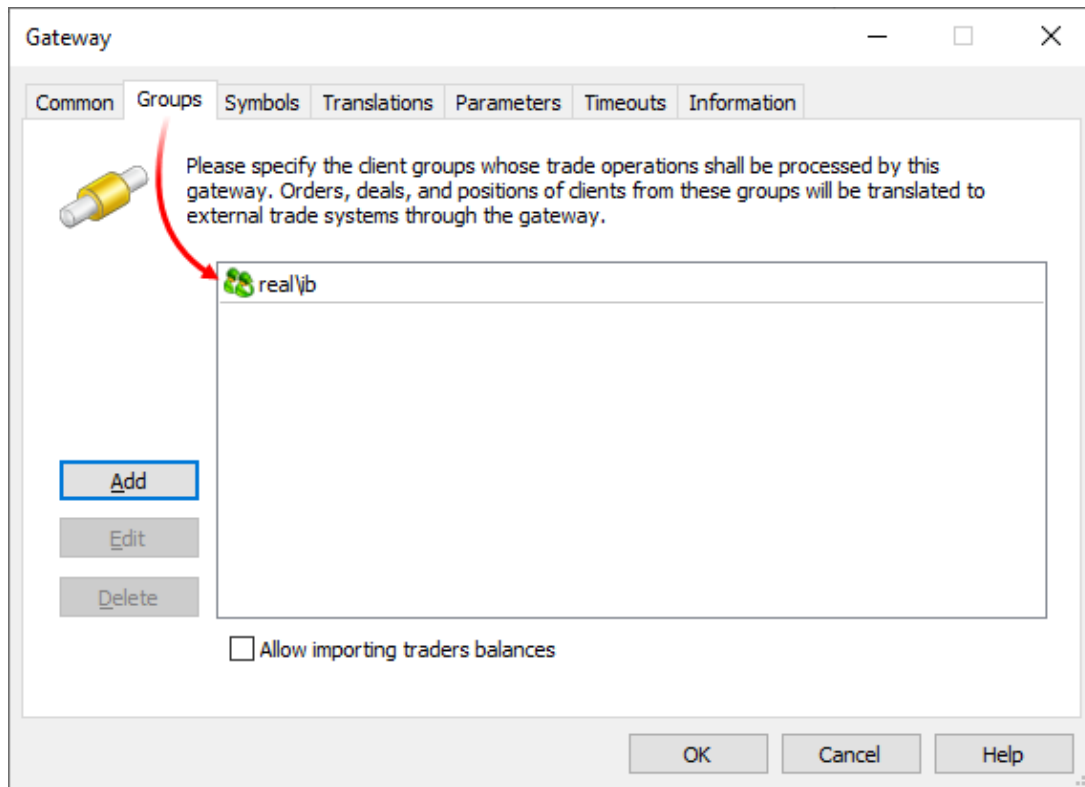
- IB Gateway version below 974 if the limit is exceeded, connection with IB Gateway (or TWS) is automatically closed.
- IB Gateway version above 974 if the limit is exceeded, connection is not closed, but request processing can slow down significantly.

You can set your own limit to the frequency of request sending by gateway, using the Rate Limit parameter. It is set to 40 by default, which is the recommended request frequency for the stable operation with IB Gateway version below 974.

Configuring groups trading accounts

Create groups for trading accounts, whose operations will be forwarded via the Interactive Brokers gateway. For example, real\ib. Then add it to the "Groups" section of gateway settings:

MetaTrader 5 trading platform



In the specified group, create accounts trade operations from which will be processed by the gateway.

The gateway trades in the one-to-one mode, i.e. each trading account on the Interactive Brokers side can correspond to only one account on the MetaTrader 5 platform side. Therefore, in the "Account" section of each account to be managed by the gateway you should specify an appropriate account code on the Interactive Brokers side. Specify your Interactive Brokers gateway configuration as the gateway ID.

MetaTrader 5 trading platform

Account: 2095, IB Demo #1, USD, 1:100

Overview Personal **Account** Security

Group: real\yb

Color: ☐ None Leverage: 1 : 100

Bank account: Agent account:

☒ Enable this account ☒ Enable trading by Expert Advisors

☒ Enable trading ☒ Enable Web API connections

☒ Enable to change password ☒ Enable daily reports

☒ Enable trailing stops ☐ Change password at next login

☒ Enable one-time password ☐ Sponsored MetaTrader Virtual Hosting

Trade accounts:

Gateway ID	Account
 114, Interactive Brokers Gateway	DU110000000

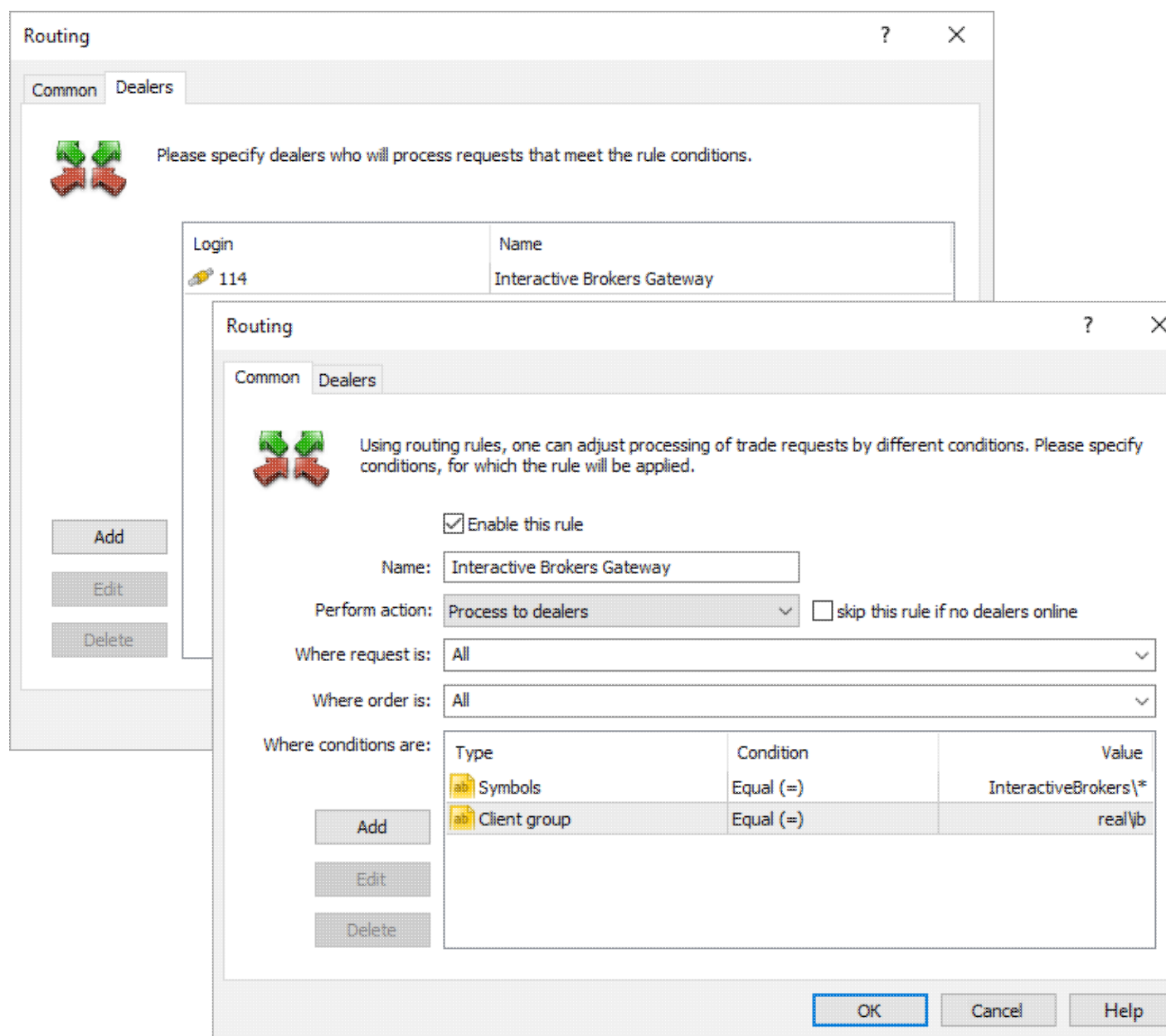


Configuring Trade Requests Routing

Configure the routing to forward client requests to the Interactive Brokers gateway. To do this, add a routing rule to the relevant MetaTrader 5 Administrator section.

Select "Process to dealers" as an action in common settings. Assign this routing rule for all requests and orders. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway.

In the screenshots below all orders created by users in the real\ib* groups for the symbols from the InteractiveBrokers* section will be forwarded to the gateway for processing.



MetaTrader 5 trading platform

Once the rule has been added, set the required priority for it relative to other rules, by moving it to the desired position in the list.

Receiving Level 1 market data

The gateway enables broadcasting of real-time financial symbol quotes to the platform. The data include the best Bid and Ask prices, as well as prices and volumes of performed trades (Last, Volume).

To start receiving symbol quotes, add them to the "Marketdata" or any nested subgroup. For example, you have added all symbols, which should be processed via Interactive Brokers, to the InteractiveBrokers* subgroup. Create the "Marketdata" subgroup under this group and add to it all the symbols for which you want to receive quotes.

Broker-Demo - MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Apply

ECN

Routing

Funds & ETF

Symbols (686)

Forex (21)

Crypto (1)

ECN (1)

Preliminary (649)

InteractiveBrokers (14)

Marketdata (5)

Spreads

Charts & Ticks

1 Minute History Charts

Bid/Ask/Last Ticks

Synchronization

Subscriptions

Live Update

Symbol	Description	Type	Execution	Digits	Expiration
Marketdata					
AAPL	AAPL	Exchange Stocks	Exchange	4	
MSFT		Exchange Stocks	Exchange	4	
LE		Exchange Stocks	Exchange	4	
ES		Exchange Stocks	Exchange	4	
NQ		Exchange Stocks	Exchange	4	
EUR		Exchange Stocks	Exchange	4	
GBL		Exchange Stocks	Exchange	4	
DAX		Exchange Stocks	Exchange	4	
EUR.USD		Exchange Stocks	Exchange	4	

For Help, press F1

1929 / 9 Kb

MetaTrader 5 trading platform

After that, the gateway will subscribe to appropriate Interactive Brokers data and will start delivering them to the platform. In the above example, the gateway will provide quotes for 5 symbols of the 14 symbols operated via Interactive Brokers. To stop receiving symbol quotes, move it away from the Marketdata subgroup.

Please note that the "Marketdata" symbol group must be accessible to the gateway. For example, if you create it under the InteractiveBrokers* section which has earlier been added to the gateway settings, no additional actions will be required. If you create it under a different section, you should add an appropriate path to the gateway settings.

- You can enable or disable the delivery of quotes at any time, by moving the symbols to/from the "Marketdata" subgroup. The gateway will immediately subscribe or unsubscribe from the relevant data.
- By default, Interactive Brokers enables the delivery of quotes for up to 100 financial instruments simultaneously. If quotes for more symbols are required, please contact Interactive Brokers and request an appropriate subscription.
- Receiving quotes from Interactive Brokers is optional. You can use the gateway only for working with trading operations, while enabling data delivery via separate datafeeds, such as IQ Feeder.

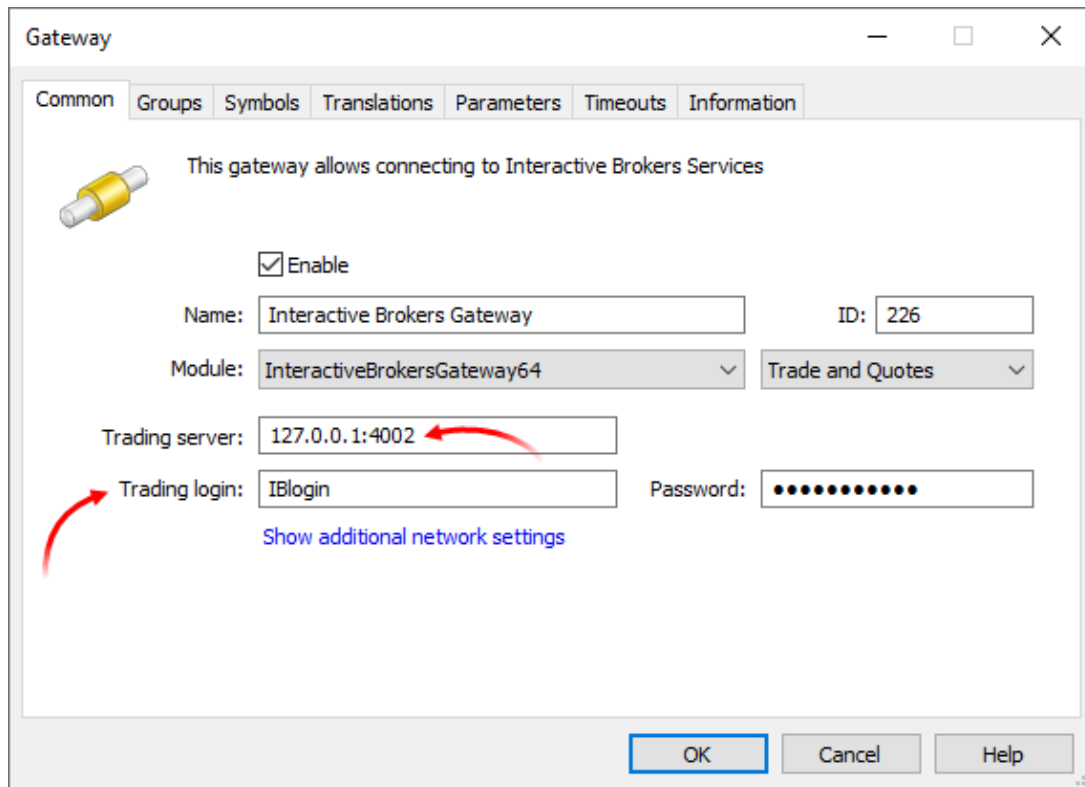
Using Multiple Gateways Simultaneously

The platform does not impose restrictions on the simultaneous use of multiple copies of Interactive Brokers gateways, neither technically nor from the point of view of license permissions. It means you can forward trading operations for execution via several Interactive Brokers accounts.

Please pay attention to the following requirements when configuring the gateways:

1. All gateways must use different logins for connection to InteractiveBrokers. They must use different ports for establishing an API connection with the IB Gateway application:

MetaTrader 5 trading platform



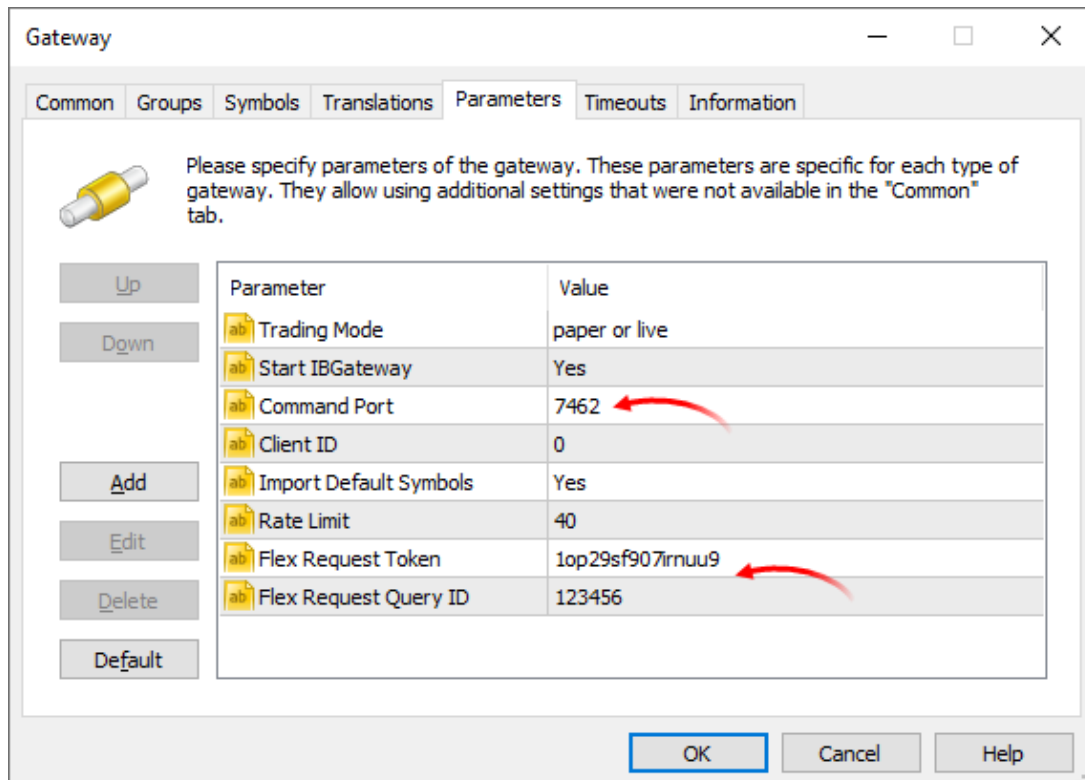
The Gateway dialog box, Common tab, is shown. It contains the following fields and controls:

- Enable:** A checked checkbox.
- Name:** A text field containing "Interactive Brokers Gateway".
- ID:** A text field containing "226".
- Module:** A dropdown menu showing "InteractiveBrokersGateway64".
- Trade and Quotes:** A dropdown menu showing "Trade and Quotes".
- Trading server:** A text field containing "127.0.0.1:4002".
- Trading login:** A text field containing "IBlogin".
- Password:** A text field containing a masked password (dots).
- Show additional network settings:** A blue link.
- Buttons:** OK, Cancel, and Help.

Red arrows point to the "Trading server" and "Trading login" fields.

2. Command Port parameter values must be different.

3. Each gateway must use Flex token and Flex query ID specifically configured for its login.



The Gateway dialog box, Parameters tab, is shown. It contains the following fields and controls:

- Buttons:** Up, Down, Add, Edit, Delete, and Default.
- Table:** A table with two columns: Parameter and Value.

Parameter	Value
Trading Mode	paper or live
Start IBGateway	Yes
Command Port	7462
Client ID	0
Import Default Symbols	Yes
Rate Limit	40
Flex Request Token	1op29sf907imuu9
Flex Request Query ID	123456

Red arrows point to the "Command Port" and "Flex Request Query ID" values.

MetaTrader 5 trading platform

4. Appropriate gateway configurations must be specified in trading accounts on the MetaTrader 5 side. The specified Interactive Brokers accounts must be available to the selected gateways.

The screenshot shows the 'Account' window in MetaTrader 5. The title bar reads 'Account: 2095, IB Demo #1, USD, 1:100'. The window has four tabs: 'Overview', 'Personal', 'Account' (selected), and 'Security'. The 'Account' tab contains the following settings:

- Group: real\yb
- Color: ☐ None
- Leverage: 1 : 100
- Bank account:
- Agent account:
- ☒ Enable this account
- ☒ Enable trading
- ☒ Enable to change password
- ☒ Enable trailing stops
- ☒ Enable one-time password
- ☒ Enable trading by Expert Advisors
- ☒ Enable Web API connections
- ☒ Enable daily reports
- ☐ Change password at next login
- ☐ Sponsored MetaTrader Virtual Hosting

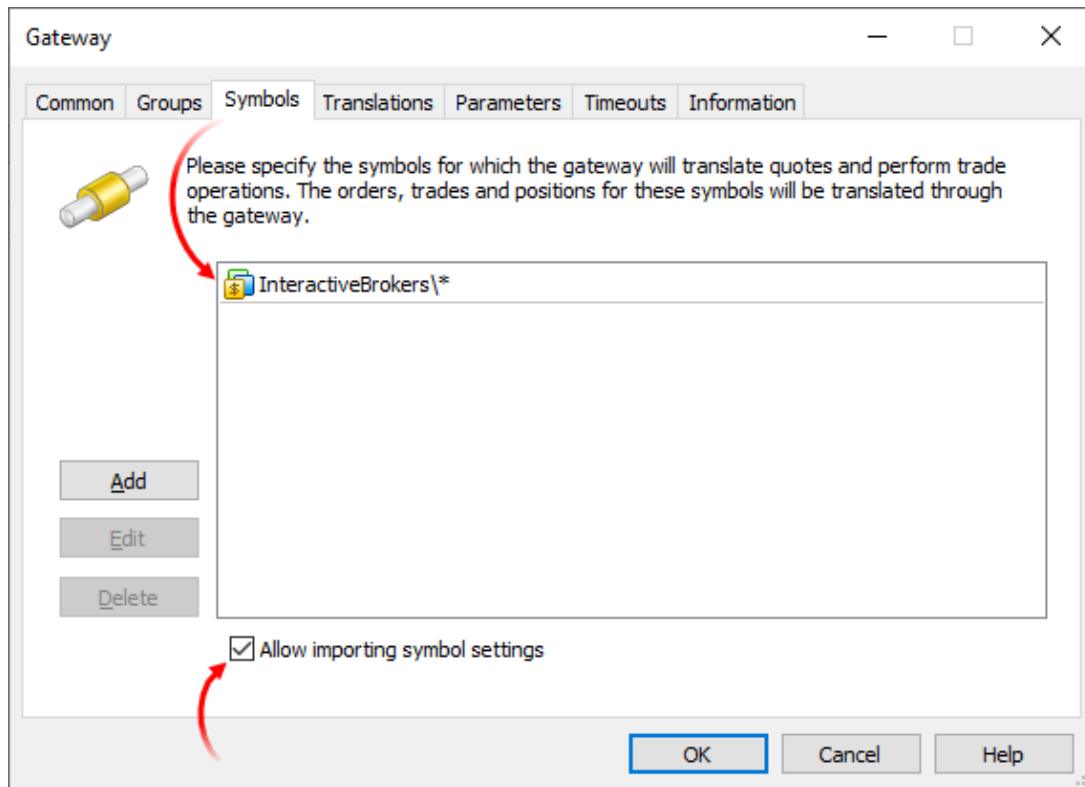
Below these settings is a table for 'Trade accounts':

Gateway ID	Account
114, Interactive Brokers Gateway	DU110000000

To the left of the table are buttons: 'Synchronize', 'Add', 'Edit', and 'Delete'. A red curved arrow points from the 'Add' button to the first row of the table. At the bottom of the window are buttons: 'New Client...', 'Update' (highlighted with a blue border), 'Cancel', and 'Help'.

MetaTrader 5 trading platform

5. If several gateways have access to the same symbols in the MetaTrader 5 trading platform, it is recommended to disable the "Allow importing symbol settings" option for all but one of the gateways.

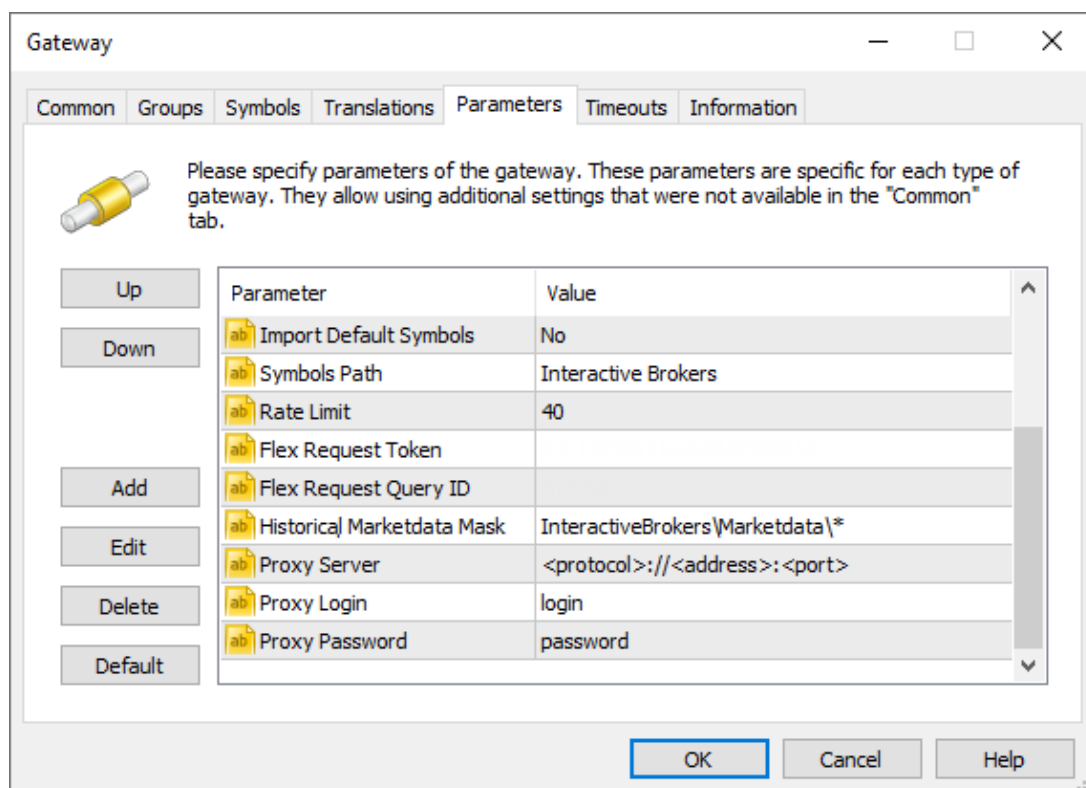


Configuring operation through a proxy server

In case the Internet access is provided through a proxy on the machine where the gateway is running, specify the relevant parameters:

- **Proxy Server** the proxy server address in the format of `<protocol>://<address>:<port>`. For example, `socks4://192.168.0.1:4145`. Supported protocols include `http`, `socks4` and `socks5`. Gateway queries, such as flex report requests, will be sent through the specified server.
- **Proxy Login** a login for authentication in the proxy server.
- **Proxy Password** a password for authentication in the proxy server.

MetaTrader 5 trading platform



MetaTrader 5 Trading Platform / Platform Components / WebTerminal

MetaTrader 5 WebTerminal

The MetaTrader 5 WebTerminal enables financial market trading using any web browser. It works in all operating systems and browsers, while requiring no extra software installations. All transmitted data is securely encrypted.

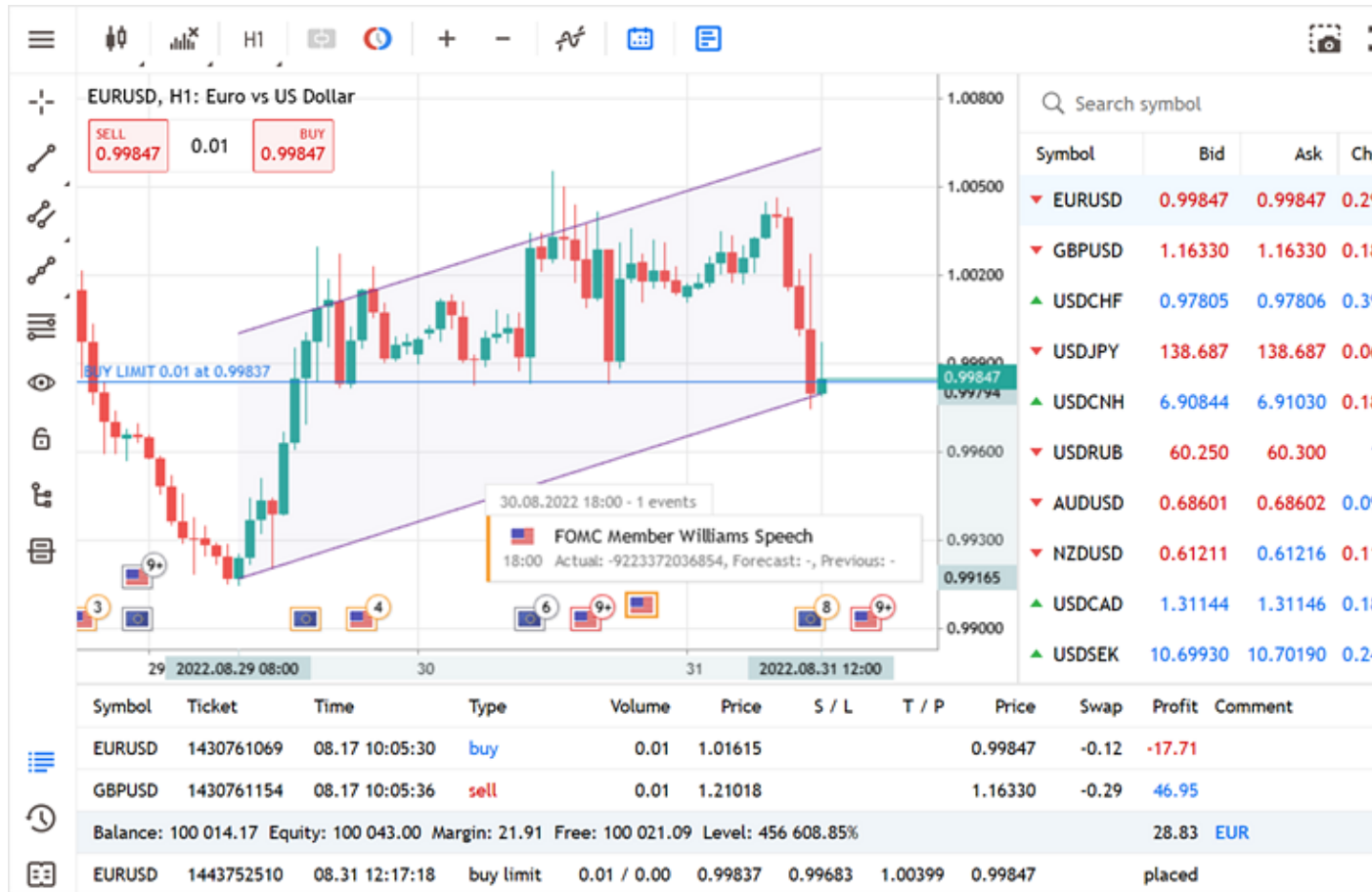
The web terminal supports all types of market and pending orders, as well as one-click trading. Traders can view real-time quotes and analyze charts using basic graphical objects. Charts can be analyzed using 30 technical indicators.

The terminal is a modern HTML5 application that can be easily integrated into any website via a simple iframe widget.

The web terminal is located entirely on the broker's access servers. It operates as an individual web terminal, which only works with your platform. This provides maximum security and full control for the broker.

When connecting to the web terminal via your website, the trader should only enter the login and password, without having to select a server. The platform will determine which server (demo, real, main or additional trading server) the account belongs to. For the seamless web terminal integration with the clients area on the broker website, the user login can be pre-selected to let your client enter a password only. If a user saves a password in a browser storage, the account will be connected automatically on the next run.

MetaTrader 5 trading platform



- The web terminal only operates with the trading platform version 3440 or higher.
- If your platform license does not include the web terminal, order it via the App Store section of the Support Center website.

How to set up the platform

The web terminal operates on an access server which acts as a web server.

Register a domain (or a subdomain for your existing domain) where the web terminal will run. Below we use the domain webtrading.broker.com as an example you should replace it with your own domain.

Associate your domain with the public IP address of your access server. For this purpose, the appropriate record must be specified on the DNS hosting. For example, for webtrading.broker.com host, the following entry should be registered in DNS: "XXX.XXX.XXX.XXX webtrading.broker.com", where XXX.XXX.XXX.XXX is the public address of the access server.

Next, open Integration \ Web services in MetaTrader 5 Administrator and add an SSL certificate for the domain. The certificate will enable connections to the access server using the HTTPS protocol.

You can add extra certificates for your White Label partners to provide a separate web terminal for each required company's domain. After that, in the relevant field, specify which company the domain belongs to.

SSL Certificates

Please specify server SSL certificates for all access servers. They will be used for connections made through them.

Up	Domains	Company	Thumbprint	Expiration
Down	✓ webtrading.label.com	Partner Broker	93ap92189sixck9191611lapx11oqa9815a	2023.05.08
	✓ webtrading.broker.com	ABC Broker	3b7a99awplj017q433492zx194qpj1jh8	2023.05.08
	+ click to add...			
Add				
Edit				
Delete				

MetaTrader 5 trading platform

It is recommended to have port 443 available on the server for HTTPS connections. In this case, your clients will not need to explicitly specify the port in the address bar to open the web terminal page.

No other settings are needed. Your web terminal will be available at <https://webtrading.broker.com/terminal/>.

If you have multiple access servers

For each platform we recommend using at least two access servers: for the main and backup configuration. As your customer database grows and your business geography expands, the number of servers will increase to ensure high-quality service.

To enable connections to your web terminal via any of your access servers, add A-records with their public addresses to the DNS server. Thus, you will have several records with different addresses but with the same domain:

XXX.XXX.XXX.XXX webtrading.broker.com

YYY.YYY.YYY.YYY webtrading.broker.com

ZZZ.ZZZ.ZZZ.ZZZ webtrading.broker.com

XXX, YYY and ZZZ are the addresses of your access servers. There can be any number of them.

No additional actions are required on the platform side. The domain certificate uploaded via the Integration \ Web Services section is used for all access servers.

To balance the load by distributing connections to the web terminal between access servers, use GeoDNS or GLSB services. Please check with your hosting provider whether this service is available.

If you hold multiple licenses

If you have multiple MetaTrader 5 platform licenses, for example, the main cluster for live accounts and an additional cluster for demo accounts, you should configure web terminals separately for each of them. The web terminal runs on access servers which belong to a particular platform, and thus they do not know about the existence of other platforms.

With such a configuration, you will have different web terminal links for each platform. You need to provide the option to select the desired platform for your customers. For example, you can implement a drop-down list on your website.

How to add the web terminal to your site

To install the web terminal on your site, simply place the `<iframe>` tag on the desired page. The address of your web terminal should be specified as the source in this tag:

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Web Terminal</title>
</head>
<body>
  <iframe src="https://webtrading.broker.com/terminal?mode=connect&marketwatch=EUR"
</body>
</html>

```

Additional web terminal settings can be specified as URL parameters:

```

<iframe src="https://webtrading.broker.com/terminal?parameter1=value1&parameter2=value2"

```

The following parameters are supported:

first_name, second_name, email first name, last name and email to be inserted into the demo and real account registration form. By using these parameters, you can make account opening easier for traders who have already registered on your site and provided the required data. Example:

```

https://webtrading.broker.com/terminal?mode=demo&first_name=John&second_name=Smith

```

Please note that the example includes the "mode=demo" parameter to automatically open the demo account registration form upon the web terminal launch.

utm_campaign, utm_source utm parameters that will be added to the relevant fields of accounts opened via the web terminal. Using them, you can analyze how efficiently you attract traders via web terminals. For further details please see "How to track accounts opened via the web terminal". Example:

```

https://webtrading.broker.com/terminal?utm_campaign=webterminal&utm_source=site

```

The parameters are used only at the first start. After that the web terminal will use the user-specified data.

mode the web terminal page that opens on first launch. Supported values:

- demo demo account registration form
- real real account registration form
- connect existing account connection form

```

https://webtrading.broker.com/terminal?mode=connect

```

The parameter is used only if the user has not previously connected to the account. If the web terminal has a saved account, it will be connected immediately upon startup.

login trading account for connection. If this parameter is specified, the web terminal will be launched with a pre-filled account connection dialog. The user will only have to enter a password. Use this parameter to create convenient client areas.

<https://webtrading.broker.com/terminal?login=123456>

If an account is specified in the URL, no demo account will be created at the first launch (the option is managed via platform settings). This also prevents from connecting using a previously saved account.

marketwatch the list of symbols to be displayed by default in Market Watch is specified by the marketwatch parameter. Example:

<https://webtrading.broker.com/terminal?marketwatch=EURUSD,GBPUSD,AUDCAD,USDJPY>

The parameter also determines the order in which the symbols are displayed. In the example above, on the first web terminal launch the user will see only four specified symbols in the Market Watch window, EURUSD will be the first and USDJPY will be the last. Later the user can re-configure the list of symbols, and the new settings will be saved in the browser. The maximum number of symbols in the parameter is 300.

theme-mode theme for the application interface. Possible values:

- 0 light (default)
- 1 dark

<https://webtrading.broker.com/terminal?theme-mode=1>

The parameter is only used for the first start. After that the web terminal will use user-specified settings. If necessary, you can force a user-defined theme to be overridden. This may be useful, for example, if your site supports light and dark themes. This will allow the synchronization of interface switching modes of the site and the web terminal.

To switch the color scheme of the web terminal, call `postMessage` on the `<iframe>` element, through which the web terminal is loaded. Specify the parameters type: 'mt5_update' and theme-mode: 0 (for the light theme) or 1 (for the dark theme):

```
const iframe = document.getElementById('iframe');
```

```
iframe.contentWindow.postMessage({
  type: 'mt5_update',
  'theme-mode': 1,
}, '*');
```

theme color scheme for the application interface. Possible values:

- greenRed red and green (default)
- blueRed blue and red
- blackWhite black and white
- neutral neutral

<https://webtrading.broker.com/terminal?theme=blueRed>

The parameter is used only at the first start. After that, the web terminal will apply user-specified settings.

MetaTrader 5 trading platform

The parameter is only used for the first start. Then the web terminal will apply user-specified settings.

lang web terminal interface language. The following languages are currently supported:

- English (en), Arabic (ar), Bulgarian (bg), Czech (cs), Chinese Simplified (zh), Chinese Traditional (zt), Hebrew (he), Hungarian (hu), Dutch (nl), French (fr) , German (de), Greek (el), Hindi (hi),
- Indonesian (id), Italian (it), Japanese (ja), Korean (ko), Malay (ms), Persian (fa), Polish (pl), Portuguese (pt), Romanian (ro), Russian (ru), Spanish (es), Thai (th), Turkish (tr), Ukrainian (uk), Uzbek (uz), Vietnamese (vi)

<https://webtrading.broker.com/terminal?lang=de>

The parameter is used only at the first start. After that, the web terminal will apply user-specified settings.

The size of the web terminal window on the page can be set using standard "width" and "height" parameters in the <iframe> tag, for example:

```
<iframe src="https://webtrading.broker.com/terminal" width="100%" height="1000"></iframe>
```

Generally we recommended to set the width and height to 100% so that the web terminal block takes up all the available web page space. If you specify an absolute size in pixels, some elements may be out of the visible window on some devices.

How to set up the account opening form

To configure the account opening form, use MetaTrader 5 Administrator. Go to the Allocations section and set available account types, leverages and balance. These settings apply to all terminals, including desktop, mobile and web.

Security settings

Prevent the web terminal page from loading in IFRAME. An attacker can place an invisible IFRAME containing the page of your website with the web terminal and combine the web terminal control element (such as a button) with another link on their website. Thus, when clicking a link, a user may actually perform an action necessary to the attacker.

Add X-Frame-Options: DENY HTTP header to your page containing the web terminal in order to disable page loading in IFRAME and protect users.

Only use the web terminal widget on https pages. The page on which the widget is installed must work over a secure https protocol (not http). Otherwise, the operation of the web terminal will not be possible in some browsers (for example Chrome version 60 and higher).

How to track accounts opened via the web terminal

Special UTM parameters are added to all accounts opened via the web terminal. Such UTM parameters inform the broker that the potential client has come from a web terminal operating on the broker's website. The UTM tags are added to the trading account parameters:

- The Comment field will contain "WebTerminal [short name of the domain from which the account was opened]". Example: "WebTerminal mysite.com". The "www" part is removed from the address.
- The domain name with 'www' is also added in the 'Lead source' field. Example: "www.mysite.com". The value can be overridden by adding utm_source to the widget parameters.
- The 'Lead campaign' field is not filled by default. You may add utm_campaign to widget parameters in order to write the name of your marketing campaign to this field.

To use your own UTM parameters for tracking clients, add the utm_source and utm_campaign parameters to the web terminal widget.

https://webtrading.broker.com/terminal?utm_campaign=webterminal&utm_source=site

For the widget added to the www.abcbroker.com site, account parameters will be filled as follows:

- Comment = WebTerminal
- Lead source = www.abcbroker.com
- Lead campaign = web.demo

MetaTrader 5 trading platform

Account: 3604613, Miguel Murray, USD, 1:100, Hedge

Overview Personal Account Limits Profile Subscriptions Security

Name: Miguel

Last name: Murray Middle name:

Company: Individual

Registered: 2021.11.08 15:32:27

Language: English Status: RE

ID number: 12300997 Lead campaign: web.demo

MetaQuotes ID: Lead source: www.abcbroker.com

Email: miguel-murray@mail.com

Phone: +112294324999

Country: United States State:

City: New York Zip code: 010101

Address: New York

Comment: abcbroker.com

Client Properties... Update Cancel Help

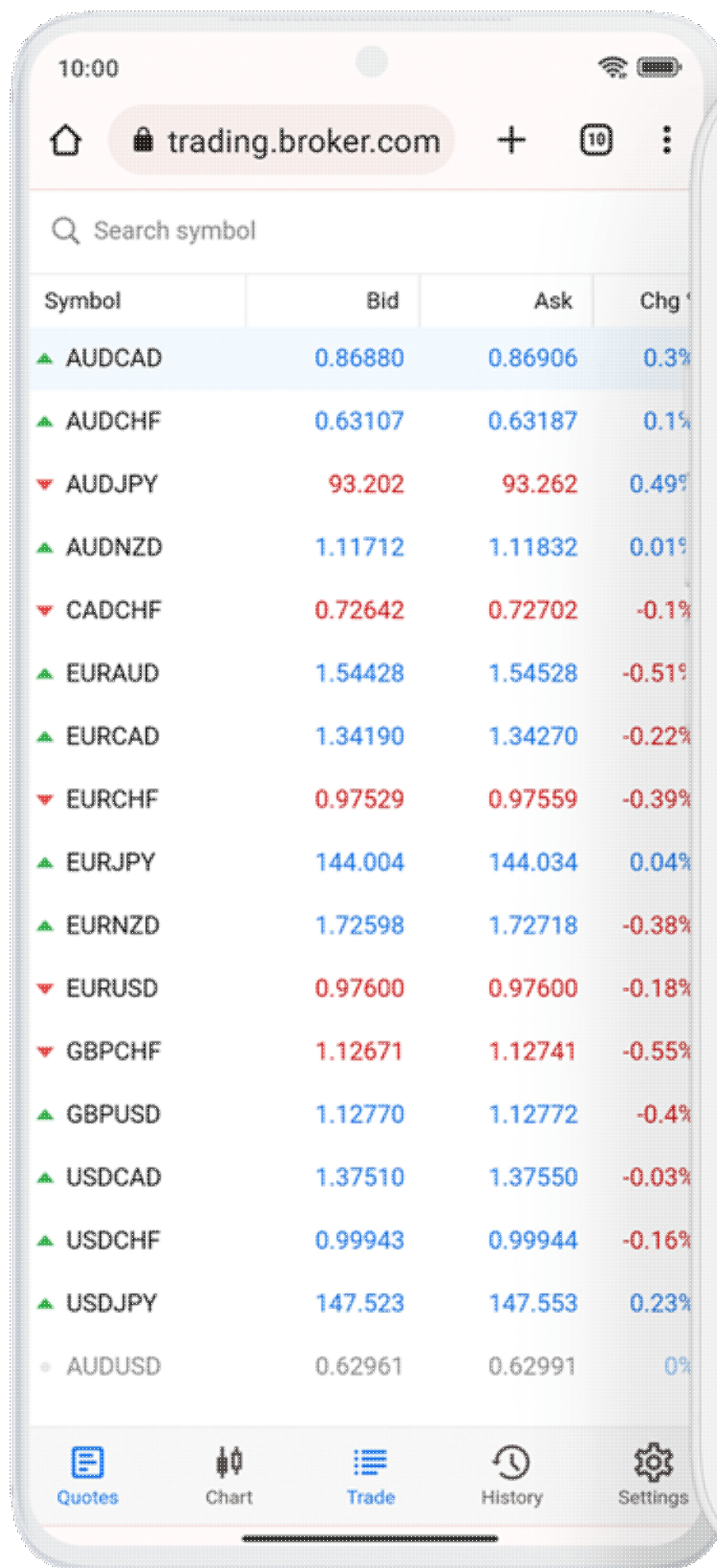
MetaTrader 5 trading platform

Mobile version of the web platform

The MetaTrader 5 web platform includes a special version adapted for iOS and Android smartphones and tablets. This enables convenient trading from mobile browsers in addition to desktop ones.

No additional settings for the mobile version are needed. The web terminal will automatically detect the user's device by the browser's "user-agent" and will adapt the interface.

MetaTrader 5 trading platform



MetaTrader 5 trading platform

You can also embed a web terminal in your mobile app. To do this, create a web form in it with the web terminal `<iframe>`.

Minimum browser versions

The web terminal supports the following web browser versions and above:

- Microsoft Edge 15
- Mozilla Firefox 54
- Google Chrome 51
- Safari 12
- Opera 38

Troubleshooting

If a trader receives a 404 error when trying to open the web terminal page, check the following:

- The SSL certificate you uploaded in the Integration \ Web Service section is valid and has not expired
- Your company name is specified correctly in the SSL certificate it must match the company name specified in the license/White Label
- Your platform does not run under a test license in this case the web terminal is not available
- The web terminal is included in your platform license. Contact the [support team](#) to check the license

Web terminal widget example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
  <title>WebTerminal for the MetaTrader 5 platforms</title>
  <style type="text/css">
    body {margin: 0; padding: 0; font-family: Arial, Tahoma; font-size: 16px; color: #000; bac
    .top {background-color: #0055A7;}
    .top h1 {margin: 10px 20px 10px 10px; font-size: 25px; font-weight: normal; color: #FFF
    .top .menu, .top .menu li {margin: 0; padding: 0; list-style: none; display: inline-block; ver
    .top .menu li {margin: 0; padding: 0; list-style: none; display: inline-block;}
    .top .menu li a {padding: 20px; font-size: 16px; color: #FFF; text-decoration: none; text-a
    .top .menu li a:hover {background-color: #0B6ABF;}
    .top .menu li a.selected {background-color: #2989DF; color: #FFF;}
    .content { box-shadow: 0 0 20px rgba(0,0,0,0.5); position: relative; }
    .footer {text-align: center; padding: 20px; color: #0A0A0A; font-size: 14px}
  </style>
</head>
<body>
  <div class="top">
    <h1>BROKER</h1>
```

```
<ul class="menu">
  <li><a href="#">Analytics</a></li>
  <li><a href="#" class="selected">WebTerminal</a></li>
  <li><a href="#">News</a></li>
  <li><a href="#">Contacts</a></li>
</ul>
</div>
<div class="content">
<!-- Web Terminal Code Start -->
<iframe src="https://webtrading.broker.com/terminal?mode=connect&marketwatch=EUR"
<!-- Web Terminal Code End -->
</div>
<div class="footer">
Copyright 2000-2015, Broker
</div>
</body>
</html>
```

MetaTrader 5 Trading Platform / Platform Components / Old WebTerminal

MetaTrader 5 WebTerminal

This section describes the old version of the web terminal. It will be supported for a limited time. We recommend upgrading to the new, improved Web Terminal version the soonest possible.

The major updates enables the web terminal to run on each broker's access server rather than on one common server. It operates as an individual web terminal, which only works with your platform. This provides maximum security and full control for the broker.

In addition, the new web terminal provides a plethora of new features:

- Convenient configuration of instruments in the Market Watch, along with the daily price change data
- Ability to request real accounts with a detailed registration form and document submission options
- More analytical objects with convenient management options
- Market entries and exits displayed on charts
- Economic Calendar events displayed on charts
- Support for price data subscriptions and the ability to receive delayed quotes
- Simplified interface to assist beginners in getting started with the terminal: removed chart context menu and top menu; all chart control commands, objects and indicators are available on the left-hand and top panels, while other commands can be accessed through the hamburger menu
- Interface dark mode

Special instructions are provided for a smooth switch to the new version.

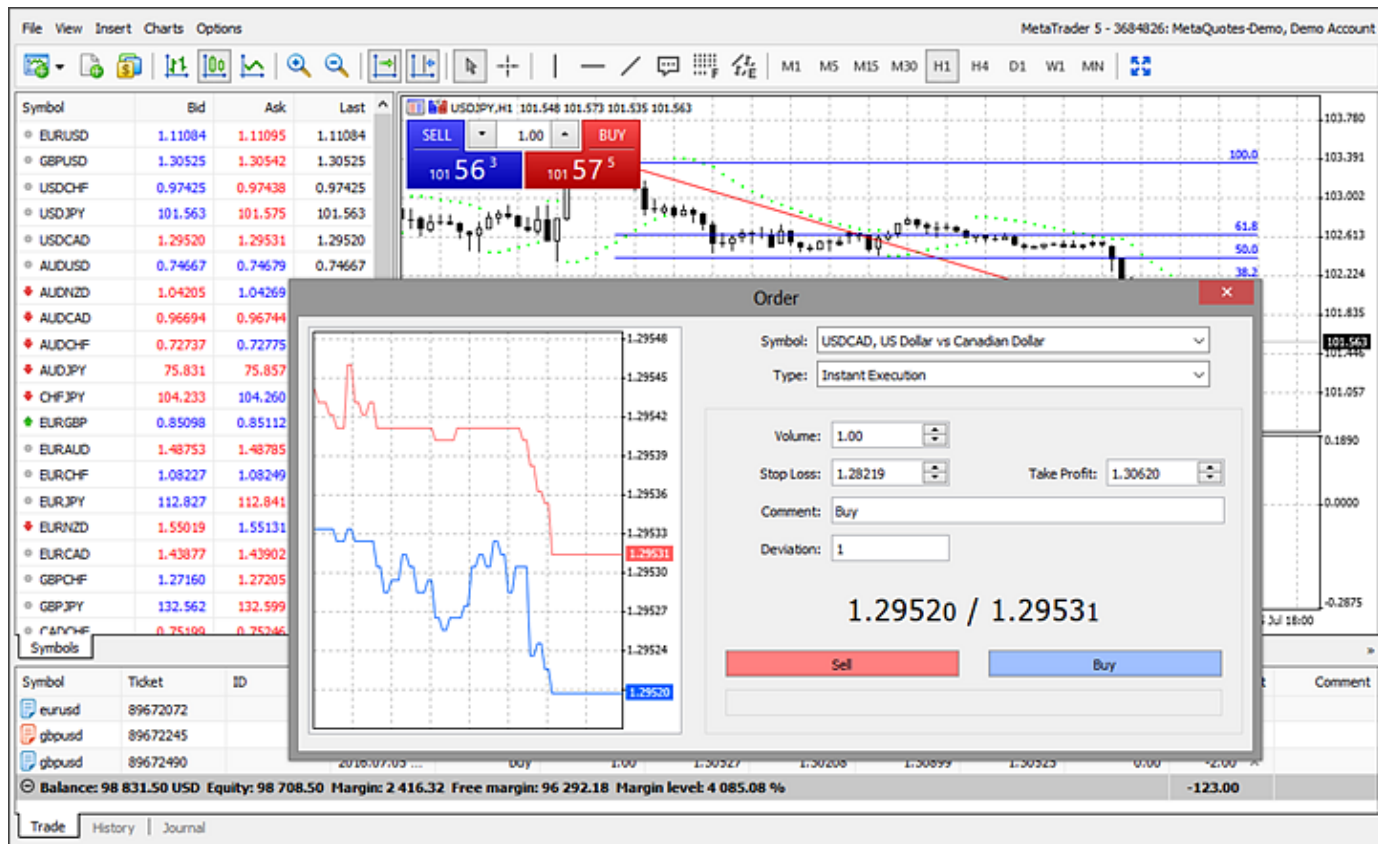
The MetaTrader 5 WebTerminal enables financial market trading using any web browser. It works in all operating systems and browsers, while requiring no extra software installations. All transmitted data is securely encrypted.

The web terminal supports all types of market and pending orders, as well as one-click trading. Traders can view real-time quotes and analyze charts using basic graphical objects. Charts can be analyzed using 30 technical indicators.

The terminal is a modern HTML5 application that can be easily integrated into any website via a simple iframe widget.

MetaTrader 5 trading platform

Similar to the desktop platform, traders should select a server from the list, enter login and password when connecting to the web terminal via your website. For the seamless web terminal integration with clients area on your website, server and login can be pre-selected to let your client enter a password only. If a users saves a password in a browser storage, the platform will login to the trading account automatically on the next run.



To get the terminal, [order it in the App Store section](#) of the Support website.

Order WebTerminal Configure WebTerminal

- The trading platform should be updated to at least build 1325 for the web terminal operation.
- [MQL5.community](#) account is not required.

How to Add the WebTerminal Widget to Your Website

To install the WebTerminal widget on your site, visit the "[App Store \ White Labels \ MetaTrader 5](#) " section of the technical support site. Then click "Show widget code \ Customize":

News Articles Documentation Workshops Certification App Store Billing ^{NEW} Analytics

MetaTrader 5 White Labels of MetaQuotes Software Corp.

MetaQuotes Software Corp. [↗](#)

Desktop Terminal <https://download.mql5.com/cdn/web/metaquotes.software.corp/mt5/mt5setup.exe>
Mobile iOS <https://download.mql5.com/cdn/mobile/mt5/ios?server=MetaQuotes-Demo>
Mobile Android <https://download.mql5.com/cdn/mobile/mt5/android?server=MetaQuotes-Demo>

WebTerminal [Hide widget code](#)

```
<div id="webterminal" style="width:100%;height:600px;"></div>
<script type="text/javascript" src="https://metatraderweb.app
/trade/widget.js"></script>
<script type="text/javascript">
  new MetaTraderWebTerminal( "webterminal", {
    version: 5,
    servers: ["MetaQuotes-Demo"],
    server: "MetaQuotes-Demo",
    demoAllServers: true,
    utmSource: "www.metaquotes.net",
    startMode: "create_demo",
    language: "en",
    colorScheme: "black_on_white"
  } );
```

[↗ Customize](#)

[Documentation: Adding a WebTerminal Widget](#)

MetaTrader 5 trading platform

Set the WebTerminal operation parameters. After that, a special code will be generated for you, which should be inserted into your website.

The following parameters are supported (the corresponding parameters in the widget code are displayed in brackets, you can change them manually if necessary).

- Version (version) default web terminal version: MetaTrader 4 or MetaTrader 5. The parameter is important only if you use both versions of the platform at the same time if the widget settings have no restrictions on used servers or the servers of both versions are present in the server list (servers). The parameter is used only for the first launch of the web terminal. Further on, the platform version will be defined based on the last used account. The detailed description is given in the section If you have two platforms: MetaTrader 5 and MetaTrader 4.
- Restrict trade servers \ Trade server list (servers) limit servers available for use in your web terminal. All servers of all brokers are available by default. Simply enter a name in the new account opening or in the existing account login dialog.
If you enable the "Restrict trade servers" option and set the list of available servers, the list of certain servers instead of the server name input box will be available in the appropriate dialogs.

Server names are case sensitive. Be sure to specify the exact name.

- Default login (login) default trading account selected in the login dialog. Use this option to create convenient client areas by immediately substituting the necessary account into the web terminal.
- Default trade server (server) default server selected in the connection dialog. Use this option to create convenient client areas by immediately substituting the necessary server into the web terminal.
- Allow opening demo accounts on any servers (demoAllServers) disable the option if you want to allow opening demo accounts via the web terminal only on demo servers from the "Trade server list" parameter. If enabled, demo accounts can be opened on any servers (regardless of the "Restrict trade servers" parameter). The server name input box is displayed instead of the list of available servers in the appropriate dialog.
- Demo account types (demoType) demo account type, multiple comma-separated values can be added. To open demo accounts in the "demo forex" group, specify "forex" (the "demo" prefix is added automatically). If not specified, all account types configured in the "Allocations" are used. At the moment, you can add this parameter to the terminal widget only by specifying it in the code manually.
- Leverage (demoLeverage) the leverage for the demo account, multiple comma separated options can be added. If not specified, all options configured in the "Allocations" are used. At the moment, you can add this parameter to the terminal widget only by specifying it in the code manually.
- First Name and Second Name (demoFirstName, demoSecondName) the first name and the second name to be automatically inserted into the demo account registration form in the web terminal. These parameters can be added to the embed code only manually. Use the parameters to display the web terminal with

- appropriate data to the users who are authorized on the site and whose first and second name are known.
- Email (demoEmail) the email address to be automatically inserted into the demo account registration form in the web terminal. This parameter can only be added to the embed code manually. Use the parameter to display the web terminal with appropriate data to the users who are authorized on the site and whose emails are known in advance.
 - Allow the Phone field in the demo account opening dialog (demoAllowPhone) show the Phone input box in the demo account registration form ("true" value).
 - UTM campaign \ UTM Source (utmCampaign \ utmSource) UTM tags to be added to accounts opened via the web terminal. The tags allow you to analyze the efficiency of this tool. For more details, see ["How to track accounts opened via the web terminal"](#).
 - Width (width) terminal widget width in % or pixels. The recommended value is 100% to allow the web terminal to automatically adjust to the maximum available width on the web page.
 - Height (height) terminal widget height in % or pixels. The recommended value is "600px" to make the entire widget visible even on small screens without the need to scroll.
 - What to do at the start for new visitors (startMode) web terminal launch mode:
 - Open the demo account creation dialog (open_demo) display a demo account opening window (instead of the login window) for users who do not have accounts stored in the web terminal. If accounts exist in the local storage, connection to the last used account is established.
 - Create a demo account automatically (create_demo) open a demo account automatically for a user when launching the web terminal. The account is opened only if the user's web terminal has no previously saved accounts.
 - Show login dialog (login) show account login window for users when launching the web terminal. Login to the last used account is not conducted automatically even if a user saved the account password in the browser storage previously.
 - Symbols (symbols) list of symbols to be displayed by default in the "Market Watch" window of the web terminal (optional). The parameter also defines the display order of the symbols. For example, if you add ["EURUSD", "EURGBP", "AUDUSD", "EURRUB"] to the widget, a user will see only the four specified symbols in the "Market Watch" window after the first launch of the web terminal. EURUSD will be displayed first, while EURRUB will be last. Subsequently, the user can re-configure the list of symbols and the settings will be saved in the browser. The maximum number of symbols in the parameter is 300.
 - Chart color scheme (colorScheme) default color scheme applied to charts. Possible values: black_on_white, yellow_on_black and green_on_black.
 - Language (lang) default web terminal interface language. Users can select a necessary language in the web terminal menu: View -> Languages. In this case, the value of this parameter is ignored. Currently, the following languages are supported:
 - Arabic (ar)
 - Bulgarian (bg)

- Chinese (zh)
- Croatian (hr)
- Czech (cs)
- Danish (da)
- Dutch (nl)
- English (en)
- Estonian (et)
- Finnish (fi)
- French (fr)
- German (de)
- Greek (el)
- Hebrew (he)
- Hindi (hi)
- Hungarian (hu)
- Indonesian (id)
- Italian (it)
- Japanese (ja)
- Korean (ko)
- Latvian (lv)
- Lithuanian (lt)
- Malay (ms)
- Mongolian (mn)
- Persian (fa)
- Polish (pl)
- Portuguese (pt)
- Romanian (ro)
- Russian (ru)
- Serbian (sr)
- Slovak (sk)
- Slovenian (sl)
- Spanish (es)
- Swedish (sv)
- Tajik (tg)
- Thai (th)
- Traditional Chinese (zt)
- Turkish (tr)
- Ukrainian (uk)
- Uzbek (uz)
- Vietnamese (vi)

Sample web terminal code to be inserted into a website:

```
<div id="webterminal" style="width:100%;height:600px;"></div>
<script type="text/javascript" src="https://metatraderweb.app/trade/widget.js"></script>
<script type="text/javascript">
  new MetaTraderWebTerminal( "webterminal", {
    version: 5,
    server: "MetaQuotes-Demo",
    demoAllServers: true,
    startMode: "create_demo",
```

```

    lang: "en",
    colorScheme: "black_on_white"
  } );
</script>

```

Configuring the account opening form (type, leverage, balance)

To configure the account opening form, use MetaTrader 5 Administrator. Go to the [Allocations](#) section and set available account types, leverages and balance.

Disable downloading the web terminal page in IFRAME

An attacker can place an invisible IFRAME containing the page of your website with the web terminal and combine the web terminal control element (such as a button) with another link on their website. Thus, when clicking a link, a user may actually perform an action necessary to the attacker.

Add X-Frame-Options: DENY HTTP response header to your page containing the web terminal in order to disable downloading the page in IFRAME and protect users.

The web terminal widget should only be used on https pages

The page on which the widget is installed must work over a secure https protocol (not http). Otherwise, the operation of the web terminal will not be possible in some browsers (for example Chrome version 60 and higher).

How to track accounts opened via the web terminal

Special UTM parameters are added to all demo accounts opened from the web terminal. Such UTM tags inform the broker that the potential client has come from a web terminal operating on the broker's website. The tags are added to the [trading account parameters](#):

- The Comment field contains the following details: "WebTerminal [the short name of the domain from which the account was opened]". Example: "WebTerminal mysite.com". The "www" part is removed from the address.
- The domain name with 'www' is also added in the 'Lead source' field. Example: "www.mysite.com". The value can be overridden by adding utm_source to the widget parameters.
- The 'Lead campaign' field is not filled by default. You may add utm_campaign to widget parameters in order to write the name of your marketing campaign to this field.

To use your own UTM parameters for tracking clients, add the [utmSource and utmCampaign parameters](#) to the web terminal widget.

```
<div id="webterminal" style="width:100%;height:600px;"></div>
<script type="text/javascript" src="https://metatraderweb.app/trade/widget.js"></script>
<script type="text/javascript">
  new MetaTraderWebTerminal( "webterminal", {
    version: 5,
    server: "MetaQuotes-Demo",
    utmCampaign: "www.abcbroker.com",
    utmSource: "web.demo",
    demoAllServers: true,
    startMode: "create_demo",
    lang: "en",
    colorScheme: "black_on_white"
  } );
</script>
</script>
```

For the widget added to the www.abcbroker.com site, account parameters will be filled as follows:

- Comment = abcbroker.com
- Lead source = www.abcbroker.com
- Lead campaign = web.demo

MetaTrader 5 trading platform

Account: 3604613, Miguel Murray, USD, 1:100, Hedge

Overview Personal Account Limits Profile Subscriptions Security

Name: Miguel

Last name: Murray Middle name:

Company: Individual

Registered: 2021.11.08 15:32:27

Language: English

ID number: 12300997

MetaQuotes ID:

Email: miguel-murray@mail.com

Phone: +112294324999

Country: United States

City: New York

Address: New York

Comment: abcbroker.com

Status: RE

Lead campaign: web.demo

Lead source: www.abcbroker.com

State:

Zip code: 010101

Client Properties... Update Cancel Help

Detecting the WebTerminal Language

The web terminal determines the interface language based on the following priorities:

- The language selected by the user in the web terminal
- The language specified in lang (if the language is supported in the web terminal)
- The preferred language in the user's web browser (if the language is supported in the web terminal)

If the web terminal cannot determine the language, English will be used.

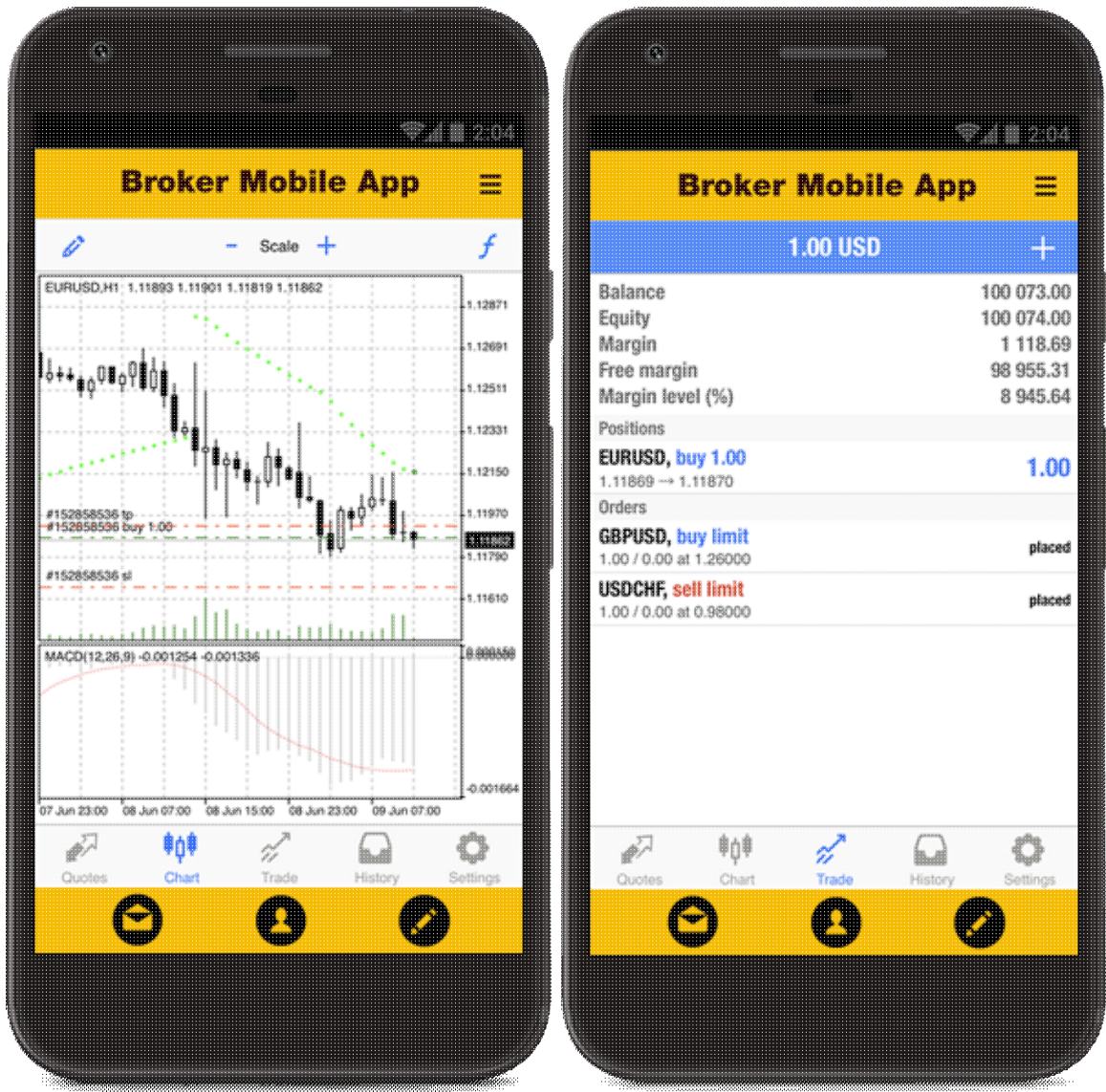
Operation Features

The maximum number of symbols a client can enable in the Market Watch window is 300 for all browsers, except for Internet Explorer/Edge 50. A greater number of symbols in Internet Explorer/Edge slows down the web terminal operation.

Mobile version of the web platform

The MetaTrader 5 web platform includes a special version adapted for smartphones. This allows for convenient trading from mobile browsers in addition to desktop ones. If your website has a mobile version, make sure the appropriate version of the web platform is used there, so that your traders are able to comfortably work from their smartphones. This web platform version can also be embedded directly into mobile Android and iOS applications using the WebView component.

MetaTrader 5 trading platform



To launch the mobile version of the web platform, add the "mobile: 1" parameter to the widget. Since the mobile version of the web platform is available only for the MetaTrader 5 platform, we also recommend setting the "version=5" parameter to ensure its launch.

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0, minimum-scale=
  <title>Web Terminal</title>
</head>
<body>
  <div id="webterminal" style="width:100%;height:600px;"></div>
  <script type="text/javascript" src="https://metatraderweb.app/trade/widget.js"></script>
  <script type="text/javascript">
    new MetaTraderWebTerminal("webterminal", {
      version: 5,
      server: "MetaQuotes-Demo",
      demoAllServers: true,
      startMode: "create_demo",
      lang: "en",
      mobile: 1,
      colorScheme: "black_on_white"
    });
  </script>
</body>
</html>
```

To call the web platform in your mobile application, create a web form and load the web platform address with the "m=1" and "version="5" parameters to it (similar to the iframe widget). In order to simplify the task, we have prepared examples in the form of simple iOS and Android projects:

- [Sample Android \(Java\) project](#)
- [Sample iOS \(Swift\) project](#)

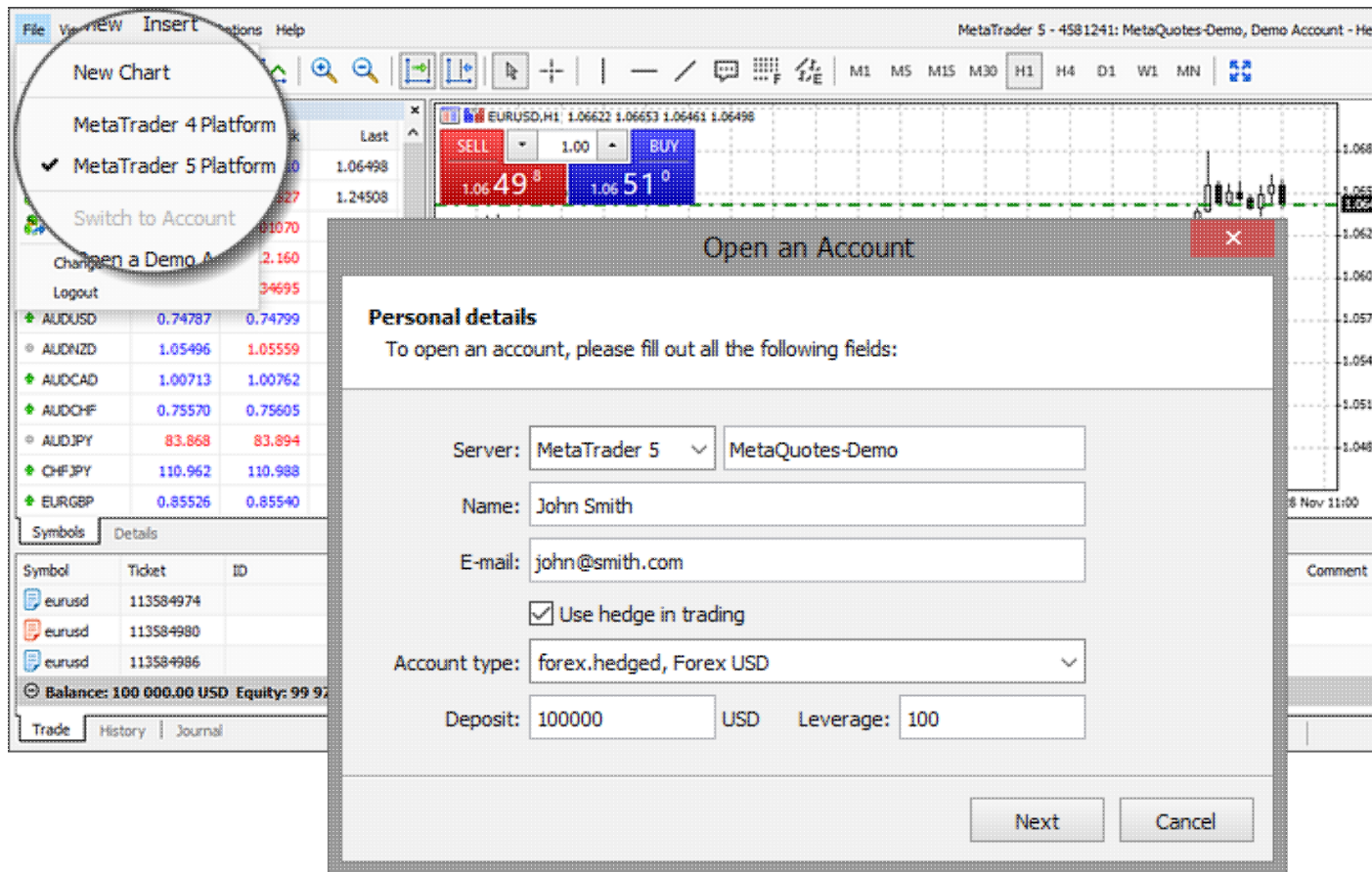
The mobile version is supported only in the MetaTrader 5 web platform.

If you have two platforms: MetaTrader 5 and MetaTrader 4

A single web terminal is used for both platform versions. If you only use one platform (MetaTrader 4 or MetaTrader 5), no additional actions are required. Simply select the necessary one when receiving the code for insertion.

If you simultaneously use two versions of the trading platform, you do not need to configure WebTerminals separately. If the web terminal finds servers of both versions in the "[Trade server list](#)" parameter, a switch between the two versions appears in the web terminal interface. It is available in the account connection dialog, in the account opening dialog and in the File menu.

MetaTrader 5 trading platform



Setting the default version

In order to set a default version that will be selected during the web terminal launch, use the Version parameter:

```
<div id="webterminal" style="width:100%;height:600px;"></div>
<script type="text/javascript" src="https://metatraderweb.app/trade/widget.js"></script>
<script type="text/javascript">
  new MetaTraderWebTerminal( "webterminal", {
    version: 5,
    servers: ["ForexBroker4-Live","ForexBroker5-Live"],
    demoAllServers: true,
    startMode: "create_demo",
    lang: "en",
    colorScheme: "black_on_white"
  });
</script>
```

In this example, the versions switch is set to MetaTrader 5 by default.

If a user switches to another platform, the selection will be remembered. During the next launch of the WebTerminal, it will be switched to the latest used version of the platform.

Supported Browser Versions

The web terminal supports the following web browser versions and above:

- Internet Explorer 11
- Microsoft Edge 12
- Mozilla Firefox 34
- Google Chrome 43
- Safari 8
- Opera 32

Example of Adding a WebTerminal Widget

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0, minimum-scale=1.0">
  <title>WebTerminal for the MetaTrader 4 and MetaTrader 5 platforms</title>
  <style type="text/css">
    body {margin: 0; padding: 0; font-family: Arial, Tahoma; font-size: 16px; color: #000; background-color: #0055A7;}
    .top {background-color: #0055A7;}
    .top h1 {margin: 10px 20px 10px 10px; font-size: 25px; font-weight: normal; color: #FFF;}
    .top .menu, .top .menu li {margin: 0; padding: 0; list-style: none; display: inline-block; vertical-align: top;}
    .top .menu li {margin: 0; padding: 0; list-style: none; display: inline-block; vertical-align: top;}
    .top .menu li a {padding: 20px; font-size: 16px; color: #FFF; text-decoration: none; text-align: center;}
    .top .menu li a:hover {background-color: #0B6ABF; color: #FFF;}
    .top .menu li a.selected {background-color: #2989DF; color: #FFF;}
    .content { box-shadow: 0 0 20px rgba(0,0,0,0.5); position: relative; }
    .footer {text-align: center; padding: 20px; color: #0A0A0A; font-size: 14px;}
  </style>
</head>
<body>
  <div class="top">
    <h1>BROKER</h1>
    <ul class="menu">
      <li><a href="#">Analytics</a></li>
      <li><a href="#" class="selected">WebTerminal</a></li>
      <li><a href="#">News</a></li>
      <li><a href="#">Contacts</a></li>
    </ul>
  </div>
  <div class="content">
    <!-- Web Terminal Code Start -->
    <div id="webterminal" style="width:100%;height:600px;"></div>
    <script type="text/javascript" src="https://metatraderweb.app/trade/widget.js"></script>
    <script type="text/javascript">
      new MetaTraderWebTerminal( "webterminal", {
        version: 5,
        server: "MetaQuotes-Demo",
        demoAllServers: true,
        startMode: "create_demo",
      })
    </script>
  </div>
</body>
</html>
```

```
        lang: "en",
        colorScheme: "black_on_white"
    } );
</script>
<!-- Web Terminal Code End -->
</div>
<div class="footer">
Copyright 2000-2015, Broker
</div>
</body>
</html>
```

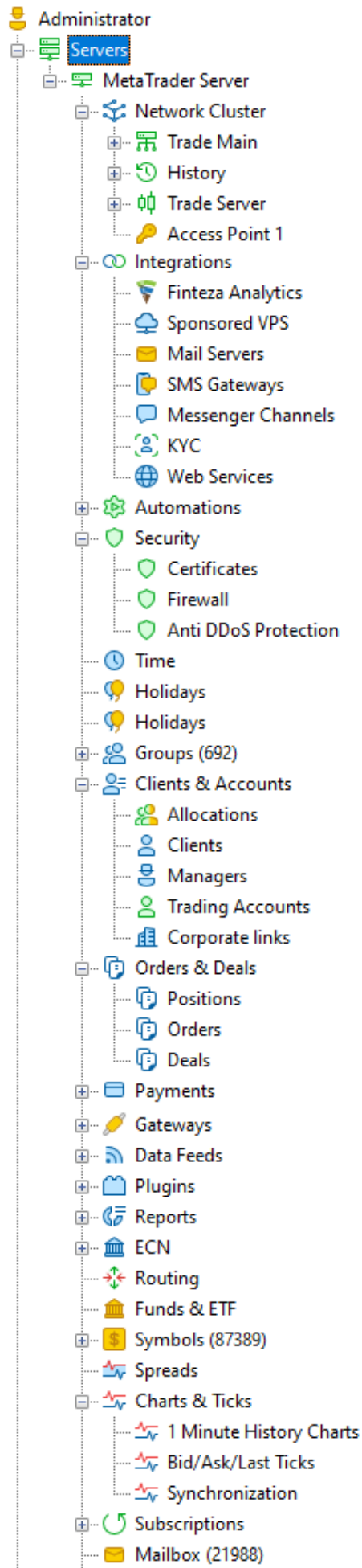
MetaTrader 5 Trading Platform / Platform Setup

Administration

The administration of the online trading platform includes managing of not only the trade server but also of the intermediate components (access points, backup servers, history server, etc.). The administration is divided into categories represented as a tree-like list in the left part of the MetaTrader 5 Administrator:

- Start Page general information about the platform, live update settings
- Network Cluster settings of platform components
- Integrations configuring mail servers, messengers and access to Finteza analytics
- Automations configuring scenario-based automated actions
- Certificates configuring certificates for extended authentication
- Firewall configuring access by IP addresses
- Anti DDoS Protection integration with external protection providers
- Time general time settings of the server and setting of trade time
- Holidays holiday settings
- Leverage floating margin settings
- Groups account group settings
- Allocations configuring the opening of demo and preliminary accounts through client terminals
- Clients working with customer data, documents and accounts
- Managers administration of manager accounts
- Trade Accounts working with accounts
- Corporate links adding custom links to the client terminals
- Positions working with positions
- Orders working with orders
- Deals working with deals
- Payments integration with payment systems
- Gateways administration of gateways that provide interaction with exchanges
- Data Feeds configuring quotes and news

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sources

- **Plugins** modules of expanding the platform functionality
- **Reports** report settings
- **ECN** configuring the built-in electronic communication network
- **Routing** rules of processing of coming trade requests
- **Funds & ETF** controlling fund settings: commissions, managers, investors
- **Symbols** configuring financial instruments
- **Spreads** configuring margin requirements for positions in spread
- **1 Minute History Charts** working with price data stored at the server
- **Bid/Ask/Last Ticks** viewing tick data
- **Synchronization** configuring synchronization with other MetaTrader servers
- **Subscriptions** configuring subscriptions for additional trader services
- **Mailbox** built-in email system for communicating with traders and colleagues
- **Live Update** configuring automatic updating of the system components
- **Support Center** this section offers help for the MetaTrader 5 platform users: articles, news, answers to frequently asked questions and a possibility to contact the technical support team directly
- **App Store** store of additional components for the trading platform: gateways, liquidity, CRM and others

MetaTrader 5 Trading Platform / Platform Setup / General Information

General Information

The series of articles of this section contain the general information about working with the administrator terminal:

- **Working with Instructions**

Useful information about working with various administration elements.

- **Specifying Symbols and Groups**

Information about specifying symbols and groups in different administration sections.

- **Data Export**

Description of process of exporting information.

MetaTrader 5 Trading Platform / Platform Setup / General Information /
Trading System

Trading System

Before you proceed with administration of the platform, please get acquainted with the basics of its trading system.

Trade Operation Types

The platform has three types of trade operations: order, deal and position. There is a separate section to manage each of them in the administrator terminal.

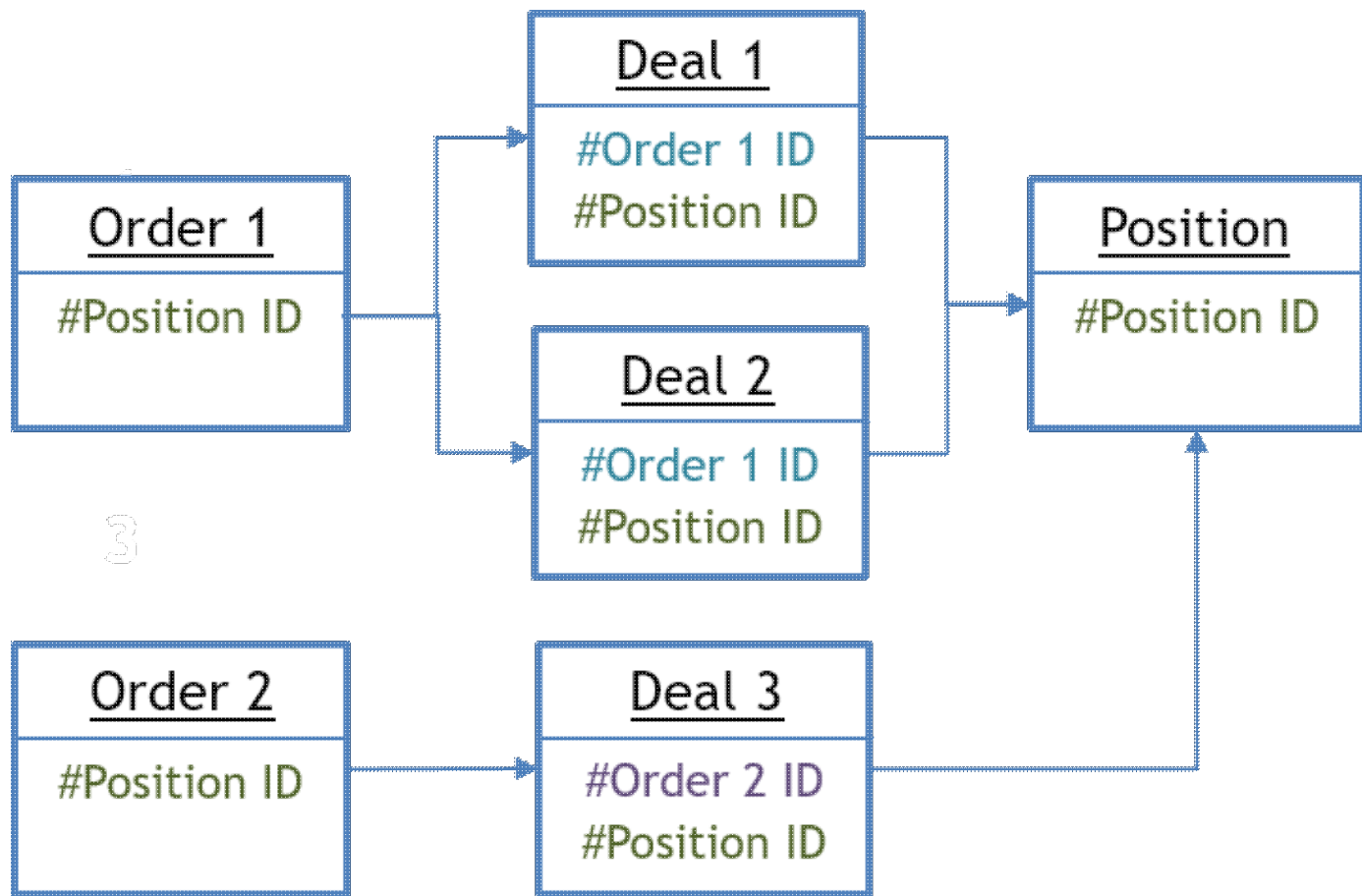
- An **order** is an instruction given to a broker to buy or sell a financial instrument. There are two main types of orders: Market and Pending. In addition, there are special Take Profit and Stop Loss levels.
- A **deal** is the commercial exchange (buying or selling) of a financial security. Buying is executed at the demand price (Ask), and Sell is performed at the supply price (Bid). A deal can be opened as a result of market order execution or pending order triggering. Note that in some cases, execution of an order can result in several deals.
- A **position** is a trade obligation, i.e. the number of bought or sold contracts of a financial instrument. A long position is financial security bought expecting the security price go higher. A short position is an obligation to supply a security expecting the price will fall in future.

Interrelation of orders, deals and positions

The platform allows you to easily track how a position was opened or how a deal was performed. Each trading operation has its unique ID called a "ticket". Each order and deal receive a ticket relating to their relevant position. Each deal receives a ticket of an order, by which it was concluded.

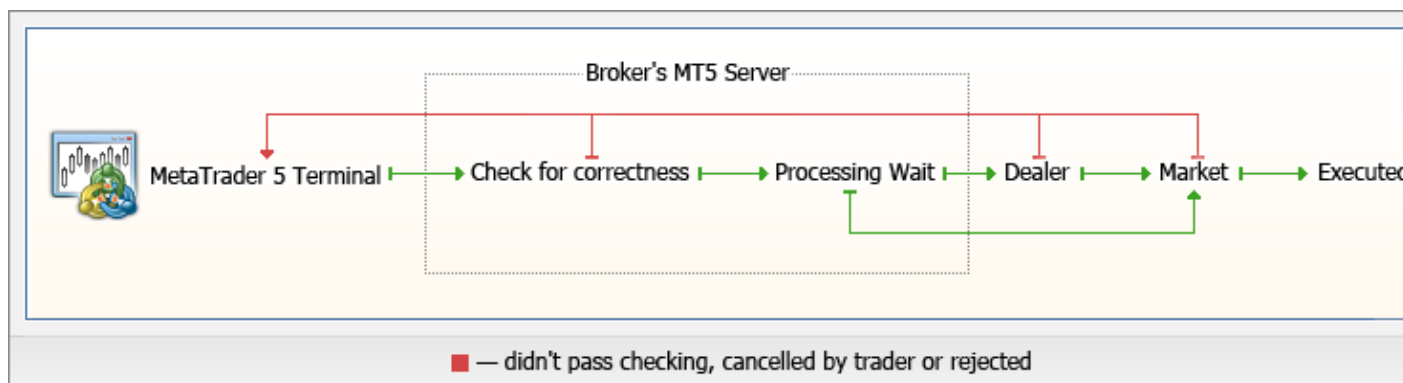
If a position was affected by multiple deals, for example in the case of a partial closing or increasing volumes, each of the deals feature the position's ticket. This makes it easy to track the entire history of the position as a whole.

If trading operations are sent to an exchange or a liquidity provider, they additionally feature an ID from an external system. This allows additional tracking of the interrelation of operations away from the platform.



A General Scheme of Trading Operations

- From the client terminal, an order is sent to a broker to execute a deal with the specified parameters;
- The correctness of an order is checked on the server (correctness of prices, availability of funds on the account, etc.);
- Orders that have passed the check wait to be processed on the trade server. Then the order can be:
 - executed (in one of automatic execution modes or by a dealer)
 - canceled upon expiry
 - rejected (e.g. when money is not enough or there is no suiting offer in the market; or rejected by the dealer)
 - canceled by a trader;
- A deal is the result of the execution of a market order or triggering of a pending order;
- If there are no positions for a symbol, conclusion of a deal results in opening of a position. If there is a position for the symbol, the deal can increase or reduce the position volume, close the position or reverse it.



Position Accounting System

Two position accounting systems are supported in the trading platform: Netting and Hedging. The system used is determined by the client group settings.

Netting System

With this system, you can have only one common position for a symbol at the same time:

- If there is an open position for a symbol, executing a deal in the same direction increases the volume of this position.
- If a deal is executed in the opposite direction, the volume of the existing position can be decreased, the position can be closed (when the deal volume is equal to the position volume) or reversed (if the volume of the opposite deal is greater than the current position).

It does not matter, what has caused the opposite deal — an executed market order or a triggered pending order.

MetaTrader 5 trading platform

The below example shows execution of two EURUSD Buy deal 0.5 lots each:

Toolbox											
Symbol	Ticket	Time	Type	Volume	Price	S / L	T / P	Pri			
 eurUSD	207796062	2018.02.12 16:50:55	buy	2.00	1.225935	1.22104 ×	1.23104 ×	1.225			
⊖ Balance: 1 000 000.00 USD Equity: 999 959.00 Margin: 2 451.87 Free Margin: 997 507.13 Margin Level: 40 783.52 %											
Trade	Exposure	History	News	Mailbox7	Calendar	Company	Market12	Alerts	Signals	Code Base	Expert

Execution of both deals resulted in one common position of 1 lot.

Hedging System

With this system, you can have multiple open positions of one and the same symbol, including opposite positions.

If you have an open position for a symbol, and execute a new deal (or a pending order triggers), a new position is additionally opened. Your current position does not change.

The below example shows execution of two EURUSD Buy deal 0.5 lots each:

Toolbox											
Symbol	Ticket	Time	Type	Volume	Price	S / L	T / P	Pri			
 eurUSD	207769623	2018.02.12 15:58:48	buy	1.00	1.22439	1.21939 ×	1.22939 ×	1.225			
 eurUSD	207795600	2018.02.12 16:50:12	buy	1.00	1.22554	1.22054 ×	1.23054 ×	1.225			
⊕ Balance: 1 000 000.00 USD Equity: 1 000 121.00 Margin: 2 449.93 Free Margin: 997 671.07 Margin Level: 40 822.43 %											
Trade	Exposure	History	News	Mailbox7	Calendar	Company	Market12	Alerts	Signals	Code Base	Expert

Execution of these deals resulted in opening two separate positions.

Impact of the System Selected

Depending on the position accounting system, some of the platform functions may have different behavior:

- Stop Loss and Take Profit inheritance rules change.
- To close a position in the netting system, you should perform an opposite trading operation for the same symbol and the same volume. To close a position in the hedging system, explicitly select the "Close Position" command in the context menu of the position.
- A position cannot be reversed in the hedging system. In this case, the current position is closed and a new one with the remaining volume is opened.
- In the hedging system, a new condition for margin calculation is available Hedged margin.

Types of Orders

The trading platform allows to prepare and issue requests for the broker to execute trading operations. In addition, the platform allows to control and manage open positions. Several types of trading orders are used for these purposes. An order is a trader's instruction to the broker to perform a trade operation. In the platform, orders are divided into two main types: market and pending. In addition, there are special Stop Loss and Take Profit orders.

Market Order

A market order is an instruction given to a brokerage company to buy or sell a financial instrument. Execution of this order results in the execution of a deal. The price at which the deal is executed is determined by the type of execution that depends on the symbol type. Generally, a security is bought at the Ask price and sold at the Bid price.

Pending Order

A pending order is the trader's instruction to a brokerage company to buy or sell a security in future under pre-defined conditions. The following types of pending orders are available:

- **Buy Limit** a trade request to buy at the Ask price that is equal to or less than that specified in the order. The current price level is higher than the value specified in the order. Usually this order is placed in anticipation of that the security price will fall to a certain level and then will increase;
- **Buy Stop** a trade order to buy at the "Ask" price equal to or greater than the one specified in the order. The current price level is lower than the value specified in the order. Usually this order is placed in the anticipation that the price will reach a certain level and will continue to grow;
- **Sell Limit** a trade order to sell at the "Bid" price equal to or greater than the one specified in the order. The current price level is lower than the value specified in the order. Usually this order is placed in anticipation of that the security price will increase to a certain level and will fall then;
- **Sell Stop** a trade order to sell at the "Bid" price equal to or less than the one specified in the order. The current price level is higher than the value in the order. Usually this order is placed in anticipation of that the security price will reach a certain level and will keep on falling.
- **Buy Stop Limit** this type is the combination of the first two types, being a stop order to place a Buy Limit order. As soon as the future Ask price reaches the stop-level indicated in the order (the Price field), a Buy Limit order will be placed at the level, specified in Stop Limit price field. A stop level is set above the current Ask price, while Stop Limit price is set below the stop level.
- **Sell Stop Limit** this order is a stop order to place a Sell Limit order. As soon as the future Bid price reaches the stop-level indicated in the order (the Price field), a Sell Limit order will be placed at the level, specified in Stop Limit price field. A stop level is set below the current Bid price, while Stop Limit price is set above the stop level.

- For symbols with Exchange Stocks, Exchange Futures and Futures Forwards calculation modes, all types of pending orders are triggered according to the rules of the exchange where trading is performed. Usually, Last price (price of the last performed transaction) is applied. In other words, an order triggers when the Last price touches the price specified in the order. But note that buying or selling as a result of triggering of an order is always performed by the Ask and Bid prices respectively.
- In the "Exchange execution" mode, the price specified when placing limit orders is not verified. It can be specified above the current Ask price (for the Buy Limit orders) and below the current Bid price (for the Sell Limit orders). When placing an order with such a price, it triggers almost immediately and turns into a market one. However, unlike market orders where a trader agrees to perform a deal by a non-specified current market price, a pending order will be executed at a price no worse than the one specified.
- If during pending order activation the corresponding market operation cannot be executed (for example, the free margin on the account is not enough), the pending order will be canceled and moved to history with the "Rejected" status.



-  current market state  forecast
 -  current price  order price
 -  price, reaching which a pending order will be placed
 -  expected growth  expected fall
- Take Profit

The Take Profit order is intended for gaining the profit when the security price reaches a certain level. Execution of this order results in the complete closing of the entire position. It is always connected to an open position or a pending order. The order can be requested only together with a market or a pending order. This order condition for long positions is checked using the Bid price (the order is always set above the current Bid price), and the Ask price is used for short positions (the order is always set below the current Ask price).

Stop Loss

This order is used for minimizing losses if the security price moves the wrong direction. If the security price reaches this level, the entire position is closed automatically. Such orders are always associated with an open position or a pending order. They can be requested only together with a market or a pending order. This order condition for long positions is checked using the Bid price (the order is always set below the current Bid price), and the Ask price is used for short positions (the order is always set above the current Ask price).

If during Take Profit or Stop Loss activation the corresponding market operation cannot be executed (for example, it is rejected by the exchange), the order will not be deleted. It will trigger again at the next tick corresponding to the order activation conditions.

Rules of Stop Loss and Take Profit inheritance (netting):

- When a position volume is increased or the position is reversed, Take Profit and Stop Loss are placed according to its latest order (market or triggered pending order). In other words, stop levels in each subsequent order of the same position replace previous ones. If zero values are specified in the order, Stop Loss and Take Profit of a position will be deleted.
- If a position is partially closed, Stop Loss and Take Profit are not changed by the new order.
- If a position is fully closed, the Stop Loss and Take Profit levels are deleted, because they are associated with an open position and cannot exist without it.
- When a trade operation is executed for a symbol, for which there is a position, the current Stop Loss and Take Profit of the open position are automatically inserted in the order placing window. This is aimed to prevent accidental deletion of current stop orders.
- During one click trading operation (from a panel on the chart or from the Market Watch) for the symbol, for which there is a position, the current values of Stop Loss and Take Profit are not changed.
- On the OTC markets (Forex, CFD, Futures), when a position is moved to the

MetaTrader 5 trading platform

next trading day (the swap), including swap through re-opening, the levels of Stop Loss and Take Profit remain unchanged.

- On the exchange market, when a position is moved to the next trading day (the swap), as well as when moved to another account or during delivery, the levels of Stop Loss and Take Profit are reset.

Stop Loss and Take Profit inheritance rule (hedging):

- If a position is partially closed, Stop Loss and Take Profit are not changed by the new order.
- If a position is fully closed, the Stop Loss and Take Profit levels are deleted, because they are associated with an open position and cannot exist without it.
- t0>During one click trading operation (from a panel on the chart or Depth of Market), the Stop Loss and Take Profit levels are not set.

These rules apply both when trading manually and when placing orders from Expert Advisors (MQL5 programs).

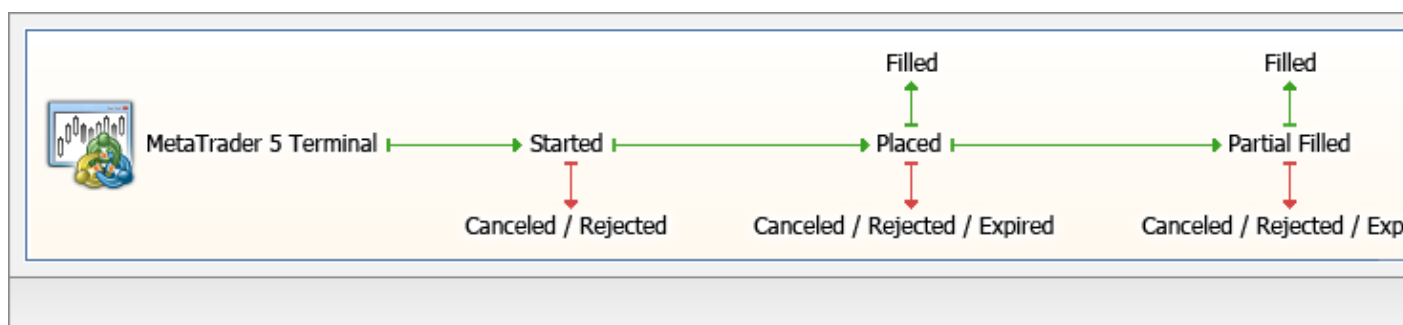
- Activation of Take Profit or Stop Loss results in the complete closing of the entire position.
- For symbols with Exchange Stocks, Exchange Futures and Futures Forwards calculation modes, Stop Loss and Take Profit orders are triggered according to the rules of the exchange where trading is performed. Usually, Last price (price of the last performed transaction) is applied. In other words, a stop-order triggers when the Last price touches the specified price. However note that buying or selling as a result of activation of a stop-order is always performed by the Bid and Ask prices.

State of Orders

After an order has been formed and sent to a trade server, it can undergo the following stages:

- **Started** the order correctness has been checked, but it hasn't been yet accepted by the broker;
- **Placed** a dealer has accepted the order;
- **Partially filled** the order is filled partially;
- **Filled** the entire order is filled;
- **Canceled** the order is canceled by the client;
- **Rejected** the order is rejected by a dealer;
- **Expired** the order is canceled due to its expiration.

You can view the state of orders in the corresponding section.



Types of Execution

Four order execution modes are available in the trading platform:

- **Instant Execution**

In this mode, an order is executed at the price offered to a broker. When sending an order to be executed, the platform automatically adds the current prices to the order. If the broker accepts the prices, the order is executed. If the broker does not accept the requested price, a "Requote" is sent the broker returns prices, at which this order can be executed.

- **Request Execution**

In this mode, a market order is executed at the price previously received from a broker. Prices for a certain market order are requested from the broker before the order is sent. After the prices have been received, order execution at the given price can be either confirmed or rejected.

- **Market Execution**

In this order execution mode, a broker makes a decision about the order execution price without any additional discussion with a trader. Sending an order in such a mode means advance consent to its execution at this price.

- **Exchange Execution**

In this mode, trade operations conducted in the trading platform are sent to an external trading system (exchange). Trade operations are executed at the prices of current market offers.

Fill Policy

In addition to common rules of order execution set by a broker, a trader can indicate additional conditions. The available fill policies are determined by the symbol settings.

- **Fill or Kill (FOK)**

This fill policy means that an order can be filled only in the specified volume. If the necessary amount of a financial instrument is currently unavailable in the market, the order will not be executed. The required volume can be filled by several offers available in the market at the moment.

- **Immediate or Cancel (IOC)**

In this case a trader agrees to execute a deal with the volume maximally available in the market within that indicated in the order. In case the order cannot be filled completely, the available volume of the order will be filled, and the remaining volume will be canceled. The possibility of using IOC orders is

determined on the trade server.

- **Book or Cancel (BOC)**

The BOC policy indicates that the order can only be placed in the Depth of Market (order book). If the order can be filled immediately when placed, this order is canceled. This policy guarantees that the price of the placed order will be worse than the current market. BOC is used to implement passive trading: it is guaranteed that the order cannot be executed immediately when placed and thus it does not affect current liquidity. This fill policy is only supported for limit and stop limit orders.

- **Return**

This policy is only used for market (Buy and Sell), limit and stop limit orders. If filled partially, an order with the remaining volume is not canceled, and is processed further. For market orders, the Return policy is used only in the Exchange Execution mode, while for limit and stop limit ones, it is applied in the Market Execution and Exchange Execution modes.

Use of fill policies depending on the execution type can be shown as the following table:

Type of Execution/Fill Policy	Fill or Kill	Immediate or Cancel	Book or Cancel	Return
Instant Execution	+			
Request Execution	+			
Market Execution	+	+		+
Exchange Execution	+	+	+	+

MetaTrader 5 Trading Platform / Platform Setup / General Information /
Price Data

Price Data in the MetaTrader 5 Platform

Three basic prices of a financial instrument are used in the trading platform:

- Bid is the highest price at which a trader can sell a financial instrument. It is the best price at which a financial symbol can be sold.
- Ask is the lowest price at which a trader can buy a financial instrument. It is the best price at which a financial symbol can be bought.
- Last is the price of the last deal executed on a financial instrument.

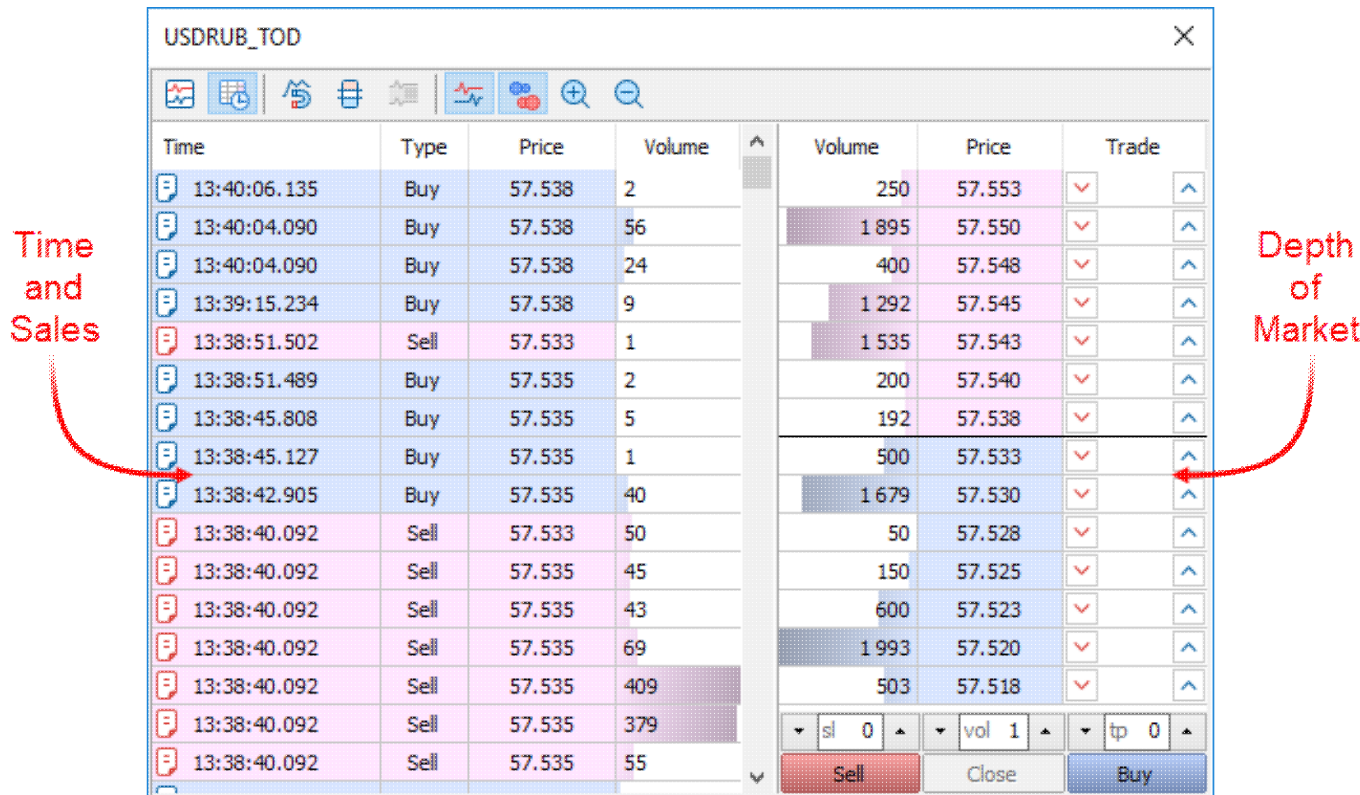
A financial symbol can be traded on the exchange and over-the-counter (OTC) market. Different approaches to symbol quote and charting are used depending on the market.

Exchange Market with the Market Depth

The only source of quotes in the exchange market is the Exchange itself. Buyers and sellers meet on the exchange, which keeps records of all executed deals. Orders of all market participants comprise a single Market Depth.

A Market Depth option featuring real orders of market participants is available in the trading platform for exchange traded symbol. Based on the best orders, the Bid and Ask prices are formed in the Market Depth (these prices are shown in the Market Watch window). Also, the exchange provides prices and volumes of last executed deals (Last and Volume). Last prices are used for creating price charts and for displaying the Time & Sales tape:

MetaTrader 5 trading platform



Although symbol charts are based on Last prices, traders execute deals at Bid and Ask prices (actual prices available in the market).

Over-The-Counter Market

Participants of the OTC market are big market players, such as banks and prime brokerages. They form networks to trade with each other. Medium market participants, such as banks, management companies and hedge funds connect to large participants. Market participants aggregate prices of their counterparties or they set their own prices based on counterparties' prices, and provide these prices to their clients.

Only Bid and Ask stream quotes are used in OTC market trading, without data on actual executed deals. Charts are based on Bid prices.

MetaTrader 5 trading platform

Symbol	Bid	Ask	!	Time
EURUSD	1.18688	1.18690	2	15:18:32
GBPUSD	1.35022	1.35024	2	15:18:37
USDCHF	0.97413	0.97416	3	15:18:36
USDJPY	112.228	112.231	3	15:18:35
USDCNH	6.61596	6.61740	144	15:18:04
AUDUSD	0.79569	0.79574	5	15:18:37
NZDUSD	0.72816	0.72819	3	15:18:37
USDCAD	1.23208	1.23211	3	15:18:37
USDSEK	8.02968	8.03046	78	15:18:37
USDHKD	7.81469	7.81484	15	15:18:05
USDSGD	1.35082	1.35087	5	15:18:37
USDMXN	17.79752	17.79979	227	15:18:32

[Symbols](#) | [Details](#) | [Trading](#) | [Ticks](#)

Over-The-Counter Market with the Market Depth

Unlike the previous variant, the broker provides traders information on volumes in addition to Bid and Ask prices, which allows displaying the Market Depth. The exchange Market Depth consists of limit orders of market participants, while OTC Market Depth is formed based on the broker's quotes. The broker provides different prices depending on the buying and selling volume.

Best buy price

Best sell price

Volume	Price	Trade
	1.19629	⬇
	1.19628	⬇
5.00	1.19627	⬇
10.00	1.19626	⬇
...		
5.00	1.19624	⬆
10.00	1.19623	⬆
10.00	1.19622	⬆
	1.19621	⬆

sl 0

vol 1.00

tp 0

Sell

Close

Buy

MetaTrader 5 trading platform

The exchange does not participate in trading and does not keep record of performed trades, therefore no Last prices are available in this mode. Charts are based on Bid prices.

How Price Charts Are Formed

One-minute bars are formed based on symbol quotes (or ticks). This bar represents a set of price characteristics of one minute:

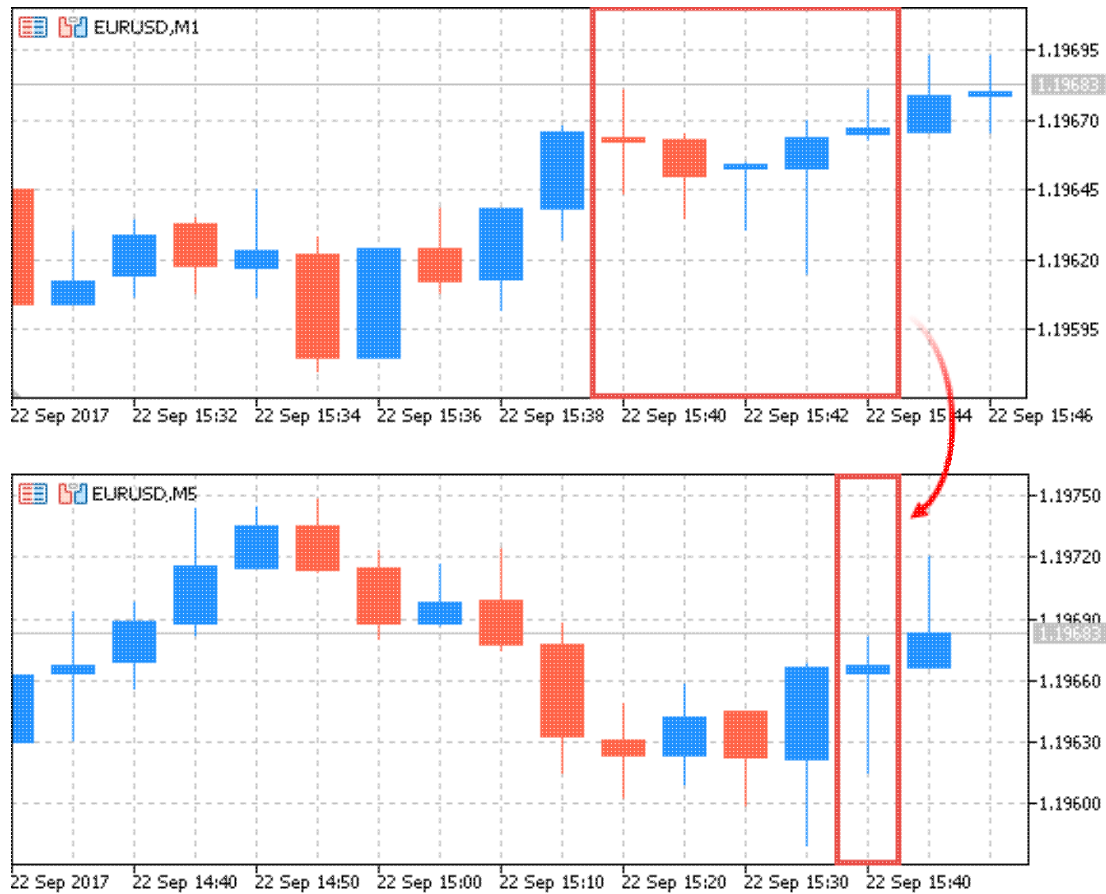
- 4 prices: Low and High price during this minute, as well as the beginning and the end of the bar, i.e. the Open and Close prices
- Spread, which is the minimum difference between Bid and Ask recorded during one minute
- Tick volume, which shows the number of ticks received during bar formation
- Volume, i.e. the real volume of deals performed during bar formation (may be not available for OTC markets)
- Date and time, i.e. the minute to which this bar corresponds

One-minute bars are based on Bid prices for OTC symbols, and are based on Last prices for exchange instruments.

In addition to one-minute bars, price charts in the trading platform can be displayed as larger time intervals. The time included in one bar or candlestick on the chart, is called a timeframe. The platform supports 21 timeframes from 1 minute to a 1-month period.

The trading platform only stores 1-minute bars. All higher timeframes are created based on these bars. This approach ensures compliance of data at all periods, as well as allows to significantly save traffic and disk space.

MetaTrader 5 trading platform



MetaTrader 5 Trading Platform / Platform Setup / General Information /
Working with Instructions

Working with Instructions

There are several methods of working with instructions in different sections of platform administration. Depending on the selected section of administering, the instruction can be an account, a group, a holiday, a directive that allows or prohibits access, etc.

Creating Instructions on the Basis of Existing Ones

In order to avoid specifying common parameters of an instruction each time it is created, one can create it on the basis of an existing one by changing its key parameter. For example, if you open the "EURUSD" symbol for editing by a double click of the mouse or using the "⚙ Edit" command, change its name to "EURGBP" and save it, you will get a new symbol with the settings similar to "EURUSD".

Key fields for different instructions are listed below:

- **IP Access** "From", "To";
- **Holidays** "Date", "Time", "Description";
- **Groups** "Name";
- **Managers** "Login";
- **Routing** "Name";
- **Gateways** "Name";
- **Plugins** "Name";
- **Data Feeds** "Name";
- **Symbols** "Symbol";
- **Synchronization** "Server".

Group Work with Instructions

The administrator terminal allows to change the settings of instructions of a selected section in bulk. To do it, one should select the necessary instructions using the mouse with "Shift" and "Ctrl" keys and start editing them by executing the "⚙ Edit" command.

Some fields may be blocked when opening multiple instructions for editing. This means that those fields cannot be changed for multiple instructions simultaneously:

Symbol

?

Common

Currency

Quotes

Trade

Execution

Margin

Margin Rates

Swaps

Sessions



The setting up of main parameters of the symbol. Please specify its name, description, and other parameters.

Symbol:

Description:

Exchange:

International:

ISIN:

Sector:

Currency

CFI:

JFTXCC

Industry:

Undefined

Basis:

Country:

Source:

Category:

Background:

☐ White

Page:

Digits:

3

Market depth:

off

Spread:

60

pt

Chart mode:

by bid price

Spread balance:

-29 bid / 31 ask

OK

Cancel

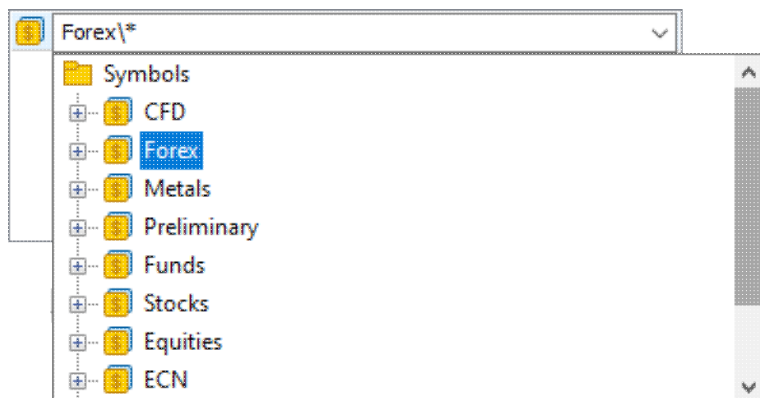
MetaTrader 5 trading platform

At that the values of the very upper instruction are displayed in the rest of the fields. Any value of edited instructions is changed only if the user changes its value. All the other fields remain unchanged.

MetaTrader 5 Trading Platform / Platform Setup / General Information / Specifying Symbols and Groups

Specifying Symbols and Groups

A standard way of specifying symbols and groups is used in different sections of the server administration:



The mask ("*" symbol) and paths to financial instruments and groups are used here, since they can have certain hierarchy in the system. For example, there is a group of instruments "Forex" that has symbols in it divided into the "Majors" and "Crosses" sections. In order to specify all the symbols from the "Majors" group, the following entry is necessary "Forex\Major*". This kind of paths are substituted automatically if a group or symbol is chosen from the list. The "*" symbol means all the instrument/groups from the specified group.

"*" Mask and "!" Negation Sign

The "*" mask and the "!" negation sign are used for making extended requests. For example, they can be used when specifying symbols to be translated from a data feed or a gateway, when specifying symbols for the synchronization of price data and when configuring holidays. Here are several examples of specification:

- **,***, all the symbols;
- **,EUR*** all the symbols where EUR is the base currency;
- **,EUR*,!EURUSD** all the symbols where EUR is the base currency except EURUSD;
- **,EURUSD,GBPUSD** only specified symbols. One can specify any number of symbols separated by comma;
- **,*USD** all the symbols where USD is the quoting currency.

To exclude symbols using "!" on data feeds and gateways, you should specify the full path of these symbols. For example, to specify all symbols from the Cboe FX subgroup, except for the EURUSD symbol located in the same subgroup, use the string "Cboe FX*,!Cboe FX\EURUSD". The string "Cboe FX*,!EURUSD" will be invalid. Also, the exception does not work for a single "*" mask. It always allows all symbols:

- **Forex*,!Forex\EURUSD** all symbols in the Forex subgroup, except EURUSD.

MetaTrader 5 trading platform

- *,!Forex\EURUSD all symbols. The EURUSD symbol will not be excluded.

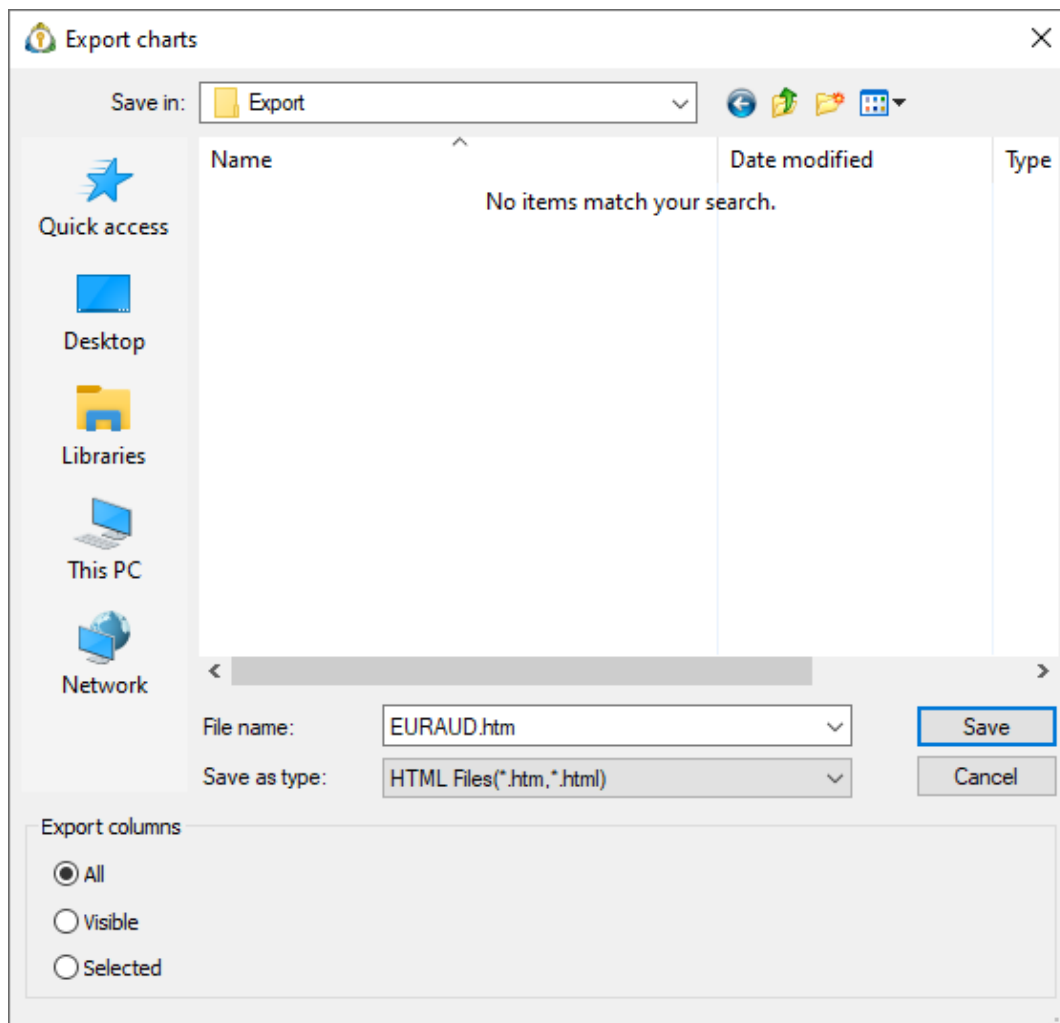
MetaTrader 5 Trading Platform / Platform Setup / General Information / Data Export

Data Export

The administrator terminal can export data to *.HTM, *.HTML and *.CSV files. Data can be exported from the following sections:

- Accounts
- Clients
- Orders
- Deals
- Positions
- Charts
- Ticks

Request data in the appropriate section to execute an export operation. Then click "Export" in the context menu of this section. After that the window where one should specify a folder to save the file to will appear. Its lower part contains the block of the export settings:

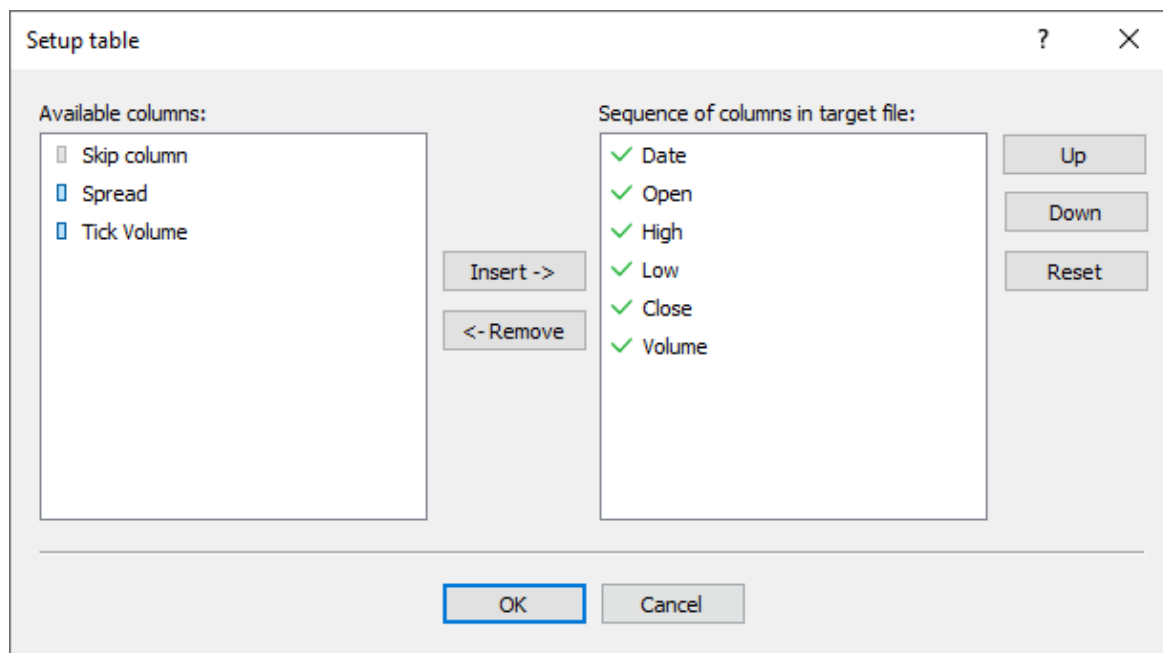


The following export variants are available here:

- **All** export all information regardless of the currently selected columns;
- **Visible** export only the columns that are enabled at the moment;
- **Selected** open the window of manual selection of columns to be exported, which is described below.

In order to export the information, one should specify the name of the end file and press the "Save" button.

Selecting Columns for Export



The left part contains the available columns and the right part contains the chosen ones. The window contains the following commands:

- **Insert** add the selected column to the list of chosen ones;
- **Remove** remove the selected columns from the chosen ones. Some fields are obligatory for exporting, they cannot be deleted;
- **Up** move the added column up relatively to the others. The order of selected columns determines the sequence of data in the exported file;
- **Down** move the added column down relatively to the others;
- **Reset** restore the default settings of columns.

It is also possible to move the columns by a double-click with the left mouse button on them.

There is a special entry "Skip column" in the list of available columns. Using it, one can create empty columns separated with the selected separator in the exported file.

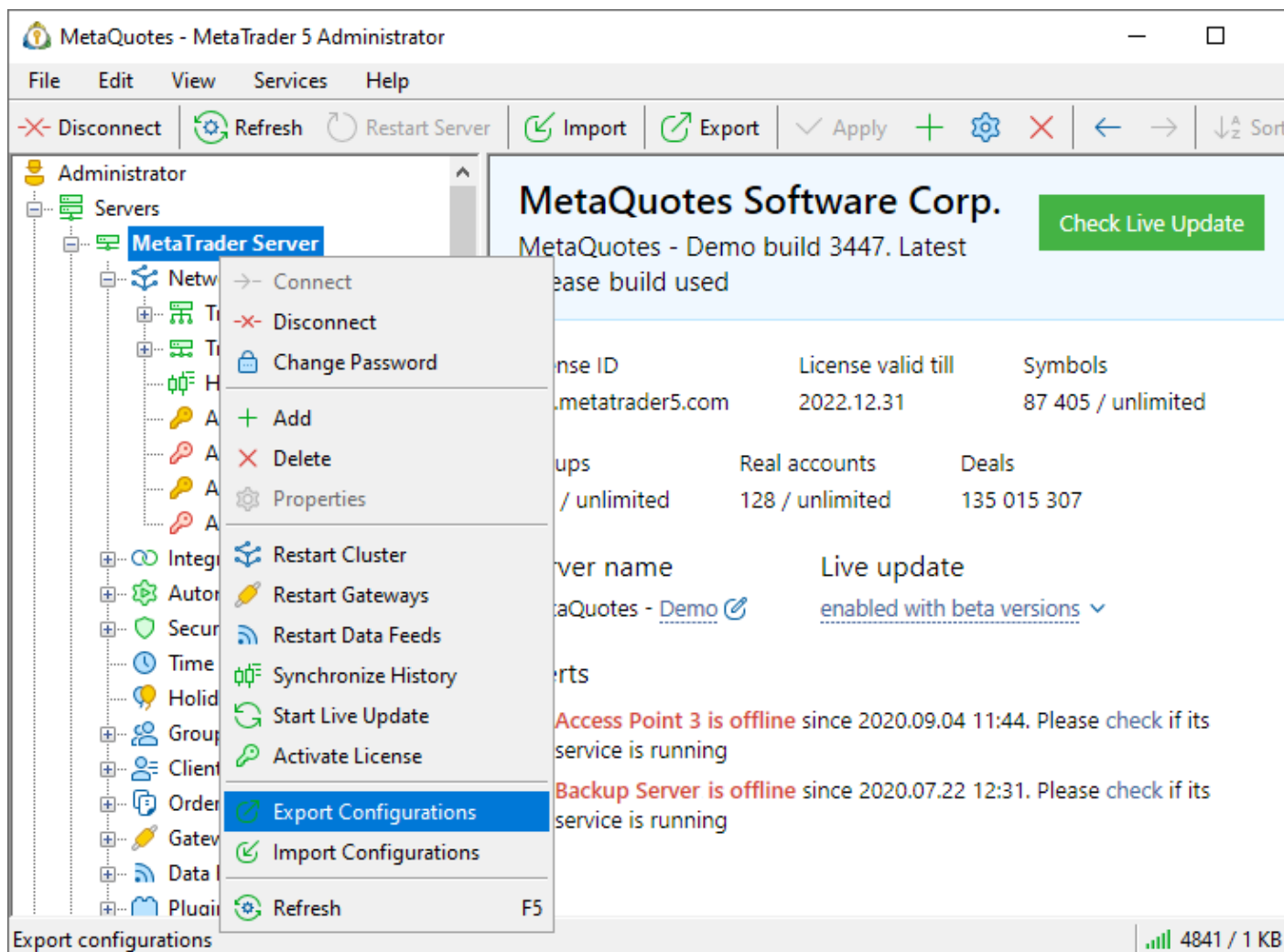
MetaTrader 5 Trading Platform / Platform Setup / General Information / Import/Export Settings

Import and Export Settings

The administrator terminal allows to quickly export and import almost all settings of the trade platform. This functionality can be useful, for example, if you need to set up a test platform with the settings similar to your work platform.

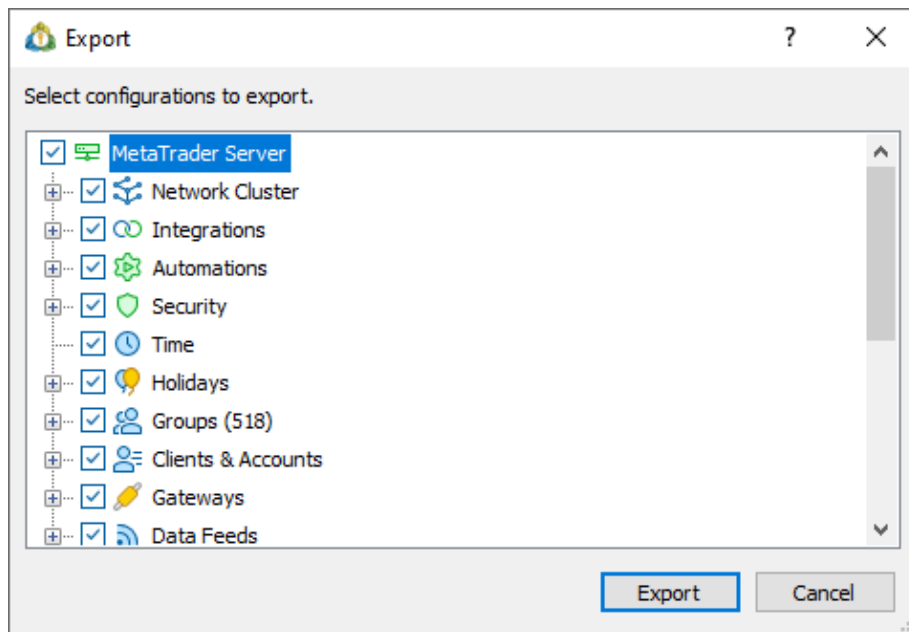
Export

The platform settings are exported in a text file in the JSON format. To start the export, open the context menu of the platform and click "Export":



MetaTrader 5 trading platform

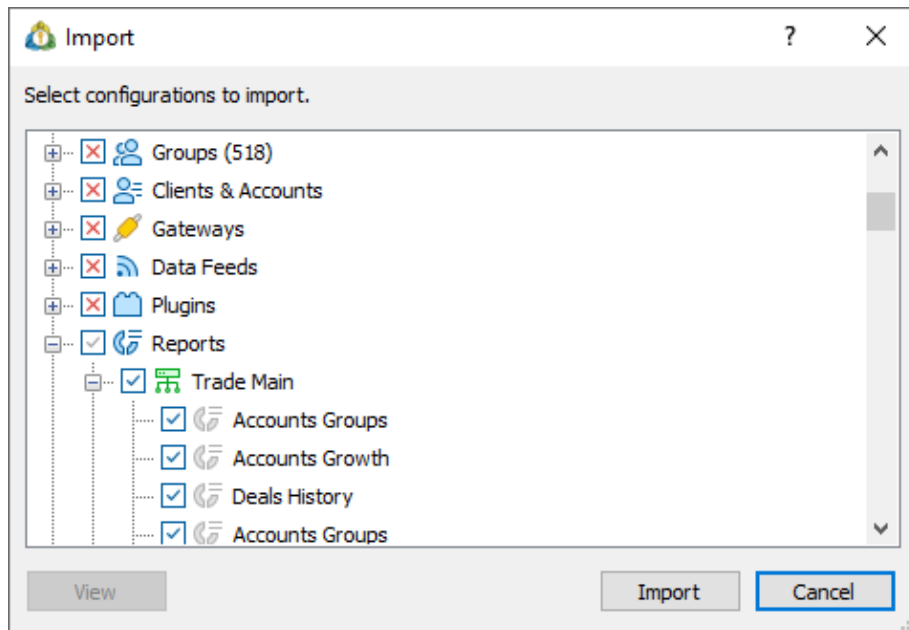
The dialog where you can select the settings to be exported appears:



Select the necessary settings and click "Export". Then specify a folder to save the file with settings.

Import

To import the settings, open the platform context menu as described above and click "Import".



MetaTrader 5 trading platform

Select settings that should be imported. All non-selected items are skipped, while the appropriate platform settings remain unchanged. Using the "View" button you can check any imported configuration and change it if needed:

Report: Accounts Groups

Common Information

 Reports are server plugins providing built-in or custom reports. Please specify report name, select report plugin, and specify additional parameters if needed.

☒ Enable

Name: Accounts Groups

Type: Accounts Groups Trade Main, 1

Parameters:

Parameter	Value
-----------	-------

Add Edit Delete

OK Cancel Help

After changing the settings in the preview mode, click "OK" and then "Import".

If an imported configuration does not exist in the platform, it will be created. If an imported configuration matches an existing configuration in the trade platform, the existing configuration settings will be updated. Configurations are compared by their key parameters, for example, group name, manager login (account number), routing rule name, etc.

Configurations of reports, plugins and funds are bound to specific servers in the cluster. When you export their settings, server IDs are also saved in JSON files. During import, the system checks the ID specified in the file and binds the configuration to the appropriate server. If the platform with the specified ID does not have a server with the specified identifier, the configuration will be bound to the main trade server. If necessary, you can change binding in the preview mode.

- Be careful when importing the settings. It can seriously affect the operation of the trade platform.
- Settings files use the UTF-16 Little Endian encoding. Files in other encodings cannot be imported.
- Starting with the platform build 1930, all volume settings are exported in two versions: with standard and extended accuracy. For example, the minimum symbol volume is exported as "VolumeMin" : "100" and "VolumeMinExt" : "1000000" (lots are obtained by dividing the values by 10000 and 100000000, respectively). Extended volume accuracy has a higher priority during import. If it is not specified (for example, if settings are exported by an older terminal version), the value with standard accuracy will be used.

Export and Import via Command Line

To automate the configuration process, as well as to automate the transfer of platform settings without having to develop additional plugins and applications in C++, you can use command line export and import.

Close the Administrator terminal before executing commands in console mode. Otherwise, the commands will not be executed because the second copy of the application cannot be run in parallel.

In order to start MetaTrader 5 Administrator in the console mode, use the /console key:

```
/console /server:<Trade Server Address:Port> /login:<Login> /password:<Password> /action
```

Specify platform connection details in the 'server', 'login' and 'password' parameters. Specify the type of performed action in the 'action' parameter:

Server Restart

Command /action:restart [/name:<server name>]. Optionally specify the server name (History, Access...). If the name is specified, the appropriate sever will be searched for (case insensitive). If the name is not specified, the command will be performed for the current connected server.

Export of a Configuration to JSON

Command /action:export /file:<path> [/type:<type> /config:<mask>]. If no optional arguments are specified, the entire configuration of the connected server will be exported.

Argument /file:<path> sets the destination file path. Argument /type:<type> is used for exporting a selected configuration branch, the argument may have the following values:

common network firewall time holidays groups managers routing gateways plugins feeders reports symbols spreads historysync
--

Argument /config:<mask> sets a search criterion to search for a particular configuration structure by a string mask (can contain *,!). Search by mask can be used for the configurations of servers, groups, managers, trade requests, gateways, plugins, data feeds, reports, and symbols. The following rule applies to the rest of configurations: if a mask is set, the configuration should be skipped, if no mask is specified, the configuration should be processed.

Import of a Configuration

Command /action:import /file:<path> [/type:<type> /config:<mask>]. All arguments are similar to those applied to exports, except for the /type argument. Branch common (common platform settings) cannot be imported.

MetaTrader 5 Trading Platform / Platform Setup / Start Page

Start Page

The start page displays important information about the platform operation: license data, the number of groups, accounts and trading operations. In addition, here you can change live update settings, as well as find a lot of useful information related to platform administration.

To open the start page, select the desired platform on the left side of the terminal.

MetaQuotes-Demo - MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Export Import Apply + Settings X Sort ↑ ↓

Administrator

- Servers
 - Forex Broker Server
 - Network Cluster
 - Trade Main
 - History
 - Trade Server
 - Anti DDoS Proxy
 - Access Point 1
 - Access Point 2
 - Integrations
 - Finteza Analytics
 - VPS
 - Mail Servers
 - Messengers
 - KYC
 - Web Services
 - Automations
 - Security
 - Time
 - Holidays
 - Groups (691)
 - Clients & Accounts
 - Orders & Deals
 - Positions
 - Orders
 - Deals
 - Gateways
 - Data Feeds
 - Plugins
 - Reports
 - ECN
 - Routing

MetaQuotes Software Corp.

MetaQuotes - Demo build 2470. Latest build used. [Check Live Update](#)

License ID	License valid till	Symbols
metatrader5.com	2020.12.31	84 217 / unlimited

Groups	Real accounts	Deals
674 / unlimited	47 / unlimited	13 824 650

Server name: MetaQuotes - Demo [Live update](#)
[enabled with beta versions](#) ▾

Alerts

Backup Server is offline since 2017.07.17 18:08. Please [check](#) if its service is running

MetaTrader 5 ForexTime is offline since 2020.06.01 00:00. Please [check its journal](#)

LP Fibo is offline since 2020.05.30 05:40. Please [check its journal](#)

Administrator best practices

- System
 - Backup Server: Upgrade your operating system. The recommended minimal version is Windows Server 2016
- Network Cluster
 - Configure the Anti DDoS Proxy Server to protect your platform from attacks
 - Install Backup Server easily using the Deploy function
- Integrations
 - Enable KYC checks to validate client personal data and documents with one click or even in a fully automated mode
- Security
 - Configure Certificates for extended authentication in the trading platform to

For Help, press F1

98.3 / 0.4 M

Server Data

The following platform data is specified in the page header:

- Name of the trading platform owner company.
- Platform name displayed in client terminals. It consists of two parts:
 - Owner company's short name indicated in the license.
 - An additional part as define by you. Here you can specify additional data, such as the server type: demo or live. The short name is set in the "Server Name" field below. After editing the data, click "Apply" on the toolbar to save settings. The additional data must not look like the domain name of the server. For example, do not specify "demo.com", "real.net", etc.

License Information

The main license details are shown at the top of the page. Here, you will be warned of approaching expiration date or account limit. Separate warnings can be shown if your platform is not activated.

- License the website of the company for which the trading platform license has been issued.
- License valid till license expiration date. For example, if 2021.12.31 is specified, the license is valid until 2021.12.30 23:59:59 trading platform time. If payments are made in a timely manner and your platform is properly connected to the license server (updates.metaquotes.net), the license will be updated automatically. If the license update fails for any reason, the license field will become highlighted in red one month before the license expiration date. In this case please contact the support team for renewal. Once the license expires, the platform operation will stop and you will not be able to provide further services to your clients.
- Symbols the total number of symbols in the platform, as well as the limitation according to your license type (if there is any).
- Groups the total number of groups in the platform, as well as the limitation according to your license type (if there is any).
- Real accounts the total number of real clients in the entire platform, as well as the limitation on the number of clients according to the license (if any). Real accounts are client records that are not included into demo*, manager*, preliminary* and contest* groups. Only enabled accounts for which trading is allowed are counted.
- Deals the number of deals executed in the platform.

Live update

The update of system components is set in this block:

- Disabled no updates to the platform components are performed, either automatically or manually.
- Enabled platform components are updated to latest release versions in the automatic mode.

MetaTrader 5 trading platform

- Enabled with beta versions when you launch live update, both release and beta versions of platform components will be installed. Beta versions precede the main release, allowing you to test the new features released in the update, as well as to test the operation of your plugins, gateways and other additional components. This option should only be enabled on test servers. Please make sure not to update your real servers with live accounts up to the beta version.

After editing the data, click "Apply" on the toolbar to save settings.

Make sure your platform is activated. Otherwise, some of its basic features will be limited. If a notification in the upper part of the page shows that the platform is not activated, launch activation using the Services menu.

Alerts

Important alerts concerning the platform components are shown in this block:

- No connection between the gateway or data feed and the source. If such an alert appears, please check the gateway or data feed journal.
- A platform component is not connected. In this case please check if the appropriate service is running in the operating system and check the latest logs in the component's log in the disk.

Please do not ignore alerts.

Administrator best practices

This block features useful tips concerning platform operation, which will assist in using the platform to its full capabilities: what components need further setup or verification, how additional symbols can be added and other tips. A click on a tip will open an appropriate Administrator terminal section. After that the advice line will be hidden from the list.

MetaTrader 5 Trading Platform / Platform Setup / Network cluster

Network Cluster

This section is intended for managing all the server components of the online trading system and for monitoring their state. All the added platform components, such as access servers, history server, backup servers, etc., are displayed here.

Type	Address	CPU	Thre...	Me...	Soc...	Conn...
Main Trade Server	14.140.80.42:1951	3%	125	34 157	91	7
Trade Server	14.140.80.42:1952	2%	127	34 157	91	0
Access Server	14.140.80.42:2000	6%	87	34 157	91	2
Access Server	14.140.80.42:1950	1%	97	34 157	91	2
History Server	14.140.80.42:1953	3%	89	34 157	91	8
Backup Server	14.140.80.42:1954	4%	29	34 157	91	
Backup Server	14.140.80.42:1955	3%	37	34 157	91	

For Help, press F1

The following information about the platform components is displayed here:

- **Type** type of the server (main trade server, trade server, history server, access server or backup server);
- **Server** name of the server;
- **Address** IP address and port separated by colon;
- **Public Addresses** the list of public IP addresses;
- **ID** internal identifier of the server that is used for recognizing it by the other platform components;
- **Connections** number of connections to the server;
- **Base Priority** priority, that is set for the server in its settings;
- **Current Priority** current priority of the access server;
- **CPU** processor loading (in percentage terms).

To add a new server, one should execute the "+ Add" command in the "Edit" menu or the same command in the toolbar or in the context menu. To change the settings of a server, it is necessary to select it and press the "⚙ Edit" button or double-click on it with the left mouse button. To delete a server press the "✗ Delete" button.

- The process of adding and editing servers is described in the [separate section](#).
- The interaction between the servers inside the platform is described in the ["Platform components"](#) section.
- If a server in the list or in the tree has a red icon, for example , it means that it is not connected to the main server. The possible reason can be the incorrect authorization details as well as the stopping of the corresponding service in the operating system.

Context Menu

The context menu of this window allows to execute the following commands:

-  **Restart Server** [restart](#) a selected server.
-  **Deploy Server** start [deploying](#) a selected server with specified configuration. This command is only available for servers that are not started yet.
-  **Switch to Backup Server** this procedure allows to make a selected backup server an active trade or history server depending on what type of server is being backed up. At that, the backed up server will become a backup server. This procedure is described in more details in ["Switching to Backup Server"](#).
-  **Restart Data Feeds** restart [data feeds](#).
-  **Restart Gateways** restart [gateways](#).
- **Manage Server Machine** a menu with commands for [managing the computer](#) on which the platform server is running.
-  **Add** [add](#) a new server.
-  **Edit** [edit](#) a selected server.
-  **Delete** delete a selected server.
-  **Move Up** move a selected server up relatively to the others.
-  **Move Down** move a selected server down relatively to the others.
- **Automation triggers** create an [automation](#) task for the selected event or edit an existing one. The menu displays only the triggers and tasks associated with the current section.
- **Automation actions** add an automation action to an existing task or create a new task based on the action. The menu displays only the actions associated with the current section.
-  **Export to File** [export](#) network settings to a file.
-  **Import from File** [import](#) network settings to a file.
-  **Find** open the [search](#) window.
- **Auto Arrange** if this option is enabled, the size of columns is selected automatically.
- **Grid** this option shows/hides grid to separate fields of the table of servers.

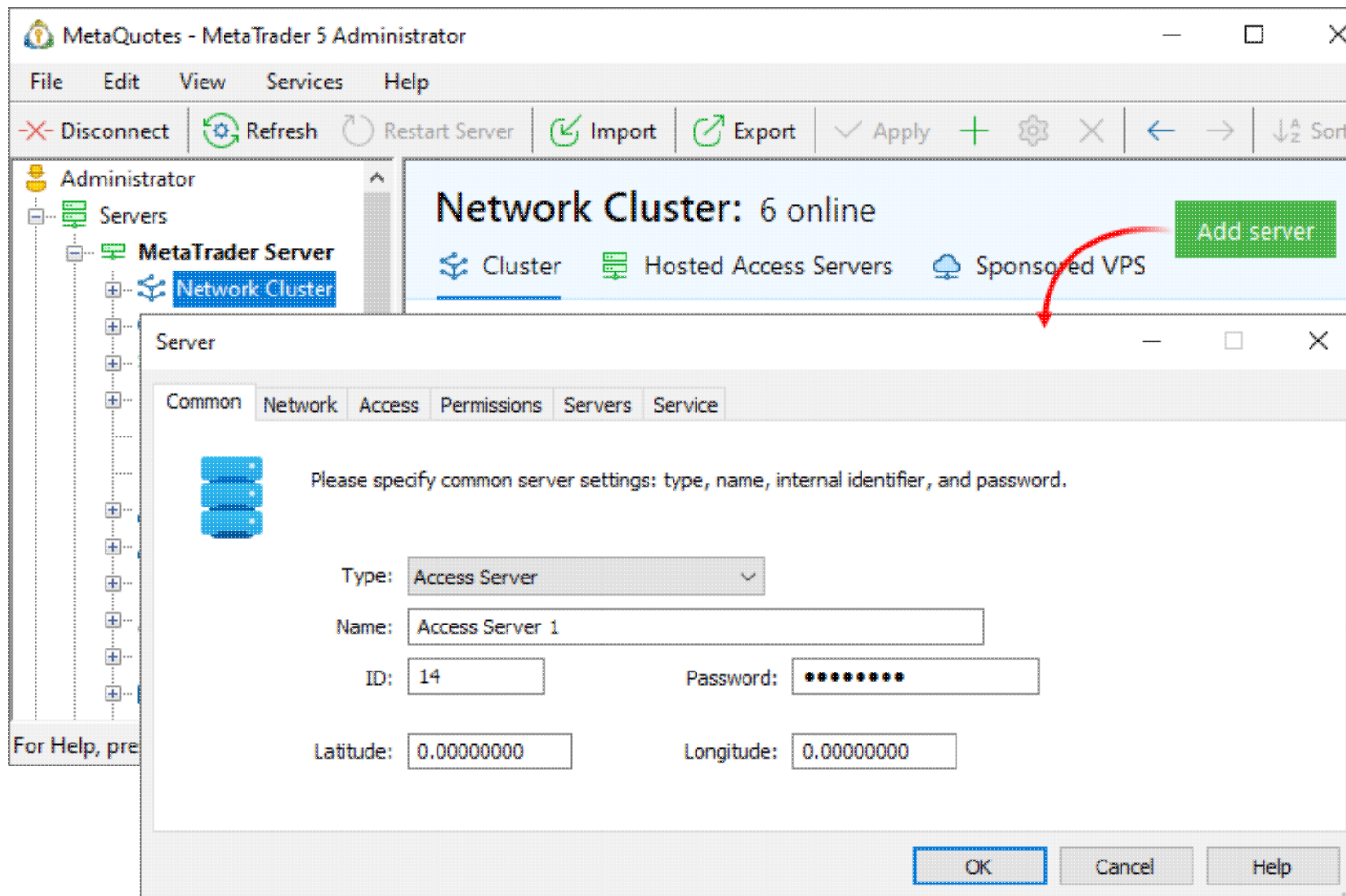
MetaTrader 5 Trading Platform / Platform Setup / Network cluster /
Configuring Servers

Configuring Servers

The trading platform cluster consists of several servers. Each server performs specific functions and thus the overall load is distributed among different components.

- Trade server
- History server
- Access server
- Backup server
- AntiDDoS server

To add a server to the system, install it and add to the "Network section" of the Administrator terminal. The order of operations does not matter.

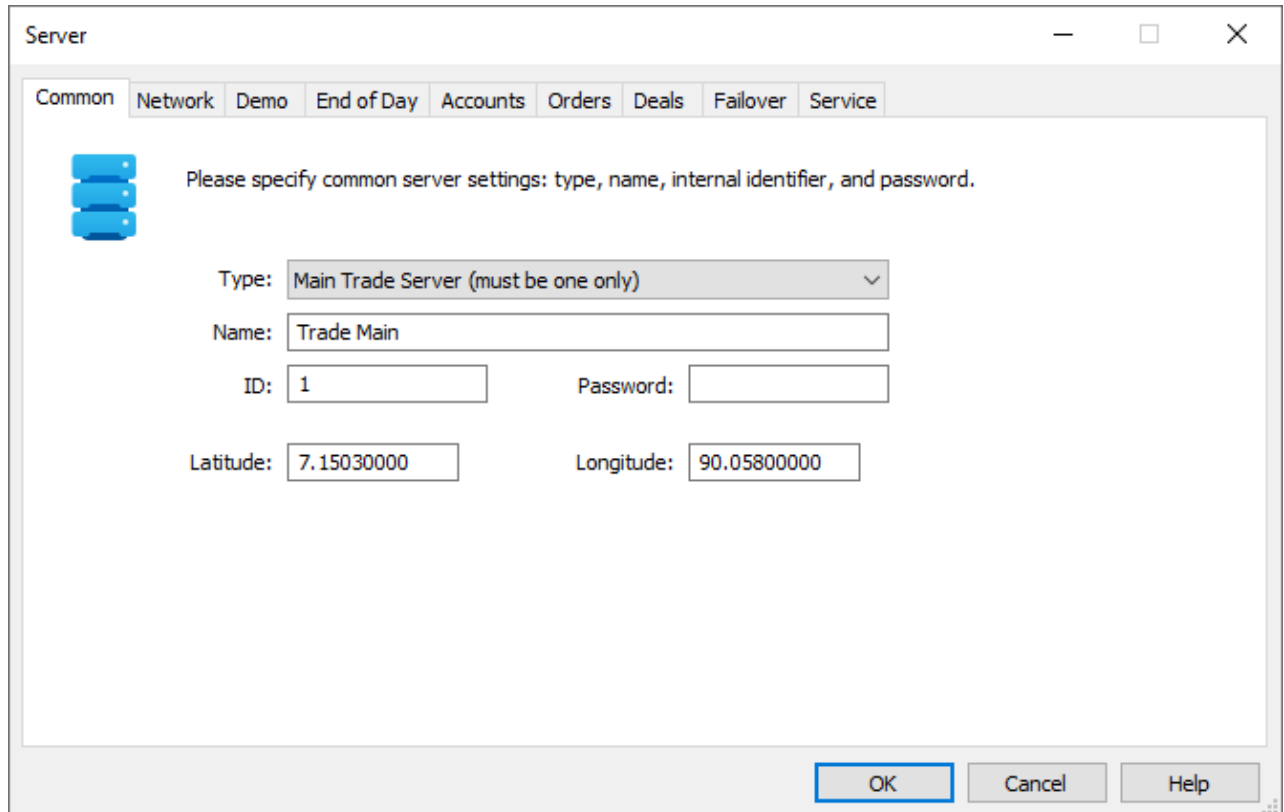


MetaTrader 5 trading platform

After completing these steps, reboot platform (the main trade server). Only after that the newly installed server will be activated.

Servers have individual sets of settings. They are explained in relevant subsections. Below is a description of the general settings: sections "General", "Network" and "Service" are the same for all types of servers.

Common



The screenshot shows the 'Server' configuration window with the 'Common' tab selected. The window title is 'Server'. The tabs are: Common, Network, Demo, End of Day, Accounts, Orders, Deals, Failover, and Service. The 'Common' tab contains a blue server icon and the text: 'Please specify common server settings: type, name, internal identifier, and password.' The settings are as follows:

Field	Value
Type:	Main Trade Server (must be one only)
Name:	Trade Main
ID:	1
Password:	
Latitude:	7.15030000
Longitude:	90.05800000

At the bottom right, there are three buttons: OK, Cancel, and Help.

The "Common" tab contains the following settings:

- **Type** type of the server (main trade server, trader server, history server, access server or backup);
- **Name** name of the server;
- **ID** identifier that is used for internal recognizing of the servers;
- **Password** internal password used for connection of other servers of the platform to this server. The password is set during server installation. If the deployment function is used, the password should be set in advance, before installation. The password must be 7 to 15 characters long.
- **Latitude/Longitude** geographic coordinates of the server to enable its precise positioning on the interactive cluster map.

- Only one main trade server can be added.
- If the platform is installed in one local network then it is necessary to indicate the internal IP addresses. If the components are located in different networks then one should specify external addresses.

Network

Network settings of the server are specified on this tab.

Server

Common Network Demo End of Day Accounts Orders Deals Failover Service

Please specify network settings of the server: outgoing address, public addresses, and bindings.

The following IPv4 and IPv6 addresses are used for outgoing connections from this server.

Address: 18.140.18.42:2000 [2a00:1111:1:20::2a]:2000

Listen on these IP addresses

☒ Automatically use as public addresses

18.140.18.42:2000 [2a00:1111:1:20::2a]:2000

Add Edit Delete

OK Cancel Help

IPv4 and IPv6 support

IPv4 which is used in every network was created more than 30 years ago. It contains IP addresses of 32 bits, which are represented as four 8-bit numbers separated by dots. This produces more than four billion unique IP addresses. However, the rapidly growing number of users and devices has accelerated the depletion of the pool of available addresses.

To avoid the depletion problem, products started to provide additional support for the modern IPv6. This protocol uses a 128-bit address, represented as x:x:x:x:x:x:x, where each x is a hexadecimal value of six 16-bit address elements. Theoretically, this format allows 5×10^{28} unique addresses. In addition to an extensive address space, this protocol has other advantages over the older version. You can read about them in specialized articles.

The platform supports two network protocols: IPv4 and IPv6. The corresponding addresses can be specified as outgoing, listen and public.

To use IPv6, please make sure this protocol is supported by your network infrastructure.

Outgoing Address

The outgoing address is used for the following purposes:

- For performing outgoing connections to other servers;
- For authentication of the server by IP address. During the connection the address specified in the settings is compared to the address of a real connection.

Use the left field to bind to an IPv4 address. The address is specified as IP-address:port, for example: 192.168.0.1:3128.

Use the right field to bind to an IPv6 address. If a port is specified for the address, the main part must be enclosed in square brackets: [address]:port.

If no outbound address is specified, the server will connect to the cluster without authentication by IP address.

Listen Addresses

A listen (binding) address is an IP address and port that will be listened by the server for incoming external connections. It is recommended to specify an IP address, not domain name, to bind to

Here one can indicate a local IP address as well as an external one, depending on the network configuration:

- If the server has direct access to the Internet and is not closed with a firewall, it is necessary to specify the external IP address;

MetaTrader 5 trading platform

- If the access to the Internet is performed through a firewall in a local network, then one should properly set up the routing of incoming connections from the external IP address to the internal IP address of the access server. In this case, the local address of the server is specified on the tab.
- To set up listening of a certain port at all available addresses, specify 0.0.0.0:port number. For example, 0.0.0.0:443.

Public Addresses

In this section you should specify the list of access points (IP or DNS addresses) that are used to accept connections from other components of the platform and client connections from terminals and API (for access servers). In order to create an access point, press "Add" and enter the IP address and port number separated by a colon. E.g. 88.168.10.91:443.

When other cluster components connect to the server, they sequentially iterate over its public addresses in the same order in which they are listed. If a domain name is specified as a public address, servers will iterate over all associated IP addresses, while preference is given to IPv6.

If you enable the "Automatically use as public addresses" option, then the specified listen addresses will be used as public.

Connection order between cluster components

LAN is preferred when one cluster component connects to another.

1. The system checks outgoing connections of both servers: whether they belong to the local subnet 10.*, 172.16.* 172.31.* or 192.168* and whether they have the same first three octets. Example: the backup server has an outgoing address of 192.168.0.100:1951 and the access server has 192.168.0.105:1950. If this condition is met, then connection will be performed using all local addresses from the list of listening addresses (from the server to which they are connecting).
2. If only the port (0.0.0.0:port number) is specified as the listening address, the connection address is formed as [outgoing address]:[listening port].
3. If the connection to the above addresses fails, all public server addresses will be searched in the order in which they are specified.

Consider the following example of a backup server connection to the main server.

- Backup Server: outgoing address= 192.168.0.104:1954
- Main Server: outgoing address = 192.168.0.101:1954; listening address = 72.15.96.137:1000, 0.0.0.0:2000; public address: forex.broker.com:443.

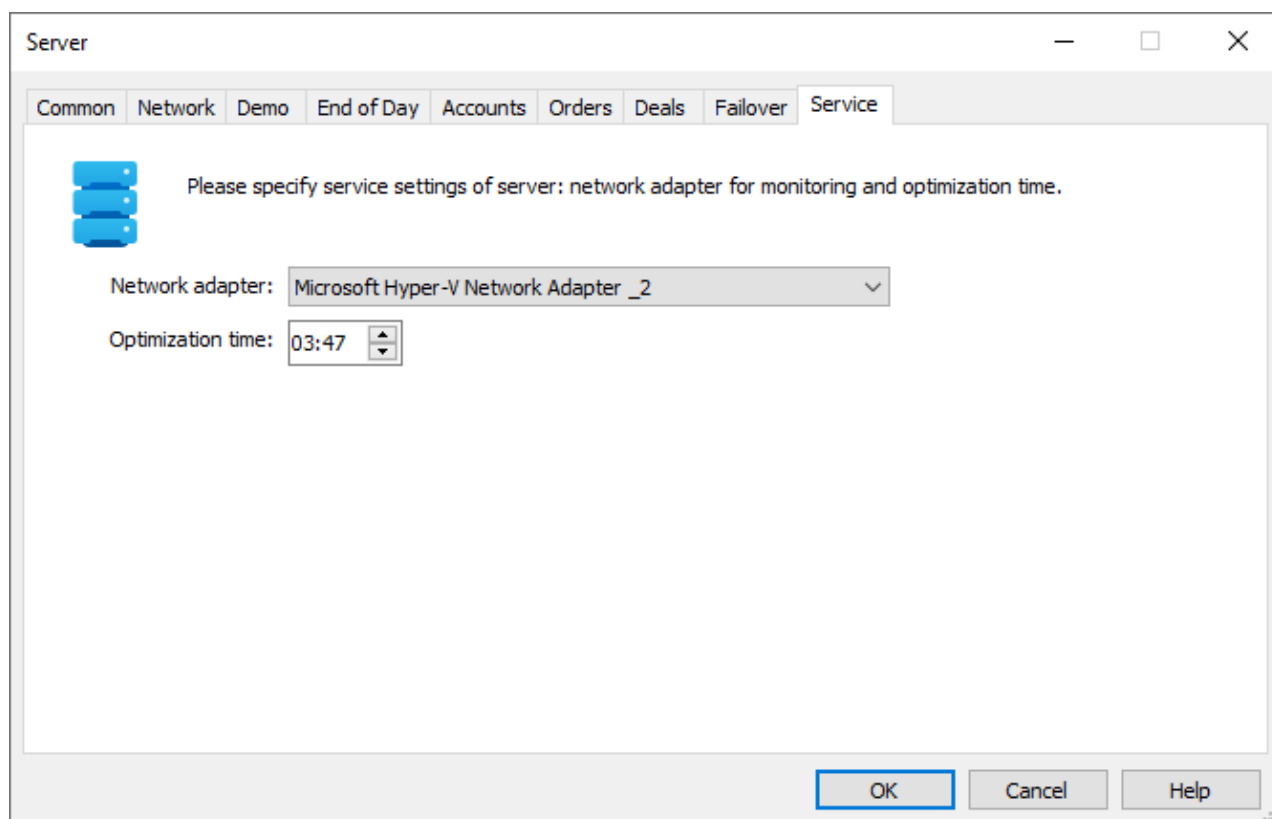
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The screenshot shows the 'Server' window with the 'Network' tab selected. The window has a title bar with standard minimize, maximize, and close buttons. Below the title bar is a tabbed interface with tabs for 'Common', 'Network', 'Demo', 'End of Day', 'Accounts', 'Orders', 'Deals', 'Failover', and 'Service'. The 'Network' tab is active, displaying a server icon and the instruction: 'Please specify network settings of the server: outgoing address, public addresses, and bindings.' Below this, a text label states: 'The following IPv4 and IPv6 addresses are used for outgoing connections from this server.' There are two dropdown menus for 'Address:'. The first dropdown shows '192.168.0.101:1954'. Below these are two list boxes. The left list box is titled 'Listen on these IP addresses' and contains two entries: '72.15.96.137:1000' and '0.0.0.0:2000'. The right list box contains one entry: 'forex.broker.com:443'. To the right of the list boxes is a checkbox labeled 'Automatically use as public addresses'. Each list box has 'Add', 'Edit', and 'Delete' buttons next to it. At the bottom right of the window are 'OK', 'Cancel', and 'Help' buttons.

In this example, address 192.168.0.101:2000 will be used for connection in the first place. If connection fails, the server will check all IP addresses associated with the forex.broker.com domain.

Service

The service settings of servers are specified at this tab.



The following parameters are specified here:

- **Network adapter** network adapter using which the monitoring of the server state will be performed;
- **Optimization time** time of optimization of the server. Different operations for increasing the performance of server and reliability of the platform are performed at that time: deletion of expired demo accounts, optimization of databases and updating of components. It is necessary to choose the time of minimum server load. For example, 03:47 by server time.

Cluster Optimization Process

A large number of operations are performed one by one during optimization. Some of them are performed daily, others are executed only on Sundays.

If the optimization start time is set to 1:00, the process will be performed as follows:

Trading Server

- 1:00 Daylight Saving Time switch is checked. If the time has switched, the server is restarted to apply the changed.
- 1:00 +/- a few seconds computer performance data is collected.

Further, no other operations are performed on all days except Sunday. The following operations are performed on Sunday:

MetaTrader 5 trading platform

- 1:10 compacting of databases and configurations (unnecessary and empty entries are deleted), deletion of expired demo accounts, server restart.
- 2:00 check for updates; if necessary the server is updated and restarted.
- 3:00 launch of compression of previous week's server logs to ZIP archives.

History Server

- 1:00 +/- a few seconds computer performance data is collected.
- 1:10 compacting of databases and configurations (unnecessary and empty entries are deleted)

Further, no other operations are performed on all days except Sunday. The following operations are performed on Sunday:

- 2:00 check for updates; if necessary the server is updated and restarted.
- 2:20 launch of tick and minute history optimization. The data of financial instruments that are not available in the platform are deleted from the \history\ directory.
- 3:00 launch of compression of previous week's server and gateway logs to ZIP archives.

Access Server

- 1:00 +/- a few seconds computer performance data is collected.
- 1:10 compacting of databases and configurations (unnecessary and empty entries are deleted)

Further, no other operations are performed on all days except Sunday. The following operations are performed on Sunday:

- 2:00 check for updates; if necessary the server is updated and restarted.
- 2:20 launch of tick and minute history optimization. The data of financial instruments that are not available in the platform are deleted from the \history\ directory.
- 3:00 launch of compression of previous week's server logs to ZIP archives.

Backup Server

- 1:00 +/- a few seconds computer performance data is collected.

Further, no other operations are performed on all days except Sunday. The following operations are performed on Sunday:

- 1:10 compacting of databases and configurations (unnecessary and empty entries are deleted)
- 2:00 check for updates; if necessary the server is updated and restarted.
- 3:00 launch of compression of previous week's server logs to ZIP archives.

In addition to the above operations, the server periodically optimizes databases loaded into RAM. These operations can be performed daily as needed and do not significantly impact the platform performance. During optimization, log entries of the following

format are recorded:

2025.01.24 09:50:03.469 TradeDeals	index compact started
2025.01.24 09:50:03.469 TradeHistory	history_2019.08.dat: index compact started
2025.01.24 09:50:03.469 TradeHistory	history_2019.08.dat: index compact finished [10635

MetaTrader 5 Trading Platform / Platform Setup / Network cluster / Configuring Servers / Trade Server

Trading Server

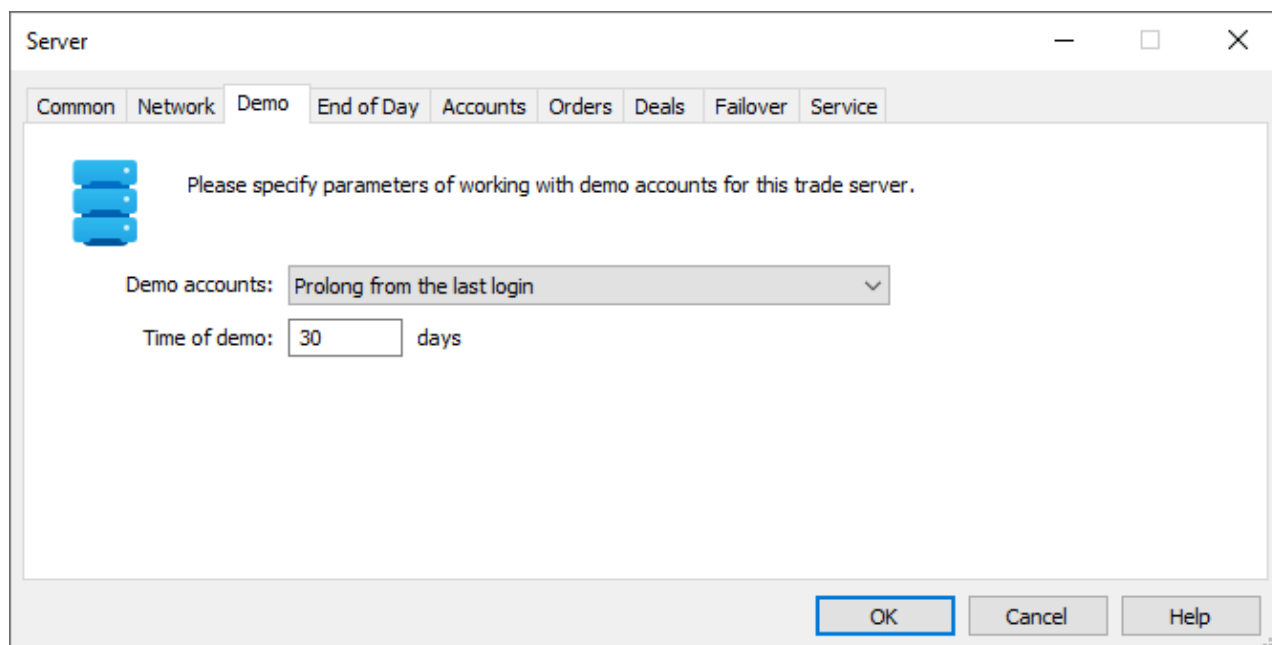
There are two types of trade servers in the system - the main one and all other additional ones. Plus to trade server functions, the main server implements system configuration management and management of other servers. The trade server performs the following functions:

- Storage and management of client records
- Authentication and authorization of client connections
- Storage and management of trade records
- Checking, management and execution of trade requests
- Management of the internal mailing system

The procedure of setting up the main and other trade servers are similar. Settings on the "Common", "Network" and "Service" tabs are the same for all server types. The trade server setup window contains four more tabs.

- Only one main trade server can exist in a system.
- The installation of the main and additional trade servers is described in the corresponding section.
- There is a number of special features of adding secondary trade servers to the system.

Demo



This tab is intended for setting up the parameters of working with demo account on the trade server:

- **Demo accounts** type of working with demo accounts. Three variants are available:
 - ♦ **Disable** opening new demo accounts via client terminals will be impossible on this server. At that demo accounts can still be opened via administrator and manager terminals and via API.
 - ♦ **Prolong from the last login** with each connection to a demo account, its expiration date will be increased by the time specified in "Time of demo" if the account hasn't expired by the time of connection. Only connections using the Master password (not the Investor one) are considered.
 - ♦ **With fixed period** demo accounts will be active for the time period specified in "Time of demo";
- **Time of demo** number of days when the demo accounts opened on this server will be active.

If you disable opening new demo accounts (select the option "Disable"), then demo accounts opened earlier will work according to the option "Prolong from the last login".

Expired demo accounts are deleted every Sunday during optimization. The following entries appear in the trade server journal during deletion:

2017.06.04 03:57:02.626 ClientBase	prepare demo accounts for clean
2017.06.04 03:57:02.626 ClientBase	'1008': demo account added to clean [80 days]
2017.06.04 03:57:02.626 ClientBase	prepare demo accounts for clean finished [1 users]
2017.06.04 03:57:02.626 TradeCenter	demo clean started [1 logins]
2017.06.04 03:57:02.626 TradeCenter	demo clean finished

The deletion of a demo account also deletes its trading operations.

End of Day

Server: Trade Main - ID 1

Common Network Demo **End of Day** Accounts Orders Deals Failover Service

Please specify trade server settings: end of day time, schedule, statements generation modes.

End of day time: 23:59

End of day schedule: Monday, Tuesday, Wednesday, Thursday, Friday

Rollover schedule: Monday, Tuesday, Wednesday, Thursday, Friday

Daily statements: End of day

Last end of day: 2022.09.22 23:59:59 Previous end of day: 2022.09.21 23:59:59

Monthly statements: Last day of month

Last end of month: 2022.08.31 23:59:59 Previous end of month: 2022.07.31 23:59:59

OK Cancel Help

This tab is intended for setting up the end of day time and the generation of daily and monthly statements.

- **End of day time** time of the trading day end, when the generation of statements, charging of swaps, interest rate and commissions is performed. The execution of the procedures is started on the 59th second of the specified time.
- **End of day schedule** in this field, specify days when the operations connected with the end of a trading day are performed. The generation of reports, charging of swaps, interest rate and commissions is performed on these days only. This option does not affect the generation of monthly statements.
- **Rollover schedule** by default, rollover calculations are not performed on Saturday and Sunday (even if these days are specified in the "End of day" parameter). When work days are shifted to weekend, you may adjust the operation of the trading platform accordingly. For example, if a work day is shifted from Monday to Saturday, you can enable rollover calculation on Saturday. However, note that traders' rollover will be calculated eight times for that week, instead of seven: five simple swaps and one three-day swap. To compensate for the extra rollover, explicitly disable rollover calculation on Monday. Otherwise, rollover will be charged even if you set up a holiday on that day and disable trading in the server schedule. On the next day (Tuesday), rollover settings should be reset to the original state. In addition to enabling rollover calculation, make sure to enable "End of Day" for that day. Otherwise, rollover will not be charged.
- **Daily statements** time when daily statements are generated: at the end of a trade day (before swaps, interest rate and commissions are charged) or at the beginning of a trade day (after swaps, interest rate and commissions are charged).

- **Last end of day** the date and time of the last end of day. When generating the last daily statement, the server uses trading data beginning from the specified date.
- **Previous end of day** the date and time of the previous end of day. Daily statements include a client's trading status for the previous trading day. The server uses the specified date to calculate it.
- **Monthly statements** month closing date on which the monthly reports should be generated: the last day of the current month of the first day of the next month.
- **Last end of month** the date and time of the last end of month. When generating the last monthly statement, the server uses trading data beginning from the specified date.
- **Previous end of month** the date and time of the previous end of month. Daily statements include a client's trading status for the previous trading month. The server uses the specified date to calculate it.

It is strongly not recommended to change the dates of the last and the previous ends of day and month. Edit them only in case of emergency (for example, if the server emergency shutdown during the generation of daily/monthly statements has led to incorrect dates).

The order of end-of-day/month service operations

During the end-of-day, clients and managers are forbidden to perform any trading operations, including balance operations. On any attempt, the server will return the "Market closed" error.

The order of service operations depends on the report generation mode: at the end or at the beginning of the day. The performed actions also depend on whether the end of the trading day (as determined by the "End of day" parameter) and end of month (as determined by the actual end of month) fall on this day.

When generating reports at the end of the day:

Condition	Operation
If there is end of the trading day	1. Beginning of the 'end of day' at the time specified in the " <u>End of day</u> " parameter under server settings
If there is end of the month	2. Beginning of the 'end of month' at the time specified in the " <u>End of day</u> " parameter under server settings
	3. Beginning of operations for each group:
If there is end of the trading day	4. Calculation and settlement of commissions (including agent commission) with the " <u>Daily</u> " calculation mode
If there is end of the month	5. Calculation and settlement of commissions (including agent commission) with the " <u>Monthly</u> " calculation mode
If there is end of the trading day	6. Calculation of interest on available funds and saving them in the client record

MetaTrader 5 trading platform

Condition	Operation
If there is end of the month	7. Calculation and of the accumulated interest on available funds and depositing it to the balance
If there is end of the trading day	8. Generation of daily reports
If there is end of the trading day	9. Updating of the previous day balances (used in daily/monthly reports)
If there is end of the month	10. Generation of monthly reports
If there is end of the month	11. Update of previous month balances (used in daily/monthly reports)
If there is end of the trading day	12. Calculation of swaps
	13. End of operations for each group
If there is end of the trading day	14. End of "end of day"
If there is end of the month	15. End of "end of month"

When generating reports at the beginning of the day:

Condition	Operation
If there is end of the trading day	1. Beginning of the 'end of day' at the time specified in the " <u>End of day</u> " parameter under server settings
If there is end of the month	2. Beginning of the 'end of month' at the time specified in the " <u>End of day</u> " parameter under server settings
	3. Beginning of operations for each group:
If there is end of the trading day	4. Calculation and settlement of commissions (including agent commission) with the " <u>Daily</u> " calculation mode
If there is end of the month	5. Calculation and settlement of commissions (including agent commission) with the " <u>Monthly</u> " calculation mode
If there is end of the trading day	6. Calculation of interest on available funds and saving them in the client record
If there is end of the month	7. Calculation and of the accumulated interest on available funds and depositing it to the balance
If there is end of the trading day	8. Calculation of swaps
	9. Generation of daily reports

MetaTrader 5 trading platform

Condition	Operation
If there is end of the trading day	
If there is end of the trading day	10. Updating of the previous day balances (used in daily/monthly reports)
If there is end of the month	11. Generation of monthly reports
If there is end of the month	12. Update of previous month balances (used in daily/monthly reports)
	13. End of operations for each group
If there is end of the trading day	14. End of "end of day"
If there is end of the month	15. End of "end of month"

Accounts

Server: Trade Main - ID 1

Common Network Demo End of Day **Accounts** Orders Deals Failover Service

Please specify range of accounts that will be processed by this trade server.

From	To
 1000	1000000
 3000000	3999999

Add Edit Delete

OK Cancel Help

MetaTrader 5 trading platform

Here you set up the range from which logins will be generated at the opening of new accounts. In order to add a range, press "Add". After that a new line will appear in the window. In "From" column, specify the first account that will be serviced in this range, and specify the last one in "To". In order to modify an already existing range, double click on the necessary field or select it and press "Edit". To delete a selected range, press "Delete".

- Account ranges on different trade servers must not coincide. We strongly recommend to strictly plan the distribution of accounts between servers.
- The range indication is obligatory. Otherwise, it will be impossible to create accounts on the server.
- The last value of the range must be greater by 1 000 than the highest number of account already allocated on the server.
- The ranges should not be changed during trading time. After changing them, restart the platform.

Orders

Server: Trade Main - ID 1

Common Network Demo End of Day Accounts **Orders** Deals Failover Service

Please specify range of orders that will be processed by this trade server.

From	To
1	100000000
5000000000	9000000000

Add Edit Delete

OK Cancel Help

MetaTrader 5 trading platform

Here you should set up the range, from which numbers of orders set on this trade server will be allocated. In order to add a range, press "Add". After that, a new line will appear in the window. In "From" column, specify the first order that will be serviced in this range, and specify the last one in "To". In order to modify an already existing range, double click on the necessary field or select it and press "Edit". To delete a selected range, press "Delete".

- Orders ranges on different trade servers must not coincide. We strongly recommend to strictly plan the distribution of orders between servers.
- The range indication is obligatory. Otherwise, it will be impossible to set orders on the server.
- The last value of the range must be greater by 1 000 than the highest number of order already placed on the server.
- The range of orders is also used for assigning tickets to positions.
- The ranges should not be changed during trading time. After changing them, restart the platform.

Deals

Server: Trade Main - ID 1

Common Network Demo End of Day Accounts Orders Deals Fails over Service

Please specify range of deals that will be processed by this trade server.

From	To
1	1000000000
5000000000	9000000000

Add Edit Delete

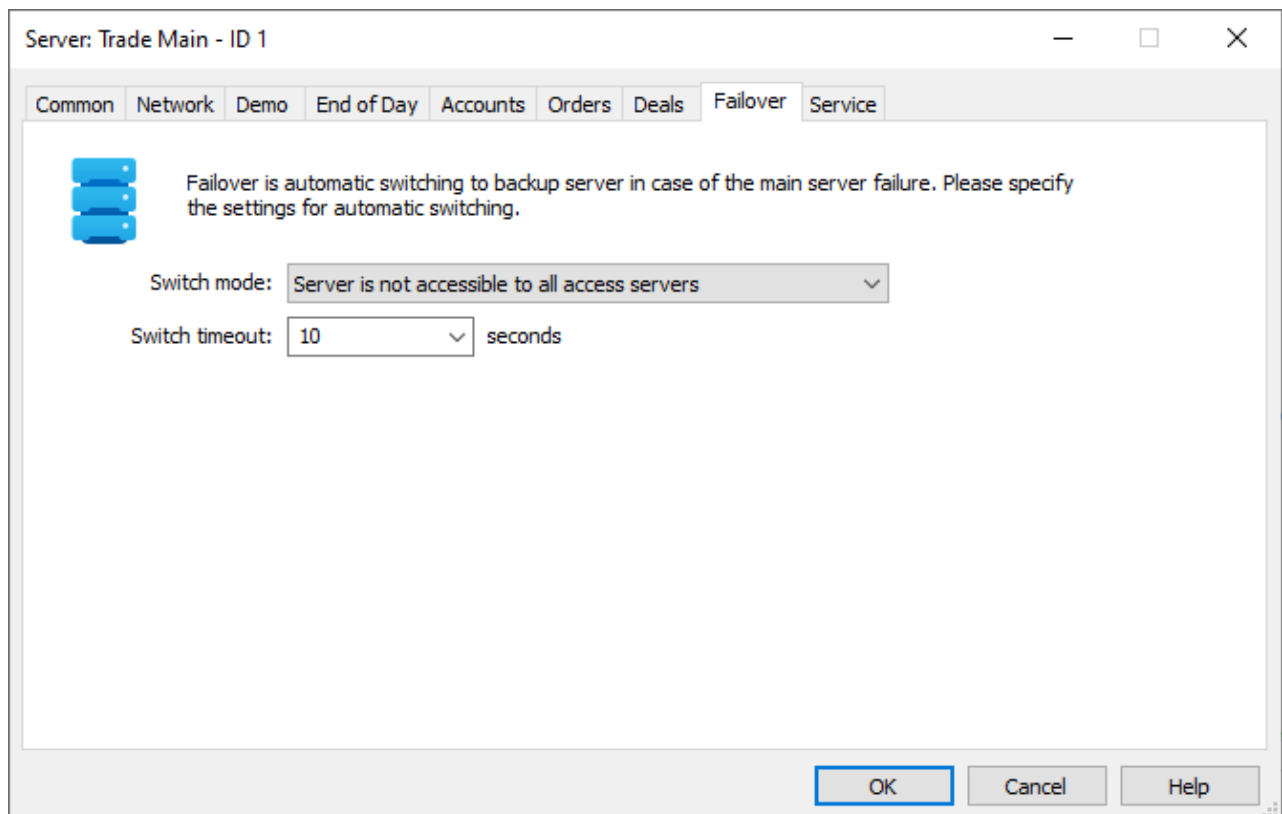
OK Cancel Help

MetaTrader 5 trading platform

Here you should set up the range, from which numbers of deals executed on this trade server will be set. In order to add a range, press "Add". After that, a new line will appear in the window. In "From" column, specify the first deal that will be serviced in this range, and specify the last one in "To". In order to modify an already existing range, double click on the necessary field or select it and press "Edit". To delete a selected range, press "Delete".

- Deals ranges on different trade servers must not coincide. We strongly recommend to strictly plan the distribution of orders between servers.
- The deals range indication is obligatory. Otherwise, it will be impossible to set orders on the server.
- The last value of the range must be greater by 1 000 than the highest number of deal already performed on the server.
- The ranges should not be changed during trading time. After changing them, restart the platform.

Failover



Here you can specify the parameters of the automatic switching to the backup server in case the current one fails. The necessity to switch to the backup server is defined by the monitoring ("witness") servers. The backup server itself and access servers (with monitoring mode enabled) act as the monitoring ones. The backup server monitors the availability of the master server in real time mode and checks if it is available for the access servers as well.

- **Switch mode** mode of switching to the backup server:
 - ♦ **Off** automatic switch to the backup server is disabled.
 - ♦ **Server is not accessible to most access servers** the number of the monitoring servers unable to access the master server should exceed the ones able to access it at least by one for the switch to occur.
 - ♦ **Server is not accessible to all access servers** the master server should be unavailable for all monitoring servers for the switch to occur.
- **Switch timeout** here you can specify the time (in seconds) during which the server should be unavailable for monitoring servers to start switching to the back-up server. Also, after this time period, other platform components start their attempts to connect to the access points of the current backup server (trying to connect to it as to the main one).

More detailed information about switching to the backup server can be found in the separate section.

To complete creating or editing a trade server press "OK". If you press "Cancel", the window will be closed, while settings will not be saved.

MetaTrader 5 Trading Platform / Platform Setup / Network cluster / Configuring Servers / History Server

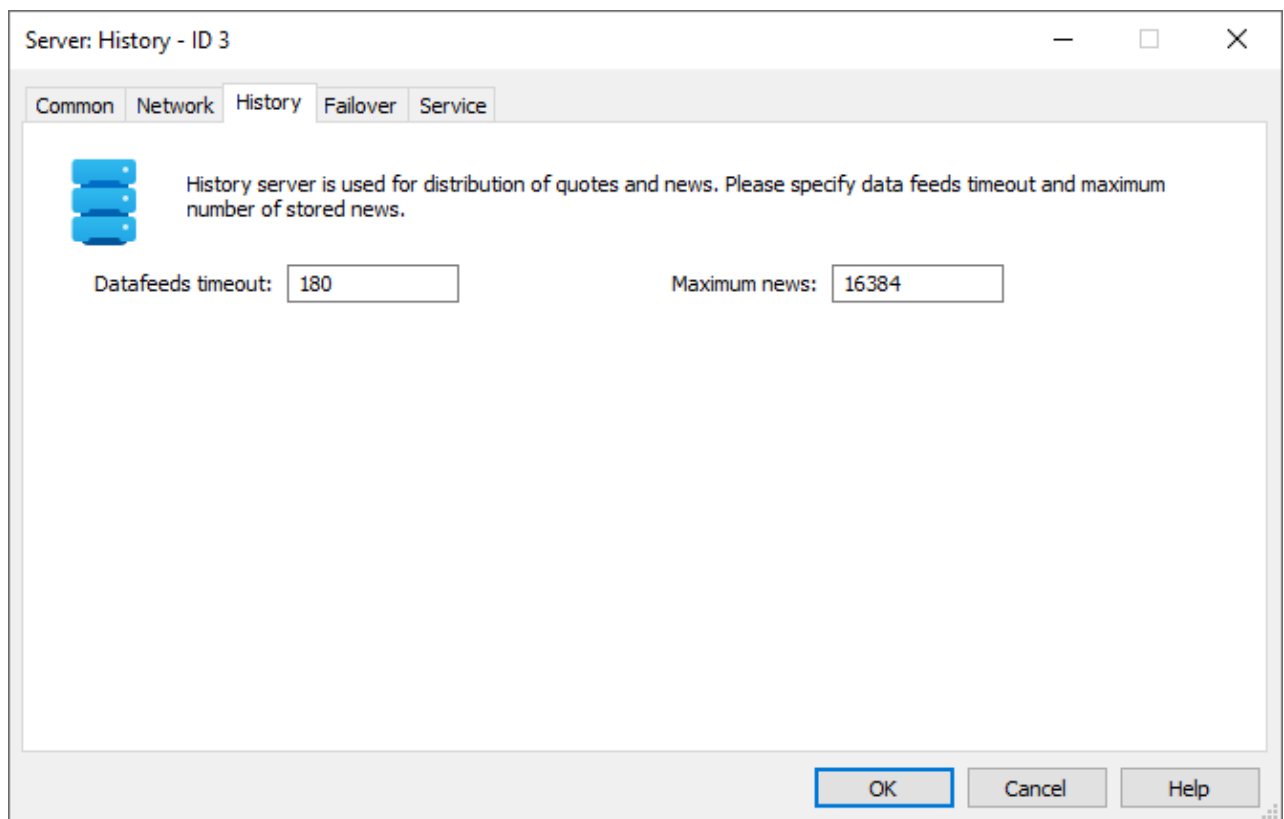
History Server

History server is a part of the online trading platform. It is used for:

- Receiving, filtering and packing price and news data
- Storing and providing price history as minute bars and ticks
- Storing and providing the price thread
- Receiving, checking and distributing Live Updates among MetaTrader 5 servers

Information that comes from data feeds is passed to the history server and is then resent to access servers and trade servers. Settings on the "Common", "Network" and "Service" tabs are the same for all the server types. The history server setup window contains one more tab - History.

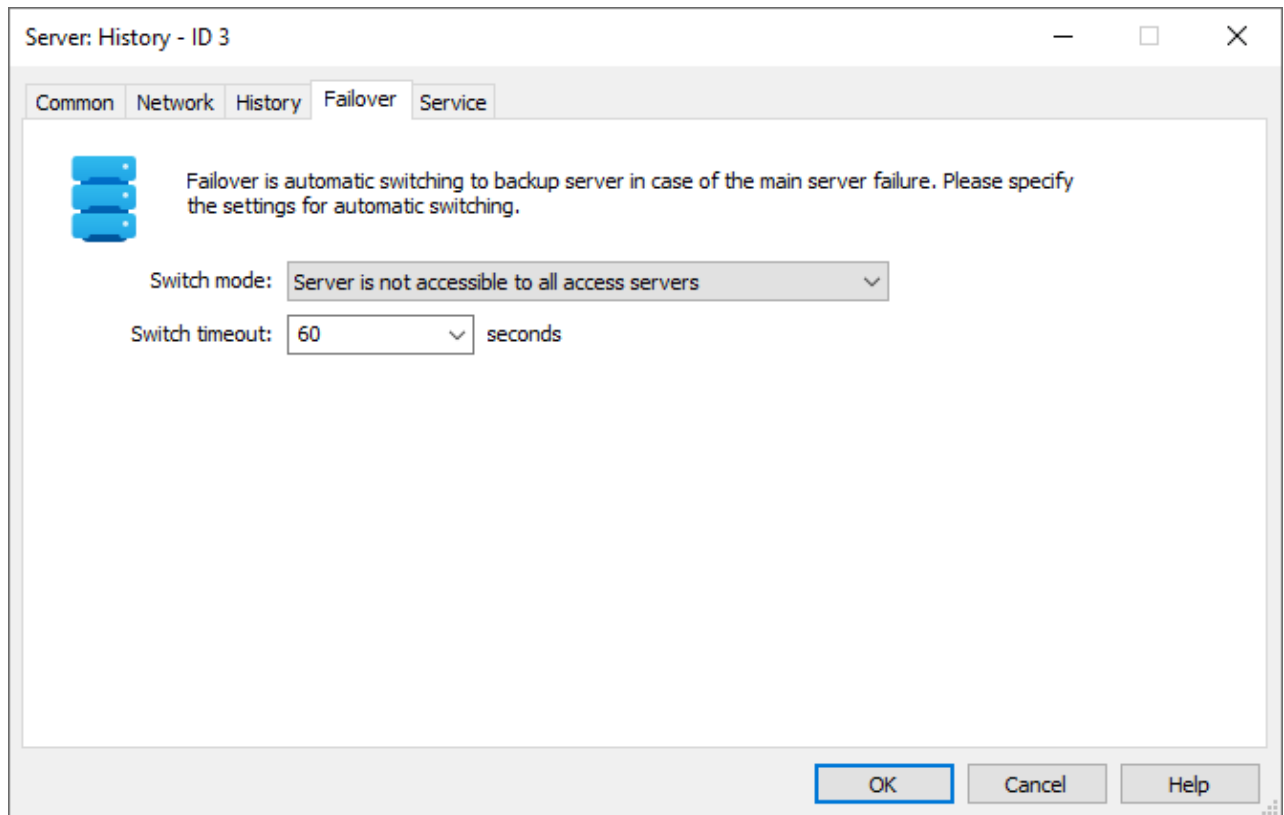
History



The following parameters are set up on this tab:

- **Datafeeds timeout** the period of time (in seconds), within which a server is waiting for history data (quotes and news) from a data feed or a gateway. If no data have been received within this period, the server switches over to another data feed. Minimum timeout is 10 seconds. Changing this setting requires a server restart.
- **Maximum news** the maximum number of news messages that can be stored on the history server.

Failover



Here you can specify the parameters of the automatic switching to the backup server in case the current one fails. The necessity to switch to the backup server is defined by the monitoring ("witness") servers. The backup server itself and access servers (with monitoring mode enabled) act as the monitoring ones. The backup server monitors the availability of the master server in real time mode and checks if it is available for the access servers as well.

- **Switch mode** mode of switching to the backup server:
 - ♦ **Off** automatic switch to the backup server is disabled.
 - ♦ **Server is not accessible to most access servers** the number of the monitoring servers unable to access the master server should exceed the ones able to access it at least by one for the switch to occur.
 - ♦ **Server is not accessible to all access servers** the master server should be unavailable for all monitoring servers for the switch to occur.
- **Switch timeout** here you can specify the time (in seconds) during which the server should be unavailable for monitoring servers to start switching to the back-up server. Also, after this time period, other platform components start their attempts to connect to the access points of the current backup server (trying to connect to it as to the main one).

More detailed information about switching to the backup server can be found in the separate section.

To complete creation or editing of a history server, press "OK". If you press "Cancel", the window will be closed, while changes won't be saved.

MetaTrader 5 Trading Platform / Platform Setup / Network cluster /
Configuring Servers / Access Server

Access Server

Access servers are used as intermediate links between clients and a trade server. The access server performs the following functions:

- Processing of incoming client connections;
- Packaging authorization requests and passing them to a trade server;
- Checking activity of client connections, protecting the trade server from attacks and overloading;
- Saving history data, depth of market and news, providing them to clients and thus reducing the load to the history server;
- Providing client terminals with LiveUpdate.

Settings on the "Common", "Network" and "Service" tabs are the same for all the server types. The access server setup window contains three more tabs.

To quickly add an access point in a high trading activity region, order Access Server Hosting from MetaQuotes directly via the Administrator terminal. The new point will become available within 30 seconds.

Access

Server: Access Point 2 - ID 6

Common Network **Access** Permissions Servers Service

Access server is used as access point for connection of clients to trade server. Please specify its priority and antiflood control settings.

Priority: 0

☒ Enable antiflood control

Connections: 100 Errors: 10

Maximum news: 16384

☒ Use this server for monitoring the cluster and failover

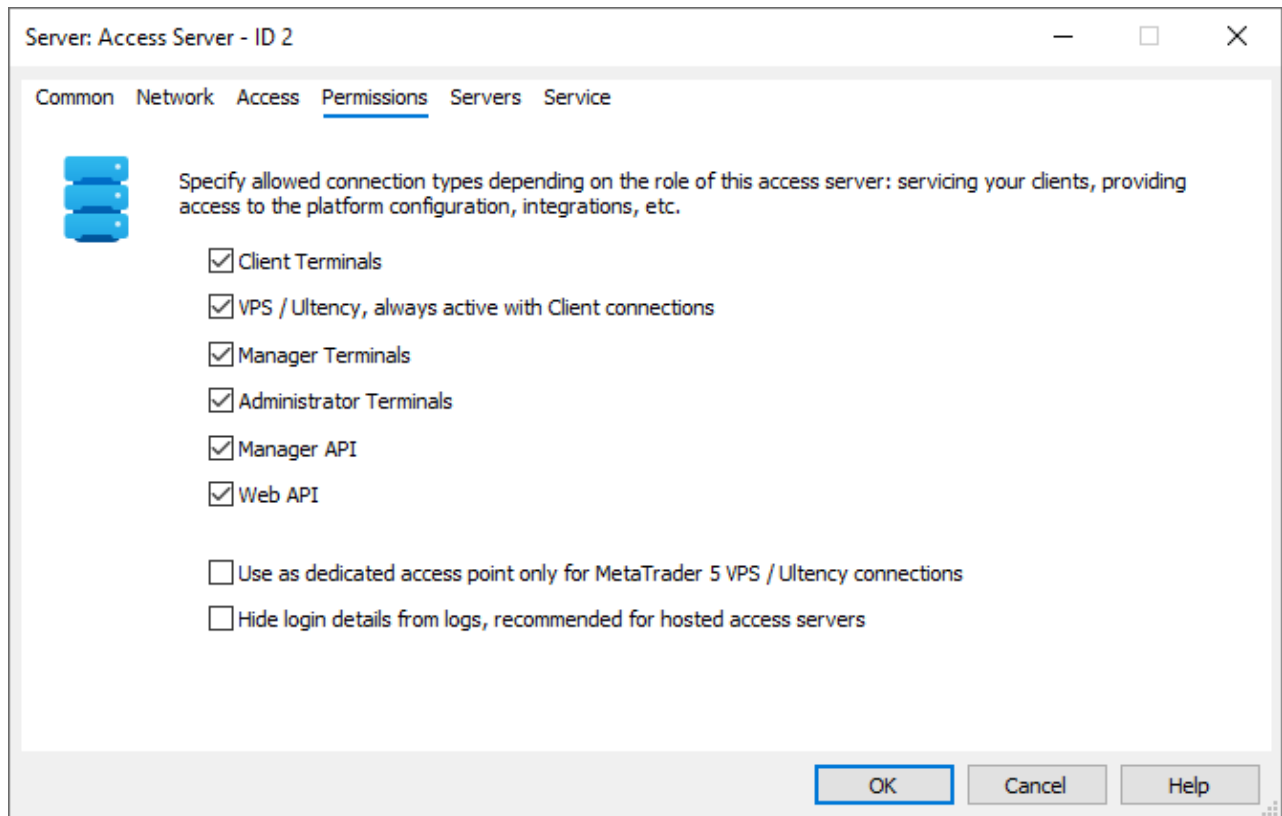
OK Cancel Help

The following parameters are set up on the "Access" tab:

- **Priority** is the basic priority of the access server. A server can have basic priority from 0 to 15. The access server preference for client connections is calculated based on the current priority and quality of connection to the server. The current priority depends on the basic priority and the number of current connections to the server. The lower the priority number is, the higher its preference for client connections is. Find more details about the calculation of the current priority in a [separate section](#).
- **Enable antiflood control** enable [defense from DoS attacks](#). The server analyses the network activity of all clients in real time mode by selecting connections with most frequent operations that come from one IP address. If such an attack is detected, the attacking IP address is moved to a temporary list of banned addresses. At a repeated attack, the ban period is extended. This mechanism allows to easily block attack attempts. The control has the following parameters:
 - **Connections** the maximum number of server connections from one IP address within a certain time period, after which the address will be temporarily banned.
 - **Errors** number of unsuccessful connection attempts, after which the address will be temporarily banned.
 - **Maximum news** maximum number of news messages that are stored on the access server.
 - **Use this server for monitoring the cluster and failover** when enabled, the backup servers use this server to check the availability of the main (backed up) ones. This is used to automatically switch to the backup server if the main one fails. More detailed information can be found in the [appropriate section](#).

- Enabling of the antiflood control is required.
- If you need to create a virtual, temporarily not existing servers, use the "Idle" priority. Such servers are standby ones. They are connected to only if all other servers do not operate.

Permissions



Here you can set up types of connections allowed for the access server:

- Client terminals – desktop, mobile and web.
- Terminals operating on a built-in VPS, as well as Ultency servers.
- Manager terminals.
- Administrator terminals.
- Applications developed using Manager API.
- Services operating through Web API.

Creating fast dedicated access points for built-in VPS

You can create fast, dedicated access points to connect VPS terminals. This will significantly improve the high frequency trading capabilities on your server. By using a widespread distributed network and automatically identifying the server closest to the broker, VPS terminals achieve exceptionally low latency when connecting. Setting up a dedicated access server for these terminals will further enhance speed and enable traders to engage in true high-frequency trading.

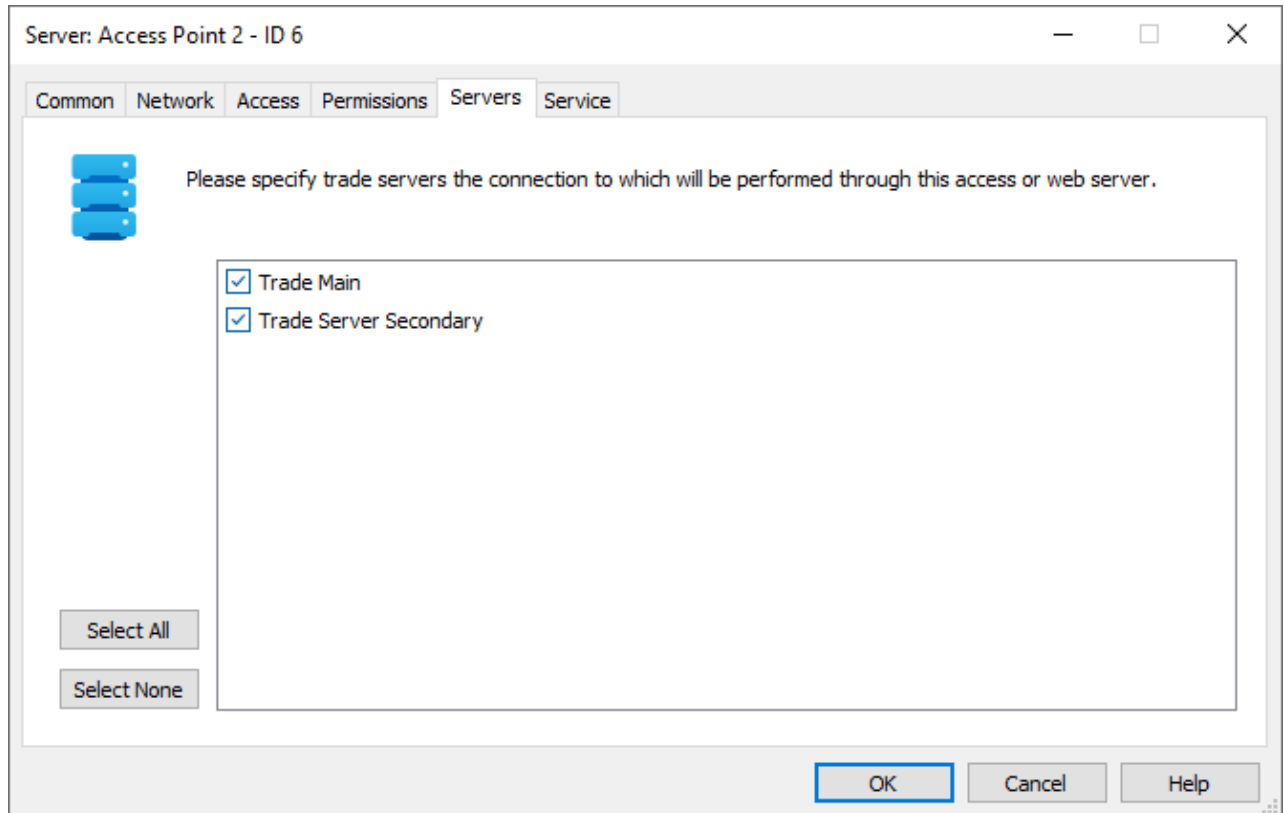
To set up a dedicated access point:

- Restrict standard client connections, allowing connections only for VPS / Ultency.
- Enable the "Use as a dedicated access point only for MetaTrader 5 VPS / Ultency" option. This prevents the access server from appearing to client terminals, while physical connections remain technically possible. As a result, standard client terminals will not attempt to connect since they will not know the server address.

Additional security for hosted servers

Hosting access servers is entirely secure. All transmitted data is encrypted, and the access server does not store account databases, passwords, or any other trading account information. For added protection, you can enable the "Hide login details from logs" option. This prevents account numbers from being displayed in the access server log. If necessary, this option can be temporarily disabled for troubleshooting, but we recommend keeping it enabled for maximum security.

Servers



MetaTrader 5 trading platform

On this tab you should select trade servers to connect to through the selected access server. To allow connections to certain servers, tick  near them. You can select all or select none of servers using the corresponding buttons.

MetaTrader 5 Trading Platform / Platform Setup / Network cluster / Configuring Servers / Backup Server

Backup Server

Backup servers allow creating backup copies in case of failure of a history or trade servers. The backup server performs the following functions:

- It provides real-time backup of a trade server and a history server. Each server is associated with one or more separate backup server instances which can replace it at any time.
- It creates backup copies of all data bases every day, and regularly creates backups of client and trade databases.
- When a history server is backed up it creates real time backups of data required for correct restoring of gateways: custom settings that can be stored (at developer's discretion) in the settings.dat file in a gateway work folder, and trade executions database.
- They provide automatic failover in case the primary server becomes unavailable.
- Using backup servers, you can easily migrate servers to new hardware.
- Backup servers enable you to quickly restore server operation in the manual mode even if the main trade server is unavailable, and connection to the platform via MetaTrader 5 Administrator is not possible.
- They replicate information to database managed by MySQL, MariaDB, PostgreSQL, Firebird, MSSQL or Oracle.

The process of restoring servers is described in a <u>separate section</u> .
--

Settings on the "Common", "Network" and "Service" tabs are similar for all types. The backup server setup window contains three more tabs.

Backup

Server

Common Network **Backup** SQL Export SQL Options Folders Service

Backup server is used for backup of trade and history servers. Please specify the name of server to be reserved and backup settings.

Master server: Trade Main, 1

☒ Use this backup server for failover

☒ Synchronize journals with master server

☒ Enable backups

Backups path: F:\MT5Backups\Main

Full backup time: 00:17 Additional backups: Every hour

Keep backups: 1 day

OK Cancel Help

The following parameters are available on this tab:

- **Master server** name of the server whose backups should be created, and the login (ID) of this server;
- **Use this backup server for failover** the trading platform features an automated failover system, which switches to a backup server in case of main server failure. By default, the platform can switch to any of existing backup servers. If a certain backup server should not be used by the automatic failover system, disable this option for that server. This may be needed if the backup server is used only for exporting data to SQL database. You can disable the "Enable backups" option for such servers - this will save resources required for the creation of backup copies in a disk.
The option only disables switching to that server in the automated mode. If necessary, you can switch to this backup server in manual mode.
- **Synchronize journals with master server** the backup server is a copy of the backed up one since it stores the same data as the main server, including log files. If you only use the backup server for data export to SQL, you can disable log synchronization with the primary server to save disk space.
- **Enable backups** apart from synchronizing with the main server in real time, the backup server creates database file copies on the disk on a daily basis. Such copies allow restoring the main server status on a specific day. Disable this option to avoid creating file copies. It does not affect data synchronization with the main server in real time. The backup server will still remain a full copy of the main one. Creating file copies can be disabled if the server is used only for exporting data to SQL.
- **Backups path** the path on the computer where the file copies of the databases will be saved. Do not specify the trading platform installation directory here.
- **Backup time** time of creating file copies.

- **Additional backups** frequency of creating additional file copies. By default, file copies are created every 24 hours, but you can set this to happen more often – once per hour or once per 4 hours. Keep in mind that this will require more server resources and more disk space. This option is only available for the backup of trading servers; backup copies of history servers are always created once every 24 hours.
- **Keep backups** file copies storage period. All backup copies older than a period specified in this field are automatically deleted. The parameter is not available for backup servers replicating history servers. They keep only one last state of price databases. It is impossible to store several states for different days due to the large size of the databases.
- **Enable ticks backups** copies of ticks are created once a day during a full backup. The backup is performed incrementally – the first copy stores the full data cast, while subsequent ones contain only the changes relative to the first copy, which significantly reduces the volume of files. The copies are stored in the \ticks directory where they are arranged in subdirectories by days, for example, ticks\2015.11.17, ticks\2015.11.18, etc. By default, the option is disabled since tick data takes much space.

For manual tick data recovery, copy the folders with backup copies to the new history server consistently beginning with the oldest. For example, suppose that the backup server has the folders: ticks\2015.11.17, ticks\2015.11.18, and ticks\2015.11.19. First, copy the contents of ticks\2015.11.17 to the \ticks directory of the history server. Then, overwrite the contents with the data from the ticks\2015.11.18 folder, and after doing that - by the contents from the ticks\2015.11.19 folder.

In addition to creating periodic and full databases backups, the backup servers make starting database copies. Such copies are created before the initial synchronization with the primary server in case its database is damaged. Starting backups have the *start postfix. Initial synchronization is performed during the first connection to the primary server after the backup server start or restart. After the initial synchronization, the backup server will receive data on changes made in the primary server, in real time.

SQL Export

Server

Common Network Backup **SQL Export** SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: FireBird 2.x - 3.x

Server: 192.168.0.100:500

Login: root Password: ●●●●●●●●

Data Folder: c:\FB\mt5.fdb

OK Cancel Help

A backup server can export its databases to an SQL database in real-time mode. Considering that data backup is also performed in real-time mode, all data stored in the backup server is sent to SQL database almost immediately.

The following database management systems (DBMS) can be used to export data:

- Microsoft SQL Server 2005, 2008, 2008 R2, 2012, 2014, 2016, 2017, 2019, 2022
- MySQL 5.1, 5.5, 5.6, 5.7, 8.0, 8.1
- MariaDB (all versions)
- Oracle 11g, 12c, 18c, 19c, 21c, 23ai
- PostgreSQL 8.4 - 17

The backup server supports the export of the following data:

- configurations: trading symbols and groups of clients
- client bases: total client base and client trading account status
- trading bases: orders, deals, positions
- trading history bases: orders and deals history

- A full description of the exported data is in the "SQL Export" section.
- If you use a backup server only for exporting data to SQL, disable the "Use the backup server for failover" option. Otherwise the server can switch to the primary server mode and will stop exporting data. You may also disable the "Enable backups" option to save resources required for the creation of backup copies in a disk.

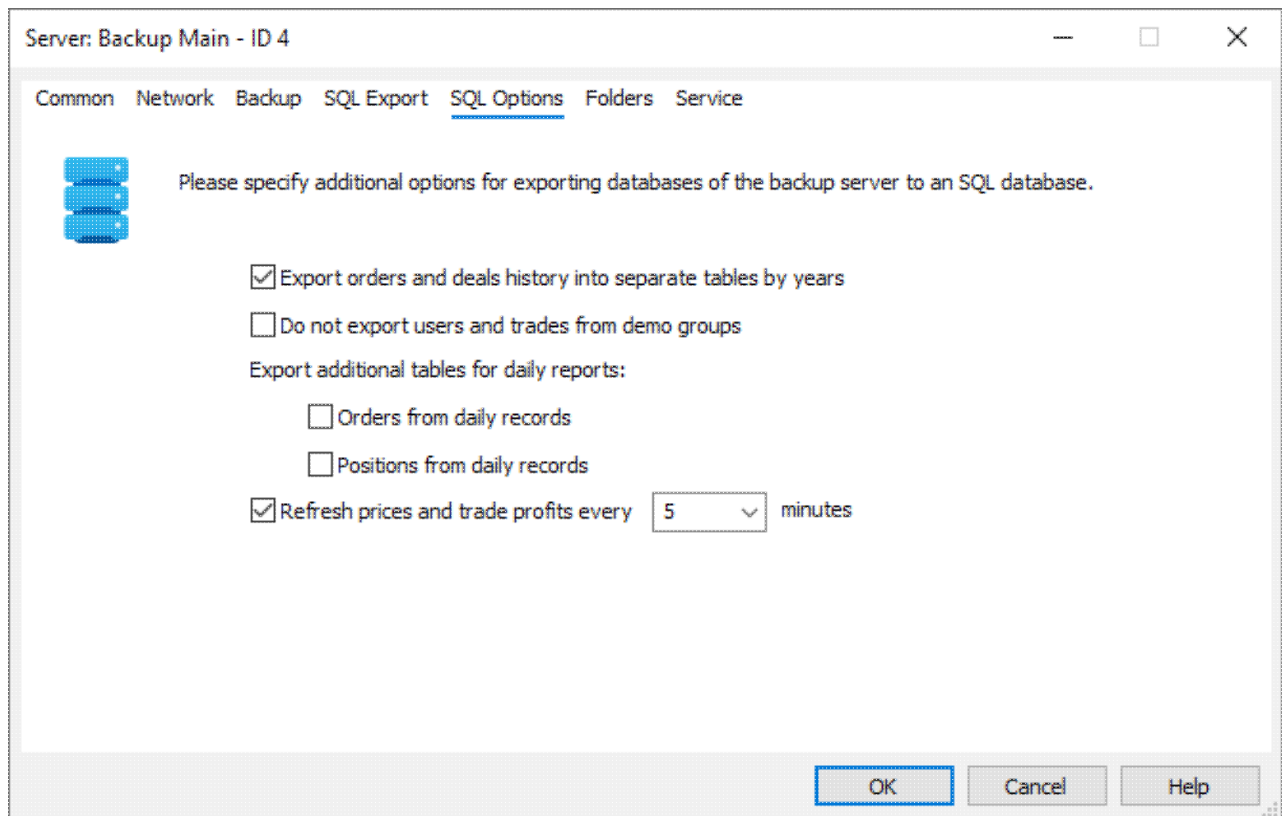
The following parameters should be specified for export setup:

MetaTrader 5 trading platform

- **Type** type of DBMS, to which data will be exported;
- **Server** address of the server with installed DBMS. Address format may vary for different DBMS. Examples for each type of DBMS are given below;
- **Login** login for connection to a database;
- **Password** password for connection to a database;
- **Data Folder** database name in DBMS, to which the data will be exported. A database should be created in advance. Examples of how to set up this parameter are given below.

Data export process can be monitored via a backup server Journal using SQL keyword to request the entries.

SQL Settings



Data export to a database using SQL can be configured on this tab.

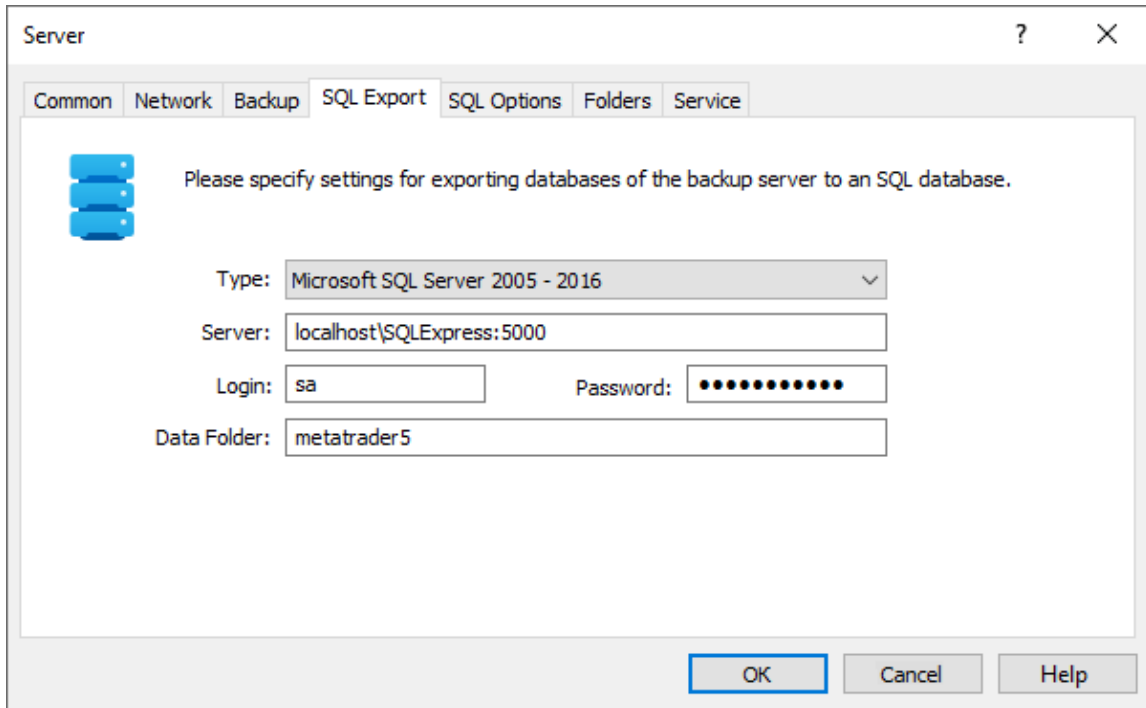
- **Export history orders and deals into separate tables by years** by default, historical orders, deals and daily reports are exported into three separate tables. However, over time, the platform's databases can grow significantly, causing the tables in the SQL database to grow accordingly. If this option is enabled, the data will be additionally split into separate tables by years. You will have separate tables for each year for historical orders, trades and daily reports. This will reduce the load on the server by reducing the table sizes, as well as speed up queries to the database. When enabling this option, modify your SQL queries against the database accordingly. For more information about exported tables, please see "[SQL Export](#)".
- **Export additional tables for daily reports** [daily reports](#) store information about the end-of-day status of the trader's open positions and pending orders. You can export this data to an SQL database. Please note that the export of these tables will only begin after the relevant options are enabled. For previously exported daily data, position and order information will not be updated. If you need to export all data, save data from the custom columns (if any) of the mt5_daily tables and remove these tables from the SQL database. In this case, the backup server will re-export all daily data, including position and order information. Full data replication may take a considerable amount of time, so we recommend performing this operation only on weekends.
- **Do not export users and trades from demo groups** when the option is enabled, accounts from [demo groups](#), as well as trade operations (orders, deals and positions) of these accounts will not be exported to SQL base. This helps to reduce the load on database and increase the speed of queries.
- **Refresh prices and trade profits every N minutes** to reduce the consumption of resources you can limit the rate of exporting the price and profit data. The data include [symbol prices](#), current prices of [positions](#) and [orders](#), floating profit of positions and almost all information in the table of the [state of trade account](#).

Click "OK" to complete a backup server creation or editing. If you click "Cancel", the window will be closed without saving changes.

Sample Data Export Settings for Various DBMS Types

Microsoft SQL Server

MetaTrader 5 trading platform



Server

Common Network Backup **SQL Export** SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: Microsoft SQL Server 2005 - 2016

Server: localhost\SQLEXPRESS:5000

Login: sa Password: ●●●●●●●●

Data Folder: metatrader5

OK Cancel Help

Sample parameter settings when configuring data export to Microsoft SQL Server:

- **Type** Microsoft SQL Server 2005 - 2022.
- **Server** address of the server (IP address or domain name) with the installed DBMS, MSSQL copy name and a port for connection (optional). For example, localhost\SQLEXPRESS:5000, 192.168.0.1\SQLEXPRESS.
- **Login** sa. Account used for connection should have the rights to create/edit/delete tables and data.
- **Password** some_password.
- **Data Folder** specific name of the database where the data is exported. For example, metatrader5. The database should be created in advance.

MySQL / MariaDB

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The screenshot shows the 'Server' dialog box with the 'SQL Export' tab selected. The dialog contains the following fields and values:

- Type: MySQL 5.x & MariaDB 5.x - 10.x
- Server: 192.168.0.100:500
- Login: sysdba
- Password: (masked with dots)
- Data Folder: mt5

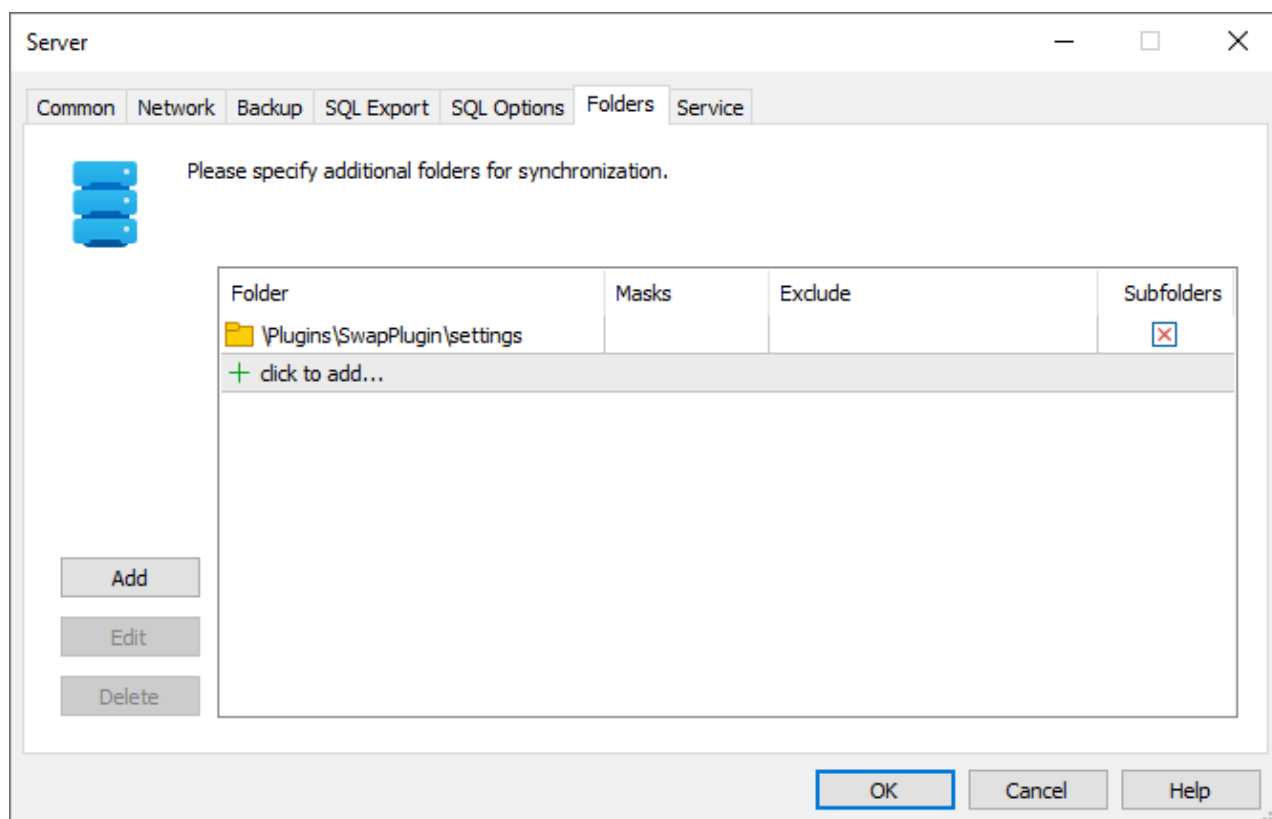
Buttons at the bottom: OK, Cancel, Help.

Sample parameter settings when configuring data export to MySQL (similar parameters are used for MariaDB):

- **Type** MySQL 5.x & MariaDB 5.x - 11.x.
- **Server** address of the server (IP address or domain name) with the installed DBMS and a port for connection (optional). For example, 192.168.0.100:5000, sqlserver.company.net.
- **Login** sysdba. Account used for connection should have the rights to create/edit/delete tables and data;
- **Password** some_password;
- **Data Folder** name of the database (schema) where the data is exported. For example, metatrader5. The database should be created in advance.

Folders

The server allows synchronizing not only the standard files and the platform databases, but also any custom directories and files as well. For example, many data feeds and plugins require LIC, DAT and other files for their operation. These files should also be backed up.



Specify custom directories for backup on the Folders tab.

- **Folder** path to the folder files from which should be backed up. Paths are specified relative to the installation directory of the primary server.
- **Masks** comma-separated list of copied files. File names can be defined using masks. If the list is specified, only listed files matching the mask are copied. Otherwise, all files from the directory will be copied.
- **Exclude** comma-separated list of ignored files. File names can be defined using masks. If the list is specified, only files from the list and the ones matching the mask will be skipped.
- **Subfolders** whether or not to back up subdirectories in the specified folder.

Example:

```
Folder=plugins\\HistoryPlugin
Masks=EURUSD*.*,*.lic,*.dat;
Exclude=*.log,*.tmp
```

.lic and .dat files, as well as files beginning with EURUSD are synchronized in the <server_installation_directory>\\plugins\\HistoryPlugin folder, while .log and .tmp files are ignored.

MetaTrader 5 Trading Platform / Platform Setup / Network cluster / Hosted Access Servers

Secure MetaTrader 5 Access Server hosting from MetaQuotes

Access points in regions with high trading activity can improve your customer service quality by ensuring fast and reliable connection to the platform. You can deploy access servers easily in a much more cost-efficient manner by ordering their hosting from our company.

We have been providing VPS for traders over many years, gaining much experience along the way. The huge popularity of the service (thousands of orders every month) confirms the validity of our approach, which is focused on simple renting and hosting management. The service is available directly from the client terminals, while the purchase and environment migration take just a few clicks. Traders do not have to worry about anything else.

The same simple and convenient solution is available to brokerage companies.

We have already made all the necessary arrangements and have signed contracts with the largest data centers. You can place your order via the Administrator terminal. Your traders will be able to connect to your platform through a new access point within 30 seconds.

Service advantages

- Hi-end powerful servers with 10 Gbps transfer speeds.
- The auto launch of a new access point takes about 10 seconds.
- Monthly subscription which can be terminated any time. You do not need to plan things ahead and purchase the hosting service for the period of one year or longer. If the activity of your traders shifts from one region to another, just cancel the previously ordered hosting and deploy a new one as needed.
- No need for the previous global server rent experience. We provide a wide range of data centers in geographically important locations. Large brokerage companies and banks rent their servers in the same data centers.
- No need to worry about server maintenance. We deal with all technicalities.

Your data is secured

We offer the hosting of Access Servers which serve as the access points of your platform. Their objective is to maintain client connections:

- Handling connections, packing and forwarding authorization requests to a trade server.
- Controlling the activity of connections to protect the server from attacks and overloads.
- Caching price and news data to decrease the load on the history server.

The access servers store no data containing trading account passwords, keys or trades of any form. All account information (personal data, trading operations, transaction

MetaTrader 5 trading platform

histories, etc.) is encrypted on the side of the client terminals and broadcast by the access servers to the trade server as is. The data is decoded only on the trade server side.

Therefore, the access server hosting is completely secure in terms of broker data access.

Our equipment

The deployment is performed via our own equipment in Equinix LD4 (London), Equinix NY4 (New York), Equinix HK1 (Hong Kong) and Servers.com (Amsterdam) data centers. All servers are up-to-date and feature NVMe disks.

Since access servers maintain client connections, the network speed is of utmost importance. In this regard, our hosting provides great possibilities up to 10 Gbps.

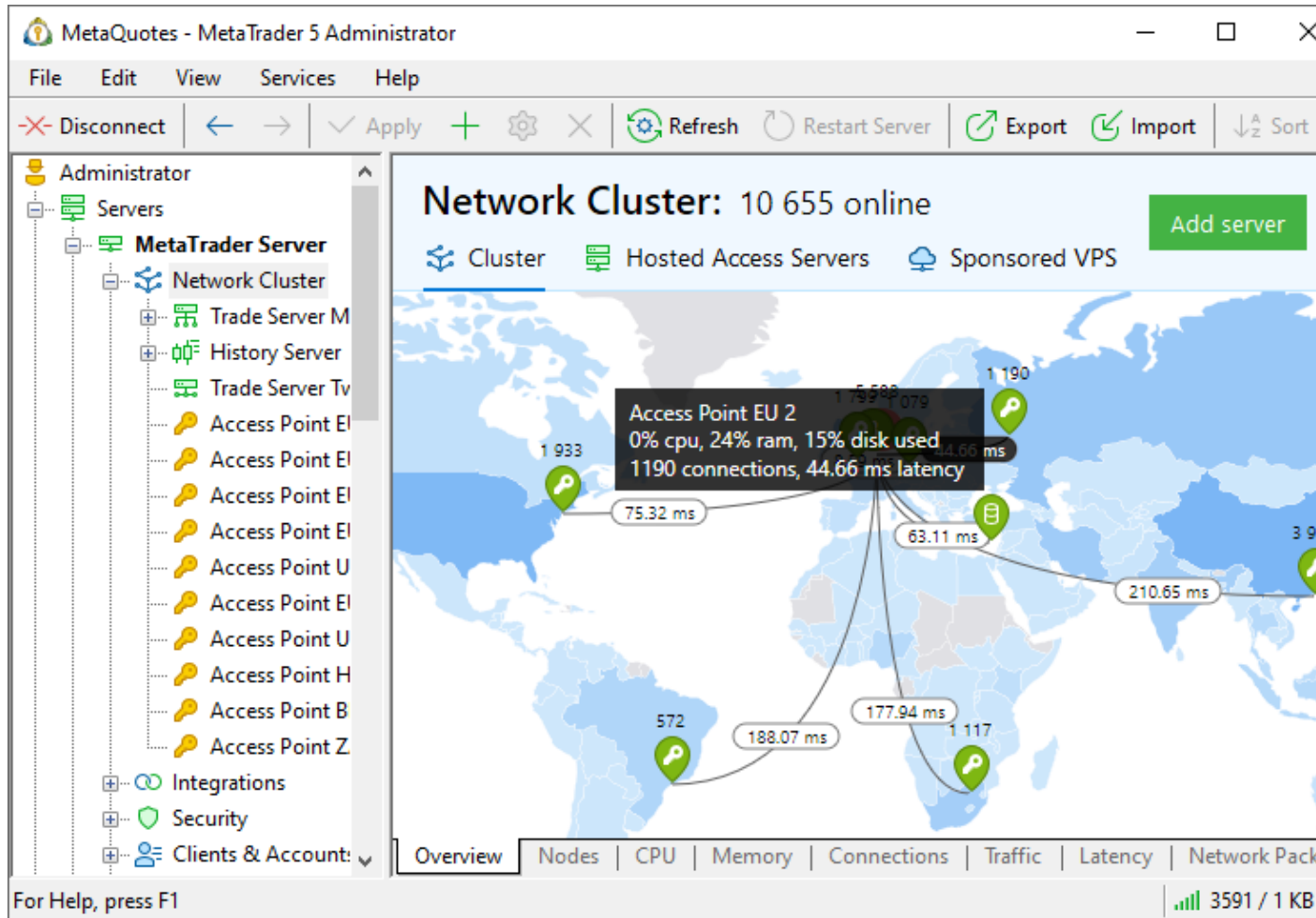
IPv4 and IPv6 addresses are allocated.

You can view server configuration details on the cluster map in the "Hosted Access Servers" section. We do not provide remote access to rented servers, since we fully undertake all maintenance.

How to deploy an access server

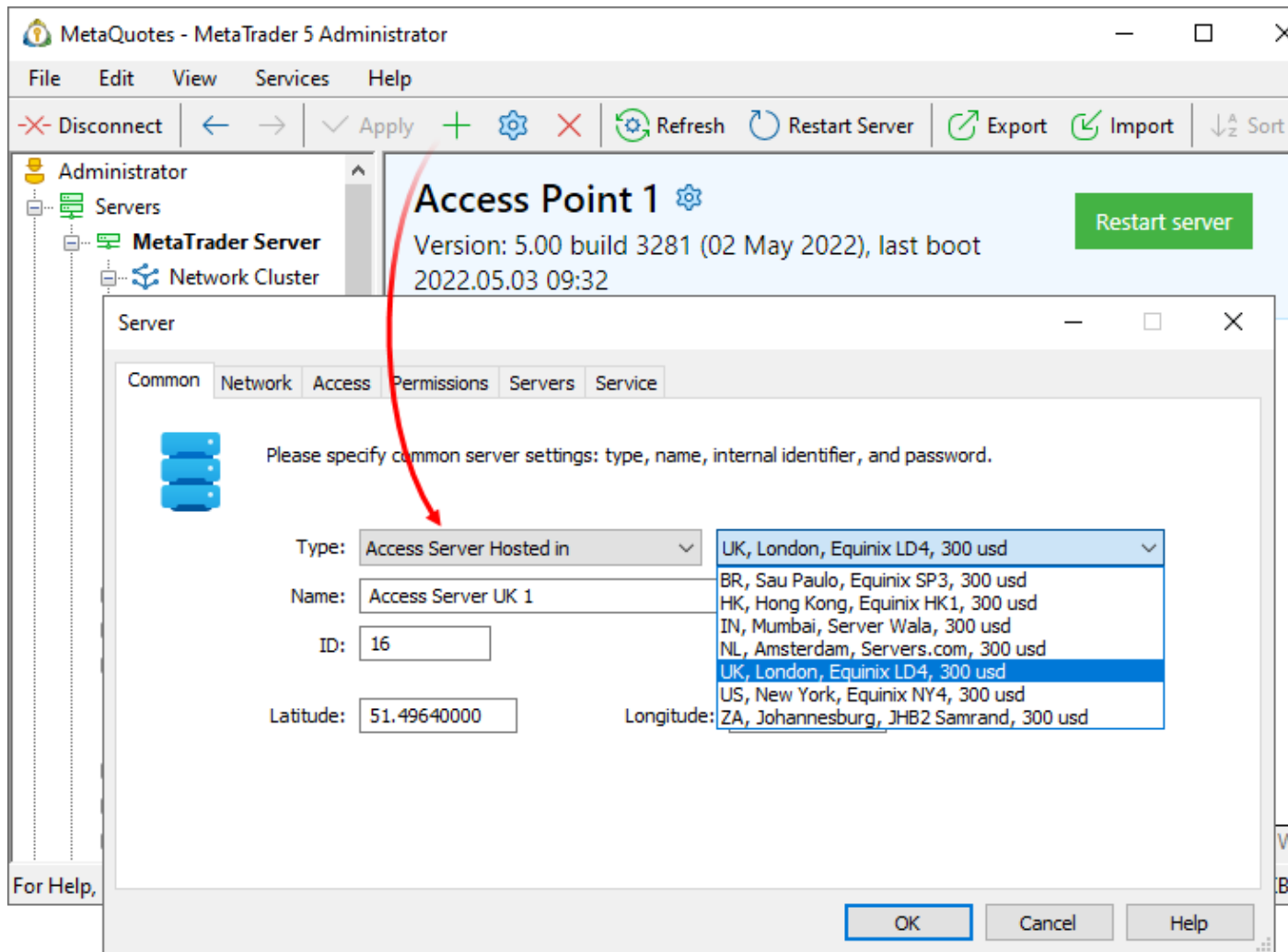
Use the [interactive cluster map](#) to view traders' activity by region, as well as the location of your access servers along with the network latency data. You can quickly detect the regions where additional access points are necessary.

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After analyzing your infrastructure, order the hosting service directly via the Administrator terminal. Add a new access server of the 'Access Server Hosted in US/UK/NL/HK' type.



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You can select the required location for your server during the purchase process. Currently, we offer deployments in the following data centers:

- Equinix LD4 London, UK
- Equinix NY4 New York, USA
- Equinix HK1 Hong Kong
- Equinix SP3 São Paulo, Brazil
- Servers.com AMS1 Amsterdam, Netherlands
- Server Wala Mumbai, India
- JHB2 Samrand Johannesburg, South Africa

Once you select a location on the Network tab, IPv4 and IPv6 addresses of the future access server will be added automatically. Other settings are specified just like for ordinary access servers. Save changes and a new server will be allocated in the selected data center. The component connects to your cluster without additional configuration. The allow rule for the allocated IP address is automatically added to the Windows firewall on trade and history servers.

You will be able to use the hosting capabilities almost immediately. Your traders will be able to connect to your platform through a new access point within 30 seconds.

Make sure that the correct network settings are used on the machine where the main trade server is installed. Otherwise, the access server will not be able to connect to the cluster.

Try MetaTrader 5 Access Server the first week is free of charge

The cost of hosting one access server is USD 300 per month. No minimal rental period of six months or a year, which is a typical requirement in case of large data centers, is required. You can cancel your subscription at any time.

Try the service free of charge for a week and establish its quality and efficiency. Order the hosting service in the Administrator terminal and obtain a trial subscription. No need to pay for the service if you change your mind during this one-week period.

MetaTrader 5 Trading Platform / Platform Setup / Network cluster /
Restarting and Stopping Servers

Restarting and Stopping Servers

A number of operations on the trading platform setup require servers being restarted afterwards. To restart a server, select the necessary one in the "Network" section and execute command " Restart server" in the Service menu, in the Standard toolbar or in the context menu.

When the main trade server is restarted, all other components are restarted automatically.

The platform (main trade server) needs to be restarted after the following operations:

- Addition of a new server;
- Changing of the platform name visible to clients;
- Changing of parameter "Risk management" in group settings;
- Changing of network parameters of the server (address, port, login or password);
- Changing of access points, bindings and of the list of serviced trade servers for access servers;
- Changing of parameters "Digits", "Tick size", "Tick value" and "Market depth" in symbol settings;
- Changing of the server time zone;
- Changing of the mode of switching to Daylight Saving Time;
- Changing of the ranges of accounts, orders and deals;

- Changing if the "Datafeeds timeout" parameter in the history server settings;

- Appearance of faults.

- Servers should be restarted only on weekends and holidays, or at night when the trading activity is minimal. The restart of servers can take from several seconds to a minute. During this time connection to this server is impossible.

- One should not confuse the restart of a server with the restart of a server's system service. After changing the platform settings, restart the server using the corresponding command in MetaTrader 5 Administrator. If you just restart the trade server's service, the other components of the platform may not be notified of changes in the settings.

Stopping and Starting Servers

All MetaTrader 5 servers work as services in the operating system. If you need to temporarily stop them, you need to stop the corresponding services. This can be done in several ways:

MetaTrader 5 trading platform

- in the command line execute the command [server executable file] /stop. For example, mt5history.exe \stop.
- in the command line execute the command sc stop [service name]. For example, sc stop mt5srv.
- in the Start menu execute the command MetaTrader 5 Platform\[server name]\Stop [server name]. For example, MetaTrader 5 Platform\History Server\Stop History Server.

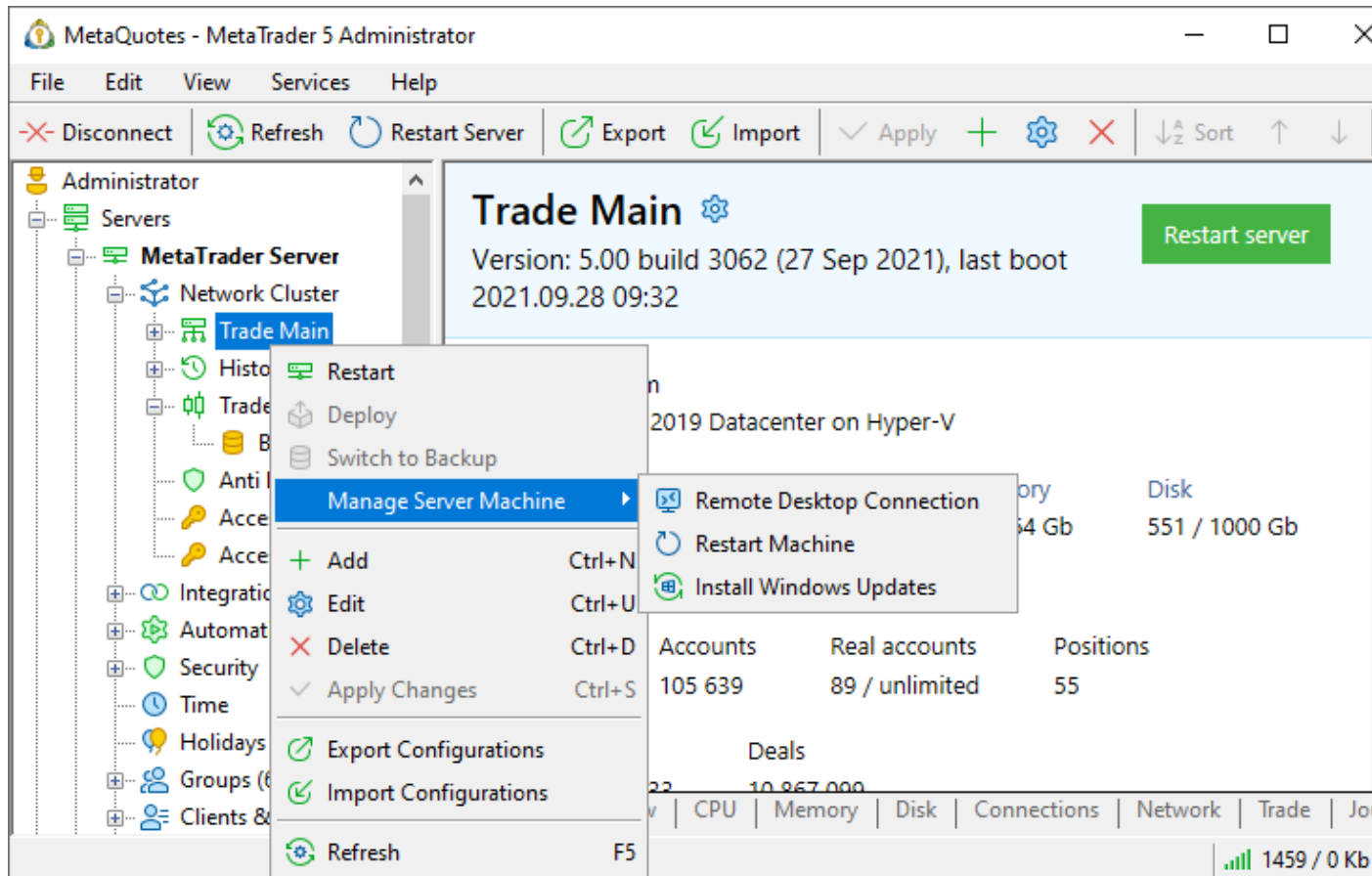
In the same way the servers can be started:

- in the command line execute the command [server executable file] /start. For example, mt5history.exe \start.
- in the command line execute the command sc start [service name]. For example, sc start mt5srv.
- in the Start menu execute the command MetaTrader 5 Platform\[server name]\Start [server name]. For example, MetaTrader 5 Platform\History Server\Start History Server.

MetaTrader 5 Trading Platform / Platform Setup / Network cluster / Managing Machines

Managing Cluster Machines

The Administrator terminals enables convenient management of the machines on which the trading platform servers are running. With these commands, you do not need to connect to the server machine manually in order to install updates or to restart it. Use the relevant command from the server menu:



If computer drives are BitLocker-protected, a reboot requires a password to unlock them. To enable automated system restart after the installation of updates, BitLocker protection is suspended until the next system restart:

- Protection is disabled before installing the updates
- The system installs the updates
- The computer is restarted without entering the BitLocker password
- BitLocker protection is immediately enabled

A similar process is applied during the execution of the computer restart command.

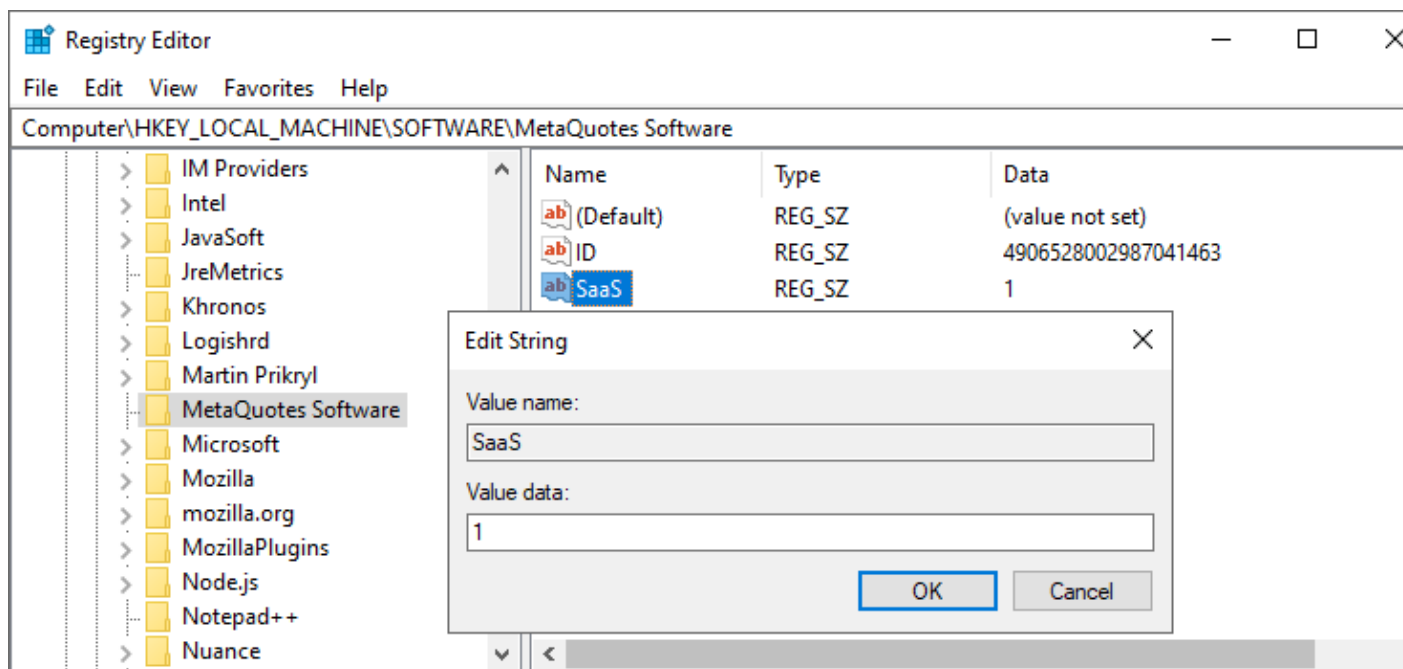
Another command enables quick connection to the server machine via remote desktop. To connect, specify only the password, while the system will automatically substitute the required IP address.

Security

A separate manager account permission regulates access to computer administration commands "Manage server machines".

For access servers, you can additionally set a physical restriction on administration access.

If your company policy or your hosting provider's policy do not allow access to update installations or reboots, you can disable these options at the machine level via the registry. Use the SaaS string key with the path "HKEY_LOCAL_MACHINE\SOFTWARE\MetaQuotes Software" and set it to a non-zero value. In this case, even if the relevant permissions are granted in the platform, the manager will not be able to use these commands.



MetaTrader 5 trading platform

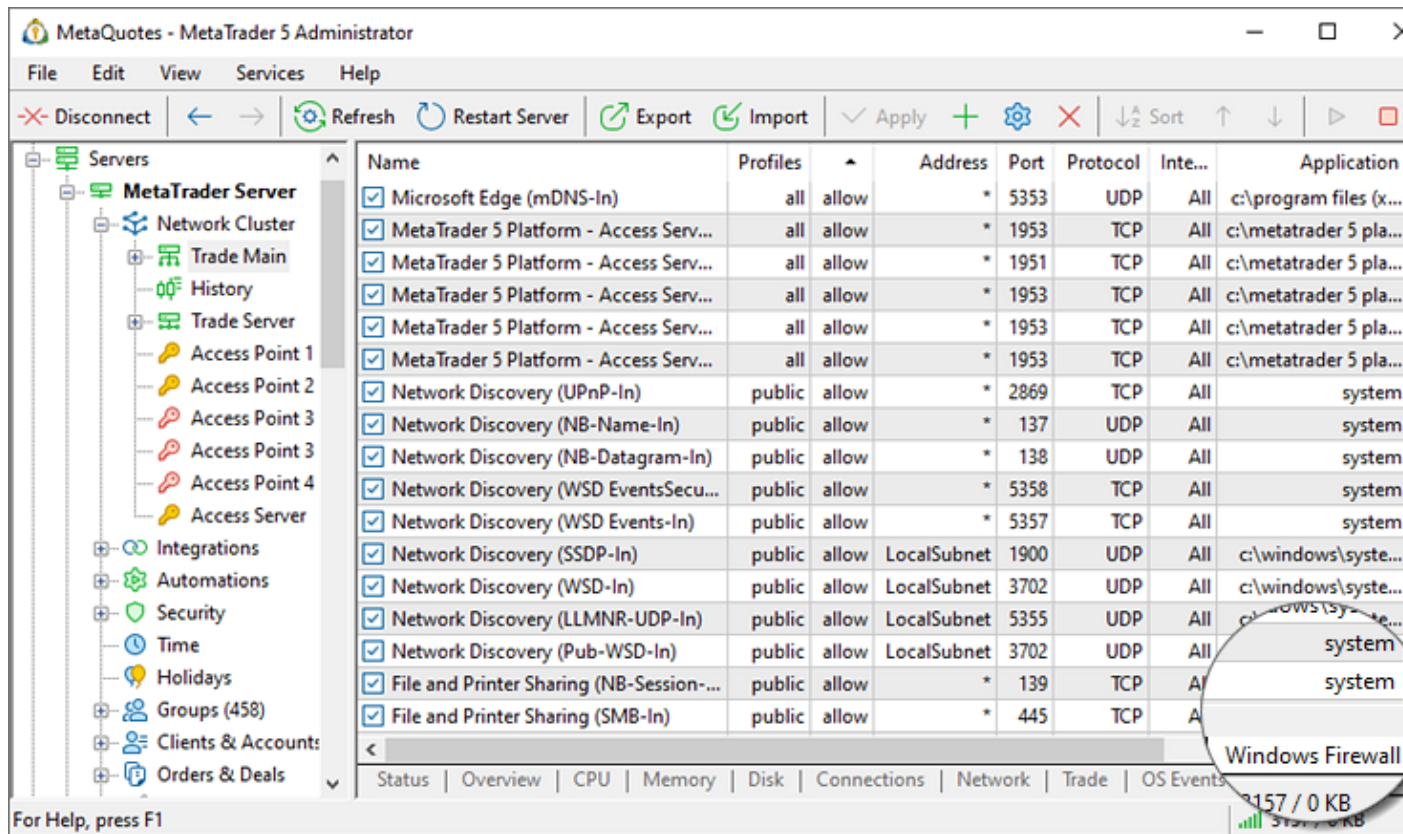
Restart a computer after adding a key or changing its value.

SaaS key also disables firewall management and viewing event logs.

Managing operating system firewall

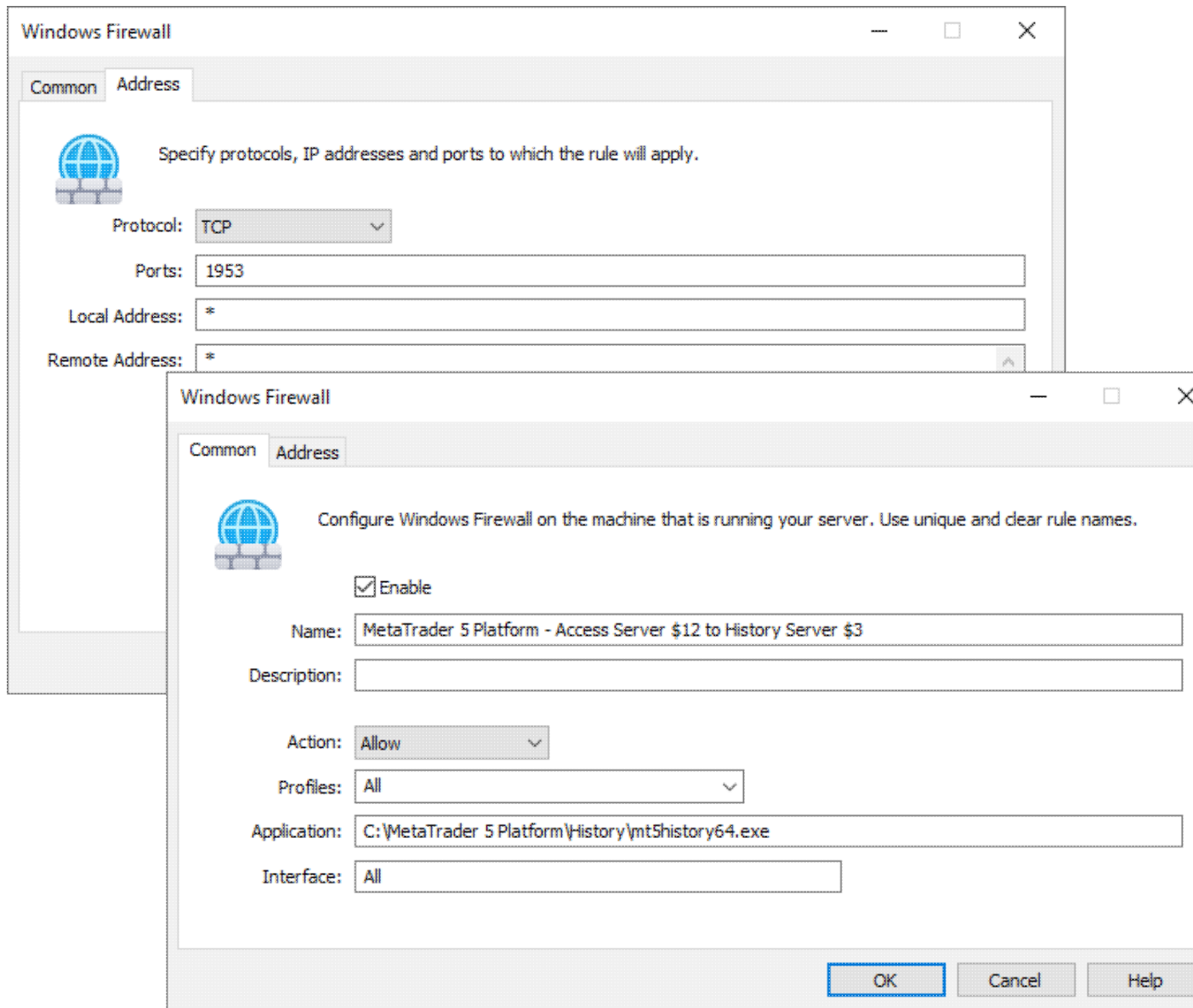
All platform servers feature the ability to remotely manage the operating system firewall on the computer they are installed on.

Administrators do not need to remotely connect to the computer to view and edit the list of current incoming rules. Simply open the Windows Firewall section in the network cluster.



MetaTrader 5 trading platform

To create a new rule, click **+** on the toolbar. To change the existing rule, double-click on it in the list.



The standard rule parameters are available for a setup:

- **Enable** if the rule is currently not needed but may be of use later, it can be temporarily disabled.
- **Name** rules should have concise and comprehensible names. All names should be unique.
- **Description** optional parameter allowing you to add some details about the rule purpose.
- **Action** action applied to a specified application\address\port: Allow or Block.
- **Profiles** system network profiles the rule is to be applied to: Domain, Private, Public. All profiles are enabled by default.
- **Application** path to the application the rule is to be applied to.
- **Interface** type of the connection the rule is to be applied to: Remote access, Wireless, LAN. All profiles are enabled by default.

In the address section, specify:

- **Protocol** the protocol the rule is to be applied to: TCP, UDP, etc.
- **Ports** the ports the rule is to be applied to.
- **Local Address** local addresses and networks the rule is to be applied to.
- **Remote Address** IP addresses and networks the rule is to be applied to. Each entry should be placed on a separate string with no comma separation.

Bulk rule configuration

The section context menu features the commands for quick bulk rule setup:

- **Allow \ Fix all** perform all actions described below.
- **Allow \ IPv4 functionality** enable the rules for handling IPv4 connections.
- **Allow \ IPv6 functionality** enable the rules for handling IPv6 connections.
- **Allow \ Remote Desktop** enable the rules allowing remote connection.
- **Allow \ MetaTrader 5 Cluster** when installing the platform, as well as when installing separate servers, the rules allowing the connection of all components to one another are added to the Windows firewall rules. For example, the incoming allowing rules for all the remaining components are created on the main trade server machine. The MetaTrader 5 Cluster command checks the presence of all rules for the cluster on the machine and creates them anew if necessary.
- **Disallow \ Useless OS rules** disable the rules not needed for the cluster operation.

- The tool works only with incoming firewall rules. Outgoing rules cannot be viewed and edited.
- An account should have the "Connect via MetaTrader 5 Administrator" and "Manage server machines" permissions to access the Windows firewall section. The rules cannot be managed on machines with the "SaaS" key.

Operating system events

MetaTrader 5 trading platform

All platform servers allow the request of the operating system event logs from the computer on which the servers are installed.

You can monitor and handle arising issues without accessing the computer via remote access solutions. System events can be viewed directly from the Administrator terminal, which offers a more convenient presentation form than the standard Windows interface. The terminal provides the full description of events, with some of the unnecessary details disabled.

Select a server, go to the "OS Events" section and execute the following request:

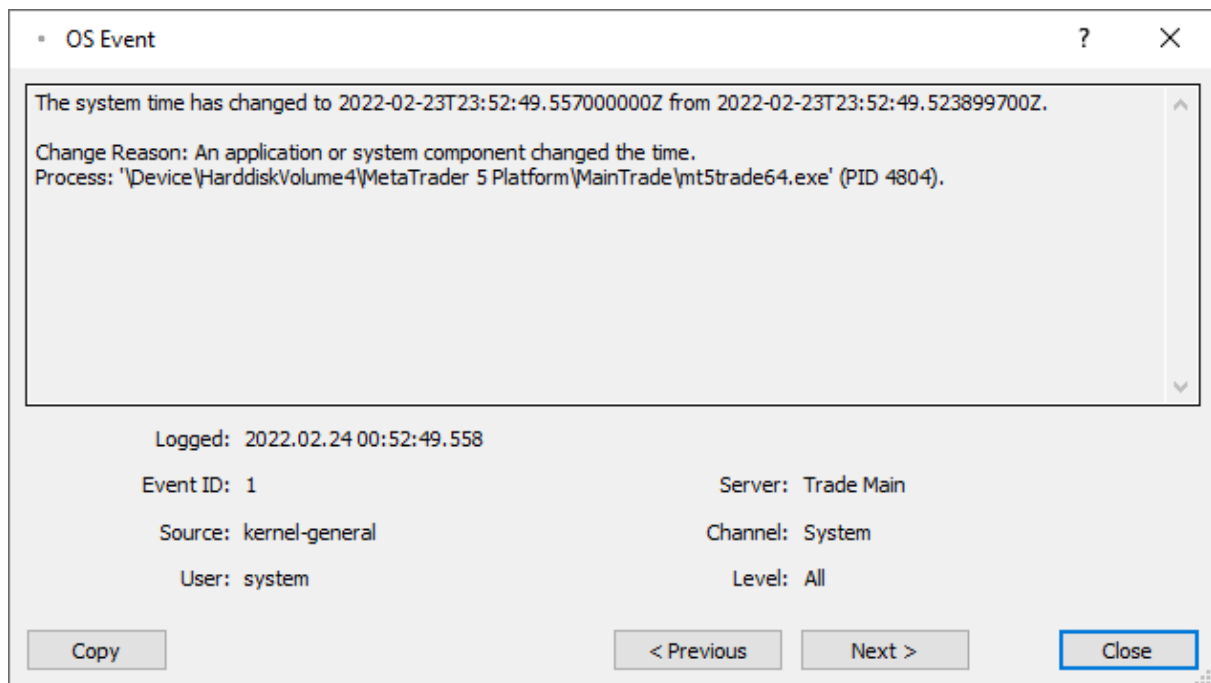
The screenshot displays the MetaTrader 5 Administrator window. On the left, a tree view shows the hierarchy: Administrator > Servers > MetaTrader Server > Network Cluster > Trade Main > History > Trade Server > Access Point 1-4 > Access Server NL > Integrations > Automations > Security > Time > Holidays > Groups (414) > Clients & Accounts > Allocations > Clients > Managers > Trading Accounts > Orders & Deals > Gateways > Data Feeds. The right pane shows a list of system events with columns: Time, ID, Source, User, and Message. The events are filtered by 'OS events filter, leave empty to request all events' and 'System'. A dropdown menu is open, showing 'All' and 'OS Events' options. The 'OS Events' option is selected, and the list of events is displayed below it.

Time	ID	Source	User	Message
2022.02.23 08:25:18.639	7040	service control manager	system	The start type of the Windows Modul
2022.02.23 08:26:40.312	7036	service control manager		The Network Setup Service service en
2022.02.23 08:27:18.757	7036	service control manager		The Windows Modules Installer servic
2022.02.23 08:29:41.607	7036	service control manager		The Network Setup Service service en
2022.02.23 08:58:40.823	1	kernel-general	system	The system time has changed to 2022
2022.02.23 09:26:40.345	7036	service control manager		The Network Setup Service service en
2022.02.23 09:29:41.642	7036	service control manager		The Network Setup Service service en
2022.02.23 09:58:40.850	1	kernel-general	system	The system time has changed to 2022
2022.02.23 10:26:40.375	7036	service control manager		The Network Setup Service service en
2022.02.23 10:29:41.671	7036	service control manager		The Network Setup Service service en
2022.02.23 10:58:40.890	1	kernel-general	system	The system time has changed to 2022
2022.02.23 11:26:40.501	7036	service control manager		The Network Setup Service service en
2022.02.23 11:29:41.708	7036	service control manager		The Network Setup Service service en
2022.02.23 11:58:40.928	1	kernel-general	system	The system time has changed to 2022
2022.02.23 12:00:00.929	6013	eventlog		The system uptime is 365762 seconds
2022.02.23 12:24:33.055	7036	service control manager		The Windows Update Medic Service s
2022.02.23 12:24:33.179	7036	service control manager		The Software Protection service enter
2022.02.23 12:24:33.523	7036	service control manager		The WinHTTP Web Proxy Auto-Disco
2022.02.23 12:24:33.783	7036	service control manager		The Windows Update service entered
2022.02.23 12:24:38.391	7036	service control manager		The Windows Modules Installer servic
2022.02.23 12:25:10.809	7036	service control manager		The Software Protection service enter
2022.02.23 12:25:33.103	7036	service control manager		The Windows Up

The requested events can be filtered:

- By contents type the desired phrase
- By level, at which the event occurred (channel):
 - System the events generated by Windows components, for example failed attempts to launch services or to initialize drivers
 - Application program operation events and errors
 - Security system security events (login, changes in folder access permissions, etc.)
 - Installation events related to the installation of applications
- By event importance (descending):
 - Critical
 - Errors
 - Warning
 - Information
 - Details
 - By date

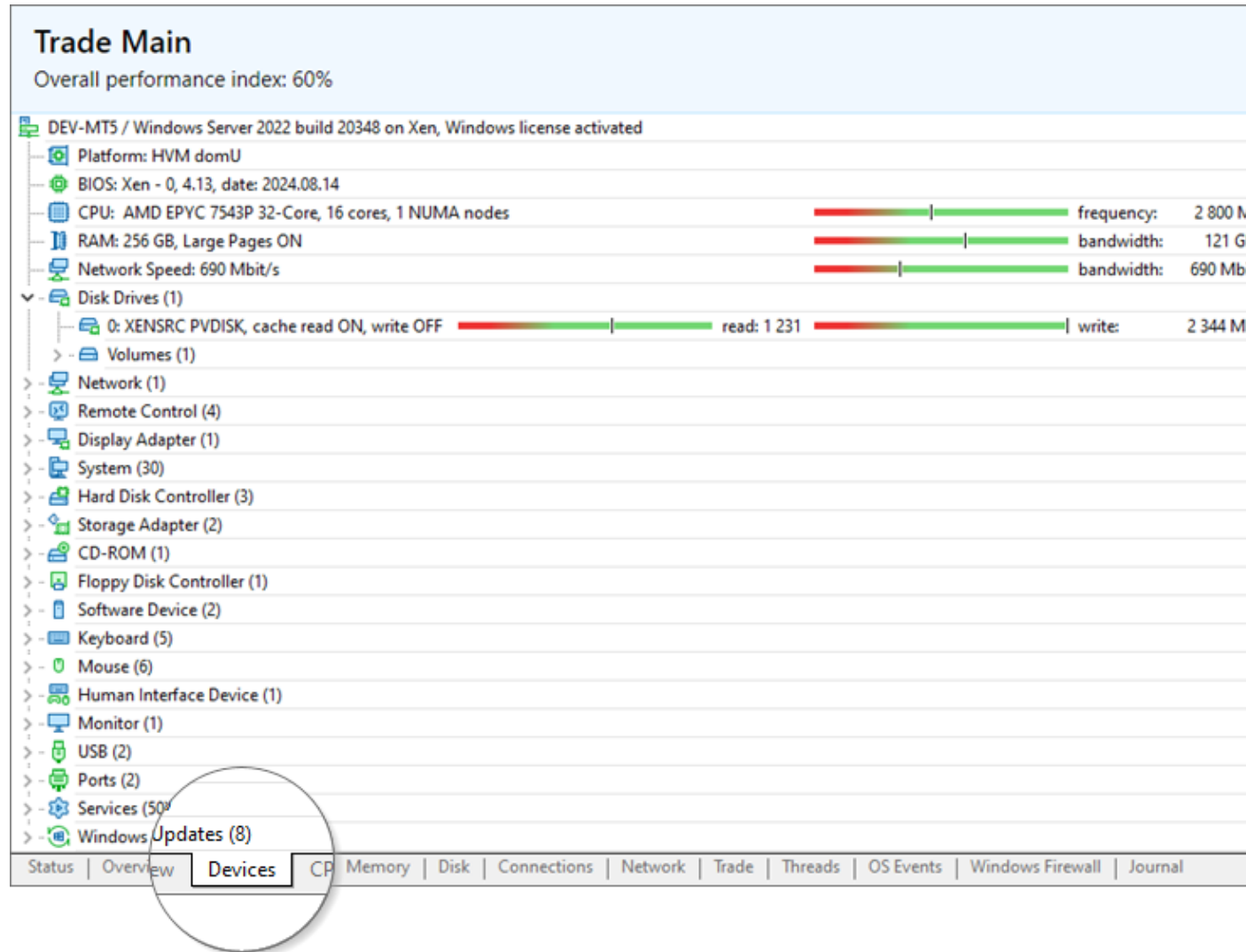
To view event details, click on the event line:



- To access the OS events section, the account needs the following permissions: "Connect via MetaTrader 5 Administrator" and "Access server logs".
- System events cannot be viewed on machines with the "SaaS" key.

Devices

The platform collects detailed information about cluster server computers to help administrators keep the infrastructure up to date. In the new "Devices" section, we have created an analogue of the Windows task manager, adapted for maximum utility and optimized by removing unnecessary information.



The section provides the following details:

- Information about installed devices, drivers, and settings that are critical to performance.
- Processor, disks and RAM performance metrics.
- Availability of the Large Page technology for RAM.
- Available Windows updates that can be installed with one command.
- Use of a virtual machine.

Large Pages

The platform Large Pages for memory management in Windows. This technology has been introduced to optimize memory management on Windows. It reduces overhead associated with memory management in systems requiring large data volumes, including trading platforms.

Operating systems typically divide physical memory into small blocks called pages, usually 4 KB in size. When a process needs memory, the OS allocates several pages and maintains a Page Table that maps virtual addresses used by the process to physical addresses in RAM.

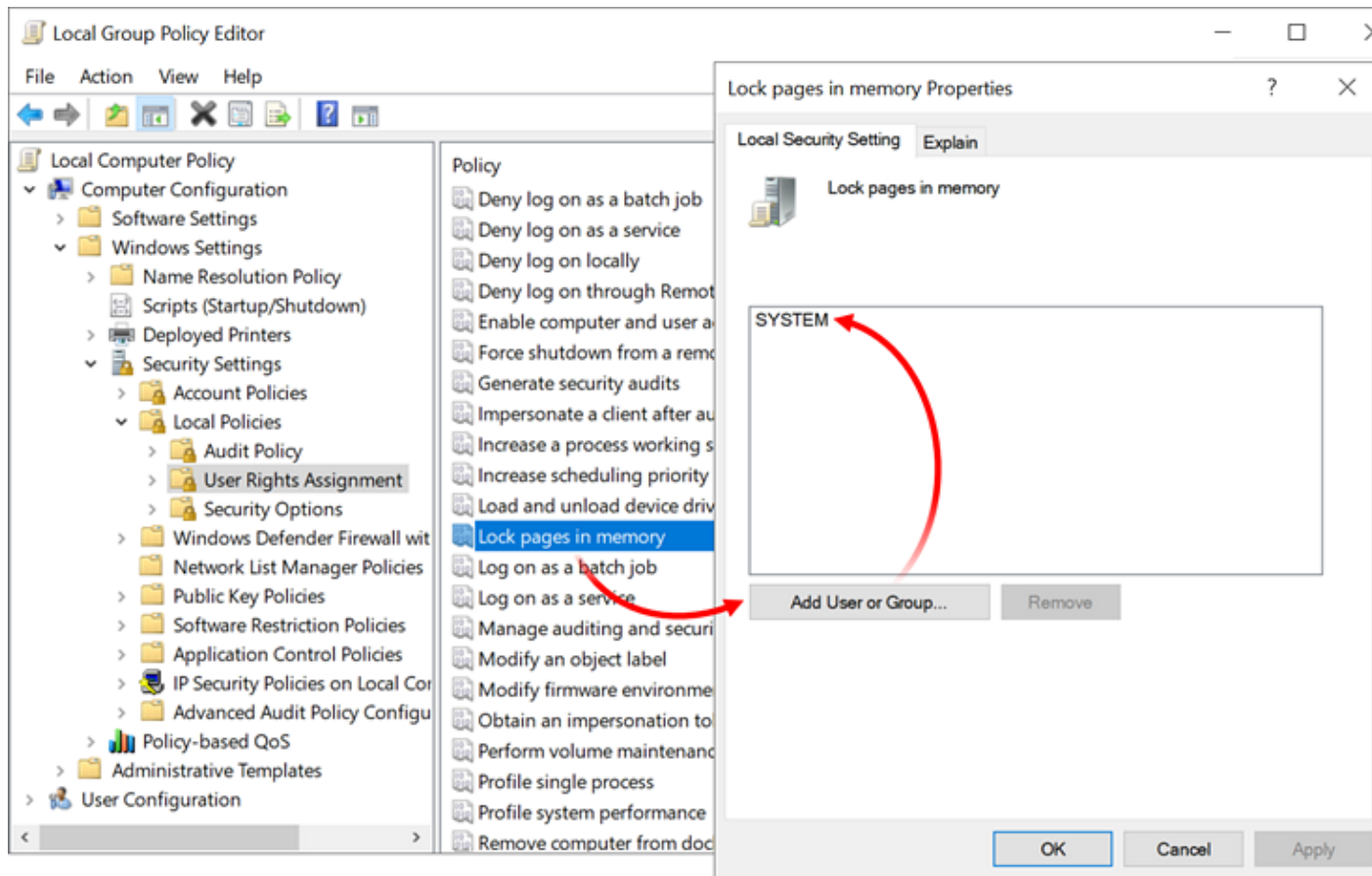
Each time a process accesses memory, the system translates the virtual address to a physical one using the Page Table. For processes that use large amounts of memory, the Page Table can become large and frequently accessed, adding extra load to the processor.

Large Pages technology allocates larger memory blocks instead of the standard 4 KB pages. This reduces the number of entries in the Page Table and decreases the frequency of accesses to it.

How to Enable Large Pages

Large Pages technology is memory-intensive. It is recommended to enable it only if you have at least 32 GB of RAM. Ensure you have sufficient memory for the platform operations.

- Open the Command Prompt and type gpedit.msc
- In the window that opens, navigate to "Local Computer Policy \ Computer Configuration \ Windows Settings \ Security Settings \ Local Policies \ User Rights Assignment".
- In the list of policies, find 'Lock pages in memory' and add access to it for the SYSTEM user.
- Save your settings and restart your computer.
- Verify that the technology is enabled by checking the "Devices" section in the server settings in MetaTrader 5 Administrator. The status will be displayed in the RAM property.



For further details about the Large Pages and their configurations, please refer to the article at <https://mahdytech.com/large-pages-how-when/>.

MetaTrader 5 Trading Platform / Platform Setup / Network cluster /
Status

Status

The section displays important component operation characteristics: hardware load, database sizes, network speed and much more. These metrics should be monitored closely: hardware should always have a power reserve to ensure the maximum platform performance.

MetaTrader 5 trading platform

MetaQuotes - MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Import Export Apply + Settings X Back Forward Sort

Administrator

- Servers
 - MetaTrader Server**
 - Network Cluster
 - Trade Main**
 - Trade Server
 - History
 - Access Point 2
 - Access Point 3
 - Access Point 1
 - Access Point
 - Integrations
 - Automations
 - Security
 - Time
 - Holidays
 - Groups (518)
 - Clients & Accounts
 - Orders & Deals
 - Gateways
 - Data Feeds
 - Plugins
 - Reports
 - ECN
 - Routing
 - Funds & ETF
 - Symbols (87405)
 - Spreads
 - Charts & Ticks
 - Subscriptions
 - Mailbox (21991)
 - Live Update
 - MetaTrader Server

Trade Main ⚙️

Version: 5.00 build 3447 (26 Sep 2022), last boot 2022.09.26 16:04 [Restart server](#)

Operating system
Windows Server 2022 Standard on Xen

Processor	Memory	Disk
8 x Intel Xeon E5-2670 v2 @ 2.50GHz	53 / 64 GB	977 / 1200 GB

Databases

Groups	Accounts	Real accounts	Positions
137	69 965	97 / unlimited	29 096

Orders	Deals
25 / 43 479 488	119 289 930

Maximum, last 24hrs

Connections	Active sockets	Network load	CPU load
68	165	5.6 Mbps	96 %

Memory
56 / 64 GB

Ping

Trade Server Secondary:	0.07 ms	Access Point 3:	0.05 ms
History:	0.07 ms	Backup Second:	0.05 ms
Access Point 1:	0.06 ms	Backup Main:	0.06 ms
Access Point 2:	0.06 ms		
Access Point 3:	0.07 ms		

Status Overview CPU Memory Disk Connections Network Trade Threa

For Help, press F1 2096 / 0 KB

Basic Information

Basic information about the selected server is shown at the top of the window:

- **Version** server version, build number and its date.
- **Last boost** last system boot time. The server restart button is located in the right part.
- **Operating system** name of the operating system. The minimum recommended system is Windows Server 2016 Standard x64.
- **Processor** CPU type and its clock rate. The minimum recommended CPU type is Intel i7 4xxx 4 quad-core and above.
- **Memory** free RAM amount/total RAM.
- **Disk** free hard drive space in gigabytes. Make sure there is always enough free disk space.

Databases

This block features information about databases on the selected server. It is only available for trading servers.

- **Groups** the total number of groups on the server.
- **Accounts** the total number of accounts on the server.
- **Real accounts** the total number of real clients on the server, as well as the limitation on the number of clients according to the license (if any). Real accounts refer to the client records that are not included in demo*, manager*, preliminary* and contest* groups. The statistics only include enable accounts from which trading is allowed.
- **Positions** the number of positions on the trade server.
- **Orders** the number of active orders and history orders on the selected server.
- **Deals** the total number of deals executed on the selected server.

Maximum load

This block shows the maximum server load registered during the current session:

- **Connections** statistics on the total amount of all standard operating terminal connections and temporary connections made for executing trades, downloading history or news. In the quiet market, this parameter allows determining the number of online users. It is shown for the cluster and for individual servers.
- On access servers, connections show the number of client connections.
- On trade server, the figure means virtual connections (clients actually connect via access servers).
- On history servers, the metric shows connections of other components of the cluster.
- **Active sockets** the total number of TCP endpoints (with a specific IP address and port number) connected throughout the operating system. This also takes into account half-open connections when there is no connection, but the socket is still closed by the operating system.

- **Network load** total incoming and outgoing traffic, in Mbit/s. The load of the selected network interface is measured in terms of the incoming and outgoing traffic (average value per minute). Takes into account the traffic of all programs that run on the server computer. Unexpected load spikes can easily point to DDoS attacks.
- **CPU load** total CPU load in percent. This parameter affects on how fast users and their trade operations are serviced. If the processor load regularly exceeds 50% in the middle of a work day, it is time to think of the computer upgrade. Over 85% load is critical. If the processor load is 100%, it means that the processing power is not enough for processing of tasks. However, rare spike loads are no reason to worry.
- **Memory** free RAM amount/total RAM. Availability of a large amount of free memory is extremely important for a server. This enables you to serve more users currently connected, and handle large databases. If the available memory becomes less than 100MB, the graph is colored in red.

To view the statistics in dynamics, use monitoring section.

Ping

This section shows network delays between the selected server and all other components of the cluster.

Server operation statistics (network load, CPU and others) are updated during optimization.

Backup last time

This section is only displayed for backup servers. It indicates the last time the backups were made.

- **Startup** at startup the backup server creates a file copy of its databases which were synchronized with the main trade server in real time. This happens before the start of synchronization with the main server in case its databases are already damaged. This allows the rolling back to the previous state using the file copy.
- **Full** the time when the file copies of all databases were created.
- **Archive** the time when additional file copies were created.
- **Databases synchronization** the time of the last backup of non-core data (the initial synchronization at server startup or periodic synchronization during operation).
- **SQL synchronization** time of the last full synchronization of databases and platform configurations with the SQL database. It is launched when a backup server is started or after connection to the SQL database or to the trading server is lost. After synchronization, the SQL database is updated in real time in accordance with the transactions of changes in the platform databases.

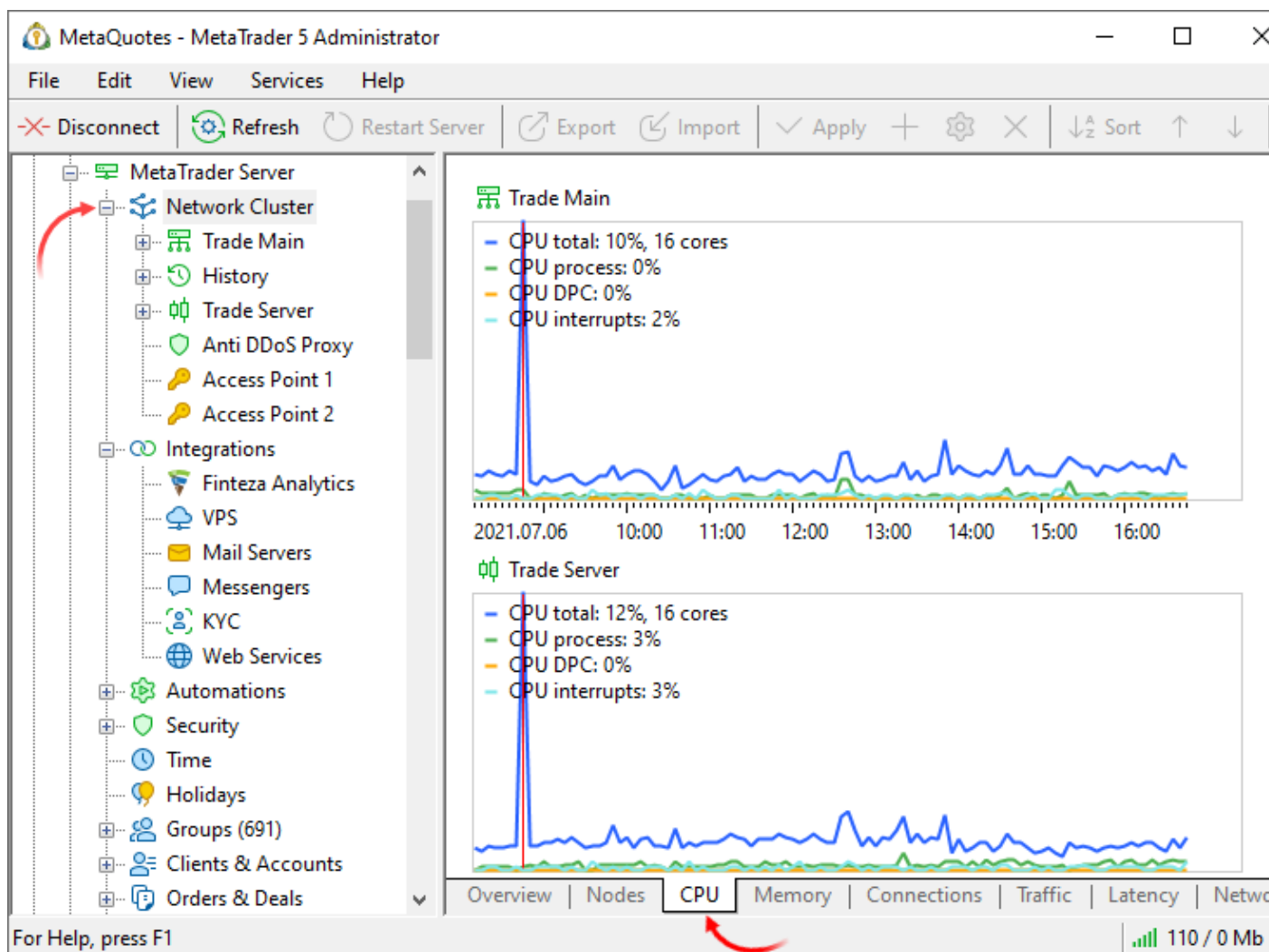
MetaTrader 5 Trading Platform / Platform Setup / Network cluster / Monitor

Monitoring

Each minute, the platform measures multiple parameters of its own operation and maintains the performance database allowing an administrator to quickly assess the status of the cluster components and respond to potential issues.

The data is conveniently presented as graphs for the entire cluster and each platform component separately.

To see the data on the entire platform, select the "Network Cluster" and then one of the tabs: CPU, Traffic, Network Packets, Memory, Connections or Trade.

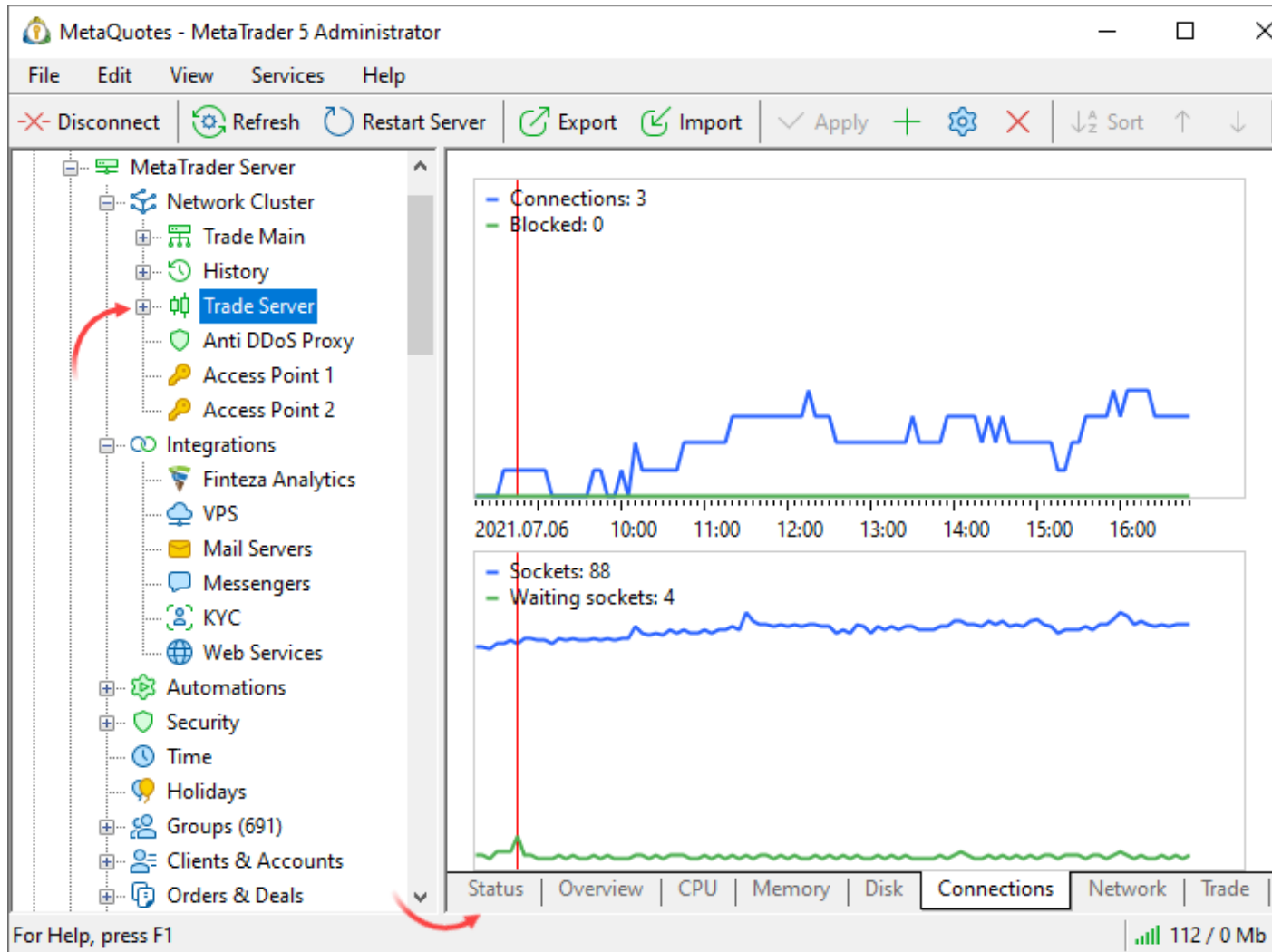


Server restarts are shown on the graphs as vertical red lines. To view exact graph values, enable the crosshair mode in the context menu.

To view more detailed information about any component, select it in the tree.

- The variables are measured once a minute. Most of them show a one-time instant state of the server (for example, CPU load at a particular moment in time). The state can be different between measurements.
- Performance measurements of different platform components are not synchronized (they are performed in different times). Therefore, common parameters (such as CPU) can differ in monitoring results of different components, even if they are installed on one physical server.
- On higher chart timeframes (5 minutes, 15 minutes, 1 hour, 1 day), the displayed variables represent the sum, the highest and the lowest values over the selected period. For example, the 15-minute connections diagram displayed the maximum number of connections for every 15 minutes.
- Current metrics (in the upper left corner of the graph) always display the last minute values regardless of the selected timeframe.

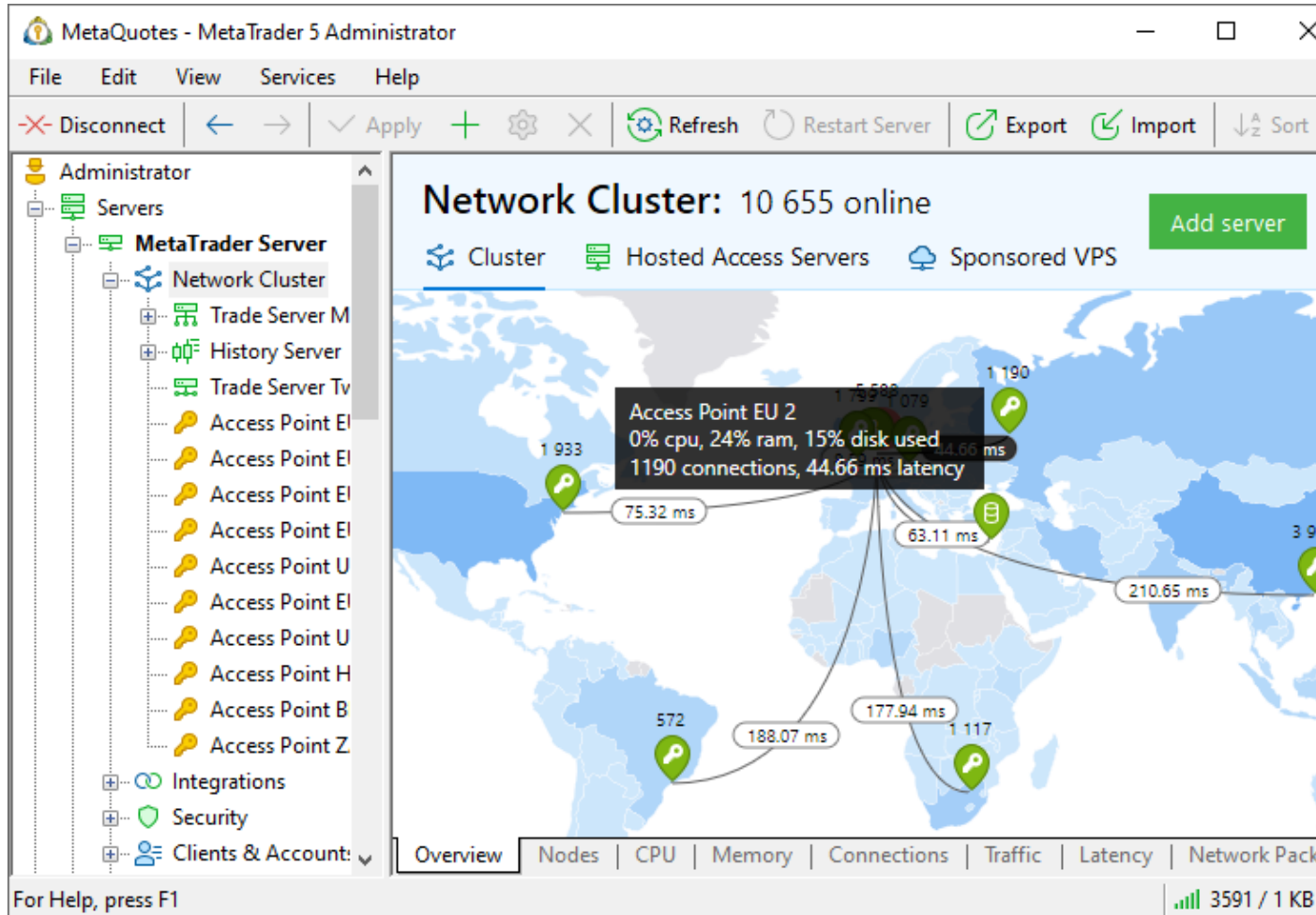
MetaTrader 5 trading platform



MetaTrader 5 trading platform

Cluster map

The Overview section of the network cluster displays an interactive map of platform servers. The map enables easy platform state monitoring in terms of server loads, network delays and other metrics.



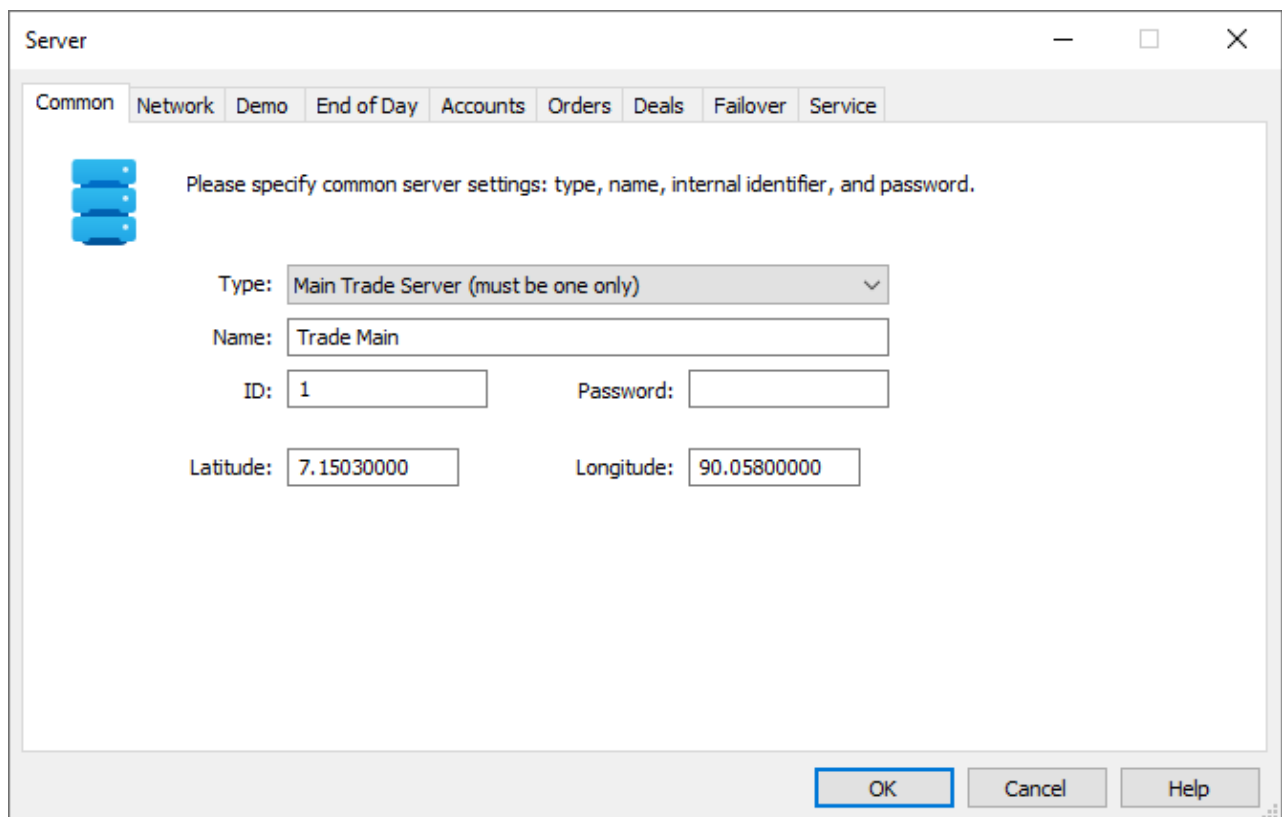
MetaTrader 5 trading platform

The number of connections is shown for each server. Additionally, map regions are colored depending on the number of connections from them: more intense colors are used for higher connection numbers. Thus, you can spot regions with an extremely high load which require additional access points.

Hover the mouse over a server to view monitoring details: CPU load, Memory and disk usage, and the number of connections. Disconnected servers are shown in red on the map.

Use the mouse wheel or the "+" and "-" keys to change the map scale.

Server position on the map is determined by the IP-addresses automatically, through the built-in GeoIP database. If the coordinates could not be accurately determined automatically, you can specify them manually under the server properties:



The screenshot shows the 'Server' dialog box with the 'Common' tab selected. The dialog has a title bar with standard window controls. Below the tabs, there is a blue server icon and a prompt: 'Please specify common server settings: type, name, internal identifier, and password.' The form contains the following fields:

- Type: A dropdown menu showing 'Main Trade Server (must be one only)'.
- Name: A text box containing 'Trade Main'.
- ID: A text box containing '1'.
- Password: An empty text box.
- Latitude: A text box containing '7.15030000'.
- Longitude: A text box containing '90.05800000'.

At the bottom right, there are three buttons: 'OK' (highlighted with a blue border), 'Cancel', and 'Help'.

The Hosted Access Servers section displays the location and configuration of machines on which you can order [Access Server Hosting from MetaQuotes](#). Click on a hosting point icon to proceed to renting a server in the selected location.

In the "Sponsored VPS" section, you can view the network infrastructure of the [hosting service](#), as well as the number of VPSs you have sponsored for your traders in each location.

Processor

- Total CPU usage in percentage. This parameter affects on how fast users and their trade operations are serviced. If the processor load exceeds 50% in the middle of a work day, it's time to think of the computer upgrade. If the CPU load is more than 85%, the chart is colored red. This value is critical. If the processor load is 100%, it means that the processing power is not enough for processing of tasks. However, rare spike loads are no reason to worry.
- CPU usage by the server process. If the total CPU load is high, while server process load is not very high, the computer resources must be consumed by some third-party application.
- Number of CPU cores.
- CPU DPC ([Deferred Procedure Call](#)). DPCs have lower priority compared to normal calls. In a normal call, control is immediately transferred to called procedure code. With a DPC call, control is transferred only at a time "favorable" for the processor. While the processor is occupied by higher-priority tasks, lower-priority DPCs are placed in a special DPC Queue. For more information, please refer to [Microsoft documentation](#).
- CPU [Interrupts](#). This parameter provides an indirect measure of activity of connected devices: from the system clock to the mouse. A large interrupt load can mean hardware problems on the server. For more information, please refer to [Microsoft documentation](#).
- If the total CPU load stays at 85% or more, and the load by DPCs/interrupts is 15% or higher, this may indicate insufficient processor performance. In this case, we recommend that you replace the processor with a more powerful one, or allocate additional processors if you are using a virtual machine.
- If the total CPU load is less than 85% and the load by DPCs/interrupts is less than 15%, but you still have performance issues, the problem can be caused by third-party soft- or hardware. If DPC load surges unexpectedly, such load can be caused by recently installed soft- or hardware.
- If you are using a multiprocessor system and the load is distributed unevenly among CPUs, try installing additional network adapters, one for each processor. This will help improve system performance. For more information, please refer to [Microsoft documentation](#).
- The number of threads used by the server process. The values should be generally constant and may only change by a few units.
- The total number of descriptors (handles) in the system. A descriptor is a pointer, which enables a program to access a dedicated resource. The more descriptors a process uses, the more resources it consumes.
- The number of descriptors (handles) used by the server process.

When you switch to higher timeframes, the graph shows the highest values for the selected period.

Memory Usage

- The amount of free RAM, in megabytes and percent. Availability of a large amount of free memory is extremely important for a server. This enables you to serve more users currently connected, and handle large databases. If the available memory becomes less than 100MB, the graph is colored in red.
- The amount of RAM allocated by the server process.
- Page Faults – a page fault occurs when the server process tries to access a page in virtual memory but the page is not currently in physical memory and needs to be loaded from disk. This metric allows you to assess the efficiency of memory management. For more details, refer to the article [The Basics of Page Faults](#).

When you switch to higher timeframes, the graph shows the lowest values (for free memory and the memory block) and the highest values (for the memory allocated by the server) for the selected period.

Allocation of a large amount of RAM for a trade server is a normal behavior. The server allocates and uses memory depending on the total amount of available physical memory. The more memory is available, the larger share of this memory the server can allocate to speed up its performance.

Disk Usage

- Free disk space, in megabytes and percentage.
- The number of input/output operations per second ([IOPS](#)). This metric allows you to monitor the actual disk load. Track this metric carefully when using public cloud services such as AWS. They may have limits of several hundred operations per second. Since platform servers frequently execute bulk operations requiring thousands or even tens of thousands of IOPS, you may experience slowdowns in cloud environments. We recommend using NVME drives or high-speed network storage solution, as the speed of disk devices directly affects trading delays.
- Disk queue length. The values of this parameter should be close to zero. An increase in the value indicates that the disk fails to timely write data. In this case you should consider replacing the hard drive.
- Average latency. It shows the average response time of the physical disk to requests. This includes both processing time and queue time. The metric enables more efficient disk performance monitoring and assists in the identification of potential problems. High values (more than 15 ms) may indicate significant disk fragmentation, low speed, or malfunction.
- Data reading speed, in MB/s.
- Data writing speed, in MB/s.

When you switch to higher timeframes, the graph shows the lowest values (for free space) and the highest values (for the queue length and speed) for the selected period.

- Disk performance values are measured once a day.
- The measurement of data reading and writing speed is not accurate. The measures are only provided for reference.

Connections

- Connections show statistics on the total amount of all standard operating terminal connections and temporary connections made for executing trades, downloading history or news. In the quiet market, this parameter allows determining the number of online users. It is shown for the cluster and for individual servers.
- On access servers, connections show the number of client connections.
- On trade server, the figure means virtual connections (clients actually connect via access servers).
- On history servers, the metric shows connections of other components of the cluster.
- Blocked shows the number of connections blocked by the antiflood control system or by the built-in firewall.
- Sockets show the number of active sockets. This values means the total number of TCP endpoints (with a specific IP address and port number) connected throughout the operating system. This also takes into account half-open connections when there is no connection, but the socket is still closed by the operating system.
- Waiting sockets: the number of half-open connections when there is no connection, but the socket is still not closed by the operating system (sockets that do not have the TCP_STATE_LISTEN, TCP_STATE_ESTAB or MIB_TCP_STATE_CLOSED status).

When you switch to higher timeframes, the graph shows the highest values (for connections) and total values (for sockets and blocked connections) for the selected period.

Network and Traffic

- Incoming and outgoing traffic, in Mbps. The load of the selected network interface is measured in terms of the incoming and outgoing traffic (the average value per minute is used). The traffic of all programs that run on the server computer is taken into account. Unexpected load spikes can easily point to DDoS attacks.
- Packets show the total number of network packets.
- Retransmits: the number of resent packets (the absolute number and percent of the total amount). Values above 3% indicate poor network connection.
- Packet Queue: the number of network packets in the server's processing queue.
- Packet Efficiency: calculated as the number of packets in the queue divided by the number of physical connections to the server. This metric helps evaluate the efficiency of network operations.

When you switch to higher timeframes, the graph shows the highest values (for speed) and total values (for packets) for the selected period.

MetaTrader 5 trading platform

The Networks section features graphs displaying network latency between cluster components:

- The main server displays latency to the history server
- The history server shows latency to the main server
- Latency to the main and history servers are shown for additional servers
- The backup server displays latency to the main and backup servers

Threads

The Threads section displays the activity of all computational threads on the server. In particular, it shows the activity of plugin and report threads, as well as the resources they consume. Use this information monitor your server loads and to determine whether you need to upgrade your hardware. This information will also assist in identifying third-party solutions which slow down the platform.

The screenshot displays the MetaTrader 5 Administrator window. The left sidebar shows a tree view with 'Administrator' expanded, containing 'Servers' and 'MetaTrader Server'. Under 'MetaTrader Server', 'Network Cluster' is expanded, showing 'Trade Main', 'Trade Server', 'History', and four 'Access Point' entries. The main panel shows a table of threads with columns: Thread Name, CPU, %, CPU Cycles, Context ..., Prior..., and Stack, kb. The table lists various threads including 'system thread pool', multiple 'trade groups demo pool' entries, and a 'sessions dispatcher'. At the bottom, a status bar shows 'For Help, press F1' and a memory usage indicator '3280 / 2 KB'.

Thread Name	CPU, %	CPU Cycles	Context ...	Prior...	Stack, kb
system thread pool	9.44%	372 567 786 633	201 403	8 / 8	116 / 1024
trade groups demo pool	1.73%	5 005 431 013 337	36 691 267	8 / 9	1928 / 2560
trade groups demo pool	1.73%	5 101 835 053 678	35 749 841	8 / 9	72 / 2560
trade groups demo pool	1.57%	5 203 691 773 865	39 677 426	8 / 9	72 / 2560
trade groups demo pool	1.57%	5 179 712 282 766	36 080 713	8 / 9	76 / 2560
trade groups demo pool	1.57%	6 067 315 070 031	36 418 741	8 / 9	1928 / 2560
trade groups demo pool	1.42%	5 195 384 393 896	36 402 071	8 / 9	72 / 2560
trade groups demo pool	1.26%	5 008 475 278 557	36 682 505	8 / 9	1928 / 2560
trade groups demo pool	1.26%	5 208 425 774 001	36 230 097	8 / 9	72 / 2560
sessions dispatcher	0.94%	35 810 378 741 735	37 101 298	8 / 8	4 / 2048
trade groups demo pool	0.94%	4 251 064 133 019	34 921 777	8 / 9	1928 / 2560
trade groups demo pool	0.94%	4 863 991 719 620	35 157 262	8 / 9	72 / 2560
trade groups demo pool	0.94%	4 804 957 959 538	35 417 002	8 / 9	76 / 2560
trade groups demo pool	0.94%	4 897 485 990 626	35 286 565	8 / 9	76 / 2560
trade groups demo pool	0.79%	4 479 510 164 315	35 022 744	8 / 9	1928 / 2560

For Help, press F1

3280 / 2 KB

Examples of threads utilized by the server:

- automation task execution of tasks in the Automations service
- logger operation logs
- trade group tick apply pool application of new ticks to a group: updating trading account states, updating prices, and recalculating position floating profits
- trade groups demo pool processing of demo accounts
- trade groups real pool processing of real accounts
- trade request check validation of incoming requests
- trade request process - processing of trade requests.
- trade request route distribution of trading requests in accordance with the routing rules

The following information is available for each thread:

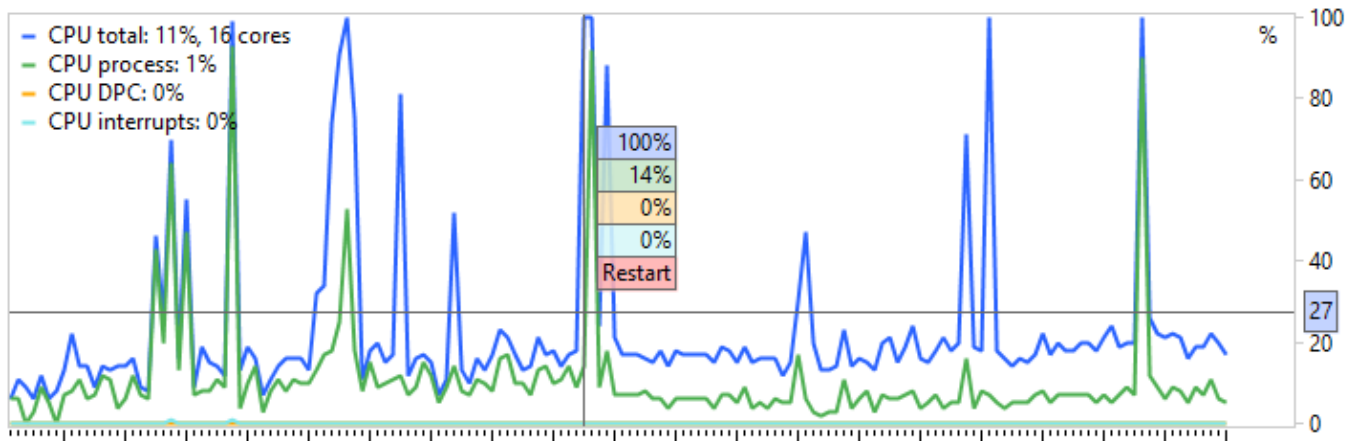
- CPU CPU usage by the specified process
- CPU Cycles the total number of computational cycles spent by the processor to service the process, per second. The higher this metric, the more actively the processor is used.
- Context switches the total number of context switches. High values (1000 or more) may indicate too many active threads in the system. They are trying to access the CPU time, and the system has to switch too often between them, thus wasting resources. For further details please read the Microsoft Documentation.
- Priority base and dynamic process priority
- Stack the amount of used and allocated memory stack
- Kernel Time kernel mode operating time An increase in this metric compared to the time spent in user mode can indicate system-level issues: problems in drivers, hardware errors or slow hardware. For further details please read the Microsoft Documentation.
- User Time user mode operating time
- ID thread identifier.

Use the context menu to set the refresh frequency for thread information. Three modes are available: slow (500 ms), normal (1 s), and pause.

Restarts and crashes

When restarting the server, some load metrics may increase significantly because a lot of service operations occur at this time. To better understand the reasons for such load surges, restart moments are displayed on the monitoring graphs as red vertical lines. If you do not need this information, you can hide it using the context menu.

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The monitoring graphs also display the moments of server crashes, which are shown as purple vertical lines. Since this is critical information, crash data cannot be disabled. You should track such crashes and find out the causes. One of the most common reasons is the incorrect operation of plugins.

For all servers, a log entry with information about the previous crash is added to the journal at startup:

Startup	last session that started 2024.05.21 09:57:31 crashed
---------	---

Trade (for Trade Servers)

- **Users** the total number of accounts on the server.
- **Real** the number of live accounts on the server.
- **Online** the number of accounts, which are currently connected to the server (in the minute of measurement).
- **Errors** the number of errors in connecting to the server during the minute of measurement (such as invalid password and others).
- **Deals** the total number of deals executed on the server per minute.
- **Real deals** the number of deals performed on real accounts per minute.
- **Requests** the total number of trade requests executed on the server per minute.
- **Real requests** the number of trade requests from real accounts executed on the server per minute.
- **Average and maximum request time** time in milliseconds between the addition of a trade request into a server queue and its execution. Statistics on all requests.
- **Average and maximum real request time** time in milliseconds between the addition of a trade request into a server queue and its execution. Statistics on requests sent from accounts belonging to real groups.

A separate group of diagrams show the statistics of trade request failures. Analyze them to improve operational control and customer service quality. For example, by analyzing the diagrams, you can easily see if your routing rules or gateways began to work incorrectly.

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- **Rejects on request checks** the request did not pass the validation check: incorrect volume, stop levels, etc. The full list of checks is available in the [MetaTrader 5 API documentation](#) (clause 4).
- **Rejects on requotes** the request was rejected due to a significant price change. Such rejections can only appear during [Instant Execution](#).
- **Rejects on processing** an error occurred while processing the request by the server.
- **Rejects on funds shortage** the client did not have enough money to fulfill the request.
- **Rejects on routing rules** the request was rejected according to the [routing rule](#).
- **Rejects on dealers** the request was rejected by a dealer via the Manager terminal or Manager API.
- **Rejects on gateway** the request was routinely rejected by the gateway, for example, if an external system refuses to execute it.
- **Rejects on gateway processing** an error occurred while processing the request by the gateway.

When you switch to higher timeframes, the graph shows the highest values (for accounts) and total values (for errors, deals and requests) for the selected period.

History (for History Server)

- **Ticks** the number of ticks received per minute.
- **Depth of Market** the number of received changes in the Market Depth.
- **Trade statistics** the number of changes in the trade statistics received per minute.
- **Data sources** the number of data sources running at the time measurement.
- **Gateways** the number of gateways running at the time measurement.

When you switch to higher timeframes, the graph shows the highest values (for data feeds and gateways) and total values (for price data) for the selected period.

Performance parameters in the servers journal

Apart from the graphs, the performance parameters are also displayed in [each server's journal](#). Such entries can be requested using "Monitor" keyword.

Sample entries:

```
2018.11.23 02:51:01.657    Monitor
connections: 0, cpu: 12%, process cpu: 0%, threads: 1369, process threads: 94, handles: 233
2018.11.23 02:51:01.657    Monitor
net in: 272 kb/s, 16366 kb, net out: 17 kb/s, 1077 kb, retransmit: 0.109% (22 of 20081 packe
2018.11.23 02:51:01.657    Monitor
ticks: 17755, requests: 0 (real: 0), executions 0 (processed: 0), deals: 0 (real: 0, auto: 0, man
2018.11.23 02:51:01.657    Monitor    users: 6999, real: 8, online: 0
2018.11.23 02:51:01.657    Monitor
transactions symbols: 0, groups: 0, users: 0, orders: 0, history: 0, deals\positions: 0, request
2018.11.23 02:51:01.657    Monitor    trade request processed income: 0, checked: 0, lo
```

MetaTrader 5 trading platform

2018.11.23 02:51:01.657	Monitor	trade request queue income: 0, checked: 0, waiting:
2018.11.23 02:51:01.657	Monitor	base cache miss users: 0, deals: 0 months: 0, history:
2018.11.23 02:51:01.657	Monitor	total packets: 20, external contexts: 8, internal contexts:
2018.11.23 02:51:01.657	Monitor	memory available: 40697 Mb, memory commit: 3362 Mb, max region: 40697 Mb, system commit:
2018.11.23 02:51:01.657	Monitor	allocators: 102216 Kb (fragments: 59, used 92270 Kb)
2018.11.23 02:51:01.657	Monitor	clients index: 28 kb, search index: 15 Kb, cache: 0 Kb
2018.11.23 02:51:01.657	Monitor	documents index: 26 kb, search index: 26 Kb, cache: 0 Kb
2018.11.23 02:51:01.657	Monitor	comments index: 5 kb, search index: 4 Kb, cache: 0 Kb
2018.11.23 02:51:01.657	Monitor	files index: 26 kb, search index: 15 Kb, cache: 0 Kb

The entries contain the following data:

- **connections** number of the current server connections.
- **trades** number of trades since the last time the status data has been collected.
- **blocked** the number of connections that were blocked by the Antiflood Control system.
- **cpu** current CPU load in %.
- **process cpu** current CPU usage by the server process in %.
- **dpc cpu** CPU load when handling deferred procedure calls. Please see the the Processor section for further details.
- **interrupt cpu** CPU load when handling interrupts from devices. Please see the the Processor section for further details.
- **threads** number of threads.
- **process threads** number of threads used by the server process.
- **handles** number of system handles.
- **process handles** number of system handles used by the server process.
- **disk queue** disk queue length.
- **avg. transfer** disk average latency.
- **net in** incoming traffic volume (in kilobits) on the adapter selected for measurement.
- **net out** outgoing traffic volume (in kilobits) on the adapter selected for measurement.
- **retransmit** number of resent packets.
- **sockets** total number of sockets in the operating system.
- **ticks** number of ticks received per minute.
- **requests** the total number of trade requests executed on the server per minute (of which, requests from real customers).
- **executions** number of trade executions received per minute (of which processed).
- **deals** the total number of deals executed on the server per minute (of which, deals from real clients, those processed automatically, processed manually and processed by the gateway).
- **request time avg** average request processing time in milliseconds: for all requests and for requests from real accounts).
- **single tick apply** tick processing data:
 - ♦ **avg: W / X us calc, max: Y / Z us calc** average and maximum time for applying one tick / time associated with the recalculation of the account trading status and of floating profit for orders and positions. Specified in

microseconds.

- ◆ **package avg: W / X us calc, max: Y / Z us calc** average and maximum time of applying a package of ticks (multiple ticks arriving simultaneously) / time associated with the recalculation of the account trading status and of floating profit for orders and positions. Specified in microseconds.
- ◆ **delay avg: X / Y us max** average and maximum tick delay. Calculated as the difference between the tick time and the current time when the tick is processed. Specified in microseconds.
- **threads tick apply** data on group tick processing threads, used for service purposes.
 - ◆ avg: X / Y us max average / maximum time for calling the tick processing method. Specified in microseconds.
 - ◆ period avg: X / Y us average / maximum period between calls of the tick processing method. Specified in microseconds.
 - ◆ ticks avg: X / Y max average / maximum number of ticks processed in the tick processing method.
- **users** the total number of accounts on the server (of which real, of which currently online).
- **transactions** number of changes performed in symbol configurations, account details, open orders, closed orders, deals\positions and trade requests.
- **trade request processed** number of processed trade requests: number of incoming, verified requests and those captured from the queue.
- **trade request queue** the state of the trade server request queue. An increase in the queue may indicate that the server does not have enough resources to process requests on time. income incoming requests which has not yet been processed; checked the requests which have been checked; waiting requests awaiting execution, locked requests captured by dealers and gateways for processing.
- **automation trigger queue** the number of triggered automation triggers. activated the number of activated tasks; action queue the number of tasks awaiting action; executed the number of executed requests.
- **users, deals, history, daily, report** size of databases of users, deals, orders, daily reports and server reports. This section also provides information about the number of entries in the databases and the cache size (entries from databases which the server keeps in memory).
- **base cache miss** number of misses when accessing the databases of users, deals, historical orders and daily reports. A miss occurs when the server accesses data in the cache, but such data is not available in the cache (for example, they were pushed out by other data). In case of a cache miss, the server is forced to access the disk and load the requested data into the cache; this reduces the overall system performance. The number of misses when accessing the appropriate monthly database is shown in 'months' (data about users and trade operations are stored on a disk, in separate files by months). In fact, this is the number of calls to monthly databases on the disk. A large number of misses may indicate an insufficient RAM amount and non-optimal access to databases. A large number of misses during server start is a normal behavior (there is no data in the cache yet at this time).
- **index cache state** the record is displayed when a database index is replaced from the cache. This happens when the server needs to load new data into the

cache (the index of the database that the user has accessed), but there is not enough space for this data. In this case, currently unused data is unloaded from memory. Statistics of the unloaded index is written to the server log: name, number of hits, last access time and size.

An example of such logs is shown below.

- **clients index** client database index size, size of its search indexes, amount of client data in the cache and number of misses (similar to 'base cache miss').
- **documents index** document database index size, size of its search indexes, amount of document data in the cache and number of misses (similar to 'base cache miss').
- **comments index** comment database index size, size of its search indexes, amount of comment data in the cache and number of misses (similar to 'base cache miss').
- **files index** file database index size, size of its search indexes, amount of file data in the cache and number of misses (similar to base cache miss).
- **total packets** number of network packets in the system at the time of data collection.
- **external contexts** external connections.
- **internal contexts** internal connections.
- **session commands** internal parameter displaying the number of unprocessed commands.
- **async requests** number of asynchronous requests.
- **memory available** the lesser of two values:
 - ◆ total size of free memory blocks available for the process,
 - ◆ size of the available (free) physical memory.
 - ◆ **memory commit** total volume of memory blocks allocated to the process.
 - ◆ **max region** size of the maximum continuous free memory block (overall memory fragmentation parameter).
 - ◆ **system commit** total volume of memory blocks allocated for the operation of the operating system.
 - ◆ **allocators** data on the memory internal status (this parameter is used only by the developers).

Performance metrics are measured and written to the journal once a minute. Performance measurements of different platform components are not synchronized (they are performed in different times).

A log example with data about server operations with the trade cache

Trading history is stored on the server disk grouped by months. When data of a certain month is accesses, the server loads the minimum month description, i.e. the index, to its memory. The collection of monthly indexes loaded into server memory is the monthly data cache. The cache information is periodically printed to the server log:

07:00:05.119 Monitor deals months: 36646 Mb / 33 records, month miss: 0, cache: 15077

This record shows information on the database itself (the first part) and on the cache (the second part): size, number of months and number of missed access attempts.

Cache misses occur when the required month is not in memory, and its index has to be loaded from disk - this can possibly replace another month due to the limited memory.

When data is replaced, statistics of each month is printed to log:

```
index cache state: deals_YYYY.MM.dat: index cache state: XXX hits, last access: XXX, size: X
```

The log displays the name, the number of hits, last access date and size.

The presence of three or more misses in a monthly cache ('months miss') or 500 or more misses in the read cache ('cache miss') may indicate:

- The need to increase memory for the server index data take up a lot of space, therefore, the cache can contain few months. Before making a decision, you should additionally check how many months are loaded into memory.
- A client (plugin\report\manager\terminal), who accesses deep trading history or requests a bulk of deals from different months. In this case, you should find the source of such requests.

Context Menu

Using the context menu the following commands can be executed:

- **Overview** switch to the overview mode, where the four main parameters of the server operation are shown;
- **Connections** switch to viewing statistics on server connections;
- **Free memory** switch to viewing statistics of free memory;
- **CPU usage** switch to viewing statistics on CPU usage;
- **Network usage** switch to viewing statistics on network usage;
- **Others** open the submenu for selecting additional parameters;
- **5 minutes** switch the graph period to 5-minute;
- **15 minutes** switch the graph period to 15-minute;
- **1 hour** switch the graph period to hour;
-  **Save as Picture** save the current diagram as an image (*.bmp, *.gif or *.png);
- **Grid** show/hide grid in charts.

MetaTrader 5 trading platform

MetaTrader 5 Trading Platform / Platform Setup / Network cluster /
Journal

Journal

On this tab you can view logs of work of the platform components.

The screenshot shows the 'ForexBroker-Demo - MetaTrader 5 Administrator' window. The left sidebar contains a tree view with categories like Servers, Integrations, Automations, Security, Time, Holidays, Groups (41), Clients & Accounts, Orders & Deals, Gateways, Data Feeds, Plugins, Reports, and ECN. The 'Servers' category is expanded, showing 'Forex Broker Server' and 'Network Cluster'. The 'Network Cluster' is further expanded, showing 'MetaTrader 5 Main', 'History Server', and 'Access Server'. The main panel displays a log table with columns: Time, IP, and Message. The log entries show various system events, including updates, scheduler tasks, and hardware information. At the bottom, there are tabs for Status, Overview, CPU, Memory, Disk, Connections, Network, and Journal (which is selected). The status bar at the bottom indicates 'For Help, press F1' and '19.6 / 0.0 M'.

Time	IP	Message
2021.07.07 11:12:30.865	LiveUpdate	'mt5clwide64.2885' loaded [23107 kb]
2021.07.07 11:12:30.929	LiveUpdate	'mt5clwtst64.2885' loaded [10288 kb]
2021.07.07 11:12:31.024	LiveUpdate	'mt5man64.2885' loaded [14160 kb]
2021.07.07 11:12:31.024	LiveUpdate	5 updates loaded in 594 ms
2021.07.07 11:12:31.024	Scheduler	server state collection started
2021.07.07 11:12:31.024	Monitor	Operating system: Windows 10 Pro for Wor...
2021.07.07 11:12:31.024	Monitor	Processor: 20 x Intel Xeon E5-2630 v4 @ 2.2...
2021.07.07 11:12:31.024	Monitor	BIOS: DELL - 1072009
2021.07.07 11:12:31.024	Monitor	Memory: 55731 / 65457 Mb
2021.07.07 11:12:31.037	Monitor	Pagefile: automatically managed
2021.07.07 11:12:33.095	Monitor	Disk: 425971 / 488256 Gb
2021.07.07 11:12:33.097	Monitor	Network: Adapter 'Intel[R] Ethernet Connect...
2021.07.07 11:12:33.351	Monitor	Event Log: no blue screen errors
2021.07.07 11:12:33.351	Scheduler	server state collection finished
2021.07.07 11:12:33.351	Startup	initial synchronization with main server start...
2021.07.07 11:12:33.351	Startup	startup thread finished successfully

In the journal, entries are marked by different icons:

-  error message.
-  critical error message.
-  warning.
-  information message.
-  day separator in the journal. In the IP field of such entries the total size of logs generated for the day is shown.

Log files for all servers except the backup server can be found at [server installation directory]\logs. The backup server logs are available at [backup server installation directory]\logs\mtbackup.

Different types of errors returned by the server are described in the [separate section](#).

Requesting Entries

You can also request only some of journal entries. In order to do this, enter the searched word in the search field. Additionally you can specify the time period and type of log request:

- **Full** all types of journal entries;
- **Without logins** all messages except the information about the connections of users to a server;
- **Errors only** only messages about errors.

You can also select the type of events:

- **All** all types of events, except LiveUpdate;
- **Configuration** changes in the server configuration;
- **System** system events;
- **Network** events connected with the network activity;
- **History** events connected with [history data](#);
- **Accounts** events connected with [accounts](#) on the server;
- **Trades** events connected with trade operations on the server;
- **API** events connected with work via API;
- **LiveUpdate** events connected with [update](#) of the platform components;
- **Report Mailer** events connected with sending [daily reports](#) via e-mail;
- **Failover** events connected with [automatic switching to a backup server](#).

- Please note that the request of logs is performed only by the server that is chosen in the tree to the left.
- The LiveUpdate logs are not displayed when choosing the "All" type of events, because they are stored in a separate file (mt5srvupdater.log).
- For a more accurate request of journal entries, use [keywords](#).

To receive logs press "Request" or execute the same command of the context menu. After that the logs will be displayed as a table with the following fields:

- **Time** time when the event occurred;
- **IP** IP address, with which this event id connected;
- **Message** description of the event.

If the journal file was manually edited on the server, then the first edited entry and all coming after it will be highlighted red.

Request Language

To request entries from the journal of the server operation, keywords and logic operators are used. It will help to easily find the required information about different aspects of the platform functioning.

Keywords

To analyze the server journal use any words and word phrases that can be found in server logs. You should remember that if you request the full journal without the request line, you will not receive the whole server journal, because the volume of given information is limited (approximately up to 30000 lines of the server journal).

To analyze separate situations use the following key words:

- **Startup** information about server initialization and restart;
- **Exit** information about server stop;
- **Synchronization** synchronization of history. Possible errors: no connection with the server you want to synchronize with;
- **ClientBase** information on work with the base of accounts. Possible errors: incorrect base header, error writing/reading from disk, not enough memory;
- **History** information on work with history data of a symbol. Possible errors: incorrect base header, error writing/reading from disk, incorrect quotes in the base;
- **datafeed** or '**Datafeed name**' activation and work of datafeeds. Possible errors: error creating feeder, incorrect timeout stop, error reading data from the feeder;
- **Filter** filtering of quotes (quotes no passed to the server);
- **Mail** work with the mail base. Possible errors: incorrect base header, error writing/reading from disk, not enough memory;
- **Monitor** performance measuring and work with the monitoring base. Possible errors: incorrect base header, impossibility to obtain information about the PC performance;
- **News** news base. Possible errors: incorrect header or base version;
- **Access, Common, Feeders, Groups, Managers, SymbolGroups, HistorySync, Holidays** files of corresponding server configurations. Possible errors: read/write error;
- **Network** server operation. Possible errors: port already busy;
- **LiveUpdate** work of the LiveUpdate system and its base;
- **block** block setting and removing by antiflood control;

- **'...'** to analyze actions on a certain account, specify the account number in single quotes. For example: '12345';
- **#...** to analyze actions on a certain order or deal, specify this order with symbol "#". For example: #111222.
- **login** connection with an account to the server. These entries contain information about connection: time, type (client, manager, web, etc.), terminal build and other details.
- **logout** disconnection of an account from the server.

Logical Operators

You can specify key words joined by logical operators in the request line. The search line supports the following logical operators:

- **| (or)** logical operator OR allows selecting journal lines that contain either the first or the second key word;
- **& (and)** logical operator AND allows selecting only those journal lines that contain both the first and the second key words. A space between key words is considered logical operator AND;
- **^ (-, not)** logical operator of exclusion NOT allows selecting journal lines that do not contain the specified key word.

The possibility to create a precise request line using logical operators allows quickly and efficiently select required logs. For example, line "5'&login" allows selecting the journal entries about the logging in of a manager with login 5.

The request is case sensitive. For example, search by "Session" and "session" will give different results.

Context Menu

The journal context menu contains the following commands:

-  **Request** request logs;
- **Search** using this submenu, you can easily form new requests on the basis of an already found entry in the journal:
-  **Account** copy an account from the found entry to the request field of the journal;
-  **Order** copy an order ticket from the found entry to the request field of the journal;
-  **Deal** copy a deal ticket from the found entry to the request field of the journal;
-  **Symbol** copy a symbol name from the found entry to the request field of the journal;
-  **Computer ID** copy a unique computer identifier (CID) the found entry to the request field of the journal;
-  **IP** copy an IP address the found entry to the request field of the journal.
- **Copy As** this command allows copying information from the journal to the clipboard. There are three variants of how to save it:  "As Lines", "As List of Logins", "As List of IP Addresses";

- **Firewall** enables the quick addition of firewall rules without leaving the journal page. Select one or more log lines containing the required addresses in the "IP" field. For example, these can be logs about anti-flood control activation. Next, select "Allow" or "Block" in the menu. This will add a new record containing the selected addresses to the end of the list of firewall rules. The "Allow" command adds rules with the "Permit always" type. When adding the rule, the system searches for the selected addresses in existing rules. If the address to be blocked (allowed) has already been blocked (or allowed), a new rule is not created.
- **Export** export the requested logs in a CSV or HTML file;
-  **Find** search through requested logs;
- **Auto arrange** If this option is enabled, the size of columns is selected automatically;
- **Grid** show/hide grid to separate fields;
- **Columns** show/hide columns of IP and Message in the Journal.

MetaTrader 5 Trading Platform / Platform Setup / Integrations

Integrations

The "Integrations" section allows configuring platform integration with external service:

- The Finteza business analytics system which provides audience analysis and new client attraction features
- Sponsored VPS for attracting new customers and increasing trading volumes on existing accounts
- Mail servers for sending trading reports and notifications
- SMS Gateways for phone confirmation during registration and for notifications
- Instant Messengers for phone confirmation during registration and for notifications
- KYC systems for automated client data and document verification
- Web Services for integration with websites and for creating trader rooms

Integrations enable automation of various processes. You can reduce manual operations and improve the quality of services provided to traders.

Service overview

The home section contains stats and graphs revealing key data on using the services:

- Number of user action events in the platform you have tracked via Finteza
- Number of hosting spots you have offered to your traders
- Number of emails and SMS messages you have sent to your clients
- Number of documents automatically checked via the KYC provider

Here you can also see if any of the services has not been configured yet meaning you are not using all the platform features.

Integration

www.mql5.com

Finteza Analytics

✓ License active ID MasterSite ID: specified

i Data: full export

Use end-to-end brokerage analytics for MetaTrader 5.

[More about the service](#)



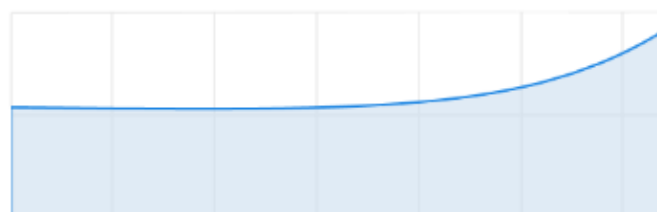
Sponsored VPS

✓ Activated 👤 Groups: 1 Balance:

Latency:

Offer your traders free virtual hosting for round-the-clock trading.

[More about the service](#)



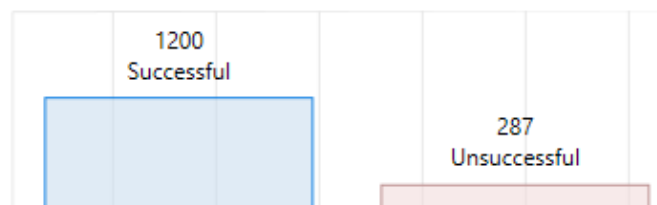
Mail Servers

✓ Used @ Mail accounts: 1

📧 Mail servers: 1

Send newsletters and personal emails to your traders.

[How to set up a mail server](#)



[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / Finteza Analytics

Finteza Analytics

Finteza is the advertising analytics system, using which you can analyze users' behavior on your site and launch advertising campaigns. The system is integrated into the MetaTrader 5 platform, so right in the platform you can analyze your audience and the efficiency of your campaigns, as well as monitor the client's entire life cycle, from his first visit to your site to the first real trading account deposit.

Finteza is integrated with MetaTrader 5, so you can track your traders' actions in terminals. Obtain statistics on which functions and services are the most popular among your traders: algo trading programs, signal copying, account opening and others.

Order MetaTrader 5 Finteza Analytics How to Set Up Analytics

Create a Finteza account

Register a Finteza account directly through the Administrator terminal and receive a full 30-day trial access. Simply specify your name, website address, and corporate email. A tracker will be created in Finteza based on the provided website, which will collect statistics on your traders. You can then integrate Finteza into your website and consolidate its data with your platform statistics to obtain end-to-end analytics.

The tracker ID will be automatically added to the platform settings. Thus, immediately after registration, you will begin collecting data on your traders' activities within the platform: deposits and withdrawals, account requests, utilization of windows, Expert Advisors and indicators, and more. The system provides data on all terminals, including desktop, mobile and web. Use comprehensive analytics to evaluate your clients' Lifetime Value and to identify the most in-demand features.

Broker - MetaTrader 5 Administrator

File

Edit

View

Services

Help

248 / 5

Disconnect

Refresh

Restart Server

Import

Export

Apply

+

⚙️

✖️

←

→

↓ Sort

Administrator

Servers

Broker Server

Network Cluster

Integrations

Finteza Analytics

Sponsored VPS

Mail Servers

SMS Gateways

Messenger Channels

KYC

Web Services

Automations

Security

Time

Holidays

Leverages

Groups (44)

Clients & Accounts

Orders & Deals

Gateways

Data Feeds

Plugins

Reports

Create main account

Company

MetaQuotes Trial Servers

Website

yourdomain.com

Full Name

John Smith

Email

You need to confirm your email to create an account

Password

The password must include an uppercase letter, a capital letter, a number, a special character (@,\$,#,&) and be no shorter than 8 characters

+ Create account

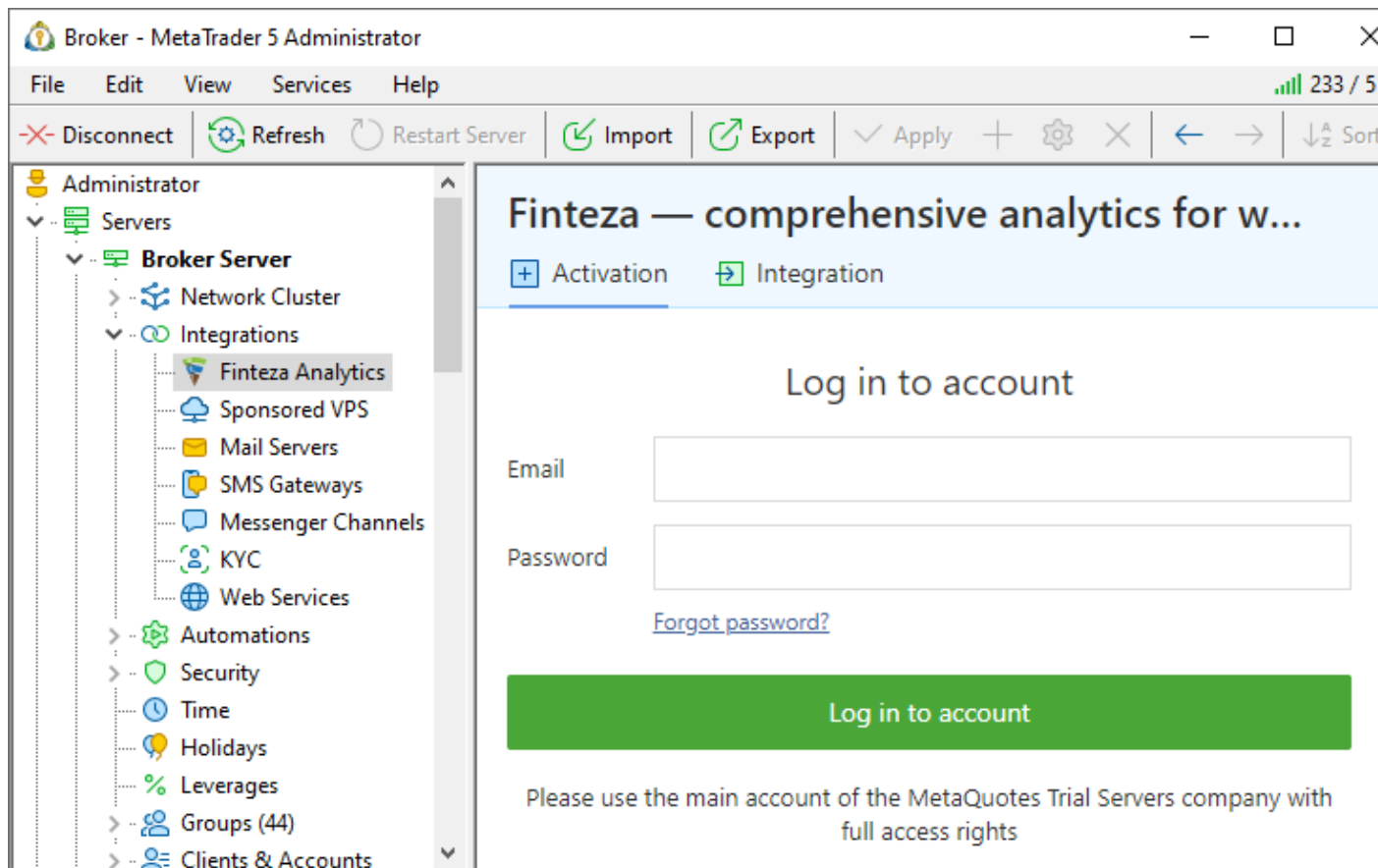
After registration, a master account will be created for your company, with the right to create subordinate accounts

To switch to the full version, [order Finteza](#) via MetaQuotes Support Center. If you wish to extend your trial period, please contact [technical support](#).

Log into your account

Log into your account in the "Activation" section. This will enable the display of primary statistics across websites and platforms directly within the Administrator terminal, without the need to switch to Finteza.

If you have multiple Finteza accounts, use the one with the highest privileges to access the most comprehensive statistics across all projects.

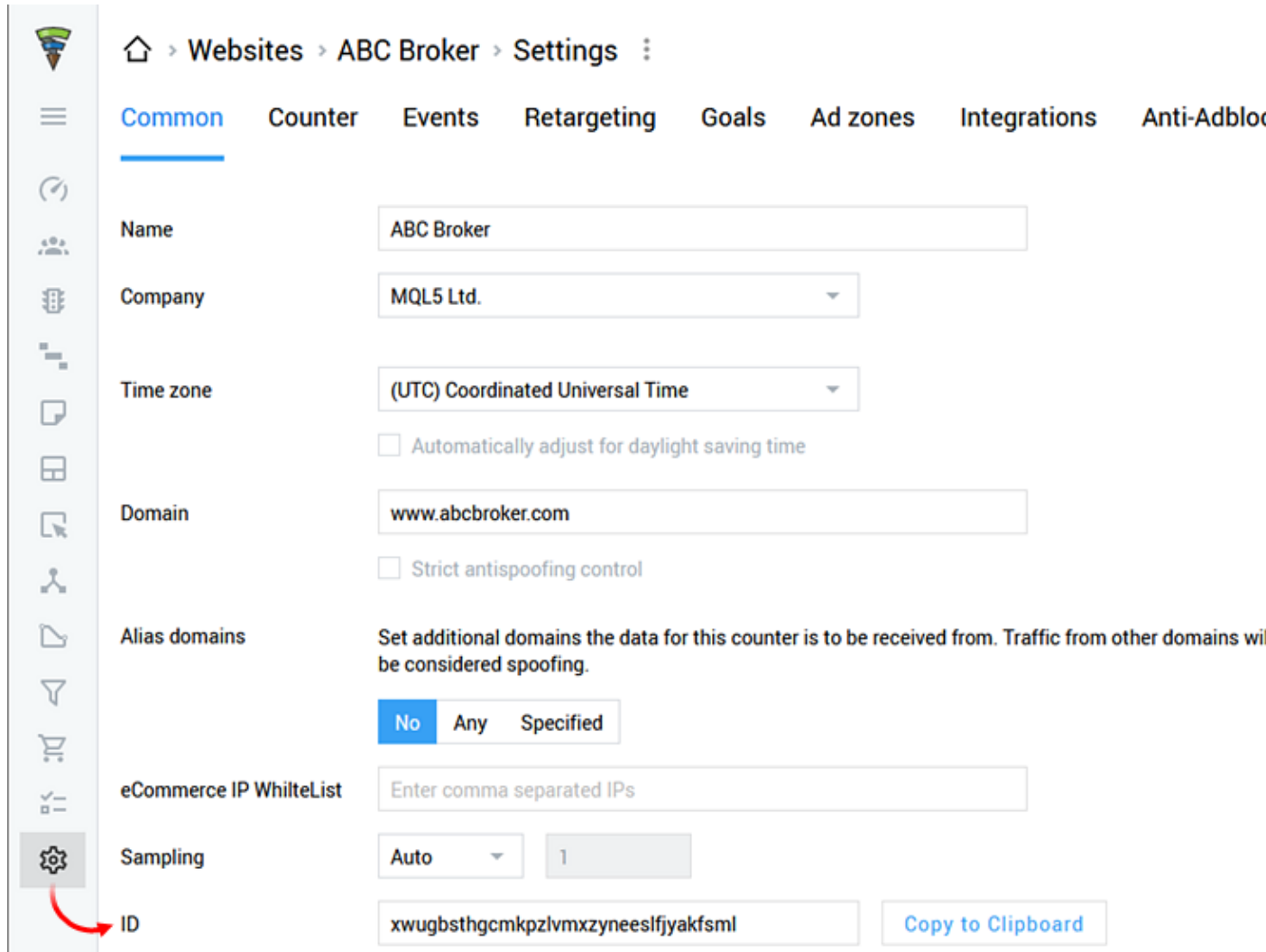


MetaTrader 5 trading platform

After that, the trading platform will start receiving data from Finteza. The main website statistics will be displayed in the Administrator terminal, and more detailed metrics will become available under the "Analytics" section of the Manager terminal. For further details please watch the video "[Finteza Integration with MetaTrader 5](#)".

Manual configuration

If you have a previously registered account with a created tracker, log in and specify the tracker ID. Open the Finteza panel and copy it from the "Settings \ Common" section:



The screenshot shows the 'Settings \ Common' configuration page for 'ABC Broker' in the Finteza interface. The left sidebar contains a menu with icons for various settings, with the 'Settings' icon (a gear) highlighted and a red arrow pointing to the 'ID' field. The main content area has a breadcrumb trail: 'Home > Websites > ABC Broker > Settings'. Below this is a horizontal tab bar with 'Common', 'Counter', 'Events', 'Retargeting', 'Goals', 'Ad zones', 'Integrations', and 'Anti-Adblock'. The 'Common' tab is selected. The configuration fields include: 'Name' (ABC Broker), 'Company' (MQL5 Ltd.), 'Time zone' ((UTC) Coordinated Universal Time), 'Domain' (www.abcbroker.com), 'Alias domains' (Set additional domains the data for this counter is to be received from. Traffic from other domains will be considered spoofing.), 'eCommerce IP WhiteList' (Enter comma separated IPs), and 'Sampling' (Auto, 1). The 'ID' field contains the value 'xwugbsthgcmkpzlvmxzyneeslfjyakfsmi' and a 'Copy to Clipboard' button.

Home > Websites > ABC Broker > Settings

Common Counter Events Retargeting Goals Ad zones Integrations Anti-Adblock

Name ABC Broker

Company MQL5 Ltd.

Time zone (UTC) Coordinated Universal Time

☐ Automatically adjust for daylight saving time

Domain www.abcbroker.com

☐ Strict antispoofing control

Alias domains Set additional domains the data for this counter is to be received from. Traffic from other domains will be considered spoofing.

No Any Specified

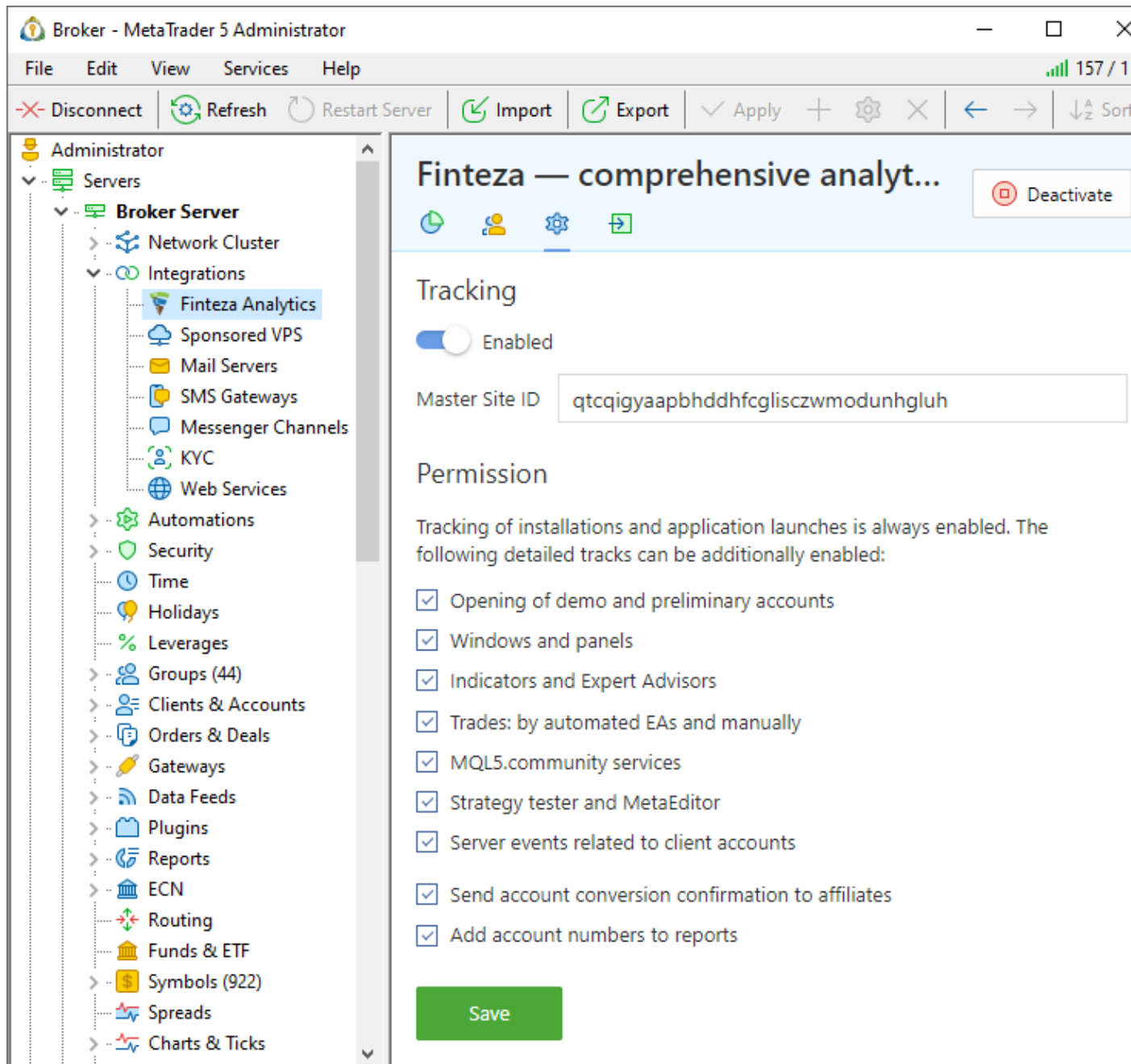
eCommerce IP WhiteList Enter comma separated IPs

Sampling Auto 1

ID xwugbsthgcmkpzlvmxzyneeslfjyakfsmi Copy to Clipboard

MetaTrader 5 trading platform

Add the ID to the "Main Site ID" field in the MetaTrader 5 Administrator settings section:



Add Finteza code to your website

To obtain comprehensive analytics for your website and link them with traders' activities in the platform, install Finteza on your website. This can be done by adding a small code to every page. Open your website in the [Finteza panel](#) and go to the "Settings\Counter" section.

Home > Websites > ABC Broker > Settings

Common **Counter** Events Retargeting Goals Ad zones Integrations Anti-Adblock

Select parameters to generate the counter installation code

- ☐ Track hash in browser address bar
- ☐ Track external links clicks
- ☐ Count exact time on site
- ☐ Proxy the script and requests

Proxy scripts through your website to get precise and secure analytics. Learn how to do that in the [user guide](#).

<--> Insert this HTML-code into all pages of your website (in body section) Send to Email Copy to Clipboard

```
<script type="text/javascript">
  (function(a,e,f,g,b,c,d){a[b]||(a.FintezaCoreObject=b,a[b]=a[b]||function(){(a[b].q=a[b].q||[])
  .push(arguments)},a[b].l=1*new Date,c=e.createElement(f),d=e.getElementsByTagName(f)
  [0],c.async=!0,c.defer=!0,c.src=g,d&&d.parentNode&&d.parentNode.insertBefore(c,d))})
  (window,document,"script","https://content.finteza.org/core.js","fz");
  fz("register","website","xwugbsthgcmkpzlvmxzyneeslfjyakfsm1");
</script>
```

To accelerate the tracking script download, it is recommended to add the [X-DNS-Prefetch-Control](#) instruction to the header of all website pages.

Copy the tracking code and paste it into all pages of your site in the `<head></head>` tags. This step can be completed by your web developer. For further details please see the [Documentation](#).

Immediately after integration, you will start receiving data on page visits, unique and non-unique users, traffic sources and quality, and more.

Add event tracking to your site

Finteza allows you to track any events on your website, such as link or button clicks, form submissions, page scrolls, and more. To do this, add a simple function call to the desired location on your site:

```
fz( "event", "{EVENT_NAME}" );
```

Instead of `{EVENT_NAME}`, specify a certain event name, for example "Registration" or "Create account". This name will be displayed in Finteza reports.

Here's an example of how to create an event for a link click and form submission:

```
<a href="https://www.example.com/" onclick="fz('\event', '\Click+Link\'); return true;">www.example.com</a>
```

or

```
<a href="https://www.example.com/"  
data-fz-event="Click+Link">www.example.com</a>
```

```
<form action="" method="get" onsubmit="fz('\event', '\Form+Order+Submit\');  
return true;">
```

...

```
</form>
```

For further details please see the [Documentation](#).

If you have your own mobile application, you can also track events within it. Finteza offers [ready-made SDKs for iOS and Android](#).

Statistics on your trackers

Directly from MetaTrader 5 Administrator, you can track key metrics for your trackers and advertising zones, including the number of visitors, page views, and events. The relevant data is available under the "Finteza Analytics \ Statistics" section.

Finteza — comprehensive analytics for websites, apps and traders							Deactivate
Statistics Users Settings Integration							
Project	Visitors	Pageviews	Events	Online			
 Broker Website www.broker.com	 1.5 M +6%	 4.6 M -8%	 18.7 M -7%				7 70
 Broker Website www.broker.com	 102.1 K +4%	 332.4 K -6%	 576.8 K +8%				34
Ad zone	Clicks	Impressions	CTR	Requests			
 Header Banner Desktop/Tablet www.broker.com	 5 913 +1%	 2.6 M -9%	 0.23% +10%				2.7 M -9%
 Footer Banner Desktop/Tablet www.broker.com	 1 267	 1.9 M -9%	 0.07% +17%				2 M -9%
 Left margin www.broker.com	 1 010 -10%	 1.5 M -9%	 0.07%				1.6 M -8%
 Mobile Installer www.broker.com	 9 149 -11%	 1 M -10%	 0.88%				1.2 M -10%
 Mobile Banner www.broker.com	 15.7 K -10%	 986 K -10%	 1.59%				1 M -10%

Manage employee access

From the "Users" section, you can manage employee access to Finteza and create new accounts. Use the context menu to open and view user profiles in Finteza, enable and disable users, and manage passwords.

User	Name	Last access
John.Smith@mail.com	John Smith	2024.09.20 16:05
jane.wilson@mail.com	Jane Wilson	
joe.black@mail.com	Joe Black	

- Open Profile
- Disable
- Change Password

To create a new account, click "Add User" at the top of the section and enter the basic information: name, position, email and password.

Finteza — comprehensive analytics for websites, apps and...

 Deactivate Statistics Users Settings Integration

Add User

Full name

John Smith

Position

Manager

Phone

+112496877563

Language

English

Email

john.smith@mail.com

Password

.....

Confirm password

.....

The password must include an uppercase letter, a capital letter, a number, a special character (@,\$,#,&) and be no shorter than 8 characters

Add

Cancel

Server and Client Events

Finteza automatically accumulates data on your terminal installations and launches. All such events are linked to the website specified in the "Master Site ID" field. These events can be viewed at any time under the "Events" section of the Finteza panel. This data is only available to you.

To enable tracking of additional events, use the "Integration \ Finteza Analytics \ Settings" section in the Administrator terminal:

- Demo and preliminary account opening
- Opening of windows and panels inside the terminal
- Launch of indicators and Expert Advisors
- Trades: performed automatically by trading robots and manual trades
- Usage of MQL5.community services
- Usage of Strategy Tester and MetaEditor
- Server events related to client accounts: live account creation, moving of a preliminary account to a real group and first account deposit

The screenshot shows the 'Broker - MetaTrader 5 Administrator' window. The left sidebar displays a tree view of the server structure, with 'Finteza Analytics' selected under the 'Integrations' folder. The main panel on the right is titled 'Finteza — comprehensive analyt...' and contains the following settings:

- Tracking:** A toggle switch is set to 'Enabled'.
- Master Site ID:** A text field containing the value 'qtcqigyaapbhdhfcglisczwmodunhgluh'.
- Permission:** A section with a descriptive text and a list of checkboxes:
 - Tracking of installations and application launches is always enabled. The following detailed tracks can be additionally enabled:
 - ☒ Opening of demo and preliminary accounts
 - ☒ Windows and panels
 - ☒ Indicators and Expert Advisors
 - ☒ Trades: by automated EAs and manually
 - ☒ MQL5.community services
 - ☒ Strategy tester and MetaEditor
 - ☒ Server events related to client accounts
 - ☒ Send account conversion confirmation to affiliates
 - ☒ Add account numbers to reports
- Save:** A green button at the bottom of the settings panel.

MetaTrader 5 trading platform

All events, except for the last point, are sent to Finteza by client terminals. When connected to your trading server, the terminals request the relevant settings and sends the appropriate events to the specified tracker. Server events are sent to Finteza by the trading server.

All event names start with "MetaTrader 5". An additional specification is used for server events: "MetaTrader 5 Trade Server".

☐ Tracked event	Value	Unit	Visitors ▼	Events
☐ MetaTrader 5 Trade Server Account Real Deposit		srv	1 541 -48%	5 648 -9%
☐ MetaTrader 5 Trade Server Account Real Withdrawal		srv	703 -59%	1 098 -9%
☐ MetaTrader 5 Trade Server Account Real Create		login	659 -50%	1 244 -9%
☐ MetaTrader 5 Trade Server Account Demo Confirm No Unique		login	403 -58%	509 -6%
☐ MetaTrader 5 Trade Server Account Demo Confirm Unique		login	354 -89%	400 -8%
☐ MetaTrader 5 Trade Server Account Preliminary Confirm Unique		login	47 -64%	48 -6%
☐ MetaTrader 5 Trade Server Account Real Link		login	39 -22%	159 -1%
☐ MetaTrader 5 Trade Server Account Real Convert		login	15 -40%	16 -4%
☐ MetaTrader 5 Trade Server Account Preliminary Confirm No Unique		login	6 -63%	7 -6%

To view detailed statistics, enable the option "Add account numbers to reports". In this case, the relevant account number will be added to the "Value" field (accordingly, "Account" will be indicated as "Units") of each tracked event. If the option is disabled, only your server name will appear as an additional parameter for each event in Finteza.

Finteza automatically associates server and client events with user actions on your site. This action chain provides an in-depth end-to-end analytics. View the entire journey of your traders, from their first visit to your website, to opening of a real account, deposits and actions performed in the terminal.

Additional settings for mobile terminals

Add the `master_site_id` parameter to the [mobile terminal download link](#). Use your Finteza tracker ID as the parameter value. For example:

```
https://download.mql5.com/cdn/mobile/mt5/ios?server=ABC-Demo,ABC-Real&utm_campaign=
https://download.mql5.com/cdn/mobile/mt5/android?server=ABC-Demo,ABC-Real&utm_campaign=
```

The parameter effect is similar to the "Main website ID" under the [Integration\Finteza Analytics](#) section. It specifies to which of your Finteza trackers the client terminal should send information about its actions. However, the server parameter only reaches the terminal after the user logs in using their trading account. Therefore, during the first launch, the terminal is unaware of the receiving Finteza tracker, and thus it cannot send the successful installation and account registration start tracks.

MetaTrader 5 trading platform

The master_site_id parameter, passed in the deep link, fills this gap.

Last event	City	Device	Application	Tracked event
2023.07.03 09:30:51	United Kingdom, London	Apple iPhone 12	MetaTrader 5 iPhone	MetaTrader 5 iOS View MarketWatch
2023.07.03 09:30:50	United Kingdom, London	Windows 10	MetaTrader 5 Trade Server	MetaTrader 5 Trade Server Account Demo Con
2023.07.03 09:30:50	United Kingdom, London	Apple iPhone 12	MetaTrader 5 iPhone	MetaTrader 5 iOS Account Real
2023.07.03 09:30:50	United Kingdom, London	Apple iPhone 12	MetaTrader 5 iPhone	MetaTrader 5 iOS Run Start
2023.07.03 09:30:50	United Kingdom, London	Apple iPhone 12	MetaTrader 5 iPhone	MetaTrader 5 iOS Install Finish

If the link does not contain master_site_id, the two events highlighted in the screenshot below will not be registered. You will not be able to track your traders' first actions which precede the account opening step. Without these events, you cannot accurately determine the number of installed terminals, which is one of the key traffic acquisition metrics. Furthermore, you will not be able to measure the app install to real account conversion rates.

Working with Affiliates

To manage affiliate programs, enable the "Send account conversion confirmation to affiliates" option. An additional utm_affiliate_site parameter will be added to all tracked events sent from your terminals to Finteza. Thus, you can track the efficiency of your affiliates in terms of client attraction, using a special Finteza panel section: "Sources / Referral".

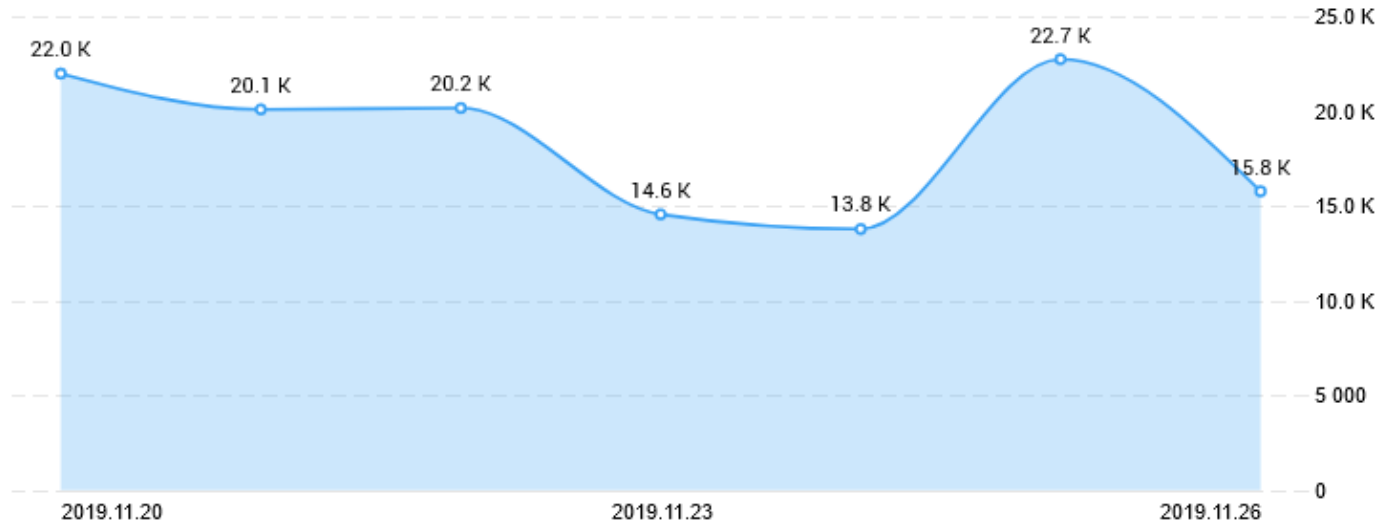
MetaTrader 5 trading platform

Home > Websites > ABC Broker > Sources > Referral

Saved filters

Today Week Month 3 months Year All < 2019.11.20 – 2019.11.26 >

Visitors 124.0 K -5% New visitors 0 Pageviews 0 Pages/Session 0.00 Sessions 0



All traffic UTM Source Campaign Medium Term Content Referrer

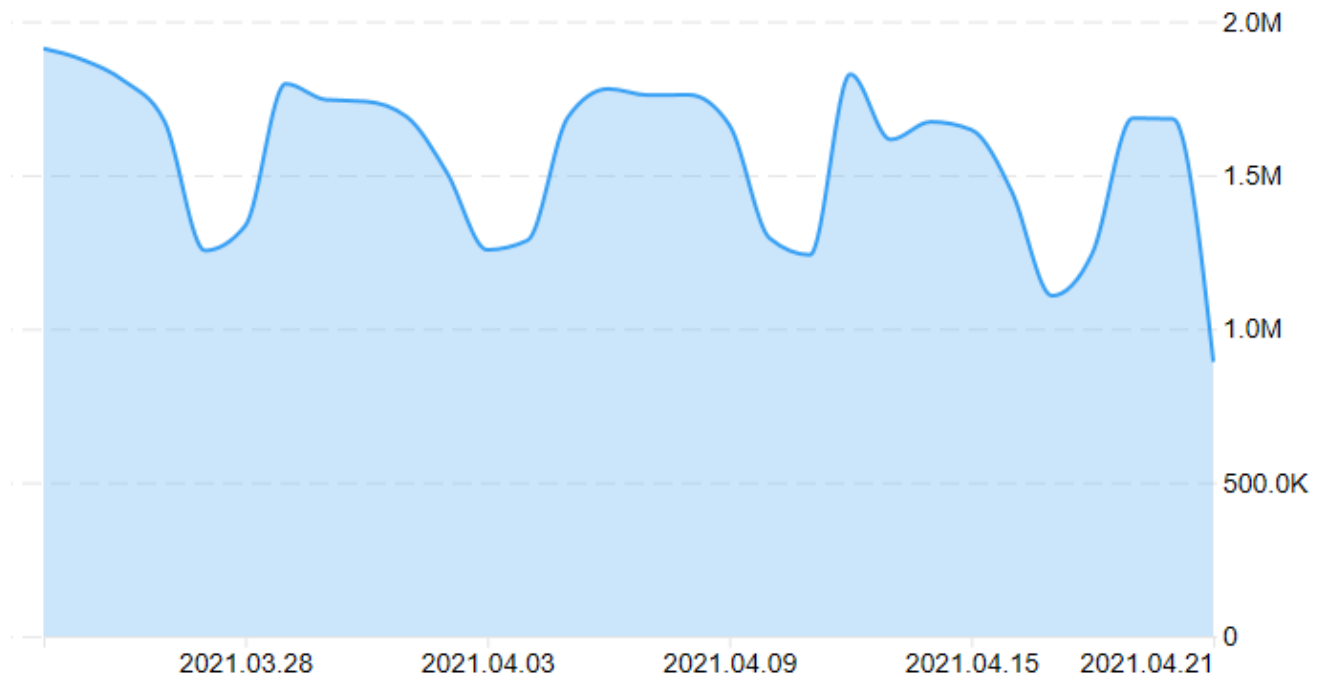


☐ Referral	Visitors ▾	%	Pageviews
☐ www.affiliate-site-1.com	28.3 K +4%	22.85%	56.1 K
☐ www.affiliate-site-2.com	19.2 K -11%	15.46%	32.3 K
☐ www.affiliate-site-3.com	13.6 K -2%	10.96%	21.6 K

Tracked Events Chart

The bottom of the section features a summary chart on the total number of events sent to your tracker by the trading server and client terminals.

Dynamics of tracks related to user actions in terminals and on your websites



Manager Access Control

To grant your managers appropriate access permissions to Finteza data, use [manager account settings](#). The following access levels are provided: access to the integration section, access to page views, marketing campaigns and reports.

Trader tracking

If you have an active [Finteza subscription](#), additional data will be available to you in the account and client dialogs:

- **Visitor ID** – a unique identifier assigned to a user when he/she installs your terminal or visits your site, if a Finteza tracker is installed in it. With this tracker, you can trace trader behavior, from the first website visit to a real account deposit.
- **Affiliate** – the name of the website from which the trading terminal installer was downloaded.

Client: 2097 - John Smith

Client

Personal

Address

Client

Type

Individual

Account: 1819052, John Smith, USD, 1:100, Hedge

Overview

Personal

Account

Profile

Subscriptions

Security

John Smith, 1819052, demo\demo forex-5-hedged, 1 : 100

India, Mumbai

+1293992297331 MetaQuotes ID: 4B676913, Visitor ID: 4930739665660928282, Affiliate: abcbroker.com

john.smith@gmail.com

Registered: 2021.01.06 Last access: 2021.01.06 05:58 Last Address: 124.123.165.82

Symbol	Type	Volume	Price	S / L	T / P	Price	Swap	Profit
eurjpy	buy	0.01	126.415	0.000	0.000	129.135	0.00	24.82
eurjpy	sell	0.01	126.372	0.000	0.000	129.175	0.00	-25.57
Balance: 3 003.13 USD Equity: 3 002.38 Margin: 36.88 Free Margin: 2 965.50							0.00	-0.75
Margin Level: 8 140.94 %								

Client Properties...

Update

Cancel

Help

Visitor ID

4930739665660928282

Copy

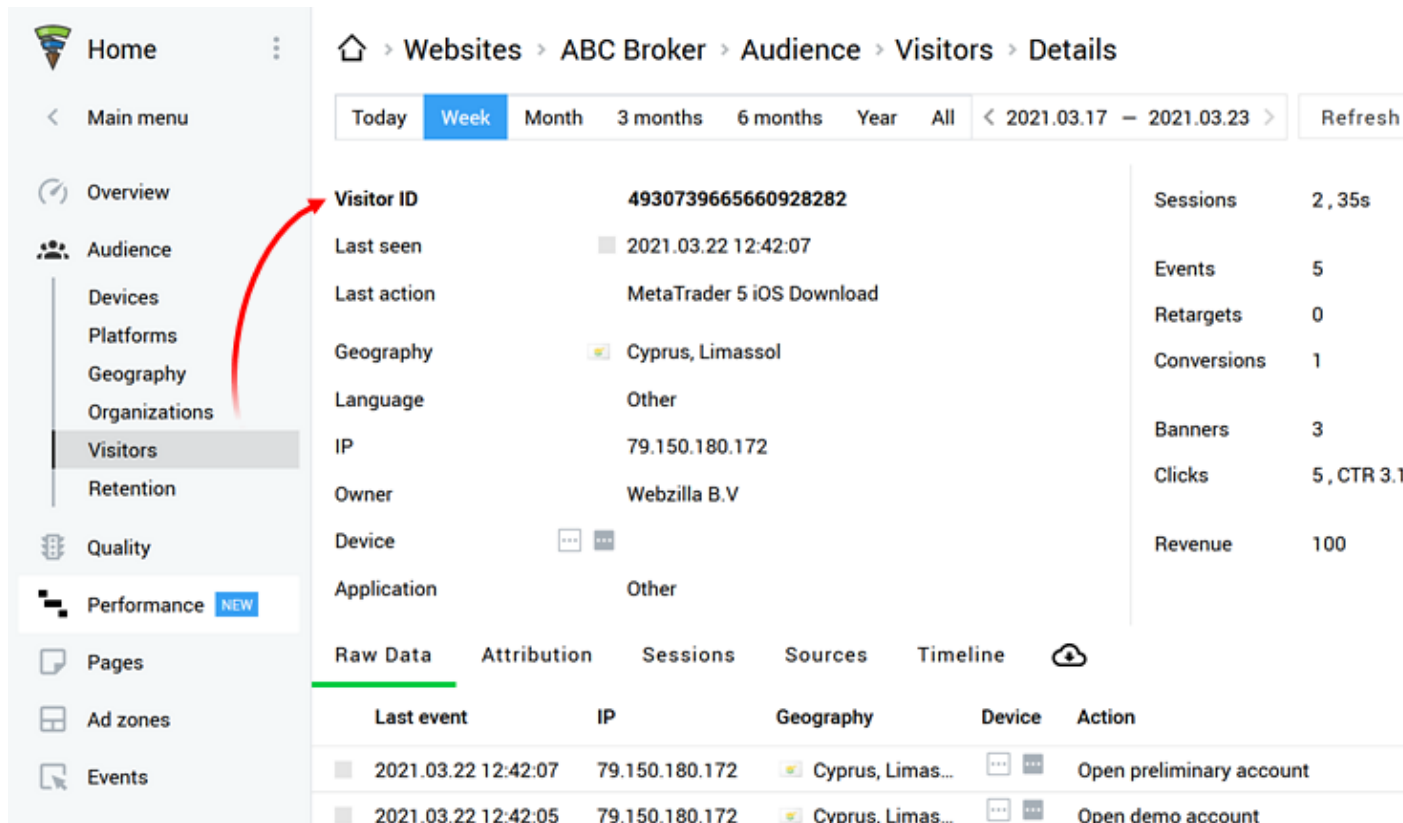
Export...

Cancel

Update

MetaTrader 5 trading platform

Find the required Visitor ID in the Finteza panel, under the visits section, and click on it to apply the ID as a filter. This will enable the generation of any report on trader actions in your site: page views, events, sales funnels, and others.



The screenshot displays the Finteza interface for the 'ABC Broker' website. The left sidebar contains a menu with items: Home, Main menu, Overview, Audience (selected), Devices, Platforms, Geography, Organizations, Visitors (highlighted with a red arrow), Retention, Quality, Performance (NEW), Pages, Ad zones, and Events. The main content area shows the breadcrumb path: Home > Websites > ABC Broker > Audience > Visitors > Details. Below the breadcrumb, there are tabs for time periods: Today, Week (selected), Month, 3 months, 6 months, Year, and All. A date range selector shows '2021.03.17 - 2021.03.23' and a 'Refresh' button. The 'Visitor ID' is displayed as 4930739665660928282. Other details include: Last seen (2021.03.22 12:42:07), Last action (MetaTrader 5 iOS Download), Geography (Cyprus, Limassol), Language (Other), IP (79.150.180.172), Owner (Webzilla B.V), Device (Mobile), and Application (Other). On the right, a summary of metrics is shown: Sessions (2,35s), Events (5), Retargets (0), Conversions (1), Banners (3), Clicks (5, CTR 3.1), and Revenue (100). At the bottom, there are tabs for Raw Data (selected), Attribution, Sessions, Sources, and Timeline. The 'Raw Data' tab shows a table with columns: Last event, IP, Geography, Device, and Action. Two events are listed: 'Open preliminary account' and 'Open demo account', both occurring on 2021.03.22 at 12:42:07 and 12:42:05 respectively, with IP 79.150.180.172 and Geography Cyprus, Limas...

Visitor ID	Sessions	Events	Retargets	Conversions	Banners	Clicks	Revenue
4930739665660928282	2,35s	5	0	1	3	5, CTR 3.1	100

Last event	IP	Geography	Device	Action
2021.03.22 12:42:07	79.150.180.172	Cyprus, Limas...	Mobile	Open preliminary account
2021.03.22 12:42:05	79.150.180.172	Cyprus, Limas...	Mobile	Open demo account

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) /
Sponsored VPS

Sponsored VPS

The Virtual Hosting service is extremely important for traders. It enables 24/7 operation of trading robots and copied signals, without the need to keep the PC turned on permanently. Furthermore, users do not have to worry about the hardware and connections. The service is available directly from the client terminals, so traders can start using it in just a few clicks. The service ensures minimal network delays between the terminal and the broker's server, as the system selects the nearest hosting server automatically.

Sponsored Hosting provides a great tool for attracting new traders. Start offering a free VPS and create a competitive advantage to attract more clients. Furthermore, by providing the ability to trade 24/7 for existing customers, you can increase overall trading volumes.

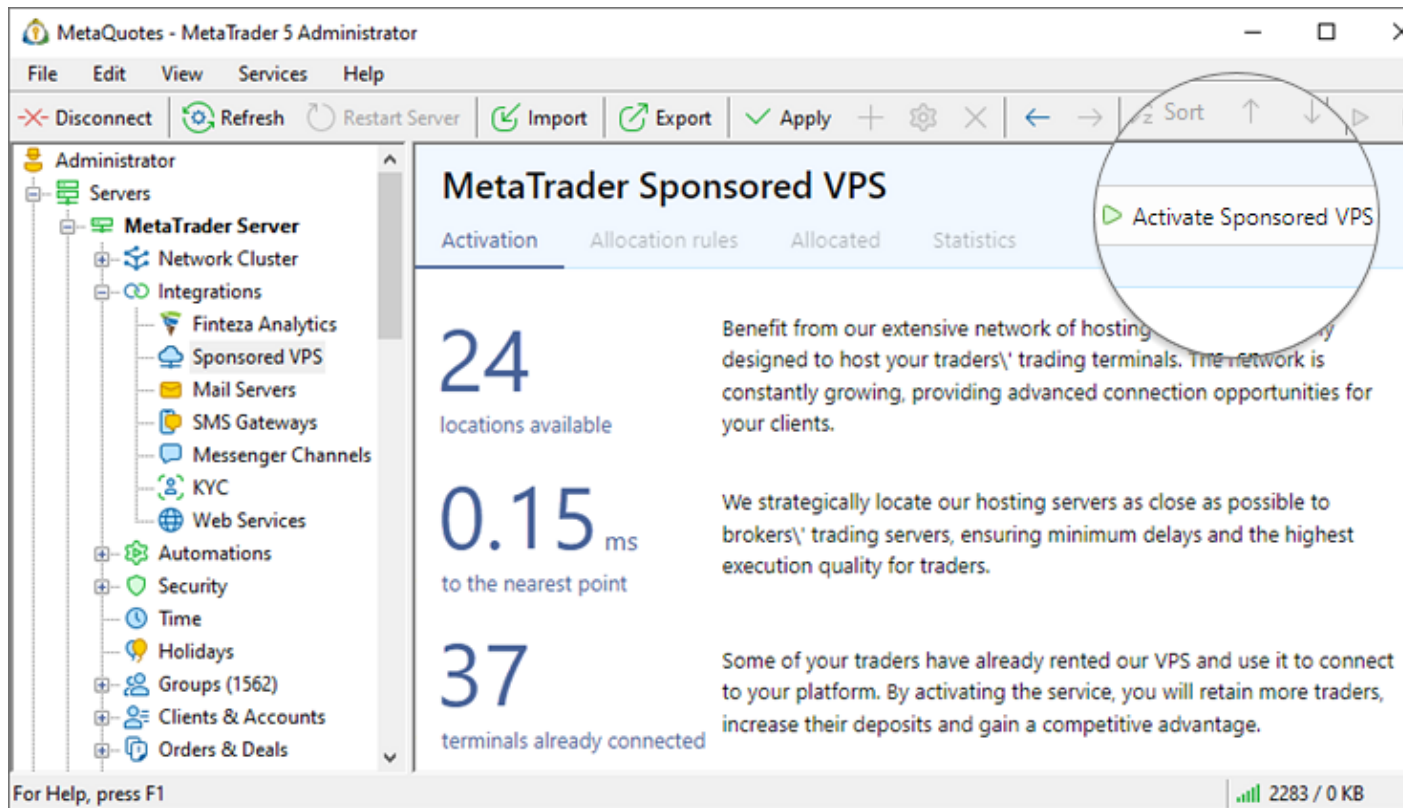
For further information about Sponsored VPS advantages please see our articles:

- [Gain a competitive advantage with MetaQuotes VPS](#)
- [4 reasons to tell your traders about VPS for MetaTrader](#)
- [Enable VPS for your traders – it pays off](#)

Activating

To manage the service, use the Integrations \ VPS section in MetaTrader 5 Administrator. Click Activate Sponsored VPS.

MetaTrader 5 trading platform



Allocation rules

Allocation of hosting to traders can be flexibly configured. You can set detailed rules using almost any account parameters: balance, country, group, last visit date, etc. For example, hosting can only be provided for real clients with a balance of at least 10,000 USD and trading activity.

Open the "Allocation rules" setting and create a new scenario:

MetaTrader Sponsored VPS

Allocation rules
Allocated
Statistics

Name	Condition
Real accounts with > 1000	Group: real*, Balance>=1 000.00

MetaTrader Sponsored VPS Scenario

☒ Enable

Name:

Type	Condition	Value
01 Accounts\Days since registration date	Less or equal (<=)	30
Accounts\Group	Match mask	real*
01 Accounts\Login	Equal (=)	United States

Add
Edit
Delete

Conditions
Schedule
Accounts
01 Login
Group
Country
City
Language
Phone
Email
Color
Comment

OK
Cancel

The following conditions are available:

Schedule

These parameters set date and time conditions. Hosting is provided if a user requests it on a specified date or time.

Accounts

These settings specify conditions based on account parameters:

- Login - hosting is allocated for a specified account.
- Group, Country, City, Language, Phone, Email, Color, Comment - hosting is allocated for an account from a specified group, country, city, etc. For example, if "real*" is specified as a group, then hosting can be allocated for real accounts only. If you specify "Argentina" as a country, then hosting will be allocated only for accounts from Argentina. The condition works similarly for other account parameters.
- Registration - the condition is set relative to the account creation date in the platform. Specify the date and the comparison type: equal to, less than or greater than. Accordingly, the condition will trigger for account registered on the specified date, earlier or later than this date. For example, if you specify "Registration > 2020.09.01", hosting allocation will only be possible for accounts created after the specified date.
- Last visit - the condition is set relative to the last connection of the account to the platform. Specify the date and the comparison type: equal to, less than or greater than. Accordingly, the condition will trigger for account connected on the specified date, earlier or later than this date. For example, if you specify "Last visit > 2023.01.01", then the hosting allocation will be available for accounts that last connected to the platform after the specified date.
- Days since registration - the condition is set relative to the number of days that have passed since the account was created. The calculated number of days is rounded down. For example, if the account was created 5 days and 23 hours ago, this will be counted as 5 days. The principle of operation and purpose of this condition are similar to the "Registration" condition.
- Days since last login - the condition is set relative to the number of days that have elapsed since the last connection to the account. The calculated number of days is rounded down. For example, if the user connected to the account 7 days and 15 hours ago, this will be counted as 7 days. The principle of operation and purpose of this condition are similar to the "Last visit" condition.
- Days since last trade activity - the condition is set relative to the number of days that have elapsed since the last operation on the account. This considers any trading operations, commission charges, etc. Balance operations are not taken into account. In addition, the system checks if the account has any open positions or active pending orders. With this condition, you can sort out accounts that connect to the server but do not trade.
- Online - condition is set relative to the status of account connection to the trade server. Possible values are "true" and "false", i.e. the account is either connected (any connection type: client terminal, mobile terminal and so on) or not.
- Leverage - the condition is set relative to the account leverage.

MetaTrader 5 trading platform

- **Balance** the condition is set relative to the current account balance; the amount is specified in the deposit currency. Using this condition, you can allocate hosting to large customers only.
- **Credit** is similar to the "Balance" condition. In this case, the amount of credit funds on the account is checked.
- **Positions total** the condition is set by the number of open positions which currently exist on the account. Using it, you can allocate hosting to accounts with specific trading activity.
- **Orders total** similar to the "Total positions" condition, it checks the number of active pending orders on the account.
- **Floating profit, Equity, Margin, Free margin, Margin level** these conditions are similar to the "Balance" condition. Appropriate trading account states are checked here.
- **Deposit currency** condition regarding account deposit currency. For example, if you enter "EUR*", sponsored hosting will only be available for EUR accounts.
- **Lead Source, Lead Campaign** lead source and lead campaign parameters specified in the account. These conditions will help provide hosting to customers attracted as a result of certain marketing campaigns.
- **Own funds percentage** when trading, clients are able to use their own funds they deposited on their accounts, as well as the credit and bonuses provided by a broker. Both types of funds increase their Equity parameter. The "Own funds percentage" condition allows configuring the automation task depending on the own funds share on a client's account. 100% means the account has client's funds only with no credit and bonuses. On the contrary, 0% means that only credit and bonuses are used for trading.
The parameter is calculated as $(1 - (\text{Credit} / \text{Equity})) * 100$.
When calculating, the fractional part of the value is rounded down: 1.5% becomes 1%, 0.9% becomes 0%. The parameter value cannot be greater than 100% or less than 0%. If an action is required after the client has spent all their funds, it is more efficient to use the "Own funds volume" condition.
- **Own funds volume** the parameter works similarly to the previous condition, although it applies an absolute own funds value rather than a relative one. Specified in the deposit currency. The calculation equation: $\text{Equity} - \text{Credit}$.

You can create an unlimited number of scenarios, covering any situation. Each scenario is tested independently. If a trader falls under the terms of any of them, they will be able to get sponsored hosting.

Additional settings

You can enable/disable access to sponsored hosting individually for each account. To do this, use the Sponsored MetaTrader Virtual Hosting in the settings:

MetaTrader 5 trading platform

Account: 1003, John Smith, USD, 1:100, Hedge

Overview Personal **Account** Profile Subscriptions Security

Group: demo\forex-usd

Color: ☐ None Leverage: 1 : 100

Bank account: Agent account:

☒ Enable this account ☒ Enable trading by Expert Advisors

☒ Enable trading ☐ Enable API connections

☒ Enable to change password ☒ Enable daily reports

☒ Enable trailing stops ☐ Change password at next login

☒ Enable one-time password ☒ Sponsored MetaTrader Virtual Hosting

Trade accounts:

Gateway ID	Account
------------	---------

Synchronize

Add

Edit

Delete

Client Properties... Update Cancel Help

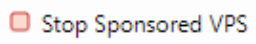
MetaTrader 5 trading platform

If this option is enabled, the service will be available regardless of the allocation rules described above. If the option is disabled, the list of rules will be checked.

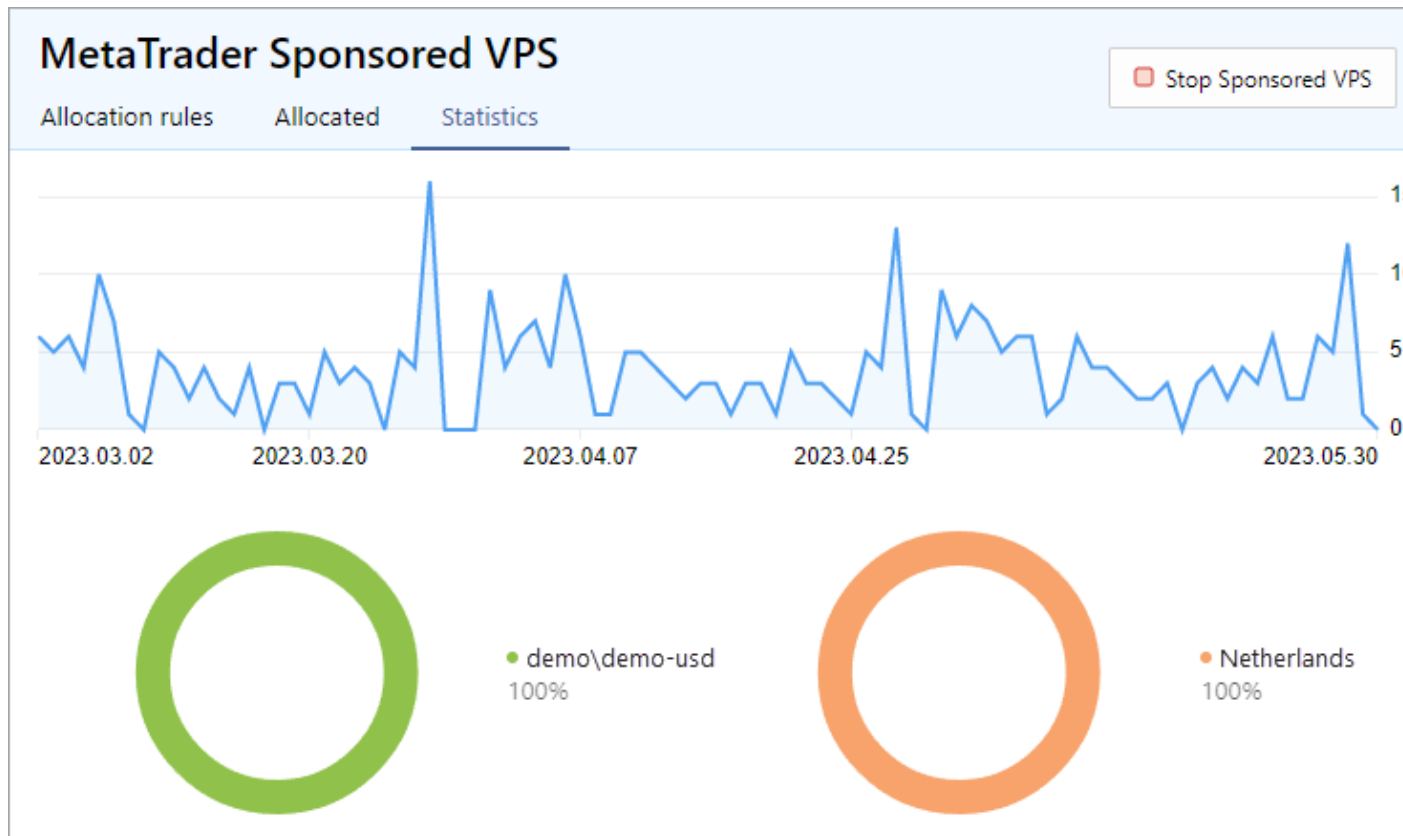
This option can also be managed via the Automations service. Create a task with the necessary trigger and conditions, and select "Enable/Disable sponsored VPS" as an action. For example, using a task with the Deposit trigger allows you to enable VPS only when replenishing an account for a certain amount.

Report on provided VPS

Open the Allocated section to view the list of accounts that have used the service. This includes the account number, group, server and duration of sponsored hosting.

MetaTrader Sponsored VPS			
Allocation rules	Allocated	Statistics	 
Account	Group	Server	Duration (days)
7726743	demo\demo-usd	MetaQuotes-Demo	31
7763120	demo\demo-usd	MetaQuotes-Demo	31
7813434	demo\demo-usd	MetaQuotes-Demo	31
7801532	demo\demo-usd	MetaQuotes-Demo	31
7743708	demo\demo-usd	MetaQuotes-Demo	31
7804871	demo\demo-usd	MetaQuotes-Demo	31
7737482	demo\demo-usd	MetaQuotes-Demo	31
7736716	demo\demo-usd	MetaQuotes-Demo	31
1 2 3 4 5 6 7 8 9 10 11 ... 16			

Statistics section features convenient diagrams for evaluating the use of the service:



MetaTrader 5 trading platform

Payment for sponsored VPS

The base cost of the sponsored hosting is USD 15 excluding VAT. This is a monthly rate. The final cost may vary depending on the VAT rate in the country the rental is performed from.

The amount for actually sponsored VPS is automatically included in your monthly platform statement. No additional settings are required.

Operation on the Client Terminal side

According to the configured access, the special payment plan will become available to the relevant users under the VPS section in client terminals. By choosing it, they receive hosting, while the payment for it will be added to the broker's monthly statement.

51190365 - MetaQuotes-Demo: Demo Account - Hedge - MetaQuotes Software Corp. - [VPS]

File View Insert Charts Tools Window Help

Algo Trading New Order

Market Watch: 16:48:09

Symbol	Bid	Ask
EURUSD	1.12988	1.12988
GBPUSD	1.32774	1.32774
USDCHF	0.92098	0.92098
USDCAD	1.28352	1.28352

Specialized VPS for MetaTrader

MetaTrader VPS
MQL5 Amsterdam 03

Ping: 0.12 ms
61.47 ms faster than the current connection

MetaQuotes-Demo
MetaQuotes Software Corp.

☒ Sponsored by broker
FREE

☐ Basic
1 month
15 USD
per month

☐ Popular
3 months
14 USD
per month
-7%

☐ Trader
6 months
13.17 USD
per month
-12%

Rent for free

EURUSD,H1 | EURCAD,H1 | CHFJPY,H1 | EURUSD,H1 | Market | Signals | VPS

Symbol	Ticket	Time	Type	Volume	Price	S / L	T / P	Price	Pro
eurusd	1226785539	2021.12.17 16:45:46	sell	0.01	1.13026	1.13226 ×	1.12526 ×	1.12988	0.34
usdcad	1124407116	2021.09.20 12:48:24	sell	0.01	1.28432	1.29432 ×	1.18432 ×	1.28352	0.55

Balance: 2 645.61 EUR Equity: 2 646.37 Margin: 18.55 Free Margin: 2 627.82 Margin Level: 14 266.15 %

Trade Exposure History News Mailbox 6 Calendar Company Alerts Articles 1 Code Base Experts Journal Market Signals VPS

For Help, press F1

Default

232 / 6

MetaTrader 5 trading platform

We have added extra protection to avoid service misuse:

- Only one sponsored hosting server can be rented for each account at a time, while the allocation of multiple servers is prohibited
- No other account can be launched on the sponsored hosting server

For further details on the Virtual Hosting service please refer to the [client terminal documentation](#).

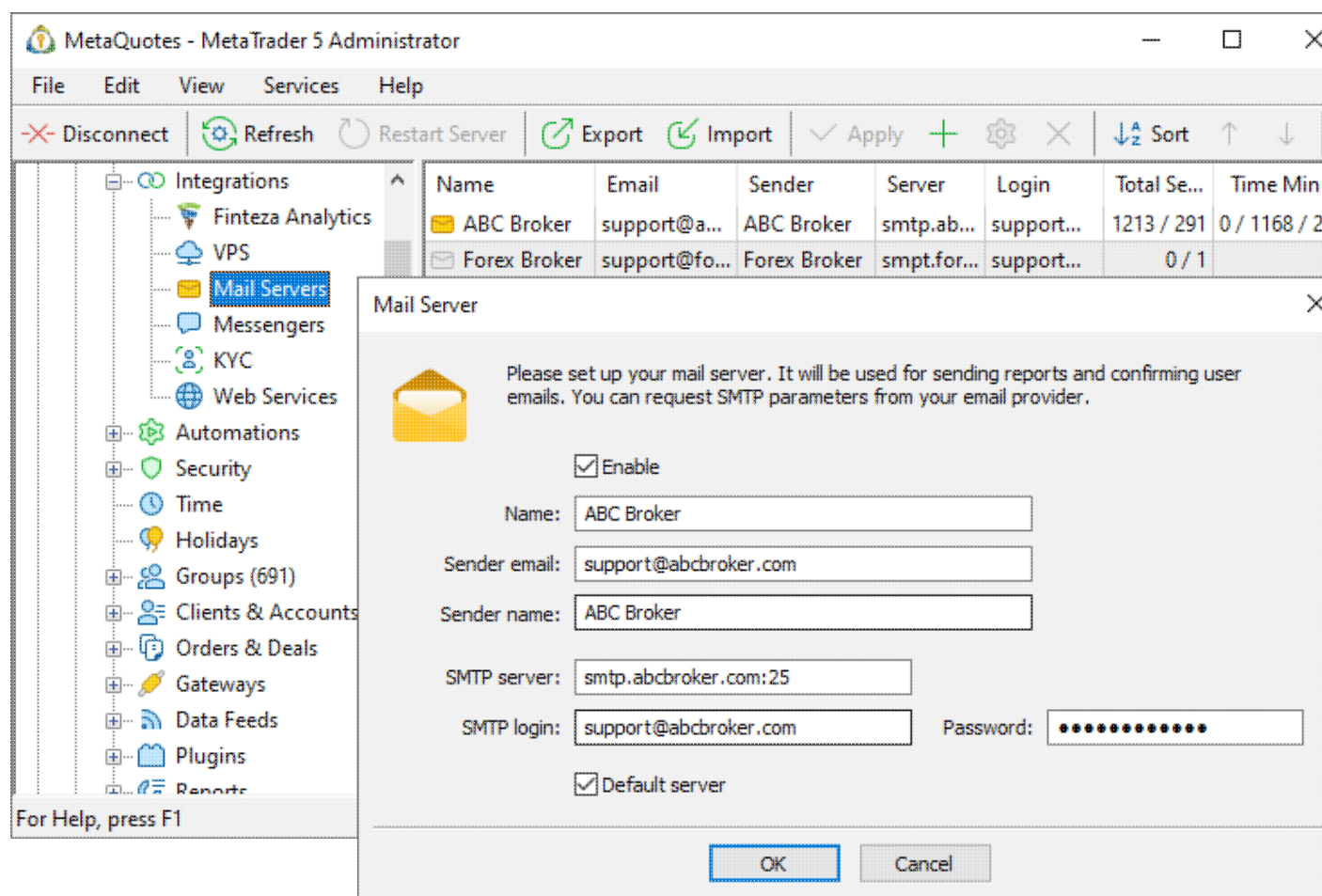
MetaTrader 5 Trading Platform / Platform Setup / Integrations / Mail Servers

Mail Servers

The MetaTrader 5 platform is provided with the built-in email service. The service is used to send trading reports to traders, for the confirmation of email addresses specified during account opening via client terminals, and for the manual email sending.

Reports and confirmation emails can be sent from different mailboxes, while a separate mailbox can be used for each client group. Thus your White Label partners can send emails to their clients from their own addresses.

Mail server configurations can be efficiently managed in the "Mail servers" section. Configure required parameters once and then you will be able to select the desired server configuration in the client group or account allocation settings.



Create a new configuration and set the mail server parameters:

- **Enable** enable/disable the mail server configuration. If the configuration is disabled, emails will not be sent via this server.
- **Name** configuration name.
- **Sender email** email address from which emails will be sent.
- **Sender name** sender name will be indicated in emails.
- **SMTP server** email server address for sending reports and port number, separated by a colon. Example: smtp.mailserver.com:80.
- **SMTP login** account login on the mail server.
- **Password** account password on the mail server. Some services, such as Gmail, Yahoo, mail.com, and others, require the use of a separate password to access the mailbox through third-party applications. It is usually called App Password. If you are using such a service for mailing, specify the App Password instead of the main password for accessing the mailbox. Additional information is usually provided on the service's website. For example, you can find information for Gmail [here](#) and [here](#).
- **Default server** the "Default" option can be selected for the used mail server in [group settings](#) and [account allocation settings](#). In this case the platform will check the list of all servers and will use the first available server with the "Default server" option enabled.

Sent email statistics is shown for each configuration in the list:

- **Total Sent / Errors** number of sent emails and number of unsent emails due to errors.
- **Queue** number of emails on the queue to be sent.
- **Time Min / Max / Avg** minimum, maximum and average time in milliseconds spent to send one email.

MetaTrader 5 Trading Platform / Platform Setup / Integrations / SMS Gateways

SMS Gateways

The trading platform features built-in integration with popular SMS providers. Use this functionality to verify phone numbers, which traders specify during account registration from terminals. Configure verification and access wider opportunities in working with potential clients by collecting valid contact information.

Smart budget saving system

Various factors affect the delivery of confirmation codes to traders, such as the provider, the trader's country and carrier, roaming, and so on. Codes can be delivered with a delay in some cases. To avoid frequent repeated quote requests, confirmation codes are valid for three hours. If the code is not entered in the field within this time frame, the trader will need to repeat the procedure.

Before sending codes, the system checks whether the specified phone/email was previously confirmed. If the trader has already passed verification from his or her computer in the last 72 hours, an account will be opened without additional confirmation. Thus, there will be no additional burden for traders and no additional costs for brokers.

The system automatically controls the correctness of specified phone numbers on the client terminal side. This prevents the sending of codes to incorrect numbers and thus saves you costs for the SMS provider services.

- The phone number must be specified in the international format: +[country code][number], for example: +79171113594. The number should be specified without spaces.
- This must be a mobile phone number, not landline.

How the messaging provider is selected

We recommend working with at least two different SMS providers at the same time, in order to ensure successful and fast delivery. How the provider is selected when sending a message:

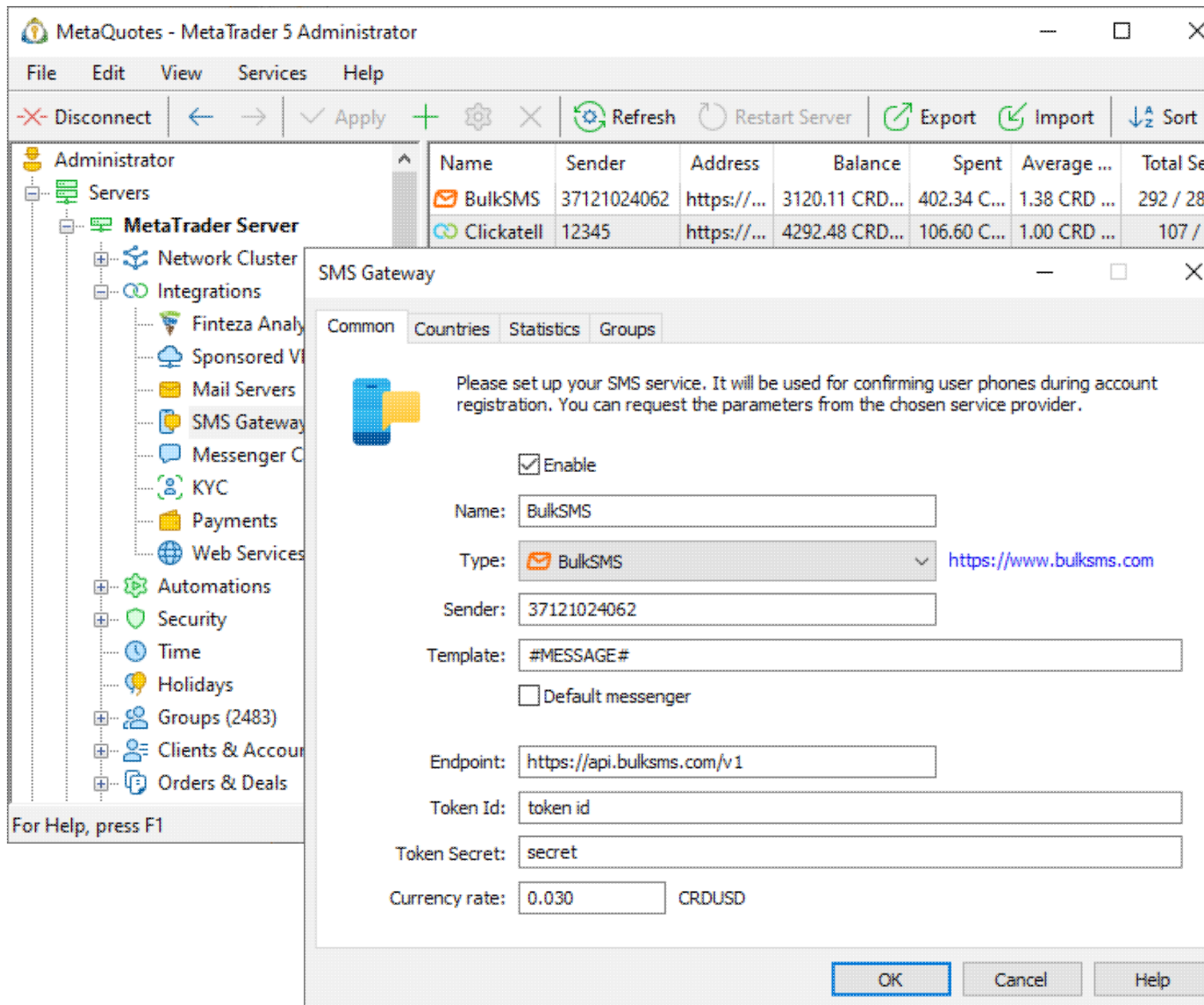
- The system checks the client group and country
- The best provider is selected from the list of providers, which meets the country and group conditions, based on the number of successfully delivered messages and message cost
- If several providers have the same statistics, the one located further up the list is selected
- If none of the providers is suitable, the default provider is utilized (as set by the "Default messenger" option in common settings)

If sending through the main provider fails, the system will send the message via the next available provider. The system can make up to 3 attempts in total.

MetaTrader 5 trading platform

The number of confirmations of different phone numbers that a trader can request from one computer is limited for security purposes. When the limit is reached, confirmation codes are no longer sent, and the following message is printed in the server log: "too many phone confirmation requests from IP".

Setting Up the Provider



Create a new configuration and set the messaging service connection parameters:

- **Enable** enable/disable provider configuration. If the configuration is disabled, the service will not be used to send messages.
- **Name** configuration name.
- **Type** messaging service provider. Each provider has an individual set of settings. Please request the required settings from the provider.
- **Sender** message sender ID. The value is provided by the service provider.
- **Template** the basic template for sending messages via this provider.
- **Default messenger** the platform uses this option when selecting a provider to send a message, if this could not be done based on the group, country or statistics.
- **Rate** exchange rate to convert message sending cost to USD. It is used to display data under the Statistics section.

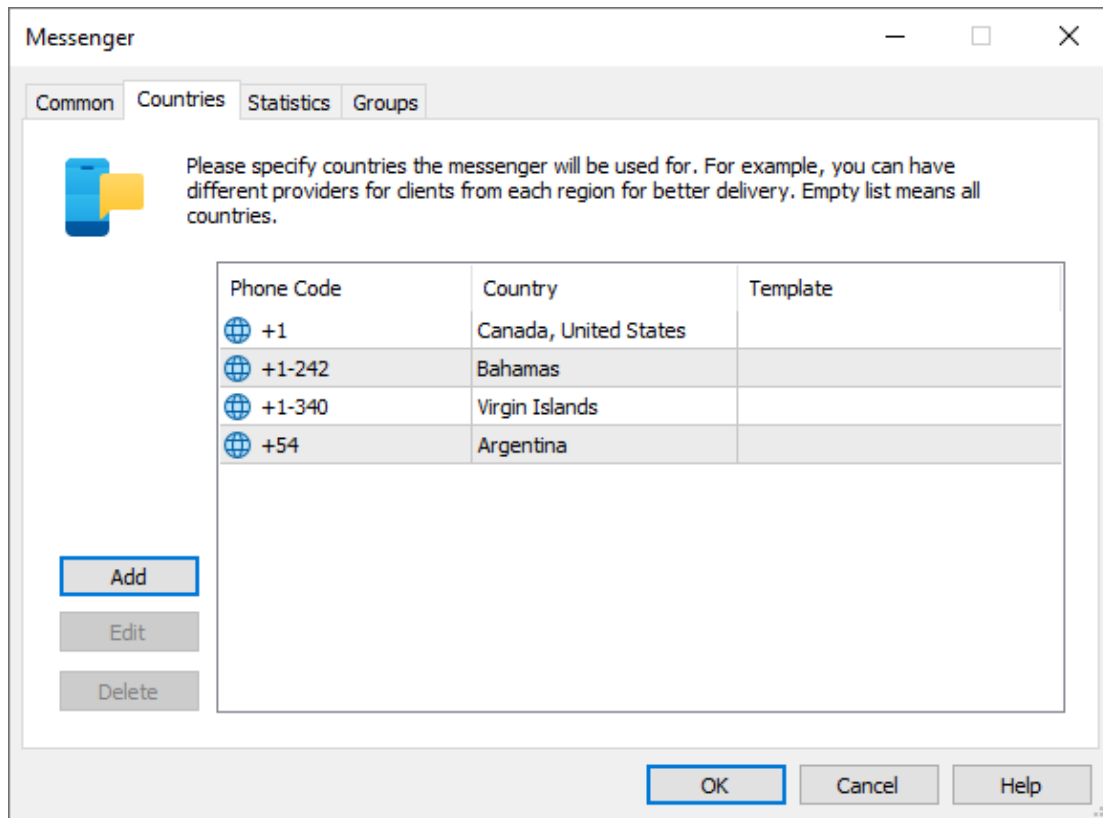
Further settings depend on the provider:

- BulkSMS
- Clickatell
- WEBSMS
- Twilio
- CM.com
- Vonage
- Foniva
- Infobip
- Surfman
- Voiso

Countries

Specify the list of countries, for which this messenger will be used. Use of different providers ensures the best delivery conditions for clients from different regions.

If no country is specified, the platform will use this messenger for clients from any country.

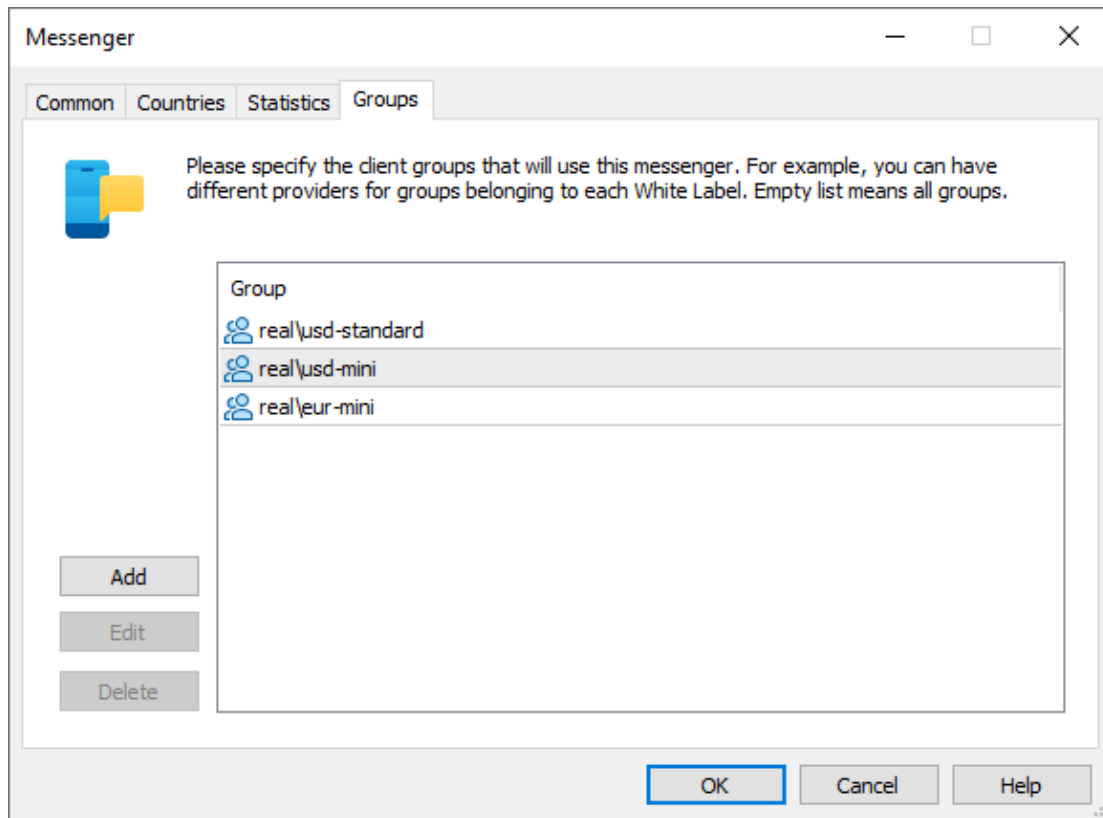


Message templates can be overridden for each specific country.

When opening a demo account, the country is determined automatically on the client terminal side. When opening a preliminary account, the country is also detected automatically, but the client can specify another country in the registration form.

Groups

Specify the list of groups, for which this messenger will be used. Thus you can organize work with your White Label partners. You partners can select providers and pay for their services, as well as send messages on their own behalf and use specific texts.



Templates

Templates can be used for messages. The text can be changed depending on the used message provider or the recipient country.

Each country has its own message sending rules: providers can filter messages by specific words, require special formats, etc. For example, in China it is recommended to indicate a signature before a message. Please check with your message provider for specific sending policies.

Use templates to customize message sending in accordance with local rules.

For example, a template from the file [main server installation directory]\templates\verify_phone is used for phone number confirmation during account registration. You can specify in this template a macro for code substitution, without a text: "<!--CONFIRMATION_CODE-->". Further, the required text can be added in the provider settings:

MetaTrader 5 trading platform

Messenger

Common

Countries

Statistics

Groups



Please set up your messenger service. It will be used for confirming user phones during account registration. You can request the parameters from the chosen service provider.

☐ Enable

Name:

Type: <https://websms.ru>

Sender:

Template:

Provider address:

Login:

Password:

Currency rate:

Messenger

Common

Countries

Statistics

Groups



Please specify countries the messenger will be used for. For example, you can have different providers for clients from each region for better delivery. Empty list means all countries.

Phone Code	Country	Template
+1	Canada, United States	Confirmation code: #MESSAGE#
+20	Egypt	
+86	China	【ABC】 #MESSAGE#

OK

Cancel

Help

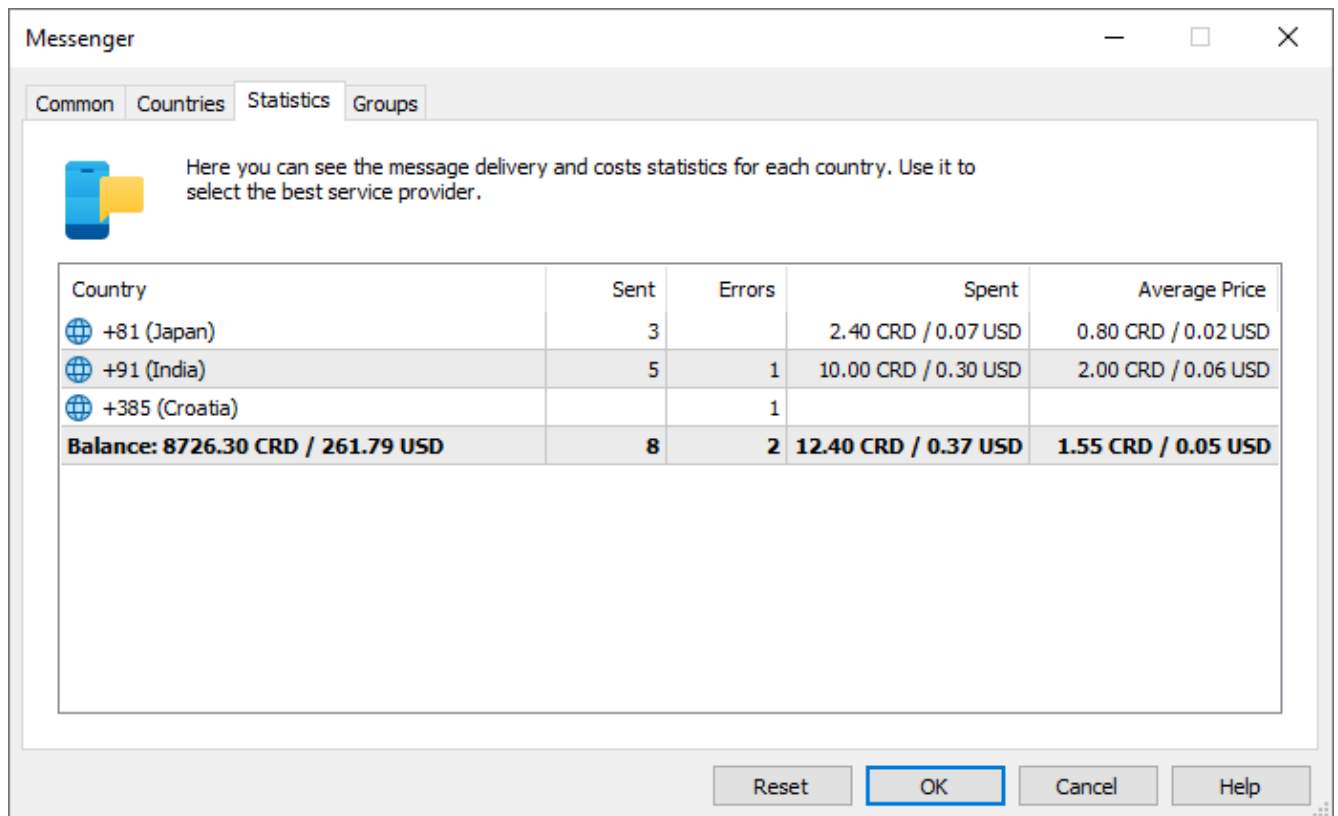
MetaTrader 5 trading platform

A basic template is set in the general provider settings. It contains the default #MESSAGE# macro which substitutes the source text. In our example, the text "Confirmation code" is added so that the resulting message will look as follows: "Confirmation code: [confirmation code]".

Country-specific templates are set in country settings. These templates have higher priority than basic ones. If a country template is specified, this template will be used. Otherwise a template from the "Common" tab will be used.

Statistics

During operation, the system collects statistics on delivered messages and related costs.



Messenger

Common Countries **Statistics** Groups

Here you can see the message delivery and costs statistics for each country. Use it to select the best service provider.

Country	Sent	Errors	Spent	Average Price
+81 (Japan)	3		2.40 CRD / 0.07 USD	0.80 CRD / 0.02 USD
+91 (India)	5	1	10.00 CRD / 0.30 USD	2.00 CRD / 0.06 USD
+385 (Croatia)		1		
Balance: 8726.30 CRD / 261.79 USD	8	2	12.40 CRD / 0.37 USD	1.55 CRD / 0.05 USD

Reset OK Cancel Help

MetaTrader 5 trading platform

Data is available for each country to which messages were sent:

- **Sent** the total number of sent messages
- **Errors** number of messages which could not be delivered
- **Spent** the spent amount in the provider's currency and in USD (the conversion rate is specified in common settings)
- **Average price** the average price spent for sending one message in the provider's currency and in USD (the conversion rate is specified in common settings)

The general statistics for all countries is available below.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [SMS Gateways](#) / BulkSMS

BulkSMS

To set up the provider, [create an SMS gateway configuration](#) and specify the following parameters in it:

- **Provider address** address for connecting to the service. The default value is <https://api.bulksms.com/v1>. In most cases, there is no need to change it.
- **Token ID** token for sending messages. It can be obtained at https://www2.bulksms.com/home/advanced_features/api_tokens/ after you authorize using your BulkSMS account.
- **Token secret** token secret for sending messages.
- **Sender** the name of the sender which will be displayed in the message received by the recipient. The fields can be left blank or it can be filled with the [Sender ID](#) value.

The token ID and token secret will be available to you at https://www2.bulksms.com/home/advanced_features/api_tokens/ after you log in using your BulkSMS account.

[My Account](#) > [Advanced](#) > [API Tokens](#) > [Credentials](#)
[Help Centre](#)
[Support](#)

New Credentials

Token name	test
Token ID	6383ECAA85014F508F96F8AEC7F2FE39-02-E
Token secret	nYQvfTigr*nLPPbloM#0zkOPEUmK

[Back to list](#)

Note: The secret is shown *once only*, so copy it now.

Sender ID is available in your account:

[My Account](#) > [Change Sender Id](#)[Help Centre](#)[Support](#)

Request new sender id

A mobile phone number including country code, or a word of up to 11 characters (e.g. John).

Warning: sender ids which contain any characters other than a-z A-Z and 0-9 will intermittently fail.

New sender Id

Manage sender ids

Default Sender ID	37137024367
Current sender ids	17979595587 37137024367

Other Settings

All other settings are standard and no different from those of other SMS gateways. You can manage the provider's availability for countries and groups, as well as customize message templates.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [SMS Gateways](#) / Clickatell

Clickatell

To set up the provider, [create an SMS gateway configuration](#) and specify the following parameters in it:

- **Provider address** address for connecting to the service. The default address is <https://api.clickatell.com/rest>. In most cases, there is no need to change it.
- **Authentication token** token for sending messages (or API key for the accounts, which were created after November 2016).
- **Sender** the sender's number which will be displayed in the message. This parameter should only be filled if the "[two way messaging](#)" option is enabled for your Clickatell account. When the option is enabled, you will receive a special number from which you will be sending messages and to which you will be able to receive replies. This is the number that should be specified in the "Sender" field.

The operation of Clickatell accounts created after November 2016 is different from earlier accounts: they utilize different API versions:

- The token for sending messages is called API key instead of Auth token.
- No message cost data is available for these accounts.
- The financial state of such accounts cannot be converted to USD, whole data in [statistics](#) is represented in the deposit, specified in the Clickatell account.

To receive the API key, log in using your account at the [Clickatell site](#) and navigate to the SMS section of your profile:

 Clickatell

 Launchpad

 SMS

 WhatsApp

 ONE API

 Reporting

 Settings

Balance:
0.00 EUR
Top up balance

Get started with SMS

10%

1

Setup your test phones

Setup your test phones to easily start testing your SMS integrations. You can register up to 3 free test phones on your Clickatell account.

2

Create and test a SMS integration

Using our easy setup wizard, you can create new API integrations. Once completed you will receive your API key

Integration name:

API key:

My first test In... **ONSGcnSSL5...**

844

MetaTrader 5 trading platform

To check whether you need to specify the "Sender" field, open your Clickatell account and navigate to SMS \ Your REST integration \ Features \ Messaging type. You can switch between message sending modes in this section.

The screenshot shows the Clickatell web interface. On the left is a dark blue sidebar with the Clickatell logo and a menu with icons and labels: Launchpad, SMS, WhatsApp, My APIs, Campaign, Reporting, and Settings. The main content area has a header 'SMS Messaging > MT5 REST Integration' and four tabs: Details, Features (highlighted in blue), Settings, and Phone numbers. Below the tabs, the 'Features' section is divided into two parts. The first part, 'Messaging type', has two radio buttons: 'One-way messaging' (unselected) and 'Two-way messaging' (selected). Below this is a note: 'Please note: Due to regulations a 2-way number is required to send messages within the United States.' The second part, 'Delivery type', has two radio buttons: 'Standard' (unselected) and 'Time Critical' (selected).

Other Settings

All other settings are standard and no different from those of other SMS gateways. You can manage the provider's availability for countries and groups, as well as customize message templates.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [SMS Gateways](#) / WEBSMS

WEBSMS

To set up the provider, [create an SMS gateway configuration](#) and specify the following parameters in it:

- **Provider address** address for connecting to the service. The default address is <https://cab.websms.ru>. In most cases, there is no need to change it.
- **Login** your system account login.
- **Password** password for sending messages via API (not password from your account).
- **Sender** the name of the sender which will be displayed in the message received by the recipient. The specified name must be added to the "[Name List](#)" of your account. If left blank, the default names from your [personal account](#) will be used.

To set the password for sending messages via API, log in to your account at the WEBSMS site and navigate to the [Password Change section in your profile](#).



личный кабинет

Текст рассылки

Телефонная книга

Дата и время рассылки

ID пользователя: 23
Логин: mq
Статус: Платный ++
На счете: 14431.60 руб.
Информер: 16

Выход

Сервисы отправки

Сервисы приема

Сервисы пополнения базы

Оплата

Персональные настройки

Профайл

Имя отправителя

Часовой пояс

Интерфейс

Безопасность

Смена пароля

Оповещения

Навигация

Смена пароля

Смена пароля доступа в личный кабинет

Введите старый пароль

Введите новый пароль

Повторите пароль

изменить пароль

Смена пароля доступа по HTTP, SMTP, SMPP

Для смены пароля необходимо указать действующий пароль.
Если пароль ни разу не менялся - установлен тот же пароль, что и для доступа в личный кабинет.
Если действующий пароль утерян - обратитесь в тех. поддержку.

Введите старый пароль

Введите новый пароль

Повторите ввод нового пароля

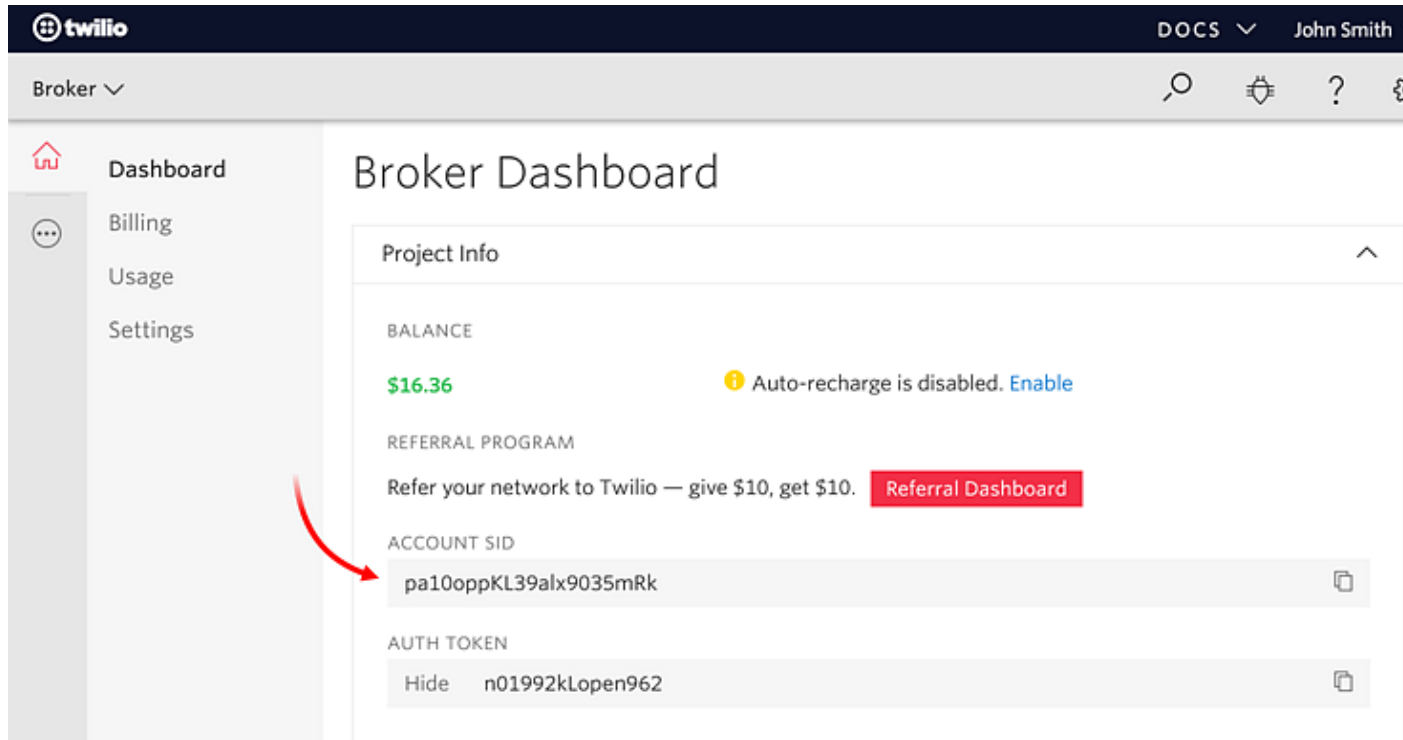
Other Settings

All other settings are standard and no different from those of other SMS gateways. You can manage the provider's availability for countries and groups, as well as customize message templates.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [SMS Gateways](#) / Twilio

Twilio

To set up the integration, sign up at [Twilio](#) website and purchase the desired service plan. Go to Dashboard and copy "ACCOUNT SID" and "AUTH TOKEN" values:



Next, [create an SMS gateway configuration](#) and specify the following parameters:

- **Provider address** address for connecting to the service. The default value is <https://api.twilio.com/2010-04-01>. In most cases, there is no need to change it.
- **Sender** your Twilio number via which the SMS messages are sent.
- **Account SID** your account identifier (ACCOUNT SID) from the Dashboard section of your account in the Twilio website.
- **Authorization Token** a token for sending SMS messages (AUTH TOKEN) from the Dashboard section of your account in the Twilio website.

Other Settings

All other settings are standard and no different from those of other SMS gateways. You can manage the provider's availability for [countries](#) and [groups](#), as well as customize [message templates](#).

MetaTrader 5 trading platform

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [SMS Gateways](#) / CM.com

CM.com

To set up the integration, sign up at CM.com and top up your account with the required amount. From the "Dashboard" section, go to "API\Messaging Gateway".

Account

My Account



John Smith
ABC Broker
Organization Owner

PLANS

 Mobile Marketing Cl
Trial available

 Sign
Trial available

 Communications Pla
Trial available

API

 Messaging Gateway
Active

 Voice API
Click to enable

 Payment API
Click to enable

Messaging Gateway API

The messaging gateway can be configured using the settings below.

General configuration

The following identifier is a unique number to access your current account.

Logical Account ID:
e9345823-35ue-5987-b68e-3b9pou6cb402

HTTP Settings

To connect to our gateway using our Business Messaging API via a HTTP client, you need to authenticate with the product token that belongs to your account.

Important: Keep your product token private and secure!

 Show settings

ProductToken: *****

MetaTrader 5 trading platform

You will need the "Logical Account ID" and "ProductToken" value from general settings. They should be specified in the corresponding fields of the messenger configuration on the MetaTrader 5 side.

Next, scroll down the page to the "Delivery Status Report Settings" section. Message delivery status notifications are configured in this section. The platform will receive this information automatically, allowing you to view [statistics](#) and to analyze delivery efficiency. Also, in case of delivery errors, the platform will try to send a message via [other providers](#).

Delivery Status Report Settings

When we sent a message to our suppliers we receive a status report, also known as delivery status report. This status report tracks the latest status of the message (examples: 'accepted', 'delivered', 'failed'). For every update of the status you will receive a new status report.

☒ **I want to receive message status reports**

Enable this if you want to receive an update about the status of your message (known as DLR or SR) on your own HTTP endpoint. [Learn more](#)

Method

POST



encoding

JSON



HTTP endpoint

<https://abcbroker.com:2000/api/callback/messengers/30163dbe9efa91ff>

Enable the "I want to receive message status report" option, specify "POST" in the "Method" field and specify "JSON" in the "encoding" field. In the "HTTP endpoint" field, copy and paste the "Callback endpoint" value from the messenger configuration on the platform side.

Next, create an SMS gateway configuration on the platform side and specify the following parameters:

- **Sender** number/name to be specified as the message sender. Up to 16 digits are allowed for a phone number or up to 11 Latin characters and numbers are allowed for the name. CM.com does not set any requirements for specifying specific values (for example, the phone number assigned to you) and thus you can specify any value. However, some mobile carriers to which messages are sent can set their own requirements. For further details please view CM.com Documentation.
- **API endpoint** the address of the CM.com server through which messages will be sent. The main server <https://gw.cmtelecom.com/v1.0/message> is used by default. It is not recommended to change the server unless absolutely necessary. For further details on how to choose a server please view CM.com Documentation.
- **Product token** the "ProductToken" value from the "Messaging Gateway" section of your profile on the CM.com website.
- **Account** the "Logical Account ID" value from the "Messaging Gateway" section of your profile on the CM.com website.
- **Callback endpoint** the endpoint for callback requests sent by CM.com. In such requests, the provider notifies the platform about the message delivery status; this information is used for generating statistics. The endpoint address is generated automatically based on the domain specified under the Web Services section. Copy and paste it to the "HTTP endpoint" field in your profile on the CM.com website.

To receive the message delivery statuses, you need to configure the platform to receive callback requests:

- Define a domain that will be used to receive requests, and then add it to the Web Services section.
- Create a DNS record for this domain to link it to a public IP address of your access sever. Next, upload a certificate for it under the SSL Certificates section. The domain specified in the certificate must be the same as the domain used to receive requests.
- Add the endpoint address from the messenger configuration to the Web Services and specify for it a list of IP addresses from which the provider will send requests. Please request this list from CM.com. At the time of this writing, the provider used the 188.94.184.1/23 range, which however may change over time.

To view the message delivery statuses, open the account menu on the CM.com website and navigate to the "Message Log" section.

Other Settings

MetaTrader 5 trading platform

All other settings are standard and no different from those of other SMS gateways. You can manage the provider's availability for countries and groups, as well as customize message templates.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [SMS Gateways](#) / Vonage

Vonage

To configure the integration, purchase the "Communications API" service access from the [Vonage](#) website. Once you receive an account, log into the dashboard at <https://dashboard.nexmo.com>. Navigate to "Settings" in your profile.

VONAGE APIs
Nexmo is now Vonage

Balance
€ 229.85

John Smith

- Edit profile
- Billing & payments
- Pricing
- Team
- Coupons
- Settings**
- Your applications

PRODUCTS

- Getting started
- SMS

Settings

API keys Notifications

API key
Master (2099c698)

Account credentials

Account name
Use this field to give your API key a name that is easier for you to identify it through.

Account secret
This is used in combination with your API key to authenticate your API requests. If you need to change your secret, create a new one to facilitate secret rotation.

API secret 1
9P1ZxinPvKyuDu8y

[Create new secret](#) ?

SMS settings

Vonage has two different APIs capable of sending SMS. You can only use one at a time because it will change the format of the webhooks you receive.

Please choose which API you want to use.

SMS API Messages API

Webhook format
POST

Delivery receipts (DLR) webhooks
Delivery status of outbound SMS will go to this webhook URL.

<https://example.com/delivery>

Inbound SMS webhooks
Inbound SMS will be forwarded to this URL. You can set a different URL for each of your Vonage numbers, manage on the [Your numbers](#) page.

<https://example.com/inbound>

MetaTrader 5 trading platform

You will need the "API key" and "API secret" values. They should be specified in the corresponding fields of the messenger configuration on the MetaTrader 5 side.

Message sending settings are provided on the same page, to the right. Select "SMS API" and "POST" value in the "Webhook format" field.

Next, create an SMS gateway configuration on the platform side and specify the following parameters:

- **Sender** number/name to be specified as the message sender. Up to 15 digits are allowed for a phone number or up to 11 Latin characters and numbers are allowed for the name. Vonage does not set any requirements for specifying specific values (for example, the phone number assigned to you) and thus you can specify any value. However, some mobile carriers to which messages are sent can set their own requirements. For further details please view Vonage Documentation.
- **API endpoint** the address of the Vonage server through which messages will be sent. The default server is <https://rest.nexmo.com/sms/json>. It is not recommended to change the server unless absolutely necessary.
- **API key** the "API key" value from the settings in your Vonage profile.
- **API secret** the "API secret" value from the settings in your Vonage profile.
- **Callback endpoint** the endpoint for callback requests sent by Vonage. In such requests, the provider notifies the platform about the message delivery status; this information is used for generating statistics. Also, in case of delivery errors, the platform will try to send a message via other providers. The endpoint address is generated automatically based on the domain specified under the Web Service section.
There is no need to configure the endpoint on the Vonage side, while the platform automatically specifies it in every message sending request. However, you need to configure the receiving of callback requests on the platform side, as described below.

To receive the message delivery statuses, you need to configure the platform to receive callback requests:

- Define a domain that will be used to receive requests, and then add it to the Web Services section.
- Create a DNS record for this domain to link it to a public IP address of your access sever. Next, upload a certificate for it under the SSL Certificates section. The domain specified in the certificate must be the same as the domain used to receive requests.
- Add the endpoint address from the messenger configuration to the Web Services and specify for it a list of IP addresses from which the provider will send requests. Please request this list from Vonage. At the time of this writing, the provider used the 169.63.86.164 address, which however may change over time.

Other Settings

All other settings are standard and no different from those of other SMS gateways. You can manage the provider's availability for countries and groups, as well as

MetaTrader 5 trading platform

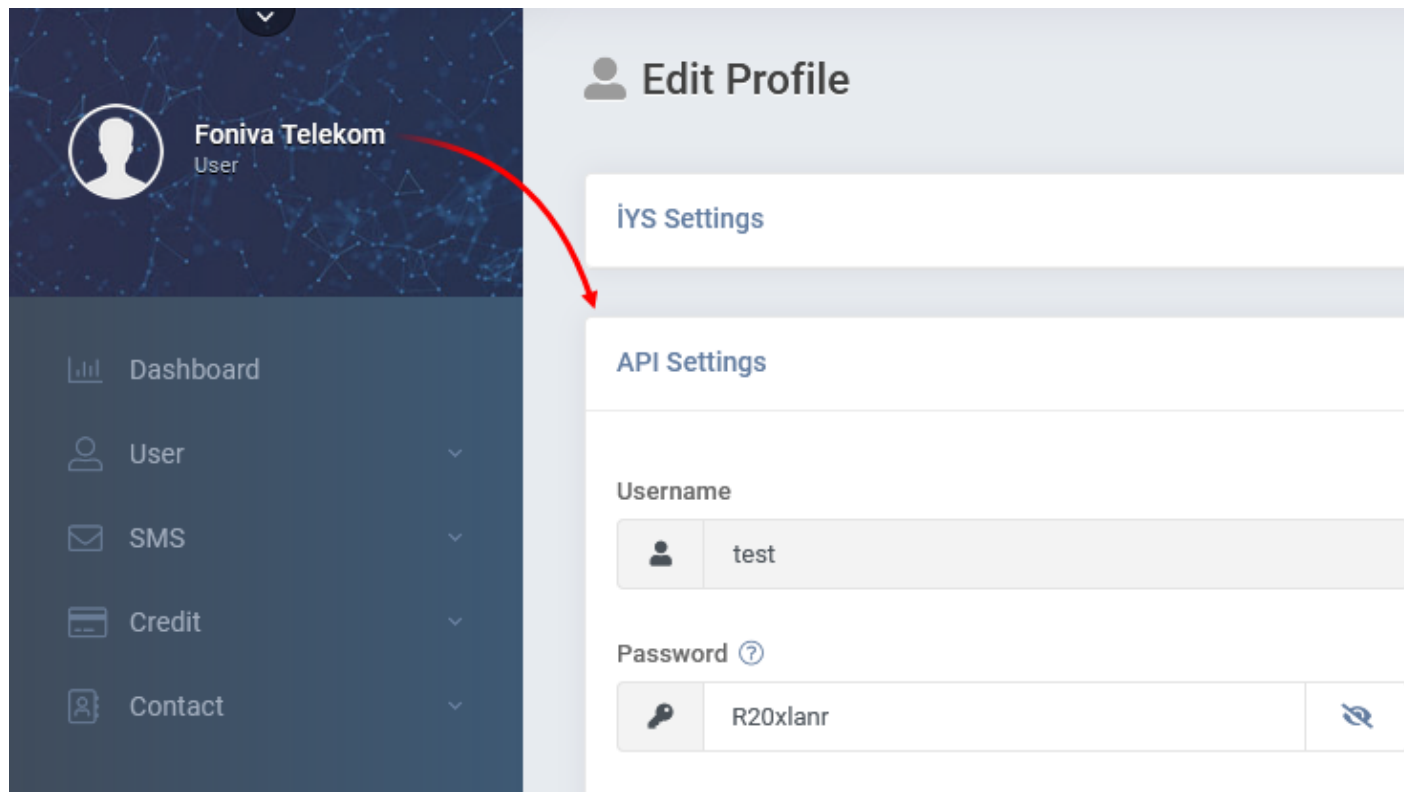
customize message templates.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [SMS Gateways](#) / Foniva

Foniva

Foniva is a Turkish platform that provides fast and high-quality SMS messaging services to subscribers throughout the country. To connect to the service, request an account on the [provider's website](#).

Then, go to your personal account at <https://sms.foniva.com.tr/login> and specify your login details. Next, click on your account name and go to the "API Settings" section. The section contains data for configuring the provider on the MetaTrader 5 side:



Create an SMS gateway configuration on the platform side and specify the following parameters:

- **Sender** the name that will be indicated as the message sender. This is a required field. The name should be discussed with the provider when you open your account.
- **Login** login for connection from the personal account.
- **Password** password for connection from the personal account.
- **Currency rate** Turkish lira to US dollar exchange rate to display cost statistics.

Other Settings

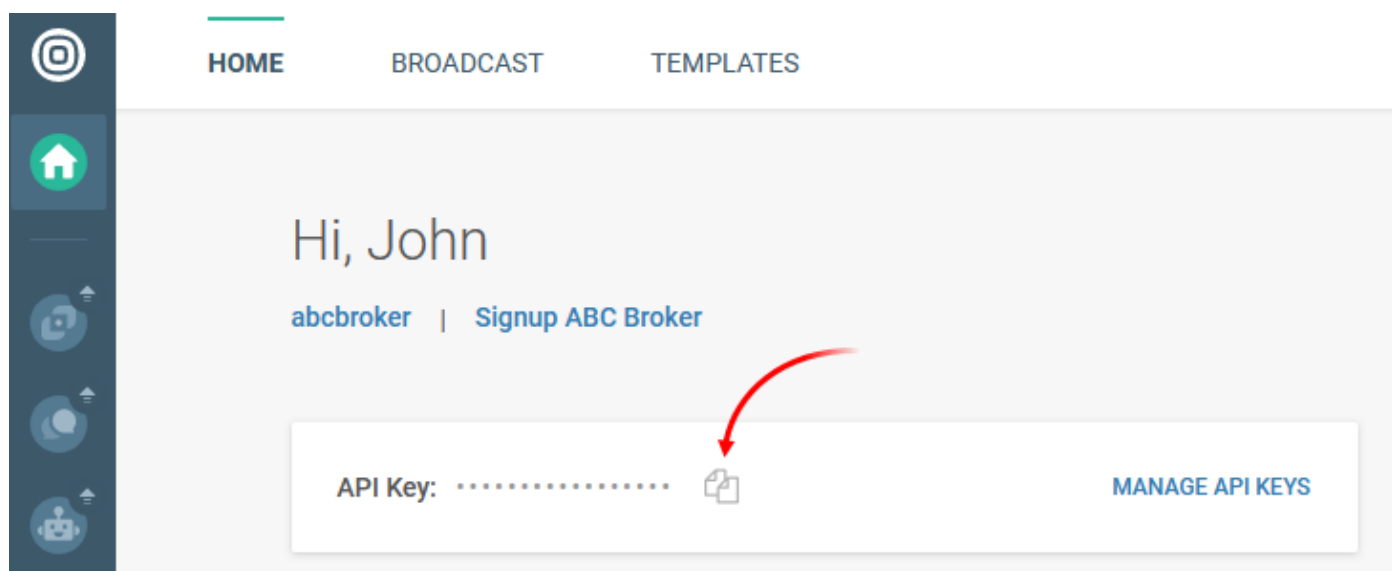
All other settings are standard and no different from those of other SMS gateways. You can manage the provider's availability for countries and groups, as well as customize message templates.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [SMS Gateways](#) / Infobip

Infobip

The service offers high speed and delivery reliability through over 700 direct operator connections and 28 data centers around the world.

To connect to the service, request an account on the provider's website. Login to your profile and copy the API key on the main page. Use the <https://portal.infobip.com/settings/accounts/api-keys> section to manage keys.



Create an SMS gateway configuration on the platform side and specify the following parameters:

- **Sender** number/name to be specified as the message sender. The field can be left blank.
- **API endpoint** the address of the Infobip server through which messages will be sent. The main server at <https://api.infobip.com> is used by default. It is not recommended to change the server unless absolutely necessary.
- **API key** your unique 69-character key. Available in your account.
- **Webhook endpoint** leave the field blank. Integration with Infobip comes with standard options for providing message delivery notifications and information about the cost to the platform, without the need to additionally set up webhooks.

Other Settings

All other settings are standard and no different from those of other SMS gateways. You can manage the provider's availability for countries and groups, as well as customize message templates.

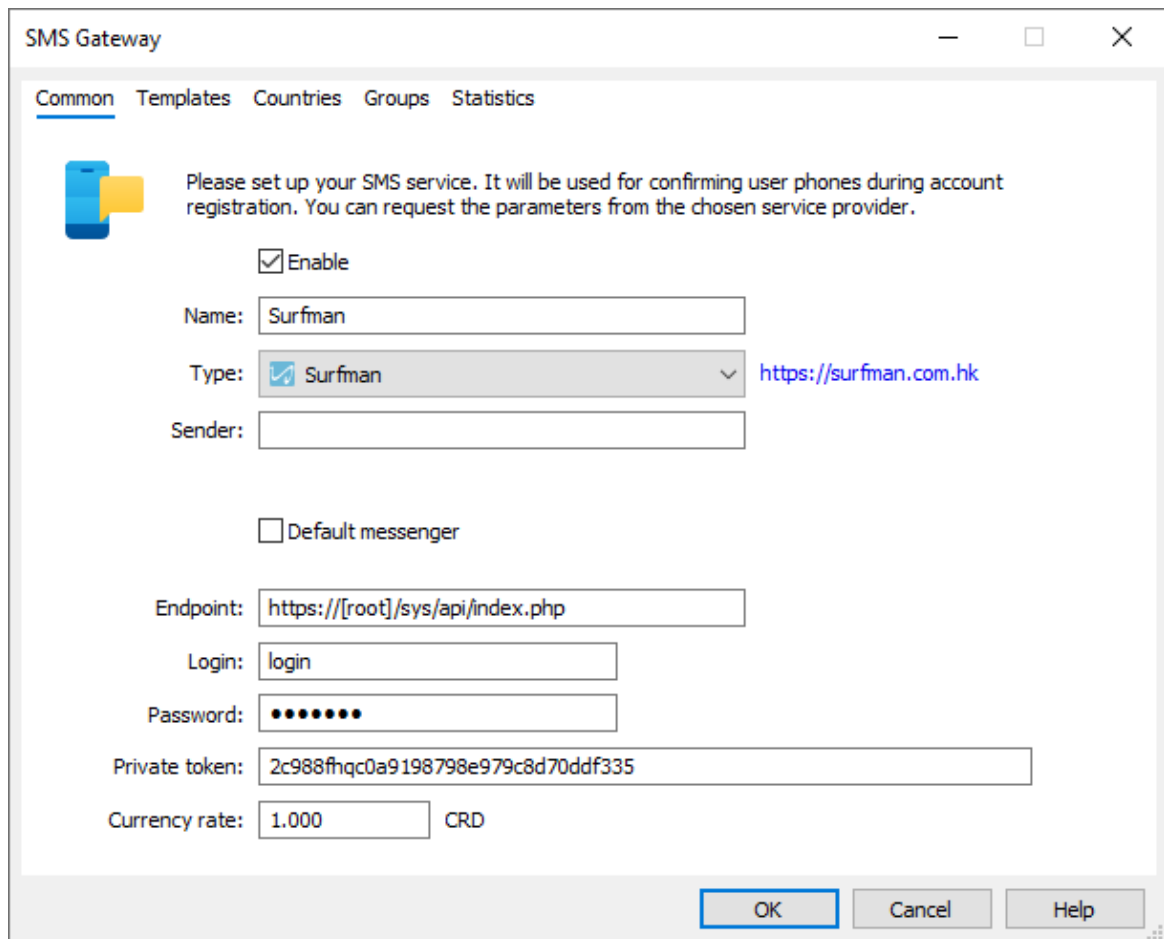
MetaTrader 5 Trading Platform / Platform Setup / Integrations / SMS Gateways / Surfman

Surfman

Surfman is an SMS messaging provider operating in China. To connect, request an account on the provider's website. You will be required to complete the KYC procedure. After signing the contract, the provider will send you the necessary credentials for configuring the connection in MetaTrader 5. Create an SMS gateway configuration and specify the provided parameters as follows:

- username enter in the "Login" field
- password enter in the "Password" field
- API Key specify in the "Private token" field

There is no need to fill out the "Sender" field as the value is specified for each message template through the Surfman personal account.



The screenshot shows the "SMS Gateway" configuration window with the "Common" tab selected. The window contains the following fields and options:

- Enable:** A checked checkbox.
- Name:** A text field containing "Surfman".
- Type:** A dropdown menu with "Surfman" selected. A link <https://surfman.com.hk> is displayed next to it.
- Sender:** An empty text field.
- Default messenger:** An unchecked checkbox.
- Endpoint:** A text field containing "https://[root]/sys/api/index.php".
- Login:** A text field containing "login".
- Password:** A text field with masked characters (dots).
- Private token:** A text field containing "2c988fhqc0a9198798e979c8d70ddf335".
- Currency rate:** A text field containing "1.000" and a unit dropdown set to "CRD".

At the bottom right, there are three buttons: "OK", "Cancel", and "Help".

SMS templates

The provider does not allow sending messages in free format. All messages must follow a strict format, and their content must be pre-approved. For each SMS verification use case within the platform (phone verification, payment status updates), create a template in your Surfman personal account.

The parameter order must strictly follow the assigned numbers. To avoid confusion, we recommend using the following templates as a reference when creating your own:

Action	Template
Phone verification	{{{REF01}}} - verification code for the account registration
Successful deposit	Deposit to {{{REF02}}} complete: {{{REF04}}} {{{REF05}}}, {{{REF03}}}
Deposit error	Deposit to {{{REF02}}} failed: {{{REF04}}} {{{REF05}}}, {{{REF03}}}
Deposit processing	Deposit to {{{REF02}}} processed: {{{REF04}}} {{{REF05}}}, {{{REF03}}}. Waiting for manager confirmation
Successful withdrawal	Withdrawal from {{{REF02}}} complete: {{{REF04}}} {{{REF05}}}, {{{REF03}}}
Withdrawal confirmation Sent when a withdrawal request is submitted for processing by the payment system.	Withdrawal from {{{REF02}}} confirmed: {{{REF04}}} {{{REF05}}}, {{{REF03}}}. Processing by the payment system
Withdrawal error	Withdrawal from {{{REF02}}} failed: {{{REF04}}} {{{REF05}}}, {{{REF03}}}
Withdrawal verification Sent upon manager approval, allowing the user to proceed with the payment. This is used for providers where all payment details are entered on the payment system's page (e.g., <u>AstroPay</u>). In	Withdrawal from {{{REF02}}} approved: {{{REF04}}} {{{REF05}}}, {{{REF03}}}. Please complete the operation in the platform

Action	Template
such cases, during manual processing, the manager only approves the transaction itself.	

After creating and approving the templates, map their identifiers (CampaignId) to the corresponding actions on the platform side under the "Templates" tab.

The screenshot displays the 'smsbroker' web application interface. The top navigation bar includes links for Home, Client, Campaign, List, Recipient, Delivery, and Report. The main section is titled 'Campaign' and features a filter area with dropdowns for 'Message Type' (set to 'All'), 'Client' (set to 'All'), and input fields for 'Campaign Name', 'Sender Name', and 'Message Text'. Below these are a 'Yes' dropdown, a 'Tag' input, and a 'Date' range selector with calendar icons, showing dates from 2024-08-07 to 2025-02-07. A 'Submit' button is present. A table lists campaigns with columns: ID, Message Type, Client, Campaign ID, Campaign Name, Sender Name, Message Text, Type, and Action. Three campaigns are visible, all with 'sms' type and 'brokersms' client. A red arrow points from the 'Campaign ID' column to the 'Campaign ID' field in the 'SMS Gateway' modal window. The modal window has tabs for 'Common', 'Templates', 'Countries', 'Groups', and 'Statistics'. It contains a warning message about message format and a list of templates. The 'Phone Confirmation' template is selected, showing a dropdown menu with options: Phone Confirmation, Deposit Completed, Deposit Failed, Deposit Processed, Withdrawal Completed, Withdrawal Confirmed, Withdrawal Failed, and Withdrawal Verified. At the bottom of the modal are 'Add', 'Edit', and 'Delete' buttons, and an 'OK' button.

Campaign

Message Type: All Client: All Campaign Name Sender Name Message Text

Yes

Tag Date 2024-08-07 2025-02-07 Submit

ID	Message Type	Client	Campaign ID	Campaign Name	Sender Name	Message Text	Type	Action
9	sms	brokersms	20250205163251465	OTP Code	Broker	{{{REF01}}} - verification code for the account registration	Real	
8	sms	brokersms	20250204203641074					
7	sms	brokersms	20250204203614642					

SMS Gateway

Common Templates Countries Groups Statistics

This provider has a strict policy regarding the format of messages, from predefined templates.

Up Down

Type

- Phone Confirmation
- Phone Confirmation
- Deposit Completed
- Deposit Failed
- Deposit Processed
- Withdrawal Completed
- Withdrawal Confirmed
- Withdrawal Failed
- Withdrawal Verified

Add Edit Delete

OK

If a template is not assigned to a specific action, the corresponding messages will not be sent. If you enable the provider for payment status notifications, ensure that the appropriate templates are configured on the provider's side.

Other Settings

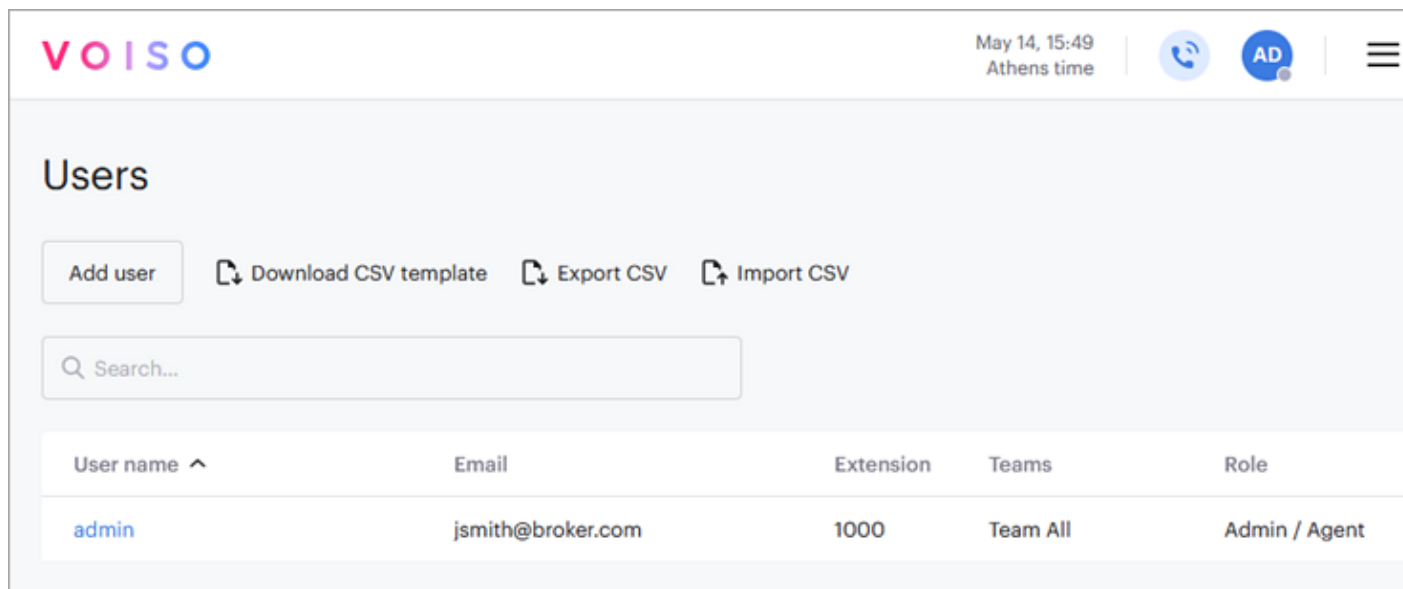
All other settings are standard and no different from those of other SMS gateways. You can manage the provider's availability for countries and groups.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [SMS Gateways](#) / Voiso

Voiso

[Voiso](#) is an international SMS messaging provider, offering service across the globe, including China and other Asian countries. To get started, request an account on the [Voiso website](#). You will need to complete the KYC process. Once your registration is approved, you will gain access to the [personal account](#) where you can find the necessary credentials for connection.

Navigate to the Users section and copy your account identifier from the 'Extension' column:



User name ^	Email	Extension	Teams	Role
admin	jsmith@broker.com	1000	Team All	Admin / Agent

Next, [create an SMS gateway configuration](#) and paste this value into the Agent ID field.

MetaTrader 5 trading platform

SMS Gateway

Common Templates Countries Groups Statistics

 Please set up your SMS service. It will be used for confirming user phones during account registration. You can request the parameters from the chosen service provider.

☒ Enable

Name:

Type:  Voiso

Sender:

☐ Default messenger

Endpoint:

Agent ID:

Contact center key:

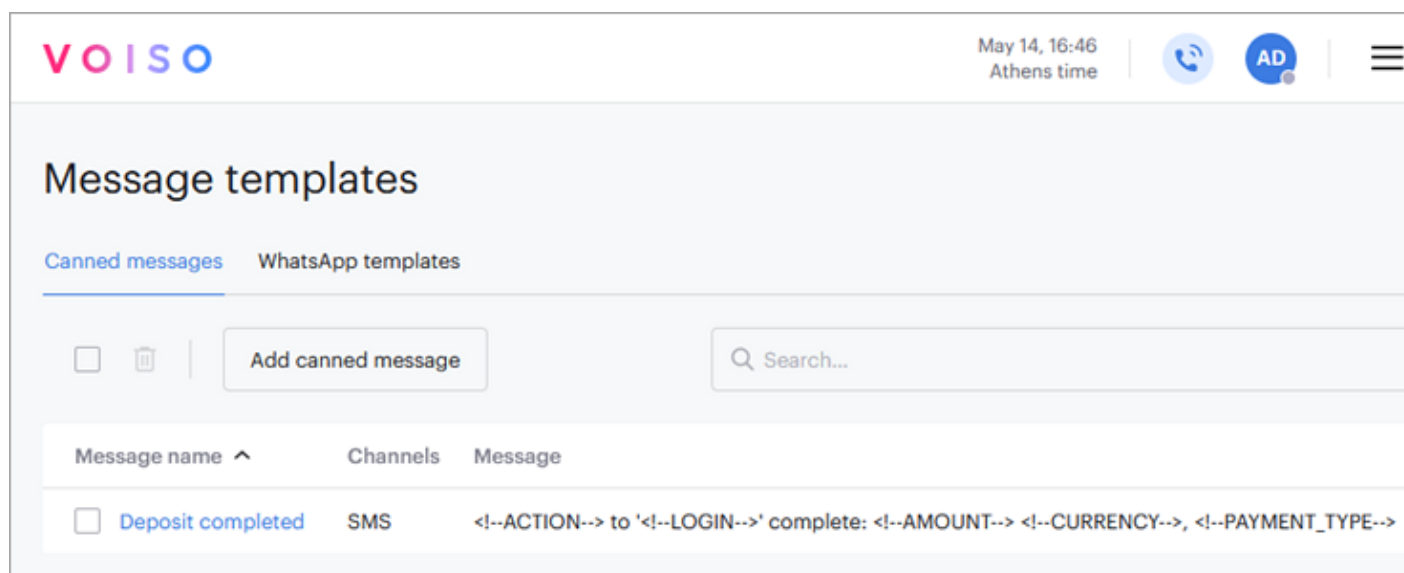
OK Cancel Help

Also, specify the 'Contact center key', which will be provided by Voiso after signing the service agreement.

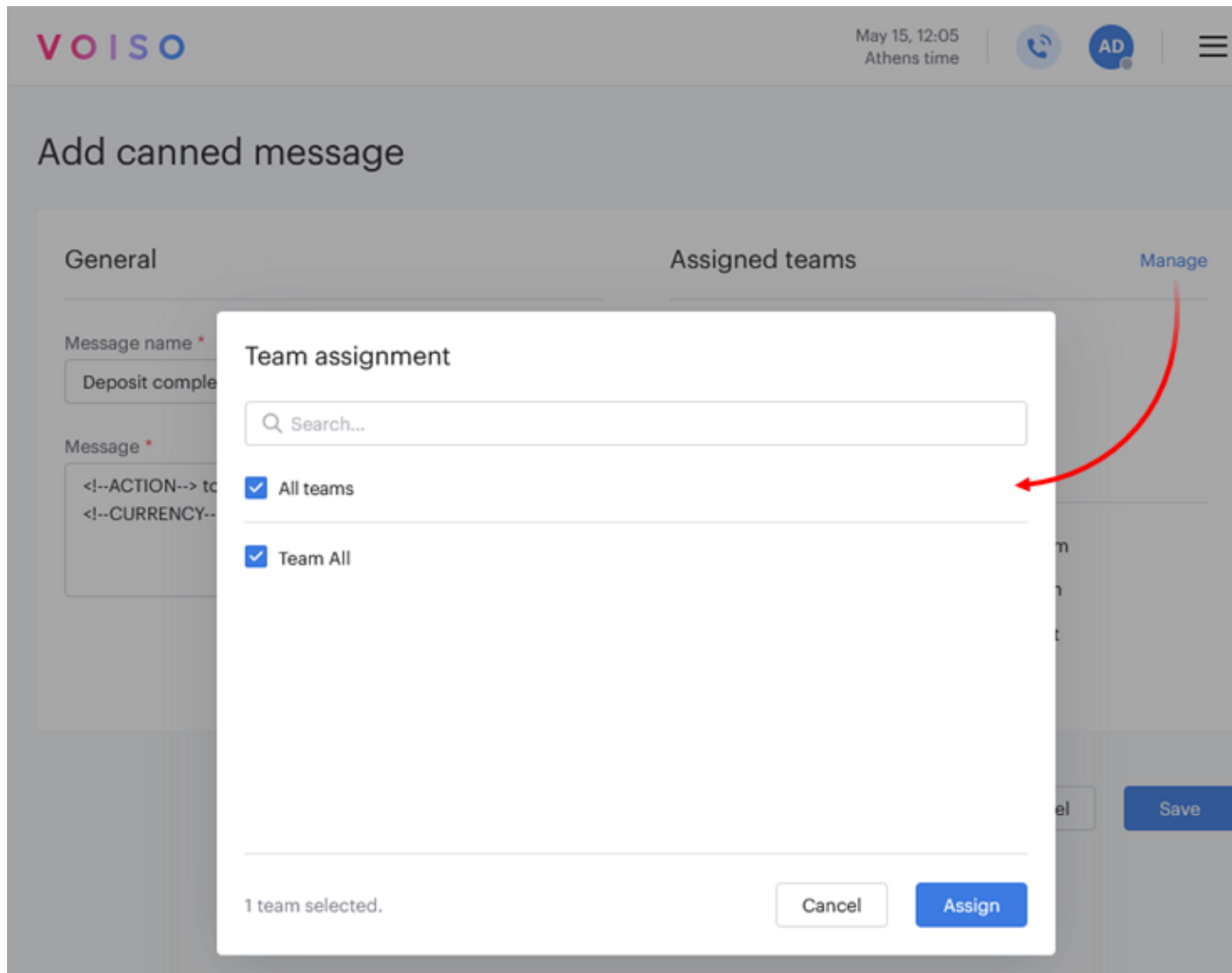
You do not need to fill in the Sender field. Sender values are configured individually for each message template via the Voiso account. The Endpoint field typically does not require changes either as it is automatically populated with the correct Voiso service address by default.

SMS templates

The provider does not support free-form SMS content. All messages must strictly follow predefined formats and must be approved in advance. For each type of SMS notification in your platform (for example, phone verification, payment status), create a template in your Voiso account, under 'Administrator \ Message templates \ Canned messages'.



Click 'Add canned message' and specify a name for the template. Next, grant access to the template to the Voiso user account designated for working with the platform. To do this, click Manage and select the group to which the user account belongs.



Next, enter the content of the template. You can use the following macros in the template if needed:

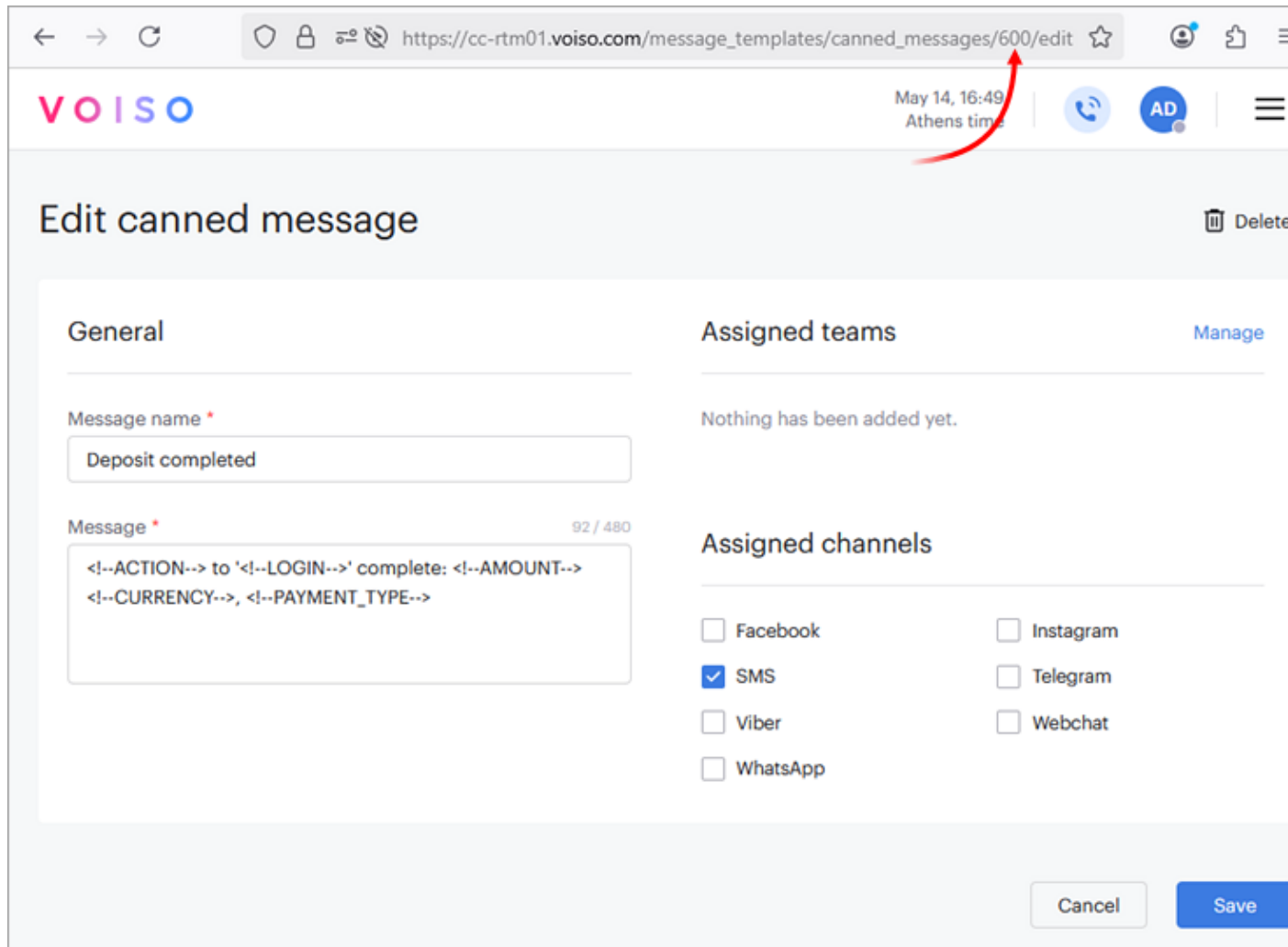
- {{CONFIRMATION_CODE}} verification code generated by platform for phone number confirmation.
- {{ACTION}} payment action: Deposit or Withdrawal.
- {{LOGIN}} user's account number.
- {{AMOUNT}} transaction amount.
- {{CURRENCY}} transaction currency.
- {{PAYMENT_TYPE}} type of payment method.

To ensure clarity and consistency, we recommend using the following templates as a basis:

Action	Template
Phone verification	{{CONFIRMATION_CODE}} - verification code for the trading account registration
Successful deposit	{{ACTION}} to '{{LOGIN}}' complete: {{AMOUNT}} {{CURRENCY}}, {{PAYMENT_TYPE}}
Deposit error	{{ACTION}} to '{{LOGIN}}' failed: {{AMOUNT}} {{CURRENCY}}, {{PAYMENT_TYPE}}
Deposit in progress	{{ACTION}} to '{{LOGIN}}' processed: {{AMOUNT}} {{CURRENCY}}, {{PAYMENT_TYPE}}. Waiting for manager confirmation
Successful withdrawal	{{ACTION}} from '{{LOGIN}}' complete: {{AMOUNT}} {{CURRENCY}}, {{PAYMENT_TYPE}}
Withdrawal confirmation Sent when a withdrawal request is submitted for processing by the payment system.	{{ACTION}} from '{{LOGIN}}' confirmed: {{AMOUNT}} {{CURRENCY}}, {{PAYMENT_TYPE}}. Processing by the payment system
Withdrawal error	{{ACTION}} from '{{LOGIN}}' failed: {{AMOUNT}} {{CURRENCY}}, {{PAYMENT_TYPE}}
Withdrawal verification	{{ACTION}} from '{{LOGIN}}' approved: {{AMOUNT}} {{CURRENCY}}, {{PAYMENT_TYPE}}. Please complete the operation in the

Action	Template
Sent when a manager approves the withdrawal, allowing the user to proceed with payment. Typically used with payment systems, where details are entered on an external page (for example, as in AstroPay). In such cases, during manual processing, the manager only approves the fact of the transaction.	platform

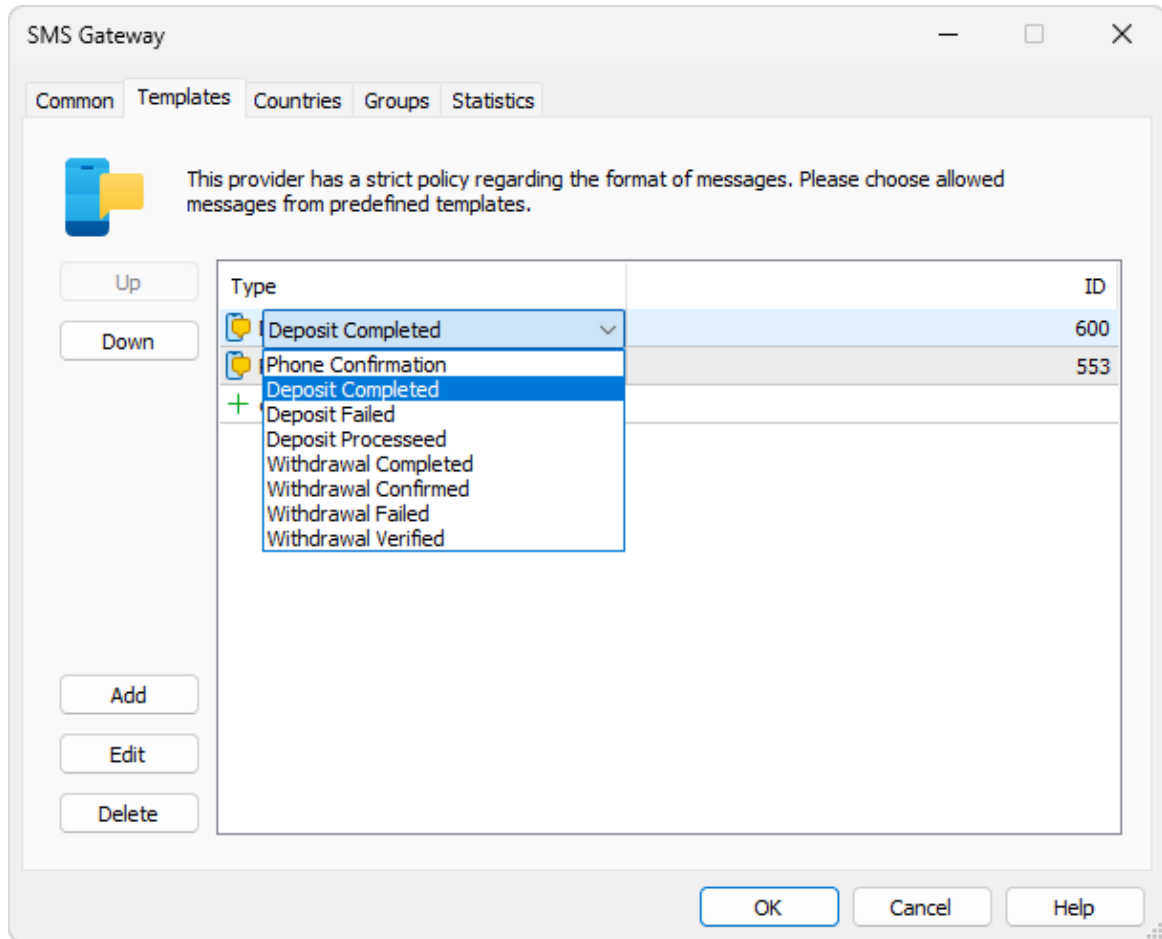
After creating a template, open it for editing and copy its identifier from your browser's address bar.



The screenshot shows the 'Edit canned message' page in the Voiso application. The browser's address bar shows the URL: `https://cc-rtm01.voiso.com/message_templates/canned_messages/600/edit`. A red arrow points to the identifier '600' in the URL. The page has a header with the Voiso logo, a date/time display (May 14, 16:49 Athens time), and user avatars. The main content area is titled 'Edit canned message' and includes a 'Delete' button. The form is divided into two columns: 'General' and 'Assigned teams'. The 'General' column contains a 'Message name' field with the value 'Deposit completed' and a 'Message' text area containing a template with placeholders: `<!--ACTION--> to '<!--LOGIN-->' complete: <!--AMOUNT--> <!--CURRENCY-->, <!--PAYMENT_TYPE-->`. The 'Assigned teams' column shows 'Nothing has been added yet.' and a 'Manage' link. The 'Assigned channels' section has checkboxes for Facebook, SMS (checked), Viber, WhatsApp, Instagram, Telegram, and Webchat. At the bottom right are 'Cancel' and 'Save' buttons.

MetaTrader 5 trading platform

Then, go to the Templates section in the SMS provider settings in MetaTrader 5. Select the appropriate template type and enter the corresponding template ID from your Voiso personal account.



If a template is not assigned for a specific action, no message will be sent for that event. If you are enabling the provider for payment status notifications, ensure that the appropriate templates are configured on the provider's side.

Other Settings

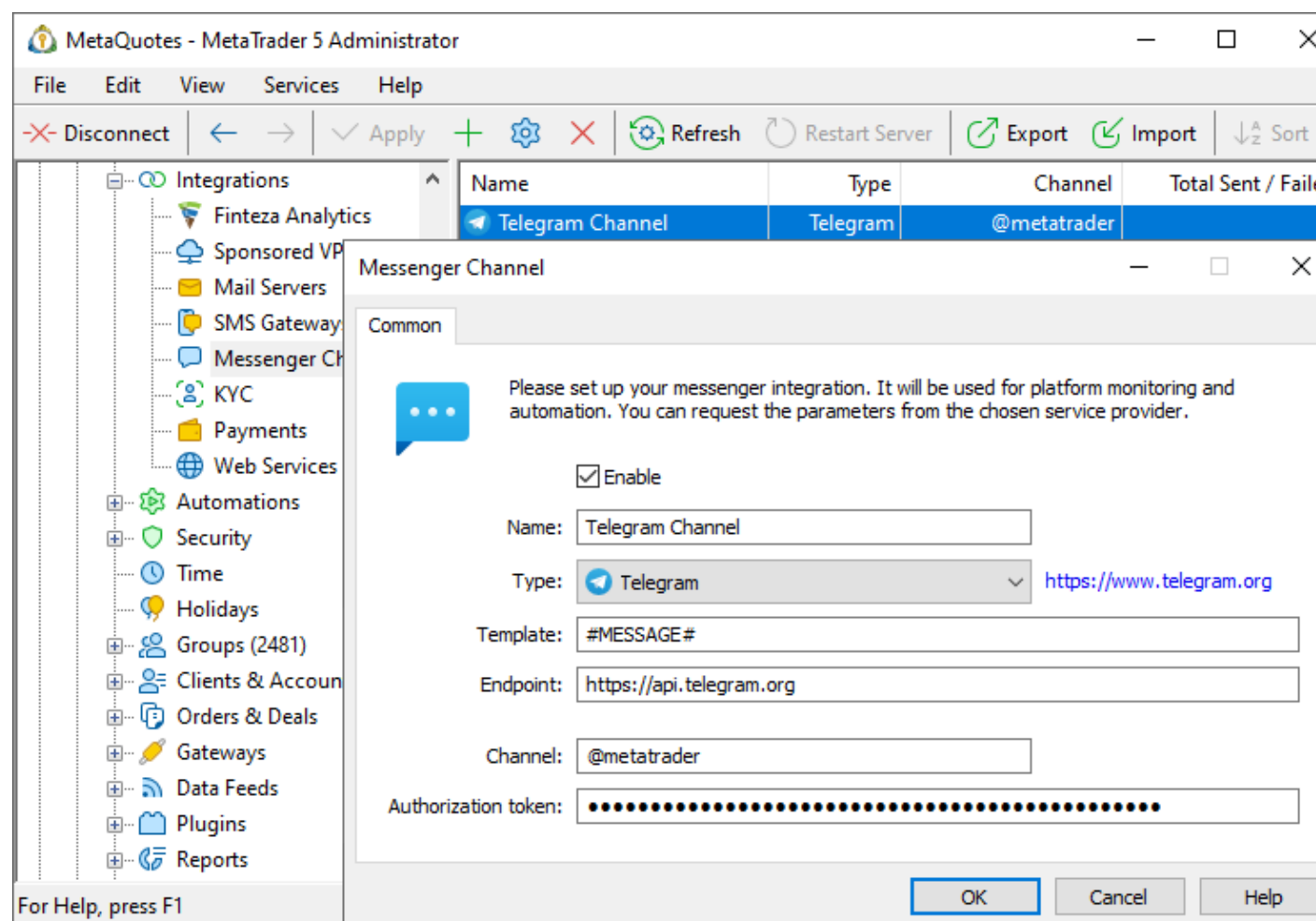
All other settings are standard and consistent with those of other SMS gateways. You can configure the provider's availability for countries and groups.

MetaTrader 5 Trading Platform / Platform Setup / Integrations / Messengers

Messengers

The trading platform features built-in integration with popular instant messengers. Using it along with the Automations service, you can configure the platform to send notifications about various system events to Telegram, Slack and other messenger channels. For example, you can promptly receive monitoring data, notifications about large financial transactions, manager connections in the platform, and other information directly on your phone.

Configuring the Provider



Create a new configuration and specify the following parameters for connecting to the messenger:

- **Enable** enable/disable messenger configuration. If the configuration is disabled, the service will not be used to send messages.
- **Name** configuration name.
- **Type** used messenger. Currently supported messengers integrations include Telegram and Slack. The list will be further expanded.
- **Template** the basic template for sending messages via this messenger.

Further settings depend on the messenger:

- Telegram
- Slack

How to use messengers after setup

To send messages to the configured channels use the "Post in messenger channel" action. Enter the text to send and select one of the available messenger configurations.

Templates

You can use message templates, while the text can be changed depending on the used message messenger or channel. For example, you can unify platform state notifications sent through the Automations service. Specify the relevant text in the "Template" field, and it will be added to all messages:

Messenger Channel

Common Statistics

Please set up your messenger integration. It will be used for platform monitoring and automation. You can request the parameters from the chosen service provider.

☒ Enable

Name: Telegram Channel

Type: Telegram

Template: Platform Monitoring: #MESSAGE#

Endpoint: https://api.telegram.org

Channel: @metatrader

Authorization token:

OK Cancel Help

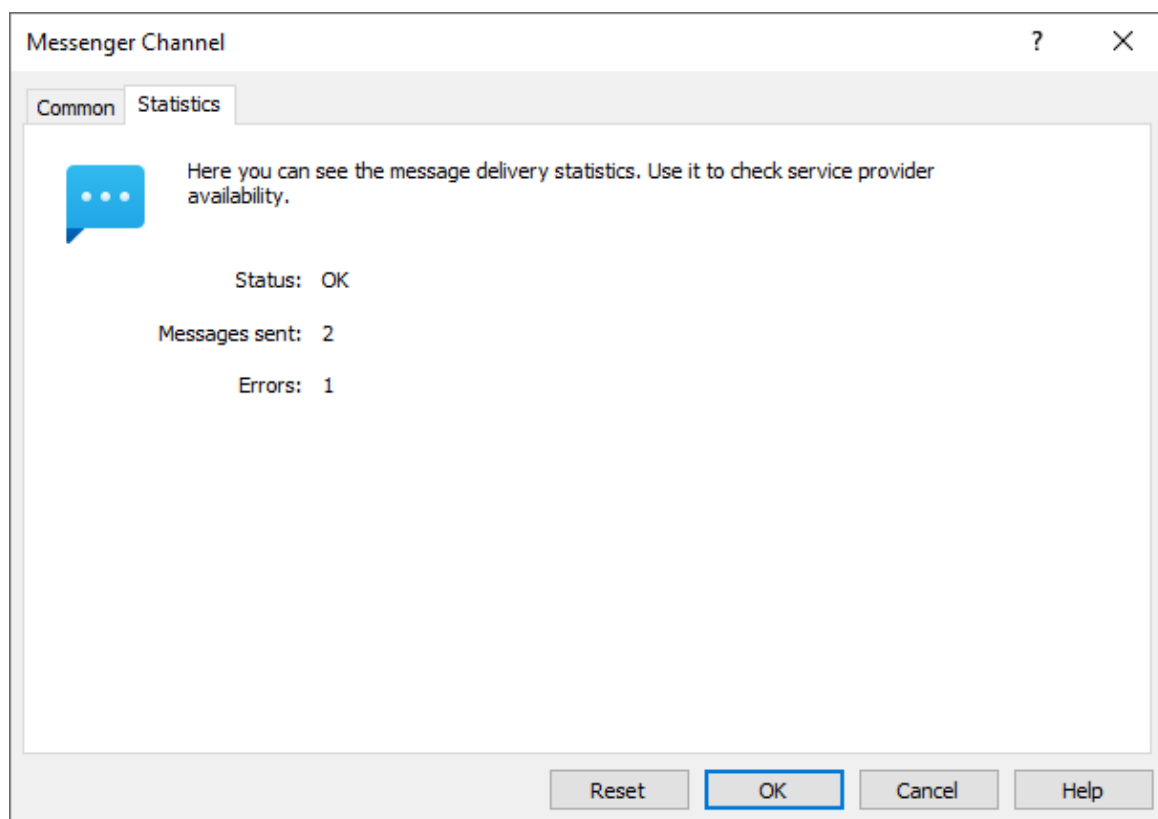
MetaTrader 5 trading platform

The field contains the default #MESSAGE# macro which substitutes the source text. In our example, the "Platform Monitoring" text is added to it, and the resulting message will be generated as "Platform Monitoring: [text from automation action]".

Statistics

During operation, the system collects statistics on sent messages. It is available in the corresponding section of each messenger, as well as in the general list of configurations.

- **Status** the availability of the server (endpoint), via which the messages are sent.
- **Message sent** the number of successfully sent messages.
- **Errors** the number of messages that could not be sent.



MetaTrader 5 trading platform

To reset the statistics and to start collecting it anew, select "Reset".

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [Messengers](#) / Telegram

Telegram

To set up a Telegram channel integration:

- [Create a channel](#) to which messages will be posted
- [Register a bot](#) which will post messages
- [Add the bot to the channel](#) and set the relevant permissions
- [Create a messenger configuration](#) in the platform

Creating a Telegram channel

Create a channel to which messages will be posted. To do this, open the Telegram app, tap the new dialog button next to the search bar and select "Create Channel". If you already have a channel, go to the next step.

Specify the channel name and privacy settings.

Back	Create Channel	Next	Back	Channel Type
 Platform Monitoring			LINK	
Work channel for platform monitoring You can provide an optional description for your channel.			☒ Public ☐ Private Public channels can be found in search, anyone can join.	
			t.me/monitoring_platform monitoring_platform is available. People can share this link with others and find your channel using Telegram search.	
			SAVING AND COPYING CONTENT	
			Restrict Saving Content Subscribers will be able to copy, save and forward content from this channel.	

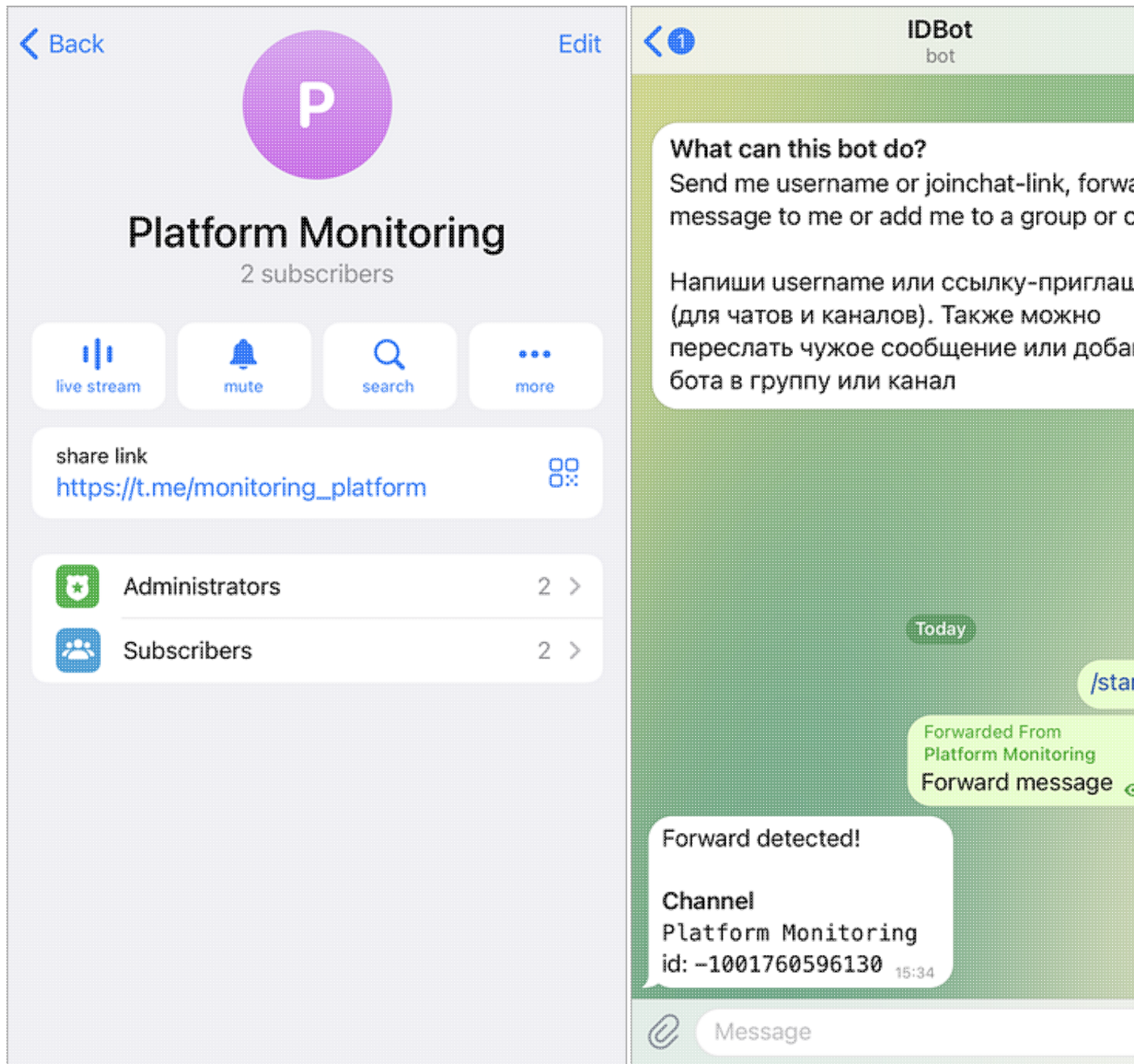
To set up the integration, you will need the name or ID of the created channel:

If the channel is public, the name is available on its properties page. Open the channel and tap on the title. In the properties window, the name is shown in the link: [https://t.me/\[channel name\]](https://t.me/[channel name]). Specify it in the platform-side messenger settings, along with the @ character. For example, @monitoring_platform.

The public name is not available for private channels. You will need to use the Channel ID instead. To receive it, forward any message from the channel to the bot at https://t.me/username_to_id_bot. Specify the ID in the platform-side messenger settings. For example, "-1001760596130" (always specify "-" before the ID number).

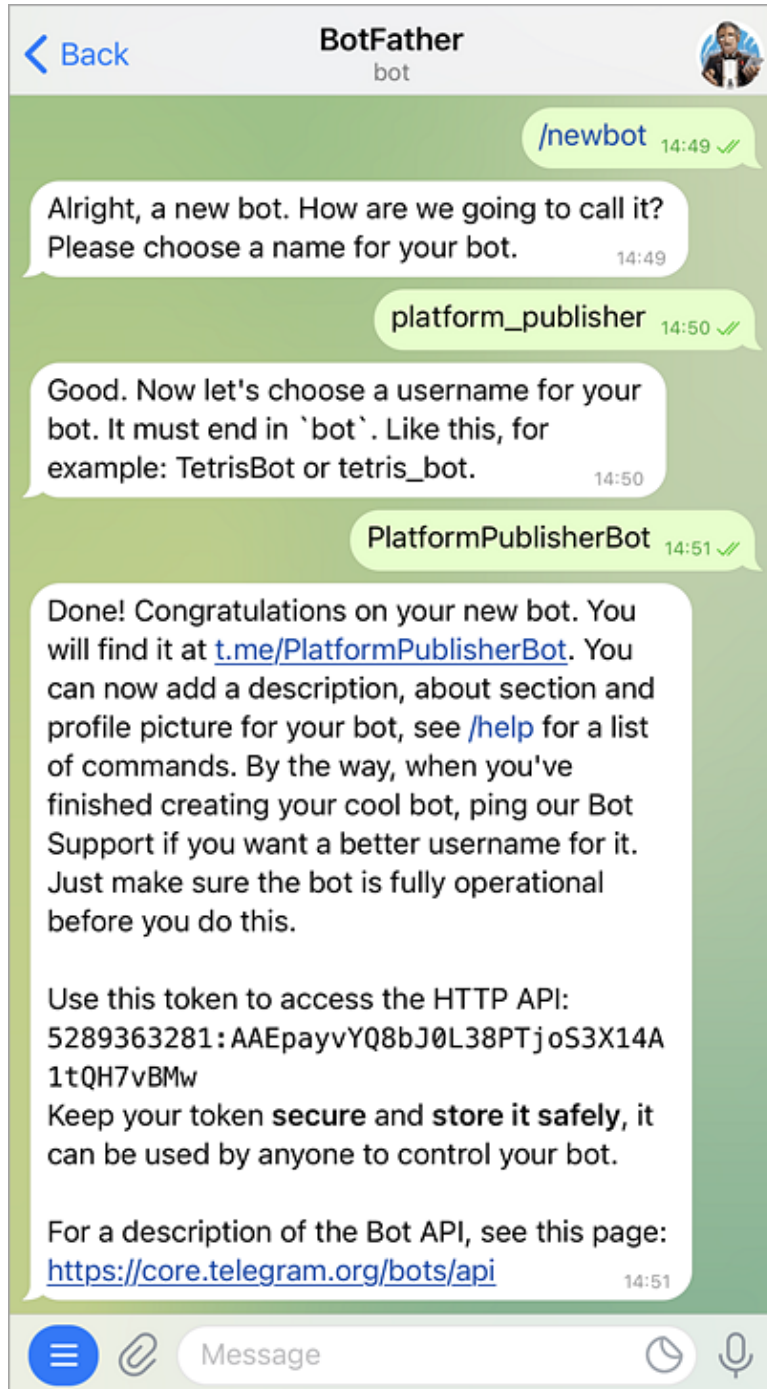
Optionally, you can use the following method to obtain the channel ID. It works for both public and private channels, but it can only be used after you have created your bot.

- Prepare a bot and a channel for integration by following the instructions from this section
- Send any message from the channel to your bot
- When setting up the configuration on the platform side, leave the Channel field empty
- The system will check the incoming message in the bot, will receive the channel ID and will add it to the configuration



Create a Telegram bot

Create a Telegram bot which will post messages to the channel. To do this, open a chat with the BotFather user at <https://t.me/botfather>. Send the /newbot command to the chat. Then set the login and the name of the bot.

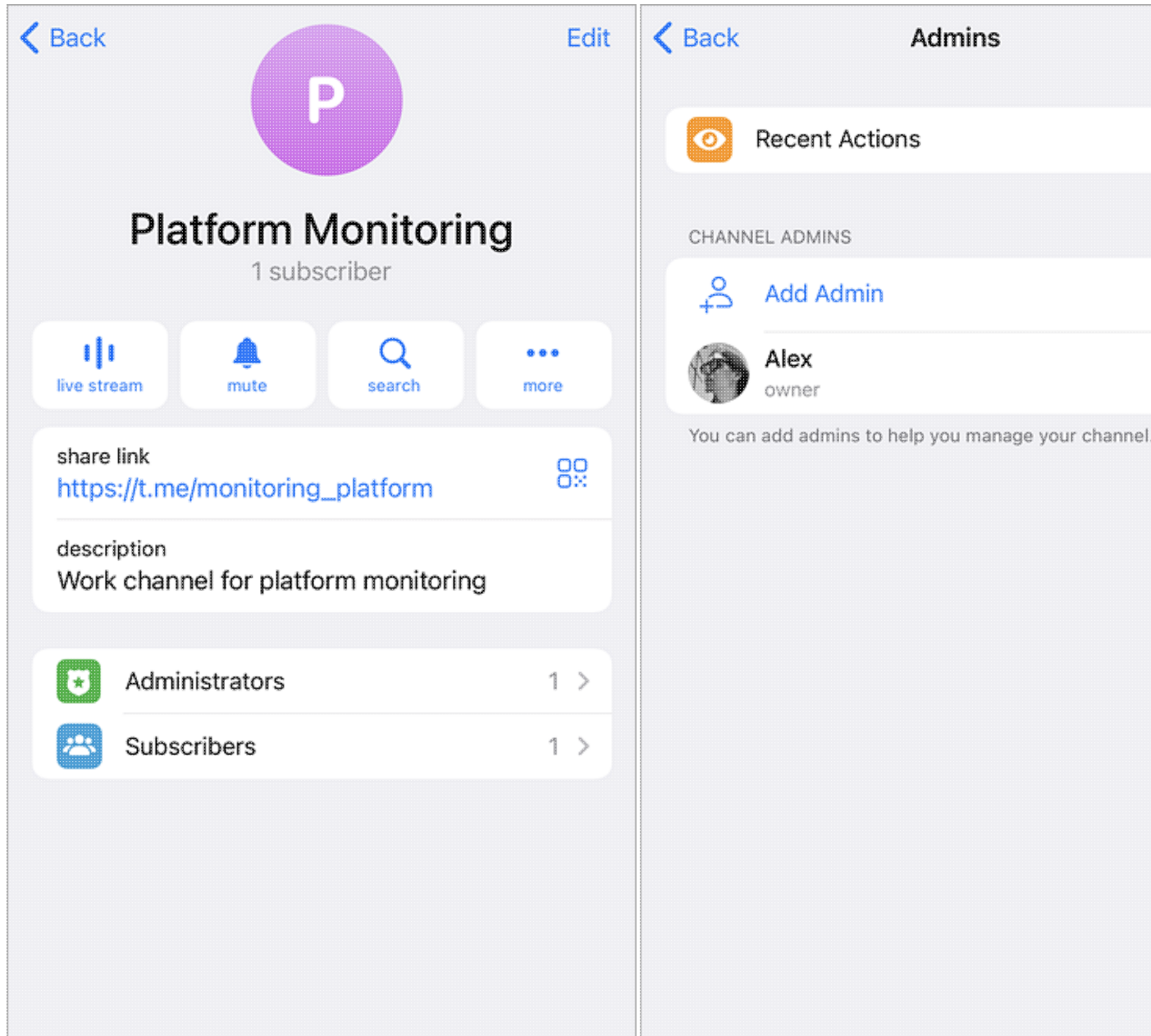


MetaTrader 5 trading platform

You will receive a message with the bot authorization token. The token is used when configuring the messenger in the platform.

Adding the bot to a channel

Next, add the bot to the channel to which you want to publish messages. Open the channel and tap on the title. Select "Administrators" and tap "Add Admin". After that, enter the bot name in the search bar.



MetaTrader 5 trading platform

Optionally, you can open the bot profile using the link obtained in the previous step and then select "Add to Group or Channel". After that, specify permissions for the bot.

[Back](#)



platform_publisher

bot

mute

search

stop

more

username

@PlatformPublisherBot



[Report](#)

[Add to Group or Channel](#)

This bot is able to manage a group or channel.

[Cancel](#)[Add Bot](#)



platform_publisher
bot

WHAT CAN THIS ADMIN DO?

Change Channel Info

Post Messages

Edit Messages of Others

Delete Messages of Others

Add Subscribers

Manage Live Streams

Add New Admins

This admin will not be able to add new admins.

[Add Bot as Admin](#)

MetaTrader 5 trading platform

The bot only needs a permission to post messages, but it must be added to the channel as Admin.

Similarly, in the "Subscribers" section of the channel properties, you can add employee accounts to the channel.

Creating a configuration

When the channel and bot are ready, create a messenger configuration in the platform, fill in the common parameters, and then specify the following details:

- **Endpoint** the address of the Telegram server to which requests are sent. The default endpoint is <https://api.telegram.org>, this value should not be changed.
- **Channel** the name of the channel to which the messages will be sent (begins with @). Optional a channel identifier, such as "-1001760596130", can be specified here.
- **Authorization token** Telegram bot authorization token.

Messenger Channel

Common

Please set up your messenger integration. It will be used for platform monitoring and automation. You can request the parameters from the chosen service provider.

☒ Enable

Name: Telegram Channel

Type: Telegram <https://www.telegram.org>

Template: #MESSAGE#

Endpoint: https://api.telegram.org

Channel: @metatrader

Authorization token:

OK Cancel Help

MetaTrader 5 Trading Platform / Platform Setup / Integrations / Messengers / Slack

Slack

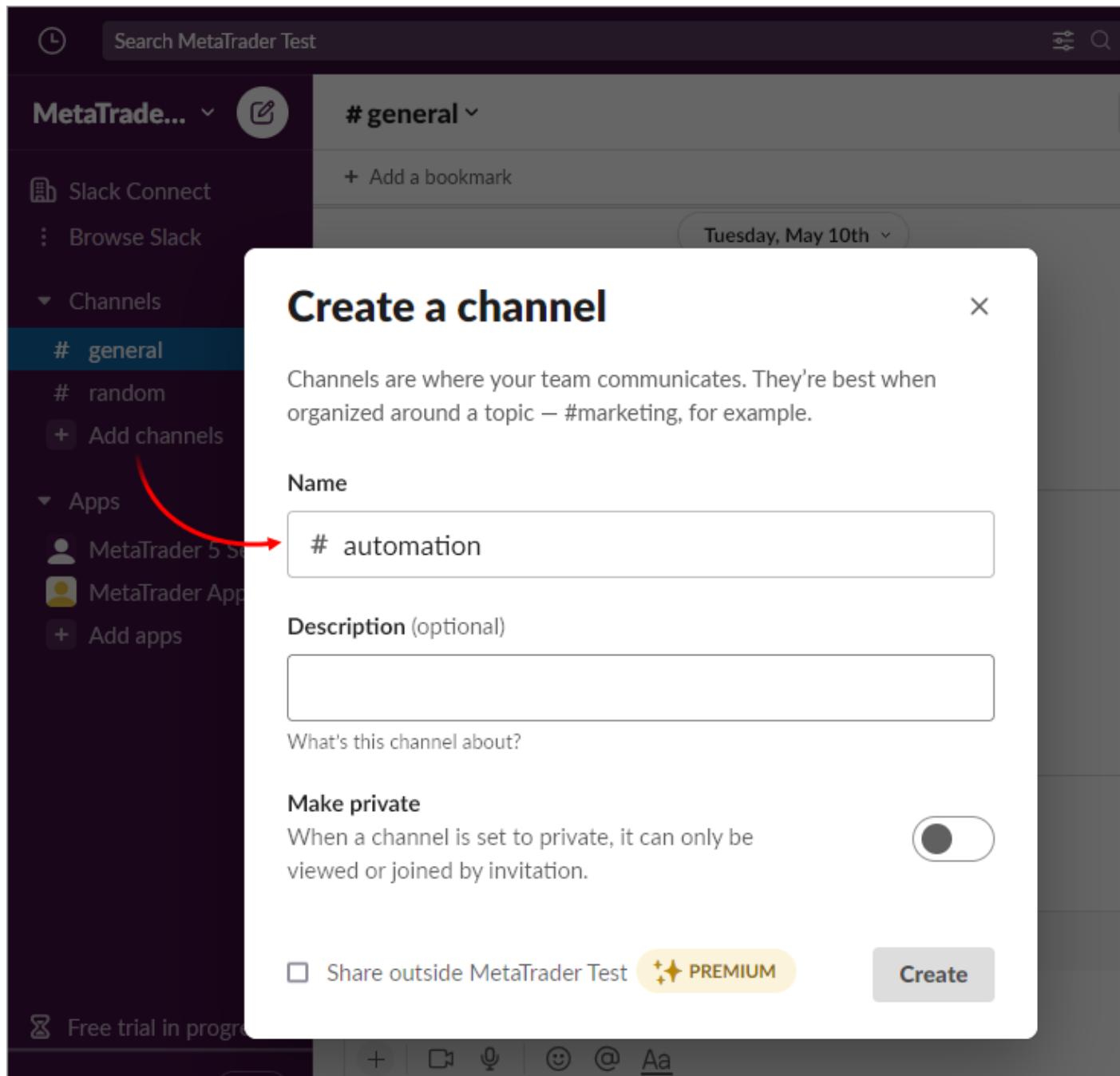
To set up a Slack channel integration:

- Create a workspace in which messages will be posted
- Create an application which will post the messages
- Connect the application to the channel in the workspace
- Create a messenger configuration in the platform

Creating a channel in the workspace

Select a workspace and create there a channel to which the system will post messages. If you do not have a workspace yet, create it following the Slack instructions.

To create a channel, click "Add channel" on the left-hand side of the workspace. Next, specify the name of the channel.

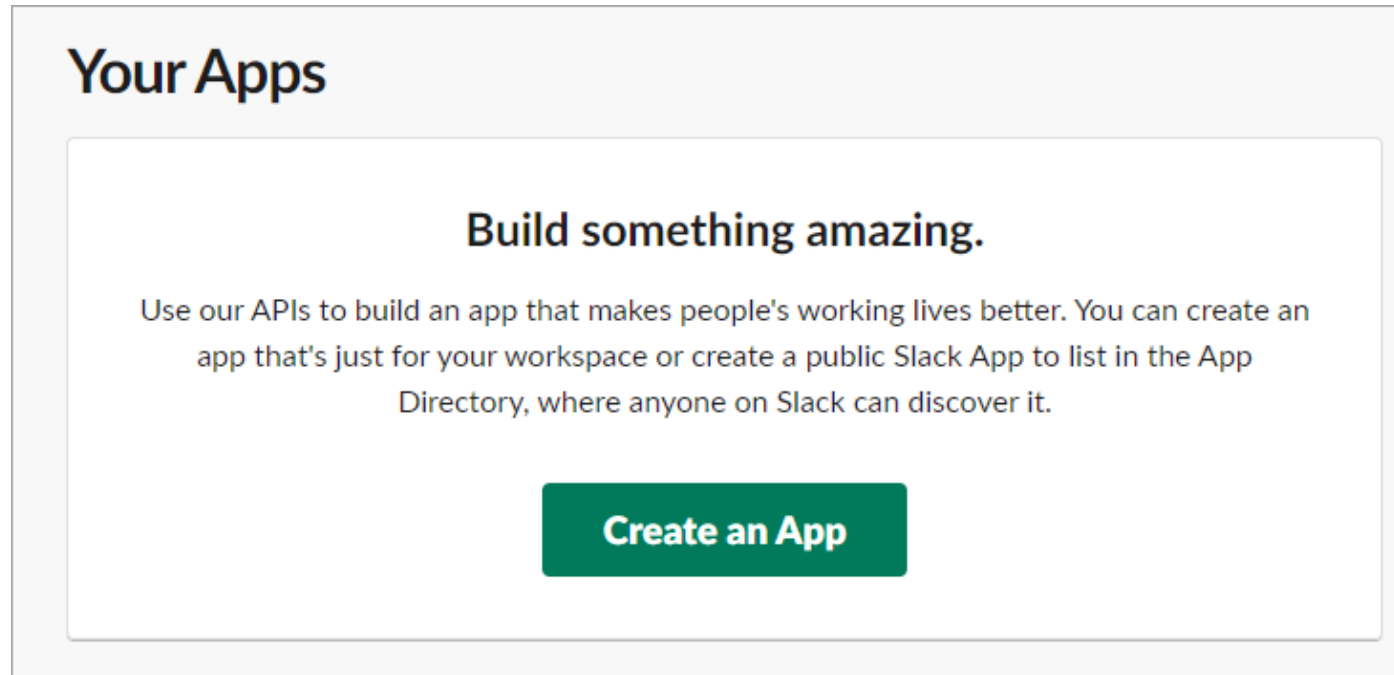


MetaTrader 5 trading platform

The channel name will be used to configure the messenger on the platform side. It is specified without #. In this example, it is "automation".

Creating an application in Slack

To create an application, follow the link https://api.slack.com/apps?new_app=1.



You can configure the created application manually or use the attached manifest a ready-made settings template, in which you should only replace a few lines.

Configuration using the manifest

To assist with the setup, we have prepared a manifest template. It sets all the necessary application parameters, and you only need to specify your name.

The manifest is available in two formats: YAML and JSON. You may use any of them at your discretion:

YAML

```
display_information:
  name: MetaTrader App
features:
  bot_user:
    display_name: MetaTrader App
    always_online: false
oauth_config:
  scopes:
    bot:
      - chat:write
```

```
- channels:join
settings:
  org_deploy_enabled: false
  socket_mode_enabled: false
  token_rotation_enabled: false
```

JSON

```
{
  "display_information": {
    "name": "MetaTrader App"
  },
  "features": {
    "bot_user": {
      "display_name": "MetaTrader App",
      "always_online": false
    }
  },
  "oauth_config": {
    "scopes": {
      "bot": [
        "chat:write",
        "channels:join"
      ]
    }
  },
  "settings": {
    "org_deploy_enabled": false,
    "socket_mode_enabled": false,
    "token_rotation_enabled": false
  }
}
```

You should change two parameters in these settings:

- name application name.
- display name the display name will be used for the messages published by the application in the workspace.

When creating an application, select "From an app manifest", then select a workspace and paste in the updated manifest text:

Create an app

Choose how you'd like to configure your app's scopes and settings.

From scratch

Use our configuration UI to manually settings, & features to your app.

From an app manifest BETA

Use a manifest file to add your app's settings & features to your app.

Pick a workspace to develop your app

Pick a workspace to develop your app

 **MetaTrader**

Keep in mind that you can't change this workspace, you won't be able to manage it. This workspace will control the app even if you change it.

[Sign into a different workspace](#)

Step 1 of 3

Enter app manifest below BETA

This is your app's manifest containing basic info, scopes, settings, and features. For help on how this works, you can check out our [documentation](#) or check out a few [examples](#).

YAML

JSON

```
1 display_information:
2   name: MetaTrader App
3 features:
4   bot_user:
5     display_name: MetaTrader App
6     always_online: false
7 oauth_config:
8   scopes:
9     bot:
10      - chat:write
11      - channels:join
12 settings:
13   org_deploy_enabled: false
14   socket_mode_enabled: false
15   token_rotation_enabled: false
```

Step 2 of 3

Back

Next

Manual configuration

When creating an application, select "From scratch", then specify the application name and select a workspace:

Create an app

Choose how you'd like to configure your app

From scratch
Use our configuration UI to manually set settings, & features to your app.

From an app manifest BETA
Use a manifest file to add your app's settings & features to your app.

Need help? Check our [documentation](#), or [contact support](#)

Name app & choose workspace

App Name
MetaTrader App

Don't worry - you'll be able to change this later.

Pick a workspace to develop your app in:
MetaTrader Test

Keep in mind that you can't change this app's workspace later. If you leave the workspace, you won't be able to manage any apps you've built for it. The workspace will control the app even if you leave the workspace.

[Sign into a different workspace](#)

By creating a Web API Application, you agree to the [Slack API Terms of Service](#).

[Cancel](#) [Create App](#)

MetaTrader 5 trading platform

After creating the application, add "chat:write" and "channels:join" permissions for the app in the "OAuth & Permissions" section:

Settings

- Basic Information
- Collaborators
- Socket Mode
- Install App
- Manage Distribution

Features

- App Home
- Org Level Apps
- Incoming Webhooks
- Interactivity & Shortcuts
- Slash Commands
- Workflow Steps
- OAuth & Permissions**
- Event Subscriptions
- User ID Translation
- App Manifest NEW
- Beta Features

Submit to App Directory

- Review & Submit

Give feedback

Slack ❤️

Help

Contact

Scopes

A Slack app's capabilities and permissions are governed by the [scopes](#) it requests.

Bot Token Scopes

Scopes that govern what your app can access.

OAuth Scope	Description
You haven't added any OAuth Scopes for your Bot token.	
Add permission by Scope or API method... ⌵ ⓧ channels:join Join public channels in a workspace channels:manage Manage public channels that MetaTrader App Test has been channels:read View basic information about public channels in a workspace chat:write Send messages as MetaTrader App Test chat:write.customize Send messages as MetaTrader App Test with a customized u	

[Add an OAuth Scope](#)

Scopes define the [API methods](#) an app is allowed to call, which information and capabilities are available on the workspace it's installed on. Many scopes are restricted to specific [resources](#) like channels or files.

Getting an OAuth token

In the "OAuth & Permissions" section, click "Install to Workspace". After you confirm access to the workspace, a token will be generated for you. Specify it in the platform-side [messenger settings](#).

OAuth Tokens for Your Workspace

These [OAuth Tokens](#) will be automatically generated when you connect the app to your workspace. You'll use them to authenticate your app.

Install to Workspace

MetaTrader App is requesting permission to access the MetaTrader Test Slack workspace

What will MetaTrader App Test be able to do?

 Perform actions in channels & conversations

Cancel

Allow

OAuth Tokens for Your Workspace

These tokens were automatically generated when you connected the app to your team. You can use these to authenticate your app [more](#).

Bot User OAuth Token

xoxb-3481870945490-3528202039889-2TXrG...

Access Level: Workspace

Reinstall to Workspace

Org-Wide Installation

Only an org admin has the option to install org-wide apps. If your app is ready to be installed on your organization, you have several options on how you can get your org admin to install it.

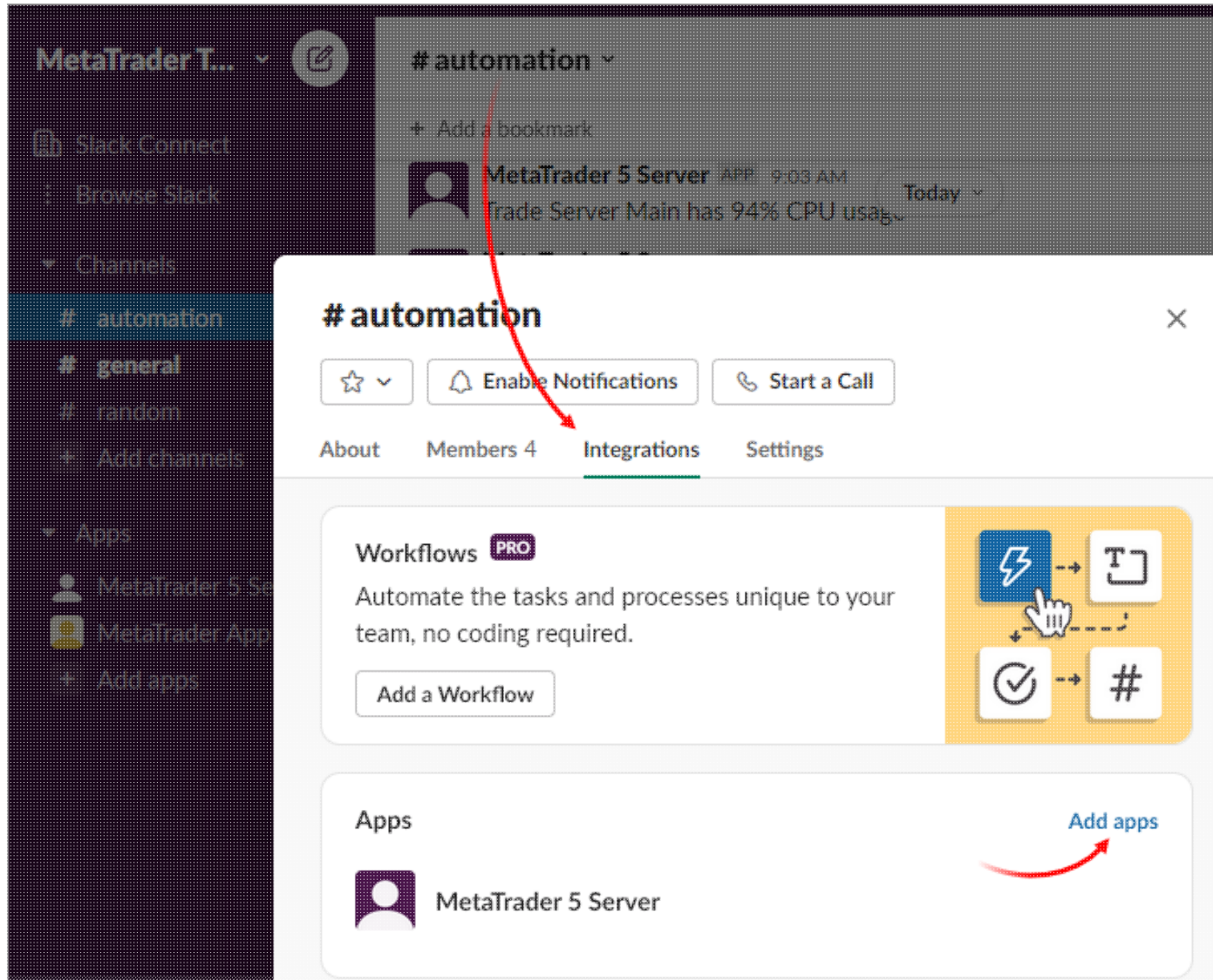
- Send your admin a [shareable URL](#), your admin can use it to authorize this app
- Add your org admin as a [collaborator](#) on your workspace. They will be able to directly install from these pages

[Learn more about org-wide app installation.](#)

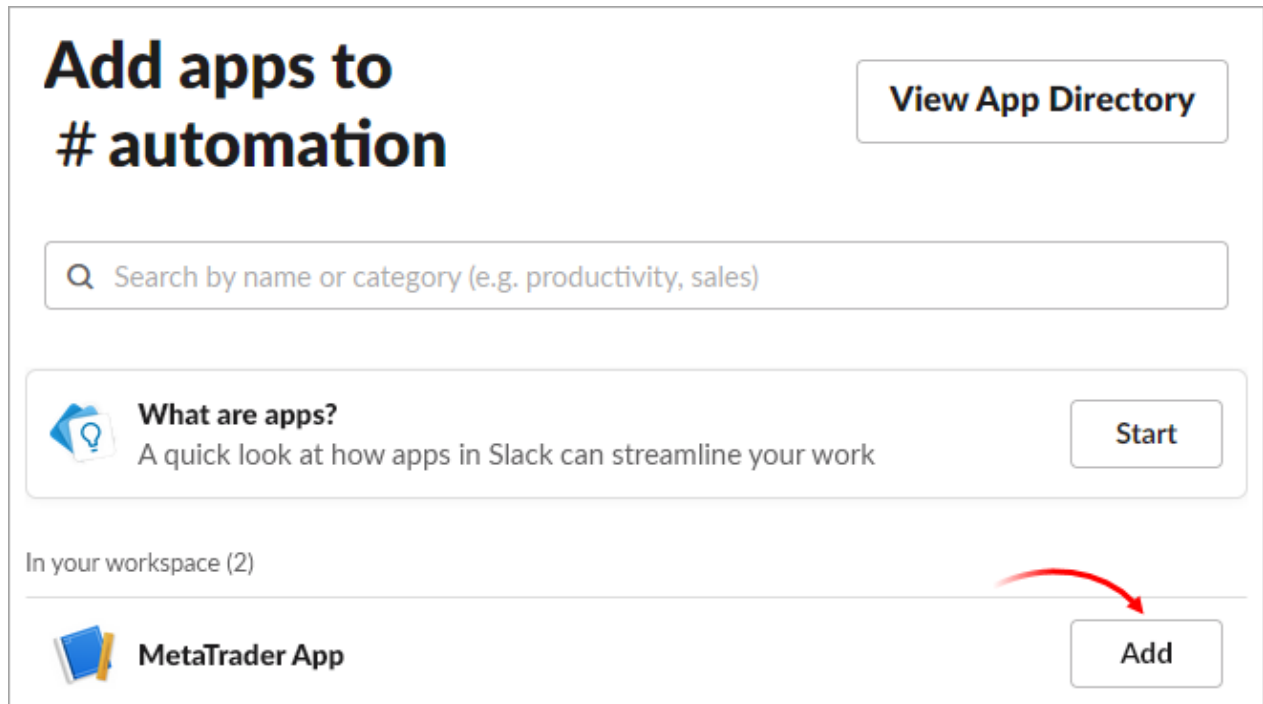
MetaTrader 5 trading platform

Connecting an application to a workspace

Go to Slack and add the application to the channel where you want to post messages. Click on the channel name and go to the "Integrations" section. Click "Add apps" in this section:



Next, select the previously created application.



Creating a configuration

When your workspace and application are ready, create a messenger configuration in the platform, fill in the common parameters, and then specify the following details:

- **Endpoint** the address of the Slack server to which requests are sent. The default endpoint is `https://slack.com/api`; this value should not be changed.
- **Channel** the name of the channel within the Slack workspace, to which the messages will be sent. The name is specified without the `#` prefix.
- **OAuth token** token for authorizing the application that will send messages to the channel.

MetaTrader 5 trading platform

Messenger Channel ? ×

Common **Statistics**

 Please set up your messenger integration. It will be used for platform monitoring and automation. You can request the parameters from the chosen service provider.

☒ Enable

Name:

Type:  Slack ▼ <https://support.metaquotes.net>

Template:

Endpoint:

Channel:

OAuth token:

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / KYC

KYC

The trading platform integration with KYC (Know Your Customer) services enables the validation of client personal data and documents with one click or even in a fully automated mode. Set up connection to an external KYC provider in just a few minutes in order to:

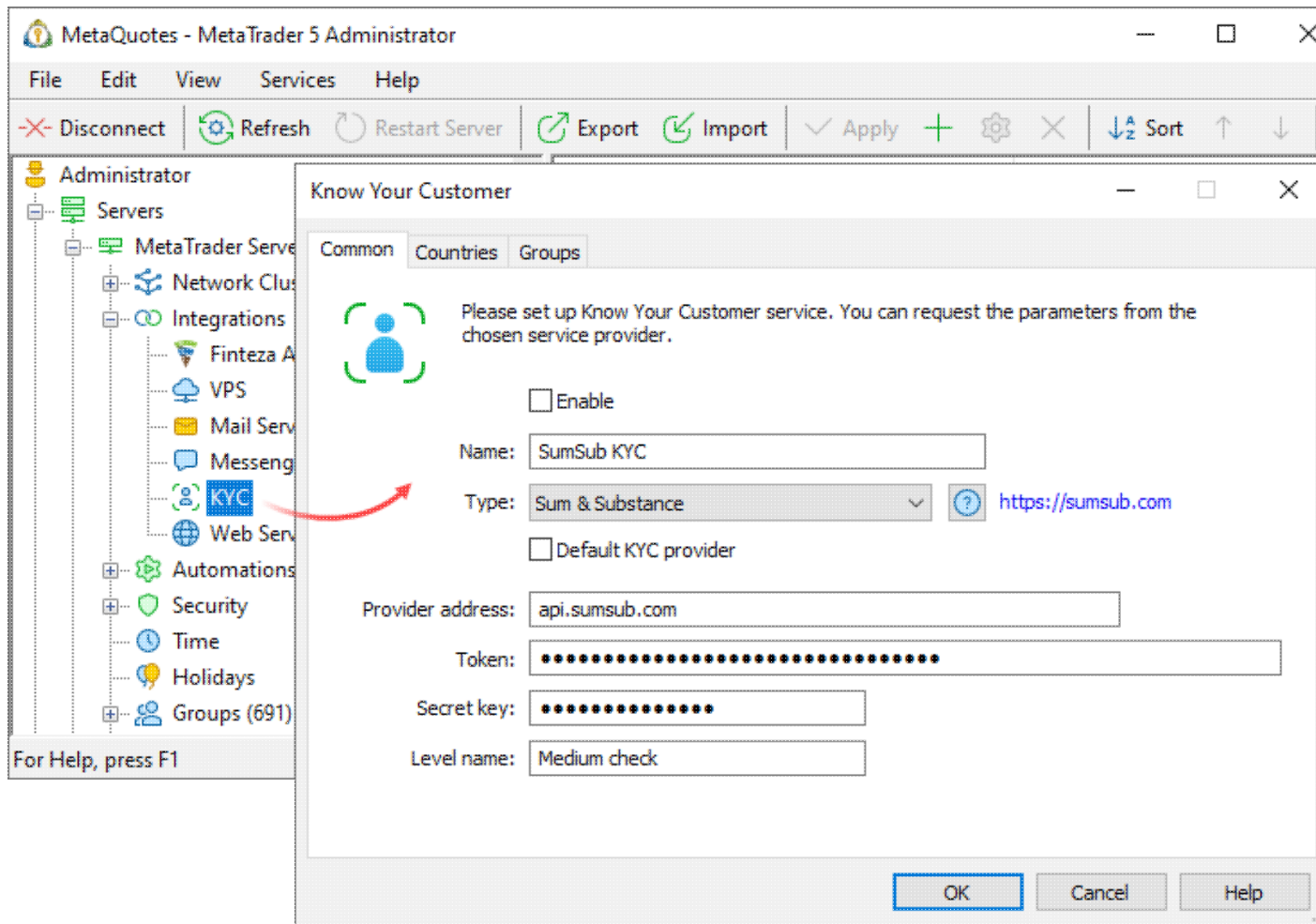
- Send data and documents for automated verification straight from the [client dialog](#). Each check results in a detailed report, which is saved in the client history. Based on the check results, an appropriate status is automatically set for [client documents](#).
- Configure [automatic data check](#) for preliminary account registration. This feature provides the further automation of client onboarding procedures.

Request trial from Sum & Substance to evaluate the KYC procedure benefits.
--

Integration Setup

Open the "Integrations KYC" section and create a new configuration:

MetaTrader 5 trading platform



Specify the following parameters:

- **Name** configuration name.
- **Type** KYC check service type. Each provider has an individual set of settings. Please request the required settings from the provider.
- **Default KYC provider** the platform uses the option when selecting a KYC provider to check client data, if the procedure could not be executed using group or country settings.

Further settings depend on the provider. Please request required settings from the provider.

Currently only the Sum & Substance provider is supported. Support for World-Check, eSpear and other providers will be added soon.

Sum & Substance

Sum&Substance provides an all-in-one technical and legal toolkit to cover KYC/KYB/AML needs. It is a single powerful platform to convert more customers, speed up the verification processes, reduce costs and detect digital fraud.

To get access to the service, please fill in the registration form at <https://sumsub.com>. Next, log into the control panel using the provided login and password. The control panel features the required integration data.

Open section "DevSpace\App Tokens" and add a new application. Add the following permissions for the application: seePersonalData, SeeBgCheckResults, managePersonalData, manageUsers, manageClientSettings, seeStats, seeSensitiveData. After adding the application, you will receive a token and a secret key which you should specify in the provider settings on the MetaTrader 5 side.



Webhooks

App Tokens

Integrations

API Health

App Tokens

Name

Abc Broke

IMPORTANT: Please, securely save the token and the s
information with untrusted parties and **don't embed** it

Token

prd:FVVUIETGwSqXlqhggXvGQDLg.vbJ8cdpYQXy1Tfj

Secret Key

VSxFqOzyaRvqNqJUpE6XLkjS6QWJpADK

All is saved

Create a new App Token

Next, create a provider configuration and specify the following parameters:

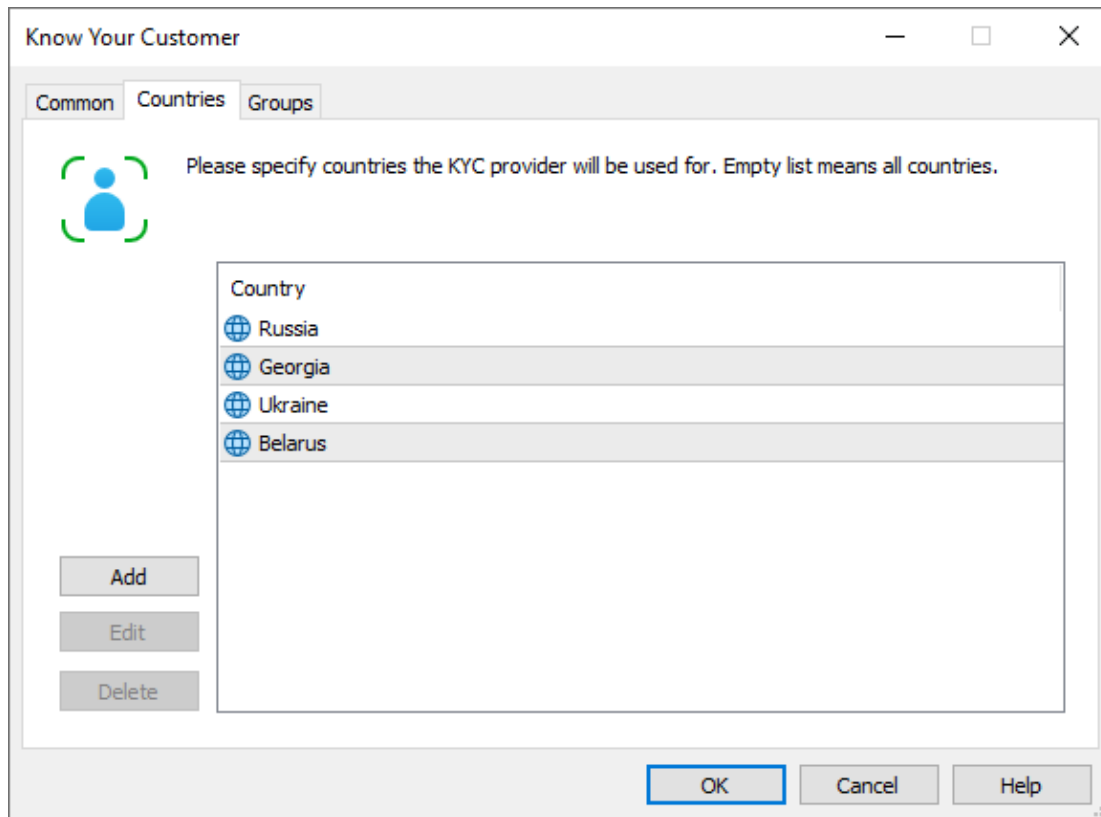
- **Provider address** address for connecting to the service. api.sumsub.com is used by default. In most cases, there is no need to change it.
- **Token** token for authorization. The token is available in the DevSpace section of the control panel.
- **Secret key** a secret key for authorization. The key is available in the DevSpace section of the control panel.

In the "Level name" parameter, specify the name of the setting which defines your client verification process. Sum & Substance allows the customization of the number of steps which the user must complete in order to obtain the verified status, as well as of the set of required documents and other parameters. This procedure is called the Verification Level. You can use different levels depending on certain conditions. For example, you can set soft verification for safer regions. You can pre-configure verification levels in the Sum & Substance control panel and create several KYC configurations on the MetaTrader 5 side. Specify in the configurations specific verification levels and define the list of countries for which they will be used.

For further details about how to set up verification levels please view the [Sum & Substance documentation](#).

Countries

Specify the list of countries, for which this KYC provider will be used, for [manual](#) and for [automatic](#) checks. Use of different providers ensures the best verification conditions for clients from different regions. You can also [adjust the verification process](#) depending on the country.



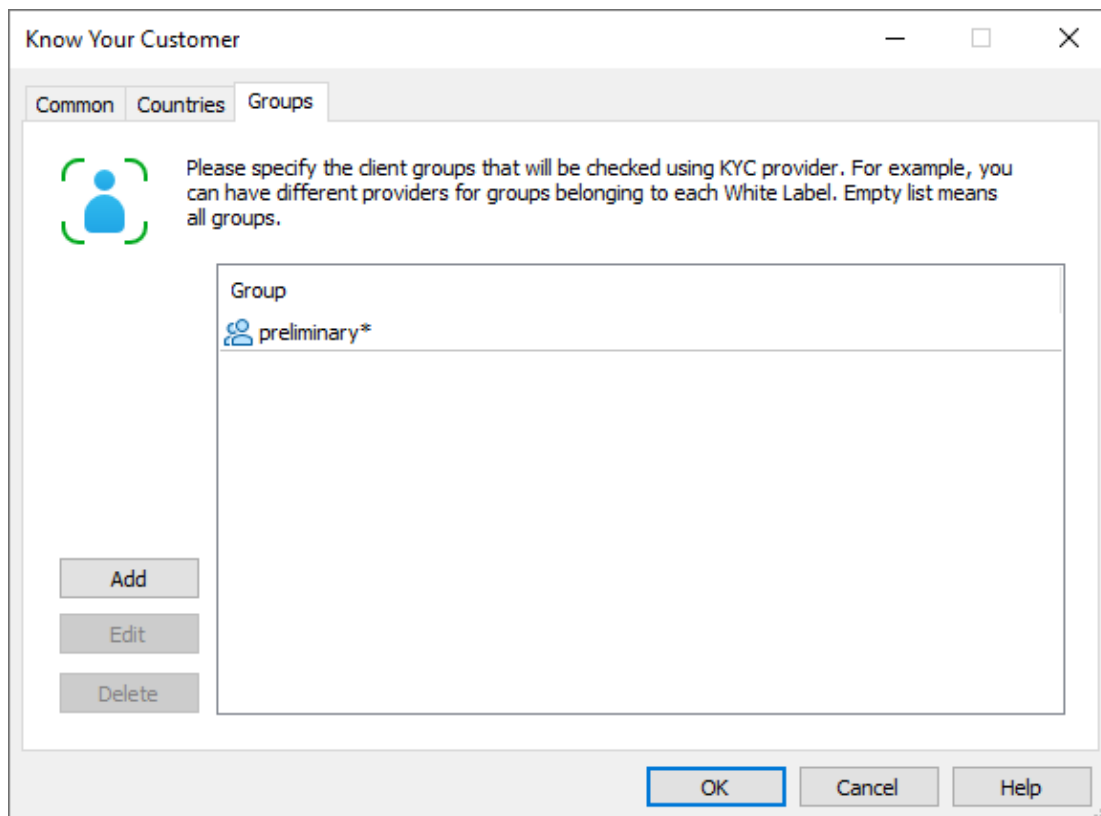
MetaTrader 5 trading platform

If no country is specified, the platform will use this provider for clients from any country.

When opening a demo account, the country is determined automatically on the client terminal side. When opening a preliminary account, the country is also detected automatically, but the client can specify another country in the registration form.

Groups

Specify the list of client groups, for which this KYC provider will be used, for manual and for automatic checks. Thus you can organize work with your White Label partners. You partners can select providers and pay for their services.



The screenshot shows the 'Know Your Customer' dialog box with the 'Groups' tab selected. The dialog has three tabs: 'Common', 'Countries', and 'Groups'. The 'Groups' tab contains a list of client groups. The list has a header 'Group' and one entry 'preliminary*'. To the left of the list are three buttons: 'Add', 'Edit', and 'Delete'. At the bottom of the dialog are three buttons: 'OK', 'Cancel', and 'Help'. The 'OK' button is highlighted with a blue border.

Know Your Customer

Common Countries Groups

Please specify the client groups that will be checked using KYC provider. For example, you can have different providers for groups belonging to each White Label. Empty list means all groups.

Group

- preliminary*

Add Edit Delete

OK Cancel Help

How the provider is selected

- The system checks the client group and country
- The provider located higher in the list is selected from all providers, which meet the country and group conditions
- If none of the providers is suitable, the default provider is utilized (as set by the "Default KYC provider" option in common settings)

Documents request from terminals and automatic verification

The KYC integration enables complete automation of client onboarding procedures.

Live accounts can be requested from desktop and mobile terminals. Configure the registration form to request supporting documents from clients during registration. This can be done by enabling the "Require identity and proof of address documents" option under account allocation section.

Account allocation group

?

×



Please specify a company and a group with description. This group will be available for opening accounts from the client terminals of this company.

Company: Default

Group: preliminary-usd

Description: Real USD

Leverages: 100,50,25,10,5,2,1

Countries: All countries

☒ Require identity and proof of address documents
 ☒ Start KYC check automatically

☒ Use advanced registration form

Confirm: Phone & Email

Mail server: Default

Agreements:

Agreement	URL	Mandatory
ab Client agreement	https://abcbroker.com/legal/client	☒
ab Risk disclosure	https://abcbroker.com/legal/risk	☒

Add

Edit

Delete

OK

Cancel

MetaTrader 5 trading platform

Enable the option "Start KYC check automatically". Once a preliminary account is opened and linked to a client, appropriate client data and documents will be automatically sent to the KYC service for verification. After the check, the system provides a report which can be viewed under the client comments section. If the option is disabled, you can send the data manually through the client's personal data section.

MetaTrader 5 Trading Platform / Platform Setup / Integrations / Event Streaming

Event Streaming

Stream trading platform events to external systems such as Apache Kafka for real-time monitoring and advanced data analysis.

Apache Kafka is a scalable and reliable platform for data stream processing that enables efficient transmission, storage, and analysis of large volumes of information. Integration of MetaTrader 5 with Apache Kafka enables you to export a variety of events to the system:

- Trading operations: all client trades, executed and unexecuted orders, position changes.
- Price data: quotes financial instruments and pricing statistics received by the platform from gateways and data feeds.
- Platform configuration changes: server settings, manager configurations, payment modules, and more.
- Trading account changes: creation, personal data updates, and changes in financial status.
- Client record updates: creation, updates to personal information, linked accounts, documents, and more.

This integration unlocks extensive opportunities for optimizing business processes:

Real-Time Analysis and Forecasting

- Detection of abnormal trading patterns
- Analysis of client behavior to identify potential risks
- Liquidity and risk management optimization

Improved Customer Service Quality

- Monitoring of client trading activity
- Prompt response to unusual situations
- Analysis of user experience to enhance service quality

Enhanced Security

- Detection of suspicious trading activity
- Monitoring of anomalous system behavior
- Automated threat alerts

Integration with Other Systems

- Connection to BI analytics and machine learning tools
- Automated reporting and real-time data processing
- Integration with CRM systems and notification services

Key Terminology

To work with Kafka, you need to be familiar with the key entities used in the system:

- **Broker** a Kafka server that processes message streams and manages their storage. A Kafka cluster can consist of multiple brokers to ensure scalability and fault tolerance.

MetaTrader 5 trading platform

- **Topic** a channel through which data is transmitted. All messages in Kafka are organized into topics, and producers publish messages to specific topics.
- **Partition** a section within a topic used to scale and parallelize data processing. Each topic consists of one or more partitions.
- **Replication Factor** the number of copies of each partition stored in the Kafka cluster to ensure data availability and fault tolerance. When Kafka writes messages to a topic, these messages are distributed across partitions. Each partition has a replication factor that determines how many copies are stored across different brokers. For example, a replication factor of 3 means each partition will have three copies on different brokers.
- **Producer** the component that publishes messages to a Kafka topic. In this case, it refers to the streaming configuration in the MetaTrader 5 platform.
- **Consumer** the component that reads (subscribes to) messages from a Kafka topic. These are external services that you can use for your business needs.
- **Controller** a server running a Kafka instance in controller mode. Controllers store metadata, broker information, and synchronize their state. Typically, three controllers are used per Kafka cluster, each located in a different availability zone to enhance fault tolerance. In earlier Kafka versions (prior to 2.8), ZooKeeper was used as the controller. It stored metadata about the cluster's state and message locations, enabling replication, fault tolerance, and sharding. In recent versions, Kafka can operate without ZooKeeper. Instead, the Kafka server in controller mode handles this. It implements a new metadata management mechanism known as KRaft. If you're planning to install a newer version of Kafka, you can use this mechanism to eliminate the dependency on a separate ZooKeeper service.

Once familiar with the terminology, proceed with system setup:

- [Kafka Installation and Setup](#)
- [Kafka Streaming Setup](#)

MetaTrader 5 Trading Platform
/ Platform Setup / Integrations
/ Event Streaming / Kafka
Installation and Setup

Apache Kafka Server Installation and Setup

There are many ways to install and configure Apache Kafka. In this article, we will demonstrate just one example. Instead of setting up your own server, you may use ready-to-use cloud-based solutions, such as:

- <https://aws.amazon.com/msk/>
- <https://yandex.cloud/en/services/managed-kafka>
- <https://www.digitalocean.com/products/managed-databases-kafka>

Before beginning the installation, we also recommend reviewing the official documentation and other helpful resources about the system:

- <https://kafka.apache.org/quickstart>
- <https://www.confluent.io/resources/ebook/kafka-the-definitive-guide/>

Kafka Cluster Installation

As an example, let's consider the installation of a Kafka cluster consisting of three brokers. All brokers will be installed on a single physical machine running Ubuntu 22.04, each configured with a different IP address. It is assumed that the server is properly configured and accessible from the Internet on ports 9092 and 9093 at the address kafka.build.srv.com.

This machine will serve both as a broker and as the controller managing the entire cluster. Run the following scripts on each instance:

Install the required packages:

```
#update and install zip and jdk

sudo apt-get update; echo Y|apt-get upgrade; echo Y|apt-get dist-upgrade; echo Y|apt-get autoremove

sudo apt-get install unzip

sudo apt-get install default-jdk -y
```

Create a user account under which Kafka services will operate:

```
#create kafka user

sudo adduser --no-create-home --shell=/sbin/nologin --disabled-password --disabled-login --gecos "" kafka
```

Create directories for cluster installation:

```

sudo mkdir /var/kafka

sudo mkdir /var/kafka/kafka1

sudo mkdir /var/kafka/kafka1/logs

sudo mkdir /var/kafka/kafka2

sudo mkdir /var/kafka/kafka2/logs

sudo mkdir /var/kafka/kafka3

sudo mkdir /var/kafka/kafka3/logs

```

Download the latest Kafka distribution from <https://kafka.apache.org/downloads> and extract it into three separate directories: kafka1, kafka2, and kafka3.

Set the Kafka user as the owner of each directory to allow modification of their contents:

```
sudo chown -R kafka:kafka /var/kafka
```

Generate a unique cluster ID. The ID is required to operate in Controller (Kraft) mode. You can use any sufficiently complex alphanumeric combination. This ID will be needed later during the configuration process.

```

/var/kafka/kafka1/bin# ./kafka-storage.sh random-uuid

[unique cluster ID]

```

Issue an SSL certificate for the domain through which Kafka will be accessible, for example, using [Let's Encrypt](#). Copy the certificate file (CRT) and the private key (PFX) to the directory: /var/kafka/kafka1/config/kraft/keys. Next, create a certificate keystore:

```

keytool -importkeystore -srckeystore [certificate.pfx] -srcstoretype pkcs12
-srcstorepass [certificate password] -destkeystore kafka.keystore.jks -deststoretype
jks -deststorepass [keystore password]

keytool -keystore kafka.truststore.jks -alias CARoot -import -file [certificate.crt]
-storepass [keystore password]

```

Replace the placeholders with your actual values:

- certificate.pfx the filename of the certificate containing the private key
- password certificate the password protecting the certificate
- keystore certificate the password protecting the certificate keystore
- certificate.crt the filename of the certificate containing the private key

Then copy the following four files to /var/kafka/kafka1/config/kraft/:

1. admin.properties this file contains the main connection parameters for Kafka.

```
bootstrap.servers=kafka1.build.srv.com:9092, kafka2.build.srv.com:9092,
kafka3.build.srv.com:9092

security.protocol=SASL_SSL

saslm.echanism=PLAIN

saslm.jaas.config=org.apache.kafka.common.security.plain.PlainLoginModule required
username="admin" password="TestAdmin";

ssl.keystore.location=/var/kafka/kafka1/config/kraft/keys/kafka.keystore.jks

ssl.keystore.password=[keystore password]

ssl.key.password=[certificate password]

ssl.truststore.location=/var/kafka/kafka1/config/kraft/keys/kafka.truststore.jks

ssl.truststore.password=hSFdoYWxJ3
```

Please note:

- bootstrap.servers=kafka1.build.srv.com:9092, kafka2.build.srv.com:9092, kafka3.build.srv.com:9092 these are the subdomains where the Kafka brokers will be running. Replace them with the appropriate addresses for your domains. These endpoints will be required when configuring the connection to Kafka from MetaTrader 5.
- username="admin" password="TestAdmin" these are the login and password of the administrator account, which you will use to connect to the Kafka server. You can replace these credentials with your own.
- security.protocol=SASL_SSL the type of authentication that the Kafka server will use. You will need to specify it when configuring the connection to Kafka from MetaTrader 5.
- ssl.keystore.password the password for the certificate keystore
- ssl.key.password the password for the certificate

2. jaas.conf this file is used to define Kafka users.

```
KafkaServer {
    org.apache.kafka.common.security.plain.PlainLoginModule required

//Cluster administrator login

    username="admin"

    password="TestAdmin"

    user_admin="TestAdmin"
```

```
//Logins in the format of user_username="Password"

    user_testconsumer="TestConsumer"

    user_testproducer="TestProducer"

    serviceName="kafka";

};
```

As an example, two users are created with the usernames testconsumer and testproducer. These accounts can be used to connect to Kafka from MetaTrader 5.

For the administrator account login and password, use the corresponding credentials from the admin.properties file.

Copy a similar file to the remaining cluster nodes: /var/kafka/kafka2/config/kraft/ and /var/kafka/kafka3/config/kraft/.

3. kafka1.service this file defines the service configuration for Kafka.

```
[Unit]

Requires=network.target

After=network.target


[Service]

Type=simple

User=kafka

Environment=KAFKA_HEAP_OPTS="-Xmx1G -Xms1G
-Djava.security.auth.login.config=/var/kafka/kafka1/config/kraft/jaas.conf"

Environment=KAFKA_JVM_PERFORMANCE_OPTS="-XX:+UseG1GC
-XX:MaxGCPauseMillis=20 -XX:InitiatingHeapOccupancyPercent=35
-XX:+ExplicitGCInvokesConcurrent"

ExecStartPre=/bin/bash -c '/var/kafka/kafka1/bin/kafka-storage.sh format -t [unique
cluster ID] -c /var/kafka/kafka1/config/kraft/server.properties --ignore-formatted'

ExecStart=/bin/bash -c '/var/kafka/kafka1/bin/kafka-server-start.sh
/var/kafka/kafka1/config/kraft/server.properties'

ExecStop=/var/kafka/kafka1/bin/kafka-server-stop.sh
```

```
Environment="LOG_DIR=/var/kafka/kafka1/logs"
```

```
Restart=on-failure
```

```
LimitNOFILE=65536
```

```
[Install]
```

```
WantedBy=multi-user.target
```

In the file, specify the unique cluster ID defined in the previous steps.

Copy a similar file to the remaining cluster nodes: /var/kafka/kafka2/config/kraft/ and /var/kafka/kafka3/config/kraft/.

4. server.properties the detailed settings of the Kafka server.

```
# Licensed to the Apache Software Foundation (ASF) under one or more
# contributor license agreements. See the NOTICE file distributed with
# this work for additional information regarding copyright ownership.
# The ASF licenses this file to You under the Apache License, Version 2.0
# (the "License"); you may not use this file except in compliance with
# the License. You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
```



```
# This configuration file is intended for use in KRaft mode, where
# Apache ZooKeeper is not present.
#
```

```
##### Server Basics #####
```

```
# The role of this server. Setting this puts us in KRaft mode
process.roles=broker,controller
```

```
# The node id associated with this instance's roles
node.id=1
```

```
# The connect string for the controller quorum
```

```
controller.quorum.voters=1@kafka1.build.srv.com:9093,2@kafka2.build.srv.com:9093,3@kafka3.build.srv.com:9093
#controller.quorum.voters=1@localhost:9093
```

```
##### Socket Server Settings #####
```

```
# The address the socket server listens on.
```

```
# Combined nodes (i.e. those with `process.roles=broker,controller`) must list the controller
```

```
# If the broker listener is not defined, the default listener will use a host name that is equal to
```

```
# with PLAINTEXT listener name, and port 9092.
```

```
# FORMAT:
```

```
# listeners = listener_name://host_name:port
```

```
# EXAMPLE:
```

```
# listeners = PLAINTEXT://your.host.name:9092

listeners=EXTERNAL://kafka1.build.srv.com:9092,CONTROLLER://kafka1.build.srv.com:9092

#listeners=PLAINTEXT://:9092,CONTROLLER://:9093

# Name of listener used for communication between brokers.
inter.broker.listener.name=EXTERNAL

# Listener name, hostname and port the broker or the controller will advertise to clients.
# If not set, it uses the value for "listeners".
advertised.listeners=EXTERNAL://kafka1.build.srv.com:9092

#advertised.listeners=PLAINTEXT://localhost:9092,CONTROLLER://localhost:9093

# A comma-separated list of the names of the listeners used by the controller.
# If no explicit mapping set in `listener.security.protocol.map`, default will be using PLAINTEXT
# This is required if running in KRaft mode.
controller.listener.names=CONTROLLER

# Maps listener names to security protocols, the default is for them to be the same. See the docs for details.
listener.security.protocol.map=EXTERNAL:SASL_SSL,CONTROLLER:SASL_SSL
#listener.security.protocol.map=CONTROLLER:PLAINTEXT,PLAINTEXT:PLAINTEXT,SSL:SSL

# SASL settings
sasl.enabled.mechanisms=PLAIN
sasl.mechanism.controller.protocol=PLAIN
sasl.mechanism.inter.broker.protocol=PLAIN
```

```
security.protocol=SASL_SSL

authorizer.class.name=org.apache.kafka.metadata.authorizer.StandardAuthorizer
allow.everyone.if.no.acl.found=false
super.users=User:admin

# Specify the service name for SASL (even if PLAIN is used; this parameter is required for c
sasl.kerberos.service.name=kafka1


# SSL/TLS Settings
ssl.keystore.location=/var/kafka/kafka1/config/kraft/keys/kafka.keystore.jks
ssl.keystore.password=[keystore password]
ssl.key.password=[certificate password]
ssl.truststore.location=/var/kafka/kafka1/config/kraft/keys/kafka.truststore.jks
ssl.truststore.password=[keystore password]


# TLS between brokers and clients
#security.inter.broker.protocol=SSL
ssl.client.auth=required


# The number of threads that the server uses for receiving requests from the network and se
num.network.threads=3


# The number of threads that the server uses for processing requests, which may include dis
num.io.threads=8
```

```
# The send buffer (SO_SNDBUF) used by the socket server
socket.send.buffer.bytes=102400
```

```
# The receive buffer (SO_RCVBUF) used by the socket server
socket.receive.buffer.bytes=102400
```

```
# The maximum size of a request that the socket server will accept (protection against OOM)
socket.request.max.bytes=104857600
```

```
##### Log Basics #####
```

```
# A comma separated list of directories under which to store log files
log.dirs=/var/kafka/kafka1/logs/kraft-combined-logs
```

```
# The default number of log partitions per topic. More partitions allow greater
# parallelism for consumption, but this will also result in more files across
# the brokers.
num.partitions=3
```

```
# The number of threads per data directory to be used for log recovery at startup and flushing
# This value is recommended to be increased for installations with data dirs located in RAID
num.recovery.threads.per.data.dir=1
```

```
##### Internal Topic Settings #####
# The replication factor for the group metadata internal topics "__consumer_offsets" and "__transaction_state"
# For anything other than development testing, a value greater than 1 is recommended to ensure high availability
offsets.topic.replication.factor=1
transaction.state.log.replication.factor=1
transaction.state.log.min.isr=1

##### Log Flush Policy #####
# Messages are immediately written to the filesystem but by default we only fsync() to sync
# the OS cache lazily. The following configurations control the flush of data to disk.
# There are a few important trade-offs here:
#   1. Durability: Unflushed data may be lost if you are not using replication.
#   2. Latency: Very large flush intervals may lead to latency spikes when the flush does occur.
#   3. Throughput: The flush is generally the most expensive operation, and a small flush interval
#   may impact overall throughput.
# The settings below allow one to configure the flush policy to flush data after a period of time
# or every N messages (or both). This can be done globally and overridden on a per-topic basis.

# The number of messages to accept before forcing a flush of data to disk
#log.flush.interval.messages=10000

# The maximum amount of time a message can sit in a log before we force a flush
#log.flush.interval.ms=1000

##### Log Retention Policy #####
```

```
# The following configurations control the disposal of log segments. The policy can
# be set to delete segments after a period of time, or after a given size has accumulated.
# A segment will be deleted whenever *either* of these criteria are met. Deletion always happens
# from the end of the log.
```

```
# The minimum age of a log file to be eligible for deletion due to age
```

```
log.retention.hours=168
```

```
# A size-based retention policy for logs. Segments are pruned from the log unless the remaining
# segments drop below log.retention.bytes. Functions independently of log.retention.hours.
```

```
#log.retention.bytes=1073741824
```

```
# The maximum size of a log segment file. When this size is reached a new log segment will be created
```

```
log.segment.bytes=1073741824
```

```
# The interval at which log segments are checked to see if they can be deleted according
# to the retention policies
```

```
log.retention.check.interval.ms=300000
```

Replace the domain addresses with your own in the parameters
controller.quorum.voters, listeners, and advertised.listeners.

Copy a similar file to the remaining cluster nodes: /var/kafka/kafka2/config/kraft/ and
/var/kafka/kafka3/config/kraft/, replacing the following lines:

```
node.id=1
```

```
...
```

```
listeners=EXTERNAL://kafka1.build.srv.com:9092,CONTROLLER://kafka1.build.srv.com:9092
```

...

```
advertised.listeners=EXTERNAL://kafka1.build.srv.com:9092
```

Replace the server number with the corresponding one: 2 and 3.

Next, create a symlink `kafka1.service -> /etc/systemd/system` and start the cluster node:

```
systemctl daemon-reload
systemctl start kafka1
systemctl start kafka2
systemctl start kafka3
```

After completing the setup, you can start working with the server: create topics, connect to brokers, and transmit data.

Any changes to the configuration files require a sequential start of the cluster nodes.

Configure Producer Permissions

To manage system access in Kafka, an Access Control List (ACL) is used. This list defines all users and their corresponding permissions.

MetaTrader 5 acts as a producer, and it requires appropriate permissions specifically, the ability to create and modify topics. Since the platform exports a wide variety of data into different topic, assigning permissions for each individual topic would be impractical. Instead, you can configure permissions once for all topics with a specific prefix. For example, MT5-. When setting up data streaming in MetaTrader 5, you can specify this prefix so that all topic names generated by the platform begin with it.

Earlier we creates the user testproducer. Now, assign permissions for this user to work with all topics that have the MT5- prefix using the following commands:

```
/var/kafka/kafka1/bin/kafka-acls.sh --bootstrap-server kafka1.build.srv.com:9092
--command-config /var/kafka/kafka1/config/kraft/admin.properties --add
--allow-principal User:testproducer --operation READ --topic MT5-
--resource-pattern-type prefixed

/var/kafka/kafka1/bin/kafka-acls.sh --bootstrap-server kafka1.build.srv.com:9092
--command-config /var/kafka/kafka1/config/kraft/admin.properties --add
--allow-principal User:testproducer --operation WRITE --topic MT5-
--resource-pattern-type prefixed

/var/kafka/kafka1/bin/kafka-acls.sh --bootstrap-server kafka1.build.srv.com:9092
--command-config /var/kafka/kafka1/config/kraft/admin.properties --add
```

```
--allow-principal User:testroducer --operation CREATE --topic MT5-  
--resource-pattern-type prefixed  
  
/var/kafka/kafka1/bin/kafka-acls.sh --bootstrap-server kafka1.build.srv.com:9092  
--command-config /var/kafka/kafka1/config/kraft/admin.properties --add  
--allow-principal User:testroducer --operation DELETE --topic MT5-  
--resource-pattern-type prefixed
```

In this example, the permissions are configured on the kafka1 server. If needed, you can change this to another server. The following options are specified:

- User the username to which the permission is granted.
- --operation the type of permission being granted. READ read access, WRITE write access, CREATE permission to create topics, DELETE permission to delete topics.
- --topic the topic prefix for which the permission is granted.

- Be sure to grant the appropriate permissions to the user account that MetaTrader 5 will use to connect to Kafka. Without these permissions, the platform will not be able to automatically create topics or publish data to them.
- Permissions apply only to topics with the specified prefix. They do not affect any other topics, so there is no risk to the security of your other data in Kafka.

Create Consumer Groups

A Consumer Group is a collection of one or more consumers that collaboratively consume messages from one or more topics. The main purpose of using consumer groups is to scale out message consumption and manage how messages are distributed among consumers in the group.

To create a consumer group and add a user to it, use the following command:

```
/var/kafka/kafka1/bin/kafka-acls.sh --bootstrap-server kafka1.build.srv.com:9092  
--command-config /var/kafka/kafka1/config/kraft/admin.properties --add  
--allow-principal User:testconsumer --operation READ --group test-consumer-group
```

In this example, the permissions are configured on the kafka1 server. If needed, you can change this to another server. The following options are specified:

- User the username to be added to the consumer group.
- --operation the type of permission being granted. In this case, only READ.
- --group the name of the consumer group.

You must also grant the user read access to the topics. As with producers, this is done by assigning permissions to topics with a specific prefix:


```
/var/kafka/kafka1/bin/kafka-acls.sh --bootstrap-server kafka1.build.srv.com:9092
--command-config /var/kafka/kafka1/config/kraft/admin.properties --add
--allow-principal User:testconsumer --operation READ --topic MT5-
--resource-pattern-type prefixed
```

Create Topics

When exporting data, the platform creates topics automatically. To create a topic manually, use the following command:

```
/var/kafka/kafka1/bin/kafka-topics.sh --create --topic test-topic --partitions 3
--bootstrap-server kafka1.build.srv.com:9092 --replication-factor 3 --command-config
/var/kafka/kafka1/config/kraft/admin.properties
```

In the example, the topics are created on the kafka1 server. If needed, you can change this to another server. The following options are specified:

- `--topic` the name of the topic
- `partitions 3` the number of partitions in the topic
- `replication-factor 3` the replication factor for partitions

For topics created automatically by the platform, Kafka's default parameters are used (e.g., replication count, compression, etc.). Modifying these parameters is the responsibility of the system administrator. However, you can predefine these settings at the topic prefix level to ensure consistent configuration across all automatically created topics.

Install a GUI for Kafka Administration

By default, Kafka is a command-line service. For easier configuration and data management, you can install an additional component that provides a graphical interface. We recommend using the open-source project Provectus Kafka UI: <https://github.com/provectus/kafka-ui>.

As an example, we will install the UI on the kafka1 server. Create a directory named kafka-ui for the installation and navigate into it:

```
mkdir -p /var/kafka/kafka1/kafka-ui
cd /var/kafka/kafka1/kafka-ui
```

Download the latest version of kafka-ui:

```
wget
https://github.com/provectus/kafka-ui/releases/download/v0.7.2/kafka-ui-api-v0.7.2.jar
```

In the kafka-ui directory, create a file named application.yml:

```
mcedit /var/kafka/kafka1/kafka-ui/application.yml
```

Add the following content into the file:

```
dynamic.config.enabled: true

logging:
  level:
  root:INFO
    com.provectus:DEBUG
    reactor.netty.http.server.AccessLog:INFO
    org.hibernate.validator:WARN
```

For convenience, create a bash script start.sh:

```
mcedit /var/kafka/kafka1/kafka-ui/start.sh
```

Add the following content into the file:

```
KAFKA_UI=kafka-ui-api-v0.7.2.jar
java --add-opens java.rmi/javax.rmi.ssl=ALL-UNNAMED -jar $KAFKA_UI
```

Add permission to execute the script:

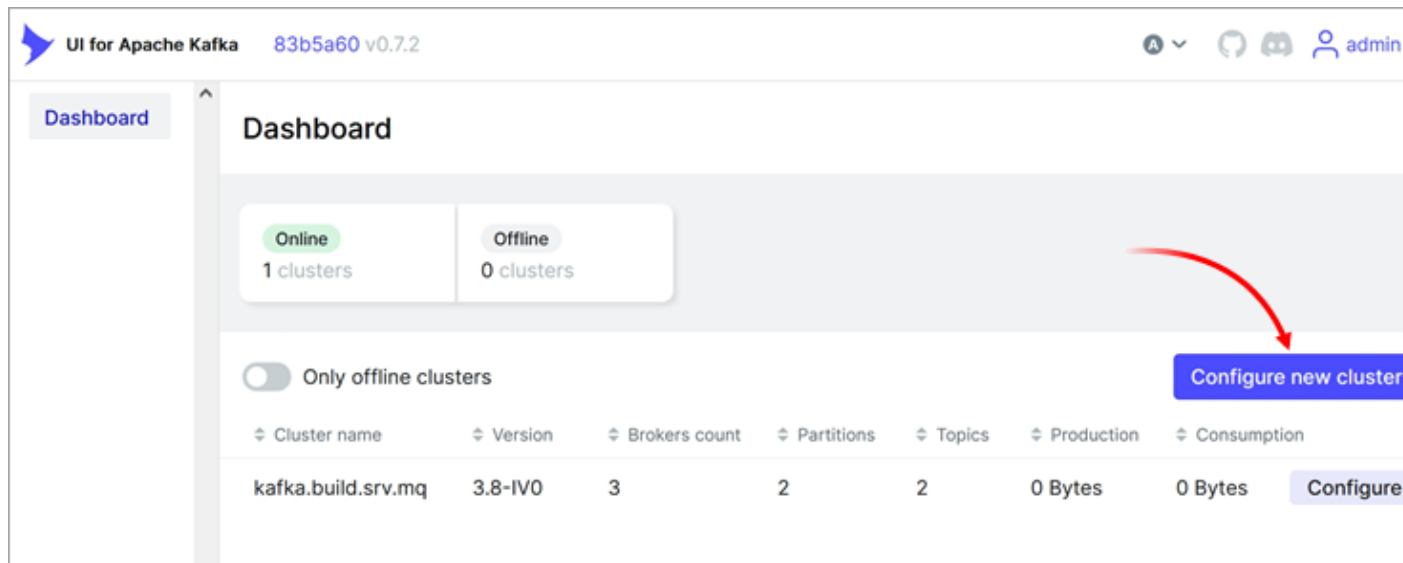
```
chmod +x /var/kafka/kafka1/kafka-ui/start.sh
```

Run the script:

```
./start.sh
```

Open your Kafka server page in the browser and specify port 8080. In our example:
<http://kafka1.build.srv.com:8080>.

Click 'Configure new cluster':



Enter the following values:

- Bootstrap servers kafka1.build.srv.com, kafka2.build.srv.com, kafka3.build.srv.com. Ports 9092.
- Truststore Location the path to the .pfx certificate file on your local machine. When you click 'Submit', the certificate will be uploaded into the truststore.
- Truststore password the password for the certificate.
- Security Protocol SASL_SSL.
- sasl.mechanism PLAIN.
- ssl.keystore.location the path to the .crt certificate file on your local machine.
- ssl.keystore.password the password for the SSL certificate issued for your Kafka servers.

When you click 'Submit', the certificates will be uploaded to the truststore.

Kafka Cluster

Cluster name *

kafka.build.srv.mq

this name will help you recognize the cluster in the application interface

☐ Read-only mode

allows you to run an application in read-only mode for a specific cluster

Bootstrap Servers *

the list of Kafka brokers that you want to connect to

kafka1.build.srv.com

9092



kafka2.build.srv.com

9092



kafka3.build.srv.com

9092



+ Add Bootstrap Server

Truststore

Remove from config

Truststore Location

/var/kafka/kafka-ui/keys/kafka.truststore.jks

Reset

Truststore Password

●●●●●●●●●●

Authentication

Remove from config

security.protocol

SASL_SSL

sasl.mechanism

PLAIN

sasl.jaas.config

org.apache.kafka.common.security.plain.PlainLoginModule required username="admin" password="TestAdmin";

After saving the settings, you will see your cluster appear in the Kafka UI.

Next, enable authentication. Stop the kafka-ui application by pressing Ctrl+C. Edit the dynamic configuration file:

```
mcedit /var/kafka/kaka1/kafka-ui/dynamic-config.yml
```

Replace the file contents with the following:

```
auth:
  type:login_form

spring:
  security:
    user:
      name:admin
      password:[password for UI access]

  kafka:
    clusters:
      - bootstrapServers:
          kafka1.build.srv.com:9092,kafka2.build.srv.com:9092,kafka3.build.srv.com:9092
        name: 'kafka.build.srv.com'
        properties:
          security.protocol:SSL
          ssl.keystore.password:[certificate password]

      ssl.keystore.location:/var/kafka/kafka1/kafkaui/uploads/kafka.keystore.jks-1724165427
      readOnly: false

      ssl:
        enabled: true
```

```
truststoreLocation:/var/kafka/kafka1/kafkaui/uploads/kafka.truststore.jks-1724165395
truststorePassword:[truststore password]
rbac:
  roles: []
webclient: {}
```

Specify here:

- The password for the admin account, which will be used to connect to the interface
- The list of servers in your Kafka cluster
- The password for the SSL certificate
- The password for the certificate truststore

Next, configure the kafka-ui service. In the /etc/systemd/system/ directory, create a file named kafka-ui.service:

```
mcedit /etc/systemd/system/kafka-ui.service
```

Copy the following content into the file:

```
[Unit]

Description = kafka-ui

Requires=network.target

After=network.target


[Service]

RemainAfterExit=true

Type=simple

User=kafka

WorkingDirectory=/var/kafka/kafka1/kafkaui/

ExecStart=/bin/bash -c '/var/kafka/kafka1/kafka-ui/start.sh'

Restart=on-failure

RestartSec=5s
```

[Install]

WantedBy=multi-user.target

Run the service:

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable kafka-ui.service
```

```
sudo systemctl start kafka-ui.service
```

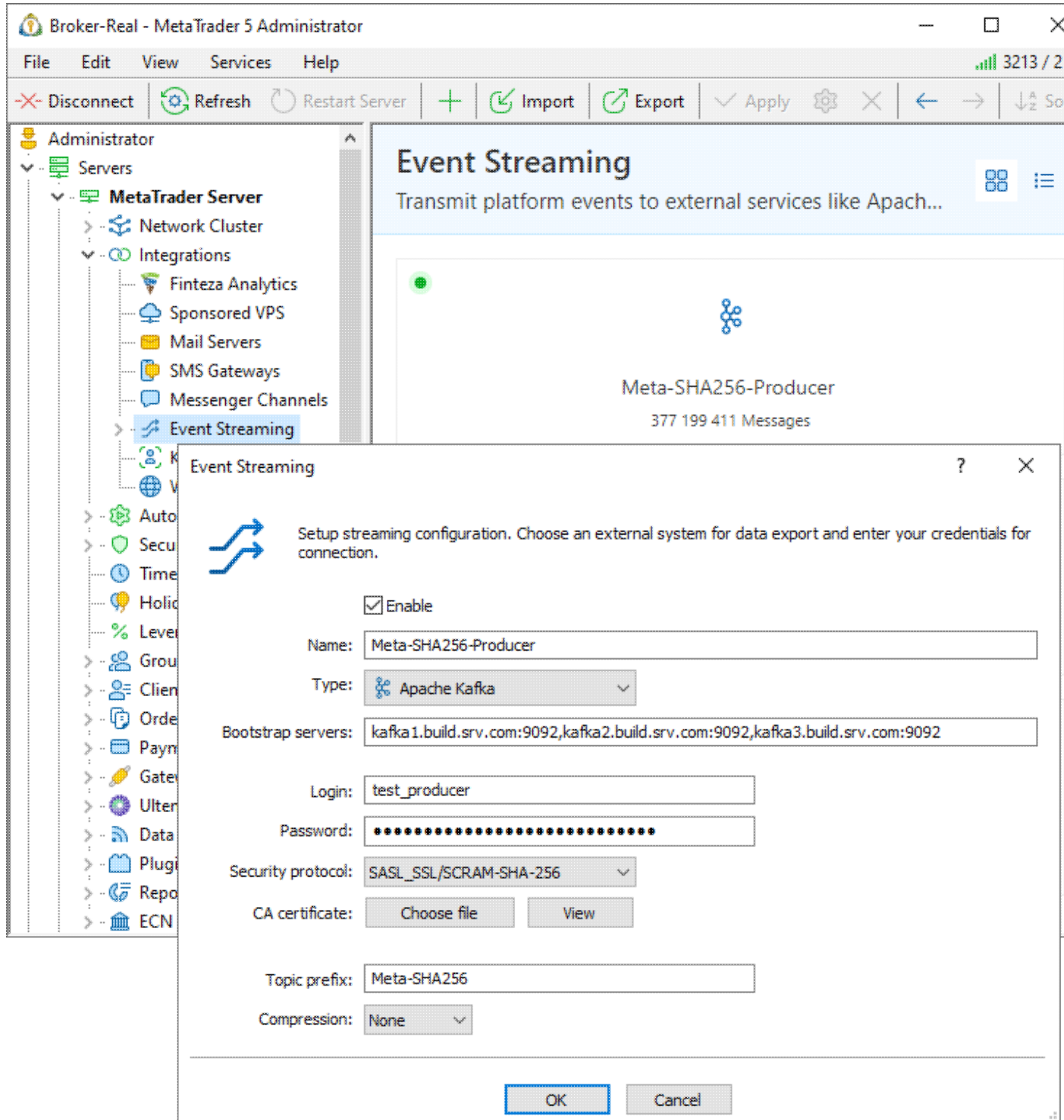
You can now connect to Kafka through the web interface at:
<http://kafka1.build.srv.com:8080/>.

MetaTrader 5 trading platform

MetaTrader 5 Trading Platform
/ Platform Setup / Integrations
/ Event Streaming / Kafka
Streaming Setup

Configuring Data Export to Kafka in MetaTrader 5

Create a new steaming configuration:



In the configuration, specify the following:

- **Enable** toggles the data streaming on or off.
- **Name** a custom name for the configuration (producer).
- **Type** the system to which data will be streamed. Currently, only Apache Kafka is supported.
- **Bootstrap servers** a comma-separated list of Kafka cluster brokers. Each address must include the port. Only ASCII characters are allowed.
- **Login** username used to connect to the Kafka cluster. This user must be pre-created and granted the necessary permissions. Used during SASL authentication. Only ASCII characters are allowed.
- **Password** password for the specified Kafka cluster user. Only ASCII characters are allowed.
- **Security protocol** security protocol and the authentication type used by the Kafka server: SASL_SSL/PLAIN, SASL_SSL/SCRAM-SHA-256, SASL_SSL/SCRAM-SHA-512.
- **CA Certificate** client certificate file that will be used for authentication when connecting to the Kafka cluster.
- **Topic prefix** prefix added to the beginning of each topic name. Only Latin letters, numbers, and the symbols '.', '-', '_' are allowed.
- **Compression** used to reduce the size of JSON text data sent to Kafka. Supported algorithms include LZ4, Gzip, and zstd. The recommended option for optimal speed and compression is zstd. Compression applies only during transmission to the Kafka cluster. Within the cluster, data is repackaged based on topic settings. Consumers must support the compression type set at the topic level.

- Do not change the configuration name while it is active. Doing so creates a duplicate configuration that begins sending messages to the same topics, resulting in data duplication.
- Changing the settings (Bootstrap servers, Login, Password, Security protocol, Certificate, Topic prefix, Compression) will recreate the producer with new parameters. Some messages in the sending queue may be lost during this process.

Full Topic Names

Platform events are streamed to Kafka cluster topics. The platform attempts to create topics automatically using the following naming convention: a prefix followed by a name specified in the settings. They form the full name of the topic. If no prefix is set, the full topic name is just the topic name from the configuration. For example, if the prefix is 'platform1-' and the topic names are 'network' and 'groups', the topics will be created as 'platform1-network' and 'platform1-groups'.

The use of prefixes simplifies management of permissions to create topic and write data to them. You can grant access by prefix instead of individual topics. Also prefixes are useful for isolating or segmenting data when streaming from multiple MetaTrader 5 platforms to a single Kafka cluster.

MetaTrader 5 trading platform

Kafka operates and supports streaming operations strictly with the full topic name. If a topic is missing, MetaTrader 5 will try to create it automatically. If you change the topic name or prefix during runtime, the platform will immediately attempt to create a new topic and continue streaming to it. If the platform's Kafka user lacks the necessary permissions for the new topic or prefix, you will see authorization errors in the trade server logs.

- Be sure to grant the appropriate prefix permissions to the user account that MetaTrader 5 will use to connect to Kafka. Without these permissions, the platform will not be able to automatically create topics or publish data to them.
- Permissions apply only to topics with the specified prefix. They do not affect any other topics, so there is no risk to the security of your other data in Kafka.

Configuring Exported Data

After creating the configuration, navigate to the "Topics" section and select the data you wish to stream:

Add a topic and specify:

- **Name** the second part of the full topic name, which is added at the end. Only Latin letters, digits, and the symbols '.', '-', and '_' are allowed.
A prefix can also be included in the name. For more details on topic naming, refer to the section Full Topic Names.
- **Description** a description of the configuration.
- **Type** the type of data to be streamed to the topic:
 - ◆ **Configurations** changes to the corresponding platform settings.
 - ◆ **Ticks** tick data and trading statistics.
 - ◆ **Trading** any changes in open orders, deals, and positions (creation, modification, deletion), changes to order history, and trade requests sent to the server.
 - ◆ **Trading accounts** creation, modification of personal details, and changes in the financial state of trading accounts.
 - ◆ **Clients** creation, modification of personal data, linked accounts, documents, and other changes to the client database.

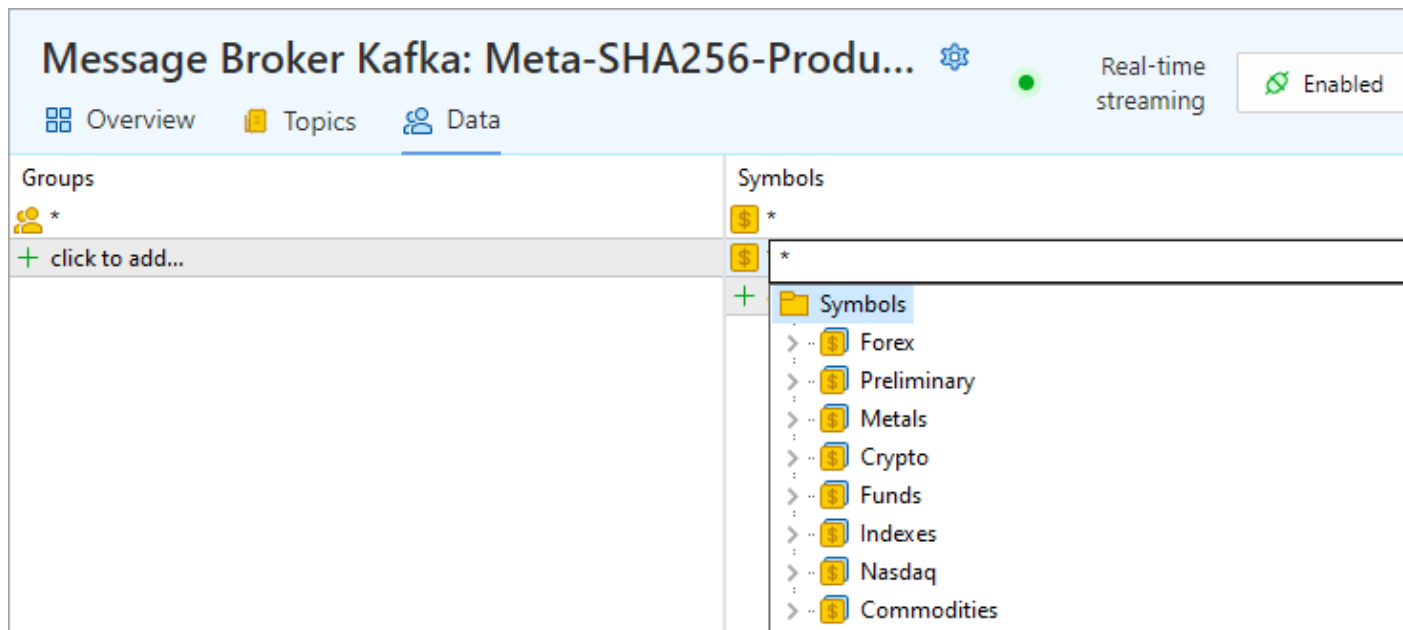
Ensure that the platform (producer) has the necessary permissions to create topics and write to them. Otherwise, authorization errors will appear in the trading server log.

Event Filtering

You can further filter the events to be streamed:

- Trading operations (orders, deals, positions, requests) by group
- Price data by instrument

To do this, configure the appropriate settings in the "Data" section:



Each producer in the platform has a fixed memory limit for storing messages prior to transmission. This limit is set to 1 GB, which typically holds around 500,000 5,000,000 messages. If messages cannot be delivered for any reason and the queue reaches this limit, the system will automatically discard the oldest messages to make space for new ones. This results in message loss.

Data Format

All events are streamed to Kafka in JSON format, encoded in UTF-8. JSON is a widely supported data format used by many systems. For example, a tick data entry would appear as follows:

```
{
  "type": "tick",
  "action": "add",
  "records": [
    {
      "Symbol": "EURAUD",
      "Time": "1743156690296",
      "Bid": "1.71045",
      "Ask": "1.71098",
      "Last": "0.00000",
      "Volume": "0"
    }
  ]
}
```

A detailed description of the exported data is available in the [SQL Export](#) section.

Recommendations

If you observe performance degradation during message transmission, check the network throughput and ensure sufficient disk space is available on each Kafka broker. If needed, configure data retention limits to enable automatic cleanup of topics in a timely manner.

Global retention settings can be adjusted via the `log.retention.hours` parameter (default is 168) in the `server.properties` file. To configure retention per topic, use the

retention.ms parameter.

Closely monitor disk space across your Kafka cluster. If it becomes fully exhausted, individual brokers will begin to fail, ultimately leading to cluster-wide failure.

Statistics and Monitoring

In the "Overview" section, each producer has detailed metrics available to help monitor performance.

Message Broker Kafka: Meta-SHA256-Produ...



Real-time
streaming



Enabled

Overview

Topics

Data

1.5 B

Total delivered messages

3.4 B

Failed messages

1 / 203 bytes

Message queue

844.87 GB

Messages traffic

76 / 15.23 KB per second

Message throughput

Trade Main

Delivered messages: 1.3 B

Failed messages: [1.9 B](#)

Message queue: 1 / 203 bytes

Messages traffic: 610.95 GB

Message throughput: 76 / 15.23 KB per second

Latency: 5 ms

Connection status: Connected

Trade

Delivered messages: 216.6 M

Failed messages: [1.4 B](#)

Message queue: 0 / 0 byte

Messages traffic: 233.92 GB

Message throughput: 0 / 0 byte per second

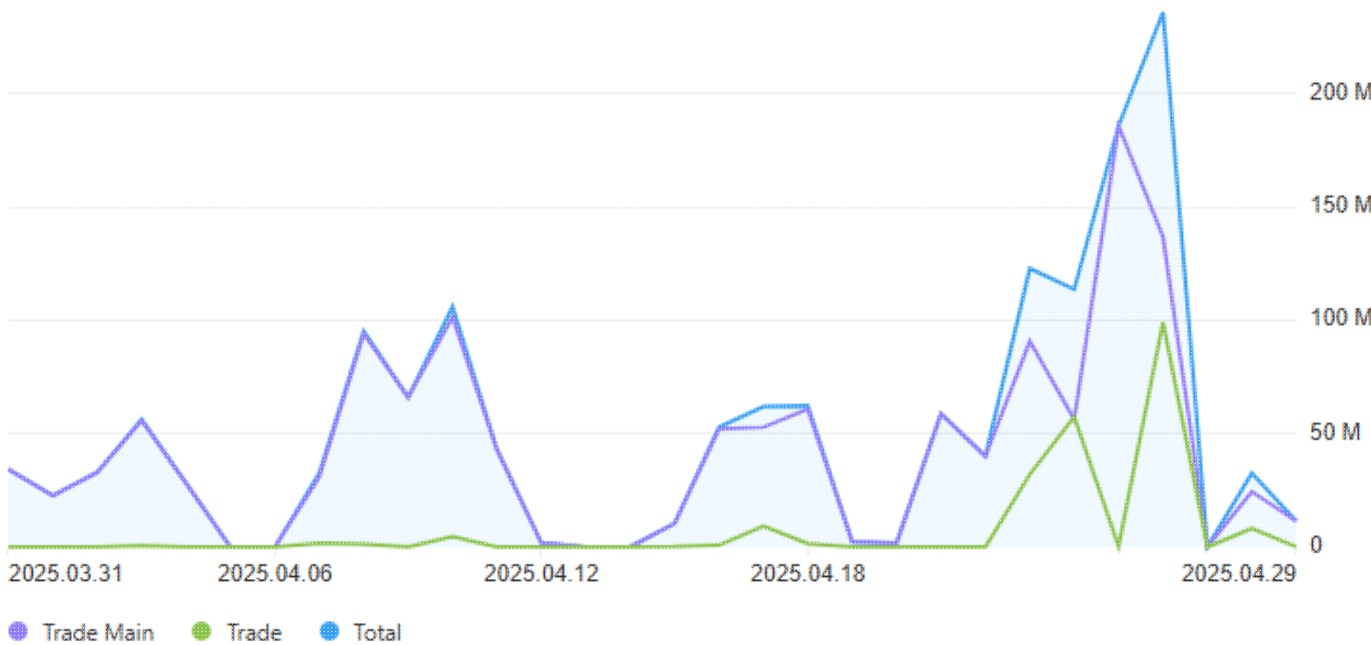
Latency: 0 ms

Connection status: Connected

Delivered messages

Failed messages

Messages traffic



The top part shows statistics for all producers.

- Total delivered messages number of messages successfully delivered from the platform to the Kafka cluster for the current calendar day.
- Failed messages number of messages that failed to be delivered during the current calendar day.
- Message queue number and size of messages (uncompressed data key and JSON) currently pending delivery. Ideally, this should remain at 0/0. If the message queue is constantly growing, check your network throughput and the performance of your Kafka cluster: this may be caused by lack of disk space or high CPU usage. Consider adding more Kafka brokers to balance the load.
- Message traffic volume of uncompressed data (key and JSON) transmitted to the Kafka cluster today.
- Message throughput number of messages and total size of uncompressed data (key and JSON) transmitted per second.

Below the general overview, statistics are displayed for each trading server in the platform cluster. Additional metrics include:

- Latency measures message delivery time with confirmation. before sending a message to the Kafka cluster, the platform timestamps it at the moment it is queued. The message is sent to the Kafka cluster, after which the platform receives delivery confirmation. The platform then calculates the difference between the confirmation time and the timestamp to determine latency. N/A indicates latency cannot be calculated due to lack of connection or traffic.
- Connection status reflects connectivity to the Kafka cluster. "Connected" means the platform has established a connection with at least one broker.

The count of message delivery errors is displayed as a clickable link. Clicking it opens the corresponding trading server log, filtered by the keyword Kafka. This will help you quickly locate relevant issues.

Each trading server's message delivery status is marked with a color:

- Green no delivery errors detected during the calendar day.
- Yellow no current errors, but some messages failed earlier in the day.
- Red unable to connect to the Kafka cluster, or delivery errors are occurring in real-time. Hover for a tooltip with details. For further investigation, request the trade server logs using the keyword Kafka.
- Gray message streaming is disabled.

All statuses reset at the end of the calendar day, and are updated based on the current state.

Below these metrics, you can find graphical visualizations of message delivery performance.

[MetaTrader 5 Trading Platform](#) / [Platform Setup](#) / [Integrations](#) / [Web Services](#)

Web Services

The trading platform offers extensive opportunities for integration with brokerage websites and web services, as well as for integration with additional service providers. This functionality is implemented via the [MetaTrader 5 Web API](#). By using integrations, you can create trader rooms on your website, to which data on trading operations will be sent from the platform, as well as implement registration, account top-up and other options.

The [App Store](#) features a plethora of ready-made web integration solutions from professional developers.

SSL Certificates

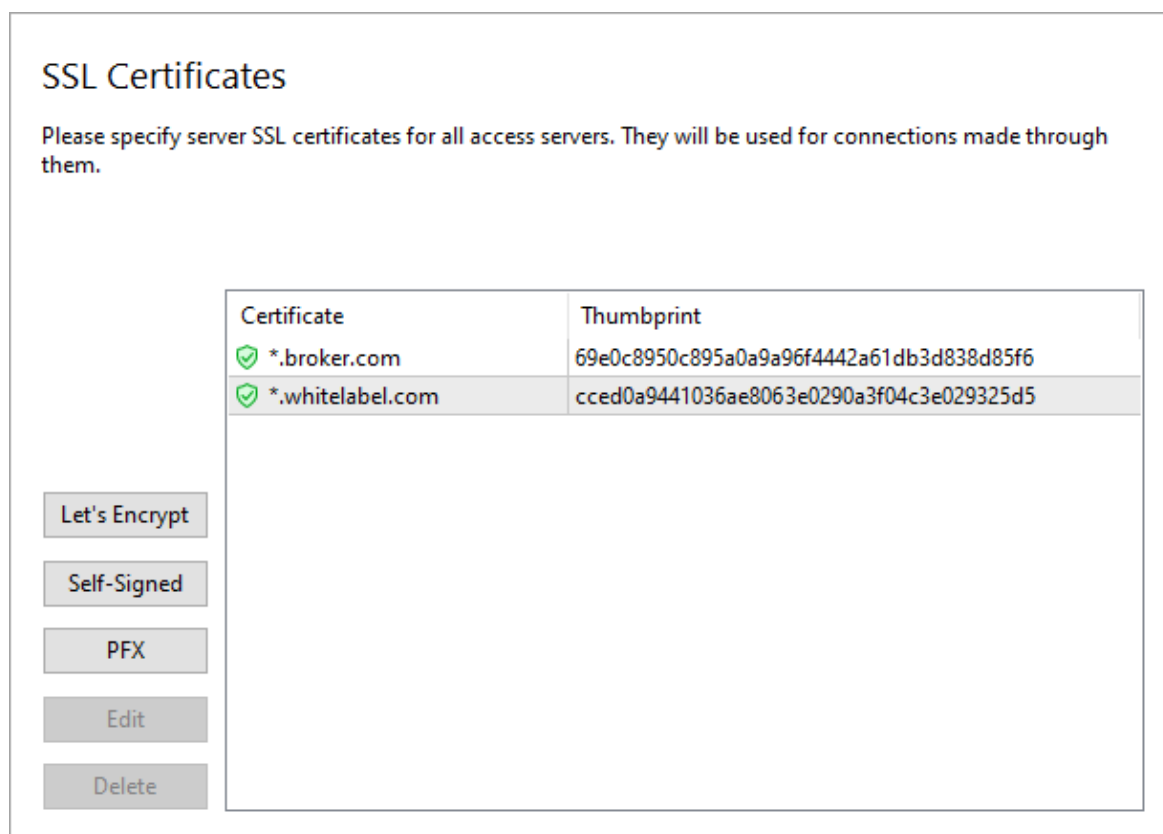
Support for connection over HTTPS is implemented on access servers (HTTP is not connected). Therefore, they can act as a web server. This allows sending to a server [Web API](#) commands as usual GET and POST requests.

To enable connection to the access server via HTTPS, at least one of its [public addresses](#) must be associated with a certain domain, i.e. the host to which POST and GET requests will be implemented. For this purpose, the appropriate entry must be specified on the DNS server. For example, if you want to implement operation via the host abc.broker.com, the following entry should be added to DNS

```
abc.broker.com 3600 IN A XXX.XXX.XXX.XXX
abc.broker.com 3600 IN AAAA XXXX:XXXX:X:XX:XX
```

Here XXX is the public IPv4 or IPv6 address of the Access server.

Since only protected TLS connections (HTTPS) are supported, an SSL certificate must be installed on the access server for the appropriate domain. Use the "Integration \ Web Services \ SSL Certificates" section to upload and manage certificates:



There are three ways to upload a certificate to the platform:

- Using a pre-made pfx file
- By generating a self-signed certificate (can only be used for testing)
- By generating a certificate via the Let's Encrypt service

The type of the certificate used is shown in a tooltip when you hover over its line.

SSL settings apply to all access servers. Changes in the list of certificates are applied instantly, so you do not need to restart the servers.

Connections from multiple domains can be supported on one server (the SNI mode). In this case, the corresponding SSL certificate must be installed for each of them. When connecting, the client specifies the address to connect to. The access server checks all installed certificates and automatically provides the necessary one to the client:

- First, the server tries to find a certificate for an exact match domain.
- If not, it tries to find the most suitable one using mask "*".
- If the certificate could not be found, the server returns the first certificate from the list.

After making changes to the list of certificates, click "Apply" on the toolbar.

One month before the certificate expiration date, its line is highlighted in red. The appropriate warning is displayed on the platform start page as well.

Adding a certificate from a PFX file

Click PFX and specify the path to the file. The certificate can be password protected. In this case, you will be prompted to enter the password.

Adding a self-signed certificate

The trading platform allows the issuing of self-signed SSL certificates. Such certificates can be used to test applications written using the [MetaTrader 5 Web API](#). Unlike a certificate issued by a trusted authority, a self-signed certificate can be obtained instantly and free of charge. However, it cannot be used for operations in a real environment since the connection will not be trusted for this reason, browsers and apps will return warnings.

Click "Add \ Self-signed" and specify the domain, your company name, certificate expiration date, and encryption algorithm:

- Elliptic Curve Cryptography (ECC) with a 256 or 384 bit key.
- RSA with a 2048 or 4096 bit key

SSL Certificate Issuance

Use a self-signed certificate for internal testing. Select the Let's Encrypt service for a free and automatically renewable certificate.

Domain:

Company:

Algorithm: for days

You need an open 443 port and a properly configured DNS records to work:

```
abc.broker.com 3600 IN A 78.140.180.42
abc.broker.com 3600 IN AAAA 2a00:1178:1:60::2a
```

☐ Issue a free and automatically renewable Let's Encrypt certificate

OK Cancel

The generated certificate will be added to the platform. A self-signed certificate must be associated with at least one public address of any of your Access servers using a DNS record. For further details please read the [SSL certificates](#) section.

Self-signed certificates are not automatically renewed. Upon expiration, you need to generate and upload a new certificate.

Adding a Let's Encrypt certificate

[Let's Encrypt](#) is a certificate authority which enables free automatic issuing of SSL certificates. You can obtain a ready-made certificate from this service right in the platform. Let's Encrypt certificates are trusted by most browsers and operating systems. Accordingly, a non-trusted domain warning will not be displayed for your clients upon connection.

How the certificate is issued

1. The platform makes the first request to Let's Encrypt and receives the initial data for issuing a certificate.
2. The platform (its main trading server) generates keys for the new certificate. The private key is not transferred to Let's Encrypt, to Access servers or to anywhere else.
3. For security purposes, Let's Encrypt verifies that the certificate is issued by the real domain owner. The service uses several confirmation methods to verify that: <https://letsencrypt.org/docs/challenge-types/>. MetaTrader 5 uses the TLS-ALPN-01 option:
 1. A domain validation certificate is generated on the platform side and is distributed to all Access servers.
 2. Using a separate SSL protocol, Let's Encrypt connects to port 443 on all IP addresses specified in the DNS for the domain name. Having connected, the service receives and checks the certificate which contains the verifiable data. It is very important that all IP addresses associated with the domain, point to the Access servers in your cluster. All Access servers must be operational and available for verification. They must be configured to listen and accept connections to port 443.
4. The platform waits for Let's Encrypt to complete the verification. After successful verification, the platform receives the issued certificate. The certificate is installed in the platform

Preparing to receive the certificate

To issue a certificate, Let's Encrypt checks the IP addresses specified for your domain. Therefore, before proceeding with the certificate issuing, correctly configure the DNS server as described in the [SSL Certificates](#) section. All addresses must point exactly to the Access servers of your cluster.

Also please note that after you add a DNS record, it takes some time for them to propagate across the Internet. You can check which IP address a domain is pointing to using the "nslookup" command. For example:

nslookup abc.broker.com

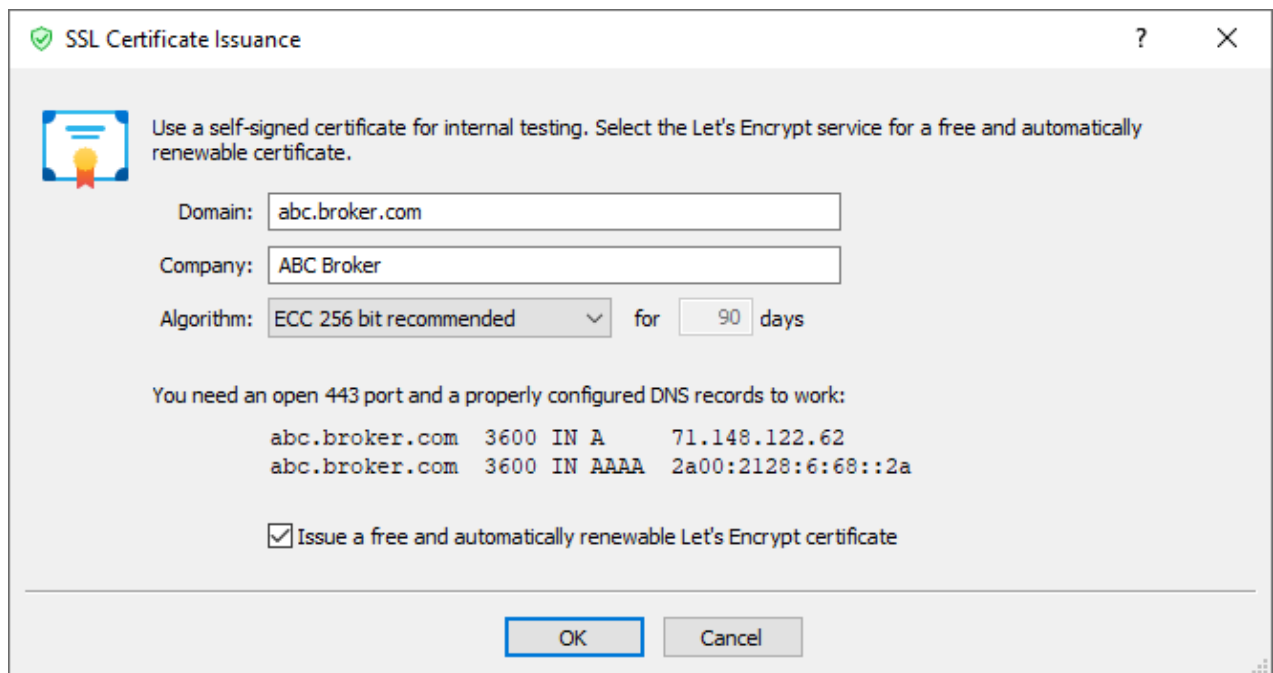
Make sure that the domain is associated with the correct address and proceed to add the certificate in the platform. Click "Add \ Let's Encrypt" and specify the domain, your company name, and the encryption algorithm:

- Elliptic Curve Cryptography (ECC) with a 256 or 384 bit key.
- RSA with a 2048 or 4096 bit key

Also make sure that the option "Issue a free and automatically renewable Let's Encrypt certificate" is enabled.

Domain name requirements:

- The name must contain only ANSI letters and numbers, dots and dashes
- Dots and dashes are allowed everywhere except at the line beginning and end
- The name must not be an IP address



SSL Certificate Issuance

Use a self-signed certificate for internal testing. Select the Let's Encrypt service for a free and automatically renewable certificate.

Domain:

Company:

Algorithm: for days

You need an open 443 port and a properly configured DNS records to work:

abc.broker.com	3600	IN	A	71.148.122.62
abc.broker.com	3600	IN	AAAA	2a00:2128:6:68::2a

☒ Issue a free and automatically renewable Let's Encrypt certificate

The generated certificate will be added to the platform.

The Let's Encrypt certificate is valid for 90 days, but the platform will automatically renew it. A week before the certificate expires, the platform will send a request to the server and then will receive and install a new certificate.

- Let's Encrypt has limits on the frequency and number of issued certificates. If the certificate generation fails, find out and fix the cause rather than trying again and again. Otherwise, the service may restrict your access.
- Let's Encrypt does not support the [Cloudflare service](#). If you are using cloud protection for your access servers, Let's Encrypt will not be able to connect to them for domain verification.

Web Services

The platform allows the receiving and sending of [callback requests](#) from external services. They are used to notify the system about various external events.

For example, SMS providers CM.com and Vonage which are available for [integration](#), can send notifications about message delivery statuses. The services need to know the address to which the messages should be sent. Such addresses are referred to as endpoints. Endpoints can be configured via the Web Services section.

Callback requests are received and processed by access servers. The access servers actually act as a web server. First, you should set a domain name for them. External services will access the platform at this address. Also, endpoint addresses for callback requests will be formed relative to this address. You should create a DNS record for the specified domain (bind the domain with a public IP address of the access server), as well as load the certificate to the platform, as described in the [SSL Certificates](#) section.

Web Services

Please specify a list of addresses, from which the platform will accept and process callbacks on /api/callback/. The list is used for all access servers.

Callback domain:

Up

Down

URL	Allowed IPv4/IPv6	Manager	Comment
✓ *	127.0.0.1		
✓ /api/callback/payments/unlimint	10.15.143.25-10.15.143.30		Unlimint payments callbacks
✓ /api/callback/automation/notify	175.25.69.173,175.25.69.174	1000	Automation callback for notifications

Add

Edit

Delete

Request URLs

To maintain a transparent structure and to ensure further extensibility, all callback endpoints are named as follows: [domain name]/api/callback/[integration section]/[unique part for a specific service]. For example:

- Endpoints for messenger and SMS gateway integrations will start with [domain name]/api/callback/messenger/.
- All endpoints for payment system integrations will start with the path [domain name]/api/callback/payments/.
- To run automation tasks through external services, all endpoints will start from the path [domain name]/api/callback/automation/.

The platform only accepts requests with the specified URLs. The platform will not process any other requests out of this list. This is necessary to ensure safety.

Authorization for requests

As an additional security measure, authorization through a manager account can be enabled for each request. To do this, specify the relevant account login in the Manager field. Requests with additional authentication should have the following information in their header:

```
Authorization: BASIC BASE64_LOGIN_PASSWORD
```

where BASE64_LOGIN_PASSWORD is a string in the BASE64 format with the login and password of the manager account specified as <login>:<password>. For example:

1000:Password

This string in the BASE64 format is shown below:

MTAwMDpQYXNzd29yZA==

Accordingly, in the request header should be as follows

Authorization: BASIC MTAwMDpQYXNzd29yZA==

If you leave the Manager field empty, the request will operate as before, without the additional authorization. The server will only check the list of allowed URLs and IP addresses.

List of allowed IP addresses for requests

Set the IP addresses from which access servers will receive callback requests to specific endpoints. This is needed for added security. The default list contains only one entry "*" 127.0.0.1". It allows requests to be received on any endpoints from the local machine.

Addresses can be specified in the following forms:

- A comma separated list: 10.12.192.1, 10.12.192.6
- A range: 10.12.192.1-10.12.192.110
- Using a netmask: 192.168.0.0/16 or 2a00::0/16

The formats can be combined. For example: 10.12.192.1-10.12.192.110, 192.168.0.1

The list of allowed addresses is valid for all access servers.

MetaTrader 5 Trading Platform / Platform Setup / Security

Security

The trading platform provides a variety of features to ensure system security. This includes protection for server and client components.

- The platform support certificate-based extended authentication. Usage of certificates provides a higher level of client and manager account security. Set up certificates to start using extended authentication.
- By configuring firewall, you can restrict access to the platform from specific IP addresses. For example, you can block addresses from which attacks were registered, or add your office address to the white list so that they are not blocked by anti-flood control.
- Furthermore, the platform provides a specialized component Anti DDoS Proxy Server. It protects from DDoS attack by using external protection providers. Unwanted connections are blocked on the distributed network of an external provider's servers rather than on your access servers.
- Components interact through secure protocols, databases are securely encrypted, while user actions are registered in logs. This information, along with other operation security details, is available under the "Authentication Protocols" section.

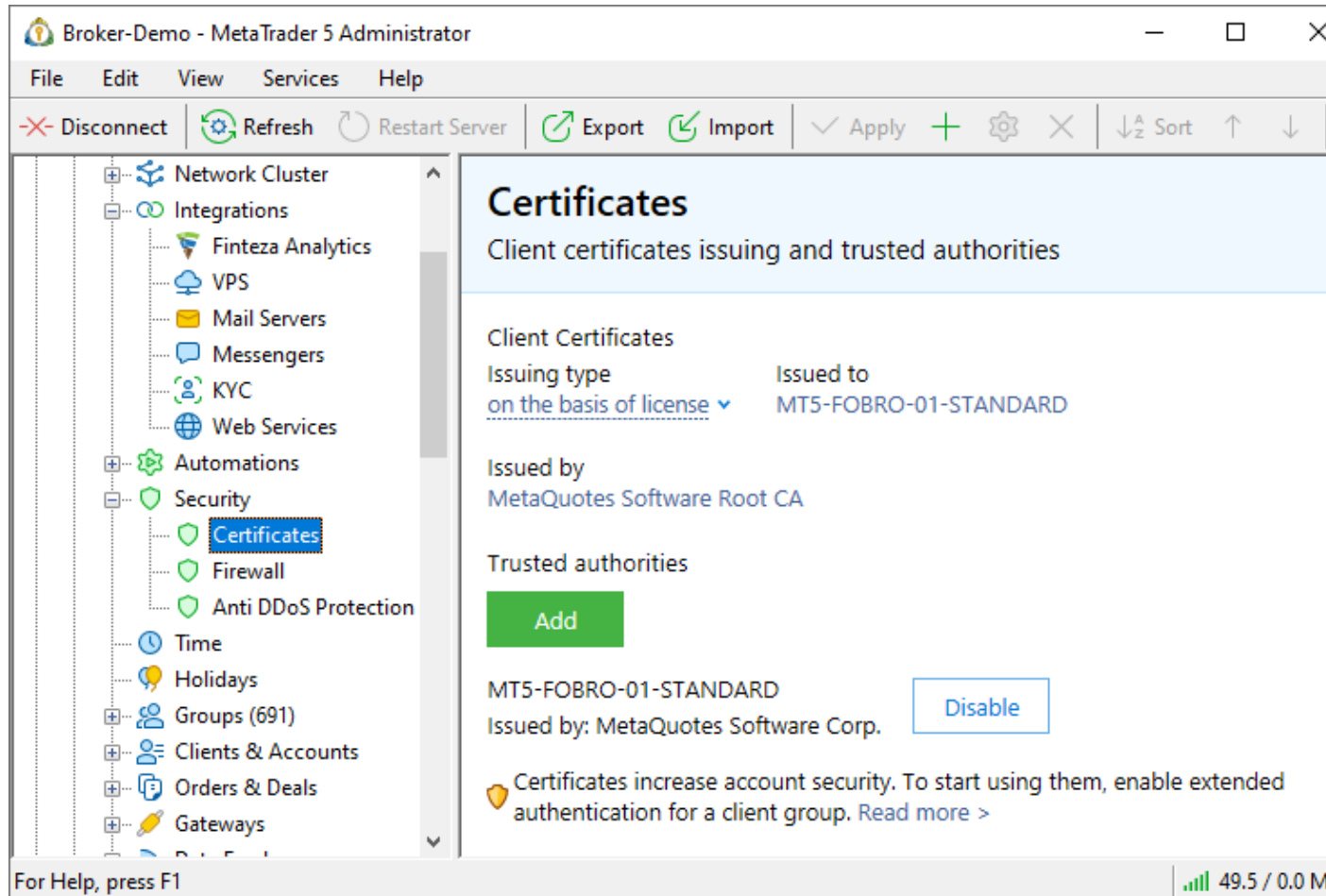
Please treat settings in this section carefully, as they affect the continuity of your business processes.

MetaTrader 5 Trading Platform / Platform Setup / Security / Certificates

Certificates

Security certificates are used for advanced authentication in the trading platform. This authentication type allows improving the system operation security. For more details please read a [separate section](#).

Here you can configure certificates.



Specify how certificates will be generated:

- **on the basis of license** by default, client certificates are generated on the basis of a license issued by MetaQuotes Software Corp. when you purchase the platform. The license file contains a unique certificate created for the brokerage company, based on which all client certificates are generated.
- **on the basis of certificate** a brokerage company can use its own certificate to generate client certificates. Upon the selection of this option a window for selecting a pfx certificate opens. The certificate file must contain a private key. The certification issuer must be added to the trusted list.
- **disabled** this option disables the generation of certificates. However, this does not mean that the extended authentication will become unavailable. In this case, the client will not be able to receive a certificate automatically when connecting to the account. The brokerage company will need to issue a certificate to the user specifically (for example, on an e-token), and then link the certificate to an account in the trading platform. To link a certificate, import it via the Security tab of this account.

Certificate Requirements

The following requirements must be fulfilled when using the company's own certificates:

- The certification authority that issued the certificate must be added to the trusted list.
- A certificate issued to a client must contain a private key.
- The client must add the issued certificate to the /certificates folder of the terminal, install it to the operating system storage or upload it to an e-token.

Trusted Authorities

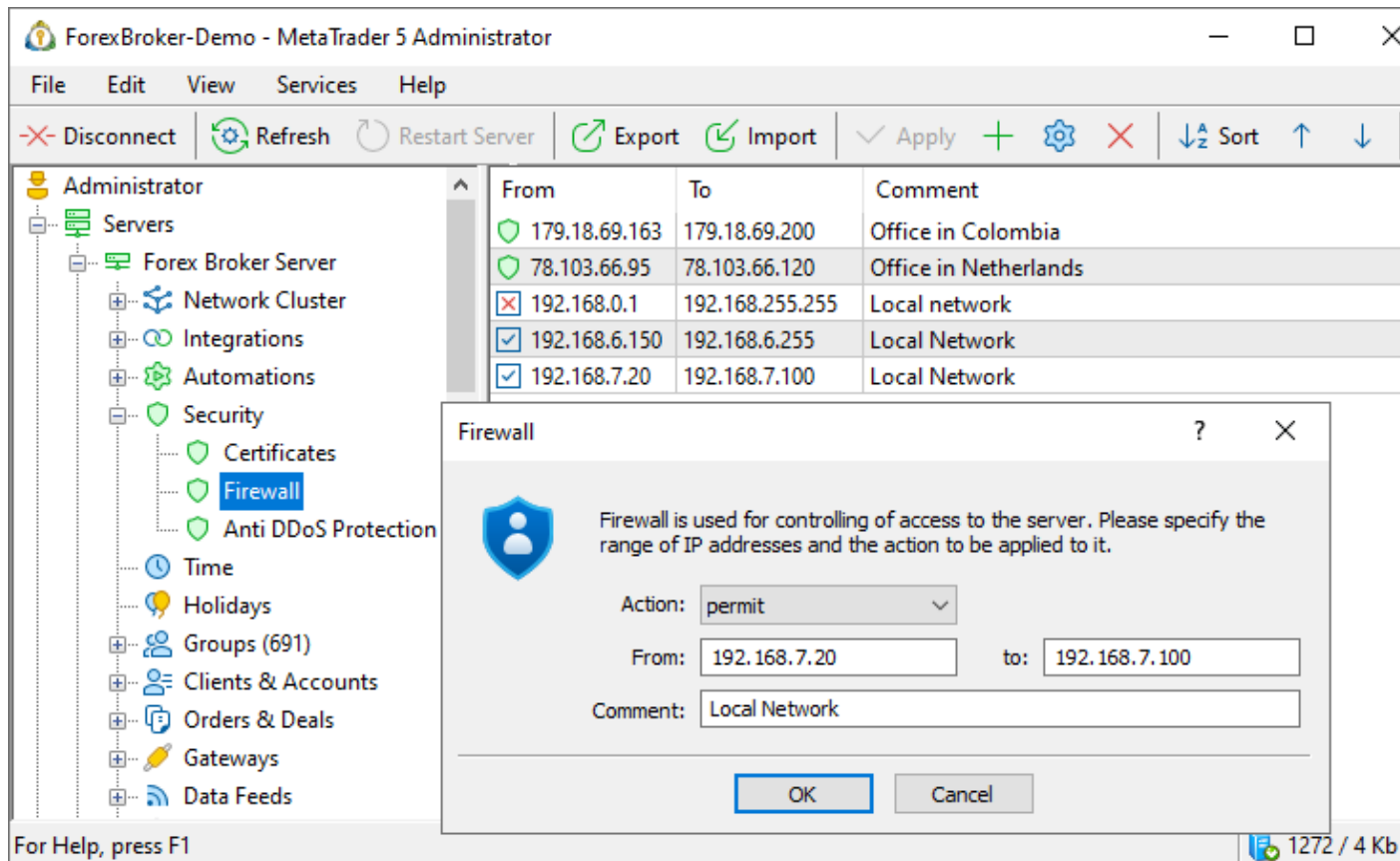
When authorizing on the server with a certificate, the certification authority is checked. Authorization is allowed only using certificates issued by trusted authorities. By default, such an authority is a certificate from the trading platform license that is used to generate client certificates.

To add a certification authority, click the appropriate button and specify the *.cer or *.crt file. If necessary, any of the previously added centers can be disconnected at any time.

MetaTrader 5 Trading Platform / Platform Setup / Security / Firewall

Firewall

The Firewall section is intended for setting up protection of the system from the access from undesirable IP addresses. If a group of addresses is blocked, no users (client, manager or administrator) from the address inside the specified range will be able to connect to the server. By default all addresses are allowed.



Checking of any address is performed top-down. Each type of instructions is displayed with a special icon:

-  blocked range of addresses;
-  allowed range of addresses;
-  range of addresses that are always allowed irrespective of blocking instructions and antiflood control.

The last instruction is always applied to an address, irrespective of previous instructions, except for the addresses that are always allowed. Thus the position of an instruction in the list is an important condition in the limitation of access from IP addresses. Let's consider an example:

From	To	Comment
 192.168.0.0	192.168.0.100	
 192.168.0.20	192.168.0.60	

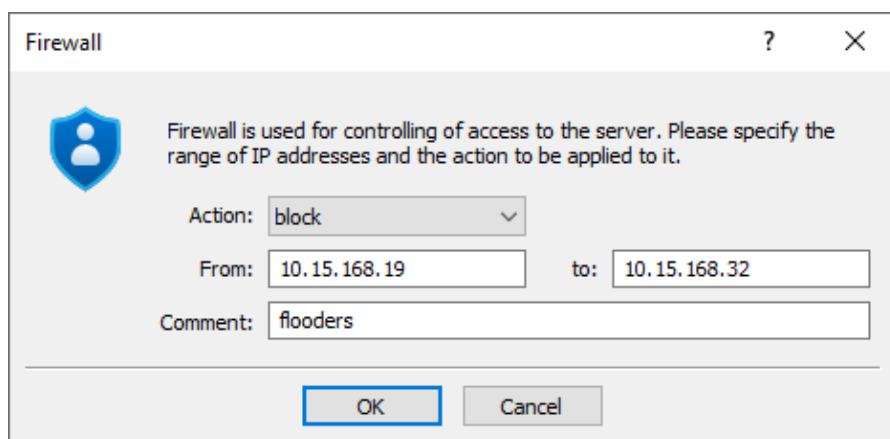
In the above example, the allowing instruction (from 192.168.0.20 to 192.168.0.60) is located below the blocking one. This allows to enable access of a certain range within blocked addresses. Only the addresses 192.168.0.0 192.168.0.20 and 192.168.0.60 192.168.0.100 stay blocked. But if you change places of these instructions, the entire range 192.168.0.0 192.168.0.100 will be blocked.

To change positions of instructions in the list, use context menu commands " Move Up" and " Move Down", ore the same command in Edit menu or on toolbar Standard.

- Do not block address ranges, which include IP used for connecting by the administrator terminal. If such instructions become effective, you won't be able to connect to the server.
- In case of blocking your own IP address and inability to connect to the server via another administrator account the only way is to stop the main server service, delete the file configs\access.ini and then start the server again. Note that this will lead to deleting all previously created firewall configurations.
- Instructions become effective as soon as you press "OK". But they are applied to new connections only. Current connections are not reset.
- You should carefully treat not only access blocking and allowing, but also the position of instructions in the list. Remember, only the last instruction concerning the IP address is applied to it, while all previous ones are ignored.

Adding and Modifying Instructions

In order to add an instruction or modify an existing one, press " Add" or " Edit", respectively. They can be found in the Edit menu, on Standard toolbar or in the context menu. After you press them, the following window will appear:



The following parameters should be specified in this window:

- **Action** select an action: always permit, permit or block the bellow range of IP addresses;
- **From** starting address of the range. The instruction will be effective starting from this address;
- **To** end address of the range. The instruction will be effective till this address;
- **Comment** a text comment to the instruction.

To finish adding or editing an instruction, press "OK". If you press "Cancel", the window will be closed while changes will not be saved.

A created instruction can be deleted using command "✗ Delete" of the Edit menu, Standard toolbar or the context menu.

- For quick addition of instructions on any of IP addresses, insert it into the Comment field from the clipboard. After you press "Ok", it will be automatically added to fields "From" and "To".
- Additional general information about working with configuration records is given in the "Working with Instructions" section.

Context Menu

The context menu of Firewall contains the following commands:

- **+ Add** add a new instruction;
- **⚙ Edit** edit the selected instruction;
- **✗ Delete** delete the selected instruction;
- **↑ Move Up** move the selected instruction up relative to others;
- **↓ Move Down** move the selected instruction down relative to others;
- **↕ Sort Alphabetically** sort configurations alphabetically. Please note that sorting is done on the server, not in the local terminal.
- **📄 Export to File** export firewall settings to a file.
- **📄 Import from File** import firewall settings to a file.
- **📄 Journal** request logs according to the selected configuration. This will open the trade server logs section, with the name of the selected configuration

MetaTrader 5 trading platform

automatically specified in the query field. You will only need to press the request button.

-  **Find** open the search window;
- **Auto Arrange** if this option is enabled the size of columns is selected automatically;
- **Grid** this option shows/hides field separators in the table.

MetaTrader 5 Trading Platform / Platform Setup / Security / Anti DDoS Protection

Protection against DDoS Attacks

The access servers of the platform are provided with the DDoS protection mechanism. The mechanism monitors the network activity of clients in real time and blocks connections from IP addresses, on which suspicious activity is registered, such as sending too many requests or invalid data, authorization errors, etc.

Unfortunately, the mechanism cannot protect the platform against all existing types of attacks. For example, it cannot defend against attacks targeting the network infrastructure, since such attacks cannot be physically dealt with at the access server application level. Additional mechanisms are required for effective protection.

The trading platform provides a special component Anti-DDoS Proxy Server. The component enables the use of external Anti-DDoS service providers. In this case, blocking of unwanted connections is performed on the distributed network of Anti-DDoS provider's servers rather than on access servers. The Anti-DDoS provider's server network actually acts as an additional component between the client and the access server.

Instead of directly connecting to access servers, all clients will be connected to the provider's public access points. Unwanted connections will be filtered in the provider's own network of servers, while legitimate connections from real clients will be forwarded to access servers.

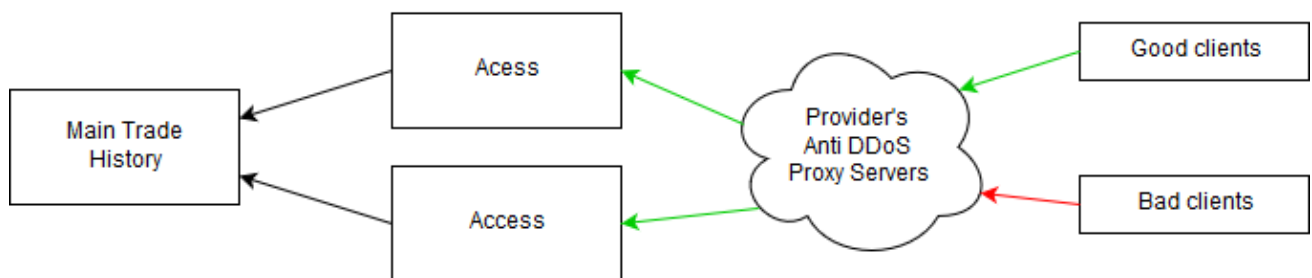
The broker's infrastructure is practically not affected.

MetaTrader 5 trading platform

General operation scheme



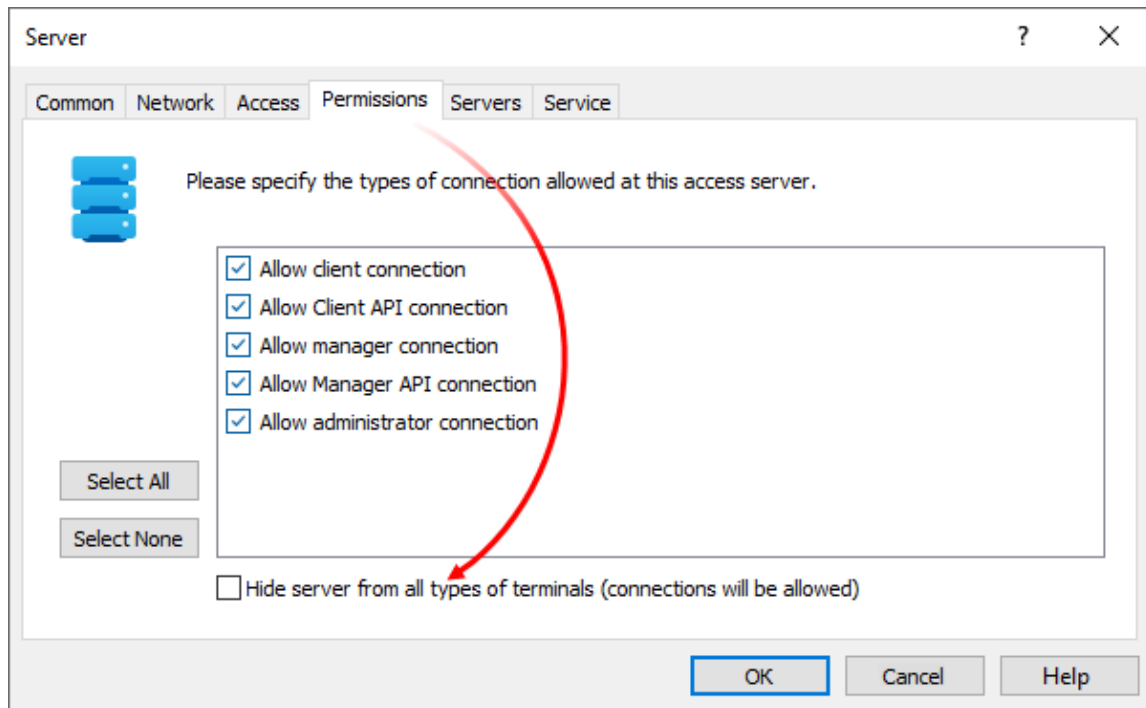
External Anti DDoS service provider



Configure your access server for the operation with an external Anti-DDoS service provider

Choose an access server, to which the connections via the Anti-DDoS provider will be redirected. It is recommended to install and configure a new access server, because its address must not be publicly known (otherwise attackers will be able to directly attack this server). If you have an access server that was never visible to clients (had the Idle priority or client connections were disabled in its settings), you may use this server.

The 'Hide server from all types of terminals (connections will be allowed)' option is available for servers that work with an external Anti-DDoS provider.



Enable this option for the access server, to which connections will be redirected. The server will be hidden from all terminals, including client, manager and administrator, and will not be displayed in the list of access points. Not knowing the address of the server, attackers will not be able to perform an attack.

By hiding a server, you do not disable the possibility to connect to it. Terminals can connect to this server if the address is specified manually, and the Anti-DDoS provider can redirect client connections to it.

Choose an Anti-DDoS service provider and receive necessary data

You should contact the Anti-DDoS provider and sign an appropriate agreement.

- You will need to provide information about the public address of your access server to your Anti-DDoS provider.
- The provide will allocate one or more public addresses, to which your clients will be connected. These addresses should be specified in the Anti DDoS Proxy Server settings in the 'Public Addresses' field.
- Also you will be provided with ranges of proxy server addresses through which client connections will be forwarded to access servers. These ranges should also be specified in the Anti DDoS Proxy Server settings (tab Source Subnets. The ranges are used as an additional protection measure: connections forwarded by the Anti-DDoS provider will only be accepted from these addresses.

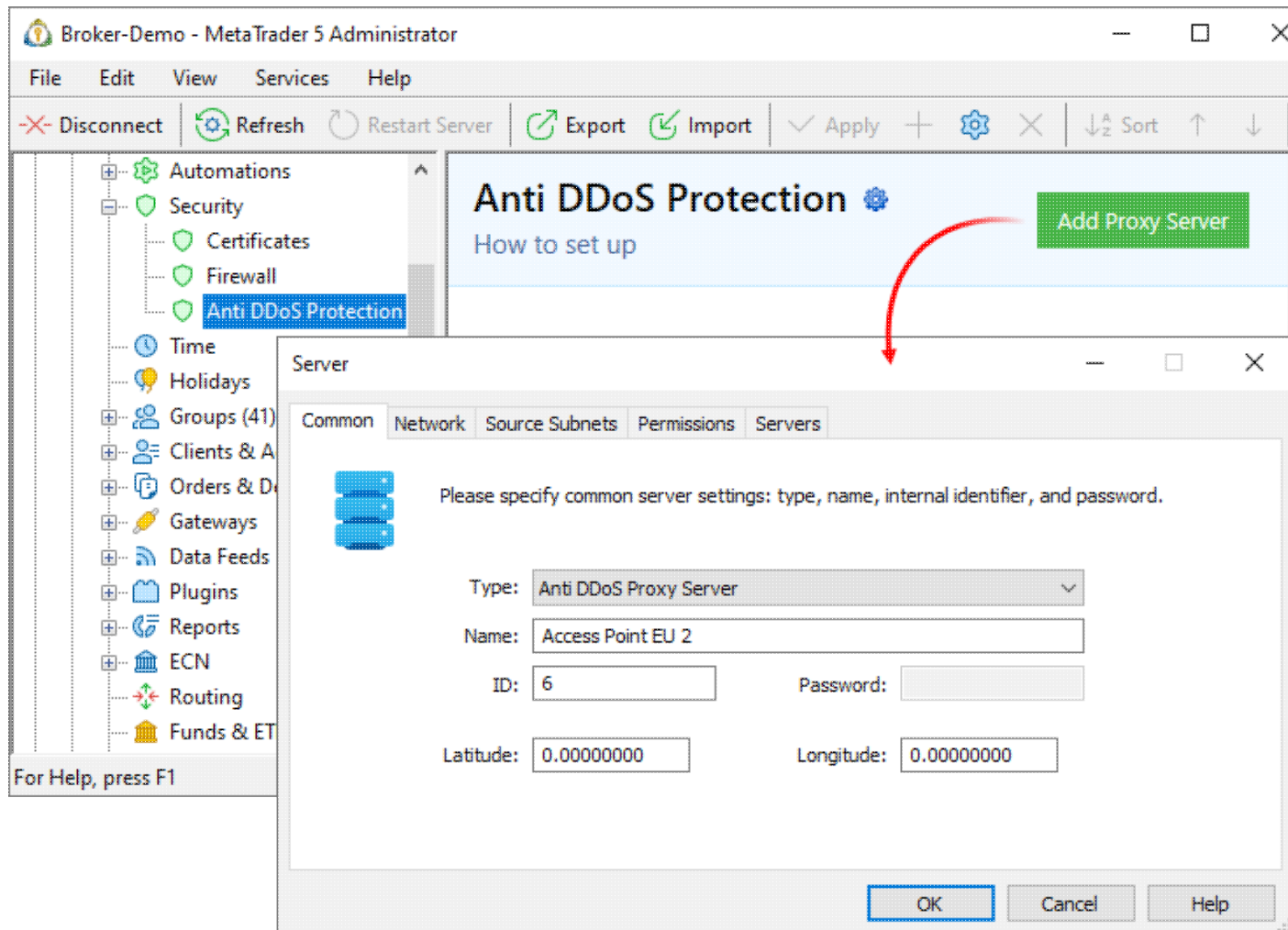
The platform supports the following service providers Akamai, Orator, Cloudflare. Support for other providers will be added soon.

Configure the Anti DDoS Proxy Server component in MetaTrader 5

MetaTrader 5 trading platform

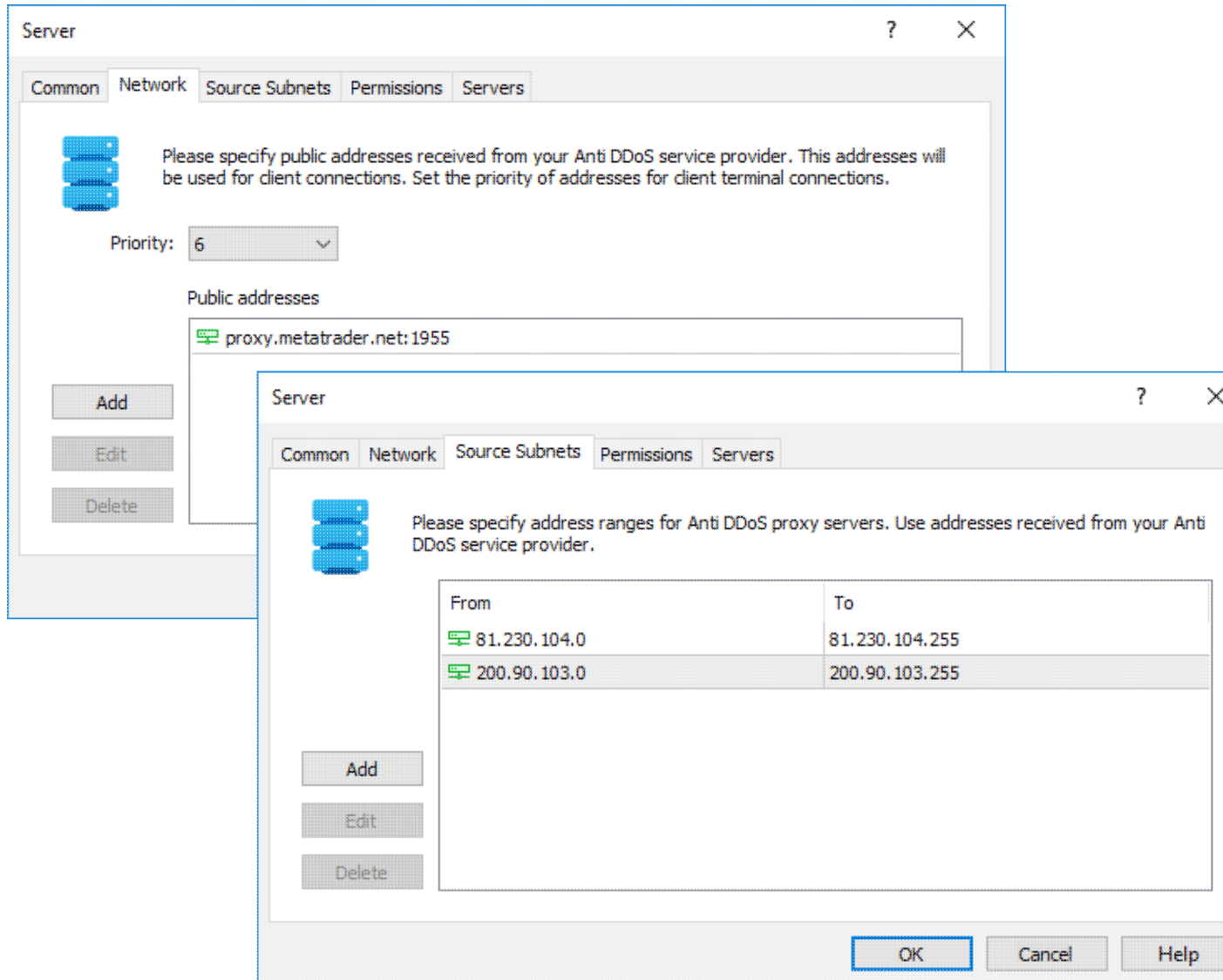
After you receive the required information, add the Anti-DDoS Proxy Server component in the Network Cluster section. Set a name for this component. Note that this name will be used for the display access points in client terminals. Do not use phrases like Anti DDoS etc., the point should look like a regular access server.

Also set the internal ID; passwords are not used for this component.



Next, on the Network tab, specify:

- **Priority** the basic priority for the access points of the Anti-DDoS provider, from 0 to 15. It is similar to priority settings of access servers. Preference of a server for client connections is calculated based on the basic priority and the quality of connection with the server (ping). The lower the priority is, the more preferred the server is for clients.
- **Public addresses** one or more public access points through which clients connect to the platform (via the provider's network). These points are provided to you by the Anti DDoS service provider.



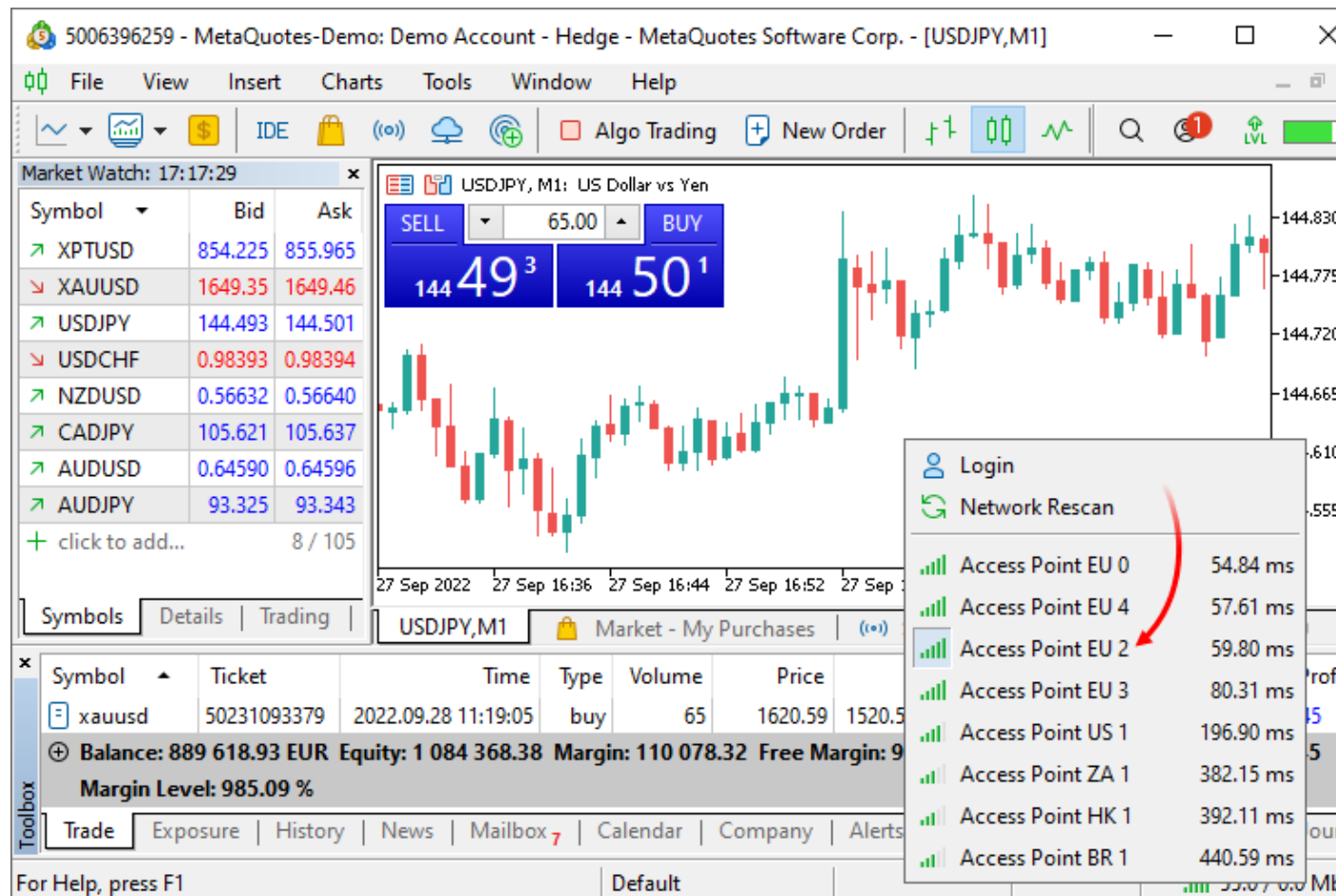
MetaTrader 5 trading platform

On the "Source Subnets" tab, set the range of addresses for the provider's proxy servers. Connections forwarded by the Anit-DDoS provider from these addresses will be accepted.

Anti-DDoS Proxy Server will work even if you do not specify provider's proxy ranges. However, we strongly recommend specifying the ranges, otherwise, connections from any addresses will be allowed in the Anti DDOS provider mode.

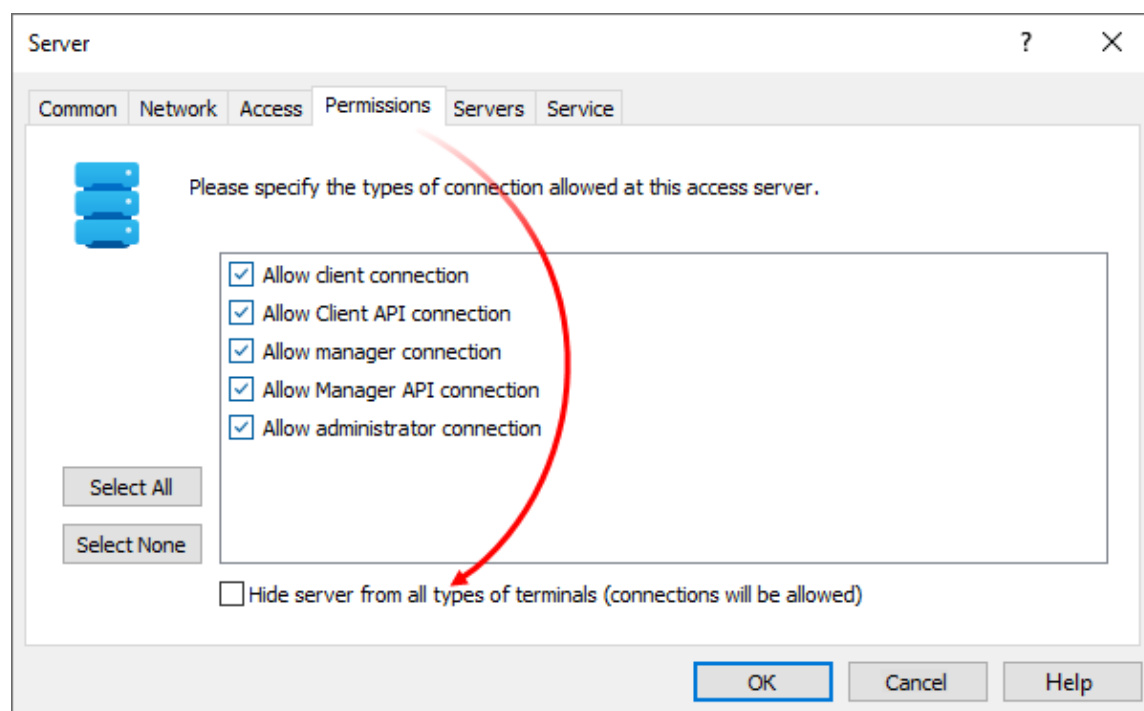
Check the accessibility of the Anti-DDoS server in terminals

After configuration Anti DDoS, the server will be available for client connections. The provider's server is displayed as a regular access point in client terminals:



How to Hide Access Servers

After the configuration of the Anti-DDoS proxy server, the provider's access point will be available for all terminals, including client, manager and administrator. However, your access points will still be available to these terminals. If a server address is publicly available, attackers can find out this address and perform an attack. In order to protect from possible attacks, you can hide access servers. This can be done by enabling the option 'Hide server from all types of terminals (connections will be allowed)'.



The server will be hidden from all terminals and will not be displayed in the list of access points. Not knowing the address of the server, attackers will not be able to perform an attack.

By hiding a server, you do not disable the possibility to connect to it. Terminals can connect to this server if it is specified manually, and the Anti-DDoS provider will be able to forward client connections to it.

How to identify connections through the Anti-DDoS provider

Forwarded client connections are received from the Anti-DDoS provider's addresses. However, the real original addresses of clients are also known, since this information is transmitted by the provider. The proper original client's address is written to the platform logs and client records (e.g. in the Last IP parameter).

Check the access server journal to find out if the client is connected via the Anti DDoS provider:

```
2017.10.13 17:42:08.441 84.101.15.116 '1814046': login (Client build 1674, cid: 13ea0a897
```


MetaTrader 5 Trading Platform / Platform Setup / Security /
Authentication Protocols

MetaTrader 5 Authentication Protocols

All client connections in MetaTrader 5 (trader, manager and administrator) can use three types of authentication:

1. Mandatory standard authentication

Standard authentication checks the client's password and is required for all types of accounts.

The client login is an unsigned integer number in the range between 1 and 18,446,744,073,709,551,615. The password is a string consisting of 8 to 16 characters. The minimum password length is configured in the properties of the group to which the client belongs.

The system controls the complexity of all used passwords. A password must contain at least two of three types of characters: lowercase letters, uppercase letters and digits.

The password is stored on the trading platform as a hash of the password and additional data.

On user terminals, the password is protected with an encryption key, which is derived from the fingerprint of the client computer (processor serial code, hard drive, OS etc.). When migrating trading account configuration files to a different computer, saved passwords are automatically deleted, because the device fingerprint changes. All password fields are protected from being spied/copied/retrieved.

The entire authentication mechanism is implemented on the server side.

Authentication consists of three stages:

1. A client is authenticated by the server. In response to an initialization command, the server sends to a terminal a random sequence, which should be signed by a client using a 128-bit key from the client password hash and then should be sent back to the server. The server performs similar actions on its side, compares signatures and verifies the client authenticity.
2. A server is authenticated by the client. After completing the first step, the client checks if the client password is known to the server by generating and sending a random sequence, which is then signed by the server and checked.
3. A channel encryption key is formed. If the first two steps are successfully completed, the server generates a 2048-bit channel encryption key as a derivative of a random value, the client password derivative and session data.

MD5, SHA256 and AES algorithms are used during standard authentication.

After a standard authentication procedure, an extended validation by SSL certificates and/or OTP (One Time Password) authentication can be performed. These three

authentication methods can be all enabled to complement each other.

2. Advanced Authentication

The server supports advanced authentication, which is completely analogous to the SSL authentication. This type of authentication uses SSL certificates in the PKCS#12 (.pfx) storage format for Microsoft Windows.

It supports both existing certificates issued by certificate authorities and certificates generated by the trade server.

On the client side, certificates can be stored in the file system, as well as installed in the standard Windows certificate store or in additional storage devices such as external eToken etc.

3. One Time Password Authentication

The use of one-time passwords can be additionally enabled. In this case, users are required to enter a special one-time password every time they connect to their accounts.

One-time passwords are generated in the MetaTrader 5 terminals for iOS or Android. The terminals generate one-time passwords using the HMAC/TOTP SHA-256 algorithm based on the OTP secret.

An issued one-time password is valid for 30 seconds. The server checks the sent one-time password using the OTP secret and the current time. Therefore, in order to provide a proper operation of the OTP authentication procedure, the mobile terminal and the trade server synchronize time with time servers.

An OTP secret is sent to a server over a 2048-bit key encrypted channel after a basic authentication, during the initial bounding of the mobile terminal to the trading account.

Protocol of interaction between the system components

All components of the system communicate with each other using their own binary compressed and encrypted data channel with a 2048-bit key.

In addition, server components can use the following protocols:

- An HTTPS protocol for receiving platform updates from the <https://updates.metaquotes.net> website. The website certificate issued by Sectigo Limited is verified. All update components are signed by MetaQuotes Ltd certificate, which is validated by COMODO CA. The signature is validated in all platform components before the update.
- TIME/NTP time synchronization protocols, if time synchronization is enabled on the server.
- SMTP and ESMTP if delivery of end-of-day client trading reports is enabled.

Audit actions in the client and administrator consoles

MetaTrader 5 trading platform

All actions of the platform servers, as well as actions of traders, managers and administrators are logged on the server side and in terminals. All logs are provided with a built-in integrity control. If any of log entries is deleted or modified, the first modified position will be highlighted in red in the log viewing mode.

For each client connection, the following details are written to the log: time, IP address, terminal type, client's login and computer ID (CID – Computer ID derived from the computer software and hardware characteristics). Details of all performed operations are displayed in logs.

Last connection time and IP are written in the client record details on the server. After a successful authentication, the time and IP address of the previous successful connection are additionally recorded in the client terminal's local journal.

Additional security options

A variety of additional security measures are available for all platform components:

1. All MetaTrader 5 program components (.exe and .dll files) are provided with:
 1. MetaQuotes Ltd digital signature to guarantee authorship and the invariability of files;
 2. Built-in protection against modification, hacking or debugging.
2. All databases of saved accounts and trade operation history on the client side are encrypted using a key derived from the client computer, making it impossible to use these databases in case of possible hacking or migration to another computer. Server databases are not encrypted due to high performance requirements.
3. In addition to using a 2048-bit encrypted data transfer channel, each trading order is additionally signed using a derivative from the client's password. During advanced authentication, a private SSL key of the client's certificate is used in the signature.
4. The server cluster is provided with a built-in anti DoS and DDoS protection based on the analysis of connection frequency and actions performed.
5. The server cluster has its own configurable firewall to control access to the cluster.
6. Connection of all server components inside the MetaTrader 5 cluster is controlled based on the IP address, login and password, similar to the standard authentication scheme.
7. Each administrator/manager of the system is provided with an advanced system of permissions with a double control on the administrator/manager and the server side.
8. Connection of the managers and administrators of the cluster can be limited to a range of addresses specified in the manager configuration settings.

MetaTrader 5 Trading Platform / Platform Setup / Automations

Automations

The Automations service enables the execution of various actions in the platform based on predefined scenarios. The service allows specifying a list of conditions and a list of actions that should trigger under these conditions. Automations allows companies to streamline numerous day-to-day operations and eventually reduce manual work. The service does not require from the administrator any extra programming skills, while all rules can be configured in a user-friendly visual mode.



The following actions can be automated:

- Communication with clients. Automatically notify clients when their balances drop below a certain value, send promotional emails to new clients and reminders to inactive customers.
- Platform maintenance. Reboot servers and gateways if performance metrics degrade, launch history synchronization according to a pre-configured schedule.
- Managing of configurations. Set dynamic time-based symbol and group parameters, rearrange routing rules after reaching threshold volumes.
- Managing of deals. Close positions and cancel orders upon expiration of futures contracts.
- Managing of accounts. Disable trading for malicious traders, conduct balance operations and add automated comments upon certain actions.
- Sending events to external systems. Back-office integration options will be added soon. For example, you can send information on opened accounts to online trader rooms or to your CRM system via the REST API.

Reduce manual work and free up resources for important business tasks, as well as improve the quality of services by eliminating human errors. Save money on programmer services and paid plugins by replacing them with simple and easily controlled automation tasks.

Test the service and purchase a subscription

You can create up to three Automation tasks for free. Any types of conditions and actions can be automated. Test the system and evaluate its efficiency for your business.

To remove trial restrictions and to add more automations, [purchase the service subscription from the App Store](#). The service costs USD 500 per month, but the savings and the additional earnings can certainly be much higher than the aforementioned cost.

Order MetaTrader 5 Automations

Task configuration

To configure an automation task, specify:

- Time and repetition periodicity for the task
- Triggers the events upon which the task should be performed
- Additional conditions, under which the task should be performed
- Action to be executed upon the specified conditions

The platform features a few ready-made service examples, which can help you in understanding the service operation and setup principles. Customize the settings and enable tasks to see how the service works.

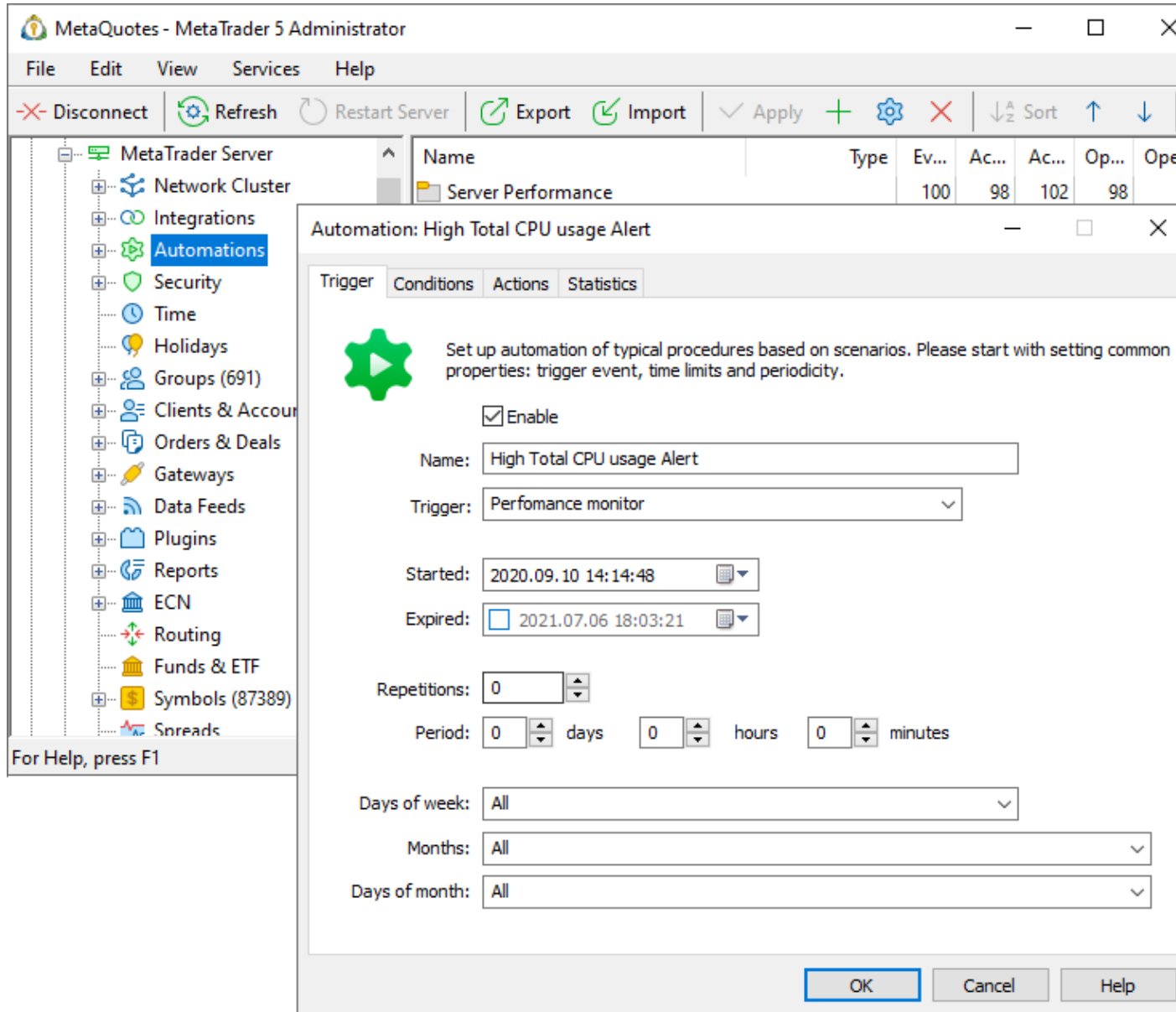
MetaTrader 5 trading platform

Detailed statistics will show how effectively the Automations service assists you in solving everyday tasks.

MetaTrader 5 Trading Platform / Platform Setup / Automations / Common Settings

Common Automation Settings

To create a task, open the "Automation" section and create a new configuration:



Set general parameters for the task:

- **Name** task name. Set descriptive names which reflect the idea of target actions. Appropriate naming will ensure efficient task managements when many tasks are accumulated in the system. While the maximum name length is 127 characters, we recommend keeping names short and concise. This will make it easier to search for automation events in server logs, among other things. It is not recommended to use special characters in the name: %, !, " etc.
- **Trigger** an event in the platform upon the occurrence of which the automation task should be executed. When a trigger fires, the correspondence of the occurred event to the relevant conditions is checked. If all the conditions are met, a predefined action is performed.
- **Started** the date and time on which the trigger check will be activated.
- **Expired** the date and time after which the trigger check will be disabled. If no value is specified in this field, the trigger check interval is considered unlimited.
- **Repetitions** the maximum number of event repetitions over the period specified in the next field. Every time a trigger fires, the system checks how many similar events have occurred during the specified period. If the threshold has been reached, no action is performed. For all triggers except for those of the "Schedule" type, a zero value means that the number of repetitions is not limited. If zero is set for a scheduled event, the trigger will fire once at the time specified in the "Started" field. For more setup details please view section "Period and repetitions setup specifics" below.
- **Period** the period of time for which the number of repetitions should be checked. If you are limiting the occurrence of an event, do not leave the period field set to zero. You should explicitly specify some value. Otherwise, the repetition limit will not work.
- **Days of week, Months, Days of month** days and months for which triggers are enabled. If an event does not meet the specified criteria, the event handling will be skipped.

If a task configuration is not ready or if you wish to suspend it, uncheck the "Enable" box.

- For convenience, tasks can be grouped in directories. Click "Add" in the context menu of the "Automation" section in the left-hand tree and specify the new directory name. Then drag and drop the desired tasks into this directory.
- To quickly create similar tasks, use the "Add copy" command of the context menu. Instead of creating each task from scratch, create a copy of the existing one and adjust the required parameters.
- When you cancel your subscription to the Automations service, only the three tasks that were activated first will continue to operate. All other tasks will stop running, even if their configurations are enabled. The number of active tasks and their names are displayed in the server log at startup:
Automation loading 130 tasks (91 disabled, 3 active)
Automation active tasks: High Process CPU usage Alert, Manager Order Modify, Geo Condition

Period and repetitions setup specifics

Repetitions and period settings have different logic for the "Schedule" triggers and for all other triggers:

- For schedule triggers, this setting indicates how many times and at which intervals the event generation shall be repeated. Therefore, the Repetitions and Period parameters cannot be zero for such events, otherwise they will never be triggered.
- For all other triggers the setting indicates how many event repetitions in the specified period shall cause the trigger to fire.

Example: Started = 10:30, Repetitions = 3, Period = 3 minutes.

For the "Scheduled event" trigger, the event will be generated three times: at 10:30, at 10:33 and at 10:36. After that the generation will stop.

For all other triggers, the first 3 events over the last 3 minutes at the event occurrence time will be used. For example, if the "Login" trigger is used and connection events occur in the specified period, then:

- Logging in at 10:30:01 will activate a trigger, as there have been no triggers in the last 3 minutes.
- Logging in at 10:30:03 will activate a trigger, as there have been only 1 trigger, at 10:30:01.
- Logging in at 10:31:00 will activate a trigger, as there have been only 2 triggers, at 10:30:01 and at 10:30:03.
- Logging in at 10:31:01 **will not** activate a trigger, as there have already been 3 triggers in the last 3 minutes: at 10:30:01, at 10:30:03 and at 10:31:00.
- Logging in at 10:33:01 will activate a trigger, as there have only been 2 triggers in the last 3 minutes: at 10:30:03 and at 10:31:00.
- Logging in at 10:33:02 **will not** activate a trigger, as there have already been 3 triggers in the last 3 minutes: at 10:30:03, at 10:31:00 and at 10:33:01.
- This logic will continue until the time specified in the "Expired" field.

Trigger activations are calculated in the context of the event source. For example, connections of users 1000 and 1001 are different and thus they will be counted separately. The following unique keys are used:

- Login for triggers from sections "Connections", "Finance" (except for first operation events, such as first deposit, first withdrawal, etc.) and "Trade".
- Symbol for triggers from the "Prices" section.
- Absent (no key) triggers for first operation events, such as first deposit, first credit, etc., "Scheduled event" triggers from the "Platform" section.

If only the period is specified and the number of repetitions is set to 0, the "Scheduled event" trigger will generate an event every specified interval, without any limitations, until the time specified in the "Expired" field. All other trigger types will activate a trigger for each event, without any limitation on the number of repetitions.

MetaTrader 5 trading platform

If only the number of repetitions is set and the period is equal to zero, the "Scheduled event" trigger will fire once, at the time specified in the "Started" field (the number of repetitions is ignored). All other trigger types shall fire only the first n times, where n is the number of specified repetitions.

MetaTrader 5 Trading Platform / Platform Setup / Automations / Triggers

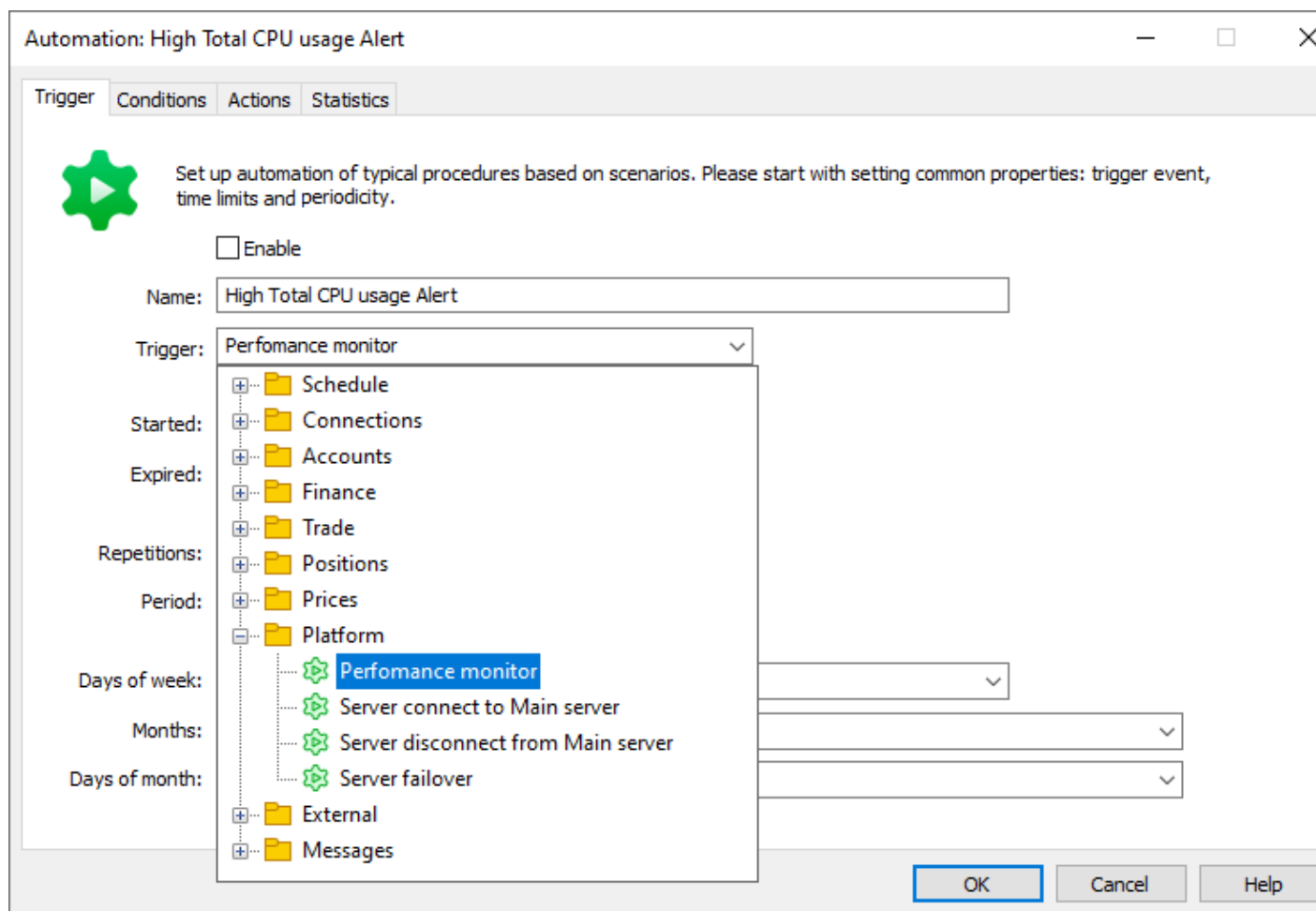
Triggers for Automation Tasks

Triggers are events in the platform for which automation tasks can be performed.

A trigger fires (the event check begins) if:

- The task is enabled.
- The type of the occurred event corresponds to the specified trigger.
- The monitored event occurs within the time interval specified in the "Started" and "Expired" fields.
- The number of event repetitions in the specified period (these two parameters are specified in general settings) has not been exceeded.
- The event occurs on the day and month specified in settings.

If all the conditions are met, the system proceeds with checking the correspondence of the occurred event to the relevant automation task conditions. The specified action is performed if all conditions are met.



All triggers are divided into several categories:

Schedule

These events enable the execution of schedule-based tasks.

Scheduled event

The action is performed at a specified frequency. The first event is performed at the time specified under "Started". Further repetitions are executed in accordance with the "Period" and "Repetitions".

Scheduled account database processing

This trigger is intended for periodic maintenance of the entire trading server account database. The service allows the selection of accounts from the database according to specified criteria and the execution of bulk actions. For example, you can:

- Archive or disable the accounts which have no active positions and have not connected for more than a specified number of days.
- Set users' balances to zero if they fall below zero.
- Create your own Stop-Out mechanisms: when the account balance is less than a specified value, close positions for specified symbols.

When a task triggers, the system checks each account to determine if it corresponds to the specified conditions. For example, you can filter accounts by the last connection time, by the balance value, by country, etc. We strongly recommend that you always specify in conditions the account groups to which the actions should be applied. Checking of a large number of accounts is resource-intensive, and eliminating of unnecessary accounts can significantly reduce the task execution time.

The specified actions are executed for each account that matches the criteria.

Below is a setup example:

- Trigger = Scheduled account database processing
- Started = 2021.03.07 22:00:00 (Sunday)
- Period = 7 days
- Repetitions = 0
- Days of week = Sunday

The following conditions are specified:

- Account group = real*
- Balance < 0

The following action is specified:

- Action = Deposit pay off
- Select by = Trigger
- Type = Balance

- Comment = Compensate negative balance

The task will be executed as follows: starting from March 1, 2021, 22:00, every Sunday at 22:00 the system will select from the database the real accounts with negative balance values, and will change the value to 0 by performing an appropriate balance correction operation, while adding the specified comment.

- Processing of an entire database and application of actions to multiple accounts can be time-consuming and resource-intensive. Therefore, we strongly recommend setting the task only for weekends and disabling repetitions.
- Only conditions from the "Accounts" list are available for the trigger.
- Actions related to configuration changes and platform management are not available for the trigger related to the processing of an account database.

Scheduled order database processing / Scheduled position database processing

These triggers work similarly to the previous one, but instead of the account database they process the the databases of orders and positions. Set the criteria for the selection of trading operations, and the system will apply the selected bulk actions to these operations. For example:

- Set a threshold volume and the system will periodically notify dealers about all larger positions.
- Set a threshold profit, and the system will periodically notify managers about clients' possible large revenue.
- Define criteria for selecting positions for a symbol approaching an expiration date and send periodic notifications to clients.
- Define criteria for selecting positions on accounts which have not been connected for a long time, and periodically remind traders about the current state of their accounts.

When a task triggers, the system checks each order or position to determine if it corresponds to the specified conditions.

- Position triggers operate with conditions from the "Accounts", "Positions" and "Platform" lists.
- Order triggers operate with conditions from the "Accounts", "Orders" and "Platform" lists.

For example, you can filter orders by symbol or expiration date. Positions can be filtered by volume or profit, etc. We strongly recommend that you always specify in conditions the account groups in which the operations should be analyzed. Checking of a large number of records is resource-intensive, and eliminating of unnecessary accounts can significantly reduce the task execution time.

The specified actions are executed for each order or position that matches the criteria. Both triggers operate with the actions from the lists "Message", "Trading account", "Finance", "Trade", "External".

Below is a setup example:

- Trigger = Scheduled position database processing
- Started = 2021.03.07 22:00:00 (Sunday)
- Period = 7 days
- Repetitions = 0
- Days of week = Sunday

The following conditions are specified:

- Accounts\Group = real*
- Position\Volume > 100

The following action is specified:

- Action = Message\Send push message
- Send to = Group
- To = managers\dealers
- Text = High trading volume on account #LOGIN#, position #POSITION_TICKET#, #POSITION_VOLUME# lots

The task will run as follows:

- Starting from March 1, 2021, every Sunday at 22:00 the system will check positions on real accounts.
- If the system finds positions with a volume of more than 100 lots, it will send a notification to the mobile phones of accounts from the "dealers" group.
- The message will contain the account number, position ticket and volume.

Processing of an entire database and application of actions to multiple trading operation can be time-consuming and resource-intensive. Therefore, we strongly recommend setting the task only for weekends and disabling repetitions.

Manually running a task

Tasks with scheduled triggers can be launched manually. Use this option to apply a one-time action to a database or platform settings. For example, you can change the spread before the release of important news and then set it back to the original value.

To manually launch a task, click "Run" in its context menu or the relevant button in the settings dialog:

MetaTrader 5 trading platform

The screenshot displays the MetaTrader 5 Administrator window. On the left, a tree view shows the 'Broker Server' structure, including 'Automations'. The 'Widen Spreads' automation is selected, and a context menu is open with the 'Run' option highlighted. A red arrow points from the 'Run' option in the menu to the 'Run' button in the configuration panel on the right.

Automation: Widen Spreads

Trigger Conditions Actions Statistics

Set up automation of typical process periodicity.

☒ Enable

Name: Widen Spreads

Trigger: Scheduled event

Started: 2024.04.02 16:29:50

Expired: ☐ 2024.04.02 17:44:03

Repetitions: 0

Period: 0 days 0

Days of week: All

Months: All

Days of month: All

Run

The task is performed exactly as if the trigger has fired. To perform the action, all of the following conditions must be met:

- The task configuration must be enabled.
- The task run time must not conflict with the "Started" and "Expired" parameters specified in common settings.
- All conditions in the task must be fulfilled as if it were a casual trigger-based execution.

Manual execution of automation tasks does not work without a valid service license.

Connections

This category includes the events related to the login and logout accounts in the trading platform.

- **Login** the event occurs every time an account is connected to the trading platform.
- **First login** the event occurs once, for each trading account, when the account connects to the platform for the first time.
- **Logout** the event occurs every time an account is logged out of the platform.
- **Authorization fail** the event occurs in case of a failed attempt to connect to an account, for instance, because of an invalid password. It allows reducing the number of technical support tickets: in case of a login issue, send your clients a message prompting to check their login data. Besides, you are able to improve account handling security. With the #USER_IP_ADDRESS# macro, you are able to specify the IP address, a connection attempt was made from, in notifications sent to your clients. In case of an unknown address, users are able to perform the necessary actions in a timely manner.
The event is not triggered if an attempt was performed using a non-existing or disabled account, as well as using an account from a disabled or non-existing group.

Accounts

This category includes the events related to the trading database.

- **New trade account** registration of a new trading account. Using this trigger, you can organize the distribution of promotional materials to new customers. The event is also generated when an account is restored from an archive or backup database.
- **Trade account deleted** this trigger can be used to notify the administrator about important changes in the account database. Also, for deleted accounts, you can automatically close positions and cancel pending orders. The deletion event is also generated when an account is archived.
- **Trade account archived** this trigger will send final notifications to inactive clients when their accounts are moved to the archive database.
- **Trade account restored** you can notify the administrator of an account restored from an archive or backup database.
- **Trade account group changed** the event is generated when an account is moved from one group to another. To filter events by the group to which the

MetaTrader 5 trading platform

account was moved, use the "Group" condition; to filter by the source group use "Previous group".

- Trade account password changed the event is generated when any of the account passwords (primary, investor or API) is changed. The trigger can be used to send security alerts. If the trading account password has changed, notify the trader via SMS or email.

Managers

These are the events associated with manager account actions. By using these triggers with the notification actions, you can implement control over the employees' work.

Triggers are fired when the specified action is performed via any client software: Manager terminal, Administrator terminal or Manager API.

- Trade order modified
- Trade order deleted
- Trade order moved to history
- Trade order reopened from history
- Trade order restored from backup
- Trade deal modified
- Trade deal deleted
- Trade deal restored from backup
- Trade position modified
- Trade position deleted
- Trade position restored from backup
- Account position fixed correction of positions based on the account's history of deals
- Account balance fixed balance correction based on the account's history of deals
- Export accounts / clients / positions / deals / orders / reports the action of saving of the corresponding data to a file executed from the Manager terminal (when the "Export" or "Save Report" command is executed). These triggers help you better monitor your employees' actions and prevent leaks of important data. For example, whenever a data export command is pressed, you can send the corresponding notification to a supervisor or higher-level employee. The login of the manager who exported the data can be included into the message using the #MANAGER_LOGIN# macro.

To get information about the modified (state after the modification) and deleted operations, use macros #ORDER_*, #DEAL_* and #POSITION_*.

Payments

These are the events related to payment transactions performed through the integrated payment system.

- Payment requested the event occurs whenever a user initiates a payment in the client terminal by entering the required details and submitting a request to

the broker or payment provider. The event is called before the request is sent to an external provider or manager for verification (if provided by the processing rules). This trigger allows you to track detailed statistics on internal payments, such as sending data to an external system via the "Send Web Request" action. To display transaction data, use macros from the "Payments" section.

- **Payment rejected** this event occurs when a request is declined due to incorrect parameters or when it is rejected based on processing rules (either automatically or manually by a manager).
- **Payment confirmed** this event occurs when a payment request is manually approved by the manager.
- **Payment done** this event occurs when a payment is successfully completed, following a confirmation notification from the payment provider and the crediting or debiting of funds from the account.
- **Payment failed** this event occurs when a payment is declined due to internal errors on the MetaTrader 5 server (e.g., missing conversion rates, provider unavailability, etc.).

To filter events, use the conditions from the "Payments" section.

Finance

These are the events related to balance operations on accounts.

- **Deposit** the event occurs for each deposit operation (the "Balance \ In" balance operation).
- **First deposit** the event occurs once, for each trading account, when the user makes the first balance operation of "Balance \ In" type.
- **Withdrawal** the event occurs for each withdrawal operation (the "Balance \ Out" balance operation).
- **Credit** the event occurs every time when credit funds are added to the account (the "Credit \ In" balance operation).
- **First credit** the event occurs once, for each trading account, when the user makes the first balance operation of "Credit \ In" type.
- **Credit out** the event occurs every time when credit funds are withdrawn from the account (the "Credit \ Out" balance operation).
- **First credit out** the event occurs once, for each trading account, when the user makes the first balance operation of "Credit \ Out" type.
- **Operation** the event occurs when service operations are performed on the account. These include balance, credit and correction operations performed by managers or API applications, as well as negative balance compensation and credit write-off operations. You can use conditions from the Finance category to additionally filter these events.

Trading

These are events related to changes in account trading states.

- **Margin call** the event occurs when a Margin Call hits on any account. Margin call status is checked for accounts on every tick. To avoid massive trigger activations, limit the number of repetitions in the tasks.

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- **Stop out** the event occurs when a Stop Out happens on any account.
- **Request timeout** deleting a trade request upon the expiration of the maximum allowable processing time.

Three minutes are given for processing each trade request. During this time, it must be processed by the gateway or dealer, or it must be automatically confirmed/rejected by the routing rule. If none of the actions happens, the request is removed. This activates the "Request timeout" trigger. Use the trigger to promptly notify the platform administrator about trading issues on the server, including those related to trade forwarding to external systems via gateways.

Positions

This category includes the events related to trading operations on accounts.

- **Position Open** the event occurs when a market entry deal ("In") is executed for the symbol for which the account does not have open positions yet.
- **Position Increase** the event occurs when a market entry deal ("In") is executed for the symbol for which the account already has an open position. The volume of the existing position increases after this operation.
- **Position Decrease** the event occurs when a market exit deal ("Out") is executed for the symbol for which the account already has an open position. The volume of the executed deal is less than the existing position volume. As a result, the existing position volume decreases, but the position is not completely closed.
- **Position Close** the event occurs when a market exit deal ("Out") is executed for the symbol for which the account already has an open position. The volume of the executed deal is equal to the existing position volume. As a result, the existing position is completely closed.
- **Position Reverse** the event occurs when a reversal deal ("In/Out") is executed for the symbol for which the account already has an open position. The volume of the deal is greater than the existing position volume. As a result, the current position is completely closed, and a new one is opened in the opposite direction.

Combine these triggers with the Web Request action to send trading information to traders' personal rooms. By using the triggers together with Message actions, you can automatically notify dealers about large operations. Use macros to obtain data about changed positions and about the deals that caused these changes.

Prices

These events are connected with trading symbol prices.

- **Prices gap started** the event occurs when a gap mode is activated on any trading instrument.
- **Prices gap finished** the event occurs when a gap mode finishes on any trading instrument.
- **Delayed** every symbol in the platform has the Max delay parameter. If no quotes are received for the symbol within the specified time, trading is automatically disabled for this symbol. This automation event is called at the same time. By using this event, you can promptly inform your platform

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administrator about issues in the delivery of quotes. To specify more details, such as symbol name and price delay value, use macros from the "Prices" section.

- **Resumed** this event is paired with the previous one. It occurs when the quoting steam for the symbols is restored after a break.

To filter events by individual instruments, use the "Prices\Symbol" condition. To filter events by a group of instruments, use the comparison type "Match mask".

Platform

These events are associated with the platform and hardware operation.

- **Performance monitoring** the event occurs every time when platform performance is measured (once a minute).
- **Server connection** the event occurs when any of the cluster servers connects to the trading server. If the platform uses multiple trading servers (additional servers, along with the main one), the event will trigger upon connection to any of them. In this case, each event will be considered unique and will not be counted as repeated events.
- **Server disconnection** the event occurs when any of the cluster servers disconnects from the trading server. If the platform uses multiple trading servers (additional servers, along with the main one), the event will trigger upon disconnection from any of them. In this case, each event will be considered unique and will not be counted as repeated events.
- **Server failover** the event occurs when the Failover System switches any of cluster components to a backup server. This can refer to the main or additional trade server, as well as a history server. To create a task related to switching of a specific component, use a additional condition "Server ID".
- **Gateway connection/disconnection** the event occurs when a gateway connects to or disconnects from the trading platform. Use these triggers with Message actions to instantly notify the administrator about issues with liquidity providers. Information about the specific component for which the event triggered can be immediately displayed in the message. This can be done using macros, such as "Gateway name" or "Gateway ID". To create a task for a specific component, use additional conditions: "Gateway name" or "Gateway ID".
The trigger is not activated when you enable/disable a gateway configuration.
- **Data feed connection/disconnection** the event occurs when a data feed connects to or disconnects from the trading platform. Use these triggers with Message actions to instantly notify the administrator about issues with quotes or news providers. Information about the specific component for which the event triggered can be immediately displayed in the message. This can be done using "Data feed name" macro. To create a task for a specific component, use the additional "Data feed" condition.
The trigger is not activated when you enable/disable a datafeed configuration.

Messages

This category includes the events related to the internal mail system of the platform.

MetaTrader 5 trading platform

- Client sent internal mail this event enables the creation of your own notification system for new customer requests. Use the trigger with Messages actions.
- Client read internal mail using this trigger, you can collect email read statistics and analyze the efficiency of your marketing actions. The trigger can be combined with the action that sends events to Finteza.
- Fail to send via SMS / Email / Messenger these events allow you to track and promptly respond to issues arising in the relevant services: integrations with SMS providers, instant messengers and mail services. For example, you can easily track if your balance with the SMS provider runs out and sending of notifications to clients stops. Just select the appropriate trigger and set additional conditions specifying the provider name and the balance below limit. For the action, you can use sending of a message to an administrator via Telegram.

Use macros to obtain email related data and to use it in actions.

KYC

These are the events related to user verification via KYC providers.

- KYC started the event occurs when user data is sent to the KYC provider: manually by the manager or automatically during account registration.
- KYC approved the event occurs when the client successfully passes verification on the KYC provider side and the check status changes to "Approved".
- KYC rejected the event occurs when the client fails to pass verification on the KYC provider side and the check status changes to "Rejected".

Using these triggers, you can promptly inform your clients about their registration stages. For example, using the "KYC rejected" trigger and the Push Notification action, you can instantly notify the client about any additionally required actions. Specify the #KYC_STATE_DESC# macro in the message to provide the reason for rejection.

External

These triggers enable the integration of your MetaTrader 5 platform with external systems. By using them, you can:

- Enable your own task trigger logic by generating events via the MetaTrader 5 Server API plugin.
- Enable your own task trigger logic by generating events using callback requests.

API event

This trigger enables the integration of the Automations service with any solutions that use MetaTrader 5 Server API. The event will emerge when the plugin running on the

trading server calls a special AutomationTrigger function. You can use any function call logic in the plugin, depending on the desired purpose.

The AutomationTrigger function allows not only triggering a task, but also passing additional data to the service: account information, trading operation and others. For example, if the event concerns a trading operation, you can pass the full description of an order or a deal; the full trading account description can be passed for balance change events. The relevant data can be used for checks in additional condition and in macros. For example, you can check a login from the account description in the "Login" condition or a ticket from the deal description in the "Deal ticket" condition.

Web Callback

With this trigger, you can run automation tasks using Web Callback requests.

This function facilitates routine operations related to accounts and trading databases. For example, you wish to remove position stop levels according to certain rules. This could be done by creating a special plugin, for which you would need to hire a qualified developer. Instead, you can simply create an automation task, describing the necessary conditions and actions. This task is then executed by a normal web request.

Preparation

To enable web requests, configure the platform in the Integration \ Web services section:

- Associate your access server's public address with a domain and upload an SSL certificate for that domain
- Specify the list of allowed callback-requests and IP addresses from which the platform will accept them

How to configure a task

Create a task with the Web Callback trigger. Specify all other parameters, similar to a regular automation task, that is, define the necessary conditions and actions.

How to run a task

To run a task, send a web request starting with /api/callback/automation to the server. The specification of the part following this prefix is arbitrary. For example, for convenience, you can add the purpose of the automation task in the request:

- /api/callback/automation/notify for tasks that send notifications
- /api/callback/automation/position for tasks relating to positions
- /api/callback/automation/account for tasks relating to accounts
- etc.

Please make sure to add the URL of your requests to the list of allowed requests in the Integration \ Web services. Otherwise, the platform will not process them. For each request, provide a list of allowed IP addresses. The platform will only accept the web requests from the addresses specified in this list. This ensures complete safety when

using the service.

Web Services

Please specify a list of addresses, from which the platform will accept and process callbacks on /api/callback/. The list is used for all access servers.

Callback domain:

Up	URL	Allowed IPv4/IPv6	Comment
Down	✓ /api/callback/automation/notify	175.25.69.173,175.25.69.174	Automation callback for notifications
	✓ /api/callback/automation/position	175.25.69.173,175.25.69.174	Automation callback for positions
	✓ /api/callback/automation/account	175.25.69.173,175.25.69.174	Automation callback for accounts
Add			
Edit			
Delete			

Add the URL of the specific request which should run the automation task as a Web callback URL condition. This will prevent the task from being triggered when calling other queries.

Automation

Trigger
Conditions
Actions


Please specify detailed conditions for the automation task to trigger. Date, account and platform-based parameters are available.

Type	Condition	Value
ab External\Web callback URL	Equal (=)	/api/callback/automation/notify

Add
Edit
Delete

OK
Cancel
Help

You can pass the account number and the order, deal or position ticket as additional request parameters. This will determine the task context: the account or operation for which it should be executed. Examples:

```
/api/callback/automation?login=11001  
/api/callback/automation?position=12476698  
/api/callback/automation?login=11001&position=12476698&order=12514593&deal=33696
```

Only one value can be passed in each parameter. The parameters work independently of each other. If you specify both a login and an order ticket, this does not mean that the automation will search for the order from the specified user. The user object and the order object will be passed separately to the task execution context.

The account number and tickets can also be passed through the request body, for example:

```
/api/callback/automation  
{  
  user: { Login: 1000 },  
  order: { Order: 11543 },  
  deal: { Deal: 961498 },  
  position: { Position: 59723 }  
}
```

Data in request parameters takes precedence over data in the body. For example, if an account login is found in the parameters, then the login specified in the body will be ignored.

You can send a request to the server directly from the command line using the [curl](#) utility. for example:

```
curl -k --request GET --url https://web.broker.com:443/api/callback/automation?login=1002
```

Here:

- Specify the -k flag if [self-signed certificates](#) are used on your server
- Specify your server's domain and port instead of https://web.broker.com:443

To get the URL of the request that triggered the automation task, use [macro #WEBCALLBACK_URL#](#). For example, you can add it to the body of a sent email.

Actions related to the clearing of stop levels

Actions "[Clear Stop Loss / Take Profit / Stop Loss and Take Profit](#)" execute differently if the "Select by" parameter is set to "Position or order from trigger":

- If a position ticket is passed in the request, actions will only process the position
- If an order ticket is passed in the request, actions will only process the order
- If the request contains the both the position and order tickets, the actions will only process **the position**

For example, the following request will only process the position with ticket 12500033:

```
curl -k --request GET --url https://web.broker.com:443/api/callback/automation/clear_sl_tp?or
```

The next request will affect the order with ticket 13111609:

```
curl -k --request GET --url https://web.broker.com:443/api/callback/automation/clear_sl_tp?or
```

This feature only applies to actions concerning the clearing of stop levels. In other actions, you can simultaneously pass and use order and position data. For example, by passing operations in a request, you can print their tickets in an email using the #POSITION_TICKET# and #ORDER_TICKET# macros.

An example of sending an internal email to the user

- Create a task with the Web callback trigger
- Add condition Web callback URL = /api/callback/automation/notify. The task will only be run by such requests.
- Add action Message\Send internal email. In the "Send by" parameter, select "Trigger". This way, the action will only apply to logins passed in the request. Specify the email text.

To run the task, send the following request:

```
curl -k --request GET --url https://web.broker.com:443/api/callback/automation/notify?login=
```

Account number 10000 will receive the email.

Example of closing a position by ticket

- Create a task with the Web callback trigger
- Add condition Web callback URL = /api/callback/automation/position_close. The task will only be run by such requests.
- Add the "Trade\Close positions" action. In the "Select by" parameter, select "Position from trigger". The action will only apply to the positions passed in the request. Additional filters can be set in the "Type" and "Symbol" parameters. For example, if the symbol of the passed position does not match the specified one, the position will not be closed.

To run the task, send the following request:

```
curl -k --request GET --url https://web.broker.com:443/api/callback/automation/position_close
```

The position with ticket 12415801 will be closed.

Overriding data in the task call context

If for some reason you need to add or edit the data in the object passed to the automation task, you can specify it in the additional request body through the 'user', 'account', 'deal', 'position' and 'order' objects. For example:


```
/api/callback/automation?login=11001
{
  user: { LastName: 'NewLastName', Balance: 0 }
}
```

A trading account with number 11001 will be passed to the automation task. The trader's last name in the account will be replaced with "NewLastName" and the balance will be set to zero.

- The information passed in this way does not make any changes to the platform databases. The information is used only in the context of the automation task.
- The list of possible fields for the 'user', 'account', 'deal', 'position' and 'order' objects is provided in the [MetaTrader 5 Web API](#) documentation.

In general, automation works with data from parameters and request body as follows:

- First, the system tries to find object identifiers (account number or ticket) in the parameters.
- If the identifiers are not found in the parameters, it tries to find identifiers in the request body.
- If the identifier is found, the automation tries to load the object data from the server database. If the record is not found, the task does not run and an error is returned in response to the request.
- If the record is found, it is updated according to the data in the request body.
- The automation task starts.

Let us look at a more complex example:

```
/api/callback/automation?login=11001
{
  account: { Equity: 100 },
  user: { LastName: 'NewLastName', Balance: 0 },
  order: { Order: 12514593, Symbol: 'AUDUSD' },
  position: { Position: 124766980, Symbol: 'EURUSD' },
  deal: { Deal: 33696169 }
}
```

The case will be processed as follows:

- Automation will receive an account object with number 11001 from the database. In the received data, the Equity value will be redefined to 100, the user's last name will be replaced with "NewLastName", and the balance will be set to zero.
- Automation will find an order with ticket 12514593, and the symbol in it will be redefined to "AUDUSD".
- Automation will find a position with ticket 124766980, and the symbol in it will be redefined to "EURUSD".
- Automation will find a deal with ticket 33696169.
- The task execution will start with the final data.

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The identifier passed as a request parameter takes precedence over the identifier in the request body. For example, the following request will process the account number 10000:

```
curl -k --request POST --url https://web.broker.com:443/api/callback/automation/notify?login=
```

The following request will process the account number 10001:

```
curl -k --request POST --url https://web.broker.com:443/api/callback/automation/notify? --hea
```

Similar rules apply when searching for tickets.

MetaTrader 5 Trading Platform / Platform Setup / Automations / Conditions

Trigger Conditions for Automation Tasks

Having configured general task triggering parameters, describe more detailed event conditions. For example, if you have set a Login trigger, you can specify which group this account should belong to, as well as the registration city, language and many other additional parameters.

Automation

Trigger
Conditions
Actions


Please specify detailed conditions for the automation task to trigger. Date, account and platform-based parameters are available.

Type	Condition	Value
 Accounts\Group	Equal (=)	real*
 Accounts\Language	Equal (=)	English
 Accounts\Last visit	Less (<)	2021.09.28 18:11:02
 Schedule\Date and time	Equal (=)	2022.09.28 18:11:11

Conditions
Schedule
Date and time
Date
Time
Accounts
Platform

Add
Edit
Delete

OK
Cancel
Help

Comparison Types

Standard comparison types are available for all conditions:

- Equal (=)
- Not equal (! =)
- Greater than (>)
- Greater than or equal (>=)
- Less than (<)
- Less than or equal (<=)

The "Match mask" condition is additionally available for string conditions, such as country name, symbol name or email address. Here you can specify strings with the "*" mask. For example:

- Condition "Symbol", value "Forex\USD*" means all symbols from the Forex group where the base currency is USD.
- Condition "Email", value "*@mailbox.com" means all email address with the mailbox.com domain.

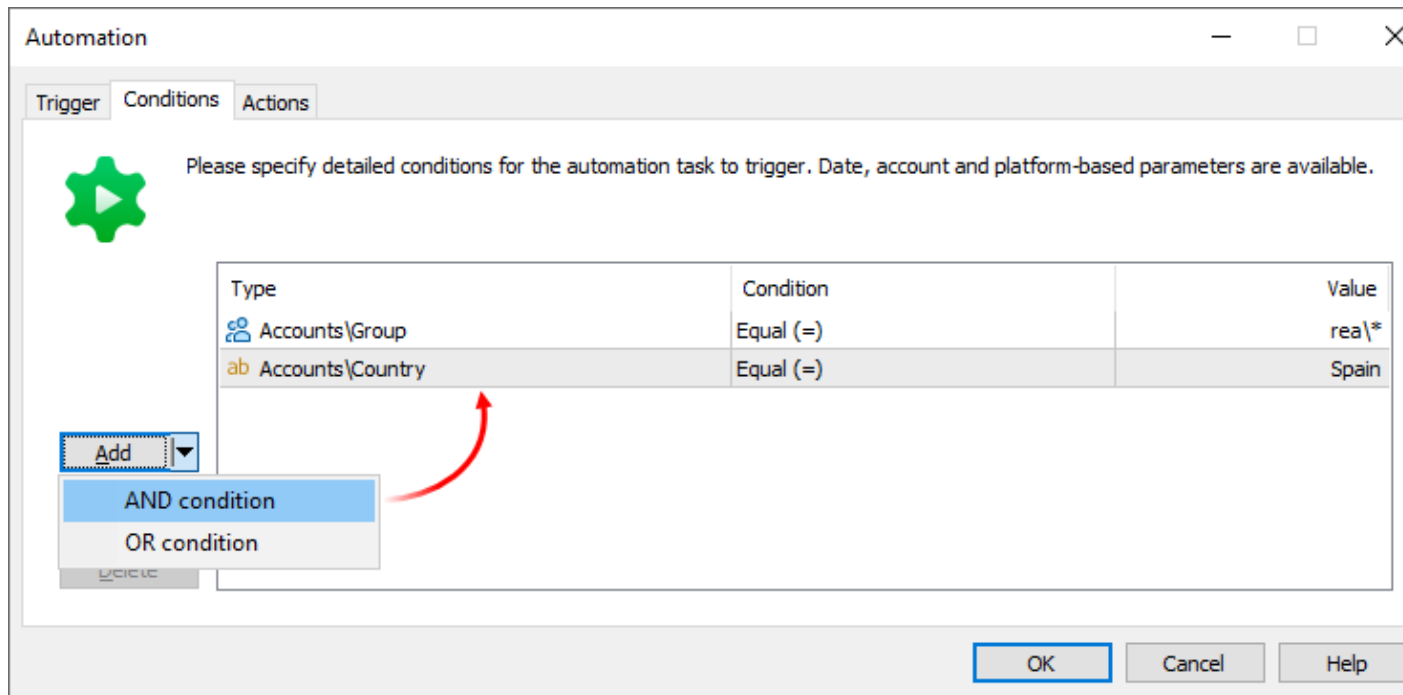
For login specifying conditions, you can specify multiple values and value ranges. For example: Account\Login = 100,102,107,120-130.

- Conditions are checked **after** triggers fire.
 - If an event does not have a characteristic which is defined in a condition, such a condition will never hit. For example, a "Platform monitoring" event has occurred, and the conditions specify the balance requirements.

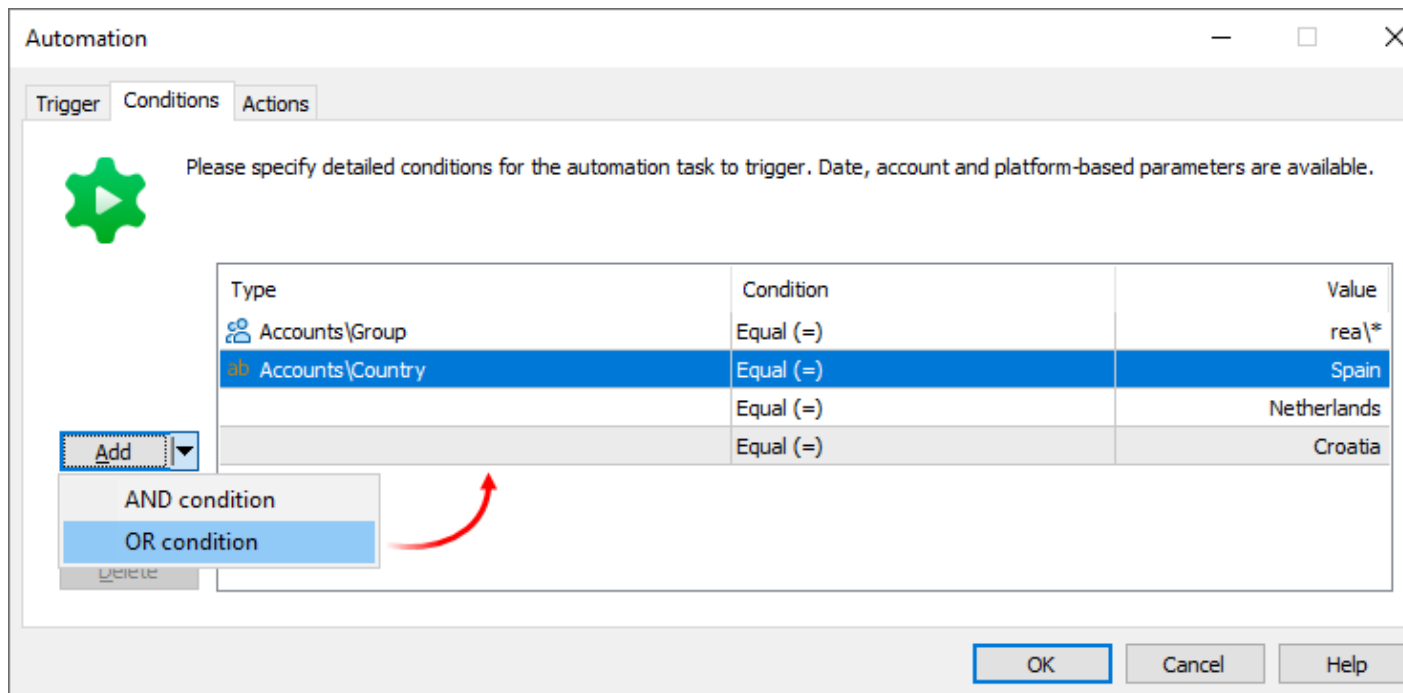
Condition types

Conditions can be of two types: "AND" and "OR". The AND condition is mandatory, while the OR condition is optional.

For example, a rule specifies two "AND" conditions: country = Spain, group = real*. In this case, the task will only trigger for real accounts from Spain.



Now let us add two additional "OR" conditions to the rule: country = Netherlands, country = Croatia. The task will trigger for real accounts from any of the three countries: Spain, the Netherlands or Croatia.



"OR" conditions can only be added in addition to the "AND" condition. In this case, the "Country" condition is mandatory, while it can have three possible values.

To add an "OR" condition, you must first add the mandatory "AND" condition. Then select it in the list and select the appropriate option in the drop-down menu of the Add button.

Time

These parameters set date and time conditions. The automation task will be launched if the event (trigger) fires on the specified date and time.

When using "more than" or "less than" comparisons, time is counted in seconds. For example, "< 00:01" means "less than 61 seconds", "> 08:15" means "more than 29700 seconds". Therefore, using a comparison with 00:00 (zero seconds) will be incorrect. If you want to set the condition to "before midnight", enter "< 23:59".

To specify time intervals within one day, use AND conditions. For example, Time > 07:00 AND Time < 23:00. For time intervals that cross over midnight, use OR conditions. For example, Time > 23:00 OR Time < 07:00. If you specify the "AND" condition here, it will never be met because the value will cannot simultaneously be greater than 23:00 and less than 07:00.

Accounts

These settings specify conditions based on account parameters:

- **Login** the event (trigger) occurred for the specified account. For example, if the "Login" event is specified as the trigger, and the account 1000 is specified as the login, then the automation task will be executed every time this account connects to the platform. Connection of other accounts will not trigger the task.
- **Group, Country, City, Language, Phone, Email, Color, Status, Company, Comment** the event occurred for the account from the specified group, country, city and so on. For example, if the "Login" event is specified as the trigger, and "real*" is indicated as the group, then the automation task will be executed every time an account from any real group connects to the platform. If the "Login" event is specified as the trigger, and "Argentina" is indicated as the country, then the automation task will be executed every time an account from Argentina connects to the platform. The condition works similarly for other account parameters.
- **Previous group** the condition is only used with the "Trade account group change" trigger. It enables filtering of events by the group from which the account was transferred.
- **Registration** the condition is set relative to the account creation date in the platform. Specify the date and the comparison type: equal to, less than or greater than. Accordingly, the condition will trigger for account registered on the specified date, earlier or later than this date. For example, if the "Deposit" event is specified as the trigger, and the condition is set to "Registration > 2020.09.01", then the automation task will be executed for all deposit operations on accounts registered after the specified day. By using Bonus as the action, you can automate promotions to attract new customers.

- **Last visit** the condition is set relative to the last connection of the account to the platform. Specify the date and the comparison type: equal to, less than or greater than. Accordingly, the condition will trigger for account connected on the specified date, earlier or later than this date. For example, if the "Login" event is specified as the trigger, and the condition is set to "Last visit < 2019.01.01", then the automation task will be executed for connections from accounts for which the previous login date was before the specified day. By using "sending of an internal email" as an action, you can automatically inform returning clients about new services and conditions.
- **Days since registration** the condition is set relative to the number of days that have passed since the account was created. The number of days is calculated based on the number of times the clock crosses midnight (00:00) since the event. The principle of operation and purpose of this condition are similar to the "Registration" condition.
- **Days since last access** the condition is set relative to the number of days that have passed since the last connection to the account. The number of days is calculated based on the number of times the clock crosses midnight (00:00) since the event. The principle of operation and purpose of this condition are similar to the "Last visit" condition.
- **Days since last trade activity** the condition is set relative to the number of days that have passed since the last operation on the account. This considers any trading operations, commission charges, etc. Balance operations are not taken into account. In addition, the system checks if the account has any open positions or active pending orders. Using this condition, you can automatically archive inactive accounts in order to reduce the server load, or you can send promotional emails. Set the trigger to "Scheduled account database processing", conditions to "Days since last trade activity >= 90", "Balance > 0" and the action to "Send SMS". The system will check the entire database and will send SMS to inactive traders who have money on their accounts. When using this condition, please note that newly registered accounts may also fall under this condition. For example, a trader may have created an account 1-2 days ago and has not yet performed any trading operations. If you set a condition based on the absence of trading activity in last 180 days, this trader will also be included in the automation action. To avoid such situations, use the trading activity condition in conjunction with other conditions, such as "Registration" or "Days since registration".
When calculating the number of days, the result is rounded down. For example, if 5 days and 23 hours have passed since the event, this will be counted as 5 days.
- **Days since last balance operation** the condition is set relative to the number of days that have passed since the last balance deal on the account. Such deals include deposits/withdrawals, credit transactions, bonuses, adjustments, dividends, taxes, commissions, annual interest accrual and compensation. You can use this condition in addition to checking trading activity on the account, for example when automatically archiving accounts. If you archive accounts with zero balances, you can additionally check if the funds were withdrawn recently.
When calculating the number of days, the result is rounded down. For example, if 5 days and 23 hours have passed since the event, this will be counted as 5 days.

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- **Days since last credit operation** the condition is set relative to the number of days that have passed since the last credit transaction on the account. Such transactions include credit transactions themselves, credit reset operations, as well as the accrual and withdrawal of bonuses.
When calculating the number of days, the result is rounded down. For example, if 5 days and 23 hours have passed since the event, this will be counted as 5 days.
- **Online** the condition is set relative to the status of account connection to the trade server. Possible values are "true" and "false", i.e. the account is either connected (any connection type: client terminal, mobile terminal and so on) or not. When combined with the web request sending action, this condition enables the generation of your own reports and the analysis account activity in third-party systems. By using the "Login" and "Online" conditions, as well as the actions from the "Messages" section, you can set up manager notifications about the connection of certain clients.
- **Leverage** the condition is set relative to the account leverage. For example, the regulator's requirements have changed and you need to disable trading with the leverage greater than 1:20 on accounts from the US. Set the "Login" trigger and specify two conditions: "Country = USA", Leverage > 1:20. As an action, specify "Disable trade". Once an account with the specified leverage connects to the platform, trading for such an account will be disabled.
- **Balance** the condition is set relative to the current account balance; the amount is specified in the deposit currency. By using this conditions, you can send automated emails to clients when the account balance drops below a certain threshold. For example, you can set the "Login" trigger, the "Balance < 100" and "Group = real*" conditions, "Action = Send internal mail". The automation action will be performed every time a real account with a balance less than 100 (in deposit currency) connects to the platform.
- **Credit** is similar to the "Balance" condition. In this case, the amount of credit funds on the account is checked.
- **Positions total** the condition is set by the number of open positions which currently exist on the account. This action allows you to control extremely active accounts. For example, you can set a trigger for "Scheduled event = once a day" and two conditions: "Total positions > 100", "Group = demo*". Specify "Close positions" as the action. Using the above task, you can reduce server load associated with a high trading activity on demo accounts.
- **Orders total** similar to the "Total positions" condition, it checks the number of active pending orders on the account.
- **Floating profit, Equity, Margin, Free margin, Margin level** these conditions are similar to the "Balance" condition. Appropriate trading account states are checked here. By using these conditions, you can implement your own system for notifying clients about issues with their account states. For example, you can set a trigger for "Scheduled event = once a day" and two conditions: "Margin level < 100", "Group = real*". Specify "Send push message" as the action. Your real clients will receive automated notifications about low margin states directly on their mobile devices.
- **Deposit currency** the event occurred for an account with the specified deposit currency. For example, if the "Login" event is specified as the trigger, and "EUR*" is indicated as the deposit currency, then the automation task will be executed every time an account from any group with the deposit currency Euro

connects to the platform.

- **Lead Source, Lead Campaign** lead source and lead campaign parameters specified in the account. These conditions can be used for the analysis of marketing campaigns. For example, when a new account with the specified parameters is opened, you can send the corresponding event to Finteza or to any other analytics system using the Web request action.
- **Enabled** the condition is specified relative to the "Enable this account" setting. Use it along with the "Scheduled account database processing" trigger to exclude disabled accounts from the selection so that the system does not waste time on such accounts.
- **Trading enabled, Algo trading by Expert Advisors enabled** the conditions are set relative to the corresponding trading account settings. Use them to fine-tune the delivery of marketing materials. For example, you can exclude the accounts with algo trading disabled from your algo trading mailing list.
- **Agent account** the condition is set relative to the agent account specified in the trading account settings. Use it along with the "Scheduled account database processing" trigger to send referral reports to agents.
- **Client ID** the unique identifier of the client with whom the trading account is associated. A client can have several trading accounts. With this condition, you can work directly with the client rather than separate accounts. For example, you can send a notification to a client if they connect with any of their accounts.
- **Own funds percentage** when trading, clients are able to use their own funds they deposited on their accounts, as well as the credit and bonuses provided by a broker. Both types of funds increase their Equity parameter. The "Own funds percentage" condition allows configuring the automation task depending on the own funds share on a client's account. 100% means the account has client's funds only with no credit and bonuses. 0% means only credit and bonuses are used for trading. Using this data, you can conduct a recurring database check and withdraw bonuses from the accounts of clients who withdrew their own funds. Alternatively, you can encourage traders using solely their own funds by granting them bonuses.
The parameter calculation equation: $(1 - (\text{Credit} / \text{Equity})) * 100$.
During the calculation, the fractional part of the value is rounded down: 1.5% becomes 1% and 0.9% becomes 0%. The parameter value cannot exceed 100% or be less than 0%. If an action is required after the client has spent all their funds, it is more efficient to use the "Own funds volume" condition.
- **Own funds volume** the parameter works similarly to the previous condition, although it applies an absolute own funds value rather than a relative one. Specified in the deposit currency. The calculation equation: $\text{Equity} - \text{Credit}$.

Manager

Set the conditions for the parameters of the manager account for which the event occurred:

- **Login** the event (trigger) occurred for the specified manager account. For example, if the "Trade order modified" event is selected as the trigger and the account 1000 is specified as the login, then the automation task will trigger every time the specified manager modifies the parameters of any order. The actions of other managers will not activate the task.

Platform

These settings specify conditions based on the cluster performance:

- Total CPU usage
- Process CPU usage
- Process threads
- Total free memory, MB
- Process memory, MB
- Disk free space, MB
- Disk read speed, MBps
- Disk write speed, MBps
- Disk queue
- Connections
- Blocked connections
- Total sockets
- Traffic in, Mbps
- Traffic out, Mbps
- Retransmits, %
- Process handles

The above parameters and their impact on the platform operation are described in detail under the "Monitoring" section.

Use these conditions to timely inform the administrator about a decrease in platform performance. For example, set the "Performance monitoring" trigger, specify the condition "Disk free space < 1000" and the action: "Send push message", "Send Email". Your administrator will be promptly notified about the need to buy an additional disk to ensure relevant data storage. "Scheduled event" can also be used as a trigger here. This allows sending periodic platform state reports to the administrator, for example such reports can be sent once a day.

Additionally, the "Platform" section provides general cluster parameters:

- **Server ID** the condition triggers when the server with the specified ID connects to the cluster. For example, the Server ID is set to 3, and the trigger is set to "Server connect to Main server". The automation task will be executed every time when a server with the ID 3 connects to the Main server.
- **Server type** similar to the "Server ID" condition. Instead of a specific server, you indicate the server type: access, history server, etc.
- **Server connected** this condition enables periodic checks of connection states of all cluster components. Combine it with the "Server type" or "Server ID" conditions. This condition is always considered fulfilled for the main server.

If a cluster has several trading servers, automation tasks are performed on each of the servers independently of each other. For example, if you create a task based on the "Server disconnection" trigger, then it will be executed in parallel on each trading server, monitoring connection to each of the servers.

The conditions described below allow you to identify the specific server on which the task is running. In some cases, they can help avoid unnecessary steps.

For example, you want to configure notifications when an access server is disconnected. In this case, you create the following task:

- Trigger: "Server disconnection"
- Conditions: "Server type = Access Server", "Server connected = false"
- Action "Send push message"

This task will cover most of your needs, but it can be further improved: disable sending of notifications during a regular reboot by the administrator. This can be especially useful if your access server operates with multiple trading servers. For example, if you have 3 trading servers, then if the main server (and, accordingly, the entire cluster) is rebooted, you will receive three notifications at once. The task will trigger on each trading server and will report that the connection with the access server has been lost.

To solve this problem, add another condition to the automation task: "Current server status = Not set". It will ignore all servers where a reboot was started.

- Current server ID the identifier of the trade server on which the automation task was triggered.
- Current server type the type of trading server on which the automation task was triggered: main or additional.
- Current server state the state of the trade server on which the automation task was triggered. Available values: No the server does not reboot; Restart a regular reboot has been started, Shutdown the server service is stopped, Live Update the server service is stopped due to a platform update.
- Current server connected the state showing the connection between the trade server on which the automation task was triggered and the main server of the platform.

A separate group of conditions is intended for use with triggers of gateway connection and disconnection from the trading platform. They enable the creation of tasks for specific components.

- Gateway name gateway configuration name.
- Gateway ID ID specified in the gateway configuration.
- Gateway connected the state of gateway connection to the platform at the trigger activation time. true the gateway is connected, false the gateway is not connected. Use this condition to determine which event happened: connect or disconnect. You can send the relevant notifications to the administrator depending on the event.

A separate group of conditions is intended for use with triggers of data feed connection and disconnection from the trading platform. They enable the creation of tasks for specific components.

- Data feed name data feed configuration name.
- Datafeed connected the state of data feed connection to the platform at the trigger activation time. true the data feed is connected, false the data feed is not connected. Use this condition to determine which event happened: connect or disconnect. You can send the relevant notifications to the administrator

depending on the event.

Conditions from the "Platform" section can only be used with the "Scheduled event" trigger and with triggers from the "Platform" section.

Connection

Platform integration with geo-services enables access to IP address-based information about clients connecting to the platform, such as their geographical location, provider and various signs of suspicious activity. You can use this data to create automation tasks. For example, you can send notifications about suspicious connections to an external system for analysis.

The following conditions are available:

- IP address client's IP address.
- Country the country in which the IP address is located.
- City the city in which the IP address is located.
- ASN the Autonomous System Number (ASN) to which the IP address belongs.
- ISP the name of the internet service provider that owns the IP address.
- Additional true/false flags:
 - ◆ Datacenter the address belongs to a data center.
 - ◆ TOR the address belongs to the TOR network.
 - ◆ Proxy the address is a proxy server.
 - ◆ VPN the address is a VPN server.
 - ◆ Attacker the address was involved in various attacks, such as spam, password brute-forcing, DDoS, etc.
 - ◆ Botnet the address belongs to a botnet.

The conditions are only used with connection triggers.

Payments

These settings specify conditions based on the parameters of the internal payment transaction, conducted on the account. Used only with the triggers from the Payments section.

- Action the type of transaction: deposit or withdrawal.
- Amount the transaction amount requested by the client. Specified in the deposit currency.
- Currency the transaction currency.
- Commission the commission charged by the broker in accordance with the settings.
- Wallet Amount the amount of the transaction on the side of the payment provider. It is specified in the currency selected by the user on the client terminal side when making the operation.
- Wallet Currency the currency of the transaction on the payment provider's side.
- Wallet Commission the payment provider's fee for the transaction.
- Description additional information about the transaction. Can be filled in by payment systems or manually by managers.

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- **Manager** the login and name of the manager who processed the transaction. It is filled only for manually processed transactions.
- **Type** the payment method type: card, bank transfer, etc.
- **Provider type** the type of the payment provider (gateway).
- **Provider name** the name of the payment provider configuration under which the payment was made.
- **Client IP** the IP address from which the payment was requested.
- **Client type** the type of client terminal from which the payment was requested: desktop, mobile, or web.
- **Deal ticket** the ticket of the balance operation through which funds are credited to or debited from the trading account.
- **External ID** the transaction identifier on the payment provider's side.
- **External error code** the error code that occurred on the provider's side.
- **External error description** a description of the error that occurred on the provider's side.
- **Error code** the error code that occurred on the platform side.
- **Error description** a description of the error that occurred on the platform side.

Finance

These settings specify conditions based on the parameters of financial transactions on the account:

- **Amount** the amount of transactions. By using this conditions, you can send an automated email to a manager if a too large deposit transaction is detected, requesting from the manager additional payment verification. For example, you can set trigger to "Deposit" and the following conditions "Amount > 100,000" and "Currency = USD".
- **Currency** the currency in which the transaction was performed. Combine this parameter with the previous one.
- **Comment** a comment to a transaction. With this parameter, you can take into account additional conditions when processing financial transactions on the account. For example, when depositing funds to a VIP client's account, you can add an appropriate comment to the transaction, and then use it in the Automations service to send appropriate notifications to the client's personal manager.
- **Action** deal type: Buy, Sell, balance or credit operation, etc. Using this parameter, you can control in detail the operations occurring on user accounts. For example, you can monitor negative balance compensations to quickly identify unscrupulous users. For this, select the trigger Finance \ Operation and add the following condition: "Action = SO Compensation". Additionally, you can use the "Amount" condition to specify the amount of compensation to be reported, for example, higher than USD 100. For the action, select sending notifications to the manager.
- **Manager** the login of the manager who performed the financial transaction. With this condition, you can control account deposits and withdrawals performed by your employees. For example, you can create a task with the trigger "Finance \ Operation" and the conditions "Finance \ Manager = 1000" and "Amount > 1000". For the action, you can enable the sending of a push

notification to a senior manager. To add the manager's login, indicate the #FINANCE_MANAGER# macro in the message.

Prices

Prices set conditions by the parameters of arriving quotes:

- **Symbol** the name of the trading instrument for which the event triggered. For example, the trigger is set to "Price gap started". If the condition is set to "Symbol = EURUSD", the action will only be performed if the gap starts for the specified currency pair. The same behavior is applied to triggers related to quoting stream delays and its restoration.
- **Last tick time** time of the last received quote of the symbol for which a trigger from the "Prices" section was activated. The condition can be used to disable the quotes delay trigger for archived symbols which are no longer used for trading.

Position

This groups sets conditions by position parameters. The conditions are used by triggers from the Positions section and the "Scheduled order database processing" trigger.

- **Login** account number the position was opened on.
- **Ticket** unique position ticket.
- **Symbol** symbol, for which the position was opened.
- **Volume** position volume in lots.
- **Type** position type: buy or sell.
- **Open price** weight-average price of the position opening: $(\text{price of deal 1} * \text{volume of deal 1} + \dots + \text{price of deal N} * \text{volume of deal N}) / (\text{volume of deal 1} + \dots + \text{volume of deal N})$.
- **Current price** the price of the financial symbol for which the position has been opened, at the trigger activation time.
- **Profit** profit of a position at the trigger activation time.
- **Reason** reason for position opening.
- **Creation time** position opening time.
- **Time since creation** the amount of time that have passed since the position was opened. 1 day corresponds to 24 hours, not the passing of a calendar day. For example, a position is opened at 21:00 on 01.04.2025 and the condition is set as "Time since creation > 1 day", while the automation task runs daily at 19:00. In this case, the condition will be triggered at 19:00 on 03.04.2025.
- **Update time** time when the position was last modified (when its volume was changed).
- **Time since modification** the amount of time that have passed since the last position modification (a change in its volume).
- **Dealer** the login of the dealer who confirmed or placed the order (deal) which caused the position to open. Use this condition to track dealers' trading activities. For example, you can create a task with the trigger "Position opened" and the conditions "Position \ Reason = Dealer" and "Position \ Dealer = 100". For the action, you can use sending of a push notification to a senior manager.

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To add the dealer login, indicate the macro #DEAL_DEALER# in the notification text.

- Expert ID identifier (magic number) of an Expert Advisor by which a position was opened in the client terminal.
- Comment a text comment to the position.

Deal

These settings specify conditions by deal parameters. The conditions are used by triggers from the Positions section.

- Login the number of the account, on which the deal was executed.
- Ticket unique deal identifier.
- Symbol financial instrument for which the deal was executed.
- Volume deal volume in lots.
- Type deal type: Sell, Buy, Balance etc.
- Direction direction of a deal relative to the current position: entry ("in"), exit ("out"), reversal ("in/out") or close by ("out by").
- Price deal price.
- Profit profit gained from the deal.
- Reason reason for deal execution.
- Time deal execution time.
- Dealer the login of the dealer who confirmed or placed the order, which caused the deal to execute. Use this condition to track dealers' trading activities. For example, you can create a task with the trigger "Position closed" and the conditions "Deal \ Reason = Dealer" and "Position \ Dealer = 100". For the action, you can use sending of a push notification to a senior manager. To add the dealer login, indicate the macro #DEAL_DEALER# in the notification text.
- Expert identifier (magic number) of an Expert Advisor by which the deal was executed in the client terminal.
- Comment a text comment to the deal.

Orders

This group sets conditions by order parameters. The conditions are used by triggers from the Positions section and by the "Scheduled account database processing" trigger.

- Login the number of the account on which the order was placed.
- Ticket the unique order ticket.
- Order ID order identifier in the external system.
- Symbol the financial instrument for which the order was placed.
- Setup time time of order placing by a client.
- Days since setup the number of days that have elapsed since the client placed the order. The calculated number of days is rounded down. For example, if the order was placed 3 days and 22 hours ago, this will be counted as 3 days.
- Expiration time order expiration date, if it was set by the client.

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- **Type** order type: "Buy", "Sell", "Buy Limit", "Sell Limit", "Buy Stop", "Sell Stop", "Buy Stop Limit", "Sell Stop Limit" or "Close By".
- **Order price** price specified by trader for execution of the order.
- **Trigger price** this field is used for the "Buy Stop Limit" and "Sell Stop Limit" orders. It sets the price level at which the orders trigger and the relevant limit orders are placed.
- **Stop Loss** the Stop Loss level.
- **Take Profit** the Take Profit level.
- **Initial volume** volume requested in the order.
- **Remained volume** if the order is not filled in the volume requested by trader this field will display the remainder volume.
- **State** current state of the order. The end states of orders (rejected, filled, etc.) are not supported since automation only works with a database of open/active orders.
- **Dealer** login of the dealer who confirmed or placed the order.
- **Expert ID** identifier (magic number) of an Expert Advisor which placed the order in the client terminal.
- **Position** ticket of the position opened, modified or closed due to this order.
- **Comment** a text comment to the order.
- **Contract size** the contract size of the symbol, for which an order was placed.
- **Currency** the deposit currency of the client who has placed the order.

Messages

These settings specify conditions by email parameters. These conditions are used with triggers from the Messages section.

- **Sender login** the account number of the user who sent the email.
- **Sender name** the name of the user who sent the email.
- **Recipient login** the account number of the email recipient.
- **Recipient name** the name of the email recipient.
- **Subject** email subject.
- **Body** email body including the HTML markup or the contents of the message sent via the SMS provider or instant messenger.
- **Address** recipient's email address or phone number.
- **Provider name** the configuration name of the SMS provider, messenger or mail server via which the message was sent.
- **Provider balance** remaining balance with your SMS provider. By using this property, you can prevent message sending issues by promptly sending notifications to administrators prompting them to top up the balance.

KYC

These settings specify conditions by KYC check parameters. Used with triggers from the KYC section.

- 'Status description' is the client verification result, such as approved, rejected, etc.

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- 'Rejection reason' is the description of the reason why KYC providers rejected the client data, such as inappropriate document files, duplicates, or other reasons.

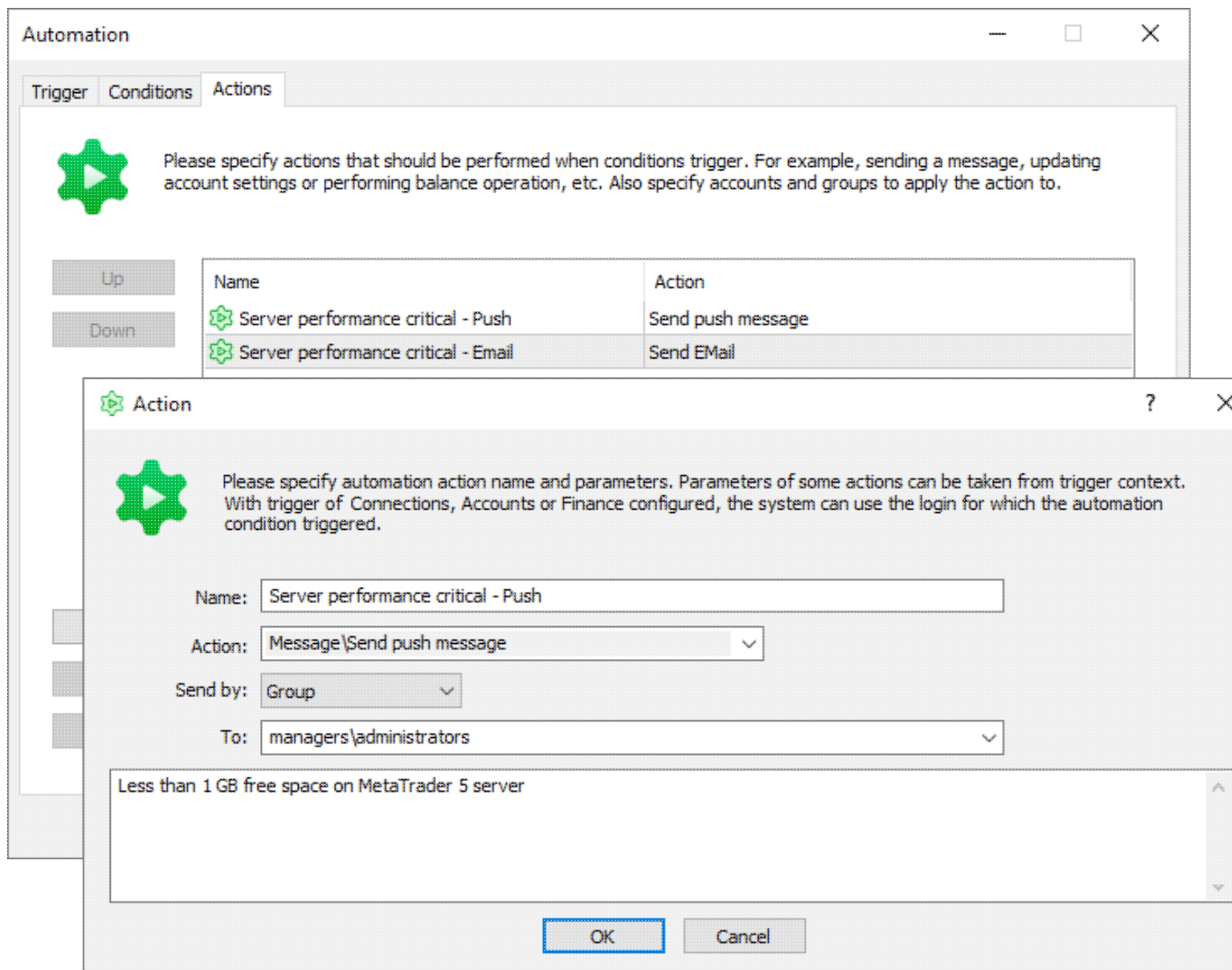
Possible statuses and rejection reasons are provided in the table.

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Actions Performed by Automations

After configuring the conditions under which the automation task should trigger, describe the actions that should be performed by this automation task.

Multiple actions can be specified for each task. They will be executed in the specified order, from top to bottom. If any action fails, execution of further actions is not terminated.



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The available actions are divided into categories, in accordance with the objects to which they apply.

- Message send a notification to users.
- Account manage account balances.
- Finance execute balance operations on accounts.
- Trade execute trading operations on accounts.
- Config change platform settings.
- Platform platform service actions.
- External platform integration with external systems.

The accounts to which actions apply, are indicated in the "Select by"/"Send by" field.

The availability of actions in the list is determined by the selected trigger. If actions are not associated with the event for which the task is being configured, such actions are automatically hidden. This makes setup easier while reducing potential errors. For example, for database processing triggers, the system does not allow the selection of actions related to platform configuration changes.

Accounts to which actions apply

The accounts to which the actions will be applied are indicated in the "Select by"/"Send by" field. Three login specification options are available for actions from categories "Message", "Account" and "Finance":

- **Login** one or more accounts separated by commas. Value ranges can also be specified here, for example: 1006-1019. It is possible to specify both an enumeration and a range, for example: 100,102,107,120-130.
- **Group** one or more groups or group masks (**all accounts** from specified groups).
- **Trigger** the action is only applied on the account on which the trigger fired. For example, if the "Login" event is specified as the trigger, the connected account will be used as the action object. This option is unavailable if a trigger event is not related to actions on the trading account. For example, "Scheduled event", "Price gap started", etc.

For actions from other categories, only the "Login" and "Group" options are available.

Message

These actions are intended to inform users:

- **Send push message** sends a notification to a mobile device via MetaQuotes ID.
- **Send internal mail** sends an email via the internal mail system. The email sender can be specified for this action, in the "From" field. The message recipient will information about the sender and will be able to send him a reply. Only the managers with a specified mailbox and appropriate permissions are shown in the list that opens in the sender selection field.
- **Send SMS** sends an SMS. The feature requires an appropriate SMS service integration configuration.

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- Send Email sends an email message. The feature requires an appropriate mail service integration configuration.
- Post in messenger channel sends a message to the selected channel via a messenger. Using this action, you can promptly receive monitoring data, notifications about large financial transactions, manager connections in the platform, and other information directly on your phone. The feature requires an appropriate messenger integration configuration.
- Send to journal prints a message to the journal of the trading server on which the automation task is running. Use this action for additional control over the platform and automation tasks.

The following data from appropriate accounts will be used to send a message:

- For push messages: MetaQuotes IDs specified in accounts.
- For internal emails: logins.
- For SMS: phone numbers specified in accounts.
- For emails email addresses specified in accounts.

You can use macros in messages. They allow substituting different data depending on a recipient (account) and a triggered event.

Trading account

These actions enable management of trading account settings:

- Disable, Enable manage account enabling/disabling options.
- Disable trade, Enable trade manage account trading permissions.
- Disable experts, Enable experts manage the permission to trade using Expert Advisors on the relevant account.
- Disable trailings, Enable trailings manage the permission to use a trailing stop in client terminals.
- Disable reports, Enable reports manage the permission to receive daily reports.
- Disable sponsored VPS, enable sponsored VPS manage the availability of the sponsored VPS option. These actions expand standard scenarios in which you can provide VPS to traders. For example, by using these actions with the "Deposit" trigger, you can provide the sponsored VPS option to the client when they deposit a certain amount to their account.
- Force to change password enables forced password change during the next account connection.
- Change group moves an account to the specified group. Please note that accounts can be moved between groups with different deposit currency only if their balance is zero and they have no open positions. Accounts cannot be moved between groups which belong to different trade servers.
- Change color changes the account color to the specified one.
- Move to archive moves an account to archive.
- Comment, Client comment adds a comment to an account or to a client record to which the account is linked.
- Change leverage changes the account leverage to the specified one. By using this action with the account balance condition, you can encourage traders to

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make larger deposits. For example, you can set the "Login" trigger, conditions "Balance > 100 000" and "Group = real*", "Action = Change leverage to 1:500". Clients with large deposits will have lower margin requirements and more trading opportunities. Accompany it with another rule, which will automatically reduce the leverage in the case of a balance drop.

Finance

These actions allow performing balance operations on accounts:

- **Deposit** a regular deposit or withdrawal from the account balance. An additional Type field is available for this action. You can specify here the deal type by which the operation will be executed: balance, correction, commission, etc.
- **Bonus** adding or deducting a bonus using a Bonus type deal.
- **Bonus, %** adding or deducting a bonus using a Bonus type deal. The action is only used with the "Deposit" and "First Deposit" triggers. It allows adding bonuses to a trader account depending on the deposit amount. For this action, specify in the "Amount" field the bonus as a percentage of the deposit amount.
- **Credit** adding or deducting credit funds using a "Credit" type deal.
- **Deposit pay off** writes off the account balance. The action will be useful when closing trading accounts. An additional Type field is available for this action. You can specify here the deal type by which the operation will be executed: balance, correction, commission, etc.
- **Credit pay off** writes off the account credit funds. Use these rules to automate promotions for new clients. For example, you can withdraw credit funds from the traders account when a month passes since the client's registration.
- **Bonus pay off** eliminates all bonus funds from the account. This action can be utilized to automate promotions for new clients. For example, you can remove bonuses from the account if the trader does not use them within a certain time.

Indicate a positive amount to add funds and a negative value to deduct the appropriate amount.

Trading

These actions allow conducting trading operations on accounts:

- **Close positions** closes positions on specified accounts.
- **Close orders** cancels active pending orders on specified accounts.
- **Clear Stop Loss / Take Profit / Stop Loss and Take Profit** delete the relevant activation levels of pending orders and open positions. Use these actions to implement your own Take Profit and Stop Loss expiration rules. For example, when combined with scheduled database processing triggers, they can clear all levels at the end of a trading day or week. Use conditions to additionally filter positions and orders to which the action will be applied. For example, you can set a specific list of symbols or groups.

As an object to which the action will be applied, you can select:

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- Position from trigger. This option is available for the "Scheduled position database processing" trigger. In this case, the action is applied to all positions which match the conditions specified for the automation task. For example, the trigger is combined with the condition "Position\Days since creation > 365". By selecting the "Close positions" action and the "Position from trigger" object, you will close all positions that were opened more than a year ago.
- Order from trigger. This option is available for the "Scheduled order database processing" trigger. In this case, the action is applied to all orders which match the conditions specified for the automation task. For example, the trigger is combined with the condition "Order\Days since Setup > 365". By selecting the "Close orders" action and the "Order from trigger" object, you will close all orders that were placed more than a year ago.
- Position or order from trigger available if the trigger is set to Web callback. In this case, the action is applied to the position or order passed in the request. Further details are provided in the Triggers section.
- Logins from trigger, specified list of logins or groups similar to any other action. The relevant description is provided in the section "Accounts to which actions apply".

Additional parameters should be specified for these actions:

- Type type of trading operations the actions will be applied to: position direction or pending order type.
- Symbol a list of trading instruments to which the action will be applied. A comma separated list of symbols or symbol masks can be specified here.
- Price only used for the "Close positions" action. Specify the price at which the positions will be closed.
- Comment a comment to be added to position closing operations (exit deals) or to canceled orders.
- Reason operation execution reason to be specified in position closing deals. This parameter enables more accurate details in the database of deals. For example, if your automation task closes positions on Stop Out, then you can specify the relevant reason in the deals. The default reason is "Dealer".

Configuration

These actions enable management of platform settings:

- Update groups configuration
- Update symbol configuration
- Move symbol configuration
- Update routing configuration
- Update server configuration
- Update gateway configuration
- Update data feed configuration

Two methods for describing required configurations are supported:

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- The Add command creates the full description of the configuration. The list includes all parameters, including those that remain unchanged. When the rule triggers, the new configuration will completely overwrite the existing one (or it will add a new configuration).
- The Add Diff command only creates a description of the parameters to be changed. When the rule triggers, the changes will be applied to an existing configuration.

Selection of any of the commands will open a corresponding configuration editing dialog:

Action

Please specify automation action name and parameters. Parameters of some actions can be taken from trigger context. With trigger of Connections, Accounts or Finance configured, the system can use the login for which the automation condition triggered.

Name: Update group settings

Action: Config\Update group config

Load Save

Parameter	Value
demo\demo forex	
Enable connections	Yes
Change password at first login	Yes
Show risk warning at login	Yes

Group: demo\demo forex

Common Company News & Mail Permissions Margin Symbols Commissions Reports

Group is a set of users that have the same permission settings and service conditions. Please specify name of group, deposit currency, trade server, and authentication type.

Name: demo\demo forex Currency: USD

Trade server: Trade Main, 1 Digits: 2

Authentication: Normal Minimum password length: 8

One-time password: Disabled ☐ Force one-time password usage

☒ Enable connections
☐ Enable certificate confirmation
☒ Change password at first login
☒ Show the risk warning window after connection
☐ Enforce country-specific regulatory restrictions for retail clients

OK Cancel Help

Change the parameters as needed and click OK. All changed parameters and their new values will be displayed in a list.

Next, add other configurations as needed.

Optionally, you can use settings files in JSON format. For example, you can save several symbol settings in advance and then add them to automation tasks using the Load command.

All settings must be specified in JSON files, not only the ones that you want to change. Export settings, make desired amendments and use them in automation tasks.

Action "Move symbol configuration" enables automatic regrouping of trading instruments. For this action, specify the path for the symbol in the field below. The symbol to be moved is identified by the name from the new path. For example, if you specify "Forex\Majors\EURUSD", the EURUSD symbol will be moved to the "Forex\Majors\" subgroup. If you specify "Delayed\EURUSD", the EURUSD symbol will be moved to the "Delayed\" subgroup.

Bulk Configuration Changes

Automation tasks allow you to change symbol and group settings in bulk. Select the appropriate action and specify the changes with the "Difference" command. Next, in the "Apply to" parameter, specify the list of groups or symbols to which the specified changes should be applied.


Action
?
✕



Please specify automation action name and parameters. Parameters of some actions can be taken from trigger context. With trigger of Connections, Accounts or Finance configured, the system can use the login for which the automation condition triggered.

Name:

Action:

Configuration\Update group configuration

▼

Load

Save

Add

Add Diff

Edit

Delete

Parameter	Value
Apply to	real\usd-standard,real\usd-mini,real\eur-s
▼  real\usd-standard	
Enable trading by Expert Advisors	

OK

Cancel

+

✕


managers

+

✕


demo

+

✕


preliminary

+

✓


real

✓


usd-standard

✓


usd-mini

✓


eur-standard

✕


eur-mini

✕


abook-usd

✕


extended

+

✕


contest

+

✕


coverage

+

✕


funds

✕


preliminary

✕


preliminary-usd

Platform

These actions enable platform management.

- Restart server restarts the server with the specified ID.
- Restart datafeed restarts the specified datafeed.
- Restart gateway restarts the specified gateway.
- History synchronization launches price data synchronization in accordance with the specified configurations.
- Live Update initiates checking for platform updates and starts installation if an update is found. This action can be combined with the "Scheduled Event" trigger to enable automatic checks and installation of platform updates on weekends.

By using these actions combined with the "Scheduled event" trigger, you can automatically execute preventive actions at a specified frequency. For example, you can automatically restart gateways and datafeeds once a week, on Sunday nights.

External

These actions will integrate your MetaTrader 5 platform into external systems. By using them, you can:

- Send standard HTTP requests to any URL
- Send events to the Finteza analytics system
- Run external files and scripts on the server where the platform is installed

Sending Web Requests

When specific events occur in the platform, standard HTTP GET and POST requests can be sent to desired addresses. The new possibility enables platform integration with any web services supporting REST API. For example, you can automatically send information about opened accounts to traders' web accounts and to your CRM systems.

In the action setup dialog, select "External \ Web request" and specify request parameters: GET or POST.

GET Requests

When using a GET method, information about an event is passed as request arguments. Therefore, such requests should only be used to transfer a small amount of event data.

After selecting this request type, specify the following parameters:

- URL the address to which the request will be sent, and additional arguments.
- Headers additional headers for the request in the "key: value" format. For example, "Content-Type: application/x-www-form-urlencoded".

Action

Please specify automation action name and parameters. Parameters of some actions can be taken from trigger context. With trigger of Connections, Accounts or Finance configured, the system can use the login for which the automation condition triggered.

Name:

Action:

Method:

URL:

Headers:

By default, the system inserts macros to URL parameters, allowing to pass information about the triggered event:

- Automation rule name
- Action name
- Trigger time
- Trigger type

```
https://?triggerName=#TRIGGER_NAME#&tiggerAction=#ACTION_NAME#&tiggerTime=#
```

The following macros are added additionally added to URL parameters for appropriate triggers:

- The user login for which the event triggered is passed for triggers
"Connections", "Accounts", "Finance", "Trade": login=#LOGIN#
- The transaction amount is added for "Finance" triggers:
amount=#ACTION_FINANCE_AMOUNT#
- The symbol name is added for "Prices" triggers: symbol=#SYMBOL#

Add to the URL a default address, to which the request will be sent. The address should be specified between characters // and ?. You can also add your own parameters to the string, separated by &.

POST Requests

Such requests accept an additional body in JSON format. They provide a convenient way to pass an extended event description.

After selecting this request type, specify the following parameters:

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- URL the address to which the request is sent.
- Headers additional headers for the request in the "key: value" format. For example, "Content-Type: application/x-www-form-urlencoded".
- Data an additional body of a POST request in JSON format.

 Action ? ✕



Please specify automation action name and parameters. Parameters of some actions can be taken from trigger context. With trigger of Connections, Accounts or Finance configured, the system can use the login for which the automation condition triggered.

Name:

Action:

Method:

URL:

Headers:

Data:

```
{
  "trigger": {
    "task": "#TRIGGER_NAME#",
    "action": "#ACTION_NAME#",
    "time": "#DATETIME#"
  },
  "platformPerformanceMonitor": {
    "cpu": #MONITOR_CPU#,
    "cpuProcess": #MONITOR_CPU_PROCESS#,
    "cpuProcessThreads": #MONITOR_CPU_PROCESS_THREADS#,
    "memoryFree": #MONITOR_MEMORY_FREE#,
    "memoryProcess": #MONITOR_MEMORY_PROCESS#,
    "diskFree": #MONITOR_DISK_FREE#,
    "diskSpeedRead": #MONITOR_DISK_SPEED_READ#,
    "diskSpeedWrite": #MONITOR_DISK_SPEED_WRITE#,
    "diskQueueLength": #MONITOR_DISK_QUEUE_LENGTH#,
    "networkConnections": #MONITOR_NETWORK_CONNECTIONS#,
    "networkBlocked": #MONITOR_NETWORK_BLOCKED#,
```

For convenience, the system automatically substitutes additional parameters in the JSON body of the request, according to the selected trigger. For example, if you configure a task related to a system monitoring event, then macros displaying measurement results will be substituted into the request. Account login macros are automatically inserted for connection triggers, login and operation amount macros are inserted for deposit operations, and so on. You can also add any other macros to requests.

Regardless of the trigger type, the following data is added to the request body by default:

- Automation rule name
- Action name
- Trigger time

```
"trigger": {  
  "task": "#TRIGGER_NAME#",  
  "action": "#ACTION_NAME#",  
  "time": "#DATETIME#"  
}
```

Sequence of Requests

The system uses one connection for all requests to the same domain. Due to this, if you set multiple tasks which send web requests, the system:

- Guarantees that requests to the same domain are executed in the same sequence, in which the events trigger
- Does not guarantee that requests to different domains are executed in the same sequence, in which the events trigger

For example, you have created tasks which execute web requests to the following addresses:

- https://example1.com/xxx
- https://example1.com/zzz
- https://example2.com/xxx
- https://example2.com/zzz

Requests to http://example1.com/xxx and https://example1.com/zzz will be executed sequentially, in the order of occurrence of associated events, regardless of request execution speed. The same will happen for https://example2.com/xxx and https://example2.com/zzz.

If requests for domain example1.com are executed slower (for example, due to slow server response), and requests for example2.com are executed faster, then the execution order may not match the order in which the events triggered.

Sending Events to Finteza

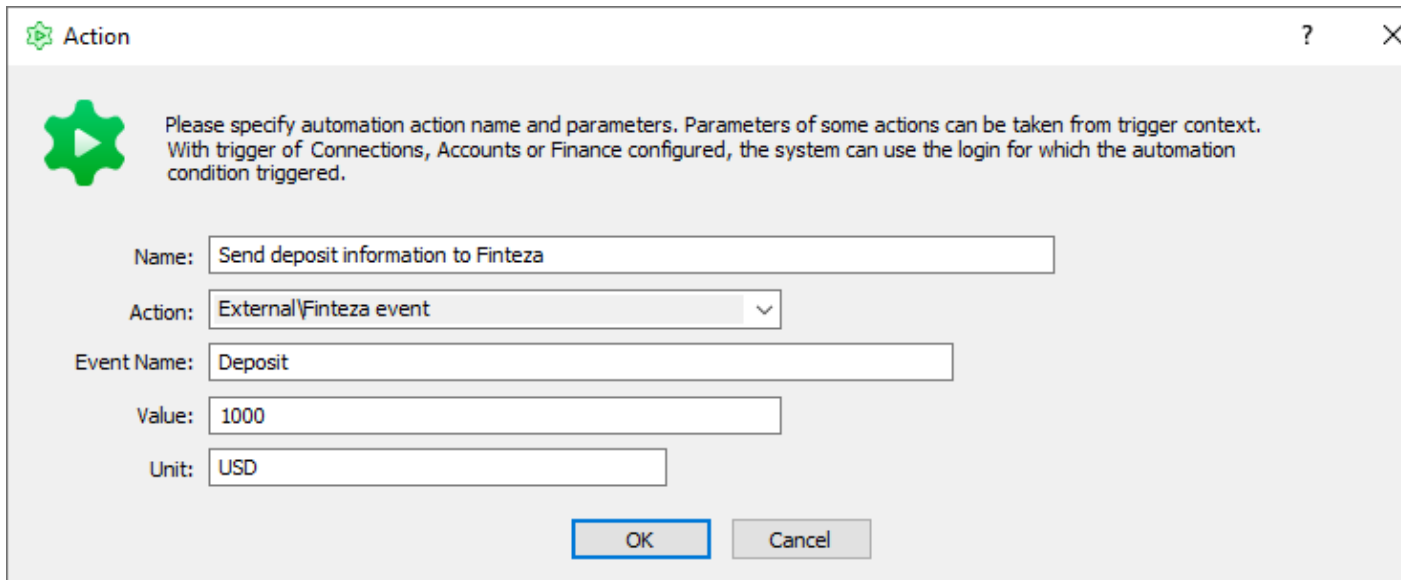
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MetaTrader 5 integration with the Finteza business analytics system enables analysis of audiences and advertising campaign efficiency, as well as tracking of client life cycles, from the first account visit to the first real account deposit.

This allows expanding the standard set of events. By using Finteza with the Automations service, you can easily track a plethora of other important platform metrics, such as account connections, deposit operations and other events.

Select "External \ Finteza event" in the action settings dialog, and specify the following parameters:

- **Event name** the name of the occurred event which will be sent to Finteza. We recommend specifying meaningful names, as they will be used in reports.
- **Value** the value of the parameter, for example, the deposit amount.
- **Units** parameter measurement units, for example, USD, items, etc.



Action

Please specify automation action name and parameters. Parameters of some actions can be taken from trigger context. With trigger of Connections, Accounts or Finance configured, the system can use the login for which the automation condition triggered.

Name: Send deposit information to Finteza

Action: External\Finteza event

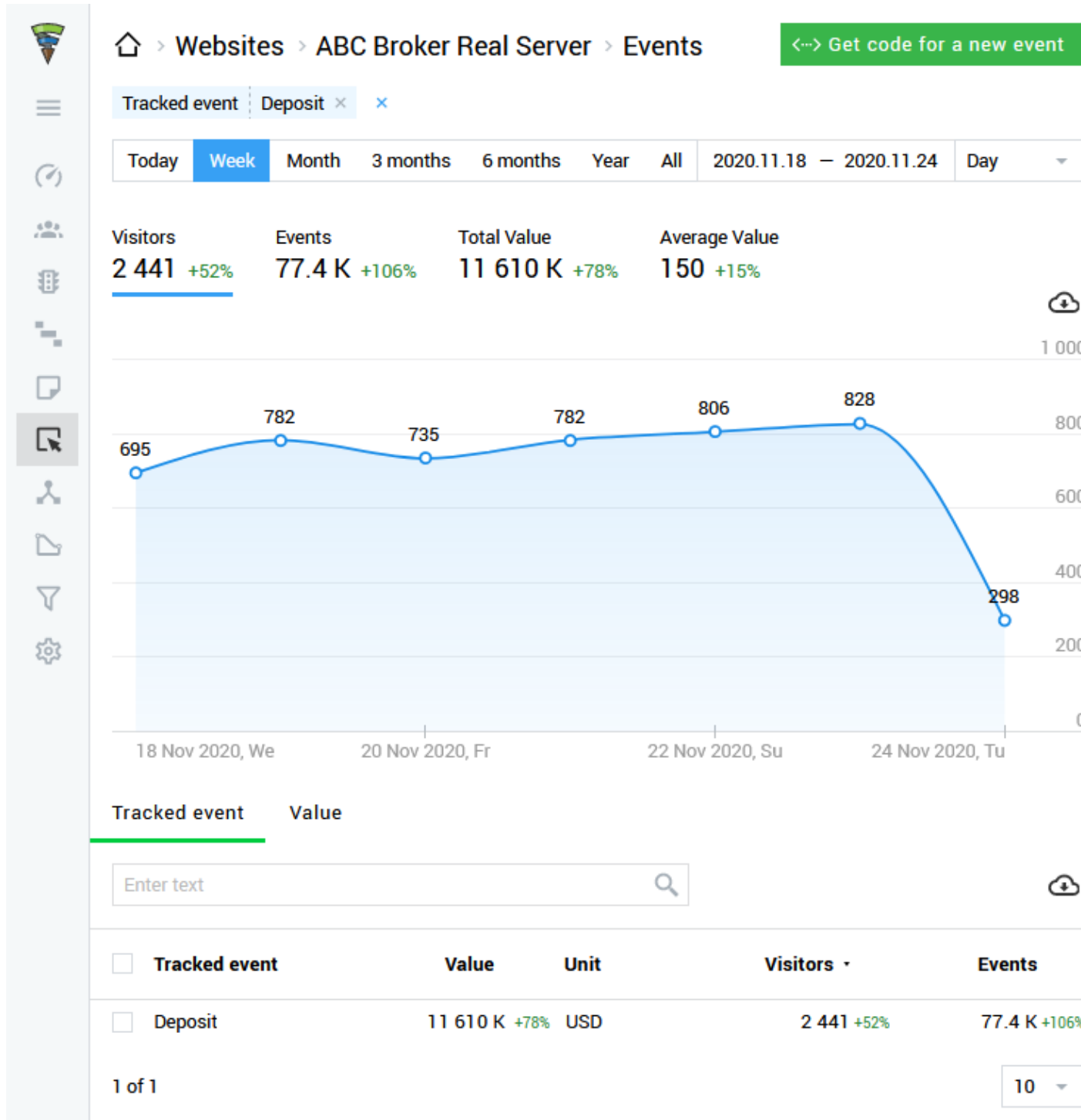
Event Name: Deposit

Value: 1000

Unit: USD

OK Cancel

The events will be collected to a website tracker specified under "Integrations \ Finteza". To view the events, open Finteza Panel, select the required tracker and navigate to "Events".



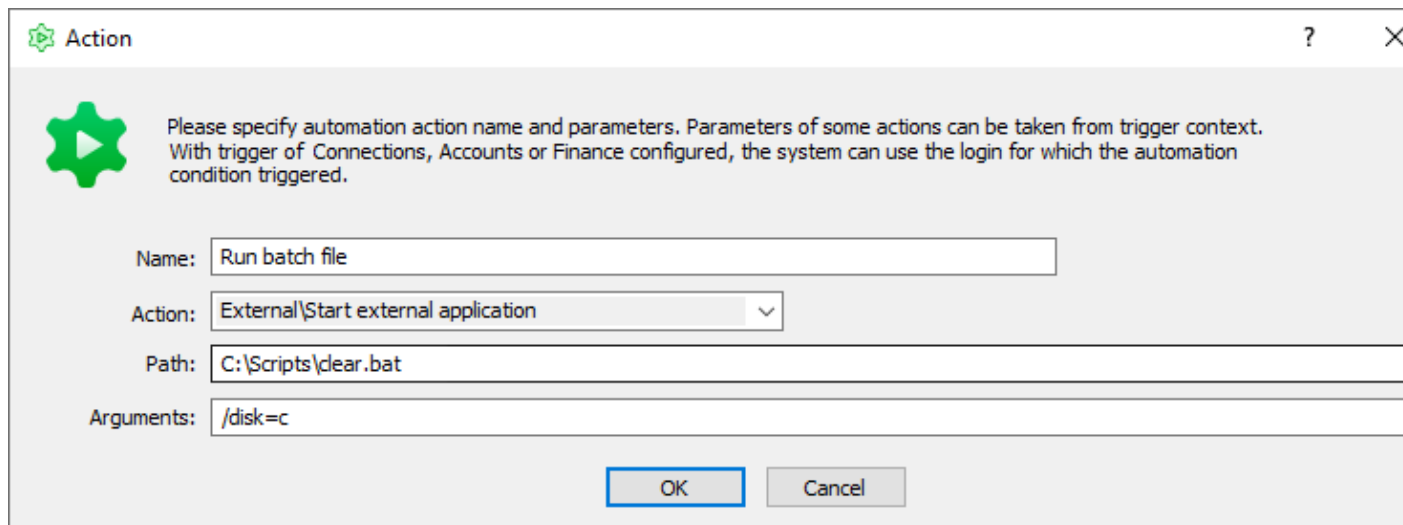
For further details about events, please read the article [How to configure events in Finteza](#).

Running External Applications

This action allows running external applications and script files on the machine where the platform's main trade server is installed. For example, when the free disk space drops below a specified threshold, you can run a BAT file which will empty temporary folders and the system's Recycle Bin.

Select "External \ Start external application" in the action settings dialog, and specify the following parameters:

- **Path** the path to the file to run, including the extension. A file can only be run on the machine where the platform's main trade server is installed. You can specify an absolute path or a path relative to the directory where the server executable file mt5trade64.exe is located. For example: "C:\Scripts\clear.bat" or "..\..\Scripts\clear.bat" (go up to levels and navigate to the "Scripts" directory).
- **Arguments** command parameters with which the executable file will be launched.



Action

Please specify automation action name and parameters. Parameters of some actions can be taken from trigger context. With trigger of Connections, Accounts or Finance configured, the system can use the login for which the automation condition triggered.

Name: Run batch file

Action: External\Start external application

Path: C:\Scripts\clear.bat

Arguments: /disk=c

OK Cancel

Macros can be used in the path line and in the arguments line. For example, you can pass the account number for which the trigger was activated (#LOGIN#), or the deal ticket (#DEAL_TICKET#).

Verification of permissions for actions

Automation tasks cover a wide range of actions, from editing of platform settings to financial operations. All these actions are performed without verifying the permissions of the manager/administrator who creates the task. Permissions are only checked when the user accesses the automation configuration section. It means that a user having access to the automation section can execute any automation action with any available parameters.

Therefore, please be extremely careful when allocating automation configuration permissions as it provides access to critical platform functions.

Task loop control

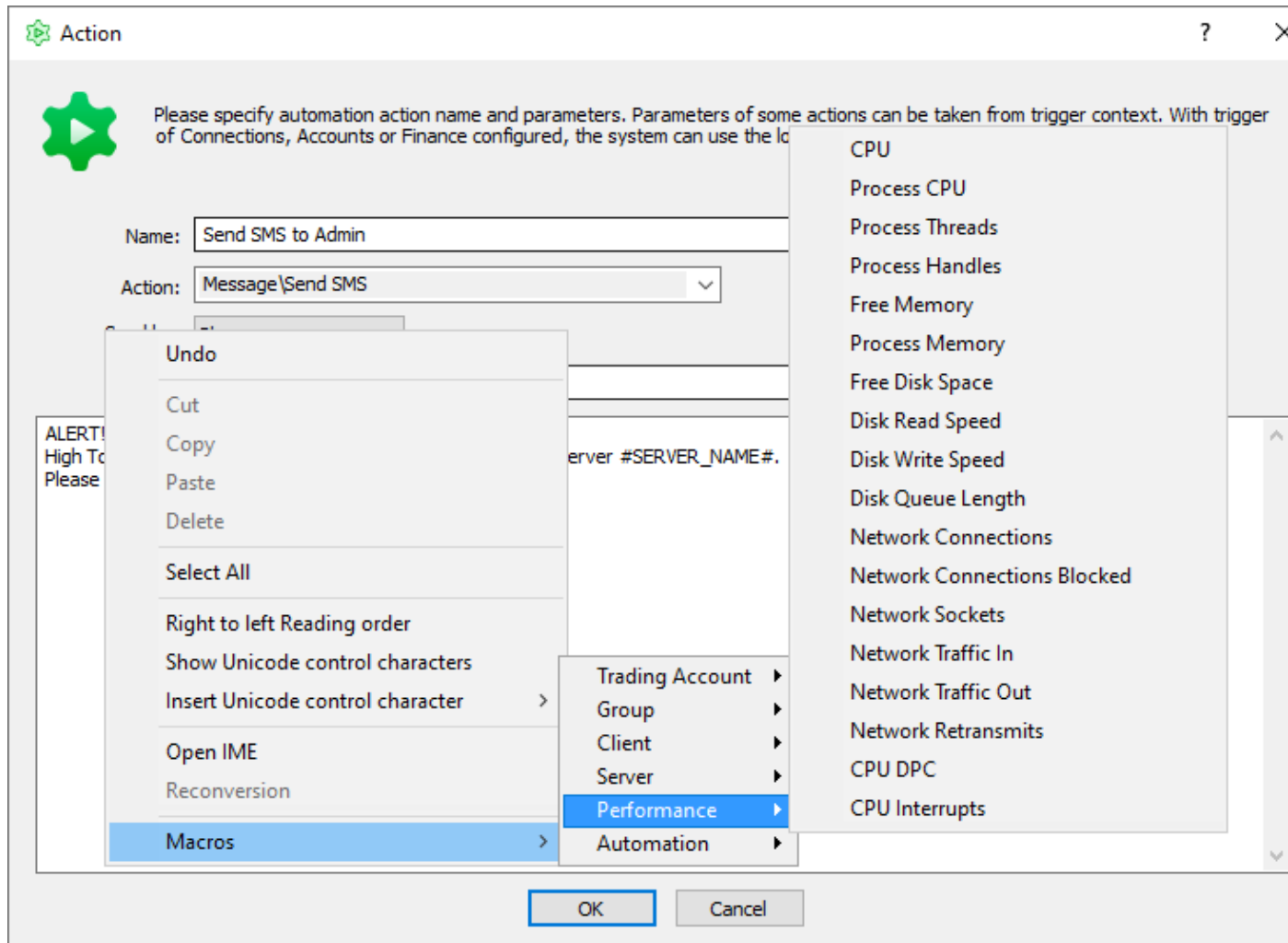
The platform does not control task looping if the task execution hits the trigger of another task. For example, if you set a chain of tasks "make a withdrawal when depositing, make a deposit when withdrawing", such a task will be executed infinitely.

Be sure to take into account possible looping and avalanche triggering when creating tasks. Pay attention to combinations of triggers from the Finance and Trade groups with actions from the corresponding groups.

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Macros

Automation actions for sending various messages (SMS, Push, emails) and web requests support macros. They allow substituting different data depending on a recipient (account) and a triggered event. For inserting, set the macro name between #..# or use the context menu.



All available macros are divided into several categories depending on what information they provide in the text.

Trading Account

These macros substitute message recipient's account data.

- Login (#LOGIN#) email/message recipient's account number.
- Full Name (#USERNAME#) recipient's first and last name.
- First Name (#USER_FIRST_NAME#) recipient's first name.
- Last Name (#USER_LAST_NAME#) recipient's last name.
- Middle Name (#USER_MIDDLE_NAME#) recipient's middle name.
- Currency (#USER_CURRENCY#) recipient's deposit currency. It is defined by the user's group.
- Company (#USER_COMPANY#) name of a company a user belongs to. It is defined by the user's group.
- Creation Date (#USER_CREATION_DATE#) account creation date.
- Registration date (#USER_REGISTRATION_ELAPSED#) the number of days that have elapsed since account creation.
- Last Access Time (#USER_LASTTIME#) last account access time.
- Days since last login (#USER_LASTTIME_ELAPSED#) the number of days that have elapsed since the last connection to the account.
- Online status (#USER_ONLINE_STATUS#) the status of account connection to the trade server: Online or Offline.
- Language (#USER_LANGUAGE#) user's language.
- Status (#USER_STATUS#) user status (resident/non-resident).
- ID Number (#USER_ID#) user's document ID.
- Email (#USER_EMAIL#) user's email.
- Group (#USER_GROUP#) user's group.
- Previous (#USER_GROUP_PREVIOUS#) the macro is only used with the "Trade account group change". Use it to get the group from which the account was transferred.
- Phone (#USER_PHONE#) user's phone number.
- Country (#USER_COUNTRY#) user's country.
- City (#USER_CITY#) user's city.
- State (#USER_STATE#) user's region.
- Zip Code (#USER_ZIP_CODE#) user's zip code.
- Address (#USER_ADDRESS#) user address.
- Comment (#USER_COMMENT#) comment to a user account.
- Bank Account (#USER_BANK_ACCOUNT#) bank account number specified in the account.
- Agent (#USER_AGENT_ACCOUNT#) agent account number specified in the account.
- Balance (#USER_BALANCE#) recipient's balance.
- Credit (#USER_CREDIT#) credit funds on the recipient's account.
- Equity (#USER_EQUITY#) account equity.
- Leverage (#USER_LEVERAGE#) account leverage.
- Margin (#USER_MARGIN#) amount of funds reserved on the account as margin.
- Free Margin (#USER_MARGIN_FREE#) amount of free margin.
- Margin Level (#USER_MARGIN_LEVEL#) margin level.

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- Position Total (#USER_POSITIONS_TOTAL#) number of open positions on the account when sending a message.
- Orders Total (#USER_ORDERS_TOTAL#) current number of active pending orders on the account when sending a message.
- Profit (#USER_PROFIT#) floating profit on the account when sending a message.
- Lead Source (#USER_LEAD_SOURCE#) lead source specified in the account.
- Lead Campaign (#USER_LEAD_CAMPAIGN#) marketing campaign which led to opening the account.
- Color (#USER_COLOR#) the color by which the client's requests to execute trading operations will be displayed in the Manager terminal.
- Full JSON (#USER_JSON#) complete data on user account in JSON format.
- Own funds percentage (#USER_OWN_FUNDS_PERCENT#) client's funds share consisting of money the client deposited to their account. The calculation equation: $(1 - (\text{Credit} / \text{Equity})) * 100$. Find out more in the [Conditions](#) section.
- Own funds volume (#USER_OWN_FUNDS_VOLUME#) client's funds volume consisting of money the client deposited to their account. The calculation equation: $\text{Equity} - \text{Credit}$.
- IP address (#USER_IP_ADDRESS#) address connection to the trade server was established from. Use the macro with triggers from the [Connections](#) section.
- Client ID (#CLIENT_ID#) the unique identifier of the [client](#) with whom the trading account is associated.

Manager

These macros are used in a text to substitute the [data of a manager account](#) with which the event is associated. Used only with triggers from the [Managers](#) section.

- Login (#MANAGER_LOGIN#) the account of the manager who performed the action.

Group

These macros substitute target account [group data](#).

- Name (#GROUP_NAME#) group name.
- Currency (#GROUP_CURRENCY#) group deposit currency.
- Company (#GROUP_COMPANY#) [name of a company](#) a group belongs to.
- Email (#GROUP_EMAIL#) company email.
- Support Email (#GROUP_EMAIL_SUPPORT#) company technical support team email.
- Site (#GROUP_SITE#) company website.
- Support Site (#GROUP_SITE_SUPPORT#) company support website.
- Margin Call (#GROUP_MARGIN_CALL#) [margin call level](#) set for the group.
- Stop Out (#GROUP_STOP_OUT#) stop out level set for the group.
- Full JSON (#GROUP_JSON#) full group description in JSON format.

Client

These macros substitute recipient's [client data](#).

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- Title (#CLIENT_TITLE#) title: Mr. or Mrs.
- First Name (#CLIENT_FIRST_NAME#) client's first name.
- Last Name (#CLIENT_LAST_NAME#) client's last name.
- Middle Name (#CLIENT_MIDDLE_NAME#) client's middle name.
- Gender (#CLIENT_GENDER#) client's gender.
- Birthday (#CLIENT_BIRTHDAY#) client's birthday.
- Language (#CLIENT_LANGUAGE#) client's language.
- Email (#CLIENT_EMAIL#) client's email.
- Phone (#CLIENT_PHONE#) client's phone number.
- Messengers (#CLIENT_MESSENGERS#) client's messenger list.
- External ID (#CLIENT_EXTERNAL_ID#) client ID in an external system.
- Country (#CLIENT_COUNTRY#) client's country.
- State (#CLIENT_STATE#) client's region.
- City (#CLIENT_CITY#) client's city.
- Zip Code (#CLIENT_ZIP_CODE#) client's zip code.
- Address (#CLIENT_ADDRESS#) client's address.
- Nationality (#CLIENT_NATIONALITY#) client's citizenship.
- Tax ID (#CLIENT_TAX_ID#) client's tax ID.
- Type (#CLIENT_TYPE#) client type.
- Status (#CLIENT_STATUS#) client status.
- Manager (#CLIENT_MANAGER#) responsible manager login.
- Lead Source (#CLIENT_LEAD_SOURCE#) lead source specified in client record.
- Lead Campaign (#CLIENT_LEAD_CAMPAIGN#) marketing campaign which led to creating the client.
- Introducer (#CLIENT_INTRODUCER#) login (trading account) of the user, who introduced the client.
- Full JSON (#CLIENT_JSON#) full client description in JSON format.

Position

These macros substitute in a text data of positions, a change in which caused the automation task trigger (triggers from the "Positions" section).

- Login (#POSITION_LOGIN#) the number of the account the position as opened on.
- Ticket (#POSITION_TICKET#) unique position identifier.
- Symbol (#POSITION_SYMBOL#) symbol the position was opened for.
- Volume (#POSITION_VOLUME#) position volume in lots.
- Type (#POSITION_TYPE#) position type: buy or sell.
- Open price (#POSITION_PRICE_OPEN#) weight-average price of the position opening: $(\text{price of deal 1} * \text{volume of deal 1} + \dots + \text{price of deal N} * \text{volume of deal N}) / (\text{volume of deal 1} + \dots + \text{volume of deal N})$.
- Current price (#POSITION_PRICE_CURRENT#) current price of the financial symbol for which the position has been opened, at the trigger activation time.
- Profit (#POSITION_PROFIT#) profit of a position at the trigger activation time.
- Reason (#POSITION_REASON#) reason for opening the position.
- Creation time (#POSITION_TIME_CREATE#) position opening time.
- Update time (#POSITION_TIME_UPDATE#) time when the position was last modified (when its volume was changed).

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- Dealer (#POSITION_DEALER#) the login of the dealer who confirmed or placed the order (deal), which caused the position to open.
- Expert (#POSITION_EXPERT#) identifier (magic number) of an Expert Advisor by which a position was opened in the client terminal.
- Comment (#POSITION_COMMENT#) a text comment to a position.
- Storage (#POSITION_STORAGE#) swap charges.
- Stop Loss (#POSITION_PRICE_SL#) position Stop Loss level
- Take Profit (#POSITION_PRICE_TP#) position Take Profit level.
- Profit rate (#POSITION_RATE_PROFIT#) profit currency to deposit currency conversion rate.
- Margin rate (#POSITION_RATE_MARGIN#) margin currency to deposit currency conversion rate.
- Full JSON (#POSITION_JSON#) full description of the position in JSON format.

Deal

These macros substitute in a text data of a deal, an execution of which caused the automation task trigger (triggers from the "Positions" section).

- Login (#DEAL_LOGIN#) the number of the account, on which the deal was executed.
- Ticket (#DEAL_TICKET#) unique deal identifier.
- Symbol (#DEAL_SYMBOL#) financial instrument for which the deal was executed.
- Volume (#DEAL_VOLUME#) deal volume in lots.
- Type (#DEAL_TYPE#) deal type: Sell, Buy, Balance etc.
- Direction (#DEAL_ENTRY#) direction of a deal relative to the current position: entry ("in"), exit ("out"), reversal ("in/out") or close by ("out by").
- Price (#DEAL_PRICE#) deal execution price.
- Profit (#DEAL_PROFIT#) the profit received from the deal.
- Reason (#DEAL_REASON#) reason for deal execution.
- Time (#DEAL_TIME#) deal execution time.
- Dealer (#DEAL_DEALER#) the login of the dealer who confirmed or placed the order, which caused the deal to execute.
- Expert (#DEAL_EXPERT#) identifier (magic number) of an Expert Advisor by which the deal was executed in the client terminal.
- Comment (#DEAL_COMMENT#) a text comment to a deal.
- Commission (#DEAL_COMMISSION#) commission charged for deal execution.
- Fee (#DEAL_FEE#) Fee commission type.
- Storage (#DEAL_STORAGE#) swap charges.
- Closed volume (#DEAL_VOLUME_CLOSED#) position volume closed by this deal.
- Stop Loss (#DEAL_PRICE_SL#) deal Stop Loss level.

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- Take Profit (#DEAL_PRICE_TP#) deal Take Profit level.
- Raw profit #DEAL_PROFIT_RAW# profit/loss obtained from the execution of the deal, in the instrument's profit currency.
- Profit rate (#DEAL_RATE_PROFIT#) profit currency to deposit currency conversion rate.
- Margin rate (#DEAL_RATE_MARGIN#) margin currency to deposit currency conversion rate.
- Market Bid (#DEAL_PRICE_MARKET_BID#) market Bid price as at the time the deal is executed by the server.
- Market Ask (#DEAL_PRICE_MARKET_ASK#) market Ask price as at the time the deal is executed by the server.
- Market Last (#DEAL_PRICE_MARKET_LAST#) market Last price as at the time the deal is executed by the server.
- Full JSON (#DEAL_JSON#) full description of the deal in JSON format.

Orders

These macros are substitute parameters of the order, with which the automation task triggering is associated ("Scheduled account database processing").

- Ticket (#ORDER_TICKET#) the unique order number.
- External order ID (#ORDER_ORDER_ID#) order identifier in an external trading system.
- Login (#ORDER_LOGIN#) the number of the account on which the order was placed.
- Symbol (#ORDER_SYMBOL#) the financial instrument for which the order was placed.
- Setup time (#ORDER_TIME_SETUP#) time of order placing by a client.
- Days since setup (#ORDER_TIME_SETUP_ELAPSED#) the number of days that have elapsed since the client placed the order.
- Done time (#ORDER_TIME_DONE#) order execution time.
- Expiration time (#ORDER_TIME_EXPIRATION#) order expiration date, if it was set by the client.
- Type (#ORDER_TYPE#) order type: "Buy", "Sell", "Buy Limit", "Sell Limit", "Buy Stop", "Sell Stop", "Buy Stop Limit", "Sell Stop Limit" or "Close By".
- Order price (#ORDER_PRICE_ORDER#) price specified by trader for execution of the order.
- Trigger price (#ORDER_PRICE_TRIGGER#) this field is used for the "Buy Stop Limit" and "Sell Stop Limit" orders. It sets the price level at which the orders trigger and the relevant limit orders are placed.
- Stop Loss (#ORDER_PRICE_SL#) the Stop Loss level.
- Take Profit (#ORDER_PRICE_TP#) the Take Profit level.
- Initial volume (#ORDER_VOLUME_INITIAL#) volume requested in the order.
- Remained volume (#ORDER_VOLUME_CURRENT#) if the order is not filled in the volume requested by trader this field will display the remainder volume.
- State (#ORDER_STATE#) current state of the order (filled, rejected, partially filled, expired, etc.).

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- Dealer (#ORDER_DEALER#) login of the dealer who confirmed or placed the order.
- Expert ID (#ORDER_EXPERT#) identifier (magic number) of an Expert Advisor which placed the order in the client terminal.
- Position (#ORDER_POSITION_ID#) ticket of the position opened, modified or closed due to this order.
- Comment (#ORDER_COMMENT#) a text comment to the order.
- Contract size (#ORDER_CONTRACT_SIZE#) the contract size of the symbol, for which an order was placed.
- Currency (#ORDER_CURRENCY#) the deposit currency of the client who has placed the order.
- Margin rate (#ORDER_RATE_MARGIN#) margin currency to deposit currency conversion rate.
- Full JSON (#ORDER_JSON#) full description of the order in JSON format.

Payments

These macros substitute data about an internal payment transaction performed on the account: Used only with the triggers from the Payments section.

- Action (#PAYMENT_ACTION#) the type of transaction: deposit or withdrawal.
- Amount (#PAYMENT_AMOUNT#) the transaction amount requested by the client. Specified in the deposit currency.
- Currency (#PAYMENT_CURRENCY#) the transaction currency.
- Commission (#PAYMENT_COMMISSION#) the commission charged by the broker in accordance with the settings.
- Wallet Amount (#PAYMENT_WALLET_AMOUNT#) the amount of the transaction on the side of the payment provider. It is specified in the currency selected by the user on the client terminal side when making the operation.
- Wallet Currency (#PAYMENT_WALLET_CURRENCY#) the currency of the transaction on the payment provider's side.
- Wallet Commission (#PAYMENT_WALLET_COMMISSION#) the payment provider's fee for the transaction.
- Description (#PAYMENT_DESCRIPTION#) additional information about the transaction. Can be filled in by payment systems or manually by managers.
- Manager (#PAYMENT_MANAGER#) the login and name of the manager who processed the transaction. It is filled only for manually processed transactions.
- Type (#PAYMENT_TYPE#) the payment method type: card, bank transfer, etc.
- Provider type (#PAYMENT_PROVIDER_TYPE#) the type of the payment provider (gateway).
- Provider name (#PAYMENT_PROVIDER_NAME#) the name of the payment provider configuration under which the payment was made.
- Client IP (#PAYMENT_CLIENT_IP#) the IP address from which the payment was requested.
- Client IP (#PAYMENT_CLIENT_TYPE#) the IP address from which the payment was requested.
- Deal ticket (#PAYMENT_CLIENT_DEAL#) the ticket of the balance operation through which funds are credited to or debited from the trading account.

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- External ID (#PAYMENT_EXTERNAL_ID#) the transaction identifier on the payment provider's side.
- External error code (#PAYMENT_EXTERNAL_ERROR_CODE#) the error code that occurred on the provider's side.
- External error description (#PAYMENT_EXTERNAL_ERROR_DESC#) a description of the error that occurred on the provider's side.
- Error code (#PAYMENT_ERROR_CODE#) the error code that occurred on the platform side.
- Error description (#PAYMENT_ERROR_DESC#) a description of the error that occurred on the platform side.

Finance

These macros substitute data about a balance operation executed on the account:

- Amount (#FINANCE_AMOUNT#) operation amount.
- Currency (#FINANCE_CURRENCY#) the currency in which the operation was performed.
- Comment (#FINANCE_COMMENT#) a text comment to a balance transaction.
- Manager (#FINANCE_MANAGER#) the login of the manager who performed the financial transaction.
- Action (#FINANCE_ACTION#) deal type: Buy, Sell, balance or credit operation, etc.

Server

These macros substitute target account server data.

- ID (#SERVER_ID#) server ID
- Type (#SERVER_TYPE#) server type as a string. For example, "Main Trade Server".
- Name (#SERVER_NAME#) server name specified in the configuration.
- Public name (#SERVER_PUBLIC_NAME#) public name of the broker's server (platform) which is displayed in client terminals.
- Connection state (#SERVER_CONNECTED#) the state of server connection to the main trade server at the time the trigger is activated. If the server is connected, the value is 'true'; otherwise it is 'false'.
- Current server ID (#SERVER_CURRENT_ID#) the identifier of the trade server on which the automation task was triggered. For further details please read the "Conditions" section.
- Current server type (#SERVER_CURRENT_TYPE#) the type of trading server on which the automation task was triggered: main or additional.
- Current server state (#SERVER_CURRENT_CONNECTED#) the state of the trade server on which the automation task was triggered. Available values: Not set the server does not reboot; Restart a regular reboot has been started, Stop the server service is stopped, LiveUpdate the server service is stopped due to a platform update.

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- Current server connected (#SERVER_CURRENT_STATE#) the state showing the connection between the trade server on which the automation task was triggered and the main server of the platform.

Gateway

These macros substitute data of a gateway, which the automation task is related with (Gateway connection/disconnection triggers).

- Gateway name (#GATEWAY_NAME#) the name of the gateway configuration.
- Gateway ID (#GATEWAY_ID#) the identifier specified in the gateway configuration.
- Connection status (#GATEWAY_CONNECTED#) the state of gateway connection to the platform at the trigger activation time. If the gateway is connected, 'true' is added; otherwise, false.

Data feed

These macros substitute data of a data feed, which the automation task is related with (Data feed connection/disconnection triggers).

- Data feed name (#DATAFEED_NAME#) the name of the data feed configuration.
- Connection status (#DATAFEED_CONNECTED#) the state of data feed connection to the platform at the trigger activation time. If the data feed is connected, 'true' is added; otherwise, false.

Prices

These macros substitute data of the symbol to which the automation task trigger is related ("Prices" triggers).

- Symbol (#PRICES_SYMBOL#) the name of the financial instrument for which the event triggered.
- Last tick time (#PRICES_LASTTIME#) time of the last received quote of the symbol for which a trigger was activated.

Performance

These macros substitute server monitoring data at the moment of trigger activation.

- Total CPU (#MONITOR_CPU#) total CPU load.
- Process CPU (#MONITOR_CPU_PROCESS#) CPU usage by the server process.
- Process Threads (#MONITOR_CPU_PROCESS_THREADS#) number of threads used by the server process.
- Process Handles (#MONITOR_HANDLES_PROCESS#) number of descriptors (handles) used by the server process.
- Free Memory (#MONITOR_MEMORY_FREE#) free RAM.
- Process Memory (#MONITOR_MEMORY_PROCESS#) amount of RAM allocated by the server process.

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- Free Disk Space (#MONITOR_DISK_FREE#) amount of free disk space.
- Disk Read Speed (#MONITOR_DISK_SPEED_READ#) speed of data reading in MB/s.
- Disk Write Speed (#MONITOR_DISK_SPEED_WRITE#) speed of data writing in MB/s.
- Disk Queue Length (#MONITOR_DISK_QUEUE_LENGTH#) disk queue length.
- Network Connections (#MONITOR_NETWORK_CONNECTIONS#) total amount of all standard operating terminal connections and temporary connections made for executing trades, downloading history or news.
- Network Connections Blocked (#MONITOR_NETWORK_BLOCKED#) number of connections blocked by the antiflood control or by the built-in firewall.
- Network Sockets (#MONITOR_NETWORK_SOCKETS#) number of active sockets.
- Network Traffic In (#MONITOR_NETWORK_TRAFFIC_IN#) incoming traffic in Mbit/s.
- Network Traffic Out (#MONITOR_NETWORK_TRAFFIC_OUT#) outgoing traffic in Mbit/s.
- Network Retransmits (#MONITOR_NETWORK_RETRANSMIT#) number of retransmitted packets.
- Total CPU DPC (#MONITOR_CPU_DPC#) CPU load when handling deferred procedure calls.
- Total CPU Interrupts (#MONITOR_CPU_INTERRUPTS#) CPU load when handling interruptions from devices.

External

These macros are used for triggers that run automation tasks upon requests received from external systems.

- Web callback URL (#WEBCALLBACK_URL#) URL of the callback request which has run the automation task.

Message

These macros substitute data of the internal email which the user interacts with. Use them with triggers from the Messages section.

- Sender login (#MESSAGE_FROM#) the account number of the user who sent the email.
- Sender name (#MESSAGE_FROM_NAME#) the name of the user who sent the email.
- Recipient login (#MESSAGE_TO#) the account number of the email recipient.
- Recipient name (#MESSAGE_TO_NAME#) the name of the email recipient.
- Subject (#MESSAGE_SUBJECT#) email subject.
- Body (#MESSAGE_BODY#) email body including the HTML markup or the contents of the message sent via the SMS provider or instant messenger.
- Address (#MESSAGE_ADDRESS#) recipient's email address or phone number.

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- Provider name (#MESSAGE_PROVIDER_NAME#) the configuration name of the SMS provider, instant messenger or mail server via which the message was sent.
- Provider balance (#MESSAGE_PROVIDER_BALANCE#) remaining balance with your SMS provider.

Connection

These macros are used to display information about the connected client based on their IP address. Use them with triggers from the Connection category.

- IP address (#CONNECTION_IP#) client's IP address.
- Country (#CONNECTION_COUNTRY#) the country in which the IP address is located.
- City (#CONNECTION_CITY#) the city in which the IP address is located.
- ASN (#CONNECTION_ASN#) the Autonomous System Number (ASN) to which the IP address belongs.
- ISP (#CONNECTION_ISP#) the name of the internet service provider that owns the IP address.
- Additional flags (#CONNECTION_FLAGS#):
 - ◆ Datacenter the address belongs to a data center.
 - ◆ TOR the address belongs to the TOR network.
 - ◆ Proxy the address is a proxy server.
 - ◆ VPN the address is a VPN server.
 - ◆ Attacker the address was involved in various attacks, such as spam, password brute-forcing, DDoS, etc.
 - ◆ Botnet the address belongs to a botnet.

Time

These macros substitute trigger activation time in the message text:

- Trading time time used in the platform, taking into account its settings.
- Local time time used in the computer on which the platform is installed.
- UTC time UTC time.

KYC

These macros substitute KYC check results in the message text: Only used with the relevant triggers.

- #KYC_STATE_CODE# check status code.
- #KYC_STATE_DESC# check status description.

The possible codes and their descriptions are as follow

#KYC_STATE_CODE#	#KYC_STATE_DESC#
STATUS_INIT	Initiated
STATUS_QUEUED	Queued

#KYC_STATE_CODE#	#KYC_STATE_DESC#
STATUS_APPROVED	Approved
STATUS_DECLINED	Declined
STATUS_EXPIRED	Expired
ERROR_GENERAL	General error
ERROR_CONNECTION	Connection error
ERROR_AUTH	Auth error
ERROR_INTERNAL	Internal error
ERROR_DOC_DUPLICATE	Duplicate document (image, video) was uploaded. Exact equality is taken into account
ERROR_DOC_TOO_MANY	Applicant contains too many documents. Adding new ones is not allowed
ERROR_DOC_TOO_BIG	Uploaded file is too big (more than 64MB)
ERROR_DOC_EMPTY	Uploaded file is empty (0 bytes)
ERROR_DOC_CORRUPTED	File is corrupted or incorrect format (e.g. PDF file is uploaded as JPEG)
ERROR_DOC_FORMAT	Unsupported file format (e.g. a TIFF image)

Automations

These macros substitute a description of an event which led to trigger activation.

Trigger Name (#CONDITION#)

This macro substitutes a description of all trigger conditions in one line. Conditions are separated by semicolons.

All comparison operations other than "equal" feature:

- Condition name
- Fixed level (actual value at the moment of trigger activation)
- Comparison type
- Condition activation threshold (condition value)

Examples:

```
Platform\CPU Total 7.5% > 1%
Platform\CPU Total 7.5% > 1%; Platform\Disk queue 100 > 10; Platform\Server ID 3
```

"Equal" comparison operations feature:

- Condition name

- Fixed level

Examples:

Platform\CPU Total 7.5% Platform\CPU Total 7.5%; Platform\Disk queue 10; Platform\Server ID 3
--

Condition Threshold (#CONDITION_LEVEL#)

The list of semi-colon separated trigger condition values. For example, the objective has two conditions:

- Platform\Disk Queue Length > 10
- Platform\CPU > 70%

The macro displays the string "10 ; 70%".

Condition Value (#CONDITION_VALUE#)

The list of actual values of tracked conditions recorded at the trigger time. For example, the objective has two conditions:

- Platform\Disk Queue Length > 10. At the time of triggering, the actual disk queue length was 50.
- Platform\CPU > 70%. At the moment of triggering, the actual CPU load was 85%.

The macro displays the string "50 ; 85%".

Action Date and Time (#DATETIME#), Action Time (#TIME#), Action Date (#DATE#)

These macros display the trigger activation date and time. Examples:

2020.10.16 15:05:14 15:05:14 2020.10.16

Trigger Name (#TRIGGER_NAME#), Trigger Type (#TRIGGER_TYPE#)

These macros display automation task name and activated trigger type.

- Schedule/Scheduled event
- Connection/Login
- Connection/First login
- Connection/Logout
- Accounts/New trade account
- Finance/Deposit
- Finance/First deposit
- Finance/Withdrawal
- Finance/First withdrawal
- Finance/Credit

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- Finance/First credit
- Finance/Credit out
- Finance/First credit out
- Trade/Margin call
- Trade/Stop out
- Price/Gap started
- Price/Gap finished
- Platform/Performance monitor
- Platform/Server connect to Main server
- Platform/Server disconnect from Main server

Action Name (#ACTION_NAME#), Action Type (#ACTION_TYPE#)

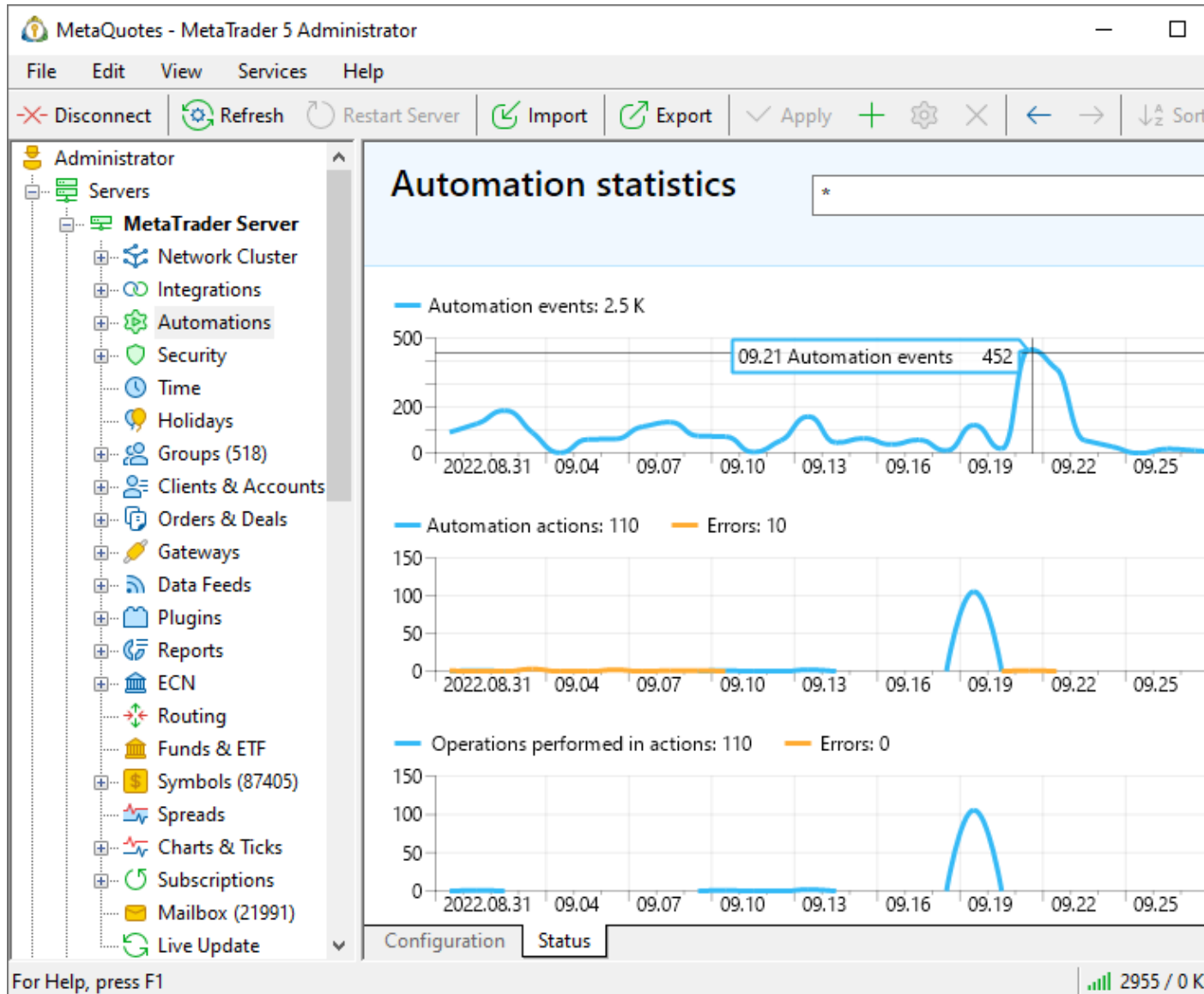
These macros display the name and type of a performed action.

MetaTrader 5 Trading Platform / Platform Setup / Automations / Statistics

Automation Task Monitoring Statistics

You can control Automations efficiency by using statistics on automation tasks: how many times each of the tasks triggered, how many actions it executed and how many actions failed for any reason.

To view the general statistics, navigate to the State section:



The following data is available here:

- Automation events the number of automation task triggers.
- Automation actions / Errors the number of successful actions and the number of execution errors. An action is one row (one setting) in the appropriate task section. Some actions require the execution of multiple operations. For example, this can be sending of emails to a range of logins. If an email is successfully sent to at least one login in the range, the whole action is considered successful.
- Operations performed in actions / Errors the number of successful and failed operations executed in accordance with the configured action. Some actions require the execution of multiple operations. For example, this can be sending of emails to a range of logins. If an email is successfully sent to only one login in the range, only one operation will be considered as successful. Other operations will be included in the Errors statistics.

Use a filter in the upper part of the section to select a specific task or a directory of tasks, for which you wish to view statistics.

Automation statistics are also available in the list of tasks and in each tasks' separate tab:

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MetaQuotes - MetaTrader 5 Administrator

File Edit View Services Help

Disconnect Refresh Restart Server Import Export Apply + Settings X Left Right Sort

Network Cluster
Integrations
Automations
Server Performance
Marketing
Trading

Name	Type	Eve...	Acti...	Acti...	Ope...
Server Performance					
Marketing					
Trading					
Christmas Bonus for deposits from ...	Deposit	92	92		92
Send special offer on Withdrawal	Withdrawal	80	80		80

Automation: test auto

Trigger Conditions Actions **Statistics**

 Automation task statistics for the last 30 days. View the number of triggers and evaluate the automation task efficiency.

Date	Last use time	Events	Actions	Action errors	Operations	Operation errors
2020.09.17	2020.09.17 09:23:59	20	20		20	
2020.09.16	2020.09.16 14:16:59	60	60		60	

For Help

Сброс OK Cancel Help

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To reset the statistics of a certain task, for example, after completing testing, click "Reset" in the "Statistics" tab.

Automation statistics are available for the last 30 days. Longer history is not stored.

MetaTrader 5 Trading Platform / Platform Setup / Time

Time

This section is intended for configuring server time settings. At non-working time clients can connect to the server, watch charts and trade history, but cannot conduct trade operations.

One can setup the trade and quotation sessions for each symbols separately at the "Sessions" tab.

The screenshot shows the 'MetaQuotes - MetaTrader 5 Administrator' window. The left sidebar contains a tree view with the following items: Administrator, Servers, MetaTrader Server, Network Cluster, Integrations, Automations, Security, Time (selected), Holidays, Groups (518), Clients & Accounts, Orders & Deals, Gateways, and Data Feeds. The main panel displays the 'Time' settings for the selected 'MetaTrader Server'. The settings are organized into two sections: 'Common settings' and 'Day settings'.

Property	Value
Common settings	
01 Time zone	GMT+01:00
Daylight Saving Time correction	Yes
ab Time synchronization with	pool.ntp.org
Day settings	
Sunday	
Monday	00:00-24:00
Tuesday	00:00-24:00
Wednesday	00:00-24:00
Thursday	00:00-24:00
Friday	00:00-24:00
Saturday	

At the bottom of the window, there is a status bar that reads 'For Help, press F1' on the left and '4277 / 1 KB' on the right.

Time settings are divided into common and day settings:

Common Settings

The following parameters are available in this block:

- **Time zone** the time zone of the trade server (e.g. GMT+1:00);
- **Daylight Saving Time correction** enable/disable DST correction;
- **Time synchronization with** address of the server that is used for synchronizing time over protocols NTP and TIME. Through this server the system will receive exact time. After setting time zones this time will be used in the system.

- At startup of the MetaTrader 5 servers (services), the Windows Time service in the operating system is automatically disabled. However, if an address of a server for the synchronization of time is not specified in the "Time synchronization with" field, that service will not be disabled.
- Time synchronization with the specified server is performed for the main trade server. All other components are synchronized with the main trade server.
- The checking and synchronization of time is performed every hour. If a time difference of more than 30ms is detected, time correction is performed. To request the journal records regarding time synchronization use "time synchronization" key phrase.
- Changes in "Time zone" and "Daylight Saving Time correction" become effective only after the server restart. It is strongly recommended not to change server time zone settings during a trade day, because this may cause failure of quotes. These settings should be changed on weekend or holidays or at night when the trading activity is minimal and the thread of quotes is slow.
- When other servers of the platform connect to the main server, they compare their time zone and DST settings. If the time zone has changed or time has switched to DST, the settings of the server that has connected to the main one are updated, and after that the server is restarted to apply changed.
- After common time settings are changed, the main server should be restarted.

Daylight Saving Time features

When switching to Daylight Saving Time (DST), the platform uses the dates stored in the operating system time zone. Therefore, we urge you to set the Windows OS time zone and DST parameters similar to the ones in the platform.

Platform and OS settings	Performed actions
DST enabled in MetaTrader 5	After Windows switches to DST, the platform does the same within an hour (during the hourly

MetaTrader 5 trading platform

Platform and OS settings	Performed actions
	synchronization). The platform uses time data stored in the OS time zone.
DST disabled in MetaTrader 5	No switch is performed.
DST enabled in MetaTrader 5 but there is no DST switch for the time zone set in OS.	No switch is performed, since the platform has no exact data on what date/time the shift is to be performed.

Day Settings

In order to start editing working time, select one of days and press button "⚙ Edit" in the Edit menu, on the Standard toolbar or in the context menu. The following window will be opened:

Time?×



The adjustment of trading time of server by days of the week. Please mark working hours with the blue color.

Sunday: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 *

Monday: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 *

Tuesday: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 *

Wednesday: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 *

Thursday: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 *

Friday: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 *

Saturday: 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 *

OK

Cancel

Working hours are colored blue. In order to change a working hour into non-working or vice versa, click it with the left mouse button. To change working hours in bulk, click on the necessary hour and drag the cursor to another hour. Button "*" at the end of the line allows switching all working into non-working and vice versa.

For changes to take effect, execute command "✓ Apply Changes" in the Edit menu, or the same command on the Standard toolbar or in the context menu.

Context Menu

The context menu of the Time section contains the following commands:

-  **Edit** edit the selected instruction;
-  **Export to File** export work time settings to a file.
-  **Import from File** import work time settings to a file.
-  **Find** open the search window;
- **Auto Arrange** if this option is enabled the size of columns is selected automatically;
- **Grid** this option shows/hides field separators.

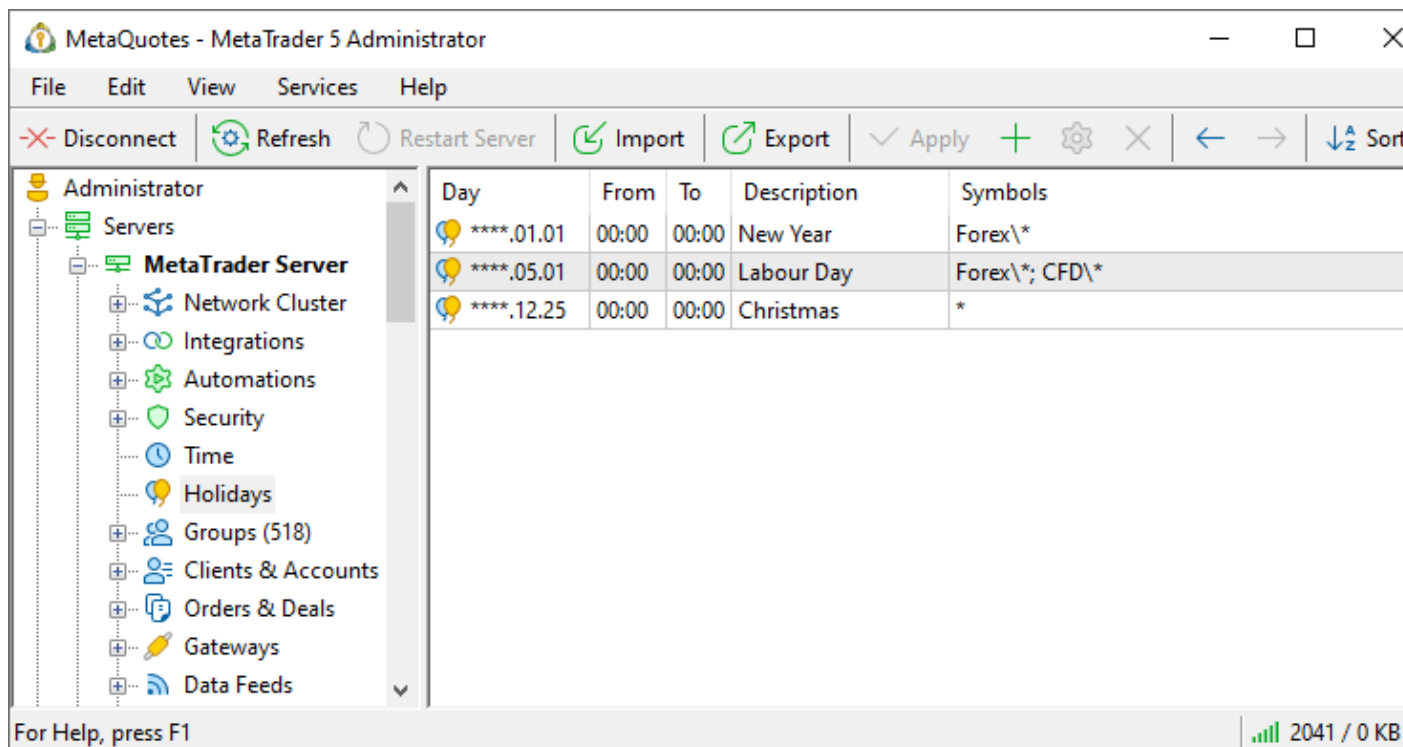
MetaTrader 5 Trading Platform / Platform Setup / Holidays

Holidays

In this section you can add holidays to the working time schedule both for symbol groups and separate symbols. During holidays clients can connect, watch charts and trading history, but cannot conduct trade operations.

- Data feeds that receive quotes only are automatically disabled on holidays. As soon as a holiday is over, the data feeds are automatically enabled. Data feeds that receive news keep working.
- Holiday settings do not affect swap calculation.

Holidays are represented like this:



The screenshot shows the MetaQuotes - MetaTrader 5 Administrator window. The left sidebar is expanded to show the 'Holidays' section under the 'MetaTrader Server' node. The main pane displays a table of holidays.

Day	From	To	Description	Symbols
****.01.01	00:00	00:00	New Year	Forex*
****.05.01	00:00	00:00	Labour Day	Forex*; CFD*
****.12.25	00:00	00:00	Christmas	*

For Help, press F1

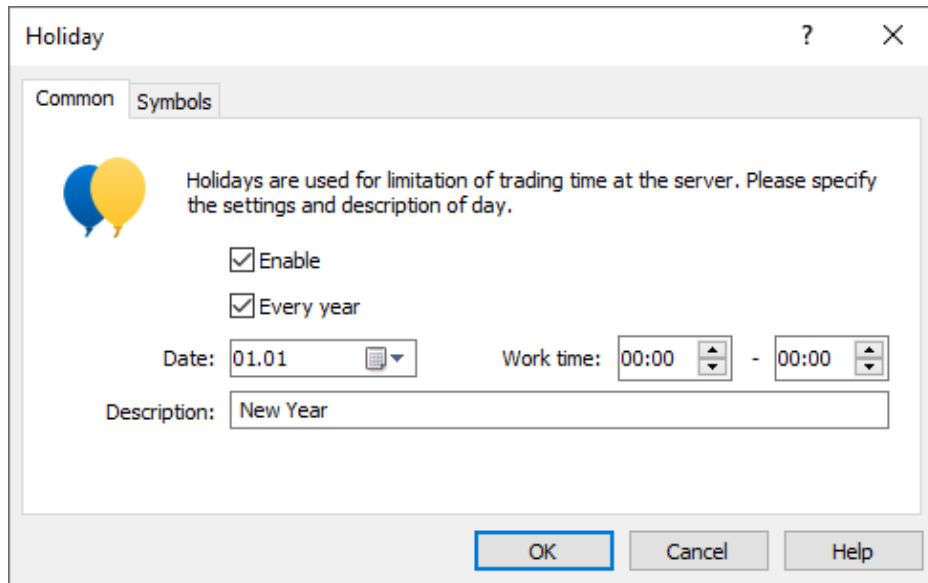
2041 / 0 KB

MetaTrader 5 trading platform

In order to add or edit a holiday, press "+ Add" or "⚙ Edit", respectively. These buttons are located in the Edit menu, on the Standard toolbar and in the context menu. The window of holiday settings will be opened as soon as you press them.

Additional general information about working with configuration records is given in the "Working with Instructions" section.

Common



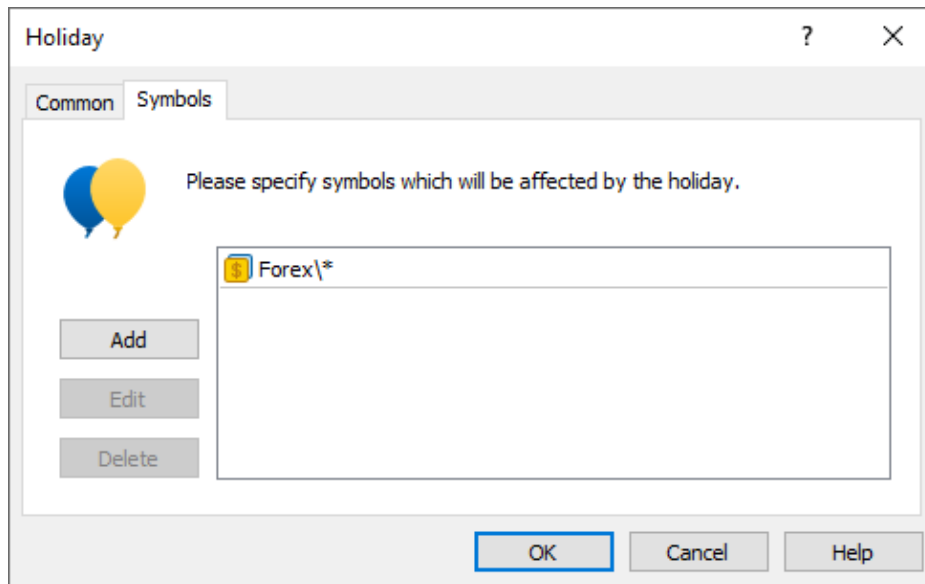
The screenshot shows the 'Holiday' dialog box with the 'Common' tab active. It includes a message about holiday settings, checkboxes for 'Enable' and 'Every year', a 'Date' field with a calendar icon, a 'Work time' field with two time pickers, and a 'Description' text field. The 'Date' is set to '01.01' and the 'Description' is 'New Year'.

The following parameters are set up here:

- **Enable** enable/disable a holiday;
- **Every year** enable holiday repeating every year. For such holidays "*****" is shown instead of the year;
- **Date** date of the holiday. A date is selected via a calendar that is opened upon pressing . The date can also be entered manually;
- **Work time** server work time on the holiday. If there is no working time on this day, leave zero values in these fields;
- **Description** text description of the holiday.

The delivery of quotes and trading continue until the last minute of the working time range inclusive. For example, if the specified range is from 00:00 to 18:00, then the actual working time is 00:00:00 18:00:59.

Symbol



here you should specify symbols or groups of symbols, for which this holiday will be effective:

- **Add** add symbols. As soon as you press this button, a new field will appear in this window. Click on this field and select a symbol or group from the appeared list of symbols available on the server. To specify all symbols, select the "Symbols" point. Read more details in ["Specifying Symbols"](#);
- **Delete** delete a selected symbol or group of symbols;
- **Edit** edit a selected symbol or group of symbols.

To complete creation or editing of a holiday press "OK". If you press "Cancel", the window will be closed, while changes will not be saved.

In order to delete a holiday, select it and press "X Delete" in the [Edit](#) menu, on the [Standard](#) toolbar or in the context menu.

Holidays Check

Holiday rules are checked from top downward. If any rule allows trading for a symbol in the specified interval, all further rules will be ignored for this symbol-interval binding. Thus:

- It is not possible to override an allowing rule by subsequent prohibiting rules.
- It is possible to override a prohibiting rule by subsequent allow rules.

Day	From	To	Description
2014.01.01	00:00	00:00	All disabled
2014.01.01	10:00	12:00	Forex
2014.01.01	13:00	22:00	Forex
2014.01.01	00:00	00:00	All disabled

Let's consider example:

- The first rule prohibits trading for all symbols on January 1, 2014
- The second rule allows trading for Forex group symbols from 10:00 to 12:00 on that day
- The third rule allows trading for the same symbols from 13:00 to 22:00
- The fourth rule prohibits trading for all symbols on January 1, 2014.

As a result, on January 1, 2014, trading will be allowed for Forex group symbols from 10:00 to 12:00 and from 13:00 to 22:00. The fourth rule is ignored for the earlier allowed intervals as the check for these intervals stopped after the rules allowing trading for them.

Thus, in order to set several trading intervals inside one time interval, you should first set a prohibitive rule followed by permissive ones.

Context Menu

The context menu of Holidays contains the following commands:

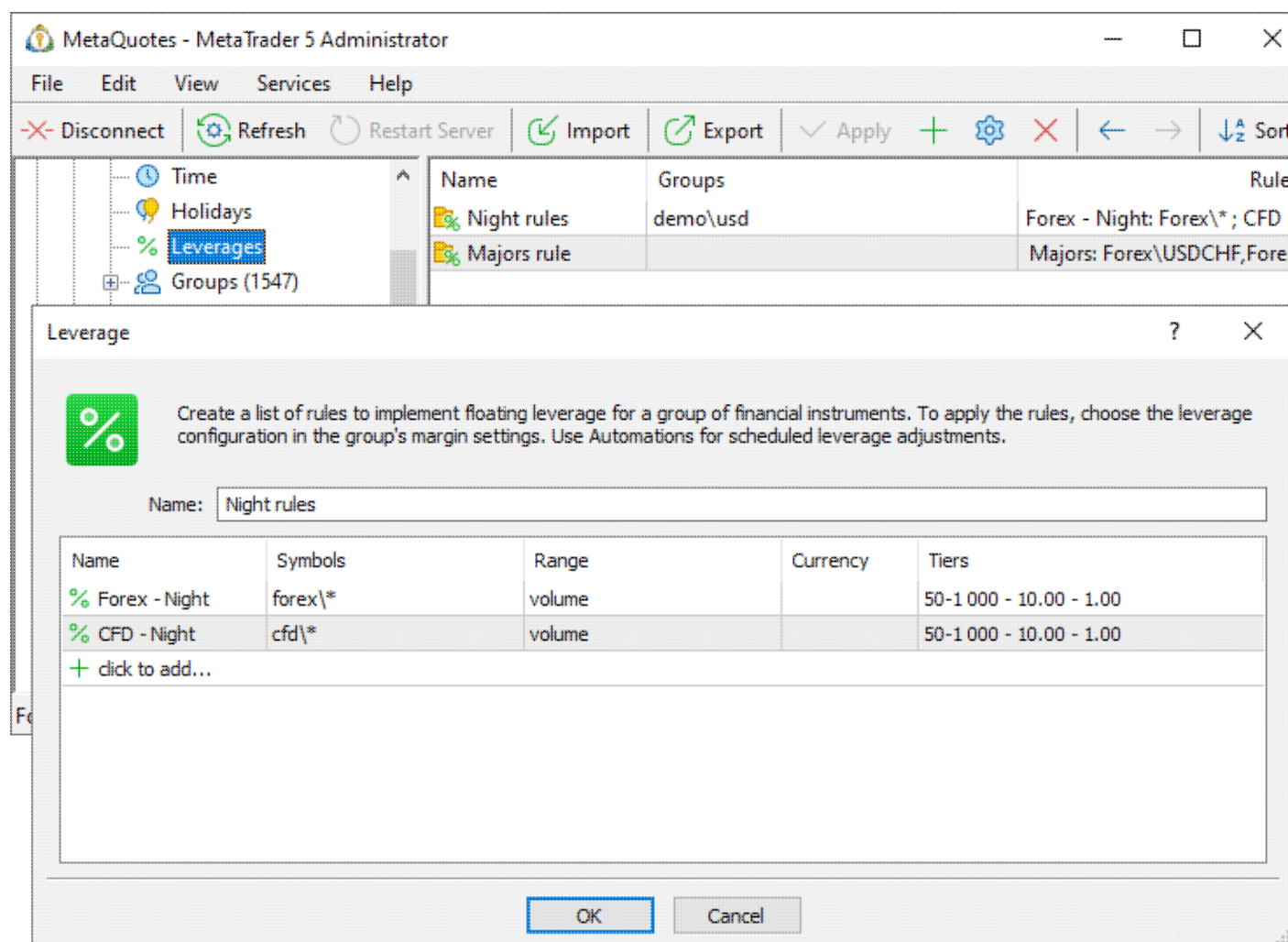
-  **Add** add a new holiday;
-  **Edit** edit a selected holiday;
-  **Delete** delete a selected holiday;
-  **Move Up** move the selected holiday up relative to others;
-  **Move Down** move the selected holiday down relative to others;
-  **Sort Alphabetically** sort configurations alphabetically. Please note that sorting is done on the server, not in the local terminal.
-  **Export to File** export holidays settings to a file.
-  **Import from File** import holidays settings to a file.
-  **Journal** request logs according to the selected configuration. This will open the trade server logs section, with the name of the selected configuration automatically specified in the query field. You will only need to press the request button.
-  **Find** open the search window;
- **Auto Arrange** if this option is enabled the size of columns is selected automatically;
- **Grid** this option shows/hides field separators in the table.

MetaTrader 5 Trading Platform / Platform Setup / Leverages

Leverages

In this section, you can configure a list of rules for quickly adjusting the leverage for clients. You can create several profiles and quickly switch between them in the group settings, as well as enable automatic schedule-based adjustments. In essence, this section enables the implementation of a dynamic leverage, often referred to as a floating leverage, which adjusts based on different conditions. For example, leverage values may vary depending on the volume of positions on the client account, on the day of the week or time of day, etc.

This system does not affect the basic margin calculation principles set for trading instruments. It allows you to change leverage by applying an additional coefficient to the initial and maintenance margin values calculated in accordance with the symbol settings.



MetaTrader 5 trading platform

Create a new configuration (profile), specify its name, and then add floating leverage rules. You can add multiple rules within one profile. For example, you can add separate rules for different types of symbols. When you enable this profile for a group, all rules will be applied.

Keep in mind that rules are checked and applied top to bottom. If the same instrument matches multiple rules, only the first applicable rule will be used.

Leverage Rule



Set up a leverage rule. Select a symbol and specify leverage levels based on position size or value. You can check the total volume/value for the selected group of symbols or a separate symbol by selecting the corresponding range option.

Name: Forex - Night

Description: Rule to be applied at night for Forex symbols

Symbol: Forex*

Range: Volume

Currency:

	To	Initial margin rate	Maintenance margin rate
%	1 000	10.00	1.00
%	∞	4.00	4.00
+ click to add...			

OK

Cancel

Set general parameters for the rule:

- **Name** the name of the rule. Set descriptive names that reflect their purpose.
- **Description** an extended description of the rule.
- **Symbol** symbol or group of symbols to which the rule will apply.

You can create multi-level rules and change the leverage in different ways depending on the volume/value of the trader's positions. Select the desired condition in the Range parameter:

- **Volume** the total volume of open positions for all symbols from the Symbols field. The volume is specified in the lots number.
- **Volume per symbol** unlike the previous option, this parameter refers to the volume of open positions for each individual symbol from the Symbols field. For example, if all symbols Forex* are specified, the volume for each instrument from this group is considered. The system calculates the volume of Forex\EURUSD positions, checks the applicable range, and applies the appropriate margin ratio. Next, it checks Forex\GBPUSD, Forex\USDJPY, and all the other symbols included in this group.
- **Notional value** the total value of open positions for all instruments from the Symbols field. The value is calculated based on the position opening price and is converted to the currency specified in the Currency field. If the Currency field is empty, the value is converted to the deposit currency of the group for which the margin is calculated. The current exchange rate is used for conversions.
- **Notional value per symbol** unlike the previous option, this parameter refers to the value of open positions for each individual symbol from the Symbols field. For example, if all symbols Forex* are specified, the system considers the notional value for each instrument from this group. The system calculates the value of Forex\EURUSD positions, checks the applicable range, and applies the appropriate margin ratio. Next, it checks Forex\GBPUSD, Forex\USDJPY, and all the other symbols included in this group.

Margin levels are configured at the bottom of the window. Specify the maximum values in the To field. The minimum value is set automatically based on the value of the previous range. Depending on the Range parameter, the field indicates the positions volume or notional value. For example, the Range field is set to "Volume". In this case, a level up to 10 means that the specified margin rates will be applicable if the total volume of the client's open positions for the specified symbols does not exceed 10 lots.

If you do not need multilevel rules, set one level covering the entire range of values for the selected condition.

Next, set the initial and maintenance margin ratios for each level. They will be applied to the margin values calculated based on the symbol types and settings. The initial margin Ratio applies to all open orders, including market orders. Set both margin ratios explicitly. Unlike symbol settings, here the zero maintenance margin does not indicate that the initial margin ratio will be used instead. The zero value means that no margin will be charged.

Floating leverage settings only work for groups with the "for Retail Forex, CFD, Futures" calculation type (hedging and netting). The settings do not apply for groups with exchange calculation type.

Applying leverage configurations

To apply a leverage configuration, open the Margin section in the group settings and select the required configuration:

Group: demo\usd

Common Company News & Mail Permissions **Margin** Symbols Commissions Reports

 Please set up the mechanism of margin calculation for the group: choose the method of calculation and margin requirements.

Risk management: for Retail Forex, CFD, Futures

Margin call level: 50.00 Stop out level: 30.00 in %, percent

☐ Stop out fully hedged accounts

☐ Compensate negative balance after stop out

☐ Withdraw credit after negative balance compensation

Floating leverage profile: Night rules

Profit/loss in free margin: Night rules

Unrealized profit: use unrealized profit/loss

Daily fixed profit: use daily fixed profit/loss

☐ Release fixed profit at the end of day

Virtual credit: (applies only to opening new positions)

OK Cancel Help

Margin calculation example based on volume

Let's look at the calculation procedure using an example. The following leverage levels are specified:

?

×

Leverage Rule

%

Set up a leverage rule. Select a symbol and specify leverage levels based on position size or value. You can check the total volume/value for the selected group of symbols or a separate symbol by selecting the corresponding range option.

Name:

Forex symbols

Description:

Rule for Forex symbols

Symbol:

Forex*

Range:

Volume

Currency:

	To	Initial margin rate	Maintenance margin rate
%	10	1.00	1.00
%	20	2.00	2.00
%	30	3.00	3.00
%	∞	4.00	4.00
+ click to add...			

OK

Cancel

The calculation is performed in the following environment:

- The account is in the group with the "for Retail Forex, CFD, Futures with hedging" type and with the deposit currency USD.
- The group has a leverage of 1:100.
- The initial and maintenance margin rates for market orders are set to 1. For pending orders, the initial margin rates are set to 1 and the maintenance margin rates are 0.
- The USDCHF contract size is 100,000; the instrument type is Forex.

The account has:

- Position 'Buy 12.00 lot USDCHF'
- Pending order 'Buy Limit 10.00 USDCHF'

Let's calculate the account margin before applying the leverage rule: $22 * 100,000 / 100 = 22,000$ USD. Since the maintenance margin rate is zero for a pending order, the initial margin rate is applied as the maintenance margin rate.

Now we apply the leverage rule, and the margin increases to 36,000. Let's look at how we got this value:

- First, the system determines how much margin is charged for each lot. In this case, it is $22,000 / 22 = 1,000$ USD.
- Next, the margin is calculated for positions. The system determines the configured margin rule ranges in which the position lots fall. The first 10 lots fall into the first range with a margin rate of 1. For these lots, the system charges a margin of $10 * 1,000 * 1 = 10,000$ USD. The remaining two lots fall into the second range with a rate of 2. For them, the system charges a margin of $2 * 1,000 * 2 = 4,000$ USD.
- Next, the system checks the orders. Since the system considers the volume of positions and orders in the aggregate, the first 8 lots of the order fall into the second range (12 lots position + 8 lots order). The charged margin is equal to $8 * 1,000 * 2 = 16,000$ USD. The remaining 2 lots of the order fall into the third range with a rate of 3. The margin is equal to $2 * 1,000 * 3 = 6,000$ USD.
- The total margin: $10,000 + 4,000 + 16,000 + 6,000 = 36,000$ USD

The approach based on per-lot margin calculations enables the transparent application of rates specified in the rule for any combination of positions. For example, the account has the following positions:

- Buy 2.00 EURUSD, margin is 1,800 USD (including conversion from EUR to USD)
- Sell 1.00 USDCHF, margin is 1,000 USD
- Buy 15.00 USDJPY, margin is 15,000 USD

The margin per lot is $(1,800 + 1,000 + 15,000) / (2 + 1 + 15) = 988.88$. Next, the total volume is split into levels from the rule. The first 10 lots will be included in the first level and the remaining 8 are in the second one. Margin = $(10 * 988.88) + (8 * 988.88 * 2) = 25,710.88$ USD. The order in which the positions are considered does not matter.

When opening positions, the initial margin is checked. For open positions, the maintenance margin is checked. For pending orders, the initial margin is always checked.

An example of calculating margin based on the notional value

Let's look at an example of calculating margin on the basis of the notional value with hedged positions present on the account. The following leverage levels are specified:

Leverage Rule?×



Set up a leverage rule. Select a symbol and specify leverage levels based on position size or value. You can check the total volume/value for the selected group of symbols or a separate symbol by selecting the corresponding range option.

Name:

Description:

Symbol:

Range:

Currency:

	To	Initial margin rate	Maintenance margin rate
%	200 000	0.10	0.10
%	1 000 000	0.20	0.20
%	2 000 000	0.50	0.50
%	4 000 000	1.00	1.00
%	10 000 000	2.00	2.00
%	∞	4.00	4.00
+ click to add...			

OK

Cancel

The calculation is performed in the following environment:

- The account is in the group with the "for Retail Forex, CFD, Futures with hedging" type and with the deposit currency USD.
- The group has a leverage of 1:500.
- The initial and maintenance margin rates for market orders are set to 1. For pending orders, the initial margin rates are set to 1 and the maintenance margin rates are 0.
- The EURUSD contract size is 100,000; the instrument type is Forex.
- The hedged margin for EURUSD is 50,000.

The account has the following positions:

- Buy 20.00 EURUSD at 1.08095
- Buy 10.00 EURUSD at 1.08095
- Buy 10.00 EURUSD at 1.08095
- Sell 15.00 EURUSD at 1.08095

For simplicity, we use the same opening price. Let's calculate the margin **before** the application of leverage rules. We have 40 lots of buy positions and 15 lots of sell positions. Consequently, the uncovered volume is 25 lots, and the covered volume is 15:

- $25 * 100,000 / 500 = \text{EUR } 5,000$
- $15 * 50,000 / 500 = \text{EUR } 1,500$

Let's convert the margin into the deposit currency (USD) at the position opening rate:
 $6,500 * 1.08095 = \text{USD } 7,026.175$.

Now we apply the margin rule:

- First we calculated the total volume. The margin is charged in full for 25 non-covered lots. For 15 hedged lots, the margin is charged at half the amount, since the hedged margin is specified as half the size of the standard contract (50,000). Thus, this volume can be reduced to the standard 7.5 lots. The total calculated volume is 32.5 lots.
- Next, we convert the volume into money, since margin levels are set as notional values. $32.5 * \text{EUR } 100,000 * 1.08095 = \text{USD } 3,513,087.5$
- Then we calculate what share of the total margin (USD 7,026.175) falls on each level, and then apply the relevant rate:
 - ◆ $7,026.175 * (200,000 / 3,513,087.5) * 0.1 = \text{USD } 40$. Unaccounted remainder: $3,513,087.5 - 200,000 = 3,313,087.5$
 - ◆ $7,026.175 * (800,000 / 3,513,087.5) * 0.2 = \text{USD } 320$. Unaccounted remainder: $3,313,087.5 - 800,000 = 2,513,087.5$
 - ◆ $7,026.175 * (1,000,000 / 3,513,087.5) * 0.5 = \text{USD } 1,000$. Unaccounted remainder: $2,513,087.5 - 1,000,000 = 1,513,087.5$
 - ◆ $7,026.175 * (1,513,087.5 / 3,513,087.5) * 1 = 3,026.175$
- Next, we sum up the values for each level and get the final margin value: $40 + 320 + 1,000 + 3,026.175 = 4,386.175 \text{ USD}$

Consideration of the conversion rate in margin calculation at notional value

MetaTrader 5 trading platform

If the margin currency of the instrument differs from the currency in which the notional value of the position is calculated, the floating margin will be recalculated in real-time based on the applicable exchange rate. Let's consider example:

The floating margin rule uses levels based on the notional value of positions. The currency is set to USD. The following levels and rates are set:

- Up to 100,000 - 0.33
- Up to 300,000 - 0.5
- Up to 500,000 - 0.75
- Up to 2,500,000 - 1
- More than 2,500,000 - 1.

A user with a leverage of 1:100 opens a 100-lot AUDCHF position. Let's calculate the value of this position in the margin currency: $100 * 100,000 \text{ AUD} = 10,000,000 \text{ AUD}$.

Let's calculate the value of this position in USD at the current rate: Ask AUDUSD = 0.63014. $10,000,000 * 0.6301400 = 6,301,400 \text{ USD}$.

Let's calculate the margin for the total notional value in accordance with the leverage-based rates:

- $100,000 * 0.33 / 100 = 330 \text{ USD}$
- $200,000 * 0.5 / 100 = 1,000 \text{ USD}$
- $200,000 * 0.75 / 100 = 1,500 \text{ USD}$
- $2,000,000 * 1 / 100 = 20,000$
- $3,801,400 * 1.1 / 100 = 41,815.40$

The total margin will be 64,645.40 USD. If the currency in the rule differs from the client's deposit currency, the resulting value will be additionally converted at the "Margin Ratio" rate, which is stored in the position. This rate does not change over time.

On the client terminal side, the margin is recalculated with each tick. Therefore, in cases as described above, you will see margin changes in real time. In manager terminals, the margin in the account properties is updated only upon request. As a result, margin changes will be visible only when reopening the corresponding dialog.

Automatic leverage adjustments

You can apply leverage profiles depending on various conditions using the Automations service. For example, you can turn on a certain profile on Friday evening and then turn it off on Monday morning before the markets open.

Create a new task and select the relevant triggers and conditions. For example, to switch a profile at a specific time, use the 'Scheduled Event' trigger. Set the action to 'Update group configuration'. To change only the leverage parameter, click 'Add Diff'. Next, select the desired group and specify the leverage profile in its settings:

Automation

Trigger
Conditions
Actions


Please specify actions that should be performed when conditions trigger. For example, sending a message, updating account settings or performing balance operation, etc. Also specify accounts and groups to apply the action to.

Up
Down

Name	Name
 Enable leverage profile	Update group configuration
+ click to add...	


Action
?
X


Please specify automation action name and parameters. Parameters of some actions can be taken from trigger context. With trigger of Connections, Accounts or Finance configured, the system can use the login for which the automation condition triggered.

Name:

Action:

Load
Save

Parameter	Value
real\real Margin leverage profile	1 changes Night rules

Group: real\real

Common
Company
News & Mail
Permissions
Margin
Symbols
Commissions
Reports


Please set up the mechanism of margin calculation for the group: choose the method of calculation and margin requirements.

Risk management:

Margin call level:
Stop out level:
in

☐ Stop out fully hedged accounts

☐ Compensate negative balance after stop out

☐ Withdraw credit after negative balance compensation

Floating leverage profile:

☐ Profit/loss in free margin

Add
Add Diff
Edit
Delete

MetaTrader 5 trading platform

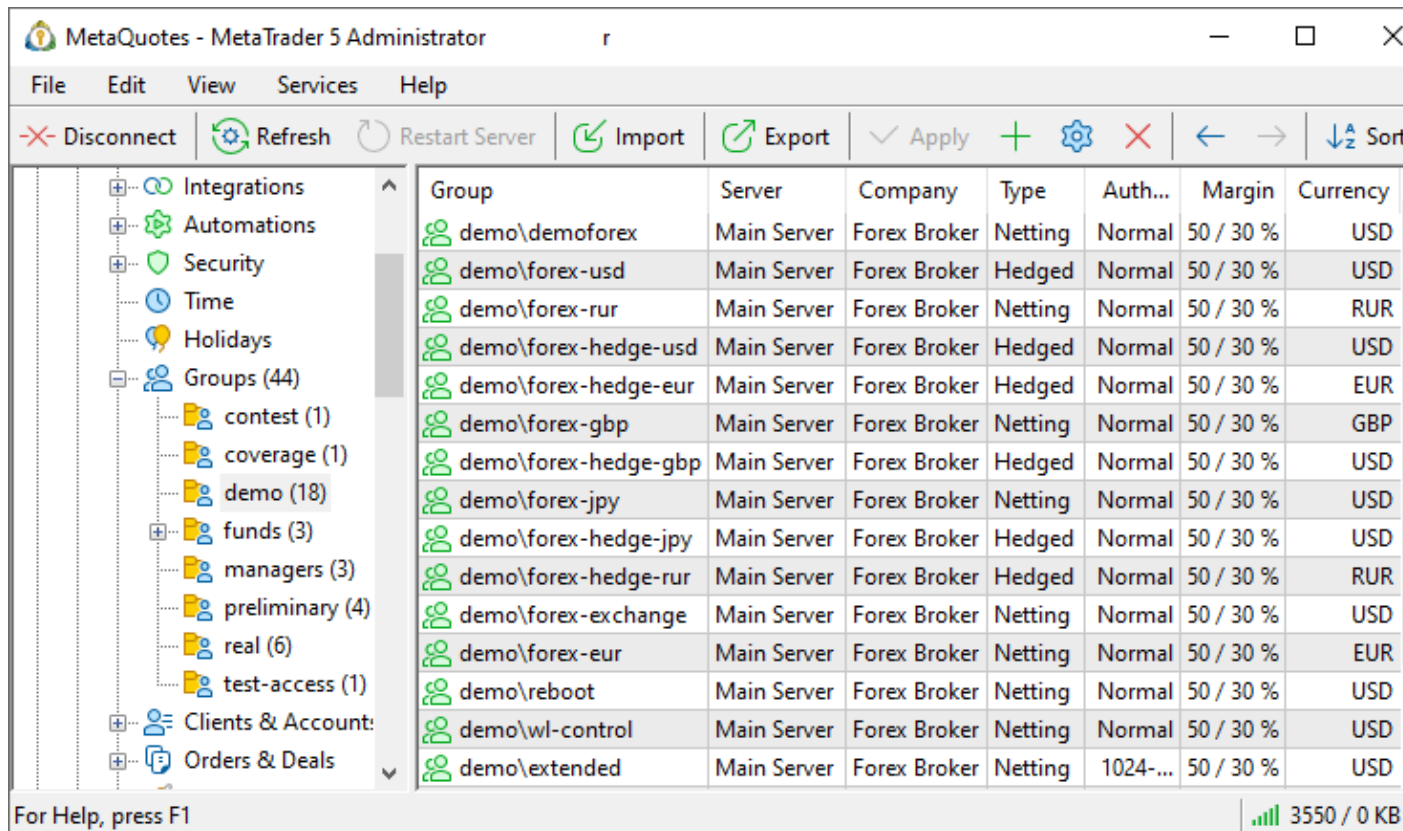
Next, create a relevant task that will disable the leverage profile in the group.

MetaTrader 5 Trading Platform / Platform Setup / Groups

Groups

All the system users are divided into groups. There can be no users on the server, who are not included into any of groups. The system contains the following default structure of groups:

- **demo** section for demo groups;
- **demoforex** groups of demo groups for FOREX
- **managers** section for manager groups;
- **administrators** group of administrators' accounts (maximal permissions in the system);
- **dealers** group of dealer accounts;
- **real** section for real groups;
- **real** group of accounts for live trading.
- **preliminary** group where preliminary accounts are created, when a request for opening a real account is sent from the client terminal.



The screenshot shows the MetaTrader 5 Administrator interface. On the left, a tree view displays the system structure, with 'Groups (44)' expanded. The main panel shows a list of groups with the following columns: Group, Server, Company, Type, Auth..., Margin, and Currency. The groups are listed as follows:

Group	Server	Company	Type	Auth...	Margin	Currency
demo\demoforex	Main Server	Forex Broker	Netting	Normal	50 / 30 %	USD
demo\forex-usd	Main Server	Forex Broker	Hedged	Normal	50 / 30 %	USD
demo\forex-rur	Main Server	Forex Broker	Netting	Normal	50 / 30 %	RUR
demo\forex-hedge-usd	Main Server	Forex Broker	Hedged	Normal	50 / 30 %	USD
demo\forex-hedge-eur	Main Server	Forex Broker	Hedged	Normal	50 / 30 %	EUR
demo\forex-gbp	Main Server	Forex Broker	Netting	Normal	50 / 30 %	GBP
demo\forex-hedge-gbp	Main Server	Forex Broker	Hedged	Normal	50 / 30 %	USD
demo\forex-jpy	Main Server	Forex Broker	Netting	Normal	50 / 30 %	USD
demo\forex-hedge-jpy	Main Server	Forex Broker	Hedged	Normal	50 / 30 %	USD
demo\forex-hedge-rur	Main Server	Forex Broker	Hedged	Normal	50 / 30 %	RUR
demo\forex-exchange	Main Server	Forex Broker	Netting	Normal	50 / 30 %	USD
demo\forex-eur	Main Server	Forex Broker	Netting	Normal	50 / 30 %	EUR
demo\reboot	Main Server	Forex Broker	Netting	Normal	50 / 30 %	USD
demo\wl-control	Main Server	Forex Broker	Netting	Normal	50 / 30 %	USD
demo\extended	Main Server	Forex Broker	Netting	1024-...	50 / 30 %	USD

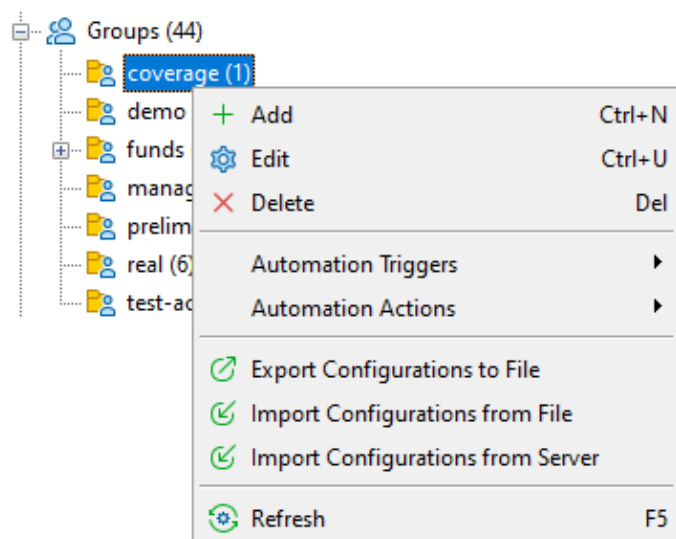
At the bottom of the window, it says 'For Help, press F1' and shows a signal strength indicator with '3550 / 0 KB'.

The groups are represented in a table with the following fields:

- **Group** group title;
- **Server** trade server that serves this group;
- **Authorization** method of authorization fro this group;
- **Currency** default deposit currency for this group. Then you can change the deposit currency for any account.

Hierarchy of Groups

Groups of accounts can be divided into sections and sub-sections depending on their types. There are three default sections: "demo", "managers" and "real". Group sections are created when groups are created.



In the "Groups" section select "+ Add" which will open a dialog window. In the "Name" field specify path to the group. For example, if you specify "*real\IB\realUSD*" in section "real" the sub-section "IB" will be created with group "realUSD" inside it. If one or several higher level sections do not exist by the moment of group creation, they will be created.

in order to delete a sections, delete all groups inside it.

Managing Groups

Groups management is performed in the right part of the administrator terminal.

• Add

In order to add a group, enter the corresponding section and execute the "+ Add" command in the Edit menu, on the toolbar or in the context menu. After that the window of group setting will be opened.

• Edit

In order to edit a group, select it and execute the "⚙ Edit" command in the Edit menu, on the toolbar or in the context menu. After that the window of group setting will be opened. The group editing window can be called by a double left click on its name in the list. You can also edit several groups together. To do

this select them using the mouse and keys "Ctrl" or "Shift", and execute command  Edit".

- **Delete**

In order to delete a group, select it and execute the " Delete" command in the Edit menu, on the Standard toolbar or the context menu. You can delete several groups at once by selecting them using the mouse and keys "Ctrl" or "Shift".

- You can't delete a group that contains at least one account.
- Additional general information about working with configuration records is given in the "Working with Instructions" section.

Context Menu

The context menu in the list of groups allows executing the following commands:

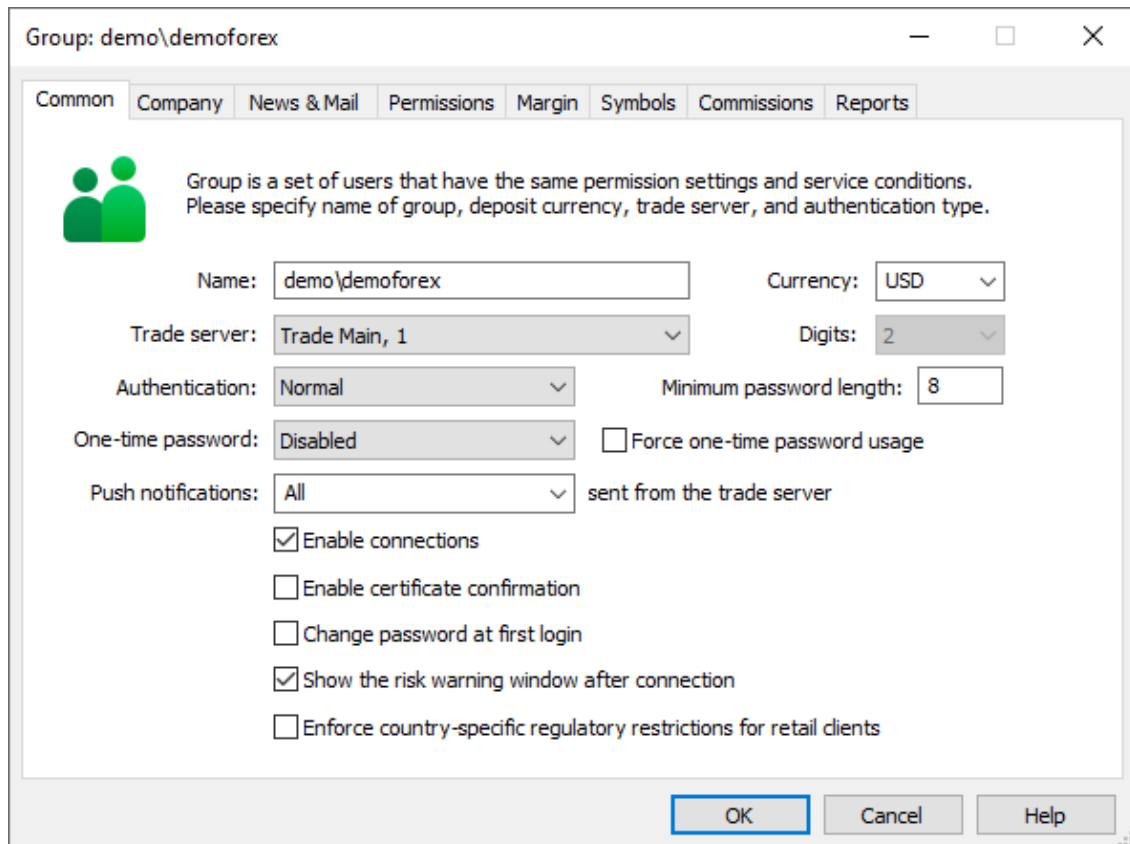
- **Servers** using this command, one can filter the groups displayed in the list by the trade server they belong to. A list containing all trade servers of the platform will be opened as soon as it is executed.
-  **Add** add a new group;
-  **Edit** edit a selected group;
-  **Delete** delete a selected group;
-  **Move Up** move a selected group up relative to others;
-  **Move Down** move a selected group down relative to others;
-  **Sort Alphabetically** sort configurations alphabetically. Please note that sorting is done on the server, not in the local terminal.
- **Automation triggers** create an automation task for the selected event or edit an existing one. The menu displays only the triggers and tasks associated with the current section.
- **Automation actions** add an automation action to an existing task or create a new task based on the action. The menu displays only the actions associated with the current section.
-  **Export to File** export group settings to a file.
-  **Import from File** import group settings to a file.
-  **Import from Server** start importing groups from a remote MetaTrader 5 or MetaTrader 4 server;
- **Request** open the sub-menu for requesting journal entries for the selected group:
 -  **Journal** request all journal entries on the group;
 - **Orders** request journal entries about the orders of the group;
 - **Deals** request journal entries about the deals of the group;
 - **Positions** request journal entries about the positions of the group;
 -  **Find** open the search window;
 - **Auto Arrange** if this option is enabled the size of columns is selected automatically;
 - **Grid** this option shows/hides field separators in the table with groups.

MetaTrader 5 Trading Platform / Platform Setup / Groups / Group Settings

Group Settings

The window of group settings is opened when you add or edit a group. It contains eight tabs:

Common



The screenshot shows the 'Group: demo\demo forex' settings window with the 'Common' tab selected. The window title bar includes standard minimize, maximize, and close buttons. The 'Common' tab is highlighted among eight tabs: Common, Company, News & Mail, Permissions, Margin, Symbols, Commissions, and Reports. On the left, there is a green icon of two people. To the right of the icon, a text box explains: 'Group is a set of users that have the same permission settings and service conditions. Please specify name of group, deposit currency, trade server, and authentication type.' The settings are as follows: Name: 'demo\demo forex', Currency: 'USD' (dropdown), Trade server: 'Trade Main, 1' (dropdown), Digits: '2' (dropdown), Authentication: 'Normal' (dropdown), Minimum password length: '8' (text box), One-time password: 'Disabled' (dropdown), and a checkbox for 'Force one-time password usage' which is unchecked. Below these, 'Push notifications' are set to 'All' (dropdown) with the text 'sent from the trade server'. At the bottom, there are six checkboxes: 'Enable connections' (checked), 'Enable certificate confirmation' (unchecked), 'Change password at first login' (unchecked), 'Show the risk warning window after connection' (checked), and 'Enforce country-specific regulatory restrictions for retail clients' (unchecked). At the bottom right, there are three buttons: 'OK' (highlighted with a blue border), 'Cancel', and 'Help'.

Group: demo\demo forex

Common Company News & Mail Permissions Margin Symbols Commissions Reports

Group is a set of users that have the same permission settings and service conditions. Please specify name of group, deposit currency, trade server, and authentication type.

Name: demo\demo forex Currency: USD

Trade server: Trade Main, 1 Digits: 2

Authentication: Normal Minimum password length: 8

One-time password: Disabled ☐ Force one-time password usage

Push notifications: All sent from the trade server

☒ Enable connections

☐ Enable certificate confirmation

☐ Change password at first login

☒ Show the risk warning window after connection

☐ Enforce country-specific regulatory restrictions for retail clients

OK Cancel Help

General settings

- **Name** group name and path to it. If a group is created/edited in the root section, then only its name should be specified. A group can contain only upper case and lower case letters, digits, symbols "_" (underline) and "-" (hyphen). The group name defines its type.
- **Currency** the deposit currency of this group. The drop-down list contains the most frequently used currencies. If the required one is not available in the list, specify it manually.
- **Digits** the number of digits after the decimal point in the deposit currency. The accuracy of the standard deposit currencies, such as USD, EUR, GBP, JPY, CHF, RUR, etc., is predetermined by the platform. It cannot be changed. When using other deposit currencies, the accuracy can be changed manually. This parameter affects the display of the accounts' trading status in the terminals, including the balance, equity, margin, etc.
- **Trade server** one of trade servers added in the Network section, which services this group.

Security settings

- **Authentication** method of authentication for this group:
 - ◆ **Normal** normal authentication via login and password.
 - ◆ **1024-bit RSA SSL certificate** extended authentication via certificates with 1024-bit encryption.
 - ◆ **2048-bit RSA SSL certificate** extended authenticator via certificates with 2048-bit encryption.
- **Custom SSL certificate** extended authentication via custom certificates.
- **Minimum password length** minimal length of passwords for accounts created in the group, and for certificates used in the extended authorization. The maximum possible password length is 16 characters.
- **Enable certificate confirmation** if you check this field, all generated client certificates will require confirmation.
- **Change password at first login** if this option is enabled, the forced change of master password will be required for each account created in this group. Any action is prohibited for the account until its master password is changed. Before enabling this option, make sure that the group is allowed to change passwords ("Allow to change password" option is enabled). Otherwise, clients will not be able to connect to the server, because they will not be able to change the password. After changing the primary password, the investor password is also reset. The user will have to specify a new password using the terminal setting.

One-time passwords

The use of one-time passwords (Time-based One-time Password Algorithm, TOTP) or two-factor authentication (2FA) provides an additional level of security for trading accounts. This technology protects a trading account from unauthorized access even if its login and password are leaked.

If 2FA/TOTP is enabled, users are required to enter a special one-time password every time they connect to their accounts, in addition to standard account login and password (as well as a certificate if the extended authentication is enabled). One-time

passwords can be generated by:

- Mobile terminals for iPhone and Android.
- The most popular 2FA applications, including Google Authenticator, Microsoft Authenticator, LastPass Authenticator and Authy. They can be downloaded to a mobile device from the App Store or Google Play.

One-time passwords are supported not only in all client terminals (Desktop, Mobile, Web), but also in Manager and Administrator terminals. This capability additionally protects platform access within your company.

Enabling: One-time passwords can be enabled for all types of connections, as well as for connections via the web terminal only. Select the necessary option in the "One-time password" parameter.

For further details about one-time passwords please visit the "2FA/TOTP" section.

Mandatory usage: In some countries, regulators require the use of additional account security measures, such as the use of OTPs. When the "Force one-time password usage" option is used, all clients in this group will need to use one-time passwords to connect. Otherwise, clients can bind their accounts to the generator or use the default authentication method. Before you enable it, please inform your clients about the new OTP option. Be extremely careful enabling this option for the only one manager group on the trading server.

Usage in the web platform: The MetaTrader 5 platform supports extended authentication via certificates. It protects users but works only on desktop and mobile terminals. For web terminal users, only one-time passwords can be used as additional means of protection.

If you enable one-time passwords for everyone, those who use certificates will experience unnecessary difficulties when connecting. They will have to pass a triple authentication: enter their standard login/password, provide a certificate and specify a one-time password.

In order to protect web terminal users without causing difficulties for other users who already apply certificates, enable the "TOTP for Web only" mode. Now, if you enable the "Force one-time password usage", one-time passwords become mandatory for web terminals (other connections are not affected).

- OTP use can be disabled for individual clients in the group.
- Accounts with OTP cannot use Virtual Hosting. The service is designed for uninterrupted operation of terminals; therefore, it is not possible to request a one-time password from the user for each authorization.
- An account can only be linked to one OTP generator.

Push notifications

MetaTrader 5 trading platform

The platform allows you to promptly notify traders of transactions taking place on their accounts. There are two standard tools:

Push notifications from the local terminal. A trader just needs to enable the corresponding option and specify the MetaQuotes ID in the terminal settings to receive messages about all transactions on the account to his or her mobile device.

Push notifications from the trade server. The advantage of this option over the previous one is that traders do not need to keep their terminals connected at all times to receive notifications since they are sent by the server. To configure them, traders just need to enable the corresponding option in their terminals and specify the MetaQuotes ID. After that, the terminal subscribes the current account to server notifications.

Brokers can enable/disable notifications for each group, as well as customize their level of details. The following types of notifications are currently provided:

- Placed orders
- Performed deals
- Balance operations (top-ups, withdrawals, credit operations, bonuses, etc.)

For example, it is possible to send notifications only about deals or only about balance operations on the account.

When subscribing, the terminal checks which notifications are available for the group the account belongs to. When enabling notifications, available types will be displayed in the terminal journal:

'1222': subscribed to deals, orders, balance notifications from trade server

To avoid unnecessary load on the trade server, notifications are sent only to real group accounts. No notifications are sent to demo accounts.

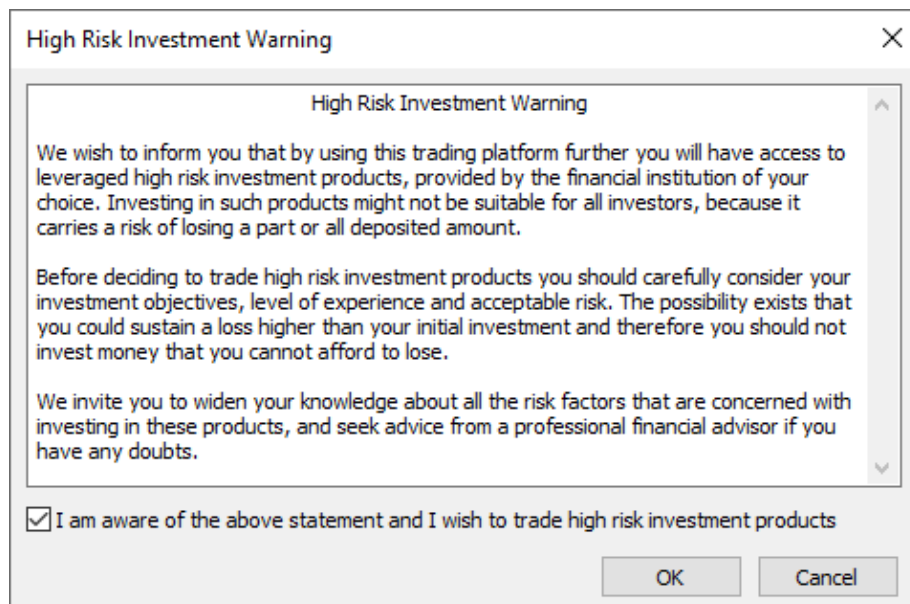
Unlike SMS messages, push notifications are sent via the Internet and are independent of cellular operators. This means brokers do not need to pay for provider services. Besides, this solves the issue of delivery to different regions. Traders only need to install the mobile terminal for iPhone or Android. Please see the details in the article "[How to get rid of SMS messages? Use MetaTrader 5 push notifications!](#)".

Other settings

Enable connections this option allows to permit/prohibit the client connection of accounts within the group. If the option is disabled none of the account within the group will be able to connect to the server. Disabled groups are shown with the  icon in the list.

Show the risk warning window after connection if this option is enabled, when a client connects in the trading terminal, a warning about the risks associated with operations on the financial markets appears. Trade operations on a client's account are not allowed until the client confirms that he or she has read the warning and is

aware of the risks. To confirm, the client should check "I am aware of the risks and I wish to trade high risk investment products". This warning is displayed once per session of the terminal. The next time it will appear after the restart of the terminal.



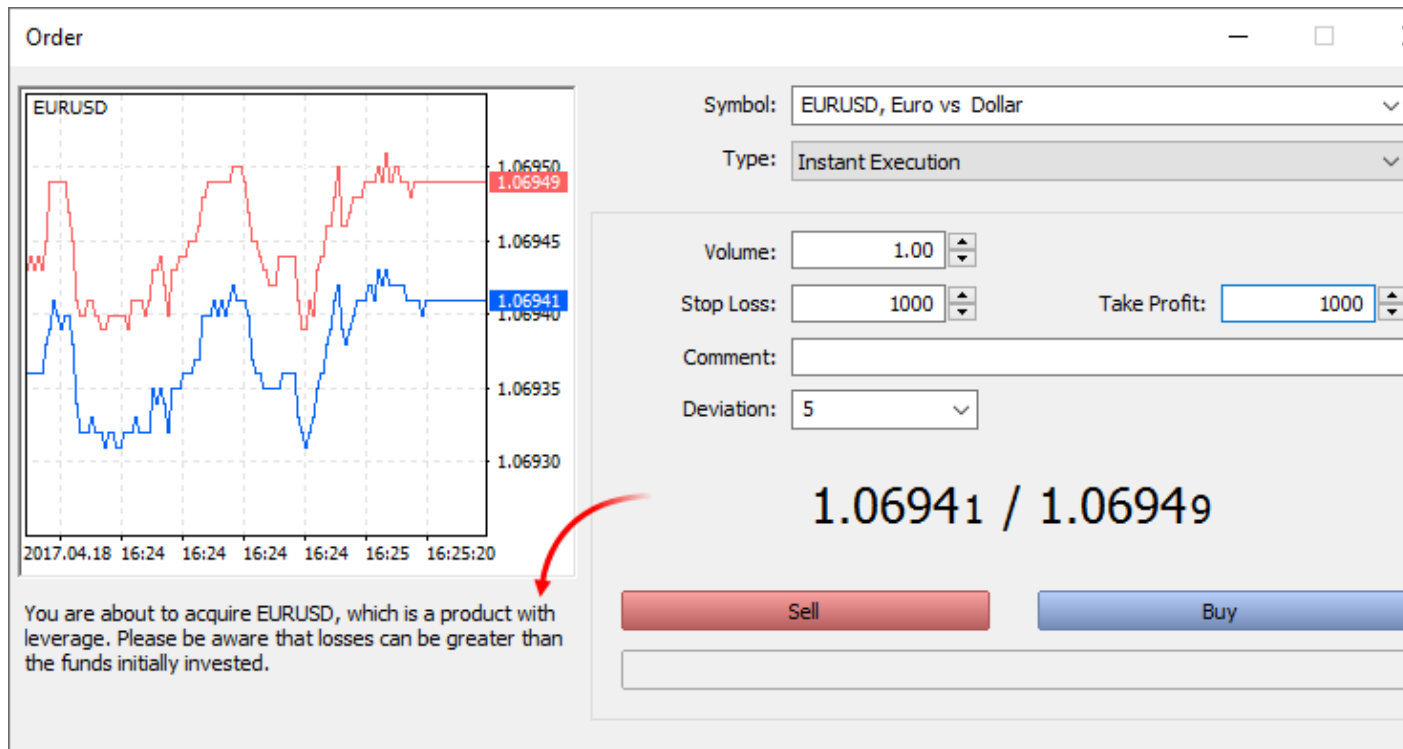
The risk warning is not displayed for demo accounts.

Enforce country-specific regulatory restrictions for retail clients. Every country has regulators – state bodies that supervise the activities of financial institutions (including brokers). Each regulator has its own set of requirements applied to trading currencies, securities and other instruments.

In most cases, clients of brokerage firms are individuals from different countries. The platform provides the means of meeting the requirements of certain regulators without interfering with the work of traders not covered with these requirements. Restrictions applied to trader accounts depend on the client's country and other conditions set by that country's regulator.

For example, the National Securities Market Commission (Comisión Nacional del Mercado de Valores) of Spain compels brokers to warn clients using the leverage of 1:10 or higher about potential risks in a special way. Therefore, if a client's country is Spain and they use a leverage of 1:10 or higher, the additional warning is displayed in the trading dialog:

MetaTrader 5 trading platform



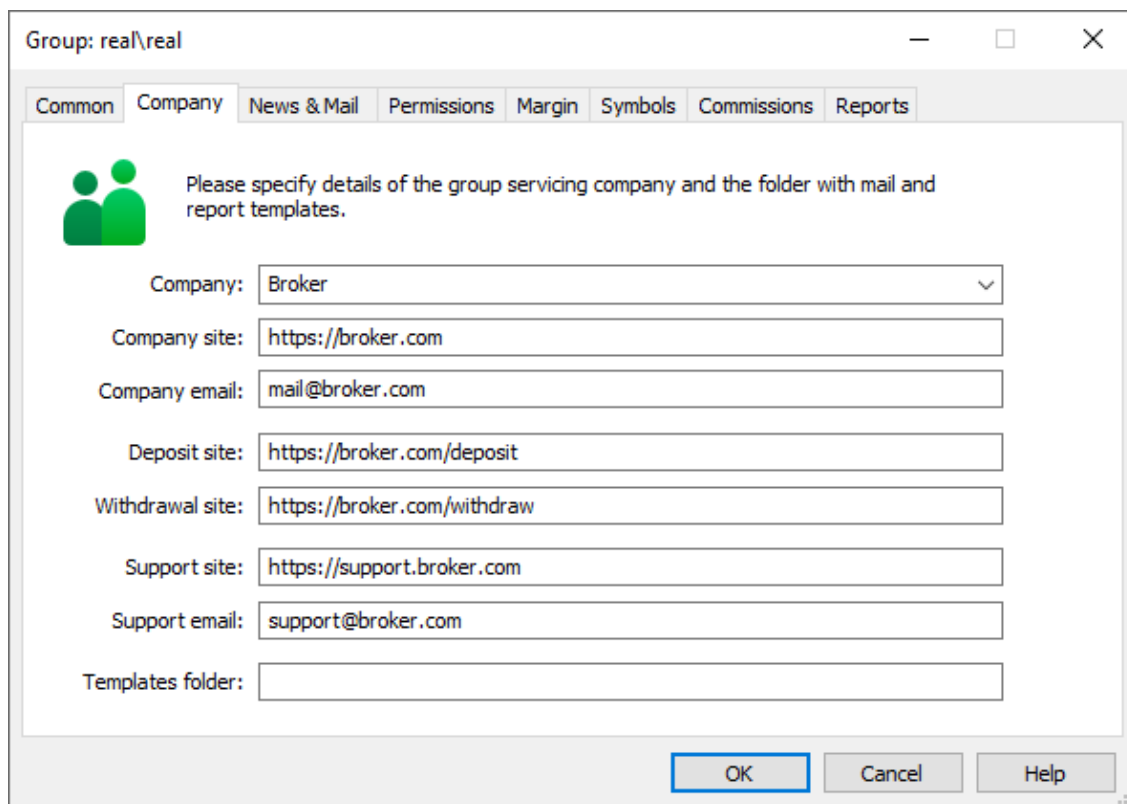
The trade dialog and the warning will appear even during one-click trading, for example when performing a trade operation from the Market Depth window or from the quick trading panel on the chart.

Currently, only one regional limitation is used in the platform. The list is to be expanded later.

- A group is only considered to be demo, if its name (including the path) contains "demo" (case sensitive). For example, "demoforex". The names of manager groups must contain "manager" characters.
- Use of 2048-bit encryption provides higher security, but slows down the process of key generation and checking, as compared to 1024-bit.
- If there is only one manager group in the system, extended authorization mode cannot be set to it. This will prevent loss of control over the platform in case the certificate is lost.
- You cannot change binding to a trade server of a group that contains accounts.

Company

MetaTrader 5 trading platform



Group: real\real

Common Company News & Mail Permissions Margin Symbols Commissions Reports

Please specify details of the group servicing company and the folder with mail and report templates.

Company: Broker

Company site: https://broker.com

Company email: mail@broker.com

Deposit site: https://broker.com/deposit

Withdrawal site: https://broker.com/withdraw

Support site: https://support.broker.com

Support email: support@broker.com

Templates folder:

OK Cancel Help

Information about the company that serves the group is specified in this tab. This information is used for generating reports, sending e-mails, etc.

- **Company** name of the company that serves the group. The field is mandatory. The list of companies available for selection is formed based on the trading platform license. This may be the company that owns either the main or additional [White Label](#).
- **Company website** website address.
- **Company email** email address.
- **Deposit site, Withdrawal site** URLs of your website pages in which traders can deposit or withdraw money from their trading accounts. If you fill these fields, the "Deposit" and "Withdraw" commands in the terminals will become available to the accounts from this group. A click on these commands opens the relevant pages of your website. Group-specific deposit and withdrawal page setting are of higher priority compared to global settings specified in the "[Account allocation](#)" settings. Links to withdrawal and deposit pages are only displayed for real accounts. These links are hidden for demo accounts. For more details, please see the "[Depositing and Withdrawing Funds](#)" section.
- **Support site** address of the website of the technical support team. The specified page will be displayed in the "Company" tab of the "Toolbox" window in the client terminals. If the address is not specified, <https://www.mql5.com> is used.

When specifying the address, one can use macros that are replaced on the client terminal side forming the final address:

- [lang:en|ru|es|..] list of languages in ISO 639-1 format. If one of the languages in the list matches the one of the client terminal's interface, the latter one is used. Otherwise, the first one from the list is applied.
- [login] client's current login.
- [uniq] unique ID of the client's PC.

- **Support email** email address of the technical support team. This address is used for receiving copies of daily reports if the relevant option is enabled for the group.
- **Templates folder** name of the sub-folder in the standard folder of templates in the server catalog where templates of reports, emails, etc. for this company are stored. If this field is empty, standard templates will be used.

News & Mail

Group: demo\demoforex

Common Company **News & Mail** Permissions Margin Symbols Commissions Reports

 Please specify the settings of news received by the group and the possibility of using the mail system.

News: full package

News categories:

News languages: Auto select Change

☒ Enable internal mail system

OK Cancel Help

In this tab you can configure parameters of news received by a group, and manage use of the inner mailing system:

- **News** setup of news receipt. A group can receive either all news messages (full package), or only news headers, or they can receive no news at all;
- **News categories** names of news categories that are received from data feeds. The group will receive news only from the categories that are specified in this field. An empty value means that all categories are allowed. Categories are indicated as a comma separated list. The backslash character "\" is used to indicate subcategories.
- **News languages** field for filtering news messages by languages. To edit the list of news, press "Change". If "Any language" is selected, the group users will receive all news items from allowed categories irrespective of their language. Window for selecting the group news language and that of selecting the language of news that are received in the terminal, are similar;
- **Enable internal mail system** if this field is checked, the group users will be allowed to use the internal mail system.

News Category and Language

- Usually, news categories are provided by data feeds. News providers supplement news with the information about categories to which the news belong. The same applies to the news language.
- A news category can be additionally specified in the News Category parameter of the data feed. This is a standard parameter supported by all data feeds. The parameter is applied as follows:
 - If a data source does not provide news categories, news items will be added to the category specified in this parameter (the category will be specified in the news).
 - If the data source provides a category, then the data feed will add the "News Category" parameter value before the original category name. The resulting category will look like "[value from News Category] \ [category from data source]". The "\" characters in a category name are interpreted as subcategory separators. Therefore, it means that all news items are added to the category specified in the parameter, within which original categories from the data source will be used.

Permissions

Group: demo\demo forex

Common Company News & Mail **Permissions** Margin Symbols Commissions Reports

Please specify the group permissions for symbols, orders, use of Expert Advisors, etc.

Maximum symbols: unlimited Available history: All

Maximum positions: unlimited Maximum orders: unlimited

Deposit by default: Leverage by default:

Annual interest rate: 0.0000 %

Trading Signals: Enable all signals from all brokers

Transfer of funds: Disable transfer of funds between accounts

Enable trading by Expert Advisors ☒ Enable position closing according to FIFO rule ☐

Enable trailing stops ☒ Prohibit hedge positions ☐

Enable charge of swaps ☒ Enable deal cost calculation ☐

Inactivity period: days, orders and positions will be deleted after inactivity period

OK Cancel Help

In this tab group permissions are set up:

- **Maximum symbols** maximal number of symbols, whose quotes an account can receive simultaneously. This option allows reducing load on the server if there are many symbols.
Client terminals only receive quotes for the financial symbols enabled in the Market Watch window. The "Maximum symbols" parameter sets the number of first instruments in the Market Watch window, for which terminals will receive quotes. For example, if you specify 10, terminals will receive quotes for the first ten symbols in the list. Traders can change the order of symbols and thus enable or disable delivery of quotes for them.
- **Maximum positions** maximum number of open position a client can have on a trade account at the same time. The limitation operates differently depending on the position accounting type:
 - ◆ **Netting** number of open positions is considered. When a limit is reached, the platform disables placing new orders whose execution may increase the number of open positions. In fact, the platform allows placing orders only for the symbols that already have open positions. The current pending orders are not considered since their execution may lead to changes in the current positions but it cannot increase their number.
 - ◆ **Hedging** pending orders are considered together with open positions, since a pending order activation always leads to opening a new position. When a limit is reached, the platform disables placing both new market orders for opening positions and pending orders.
- **Maximum orders** maximal number of pending orders that can be simultaneously placed by one account included into this group. The limit can

also be configured at the account level.

- **Available history** maximal number of days, trading history for which can be requested. If the limit is set for a group, the history of orders and deals in terminals will be available to the relevant accounts only within the specified depth.
- **Annual interest rate** annual interest on deposits of accounts included into this group. Interest is calculated at the end of each trading day on the amount of free assets (Free Margin) of a client excluding credit assets. During a month the amount of calculated interest is accumulated. Then at the end of the month (the end of the last day of the month) the accumulated amount of interest is added to the client account as a separate deal of the INTEREST RATE type;
- **Deposit by default** deposit that will be used when creating demo accounts in the group via the client terminal (regardless of the value selected in the account open dialog). The setting affects only demo groups;
- **Leverage by default** the leverage amount to be used for demo accounts created in this group from client terminals. To enable traders to choose a desired leverage, leave this field empty and specify the list of available leverage options in account allocation settings.
- **Trade signals** allow/prohibit using the "Signals" service in the client terminals.
- **Disabled** if this option is selected, the signal settings (the "Options" window) and the signal list (the "Toolbox" window) are hidden in client terminals connected using the accounts from this group. Users will not be able to subscribe to any signals both from the terminal and via www.mql5.com.
- **Enable all signals from all brokers** this option allows using the "Signals" service in trading terminals without limitations.
- **Enable signals from my servers only** if this option is selected, clients in this groups will be able to subscribe only to the signals created on the basis of accounts opened in your brokerage company. Signals created on the basis of other accounts will not be displayed in the client terminals. If no signal is left on the showcase according to this limitation, the full showcase with signals from all brokers will be displayed.
- **Transfer of funds** clients can transfer funds between accounts via the client terminal. Using this parameter, you can disable or limit the transfer of funds:
 - ◆ **Disable transfer of funds between accounts** transfer of funds is disabled.
 - ◆ **Enable between accounts with the same name** both accounts must have the same name and e-mail.
 - ◆ **Enable between account from the same group and sub-groups** the account to which the funds are transferred must be either in the same group with the source account or in its subgroups. Examples when a transfer is possible: both accounts are in the "real" group; the source account is in the "real" group, the target account is in the "real\ecn" group. Examples when a transfer is not possible: the source account is in the "real\ecn" group, the target account is in the "real\ndd" group; the source account is in the "real\ecn" group, the target account is in the "real" group.
 - ◆ **Enable between account with the same name and group** both conditions described above must be met.

Also, transfer of funds is subject to the following general requirements:

- Funds can be transferred only within the same trading server and only between the accounts of the same type. From a real account funds can be transferred only to another real account, from a demo one - only to demo.
- The accounts, between which funds are transferred, should use the same deposit currency.
- The transfer amount should not exceed the current balance and free margin on the account from which money is transferred.

- **Enable charge of swaps** allow charging swaps for this group;
- **Enable trailing stops** allow using trailing stop in the client terminal. The use of trailing stops can be set up for each account separately;
- **Enable trading by Expert Advisors** allow the group accounts to trade using Expert Advisors. This parameter can be set up for each account separately.
- **Close positions by FIFO rule** the mode of position closing according to a FIFO rule. If this option is enabled, then positions for each instrument can only be closed in the order in which they were opened: the oldest one should be closed first, then the next one, etc. The option is only valid for hedging accounts, in which traders can have multiple positions for the same financial instrument. There are three main methods to close a position; the option behavior will be different for each of the methods:
 - ◆ Closing from the client terminal: the trader closes the position manually, using a trading robot, based on the Signals service subscription, etc. In case of an attempt to close a position, which does not meet the FIFO rule, the trader will receive an appropriate error.
 - ◆ Closing upon Stop Loss or Take Profit activation: these orders are processed on the server side, so the position closure is not requested on the trader (terminal) side, but is initiated by the server. To prevent non-FIFO closing, the order in which stop levels are placed on the client side is changed. If multiple positions exist for the same instrument, the placing of stop levels for any of them causes analogous levels to be placed for all others. Accordingly, if a level triggers, all positions will be closed in a FIFO compliant order.
The same behavior is used when modifying and deleting stop levels.
 - ◆ Closing upon Stop Out triggering: such operations are also processed on the server side. In a normal mode, in which FIFO-based closing is disabled, in case of Stop Out positions are closed starting with the one having the largest loss. If this option is enabled, the open time will be additionally checked for losing positions. The server determines losing positions for each symbol, finds the oldest position for each symbol, and then closes the one which has the greatest loss among the found positions.
- **Prohibit hedge positions** prohibit opening of opposite positions and placing of opposite orders. If this option is enabled, accounts are not allowed to have oppositely directed positions and orders for the same financial instrument. For example, if the account has a Buy position, then the user cannot open a Sell position or place a pending sell order for the same symbol. If such an attempt is made, the user will receive an error stating that hedging is prohibited. The option is only valid for hedging accounts.

- **Enable deal cost calculation** calculate the cost of deal execution and display it in client terminals. This option should be enabled by NFA regulated brokers.

In accordance with the regulatory requirements, brokers must explicitly provide to traders information about the costs of each deal relative to the current mid-point price of the symbol (mid-point spread cost). Actually, this value means the amount the trader loses on traded spread.

The value is calculated according to the following formulas.

Forex:

For Buy deals: $\text{Normalize}((\text{Market Bid price} + \text{Market Ask price})/2 * \text{Contract size} * \text{Volume in lots}) - \text{Normalize}(\text{Deal price} * \text{Contract size} * \text{Volume in lots})$

For Sell deals: $\text{Normalize}(\text{Deal price} * \text{Contract size} * \text{Volume in lots}) - \text{Normalize}((\text{Market Bid price} + \text{Market Ask price})/2 * \text{Contract size} * \text{Volume in lots})$

CFD, CFD Index, CFD Leverage and Exchange Stocks, Exchange MOEX Stocks:

For Buy deals: $\text{Normalize}((\text{Market Bid price} + \text{Market Ask price})/2 - \text{Deal price}) * \text{Contract size} * \text{Volume in lots}$

For Sell deals: $\text{Normalize}(\text{Deal price} - (\text{Market Bid price} + \text{Market Ask price})/2) * \text{Contract size} * \text{Volume in lots}$

Futures, Exchange Futures, FORTS Futures, Exchange Options, Exchange Margin Options:

For Buy deals: $\text{Normalize}((\text{Market Bid price} + \text{Market Ask price})/2 - \text{Deal price}) * \text{Contract size} * \text{Volume in lots} * \text{Tick value} / \text{Tick size}$

For Sell deals: $\text{Normalize}(\text{Deal price} - (\text{Market Bid price} + \text{Market Ask price})/2) * \text{Contract size} * \text{Volume in lots} * \text{Tick value} / \text{Tick size}$

Calculation uses the Bid and Ask prices that existed in the market at the deal execution time. When the value is calculated for historical operations, the formula uses market Bid and Ask prices stored in deals. Please note that the saving of these values was introduced in platform build 2890. The relevant values are not available for the deals which were executed earlier. Therefore, cost calculation is impossible for them.

The deal size is determined depending on the trading instrument type, similarly to profit calculation. For Forex and CFD instruments, the pip level is equal to the contract size multiplied by the trade volume; for futures it is the ratio of price to tick size multiplied by the trade volume, etc.

The value derived from the formula is the deal cost in the instrument's profit currency. If it differs from the account deposit currency, the amount is converted using the "Profit rate" value of the deal.

The calculated value is displayed in the "Cost" column in the deal history in client terminals.

If the option is disabled, the calculation is not performed, and the

MetaTrader 5 trading platform

corresponding column in the terminals is hidden.

- **Inactivity period** this function is only available for demo accounts. This feature enables automatic clearing of databases from irrelevant data and thus reducing the load on the platform. If a demo account from this group was not connected to the server for the specified number of days, all its open orders and positions will be deleted from the databases. The default inactivity period is equal to 360 days. To disable deletion of operations, sets the parameter to 0.

Inactivity check is performed every Sunday at optimization time for all accounts in the group, including newly created and existing ones. For example, if the option was disabled for the group and then you set the value of 30, then at the specified time the server will check all accounts in the group. If the server finds any accounts which have not connected to the platform for the last 30 days, all operations of these accounts are deleted from the database. The accounts are not deleted this process is managed by a separate setting on the trade server.

Margin

Group: real\real

Common Company News & Mail Permissions **Margin** Symbols Commissions Reports

Please set up the mechanism of margin calculation for the group: choose the method of calculation and margin requirements.

Risk management: for Retail Forex, CFD, Futures

Margin call level: 50.00 Stop out level: 30.00 in %, percent

☐ Stop out fully hedged accounts

☐ Compensate negative balance after stop out

☐ Withdraw credit after negative balance compensation

Floating leverage profile: None

Profit/loss in free margin

Unrealized profit: use unrealized profit/loss

Daily fixed profit: use daily fixed profit/loss

☐ Release fixed profit at the end of day

Virtual credit: (applies only to opening new positions)

OK Cancel Help

Margin settings for the group are set up in this tab:

- **Risk management** the risk management model defines the type of pre-trade control and the position accounting system used. Please see the appropriate section for details. After changing this parameter the server should be restarted.
 - ♦ **for Retail Forex, CFD, Futures** used for the OTC market. Margin calculation is based on the type of instrument, as well as group settings. Netting position accounting system is used.
 - ♦ **for Retail Forex, CFD, Futures with hedging** used for the OTC market. Margin calculation is based on the type of instrument, as well as group settings. Hedging position accounting system is used.
 - ♦ **for Stock Exchange, based on margin discount rates** used for the exchange market. Margin calculation is based on the discounts specified in symbol settings. Discounts are set by the broker, however they cannot be lower than the exchange set values.
- **Margin call level** when a client reaches this level of assets, they are in the "Margin call" state. Clients under Margin call are displayed in a special window in the manager terminal. Thus a manager can send a notification to clients that they should replenish their accounts. The Margin call state does not affect the ability of opening new positions. Only used for the 'for Retail Forex*' modes. In the 'for Stock Exchange' mode, a Margin Call occurs when the initial margin on the client's positions becomes greater than the equity.
- **Stop out level** when a client reaches this level of assets his/her pending orders are deleted and open position are closed forcibly. The order of processing a Stop out is described below. Only used for the 'for Retail Forex*' modes.
In the 'for Stock Exchange' mode, a Stop Out occurs when the maintenance margin on the client's positions becomes greater than the equity.
- **In percent/money** in this field you can select have the Margin call and Stop out levels are indicated. The "In money" option means that the levels are set as a value of the "Equity" parameter of an account. The "In percent" option means that the levels are set as a value of the "Margin level" parameter ($\text{Equity/Margin} \times 100$) of an account.
- **Stop out fully hedged accounts** if this option is enabled, then Stop out will be performed on accounts having open positions, the zero margin (positions are covered) and negative equity. If the option is disabled, orders will not be forcibly deleted and positions will not be forcibly closed in above cases. The option can be only used on hedging accounts.
- **Compensate negative balance after stop out** automatically perform on a client's account the special operation, which increases the balance and sets it to zero, if the balance has become negative after a position was closed by Stop Out. The details are available below.
- **Withdraw credit after negative balance compensation** works as an addition to the previous option. If enabled, the credit funds on the account will be set to zero after a negative balance compensation operation. Credit funds are withdrawn in a separate balance operation with the "so credit compensation" type.
- **Floating leverage profile** additional margin calculation rules for clients in the group. You can create multiple margin profiles in the Leverages section and

then quickly switch between them manually or automatically.

- **Virtual credit** additional money that a broker can allocate to a client for opening a position of larger volume than allowed by his/her assets. However, the virtual credit is not used for the calculation of free margin to support open positions. Besides, a trader cannot withdraw this money.

- If a group has clients with open positions, then position accounting type cannot be changed for this group.
- It is not recommended to move accounts between groups with different position accounting systems.

Processing of Stop out by Server

When Stop outs are processed automatically by the server (not by a plugin or a manager), then orders and positions are forcedly closed on an account in the following order:

- The server analyzes client's pending orders that are not under execution at the moment.
- It deletes a pending order with the largest margin reserved.
- If Equity (or Margin level, depending on how the SO level is determined) is still under the Stop out level, the next order is deleted. Orders without a reserved margin (no margin requirements) are not deleted.
- If Equity (or Margin level) is still under the Stop out level, the server closes a position with the largest loss. If all of the positions are profitable, the position with the lowest profit is closed (excluding swaps and commissions).
- Positions are closed until Equity (or Margin level) becomes higher than the Stop out level.
- Netting accounts allow partial position closure by Stop out if the volume of the position to be closed exceeds the maximum allowable volume for the symbol. For example, a trader has a 50-lot position, while the maximum permissible deal volume is 20. In the event of a Stop out, the platform will first close 20 lots of the position. If after this the client's funds level is still below the threshold, the platform will close the next 20 lots. Otherwise, the trader will have part of the position remaining open. In this case, the position comment will indicate [so XX%], where XX is the margin level at which Stop out occurred.

For more details about how the Stop out is processed on the server side please read the MetaTrader 5 API documentation.

Compensation of a negative balance after a Stop Out

If positions are forcedly closed in the situation of a Stop Out, the balance of a client can become negative. This may happen during sharp market moves. According to the rules set out by some Regulators, for example by the European Securities and Markets Authority, clients' balances may not fall below zero. Therefore, you should enable the 'Compensate negative balance after stop out' for clients from countries where such rules apply.

MetaTrader 5 trading platform

The server forcibly closes the client's positions upon the situation of a Stop Out. If the compensation is enabled, and the client's balance has become negative after the closing of the last position, the 'so compensation' operation is immediately performed on this account to increase the balance to zero:

Account: 1814659, John Smith, EUR, 1:5000

Time	Deal	Order	Symbol	Type	E...	Volume	Price	C...	Profit	Comment
⊕ 2018.05.16 14:43:02	100315430			balance					100.00	Deposit
📄 2018.05.16 14:43:34	100315431	100293973	eurusd	buy	in	25.00	1.17772	-0.10	0.00	
📄 2018.05.16 14:44:42	100315432	100293974	eurusd	sell	out	25.00	1.17758	-0.10	-237.78	[so at -325.76%]
⊕ 2018.05.16 14:44:42	100315433			so compensation					137.98	so compensation
Profit: -100.00 Deposit: 100.00 Withdrawal: 0.00 Credit: 0.00 Balance: 0.00									-0.20 -99.80	

All history 1970.01.01 - 2018.05.17 Request

Update Cancel Help

An appropriate entry is made in a server log:

```
2018.05.16 14:55:42.834      '1814659': negative balance -137.98 compensated after stop o
```

The standard "[StopOuts Compensation Report](#)" allows controlling such compensation operations. This report includes all deals with the "so compensation" type.

The compensation function works only if the client's balance has become negative as a result of a Stop Out. If the balance becomes negative as a result of another operation, for example the Stop Loss, compensation is not performed.

Profit/loss in free margin

In this box, you can set up how profit/loss should be considered when calculating the free margin of a client:

- **Unrealized profit** in this field, select whether the floating profit/loss should be included in free margin or not:
 - ◆ **Do not use unrealized profit/loss** do not take into account open positions;
 - ◆ **Use unrealized profit/loss** in calculation use loss and profit of open positions;
 - ◆ **Use unrealized profit** use only profit;
 - ◆ **Use unrealized loss** use only loss;

- **Daily fixed profit** in this field, select whether to include the profit/loss fixed during a trade day in the free margin or not:
 - ♦ **Use daily fixed profit/loss** include both profit and loss fixed during a trade day in the free margin.
 - ♦ **Use daily fixed loss** include only loss fixed during a trade day in the free margin. During a day obtained profit is accumulated in the special field of the client record ("Accumulated" in the manager and administrator terminals, "Blocked" - in the client terminal). If the option "Release fixed profit at the end of day" is enabled, at the end of trade day the accumulated profit is released (set to zero) and added to the user balance (included in the free margin). This option only works with the netting position accounting.
- **Release fixed profit at the end of day** this option becomes available only if the option "Use daily fixed loss" is selected. If it is enable, then the profit accumulated by a client will be released (and thus included in the free margin) at the end of trade day. If this option is disabled, then it will be possible to release the accumulated only using an external application (for example, a gateway). The server will not perform this operation.

Symbols

Group: demo\demo forex

Common Company News & Mail Permissions Margin **Symbols** Commissions Reports

Please set up individual parameters of symbols trade for the group.

Up

Down

Symbol	Spread	Trade
Forex*	Default	Default
CFD*	Default	Default
Stocks*	Default	Default

Add

Edit

Delete

OK Cancel Help

MetaTrader 5 trading platform

This tab sets the availability of trading instruments for a specific group, as well as individual settings for these instruments. By default the tab contains the value of "*", which means that all instruments in the platform are available for the group.

- **Add** add a symbol or a group of symbols to the list of available instruments for the accounts from the specified group. After you press it, a window for adding and setting up symbols for this group will be opened.
- **Delete** delete a selected item.
- **Edit** edit a selected item. The same action can also be performed by clicking twice on an entry.

For the rule, you can specify multiple symbols separated by commas, or a group of symbols using the "*" mask and the negation sign "!". For example:

- Forex* all symbols from the Forex subgroup
- Forex*,!EURUSD all symbols from the Forex subgroup, except EURUSD
- Forex\EURUSD,Forex\GBPUSD only the specified symbols

The maximum length of the string with the enumeration of symbols is limited to 128 characters. The string cannot consist of only an excluding rule, such as "!Forex\EURUSD". It must contain allowed symbols, to which you can add an exception using "!". For example, Forex*, !Forex\EURUSD, !Forex\AUDCAD.

If several individual configurations are created for the same symbol (for example, there is a configuration for all symbols by mask "*", and a configuration for a specific symbol), then every configuration will be applied sequentially from the top downwards. And if a parameter in the configurations is set to "Default", it means that it won't be changed. For example:

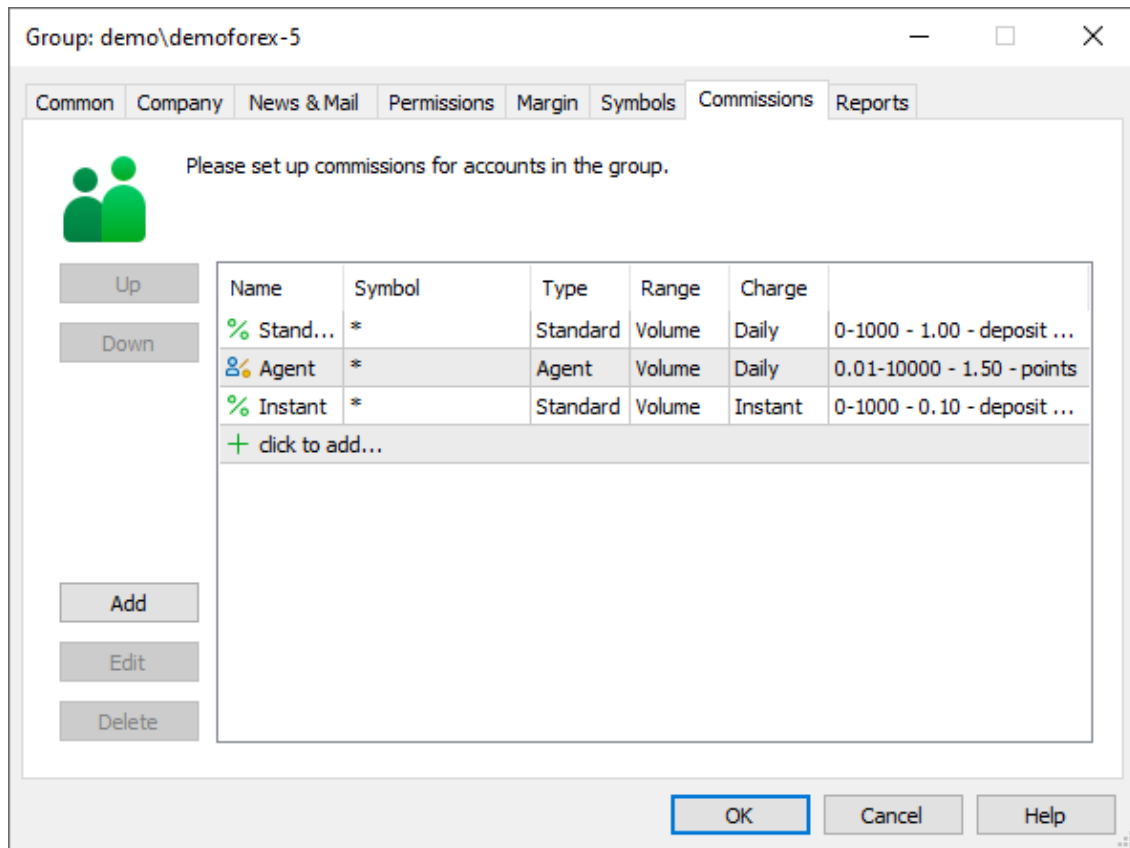
- A symbol has minimal and maximal volume set to 0.01 lot and 100 lots on default.
- A configuration is added for the group where the minimal and maximal volume is overridden to 0.1 lot and 150 lots respectively.
- Another configuration is added with settings "Use default volumes" and Spread difference = 5.

The last configuration will not reset the minimal and maximal volume to default 0.01 lots and 100 lots respectively. Values 0.1 and 150 will be used.

If the symbol row in the table is highlighted in red, then this symbol no longer exists. It may have been moved or deleted. Open the entry and edit the symbol path.

Commissions

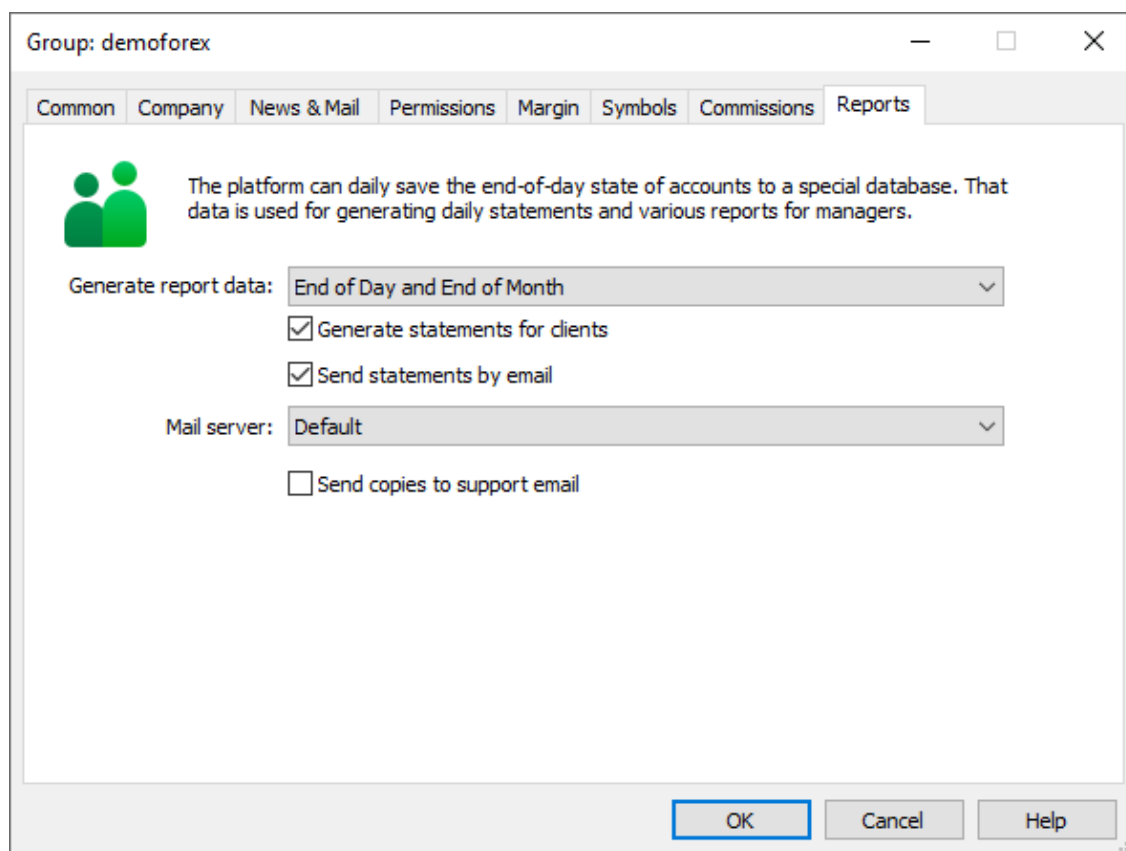
MetaTrader 5 trading platform



On this tab you can set up commissions charged from accounts included into this group. Names of commissions and symbols they apply to are specified here. The following buttons are intended for working with commissions:

- **Add** add a new commission type. A click on this button will open the commission setup window.
- **Delete** delete a selected item;
- **Edit** change a selected item. The same action can also be performed by clicking twice on an entry.

Reports



Here you can set up parameters of daily and monthly report generation. These are special reports that contain information about the state of client accounts at the end of a trading day or month.

- **Generate report data** the platform can save the end-of-day and/or end-of-month states of accounts to a special database. The information includes balance, equity, margin amount and other details. The database is kept on the trade server, in the bases\daily\daily_*.dat files. Data from the database is used in end-of-day and end-of-month trading reports sent to clients, as well as in some manager reports. If the option is enabled, the platform saves information on accounts from the selected group to the database. If the option is not enable, the below options are not available.
Three data saving options are provided: at the end of each day, at the end of each month, at the end of each day and each month.
- **Generate statements for clients** the option enables the generation of daily and monthly account state reports. The HTML reports files are created using templates available in the \templates\confirmation\ and \templates\statement\ folders. The type of reports to be generated, i.e. daily and/or monthly, depends on the "Generate report data" parameter. If there are no trading or balance operations on the account during the day, a report for that day will not be generated. However, monthly reports will be generated in this case.
- **Send statements by email** if the option is enabled the generated HTML report files will be sent to emails specified in trading accounts.
- **Mail server** configuration of the mail server to be used for sending daily and monthly reports to trader emails. If "Default" is selected, the platform will check the list of all servers and will use the first available server with the "Default

server" option enabled.

- **Send copies to support email** send copies of reports to the e-mail address of the technical support service specified at the "Company" tab.

- E-mail templates for daily and monthly reports on trade activity are located in the \templates\confirmation and \templates\statement folders of the trade server.
- To get information about the process of sending daily reports via e-mail, request trade server logs (Report Mailer).
- The report generation and sending can be disabled for separate accounts within a group.
- Trading accounts store data on the trader's balance and equity state as at the end of the previous trading day/month. When switching to the next trading day/month, the platform updates these values. To save resources, these values are not updated for demo accounts for which generation of daily/monthly reports is disabled.

To complete creating or editing a group, press "OK". If you press "Cancel", the window will be closed, while changes will not be saved.

MetaTrader 5 Trading Platform / Platform Setup / Groups / Group Symbol Settings

Symbol Settings of Groups

Group symbols are set up on the "Symbols" tab when creating or editing a group.

If several individual configurations are created for the same symbol (for example, there is a configuration for all symbols by mask "*", and a configuration for a specific symbol), then every configuration will be applied sequentially from the top downwards. And if a parameter in the configurations is set to "Default", it means that it won't be changed. For example:

- A symbol has minimal and maximal volume set to 0.01 lot and 100 lots on default.
- A configuration is added for the group where the minimal and maximal volume is overridden to 0.1 lot and 150 lots respectively.
- Another configuration is added with settings "Use default volumes" and Spread difference = 5.

The last configuration will not reset the minimal and maximal volume to default 0.01 lots and 100 lots respectively. Values 0.1 and 150 will be used.

When you add or edit symbols, a window with five tabs will appear:

- **Common**
- **Trade**
- **Execution**
- **Margin**
- **Margin Rates**
- **Swaps**

MetaTrader 5 Trading Platform / Platform Setup / Groups / Group Symbol Settings / Common

Common

Symbol: *

?

×

Common

Trade

Execution

Margin

Margin Rates

Swaps



Please set up main parameters of symbols for the group.

Symbol: *

▼

☒ Enable market depth

Market depth limit: unlimited ▼

☒ Use default spread

Spread difference: 0 ▼ pt

Difference balance:  0 bid / 0 ask

☒ Use default volumes

Minimum: 1.00 ▼

Step: 1.00 ▼

Maximum: 1000.00 ▼

☒ Use default limit

Limit: 0.00 ▼

OK

Cancel

Help

Common settings of a symbol or group of symbols are set up on this tab:

- **Symbol** select a symbol or group of symbols;
- **Enable market depth** allows disabling the depth of market for the client group. If disabled, then a scalper depth of market instead of the exchange one will be used for the symbols. The scalper depth of market shows best Bid/Ask prices as well as a range of prices above and below them generated with the same step (minimal price step), which can be used to quickly place pending orders.
- **Market depth limit** this parameter allows limiting the number of orders in the Market Depth window displayed for this particular group. If the parameter is set to "unlimited", the depth specified in symbol settings ("Market depth") will be used.
- **Use default spreads** if this option is enabled, all the below parameters will become inactive and spread on these symbols (group of symbols) will be taken from their parameters in the corresponding section;
- **Spread difference** difference of a symbol spread for a certain group of users from the basic spread of the symbol (group of symbols);
- **Difference balance** balance of spread difference distribution between bid and ask prices. The volume specified in the "Spread difference" field will be distributed between bid and ask prices in the specified correlation. For example, if you set 3 as the spread difference, then the distribution can be the following: 3 bid/ 0 ask, 2 bid/ 1 ask, etc.
- **Use default volumes** if this option is enabled, volumes for this symbols will be taken from the symbol settings in the corresponding sections. If this option is enabled, all the below parameters will become inactive;
- **Minimum** minimally allowed volume of the symbol order. Not applied when closing positions.
- **Step** step of the volume change.
- **Maximum** maximally allowed volume of a placed order for these symbols. It is considered not only when traders place orders but also when a position is closed by stop out.
- **Use default limit** if this option is enabled, volume limit for this symbols will be taken from the symbol settings in the corresponding sections. If this option is enabled, all the below parameters will become inactive;
- **Limit** maximum allowed cumulative volume of an open position and pending orders by a symbol in one direction (buy or sell). For example, if the limit is set to 5 lots, a user can have an open buy position of 5 lots and place a Sell Limit order of 5 lots. But in this case the trader cannot place a pending Buy Limit order, since the total volume in one direction would exceed the limit. Also, a client having a 5-lot Buy position cannot place a Sell Limit order with a volume exceeding 5 lots: if the pending order triggered after the closing of the initial position, the client would have a short position exceeding the specified limit.

Price transformation settings (the Spread difference and the Difference balance) for a group are applied after applying base settings of a symbol.

MetaTrader 5 Trading Platform / Platform Setup / Groups / Group Symbol Settings / Trade

Trade

Symbol: Forex*

Common Trade Execution Margin Margin Rates Swaps

 The setting of symbol trading for the group. Please specify trade mode, filling and expiration modes, etc.

☐ Use default trade settings

Trade: Full access

Filling: Immediate or Cancel

Expiration: All

Orders: All

☒ Use default trade level settings

Limit & stop level: 80 pt Freeze level: 0 pt

OK Cancel Help

Individual parameters of symbols trading are specified on this tab:

- **Use default trade settings** if this option is enabled, all the below parameters will become inactive and the trade parameters for symbols (group of symbols) will be taken from their settings in the corresponding session;
- **Trade** trade settings for the group: disabled, long only, short only, close only or full access;
- **Filling** additional rules of order filling that can be set to traders. The necessary filling type should be ticked off:
- **Fill or Kill** this fill policy means that a deal can be executed only with the specified volume.
- **Immediate or Cancel** in this case a trader agrees to execute a deal with the volume maximally available in the market within that indicated in the order. In case the order cannot be filled completely, the available volume of the order will be filled, and the remaining volume will be canceled.
- **Book or Cancel** the order can only be placed in the Depth of Market (order book). If the order can be filled immediately when placed, this order is canceled.
- **Return** this mode is not available in the list. It is always enabled for the market (Buy and Sell) orders in the Exchange Execution mode, as well as for limit and stop limit orders in the Market Execution and Exchange Execution modes.
- **Expiration** condition for the expiration of orders. Necessary expiration types should be ticked off:
- **Good till canceled** the order will stay in the queue until it is manually canceled;
- **Day** the order will be effective only during the current trading day;
- **Specified time** the order will be effective till the date specified by the trader;
- **Specified day** the order is active till 00:00 of the specified day. If that time appears to be out of a trade session, the expiration will be processed at a nearest trading time.
- **Use default trade level settings** if this option is enabled, all the below parameters will become inactive and the trade level settings for symbols (group of symbols) will be taken from their settings in the corresponding session;
- **Limit & stop level** price channel (in points) from the current market price, within which it is prohibited to place Stop Loss, Take Profit and pending orders. At the attempt to place an order within this channel, the server will return message "Invalid S/L or T/P" and will not accept the order;
- **Freeze level** level for freezing orders that are close to the current price. When an order is as close as the value specified here or less than that, its modification, closing or deletion is prohibited.

MetaTrader 5 Trading Platform / Platform Setup / Groups / Group Symbol Settings / Execution

Execution

Symbol: *

Common Trade Execution Margin Margin Rates Swaps

 Please set up parameters of orders execution by symbols for the group.

☐ Use default execution settings

Execution: Instant

Max time deviation: 7 seconds

Max profit deviation: 2 points

Max losing deviation: 2 points

☐ Fast confirmation of requotes within client deviation

Maximum volume: 0.00 before switching to Request execution

OK Cancel Help

Execution parameters of orders of a financial instrument or group of instruments specified on the previous tab, are set up here.

If you enable "Use default execution settings", execution settings for these symbols (or group of symbols) will be taken from their settings in the corresponding section. All the below parameters will become inactive.

Instant

In this mode a client places an order at prices from the Market Watch window without requesting prices before that. The following settings are available for this mode:

- **Max time deviation** the maximal difference between the time when the price specified in the client's order was received, and the time of the latest price. If this limit is exceeded, requote is performed and new prices are sent to the client;
- **Max profit deviation** maximal deviation of the order price from the current symbol price in the direction profitable for a client. This value being exceeded, new prices will be sent to the client;
- **Max losing deviation** maximal deviation of the order price from the current symbol price in the direction losing for a client. This value being exceeded, new prices will be sent to the client;
- **Maximum volume** the maximum volume of the order that can be accepted in the instant execution mode. All clients' orders larger than that will be automatically transferred to the manual execution mode.

The following options are used only if the deal volume exceeds the maximum volume that can be accepted in the instant execution mode. Such deals are processed in the "Request" execution mode.

- **Timeout** time in seconds, during which the price provided by the dealer is valid;
- **Confirm orders** if this option is enabled, then after a trader received quotes from a dealer and agrees to execute a deal at them, the dealer will have to additionally confirm the execution of this order.

Price and requote checks are described in the relevant symbols section.

Request

In this mode, a client requests the filling price before placing an order. The following options are available here:

- **Timeout** time in seconds, during which the price provided by the dealer is valid;
- **Confirm orders** if this option is enabled, then after a trader received quotes from a dealer and agrees to execute a deal at them, the dealer will have to additionally confirm the execution of this order.

Market

MetaTrader 5 trading platform

In this mode, a client places an order at the broker's price, "agreeing with it in advance".

Exchange

In this execution mode a trader's request is passed to an external gateway connected to an exchange or another liquidity provider.

MetaTrader 5 Trading Platform / Platform Setup / Groups / Group Symbol Settings / Margin

Margin

Symbol: *

?

×

Common

Trade

Execution

Margin

Margin Rates

Swaps



Please specify margin requirements for the symbols group.

☐

Use default margin values

Initial margin:

Hedged margin:

Maintenance margin:

☐

Use default margin settings

☐

Calculate hedged margin using larger leg

☐

Exclude long position PnL from free margin and margin level

☐

Recalculate margin exchange rate at the End of Day

Additional margin checks:

All

▼

OK

Cancel

Help

This tab allows setting up margin parameters for a group by the symbol (group of symbols) specified on the "Common" tab.

- **Use default margin values/settings** use margin parameters specified for the symbol in the corresponding section.
- **Initial margin** amount of money (margin) deposited for futures contracts to ensure performance of the contract to execute a deal with the volume of 1 lot;
- **Maintenance margin** the minimal amount of money (margin) that must be available on a client's account to support a position with the volume of 1 lot;
- **Hedged margin** the amount of margin charged per each hedged lot of a position. If initial margin is set for an instrument, the hedged margin is specified as an absolute value (in margin currency) terms. If the initial margin is not set (equal to 0), the contract size is specified. The margin is calculated by the appropriate formula corresponding to the type of the instrument and using the specified contract size. For more details please see the special section.
- **Calculate hedged margin using larger leg** determines the hedged margin calculation mode: basic or using larger position leg.
- **Exclude long position PnL from free margin and margin level** the option affects the display of positions purchased with own funds, on accounts with margin trading enabled. If the option is enabled, profit/loss of positions for the symbol will be ignored when calculating free margin, margin level and account equity. Also, Stop Out will not be applies to such positions.
- **Recalculate margin exchange rate at the End of Day** if this option is enabled, at the end of the trading day, the server will update the position margin conversion rate at current market prices. For further details, please see Basic Margin Calculation.
- **Additional margin check** on default, the margin is checked when any order is placed and when a pending orders triggers. This parameter allows enabling additional margin checks:
 - **check before executing orders** in this mode, another check of margin is added to those described above: the margin is checked before executing an order after it is confirmed (checked) by the server (at automated execution), by a dealer or by a gateway.
 - **check on SL-TP trigger** enables an additional check of margin before a position is closed by stop loss or take profit. If the position close results in reducing the margin to a level insufficient to maintain open positions and orders, the stop loss/take profit will not trigger, the position will stay open. This check must be enabled in case the trade operations are transmitted to an external system (exchange).

- If the initial margin is specified, then no calculations by formulas specified in the "Calculation" field are performed.
- If the maintenance margin value is not specified, the value of the initial margin is used for it.
- If the exchange risk management model is used for the client group, and a symbol has 0 margin rates, then the margin is not charged for operations by that symbol.

The margin calculation is described in details in a [separate section](#).

MetaTrader 5 Trading Platform / Platform Setup / Groups / Group Symbol Settings / Margin Rates

Margin rates

Symbol: *
?
X

Common
Trade
Execution
Margin
Margin Rates
Swaps


Please specify margin rates for different types of trade operations.

Liquidity margin rate: Default

Currency margin rate: Default

	Market Order	Limit Order	Stop Order	Stop Limit Order
Initial margin				
Buy	1.0000000	Default	Default	Default
Sell	1.0000000	Default	Default	Default
Maintenance margin				
Buy	Default	Default	Default	Default
Sell	Default	Default	Default	Default

Use default rates settings

OK
Cancel
Help

This section contains margin rates for various order types. The rates are set for the initial and maintenance margin individually. If there is no rate for the maintenance margin (equal to zero), the initial margin value is used instead.

- **Use default margin rate settings** use margin rate parameters specified for the symbol in the [corresponding section](#).
- **Liquidity margin rate** in the [exchange risk management model](#), clients can use their own assets as collateral for open positions. The liquidity margin rate determines the amount of the current value of an asset for the specified financial instrument, which will be taken into account as collateral (accounted for in client's equity). If the value is set to 0, the instrument cannot be used as collateral.
- **Currency margin rate** rate change radius of the currency, a futures contract is denominated in, relative to the Russian ruble. The parameter is used when calculating security deposit for futures contracts ([Exchange FORTS Futures](#)) traded on Moscow Exchange ([calculating variation margin and security deposit using the current USD exchange rate](#) - in Russian). The values are sent by the Moscow Exchange when using the gateway [MetaTrader 5 to MOEX Derivatives](#).
- **Margin rates** this table contains margin rates for various order types. The rates are set for the initial and maintenance margin individually. If there is no rate for the maintenance margin (equal to zero), the initial margin value is used instead.
- **Market buy order** multiplier for calculating margin requirements for long positions relative to the [main amount of margin](#);
- **Market sell order** multiplier for calculating margin requirements for short positions relative to the main amount of margin;
- **Buy limit order** multiplier for calculating margin requirements for Buy Limit orders relative to the main amount of margin;
- **Sell limit order** multiplier for calculating margin requirements for Sell Limit orders relative to the main amount of margin;
- **Buy stop order** multiplier for calculating margin requirements for Buy Stop orders relative to the main amount of margin;
- **Sell stop order** multiplier for calculating margin requirements for Sell Stop orders relative to the main amount of margin;
- **Buy stop limit order** multiplier for calculating margin requirements for Buy Stop Limit orders relative to the main amount of margin;
- **Sell stop limit order** multiplier for calculating margin requirements for Sell Stop Limit orders relative to the main amount of margin.

To avoid overriding the coefficient value for a group, leave the value set to "Default". In this case, the corresponding value from the [symbol settings](#) will be used. The "Use Default Settings" command applies these default values to all parameters in this section.

- If the initial margin is specified, no margin calculations will be performed using the formulas specified in the ["Calculation"](#) field.
- If the maintenance margin size is not specified, the initial margin value will be used instead.

- If the exchange risk management model is used for a client group and margin rates of an instrument are set to 0, such an instrument is considered to be a non-marginable security. No margin calculations are performed for such instruments (no margin is charged).

The margin rates are described in details in a separate section.

MetaTrader 5 Trading Platform / Platform Setup / Groups / Group Symbol Settings / Swaps

Swaps

Symbol: *
?
×

Common
Trade
Execution
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Margin Rates
Swaps


Please set up parameters of charging swaps by symbols for the group.

Type:
Default

Long positions:
Default
Short positions:
Default

Days in year:
Default
Consider holidays:
Default

Swap multipliers:

Day of week	Multiplier
Sunday	
Monday	
Tuesday	
Wednesday	
Thursday	
Friday	
Saturday	

Default
Forex
All week
From symbol

USDTRY, US Dollar vs Turkish Lira

Use default swap settings

OK
Cancel
Help

Position rollovers for a symbol are set up on this tab:

- **Type** swap calculation method. To disable swap calculation, select the "Disabled" option.
- **Long positions** swap for Buy positions.
- **Short positions** swap for Sell positions.
- **Days in year** the number of days in a year to be used for swap percent calculation. Depending on the country and market in which the broker operates, as well as on the financial instrument type, different number of days in a year can be used when calculating annual percent. This parameter operates with such calculation specifics. The most common option of 360 days is used by default. You can change the value to 365 or 366, as well as specify a different value manually.
- **Swap multipliers** swap multiplier for each day of the week. This multiplier will be applied to the calculated swap value before charging. Specify 1 to charge the regular amount, 3 for triple swap or 0 to cancel swap. Conveniently manage the settings using the following commands:
 - **Forex** sets standard settings for Forex instruments: standard single swaps on weekdays and triple swap on Wednesday.
 - **All week** standard single swap seven days a week.
 - **From symbol** copies swap coefficients from the selected symbol.
 - **Consider holidays** if the option is enabled, the platform will check holiday configurations and adjust swaps accordingly. The day before the holiday, the swap is doubled. No swap is charged on the day of the holiday. The calculations still use the swap multipliers specified for the corresponding days.

The "Default" value can be set for each field. In this case, the basic symbol configuration from the relevant section will be used for the group.

For further details on how swaps work, please see "Symbols \ Swaps".

MetaTrader 5 Trading Platform / Platform Setup / Groups / Commission Settings

Commission Settings

Commission for processing trading operations is the main source of revenue for brokers. The MetaTrader 5 platform provides extensive opportunities for flexible commission settings: depending on deal size, total turnover, profitability, trade direction and other parameters.

To set up commission charges, select the appropriate tab in group settings.

Commissions


Adjustment of commissions parameters of the symbol.

Name: Standard commission

Description:

Symbol: *

Range: Volume

Turnover currency:

Charge: Instant

☒ Standard commission
☐ Agent commission
☐ Fee

Deal entry: All

Deal action: All

Deal profit: All

Deal reason: All

Add

Edit

Delete

From	To	Commission	Minimal	Maximal	Mode	Currency	Type
0	10	1	0.00	0.00	deposit ccy		per trade
11	50	0.9	0.00	0.00	deposit ccy		per trade
51	100	0.8	0.00	0.00	deposit ccy		per trade

OK

Cancel

Common Settings

The following parameters belong here:

- **Name** name of commissions.
- **Description** description of commissions. This text description will be used as a comment to commission deals, performed at client accounts. Comments are only for deals related to daily and monthly charging of standard commission. Please note that the maximum length of a comment to a deal is 31 characters. If a commission description exceeds this length, part of it will be cut off when copied to a deal comment.
- **Symbol** symbols, from trading operations of which commission will be charged. As soon as you click on this field, a standard list of symbol selection will appear. Symbols can be specified using masks. The maximum line length is 127 characters.

Commission Type

There are three types of commissions available in the platform: standard, agent and fee.

Standard commission

This type of commission is charged by a broker for the clients' trades;

Agent commission

This commission is charged by agents. The payment is made from the brokerage funds. Each account has the Agent account field, in which the number of the appropriate agent account can be specified. Commissions of all trades conducted on this trading account will be transferred to the Agent account in accordance with the commission settings.

Fee

The fee is similar to standard commission: it is also charged by a broker per trader's deals and has the same settings and the calculation principle. The differences are as follows:

- Fees can only have the "Instant" calculation mode.
- The amount is written to the separate "Fee" field, while the standard commission appears in the "Commission" field.

Fees allow splitting deal commissions for a separate accounting. For example, this option can be used when part of commission is charged by the brokerage company, and the other part is payable to an external system (exchange) the broker is integrated with.

- Standard commissions and fees are charged for the profit of the brokerage company from the accounts of clients that perform trade operations.
- Agent commissions are not charged from the accounts of clients that perform trade operations. They are charged to agent accounts by the brokerage company in the form of balance deals.

Charging Commissions

Commission can be charged immediately after each trade, or it can be accumulated during the trading day or month and then charged in one operation. Select the desired option in the "Charge" field:

- **Instant** commissions are charged instantly during execution of each deal (both when entering and exiting positions). The value of the standard commission charged instantly is displayed in the Commission field of deals. Agent commissions are charged as separate balance operations ("Agent commission" deals). In case of an instant charge, commission levels can be specified only as volume. No other options are available in the Charge field.
- **Daily** the commission amount is accumulated during a day in a special field of a client record. At the end of the day the accumulated amount is charged from the account with a separate balance operation (a deal of the Daily commission or Daily agent commission type). A more detailed description of the process of commission charging is given in the "Commission calculation" section.
- **Monthly** the commission amount is accumulated during a month in a special field of a client record. At the end of the month the accumulated amount is charged from the account with a separate balance operation (a deal of the Monthly commission or Monthly agent commission type). A more detailed description of the process of commission charging is given in the "Commission calculation" section.

- Only executed (filled) volume is used for calculation of turnover or single trade operations. Unfilled volume of orders is ignored for calculation of commissions.
- Commissions are charged for deals executed in both directions (both when opening/increasing volume of a position and when closing/partially closing a position).
- Commission charging should only be changed from daily/monthly to instant when there are no accumulated commissions on accounts. Otherwise, all accumulated commissions will be reset to zero at the end of the trading day, and will not be deducted from the accounts.

Filters by deal types

In the instant mode, you can additionally specify the parameters of deals from which the commission will be charged:

- **Deal entry** direction of deals, to which the commission shall apply. This option is only available for instant commissions. If the value is "In", commission will only be charged on entry deals; if "Out" commission will apply to exit

deals; "All" (default) all deals. For reversal deals with the in/out type, "In" means that commission is only charged on the volume of the newly opened position, "Out" means commission on the closed volume. The following rules apply to Close By deals:

- ◆ No commission is charged on Close By deals when "All" or "In" is selected, since the commission is charged on two original deals. For example, a commission of USD 1 is charged on each deal. When a trader performs an entry Buy 1.00 EURUSD deal and an entry Sell 1.00 EURUSD deal, a commission of 2 USD will be charged. When the 1.00 EURUSD position is closed by the Sell 1.00 EURUSD position, the client will not be charged additional commission.
- ◆ If "Out" is selected, commission on both Close By deals will be charged, and the total commission value is written to the main exit deal (in which profit/loss is specified). For example, a commission of USD 1 is charged on each deal. When a trader performs an entry Buy 1.00 EURUSD deal and an entry Sell 1.00 EURUSD deal, no commission will be charged. When the 1.00 EURUSD position is closed by the Sell 1.00 EURUSD position, a commission of USD 2 will be charged. A commission of 2 USD will be specified in the first 'out by' deal, and a zero commission will be specified in the second deal.
- **Deal action** type of deals, to which the commission shall apply: buy, sell, or both. This option is only available for instant commissions.
- **Deal profit** filter deals by their financial result. For example, you can charge a commission only on deals which generated a profit.
- **Deal reason** filter by deal execution reasons. For example, you can reduce commissions for deals performed by trading robots to encourage algorithmic trading. Alternatively, you can charge an additional commission for deals processed through a dealer.

Commission Levels

Commissions can be multilevel, i.e. vary depending on the deal volume, turnover, profit, etc. Select the required condition in the "Range parameter":

- **Volume** deal volume (number of lots). For example, if you set levels as 0 – 10 and 12 – 20, a 15-lot deal will be subject to the second level commission.
- **Turnover Money** turnover in money over the period specified in the Charge field (day or month). For example, the levels are set as 0 – 500, 501 – 1000, and commission is charged on a monthly basis. The first-level commission will be charged until the total cost of the client's operations exceeds 500 units. As soon as the money turnover exceeds the value of 500, commission for all deals per period is recalculated according to the second level.
By default, the money turnover is calculated in the group deposit currency: the price of each trade is calculated and converted into the deposit currency. For example, the price of Buy 1 lot EURUSD with a contract size of 100,000 is 100,000 EUR. If the client's deposit currency is USD, the position price is converted at EURUSD at the time of the transaction (in this case, it is the transaction price). If commission is to be charged on the turnover in some certain currency regardless of the group deposit, set it in the Currency field. Commission ranges will be based on the turnover in the specified currency. When calculating commissions, the deal price is also converted into that